SWAFL Desktop — Complete Technical Manual

Step-by-step use, every control and trigger, every layer and simulation, and the governing equations behind every output

Naziru Halilu

Version 1.1.0

Contents

About this manual **7**

1 Installing and starting SWAFL **8**
1.1 What you receive . . . 8
1.2 Route A — the Windows program, step by step . . . 8
1.3 Route C — from source . . . 8
1.4 Requirements . . . 9
1.5 First-run check . . . 9

2 The window: a guided tour **10**
2.1 Anatomy . . . 10
2.2 Menu bar reference . . . 11
2.3 The command palette (Ctrl+K) . . . 12
2.4 Layers panel . . . 12
2.5 Processing Toolbox . . . 13
2.6 Resizing and remembering the layout . . . 13
2.7 Reading the drawings . . . 13

3 Every control, value by value **14**
3.1 Scenario . . . 14
3.2 Soil and field . . . 14
3.3 Mulch . . . 15
3.4 Weather and climate . . . 15
3.5 Irrigation . . . 16
3.6 Nutrients . . . 16
3.7 Ready-made cases . . . 16
3.8 Design controls (inside the three design layers) . . . 17
3.9 Change-by-change: what each control moves . . . 17

4 Dates, years and the clock **18**
4.1 Why the date matters . . . 18
4.2 The date control . . . 18
4.3 The year strip and the Play bar . . . 19
4.4 The 14-day look-ahead . . . 19
4.5 Detail year and sensor record . . . 19

5 The two computing engines **20**
5.1 Engine 1 — the Daisy simulations and the meta-model . . . 20
5.1.1 The simulation campaign . . . 20
5.1.2 Plant-available water and depletion . . . 21
5.1.3 The meta-model . . . 22
5.1.4 What “20-year average at these settings” means . . . 23
5.2 Engine 2 — the daily bucket model . . . 23
5.2.1 Layout and state . . . 23
5.2.2 Crop calendar and canopy cover . . . 23

5.2.3 Rainfall generator . . . 24
5.2.4 Infiltration, redistribution and drainage . . . 24
5.2.5 Roots . . . 24
5.2.6 Transpiration, evaporation . . . 25
5.2.7 Irrigation rule and calibration . . . 25
5.2.8 Scaling to the annual model . . . 26
5.2.9 Evaporation under warming (used by the E1 layer) . . . 26
5.3 The design layers and their link to the engines . . . 26
5.4 Reproducibility . . . 27

6 Irrigation control layers **28**
6.1 Sensor and trigger (V1) . . . 28
6.1.1 What it shows . . . 28
6.1.2 How to use it, step by step . . . 28
6.1.3 How the numbers are generated . . . 29
6.1.4 Reading it . . . 30
6.2 Trigger law (N1) . . . 30
6.2.1 The idea . . . 30
6.2.2 How to use it, step by step . . . 30
6.2.3 Governing relations . . . 31
6.2.4 Reading the sensor value . . . 32

7 Field water balance layers **33**
7.1 Field balance (the home layer) . . . 33
7.1.1 What it shows . . . 33
7.1.2 The control bar under the drawing . . . 34
7.1.3 How each output is generated . . . 34
7.1.4 Step-by-step: checking a budget . . . 35
7.2 Soil, plant and air (S1) . . . 35
7.2.1 What it shows . . . 35
7.2.2 Governing equations . . . 36
7.2.3 Step-by-step . . . 37
7.3 Profile evolution (V2) . . . 37
7.3.1 What it shows . . . 37
7.3.2 How it is generated . . . 38
7.3.3 Step-by-step . . . 38

8 Soil structure and tillage layers **39**
8.1 Tillage and structure (V3) . . . 39
8.1.1 What it shows . . . 39
8.1.2 Pore index and consolidation . . . 39
8.1.3 How the section is generated . . . 39
8.1.4 Controls and steps . . . 40
8.2 Soil hydraulic drivers (V7) . . . 41
8.2.1 What it shows . . . 41
8.2.2 Generation . . . 42
8.2.3 Steps . . . 43
8.3 Root structure (R1) . . . 43
8.3.1 What it shows . . . 43
8.3.2 Generation . . . 43
8.3.3 Steps . . . 43

9 Mulch and evaporation layers **44**
9.1 Mulch against bare soil (V4) . . . 44
9.2 Mulch control (V6) . . . 46
9.3 Evaporation physics (E1) . . . 46

9.4 Fate of saved water (N2) . . . 49
9.5 Regime map (N3) . . . 49

10 Feedback loops (V5) 51
10.1 What it shows . . . 51
10.2 How the seven effects are computed . . . 52
10.3 The sentence under the picture . . . 53
10.4 Steps . . . 53

11 Surface irrigation design (S2) 54
11.1 Inputs, step by step . . . 55
11.2 Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov) . . . 57
11.3 Inlet flow area (Manning) . . . 58
11.4 Advance (volume balance) . . . 58
11.5 Cut-off, opportunity time and the infiltrated depth . . . 59
11.6 Volumes, efficiencies and uniformity . . . 60
11.7 Hydrograph . . . 61
11.8 Cross-section and intake curve . . . 62
11.9 Best inflow (“Your soil” mode) . . . 62
11.10 The design table . . . 62
11.11 Sensitivity (tornado chart) . . . 63
11.12 Season and farm . . . 64
11.13 Irrigation schedule and “When to irrigate” . . . 64

12 Drip irrigation design (S3) 66
12.1 Inputs, step by step . . . 67
12.2 Lateral hydraulics: head loss and flow along the lateral . . . 68
12.3 The wetted bulb . . . 70
12.4 Spread between emitters and the efficiencies . . . 73
12.5 Layout and infrastructure . . . 73
12.6 Season and farm . . . 75
12.7 The design table . . . 76
12.8 Sensitivity . . . 77
12.9 How to use the layer . . . 77

13 Sprinkler irrigation design (S4) 79
13.1 Inputs, step by step . . . 80
13.2 Nozzle discharge and wetted radius . . . 81
13.3 The spray pattern and the overlap . . . 82
13.4 Wind drift and evaporation loss . . . 82
13.5 The soil intake, ponding and runoff . . . 83
13.6 Efficiencies and run time . . . 84
13.7 Maximum allowable application rate . . . 85
13.8 The ten textbook checks . . . 85
13.9 Kinds of sprinkler system . . . 87
13.10 Layout and infrastructure . . . 88
13.11 The design table and sensitivity . . . 88
13.12 Season and farm . . . 89
13.13 How to use the layer . . . 89

14 The Dashboard: how every number is produced 90
14.1 Results . . . 92
14.1.1 The pinned baseline . . . 92
14.2 Design . . . 93
14.3 Charts . . . 95
14.4 Messages . . . 97

15 Worked workflows **98**
15.1 Workflow 1: How much does straw mulch save? . . . 98
15.2 Workflow 2: When should I irrigate? . . . 98
15.3 Workflow 3: Design a border strip for a loam . . . 98
15.4 Workflow 4: Choose and size a drip system . . . 99
15.5 Workflow 5: Check a sprinkler against the course rules . . . 99
15.6 Workflow 6: A warmer, drier future . . . 99
15.7 Workflow 7: Save, share and reproduce . . . 99

16 Exports, files and troubleshooting **100**
16.1 What can be saved . . . 100
16.2 Troubleshooting . . . 100

17 Validity, limits and honest use **102**

18 Screenshot atlas: every screen with its governing equations **103**
18.1 The window and its controls . . . 103
18.1.1 The welcome dialog . . . 103
18.1.2 The workbench . . . 105
18.1.3 The File menu . . . 107
18.1.4 The Edit menu . . . 108
18.1.5 The View menu . . . 109
18.1.6 The Tools menu . . . 110
18.1.7 The Refresh menu . . . 111
18.1.8 The Review menu . . . 112
18.1.9 The Run menu . . . 113
18.1.10 The Help menu . . . 114
18.1.11 The command palette (Ctrl+K) . . . 115
18.1.12 The Processing Toolbox . . . 116
18.1.13 The toolbar . . . 117
18.1.14 The status bar . . . 118
18.1.15 The Dashboard, Results tab . . . 119
18.1.16 The Dashboard, Design tab . . . 121
18.1.17 The Dashboard, Charts tab . . . 123
18.1.18 The Dashboard, Messages tab . . . 124
18.1.19 The Properties panel . . . 125
18.2 S2 Surface irrigation design . . . 127
18.2.1 The whole layer: inputs and overview . . . 127
18.2.2 Layout in plan: the advance front, shaded by the depth already infiltrated 132
18.2.3 Longitudinal section: water running over your soil and soaking in . . . 135
18.2.4 Hydrograph: inflow, infiltration and runoff through time . . . 139
18.2.5 Cross-section at the inlet: the wetted bulb in your soil, and the intake curve . . . 142
18.2.6 Advance and opportunity time . . . 146
18.2.7 Infiltrated depth along the field against the required depth Zn . . . 150
18.2.8 Performance indices and the water balance of one irrigation . . . 153
18.2.9 Design table: iterations toward an application efficiency of 85 % or more . . . 156
18.2.10 What the efficiency is most sensitive to . . . 159
18.2.11 What this means for the season and the farm . . . 161
18.2.12 Irrigation schedule: when to irrigate with this design . . . 164
18.2.13 Intervals, efficiency and the advice in short . . . 166
18.3 S3 Drip irrigation design . . . 169
18.3.1 The whole layer: inputs and overview . . . 169
18.3.2 Soil moisture around the emitter: the driest, the average and the wettest emitter . . . 174

18.3.3 Uniformity along the lateral: pressure falls with friction, so far emitters deliver less . . . 177
18.3.4 How the wetted bulb grows with the run time . . . 181
18.3.5 Result for this design . . . 184
18.3.6 Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals . . . 187
18.3.7 Design table: iterations toward an application efficiency of 85 % or more . . . 190
18.3.8 What the efficiency is most sensitive to . . . 192
18.3.9 What this means for the season and the farm . . . 194
18.3.10 Irrigation schedule: when to irrigate with this design . . . 197
18.3.11 Intervals, efficiency and the advice in short . . . 200
18.4 S4 Sprinkler design and pipes . . . 203
18.4.1 The whole layer: inputs and overview . . . 203
18.4.2 Layout in plan: depth infiltrated so far, from the overlapping spray patterns . . . 207
18.4.3 Along the lateral: depth applied, depth infiltrated and where it ends up in the soil . . . 210
18.4.4 Soil intake against the application rate: when the soil cannot keep up, water runs off . . . 213
18.4.5 Result for this design . . . 216
18.4.6 Does your design pass? The course checks . . . 219
18.4.7 Kinds of sprinkler system . . . 222
18.4.8 The sprinkler: parts and pressure class . . . 224
18.4.9 The jet: trajectory angle, throw radius and droplet size . . . 227
18.4.10 The water pattern of one sprinkler, and how overlap evens it out . . . 230
18.4.11 Layouts: rectangular, square and triangular spacing for this sprinkler 233
18.4.12 What wind and soil allow . . . 236
18.4.13 Catch-can test: measuring the uniformity the way it is done in the field 238
18.4.14 Soil water as a reservoir: interval, sets per day and the flow the field needs . . . 241
18.4.15 Pipes, pump and the hydraulic grade line (the course sizing) . . . 244
18.4.16 Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals . . . 247
18.4.17 Design table: iterations toward an application efficiency of 85 % or more . . . 250
18.4.18 What the efficiency is most sensitive to . . . 252
18.4.19 What this means for the season and the farm . . . 254
18.4.20 Irrigation schedule: when to irrigate with this design . . . 257
18.4.21 Intervals, efficiency and the advice in short . . . 259
18.5 Field balance . . . 261
18.5.1 The whole layer, part by part . . . 267
18.5.2 The layer through its playback . . . 269
18.5.3 The controls of the layer, one state at a time . . . 271
18.6 Soil, plant and air (S1) . . . 275
18.6.1 The whole layer, part by part . . . 278
18.6.2 The layer through its playback . . . 280
18.6.3 The controls of the layer, one state at a time . . . 283
18.7 Profile evolution (V2) . . . 296
18.7.1 The whole layer, part by part . . . 298
18.7.2 The layer through its playback . . . 299
18.7.3 The controls of the layer, one state at a time . . . 302
18.8 Sensor and trigger (V1) . . . 305
18.8.1 The whole layer, part by part . . . 309
18.8.2 The layer through its playback . . . 311

18.8.3 The controls of the layer, one state at a time . . . 313
18.9 Trigger law (N1) . . . 316
18.9.1 The whole layer, part by part . . . 318
18.9.2 The controls of the layer, one state at a time . . . 320
18.10 Tillage and structure (V3) . . . 323
18.10.1 The whole layer, part by part . . . 325
18.10.2 The layer through its playback . . . 327
18.10.3 The controls of the layer, one state at a time . . . 329
18.11 Soil hydraulic drivers (V7) . . . 336
18.11.1 The whole layer, part by part . . . 341
18.11.2 The controls of the layer, one state at a time . . . 344
18.12 Root structure (R1) . . . 349
18.12.1 The whole layer, part by part . . . 351
18.12.2 The controls of the layer, one state at a time . . . 356
18.13 Mulch against bare soil (V4) . . . 367
18.13.1 The whole layer, part by part . . . 370
18.13.2 The controls of the layer, one state at a time . . . 371
18.14 Mulch control (V6) . . . 378
18.14.1 The whole layer, part by part . . . 380
18.14.2 The controls of the layer, one state at a time . . . 382
18.15 Evaporation physics (E1) . . . 385
18.15.1 The whole layer, part by part . . . 388
18.15.2 The layer through its playback . . . 392
18.15.3 The controls of the layer, one state at a time . . . 398
18.16 Fate of saved water (N2) . . . 406
18.16.1 The whole layer, part by part . . . 408
18.17 Regime map (N3) . . . 411
18.17.1 The whole layer, part by part . . . 413
18.18 Feedback loops (V5) . . . 415
18.18.1 The whole layer, part by part . . . 419
18.18.2 The layer through its playback . . . 421
18.19 Reverse index: where each equation appears on screen . . . 424

19 Release 1.1: design runs, weather, virtual farm, crops and seasonal records 430
19.1 The layers . . . 430
19.2 Running an irrigation design . . . 430
19.3 Weather data . . . 431
19.4 Virtual farm simulation . . . 431
19.5 Crops and C1 Crop water requirement . . . 432
19.6 Daily records that follow the settings (V1, V2) . . . 434
19.7 Seasonal records: Daisy data (Y1, W1, N4) . . . 434
19.8 Consolidation rate . . . 435
19.9 Workbench functions . . . 435
19.10 Correction of the surface layer (release 1.1.0) . . . 435

Symbols and units 437

Control value lists 439

Simulation data sets 440

Provenance ledger of the numbered equations 441

References 447

About this manual

This manual documents **SWAFL Desktop**, the Soil–Water–Atmosphere–Field–Loop decision-support workbench that accompanies the repository *Coupled Water Fluxes, Pore-Scale Dynamics, Precision Irrigation and Mulching Feedback Loops*. It is written for three readers: a farmer or adviser who wants to use the screens, a student who wants to know what each picture means, and a scientist who wants to know exactly which equation produced each number.

How the manual is organised.

1. Chapters 1–3 install the program and explain the window and every control.
2. Chapters 4–5 explain the clock (dates) and the two computing engines that sit under every layer.
3. Chapters 6–10 document each *layer* of the Layers panel: what it designs or shows, how the picture is generated, and the governing equations.
4. Chapters 11–13 document the three irrigation design layers (surface, drip, sprinkler) in full, including the irrigation schedule and the "When to irrigate" message.
5. Chapters 14–16 document the dashboard, step-by-step workflows, exports and troubleshooting.
6. Chapter 17 states the validity limits.
7. Chapter 18 is a screenshot atlas. It shows every screen of the workbench (all layers, every section of the three design layers, the menus, the toolbox, the Properties panel and the Dashboard) from the program itself, and prints under each screenshot every equation that the screen draws on, with its number, source chapter and provenance. A reverse index gives, for every numbered equation, the screens that use it.
8. The appendices give symbols, control value lists, data sets, the provenance ledger and references.

A note on honesty of the equations. SWAFL mixes three kinds of computation, and the manual labels them so that nothing is over-claimed. Every numbered equation carries its label directly beneath it, and the provenance ledger in the appendices lists them all:

Label	Meaning
[Literature]	A published relation implemented as written, cited to its source.
[Data-driven]	Obtained by interpolating or scaling the 10,190 Daisy simulations (Appendix C of the repository), not computed from a closed-form law.
[Engineered]	A parameterisation chosen for the workbench (for drawing, for speed, or because no closed form exists). Stated explicitly with its constants so that it can be audited, but not presented as a physical law.

Units. Unless stated, depths are in mm, lengths in m, suction in cm of water (1 cm $\approx$ 0.0980665 kPa), flows in L s^{-1} or m^3 min^{-1} as marked on the screen, times in min or h, and areas in ha.

Chapter 1

Installing and starting SWAFL

1.1 What you receive

SWAFL Desktop is a self-contained Electron program. The simulation engine is one HTML file (`app/engine/engine.html`) that the program serves to itself on a local port, so **no internet connection is needed once it is installed**. Three ways of running it are provided.

Route	Who it suits	Steps
A. Windows program	Most users	Download `SWAFL.exe` from the repository's *Releases* page and double-click it.
B. Online workbench	No install at all	Open the "Online workbench" link in the repository README (it points to `Software/SWAFL_Desktop/app/index.html` on GitHub Pages) in any modern browser.
C. From source	Developers	`npm install`, then `npm start` (desktop) or `npm run web` (browser).

1.2 Route A — the Windows program, step by step

1. Open the repository's **Releases** page and choose the latest release (v1.0.0).
2. Under *Assets*, click **SWAFL.exe**. Your browser downloads a single file of about 180 MB.
3. Double-click **SWAFL.exe**. If Windows SmartScreen shows "Windows protected your PC", click **More info** and then **Run anyway**. (The program is not code-signed; this warning is expected for unsigned programs.)
4. The splash screen appears for a few seconds while the engine loads, then the *Welcome to SWAFL* window opens. Click **Close**.
5. The workbench opens on the **Field balance** layer for today's date.

1.3 Route C — from source

```
git clone https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops.git
cd Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/Software/SWAFL_Desktop
npm install
npm start          # desktop window
npm run web        # the same program in a browser tab on http://127.0.0.1:5199
```

Alternatively, double-click `START_SWAFL.bat` (desktop) or `START_SWAFL_WEB.bat` (browser). `BUILD_AND_INSTALL_SWAFL.bat` builds the installer, and `get-app.js` builds, copies the program to the Desktop and starts it.

1.4 Requirements

- Windows 10 or 11 (64-bit) for `SWAFL.exe`; any current Chrome, Edge, Firefox or Safari for the online workbench.
- A screen of at least 1280 × 720 px. The layout is designed for 1600 × 900 and scales down by hiding panels you do not need (Section 2.5).
- About 400 MB of free memory. The 20-year daily simulation is computed in the program; no server is involved.

1.5 First-run check

After start-up, verify three things: (i) the status bar at the bottom reads *Ready*; (ii) the dashboard on the right shows *On* followed by today's date with numbers in the tiles; (iii) clicking **Surface irrigation design** in the Layers panel opens a drawing with a border strip. If the drawing area stays blank, see Chapter 16.

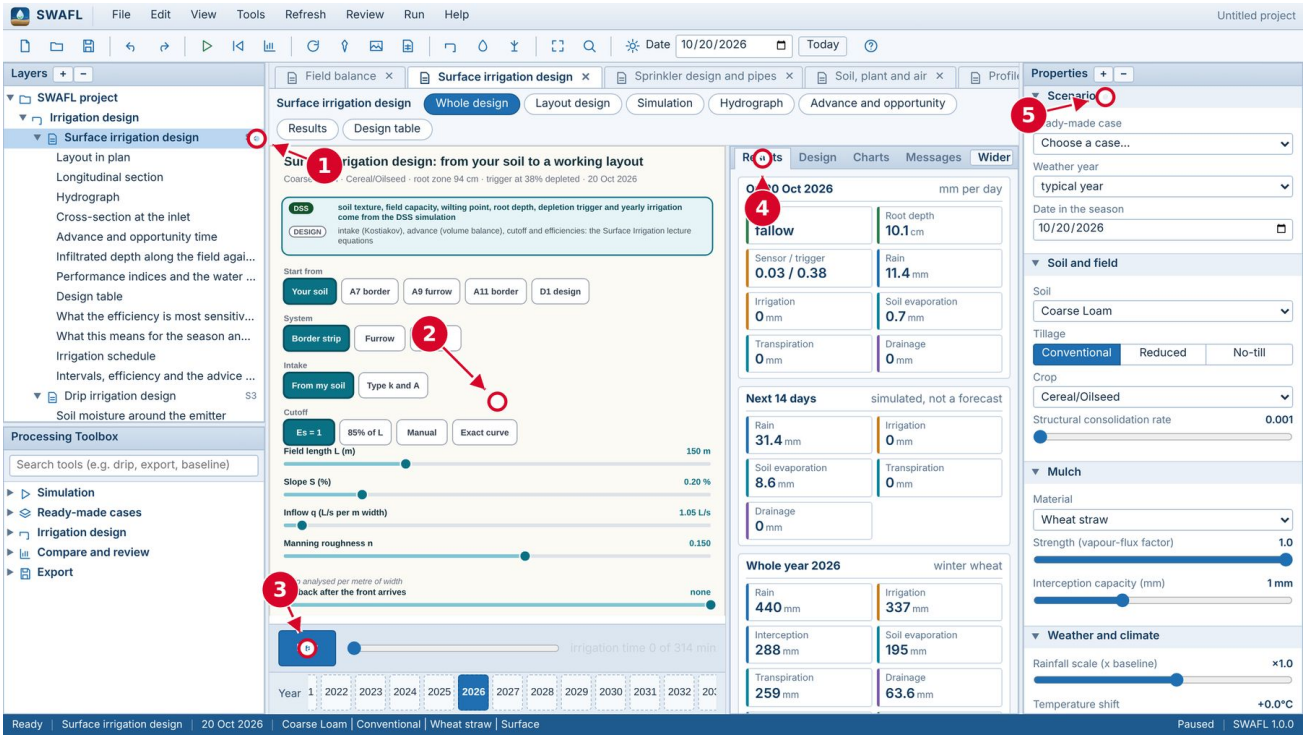


Figure 1.1: The workbench on first start (Surface irrigation design layer, today's date). Left: Layers and Processing Toolbox. Centre: the drawing. Right of the drawing: Results dashboard. Far right: Properties. The numbered arrows point to 1 the layer in the Layers tree, 2 the drawing area, 3 the Play button of the playback bar, 4 the Results tab of the Dashboard and 5 the Scenario section of Properties.

Chapter 2

The window: a guided tour

2.1 Anatomy

The window has seven regions, named here as they are named in the program (Figure 2.1).

1. **Menu bar** — File, Edit, View, Tools, Refresh, Review, Run, Help.
2. **Toolbar** — icon buttons for New, Open, Save, Undo, Redo, Play/Pause, Restart, 20-year simulation, the three design layers, fit-to-window, search, and the **Date** control with a **Today** button.
3. **Layers** panel (top left) — the project tree. Each layer is one drawing of the model.
4. **Processing Toolbox** (bottom left) — searchable list of ready-to-run actions.
5. **Drawing** (centre) — the active layer, with its tab strip above and, for the design layers, a sub-section bar (*Whole design, Layout design, Simulation, …*). The **year strip** and the **Play** bar sit under the drawing.
6. **Dashboard** (right of the drawing) — four tabs: *Results, Design, Charts, Messages*.
7. **Properties** (far right) — every model input, in collapsible groups.

The **status bar** shows, from left to right: state (*Ready*), the active layer, the working date, the soil, tillage, crop, mulch and irrigation method, and *Paused* or *Running*.

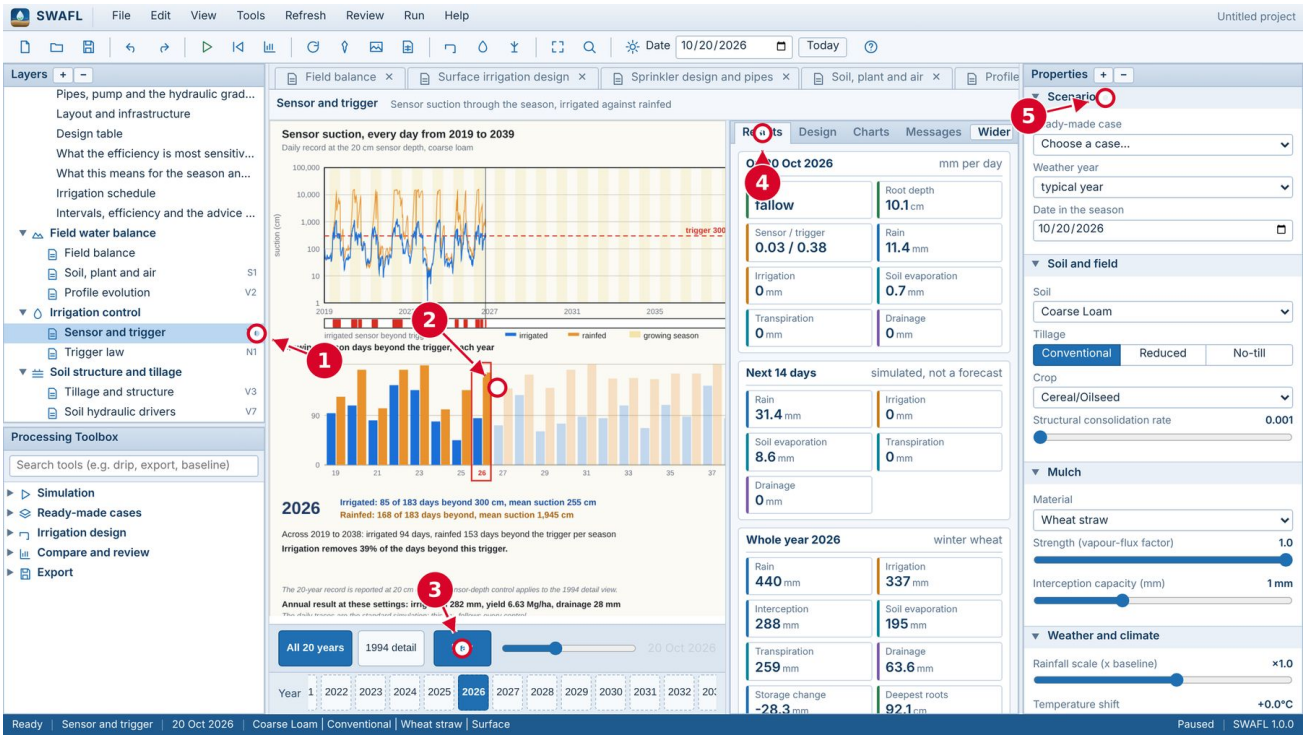


Figure 2.1: Figure 2.1. The window with the Sensor and trigger layer open. The numbered arrows point to 1 the layer in the Layers tree, 2 the drawing area, 3 the Play button of the playback bar, 4 the Results tab of the Dashboard and 5 the Scenario section of Properties.

2.2 Menu bar reference

Menu	Item	Shortcut	Action
File	New project	Ctrl+N	Resets every control to the defaults (Appendix B).
	Open project…	Ctrl+O	Loads a `.swafl` file.
	Save project…	Ctrl+S	Saves a `.swafl` file: JSON holding the application name and version, the open layer, every control, the weather year and the date.
	Export ▸		Picture of the window (PNG, Ctrl+E); yearly results (CSV); design results (CSV).
	Save full report as PDF…	Ctrl+Shift+P	Writes the complete report of all layers to a PDF file at a place you choose.
	Print / save as PDF…	Ctrl+P	Prints the active layer through the system print dialog.
	Exit	Ctrl+Q	Closes the program.
Edit	Undo / Redo	Ctrl+Z / Ctrl+Y	Steps through the history of control changes.
	Reset all settings		Same as *New project* for the controls only.
	Copy settings as text		Puts a text list of all control values on the clipboard (for reports).

Menu	Item	Shortcut	Action
	Pin results as baseline	Ctrl+B	Freezes the current 20-year averages for comparison (Section 14.4).
	Clear baseline		Removes the pinned baseline.
	Find a tool…	Ctrl+F	Moves the cursor to the Toolbox search box.
View	Layers and Processing Toolbox / Properties / Dashboard		Show or hide each dock.
	Fit drawing to window / Drawing at full width (scroll)		Two scaling modes for the drawing.
	Expand all layers / Collapse all layers		Tree display.
	Full screen	F11	
Tools	Ready-made cases ▸		The seven cases of Section 3.8.
	Irrigation design ▸		Opens the surface, drip or sprinkler layers.
	Irrigation method ▸ / Mulch material ▸		Sets the control directly.
	Run 20-year season simulation		Section 5.2.
Refresh	Refresh display (F6); Reload simulation engine; Clear message log		
Review	Results dashboard / Design results / Yearly charts / Messages		Switches the Dashboard tab.
Run	Run / pause the season (F5); Restart from day 1 (Shift+F5); 20-year season; Optimise surface layout; Ready-made cases		
Help	Manual (PDF) (F1); How to use SWAFL; Troubleshooting; Keyboard shortcuts; SWAFL on GitHub; Full documentation; Report a problem; About		

2.3 The command palette (Ctrl+K)

Press **Ctrl+K** anywhere to open a search box over *every* layer, tool and setting. Type part of a name (for example `drip`, `trigger`, `mulch`) and press **Enter** to run the first match, or use the arrow keys to choose. This is the fastest way to reach a layer without opening the tree.

2.4 Layers panel

The tree has the project root *SWAFL project* and six groups. Click a group to expand it, click a layer to open it as a tab, and click the small triangle beside a design layer to list its sub-sections, which jump the drawing to that section.

Group	Layers (ID)
Irrigation design	Surface irrigation design (`sid`, S2); Drip irrigation design (`drip`, S3); Sprinkler design and pipes (`spr`, S4)

Group	Layers (ID)
Field water balance	Field balance (`field`); Soil, plant and air (`spac`, S1); Profile evolution (`v2`, V2)
Irrigation control	Sensor and trigger (`v1`, V1); Trigger law (`n1`, N1)
Soil structure and tillage	Tillage and structure (`v3`, V3); Soil hydraulic drivers (`v7`, V7); Root structure (`r1`, R1)
Mulch and evaporation	Mulch against bare soil (`v4`, V4); Mulch control (`v6`, V6); Evaporation physics (`e1`, E1); Fate of saved water (`n2`, N2); Regime map (`n3`, N3)
Feedback loops	Feedback loops (`v5`, V5)

Right-click a layer for *Open layer*, *Go to top*, *Close tab*. Open layers stay as tabs above the drawing; closing a tab does not lose any setting, because the controls belong to the project, not to the layer. A layer can also be opened from a link of the form `index.html#open=<ID>`, for example `#open=drip`.

2.5 Processing Toolbox

Double-click (or select and press Enter) to run an action. Typing in the search box filters all tools by name and by hidden keywords.

Group	Tools
Simulation	Play/pause the season; Restart from day 1; Run 20-year season simulation; Reset all settings
Ready-made cases	The seven cases of Section 3.8
Irrigation design	Open surface / drip / sprinkler design; **Optimise surface layout**; Use overhead sprinkler / surface / drip 20–30 cm method
Compare and review	Pin current results as baseline; Clear baseline; Show results dashboard / design results / yearly charts
Export	PNG of the window; yearly results CSV; design results CSV; save project (.swafl); print / PDF

2.6 Resizing and remembering the layout

Drag the thin splitters between the docks to resize them. The program remembers the panel widths between sessions. The *Wider* button above the dashboard widens it; **View ▸ Drawing at full width** lets the drawing keep its design width and scroll, which is useful for the long design layers.

2.7 Reading the drawings

- **Hot spots.** Many drawings are clickable. Hovering shows a tooltip; clicking applies a setting (for example, clicking a year bar jumps to that year, clicking a soil on a chart selects that soil).
- **Sliders inside drawings.** In the design layers every slider is part of the picture and moves the same state as the Properties panel.
- **Colours.** Water-related blues, soil browns, crop greens. Black is not used; text is dark blue-grey on cream paper.

Chapter 3

Every control, value by value

All model inputs live in the **Properties** panel (far right). Each group can be collapsed; **+** and **−** in the panel header expand or collapse all. Every slider is an *index* into a list of discrete values, because every value in the lists corresponds to a set of Daisy simulations (Chapter 5). The number shown beside the slider is the value the index stands for.

3.1 Scenario

Control	Values	Default	What it does
Ready-made case	seven cases (Section 3.8)	—	Replaces many controls at once, as one undoable step.
Weather year	*typical year*, *dry year*, *wet year*, 1980 ... 1999	typical	Chooses which of the twenty recorded growing seasons the *detail* layers (V1, V2 “1994 detail”, Field balance year animation) draw. The 20-year totals do not change.
Date in the season	any calendar date	today	The working date. Sets the day displayed in the drawings, the Dashboard “On ...” tile, and the first day of the irrigation schedule (Chapter 4).

3.2 Soil and field

Control	Values	Default	Effect
Soil	Sand, Loamy Sand, Sandy Loam, Coarse Sand, Coarse Loam, Loam, Silt Loam, Sandy Clay Loam, Clay Loam, Silty Clay Loam (S01–S10)	Coarse Loam	Selects the soil’s measured texture, its van Genuchten parameters (α, n) (Table 5.3), the baseline Daisy run, plant-available water per cm (`AWC`), and the trigger anchor. Re-computes every layer.
Tillage	Conventional / Reduced / No-till	Conventional	Sets the initial pore index p_0 (1.00, 0.72, 0.42) and therefore the starting structure of the topsoil (Section 8.1).

Control	Values	Default	Effect
Crop	Cereal/Oilseed, Legume, Vegetable	Cereal/Oilseed	Chooses the crop calendar, the root depth limit and the yield scaling factor (1.0, 0.55, 2.3). Only the cereal rotation is a simulated crop; the other two are *illustrative scalings* and are labelled so (Section 5.1).
Structural consolidation rate	0.001, 0.0025, 0.005, 0.01, 0.02 d^{-1}	0.001	The rate τ_c at which conventionally tilled pores close (Eq. 8.1). Does not affect no-till.

3.3 Mulch

Control	Values	Default	Effect
Material	Rice husk, Wheat straw, Wood chips, Sawdust, Grass clippings, Black plastic film	Wheat straw	Sets *both* sliders below to typical values for the material and the drawn layer thickness (Table 3.1).
Strength (vapour-flux factor) φ	0.1, 0.2, ... 1.0	**1.0 (no mulch)**	The fraction of the bare-soil vapour flux that passes the mulch. 1.0 means bare; 0.1 means the flux is cut by 90 %.
Interception capacity	0.5, 1, 2, 4 mm	1 mm	Water the mulch can hold and lose before it reaches the soil.

Table 3.1. Mulch materials as defined in the engine.

Material	φ	capacity (mm)	drawn thickness (relative)
Rice husk	0.5	2	1.0
Wheat straw	0.4	4	1.35
Wood chips	0.3	2	1.7
Sawdust	0.6	1	0.9
Grass clippings	0.7	4	1.2
Black plastic film	0.1	0.5	0.25

The material buttons only *set* the sliders. Moving a slider afterwards keeps the material for drawing but uses the slider value for the calculation. Any $\varphi \geq 0.95$ is treated as "no mulch" in the drawings.

3.4 Weather and climate

Control	Values	Default	Effect
Rainfall scale	$×$0.5, 0.6, ... 1.4	$×$1.0	Multiplies the baseline annual rainfall (`rain_base`) and selects the matching Daisy runs.
Temperature shift	−1.0, −0.5, 0, +0.5, ... +3.5 °C	0 °C	Selects the warming runs (E11) and changes the evaporative demand in the engine (Eq. 5.26).

3.5 Irrigation

Control	Values	Default	Effect
Method	Surface; Overhead; Drip 10–20 cm; Drip 20–30 cm; Drip 40–50 cm; Drip 50–60 cm	Surface	Selects the irrigation run set (E4) and, in the day-by-day engine, where each dose is placed (Section 5.2).
Trigger (sensor suction)	−50, −75, −100, −150, −200, −250, **−300**, −350, −400, −500, −600, −700, −800, −1000, −1200, −1500, −2000, −3000, −5000, −10000 cm	−300 cm	Irrigation starts when the soil at the sensor depth is as dry as this suction. A *less negative* number means wetter, more frequent irrigation.
Sensor depth	5, 10, 15, … 100 cm (20 values)	20 cm	Depth of the simulated tensiometer that is compared with the trigger.

Trigger mode (Trigger law layer). The slider above applies one suction to every soil (*common* mode). The **Trigger law** layer offers an *own* mode, in which each soil gets the suction that corresponds to a soil-independent depletion fraction f^* (default 0.45): see Chapter 6.

3.6 Nutrients

Control	Values	Default	Effect
Nitrogen rate	10, 25, 40, 55, 70, 85, **100**, 115, 130, 150 % of the standard rate	100 %	Selects the E13 runs; changes yield and nitrogen leaching and, slightly, the maximum rooting depth (Eq. 5.18).

3.7 Ready-made cases

#	Case	Changes
0	Default: Coarse Loam, bare, normal climate	all defaults
1	Dry year on Sand, no mulch	Sand; rain $×$0.6; +1 °C
2	Wheat straw mulch on Loam, drip 10 to 20 cm	Loam; straw; drip 10–20
3	Wet year on Silty Clay Loam, no-till	Silty Clay Loam; no-till; rain $×$1.3
4	Hot year (+3 °C), vegetables on Sandy Loam	Sandy Loam; vegetable; +3 °C
5	Plastic film, deep drip 40 to 50 cm, Clay Loam	Clay Loam; plastic; drip 40–50
6	Low nitrogen (40 %), legume on Silt Loam	Silt Loam; legume; N 40 %

Every case first resets the controls to their defaults and then applies its changes, so cases never accumulate.

3.8 Design controls (inside the three design layers)

The surface, drip and sprinkler layers each carry their own sliders and chips (field length, slope, inflow, emitter spacing, nozzle size, pipe lengths, and so on). They belong to those layers and are documented in Chapters 11–13 with ranges and defaults. They are connected to the Properties controls above only through the *soil*, the *crop*, the *trigger*, the *sensor depth* and the *date*; changing any of those re-designs the layout automatically.

3.9 Change-by-change: what each control moves

The table lists, for each control, which outputs respond. "Run set" means that the response is read from the Daisy simulations (Chapter 5); "engine" means that the day-by-day bucket model of Section 5.3 responds as well.

Control changed	Annual outputs (Dashboard, 20-year means)	Day-by-day layers and schedules	Design layers
Soil	all (baseline run for that soil)	AWC, (α, n), trigger anchor $\Rightarrow$ dates, depths	Intake (k, A) via Saxton–Rawls and Green–Ampt; Z_n; sprinkler maximum rate row
Tillage, consolidation	drainage, evaporation, yield (E5 run set)	pore index in V3/E1/R1	S_d (surface storage) for sprinklers
Crop	yield $\times$ factor	crop calendar, $R_{\max}$, cover	peak ET_c, Z_n
Mulch φ, capacity	evaporation, irrigation, drainage, yield	evaporation term, depth of the wet-front	interception loss for sprinklers
Rain scale	all	rain amount per event, canopy catch I_c	—
Temperature	evaporation, transpiration, irrigation, yield	E_0 and seasonal weight	T_{air} in wind drift (Eq. 13.9)
Method	irrigation, drainage, yield (E4 run set)	dose placement; drip raises root density near the emitter	—
Trigger	irrigation, yield, drainage (E2 run set)	**dates of irrigation**	Z_n (as $f_T \cdot \text{AWC} \cdot R_{\max}$)
Sensor depth	irrigation, yield (E3 run set)	**dates of irrigation**	—
Nitrogen	yield, N leaching (E13 run set)	$R_{\max}$	—
Date	none	**which day is shown; first date in the schedule**	the schedule table

Chapter 4

Dates, years and the clock

4.1 Why the date matters

SWAFL does not use a weather forecast. It uses **twenty simulated weather years** (the 1980–1999 record at the study site, Chapter 5), arranged on a 365-day calendar without leap days, and it maps *any calendar date* you choose onto one of those years. The working date therefore decides (i) which day is drawn, (ii) which simulated year acts as the *analogue* for that date, and (iii) the first line of every irrigation schedule.

4.2 The date control

The **Date** box on the toolbar (and the *Date in the season* control in Scenario) accepts any date. **Today** sets the computer's date. When you change the date:

1. If the year is **1980–1999**, that exact recorded year is the analogue, and the weather-year selector follows it.
2. Otherwise (2000–2100 or earlier than 1980) the analogue is chosen by the **Weather year** control: the *typical* year (the median of the twenty by growing-season effective rain), the *driest*, or the *wettest*.
3. The engine sets the field clock to day-of-year d_{oy} of the analogue year, so that a date such as 22 Oct 2029 shows the crop state, soil water and irrigation events the analogue year had on 22 October.

Formally, with Y_0 the calendar year of the chosen date, a the analogue year index $(0, \dots, 19)$ and k the model day number $(0, \dots, 7299)$ of the twenty-year run,

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \tag{4.1}$$

Provenance (Eq. 4.1): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

and the calendar label of the first plotted year is $\text{CAL}_0 = Y_0 - a$. The choice of analogue year is

$$a = \begin{cases} \text{argsort}_y\,(P_{\text{thr},y})_{[10]} & \text{typical (median, 11th smallest of 20)} \\ \arg\min_y P_{\text{thr},y} & \text{dry} \\ \arg\max_y P_{\text{thr},y} & \text{wet} \\ Y_0 - 1980 & 1980 \le Y_0 \le 1999 \end{cases} \tag{4.2}$$

Provenance (Eq. 4.2): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

where $P_{\mathrm{thr},y}$ is the sum of throughfall (rain reaching the soil) in simulated year y (Section 5.2).

What this means for you. A date in 2029 is *not* a forecast for 2029. It shows how this soil, crop, trigger and method would behave on that day of the year in a typical (or dry, or wet) year. Chapter 17 repeats this as a validity limit.

4.3 The year strip and the Play bar

Under the drawing, the **year strip** lists the 20 plotted calendar years. Click a year to jump to the same day of that analogue. The **Play** button (F5) animates the field clock: each real second advances 15 model days times the speed setting, and the clock *dwells* for about $3.2/\sqrt{\mathrm{speed}}$ s on every day on which an irrigation, a wetting front of at least 20 cm, or drainage occurs, so you can watch each event. The clock wraps at the end of the twenty years. **Restart** (Shift+F5) returns to day 1 of the working year.

4.4 The 14-day look-ahead

The “Next 14 days” block on the Results tab sums the same simulated daily fluxes from the working date forward. It is labelled *simulated, not a forecast* because the rain on those days is the analogue year’s rain.

4.5 Detail year and sensor record

Layers V1 and V2 can show either all twenty years (*All 20 years*) or one year in detail (*1994 detail*). The detail year is the Weather-year control. The suction traces shown there are the study’s **daily suction at 20 cm for 1980–1999**, stored for the Coarse Loam and the Coarse Sand; every other soil uses the nearer of the two, and the drawing says so (*nearest simulated soil to …*). The growing season of these records is the 183 days starting 1 April.

Chapter 5

The two computing engines

Every number in SWAFL comes from one of two engines, which are tied together so that they agree on the twenty-year means.

- **Engine 1, the annual meta-model.** It reads 20-year-mean results of the 10,190 Daisy simulations and scales them for the controls. It produces the Dashboard "20-year averages" and every bar, line and arrow in the annual layers. **[Data-driven]**
- **Engine 2, the daily bucket model.** A 40-layer soil-water model run for twenty years of daily weather. It produces dates, depths, schedules, root growth, water flow arrows and the animated field. It is **calibrated** so that its mean irrigation equals Engine 1's, and its fluxes are **scaled** to Engine 1's annual means (Eq. 5.24). **[Engineered, with literature components]**

The design layers (Chapters 11–13) are a third, separate set of *closed-form hydraulic models*; they take their soil, crop, trigger and depth from Engines 1 and 2 (Section 5.3).

5.1 Engine 1 — the Daisy simulations and the meta-model

5.1.1 The simulation campaign

The repository's study ran Daisy 7.1 (Abrahamsen and Hansen, 2000; University of Copenhagen), a one-dimensional soil–plant–atmosphere model that solves Richards' equation with van Genuchten–Mualem hydraulics, coupled to the WEPP dynamic structure model for the effect of tillage and consolidation on pore structure with time. Ten soils (USDA textures from sand to silty clay loam, Ap and Bt horizons) were simulated in **10,190 runs across 17 experiments (E0–E16)**. The families that the workbench reads are:

Experiment	Runs	Varied	Used for
E2	200	Trigger suction	Trigger correction, Eq. (5.11)
E3	200	Sensor depth	Sensor-depth correction, Eq. (5.11)
E4	100	Method (overhead, surface, drip at three depths)	Method correction, Eq. (5.11)
E5	300	Tillage × structural consolidation rate	Baseline **b**, Eqs. (5.4)–(5.10)
E7	200	Mulch vapour-flux factor × interception capacity	Soil evaporation $\bar{E}$, Eq. (5.4)
E11	1,000	Temperature shift × evaporation factor × rain scale	Warming ratios, Eqs. (5.4)–(5.10)
E13	1,000	Nitrogen rate × water strategy × soil	Nitrogen correction, Eq. (5.11)

Experiment	Runs	Varied	Used for
E14	1,000	Mulch factor × capacity × rain scale	Mulch ratios, Eqs. (5.4)–(5.10)

(The other experiments are used by the repository figures and described in the repository README. The *Regime map* and *Fate of saved water* layers are computed from the E14 and E11 grids listed above.)

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

(van Genuchten, 1980), with relative hydraulic conductivity from the Mualem (1976) model,

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Equation (5.2) is used in the Soil, plant and air layer (Section 7.2); Eq. (5.1) is used for all water-retention drawings. The parameters (α, n) used by the engine are *effective* values that pass through the four anchors of the project's trigger table; for the standard textures they match the Carsel and Parrish (1988) values (Table 5.3).

5.1.2 Plant-available water and depletion

Depletion of plant-available water is defined between field capacity (suction 100 cm, pF 2.0) and wilting point (15,849 cm, pF 4.2):

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

$f = 0$ is field capacity and $f = 1$ is wilting point; $f < 0$ (wetter than field capacity) is kept as the *signed* value only in the Trigger law layer. The inverse $h(f)$ is found by bisection on $\log_{10}|h| \in [0, 4.6]$ (60 halvings).

Table 5.3. Effective van Genuchten parameters used by the engine, plant-available water, and the soil's critical suction at the soil-independent depletion $f^* = 0.45$.

Soil	α (cm^{-1})	n	AWC (mm cm^{-1})	h^* at $f^* = 0.45$ (cm)	f at 300 cm
S01 Coarse Loam	0.0961	1.353	1.5	387	0.380
S02 Coarse Sand	0.0536	1.600	0.5	262	0.496
S03 Sand	0.145	2.68	0.6	143	0.842
S04 Loamy Sand	0.124	2.28	0.9	160	0.756
S05 Sandy Loam	0.075	1.89	1.2	195	0.627
S06 Loam	0.036	1.56	1.7	284	0.468
S07 Silt Loam	0.020	1.41	2.0	401	0.367
S08 Sandy Clay Loam	0.059	1.48	1.4	311	0.439
S09 Clay Loam	0.019	1.31	1.6	503	0.314
S10 Silty Clay Loam	0.010	1.23	1.7	729	0.237

(AWC is "typical available water capacity by texture class" in mm of water per cm of soil; it is an engineered per-texture constant, not a measured value of the study soil.)

5.1.3 The meta-model

Let $\mathbf{b} = (E_b, T_b, ET_b, I_b, Y_b, D_b, N_b)$ be the 20-year-mean Daisy results for the chosen soil, tillage and consolidation rate (run set E5), and $\mathbf{b}_{\text{ref}}$ the same for conventional tillage at $\tau_c = 0.005$ d^{-1}. Let φ be the mulch strength, c the mulch capacity, ρ the rainfall scale and ΔT the warming. Define the mulch table lookups $M(\varphi, \rho)$ (E14 at capacity c) and warming table lookups $W(\Delta T, \rho)$ (E11); write

$$\text{now} = M(\varphi,\rho),\ \text{ref} = M(\varphi,1),\ \text{nm} = M(1,\rho),\ \text{bare} = M(1,1), \quad \text{tN} = W(\Delta T,\rho),\ \text{tB} = W(0,\rho),\ \text{tR} = W(0$$

Then the engine computes (the ratio $[x/y]$ is 1 where $|y| < 10^{-6}$):

$$E = \bar{E}(\text{soil},\varphi,c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

$\bar{E}(\text{soil},\varphi,c)$ is the linear interpolation, in φ, of the soil's simulated soil evaporation (run set E7) at capacity c, closed at $\varphi = 1$ by the bare-soil value. Fluxes are combined multiplicatively and drainage/leaching additively because ratios of small numbers (drainage near 0 mm) are unstable.

Four *set-specific corrections* then act in turn on the result. For each family $F \in \{$trigger (E2), sensor depth (E3), method (E4), nitrogen (E13, yield and N_L only)$\}$ with current key κ and baseline key κ_0 (trigger -300 cm; depth 20 cm; method Surface; nitrogen 100 %), with r_κ the soil's row at κ (nearest row if that exact run is absent):

$$Q \leftarrow Q\ \text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E,T,ET,I,Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.$$

(5.11)

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Finally the model enforces non-negativity, a soil-dependent yield ceiling $Y \le Y_{\max}(\text{soil})$ (the largest simulated yield), rain $P = P_{\text{base}}\rho$, and **water-balance closure**:

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

E_{other} is interception plus evaporation from the mulch layer, and ΔS collects the change in storage and small terms. IE converts Mg ha^{-1} per mm of irrigation to kg of grain per m^3 of irrigation water (1 mm $= 10\ \text{m}^3\ \text{ha}^{-1}$).

Crop yield scaling. The yield shown on the Dashboard is $Y \times f_{\text{crop}}$ with $f_{\text{crop}} = 1.0$ (cereal/oilseed), 0.55 (legume), 2.3 (vegetable, fresh weight). Only the cereal rotation is simulated; the other two factors are *illustrative*, "loosely reflecting typical relative dry-matter yield and harvest-index differences reported in general agronomy references", and carry no study-specific data.

5.1.4 What "20-year average at these settings" means

Every Dashboard tile in the last block of the Results tab is the output of Eqs. (5.4)–(5.12) for the current controls. Every slider is an index into a list of simulated values, so no value ever lies between runs; but two controls changed together (for example, mulch and warming) use the product of ratios above, which assumes the effects are **independent**. The yield ceiling and the closure window (Eq. 5.12) stop independent effects from compounding beyond what the water and the crop allow.

5.2 Engine 2 — the daily bucket model

5.2.1 Layout and state

The soil is $N = 40$ layers of thickness $\Delta z = 2.5$ cm (100 cm total). Each layer holds water w_i (mm) between 0 and the plant-available capacity $W_{\text{max}} = \text{AWC} \cdot \Delta z$. All layers start at $0.72\, W_{\text{max}}$. Relative content $s_i = w_i / W_{\text{max}}$ and depletion $f_i = 1 - s_i$. The run is 20 years of 365 days (7,300 days). The sensor sits in layer $j_s = \text{round}(z_s / \Delta z)$ for sensor depth z_s.

The trigger suction $|h_T|$ is converted to a depletion fraction $f_T = f_{\text{used}}(|h_T|)$ by Eq. (5.3). The sensor condition "soil at suction h_T" therefore becomes the equivalent statement $f_{j_s} \geq f_T$.

5.2.2 Crop calendar and canopy cover

The cereal crop follows the four-year rotation spring barley, winter barley, winter rape, winter wheat with sowing and harvest days (relative to 1 Jan of the year) $(100, 225)$, $(-90, 205)$, $(-125, 200)$, $(-80, 230)$; winter crops are sown in the previous autumn. Legume: sown day 110, harvested 215. Vegetable: sown day 100, harvested 185. Outside the window the field is *fallow*. Canopy cover $c(d) \in [0, 1]$ is

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp\!\left(-(d - d_{\text{mid}})/s\right)}, \quad s = 13 \text{ (vegetable } 8), \tag{5.13}$$

Provenance (Eq. 5.13): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12\, \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp\!\left(-(d - d_{\text{pk}} + 45)/12\right)} & \text{otherwise} \end{cases} \tag{5.14}$$

Provenance (Eq. 5.14): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

and after the peak day d_{pk} cover declines linearly to zero at harvest: $c \leftarrow c\, \max\{0,\, 1 - (d - d_{\text{pk}})/(d_{\text{harv}} - d_{\text{pk}})\}$. The mid day is $d_{\text{sow}} + 50$ (cereal), $d_{\text{sow}} + 40$ (legume), $d_{\text{sow}} + 28$ (vegetable); the peak is harvest -50 (cereal), sow $+65$ (legume), sow $+65$ (vegetable). These logistic curves are **[Engineered]**: they reproduce the shape of a canopy, not a measured leaf-area record.

5.2.3 Rainfall generator

Rain falls on a day with probability 0.28. With $\mu_0 = P/(0.28 \cdot 365)$ the mean wet-day depth of the scaled annual rain P, the depth on a wet day of simulated year y is

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0,1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

an exponential distribution capped at 70 mm, with the random sequence seeded by a hash of the soil name so every run is reproducible. The year factor $\gamma_y = \text{clip}\big((m_{\text{med}}/m_y)^{0.3};\, 0.65,\, 1.45\big)$ makes years wetter or drier according to the dryness m_y of the study's 20-year rainfed suction record at 20 cm, where $m_y = 10^{\overline{\log_{10} h}}$ is the geometric-mean suction over the 183-day growing season of year y, and m_{med} is the median over the 20 years. This is how the generated weather "follows the dryness of the real record".

The canopy and litter catch the first I_c mm of every shower:

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}. \tag{5.16}$$

Provenance (Eq. 5.16): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

P_{other} is the interception-and-mulch loss from Eq. (5.12). Only throughfall $T_f > 0.3$ mm enters the soil. The choice of I_c makes the expected interception over the exponential rain distribution equal to the annual interception of Engine 1 (the expected value of $\min(R, I_c)$ for an exponential with mean μ_0 is $\mu_0(1 - e^{-I_c/\mu_0})$, and $e^{-I_c/\mu_0} = r$ gives $\mu_0(1 - r)$ per wet day).

5.2.4 Infiltration, redistribution and drainage

Water enters the top layer and cascades down: any layer exceeding W_{max} passes the excess to the layer below; the excess leaving layer 40 is **drainage** D_d. The depth of the deepest wetted layer is the **wetting front** (cm). Between layers a slow redistribution moves water toward the drier neighbour,

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

shown as *capillary rise* arrows when $q_i > 0.03\,W_{\text{max}}$ upward. Drainage and the wetting front are therefore **tipping-bucket** results, not Richards solutions; that is why the engine scales them to Engine 1 (Eq. 5.24).

5.2.5 Roots

The maximum rooting depth is

$$R_{\text{max}} = \text{clip}\Big(R_c\,(0.85{+}0.3\,\text{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \text{cm}, \tag{5.18}$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

with $R_c = 95$ (cereal), 115 (legume), 45 (vegetable) cm, sand the sand fraction, and N the nitrogen rate in percent. (Drier triggers push roots deeper; nitrogen-rich soils root slightly shallower.) While the canopy is growing, the rooting depth R_d chases water:

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\,\mathrm{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35+\bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

where $\bar{s}_{\text{wet}}$ is the mean relative water content of the four layers beneath the root front, and $\kappa = 1.3$ if the top six layers average below 0.35 (dry topsoil) and 1 otherwise. After harvest $R_d \leftarrow \max\{3, 0.965 R_d\}$; drought prunes the roots by 0.9 cm per day when the plant-available-water-weighted root fraction is below 0.12 and $R_d > 8$ cm. Root density follows

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d + 8}\Big)\,(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

with $\lambda = 0.32$ (cereal), 0.24 (legume), 0.40 (vegetable) and z_d the drip emitter depth. The bracket is **[Engineered]**: it raises root density up to fourfold around a drip emitter. These root functions are the *modelled* roots drawn in the R1 layer, "not a scan of real ones".

5.2.6 Transpiration, evaporation

Daily soil evaporation is the annual evaporation E_{yr} of Eq. (5.4) distributed over the year with a sinusoidal weight, reduced by canopy shade,

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65 \cdot \tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

and is taken from the top four layers with weights $(0.55, 0.25, 0.12, 0.08)$, never drawing a layer below $0.05\,W_{\max}$. Crop transpiration demand is the annual transpiration spread by cover, times a **calibration factor** k_t,

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\ \text{yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

and is divided among the layers by *availability-weighted* root density,

$$q_i^{T} = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j}, \qquad a_i = \mathrm{clip}\Big(\frac{s_i - 0.15}{0.6};\,0,\,1\Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

limited so that no layer falls below $0.10\,W_{\max}$. The availability function a_i is a linear stress function between 15 % and 75 % of plant-available water; it is **[Engineered]** in the spirit of the FAO-56 water-stress coefficient (Allen et al., 1998, Eq. 84), which is also linear in depletion.

5.2.7 Irrigation rule and calibration

An irrigation of dose d (mm) is applied on day d if *all* of these hold: canopy cover $c > 0.2$; $95 \le$ doy ≤ 250; at least m days since the last irrigation ($m = 1$, or $m = \mathrm{clip}(\mathrm{round}(110\,d/\bar{I}); 2, 12)$

if $f_T < 0.1$, to space irrigations when the trigger is almost at field capacity); the yearly total would stay within the budget $6\bar{I}$ (practically no cap); the sensor layer is dry enough, $1-s_{j_s} \geq f_T$; $f_T < 0.995$; and, for drip, the layer at the emitter depth has depletion ≥ 0.12. The dose is $d = \text{clip}(\bar{I}/10; 15, 35)$ mm, and for drip $d_{\text{eff}} = \max\{8, d/2\}$ delivered as small pulses *spread over* ± 3 *layers around the emitter depth*; for other methods the dose enters at the surface. $\bar{I}$ is the annual irrigation of Eq. (5.7).

Calibration of k_t. The trigger decides *when* but the root-zone water use decides *how often*. To make the day-by-day model reproduce the annual irrigation of Engine 1, $k_t \in [0.6, 8]$ is found by eight bisection steps on the first 8 years so that the mean irrigation per year I_{sim} matches $\bar{I}$ within 6 % (if the first guess $k_t = 1$ is already within 6 %, it is kept); the result is then used for all 20 years.

5.2.8 Scaling to the annual model

The daily results are multiplied by constants so that their 20-year means equal Engine 1:

$$S_P = \frac{P}{\overline{P_{\text{sim}}}}, \quad S_{\text{thr}} = \frac{P - P_{\text{other}}}{\overline{T_{f,\text{sim}}}}, \quad S_E = \frac{E}{\overline{E_{\text{sim}}}}, \quad S_T = \frac{T}{\overline{T_{\text{sim}}}}, \quad S_D = \frac{D}{\overline{D_{\text{sim}}}}, \tag{5.24}$$

Provenance (Eq. 5.24): **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

(each set to 1 if the simulated mean is below 10^{-6}), and the daily fluxes displayed on the Dashboard are $R = P_d S_P$, $T_f = \min\{R, T_{f,d} S_{\text{thr}}\}$, $E = E_d S_E$, $T = T_d S_T$, $D = D_d S_D$ and irrigation I unscaled. The **storage change of the day** closes the balance:

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

5.2.9 Evaporation under warming (used by the E1 layer)

For the Evaporation physics layer a separate atmosphere factor is used (Section 9.3):

$$E_{0,\text{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65 \cdot \frac{1}{2}[1 + \sin(2\pi(\text{doy} - 100)/365)]}{0.675}\,\text{clip}\big(1{-}0.12(\rho{-}1);\,0.85,\,1.12\big)\sqrt{\gamma_y}\ \text{m} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

where 4.2 mm d^{-1} is the engine's season-mean bare-soil demand (the seasonal weight is divided by its mean, 0.675), the factor $1 + 0.07\Delta T$ is a 7 % per °C increase (close to the Clausius–Clapeyron rate of the saturation vapour pressure; **[Engineered]** parameter), the cloud factor reduces demand in wetter years, and $\sqrt{\gamma_y}$ links demand to the dryness of the year.

5.3 The design layers and their link to the engines

The three design layers take from the engines:

Quantity	Source	Used by
Soil texture (sand, silt, clay %)	`DATA.texture` (measured for the 10 soils)	Saxton–Rawls intake (Section 11.2)
Trigger depletion f_T	Eq. (5.3) at h_T	Required depth Z_n

Quantity	Source	Used by
Maximum rooting depth $R_{\max}$	Eq. (5.18)	Z_n
Plant-available water per cm AWC	Table 5.3	Z_n
Peak crop use ET_c	engine's peak daily transpiration plus evaporation	Interval T (drip, sprinkler)
Date and irrigation dates	Engine 2	Irrigation schedule tables
Weather year, mulch, interception	Engine 1 and mulch capacity	sprinkler interception loss

The default required depth is therefore

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \text{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

which is the net depth that refills the root zone from the trigger depletion back to field capacity ("*from the DSS*" on the screen). Any slider can override it (the label then reads *tap to reset*).

5.4 Reproducibility

All random numbers in the engine are generated by a deterministic generator seeded by the soil name (a 32-bit hash), so the same controls always give the same dates, depths and animations. There is no external data call; the engine file contains the simulation tables as JSON.

Chapter 6

Irrigation control layers

The two layers of this group answer one question: **when should irrigation start?** *Sensor and trigger* (V1) shows what a sensor sees through twenty years; *Trigger law* (N1) shows why the same suction means different things in different soils and how to express the trigger so that it works for every soil.

6.1 Sensor and trigger (V1)

6.1.1 What it shows

A simulated tensiometer at the **sensor depth** (default 20 cm) records soil-water suction h (cm of water, plotted on a log axis from 1 to 100,000 cm) every day. Two traces are drawn: *irrigated* (blue) and *rainfed* (orange). The red dashed line is the **trigger** you set. Pale bands are growing seasons. Under the trace, red ticks mark irrigated sensor days beyond the trigger, and the bar chart counts, for each year, the growing-season days (of 183) on which the sensor was beyond the trigger, irrigated against rainfed.

6.1.2 How to use it, step by step

1. Open *Irrigation control* ▸ *Sensor and trigger*.
2. Choose **All 20 years** to see the whole record, or **1994 detail** for the one year with full-profile output.
3. Move **Trigger (sensor suction)** in Properties; the red line and every count update.
4. Move **Sensor depth**. The 20-year record is reported at 20 cm only, so the depth control applies to the *1994 detail* view (and to every irrigation date and schedule), as the note on the drawing says.
5. Press **Play** to sweep a cursor through the record; click a year in the year strip to jump.
6. Read the text under the plot: *“Irrigated: 85 of 183 days beyond 300 cm, mean suction 255 cm. Rainfed: 168 of 183 days beyond, mean suction 1,945 cm. Irrigation removes 39 % of the days beyond this trigger.”*

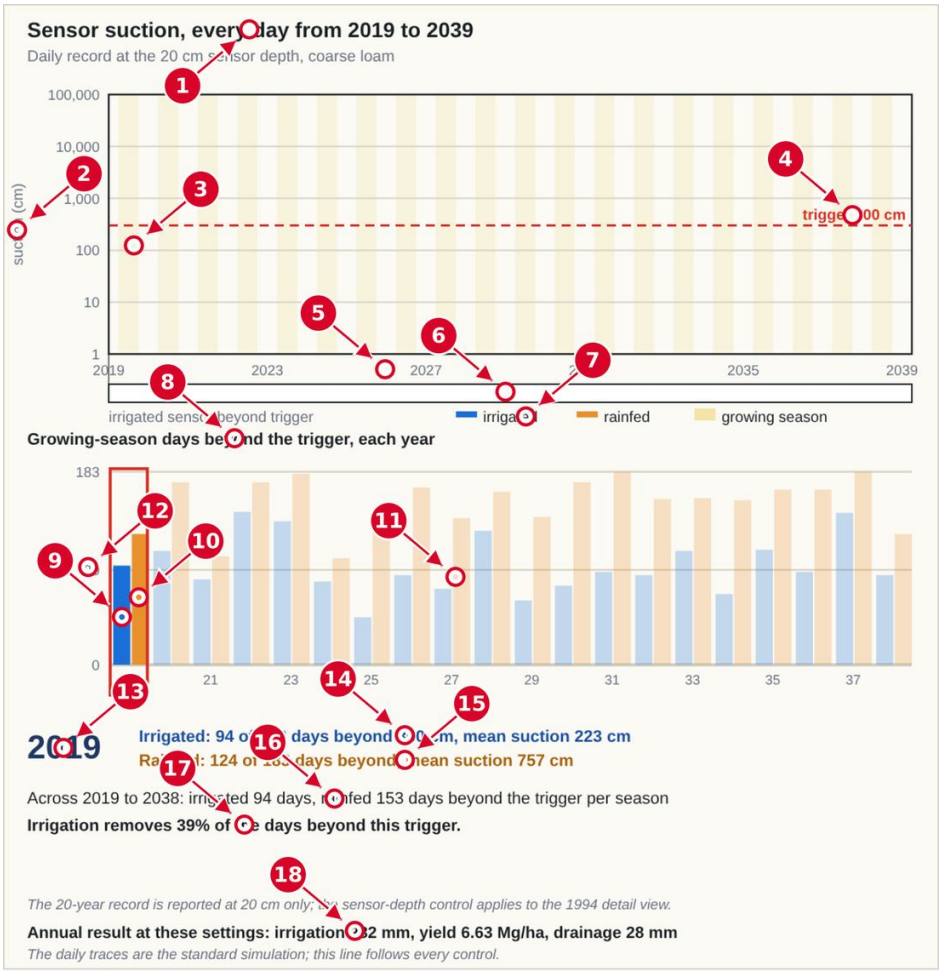


Figure 6.1: Sensor and trigger. Top: 20 years of suction at the sensor, irrigated (blue) and rainfed (orange) with the trigger (red dashed). Bottom: growing-season days beyond the trigger per year. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

6.1.3 How the numbers are generated

- The two suction traces are the Daisy-simulated daily suction at 20 cm (irrigated with the study's trigger, and rainfed) for the 1980–1999 record, stored in the engine (`D20`) for the Coarse Loam and Coarse Sand (other soils use the nearer of the two, as the drawing labels). **[Data-driven]** They are *not* produced by the daily bucket model.
- The count of days beyond the trigger is, for year y,

$$N_y(h_T) = \#\{\, d \in \text{growing season of year } y : h_d > h_T \,\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

Provenance (Eq. 6.1): **[Data-driven]**: counts over the simulated daily series.

and the "Irrigation removes X %" statement is

$$X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_{y} N_y. \tag{6.2}$$

Provenance (Eq. 6.2): **[Data-driven]**: counts over the simulated daily series.

- The annual line at the foot ("irrigation 282 mm, yield 6.63 Mg ha^{-1}, drainage 28 mm") is Engine 1 (Eqs. 5.4–5.12) at the current settings.

6.1.4 Reading it

If rainfed days beyond the trigger greatly exceed irrigated days, irrigation is doing its job; if the two bars are similar the trigger is so dry that irrigation rarely starts. A trigger of -50 cm (very wet) produces ticks on almost every day.

6.2 Trigger law (N1)

6.2.1 The idea

The same suction does not mean the same dryness in two soils. At -300 cm a sand has lost 84 % of its plant-available water while a silty clay loam has lost only 24 % (Table 5.3, last column). N1 re-expresses the trigger as the **depletion fraction** f of Eq. (5.3), the share of plant-available water already used, and shows that irrigation and yield responses from all ten soils collapse onto a common curve when plotted against f.

[Literature + study result]. The depletion form follows the management-allowed-depletion idea of FAO-56 (Allen et al., 1998, Eq. 83 and Table 22): irrigate when a fixed share p of the total available water is used. The *soil-independent* critical value $f^* = 0.45$ is the study's result (5 % yield loss), and the pooled R^2 rising from 0.48 to 0.75 (irrigation) and 0.23 to 0.70 (yield) is reported in the repository.

6.2.2 How to use it, step by step

1. Open *Irrigation control* ▸ *Trigger law*.
2. The soil column on the left is the soil by depth on the clock's day; the blue gauge beside it shows depletion from FC (0) to WP (1). **Drag the purple trigger marker** on the gauge: the soil's own trigger suction changes, and the right-hand list of ten soils moves in step.
3. Each row on the right is one soil with its trigger suction and the net irrigation depth at that depletion ("143 cm trigger, 278 mm"); the green box gives the relative yield at the soil's trigger.
4. Click a soil name to select it (it becomes the working soil).
5. **Common and own mode.** By default one suction (the Trigger slider) applies to every soil (*common* mode). **Opening the Trigger law layer switches the program to *own* mode**: each soil then uses its own critical suction for the chosen depletion, and the Trigger slider in Properties shows the nearest value on its 20-value grid (simulated runs exist only on that grid). Own mode stays on until you choose *Reset all settings* or a ready-made case, which restore the common mode.
6. Read the verdict box at the bottom: it states the soil's critical suction, the yield at the current trigger relative to the best trigger, and how many root-zone layers are beyond the trigger on this day.

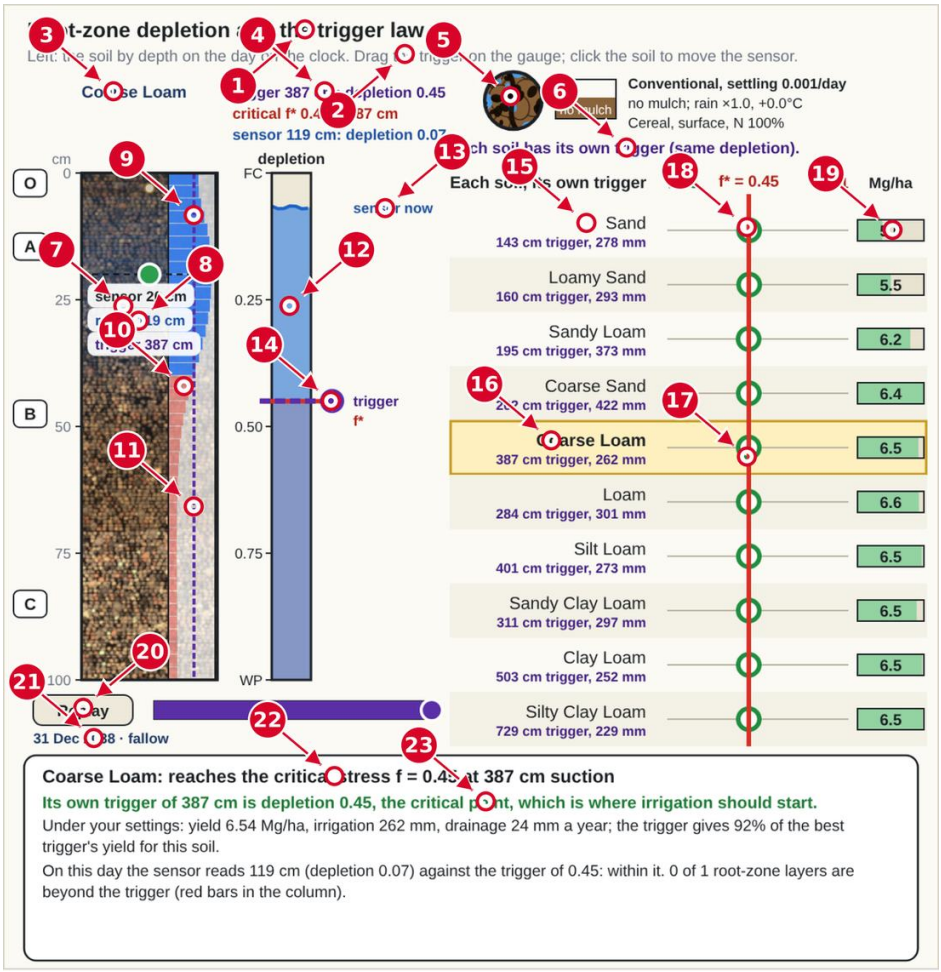


Figure 6.2: Trigger law layer: ten soils on one depletion axis, with the critical $f^* = 0.45$ (red line) and the soil's own trigger. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

6.2.3 Governing relations

The depletion of Eq. (5.3) is evaluated with each soil's effective van Genuchten pair (α, n) (Table 5.3). The suction at which a soil reaches a chosen depletion f is the inverse,

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \tag{6.3}$$

Provenance (Eq. 6.3): **[Literature]**: inverse of the van Genuchten curve, solved by bisection.
solved by bisection, giving for $f^* = 0.45$ the soil-specific sensor settings of Table 5.3 (143 cm for sand to 729 cm for silty clay loam). The net irrigation per event shown beside each soil is

$$Z_n = f \cdot \mathrm{AWC} \cdot R_{\max} \quad [\mathrm{mm}] \tag{6.4}$$

Provenance (Eq. 6.4): **[Literature]**: same relation as Eq. (5.27) without the clip.
(the same relation as Eq. 5.27 without the clip). The *relative yield at the trigger* is

$$Y_{\mathrm{rel}} = \frac{Y(h_T)}{\max_h Y(h)}, \tag{6.5}$$

Provenance (Eq. 6.5): **[Data-driven]**: ratio of Daisy run-set E2 yields.
with $Y(h)$ the soil's yield at each simulated trigger (run set E2). **[Data-driven]**.

6.2.4 Reading the sensor value

The dashboard tile *Sensor / trigger* reads two numbers: the **current depletion** f at the sensor layer and the **trigger depletion** f_T. When the first reaches the second, irrigation is due. On 18 Oct 2026 the tile reads $0.02/0.38$ for the Coarse Loam: the soil is almost at field capacity (0.02), and the trigger is at depletion 0.38 ($= f$ at 300 cm).

Chapter 7

Field water balance layers

7.1 Field balance (the home layer)

7.1.1 What it shows

One picture of the water budget of a field for one day, one year, or the twenty-year average: rain and irrigation arriving, transpiration, soil evaporation and interception leaving upward, drainage leaving downward, and the nitrogen leached with it. The soil profile is drawn to 100 cm with three depth marks A, B and C, the sensor at the chosen depth, the roots as they grow and die back, and the wetting front.

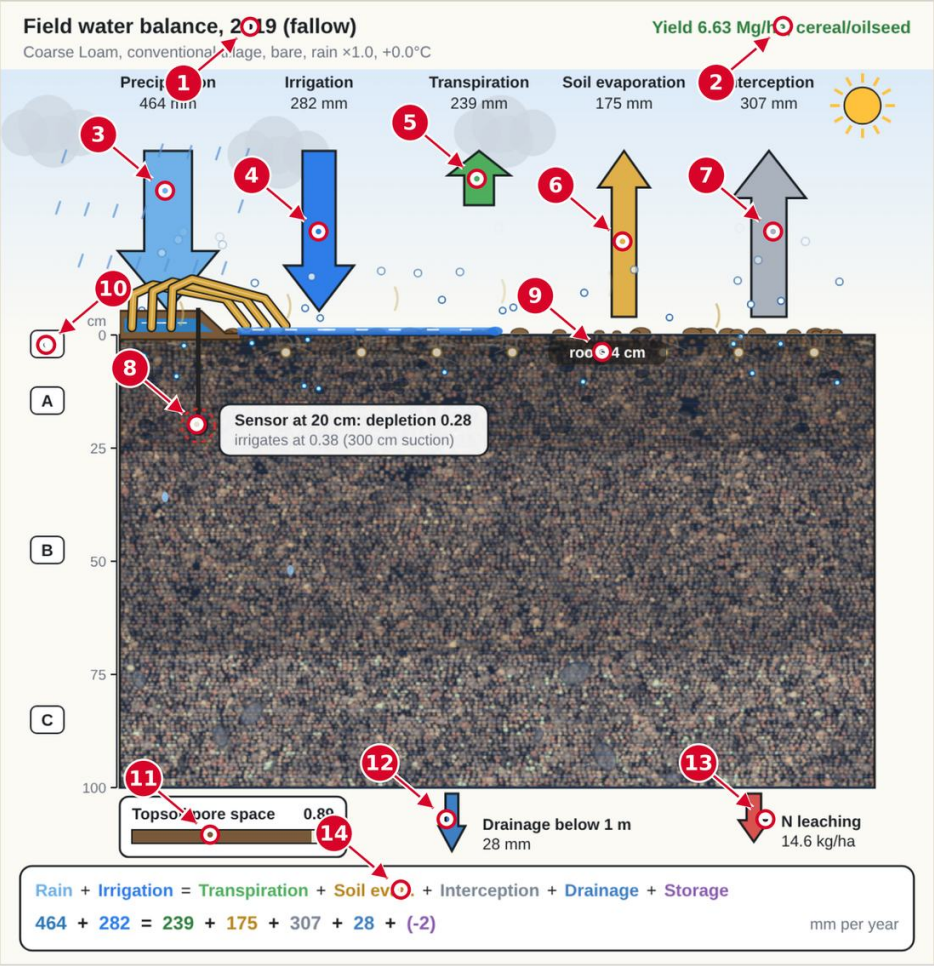


Figure 7.1: Field balance: arrows are the fluxes (mm per year in “Average year” view), the profile shows the root zone, the sensor, and drainage below 1 m. The equation strip under the picture is the closed budget. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

7.1.2 The control bar under the drawing

Button	Meaning
Day by day	The simulated day on the field clock; arrows show that day’s fluxes (T_f, I, T, E, interception, D).
All 20 years	Table and bars of every simulated year (rain, irrigation, interception, evaporation, transpiration, drainage, storage change, crop, deepest roots). Click a year.
Average year	The 20-year means of Engine 1 (Eqs. 5.4–5.12).
◀ year / year ▶	Previous or next simulated year.
1× 2× 4× 8× and speed slider	Play speed.
Play / Pause	Animates the clock.

7.1.3 How each output is generated

On the picture	Generated by
Precipitation, Irrigation, Transpiration, Soil evaporation, Interception, Drainage (mm) in *Average year*	Engine 1: P, I, T, E, E_{other}, D of Eqs. (5.4)–(5.12)
The same in *Day by day*	Engine 2 daily fluxes after scaling, Eqs. (5.15)–(5.25)
Equation strip "Rain + Irrigation = Transpiration + Soil evap. + Interception + Drainage + Storage"	The closure of Eq. (5.12): $P + I = T + E + E_{\text{other}} + D + \Delta S$
Sensor label "depletion 0.02 / irrigates at 0.38"	Engine 2 layer j_s: $f_{j_s} = 1 - s_{j_s}$ against f_T
Roots (depth, density)	Eqs. (5.19)–(5.20)
Wetting front	Depth of the deepest layer reached by the last infiltration event
Topsoil pore space bar	Pore index p of Eq. (8.1) for the chosen tillage and consolidation rate
Drainage below 1 m	D (mm), the water leaving layer 40
N leaching (kg ha^{-1})	N_L of Eq. (5.10)
Yield tag (top right)	$Y \cdot f_{\text{crop}}$ from Eq. (5.8)

7.1.4 Step-by-step: checking a budget

1. Open *Field balance* and choose **Average year**.
2. Read the five numbers over the arrows and the strip below. Check that rain plus irrigation minus the four sinks equals the storage change in brackets.
3. Change **Rainfall scale** to $×$1.3. The rain, drainage and N leaching arrows thicken.
4. Press **Pin results as baseline** (Ctrl+B) *before* each change to see the difference on the Dashboard tiles.
5. Click **Day by day**, Play, and watch irrigation events pause the clock (Section 4.5).

7.2 Soil, plant and air (S1)

7.2.1 What it shows

The soil–plant–atmosphere continuum as one water column "from the particles to the sky": a section of the soil with roots, the plant, and the air above; five irrigation layouts (**Furrow, Basin, Border strip, Sprinkler, Drip**) that you can tap, which set the irrigation method; and, on the right, the **water potential along the path** from the air to the deep soil. Two data sources are labelled with badges: *DAISY* (taken from your simulations) and *DERIVED* (computed by the workbench along the path).

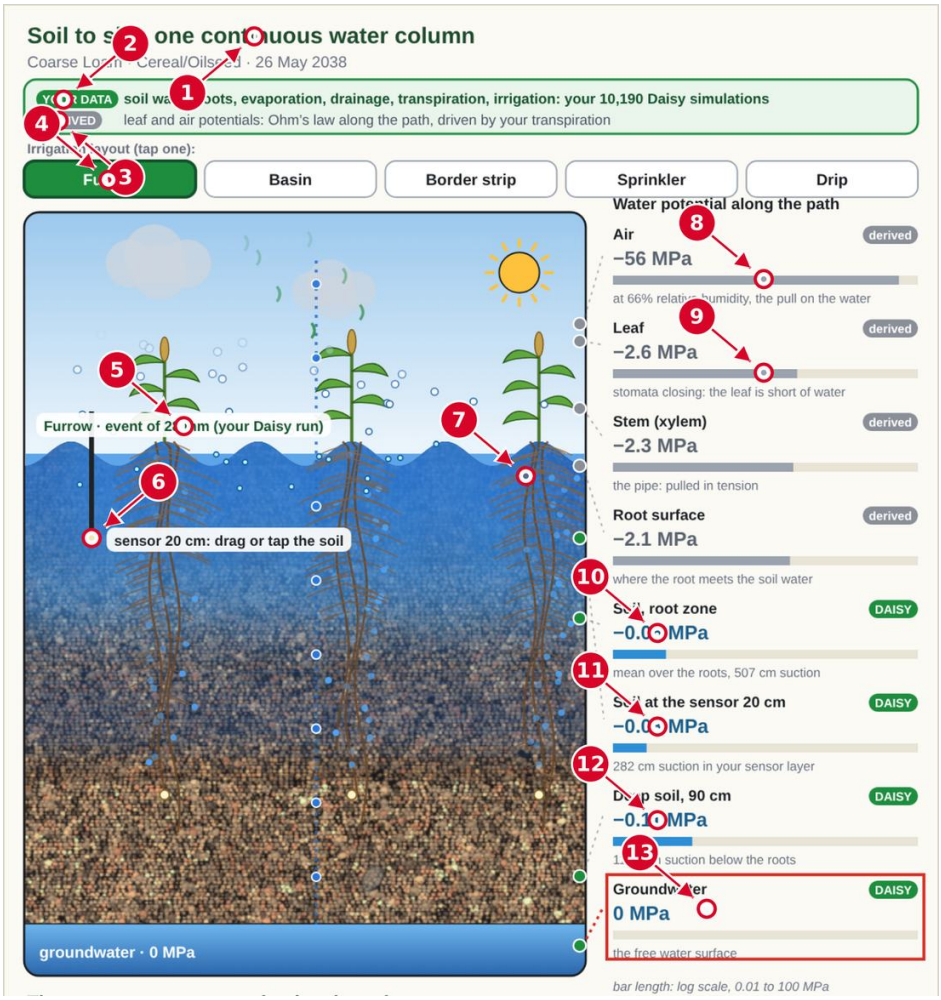


Figure 7.2: Soil, plant and air layer: potentials from the air to the deep soil. *Derived* badges mark the quantities computed along the path; *Daisy* badges mark simulation data. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

7.2.2 Governing equations

Soil. The suction in each layer is the suction that gives that layer's depletion, from the inverse of Eq. (5.3), floored at 30 cm:

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \tag{7.1}$$

Provenance (Eq. 7.1): **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

The *root-zone* suction is the root-density-weighted mean, $h_{RZ} = \sum_i \rho_i h_i \big/ \sum_i \rho_i$, and the *sensor* and *deep-soil* values are h_{j_s} and h_{N-3}.

Soil conductivity. The relative conductivity at the root zone uses the Mualem–van Genuchten form, Eq. (5.2), with $S_e = S_e(h_{RZ})$ from Eq. (5.1).

Along the path. The plant is a chain of series resistances, in the spirit of the catenary (Ohm's-law) description of water flow through the soil–plant–atmosphere continuum (van den Honert, 1948). With E the day's transpiration (mm d^{-1}, from Eq. 5.25) and R_{soil}, R_{root}, R_{xyl}, R_{leaf} resistances in MPa per (mm d^{-1}):

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

where $a = 1$ when canopy cover exceeds 0.05 (a crop is present) and $a = 0$ otherwise (fallow). Stomata are drawn as *closing* when $\psi_{\text{leaf}} < -1.6$ MPa.

The air. Water potential of the air above the canopy follows the thermodynamic relation between potential and relative humidity,

$$\psi_{\text{air}} = \frac{R\,T}{V_w}\,\ln \text{RH} \;\approx\; 135.5\,\ln \text{RH}\ \ \text{MPa} \tag{7.5}$$

Provenance (Eq. 7.5): **[Literature]**: Nobel (2009).

(Nobel, 2009), where R is the gas constant, $T \approx 293$ K and $V_w = 18 \times 10^{-6}$ m^3 mol^{-1} the molar volume of water. Relative humidity is $\text{RH} = \text{clip}(0.66 - 0.03\,\Delta T;\ 0.35,\ 0.90)$ with ΔT the temperature shift. At RH = 66 % the air is at -56 MPa, as shown.

What is literature and what is engineered. Eqs. (7.1), (7.5) and (5.2) are physical relations. The resistances in (7.2)–(7.4) are **[Engineered]** values chosen so that a typical transpiring cereal reaches ψ_{leaf} of about -1 to -1.5 MPa and so that stress develops as the soil dries; they are *not* measured resistances, and the layer labels them "derived". **Caution.** Evaluated over mid-season crop days in all ten soils (Paper 3, Section 5.1), the relative conductivity at the root-zone suction is usually below the floor of 0.002 in Eq. (7.2), so R_{soil} sits at its cap (0.92 MPa per mm d^{-1}) and the mean leaf potential is -1.8 to -2.3 MPa in every soil, below the intended range. The leaf, stem and root potentials of this layer therefore illustrate the series-resistance concept and should not be read as plant water status.

7.2.3 Step-by-step

1. Open *Soil, plant and air*.
2. Tap a layout. Furrow, basin and border strip set the **Surface** method; Sprinkler sets **Overhead**; Drip sets **Drip 20 to 30 cm**.
3. Drag or tap in the soil to move the **sensor depth** (the control snaps to the 5-cm list).
4. Change the **date** to a growing-season day and compare the potentials: with a crop present the leaf falls below -1 MPa as the soil dries; on a fallow date the path is flat (resistance terms are zero).
5. Move the **Trigger** slider and note that the *sensor* suction tile follows the same depletion as the dashboard.

7.3 Profile evolution (V2)

7.3.1 What it shows

How the pressure head profile develops through the season in a year, side by side for **irrigated** and **rainfed** soil columns, with the sensor depth shown by a dot. In *All 20 years* mode it draws the pressure head at the sensor through every year and a heat strip of the mean suction of each growing season (darker is drier). Text states the driest and wettest rainfed season and the share of the 0–60 cm root zone beyond the trigger.

7.3.2 How it is generated

- The **1994** profile is the Daisy-simulated pressure head at every depth, irrigated and rainfed (the only year with full-depth output, as the drawing says).
- In the other years only the 20 cm sensor is available; the columns are shaded only around the sensor depth.
- *Share of the root zone beyond the trigger* is

$$\text{share} = \frac{\sum_{i:\ z_i \le 60\,\text{cm},\ h_i > h_T} \Delta z_i}{\sum_{i:\ z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

Provenance (Eq. 7.6): **[Engineered]**: indicator defined for the workbench.

the layer-thickness-weighted fraction of the 0–60 cm profile whose suction exceeds the trigger, evaluated for the irrigated and the rainfed profile of 1994 (the year with full-depth output). * The mean suction of a season, used for the heat strip and for the “driest season” statements, is the geometric mean $m_y = 10^{\frac{1}{183}\sum_d \log_{10} h_d}$ over the growing season; the median across years m_{med} also drives the rainfall year factor γ_y (Eq. 5.15).

7.3.3 Step-by-step

1. Open *Profile evolution* and choose **All 20 years**.
2. Click a year in the heat strip to select it; click a column to put the sensor at that depth.
3. Move the trigger: the share beyond the trigger is recomputed.
4. Choose **1994 detail** to see the full profile through the season; press Play to step through it.

Chapter 8

Soil structure and tillage layers

8.1 Tillage and structure (V3)

8.1.1 What it shows

A modelled thin section (“microscope”) of the topsoil for the chosen soil and tillage, at three magnifications (10 mm, 2.5 mm, 0.6 mm), with pores filled by water or air according to a suction slider; the same place under the three tillage systems; the relative pore space against time; and the change in drainage and yield caused by tillage at your settings.

8.1.2 Pore index and consolidation

After tillage the loosened topsoil re-settles. The workbench uses a first-order relaxation of the **relative pore space** p (1 = freshly loosened) to a consolidated value $p_c = 0.30$,

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

with t in days since tillage and τ_c the **Structural consolidation rate** control (Table 3, 0.001–0.02 d^{-1}). **No-till** keeps a constant $p = 0.42$ (undisturbed pores). The half-life of loose pores is $\ln 2/\tau_c$: 693 d at 0.001 d^{-1}, 35 d at 0.02 d^{-1}. **[Engineered]**: the exponential form is the standard first-order consolidation relation also used in structure sub-models such as WEPP’s, on which Daisy’s tillage effect is based, but p_0, p_c and the no-till value are *presentation* constants for the drawing; the *hydrological effect* of tillage shown in the bars is read from the Daisy runs (E5), not from Eq. (8.1).

8.1.3 How the section is generated

The section is a random packing of mineral grains and matrix drawn on a raster, with these parameters, all **[Engineered]**:

- Grain-size classes follow the soil’s measured sand, silt and clay percentages (`DATA.texture`).
- **Saturated water content** $\theta_s = \phi$ and **residual content** θ_r are

$$\phi = \text{clip}\big(0.36 + 0.0016\,\text{clay} + 0.0009\,\text{silt};\ 0.34,\ 0.50\big), \qquad \theta_r = 0.01 + 0.0006\,\text{clay}, \tag{8.2}$$

Provenance (Eq. 8.2): **[Engineered]**: porosity relation with chosen coefficients.

with clay and silt in percent, and the retention curve is the van Genuchten function of Eq. (5.1),

$$\theta(h) = \theta_r + (\phi - \theta_r)\, S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

- **Cracks and aggregates.** The number of cracks is $n_b = \text{round}(2 + 6(1 - f_{\text{till}}))$ with $f_{\text{till}} = 1,\ 0.7,\ 0.4$ for conventional, reduced, no-till; the crack width (mm) is $0.3\ \max\{0,\ p - 0.3\}^{1.3}$.
- **Which pores hold water.** A pore of radius r fills with water at suction h if $r \le r_c(h)$, where the capillary radius follows the Young–Laplace (Jurin) law,

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\, g\, h} \;\approx\; \frac{0.149\ \text{cm}^2}{h} \quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

with surface tension $\sigma = 0.0728$ N m^{-1}, contact angle $\gamma \approx 0$ and $g = 9.81$ m s^{-2} (Jurin, 1718; Hillel, 1998). In the drawing a pore is water-filled when its local half-width is at most r_c **or** when it is within a film thickness $\min\{1.6,\ 0.9\,r_c/s + 0.3\}$ pixels of a grain, which draws the water films that survive drying.

- The hover probe states, for a pore, its radius, the suction at which it empties ($h = 1.49/r$) and the present suction.

8.1.4 Controls and steps

1. Open *Soil structure and tillage* ▸ *Tillage and structure*. Press **Play** to run a year of consolidation (the day appears under the picture).
2. Drag the **suction slider** (pF 0–4.2) on the microscope: at low suction the pores are water-filled; as suction rises the large pores empty first.
3. Switch the view: *Natural* (minerals), *Particle sizes* (sand/silt/clay classes), *Water and air*.
4. Change **Tillage** or the **Structural consolidation rate** and watch the curves of $p(t)$ (Eq. 8.1) and the section change.
5. Read the bars *Change in drainage at the current settings* (conventional, reduced and no-till against no-till) and the yields. These come from the Daisy runs for the selected soil: $\Delta D_{\text{conv}} = D_{\text{conv}} - D_{\text{notill}}$, $\Delta D_{\text{red}} = D_{\text{red}} - D_{\text{notill}}$, at the current τ_c.
6. The sentence at the end states whether the consolidation rate changes anything: *"In this soil the consolidation rate changes nothing; tillage is the driver"* or *"Here the consolidation rate matters as well as tillage"*.

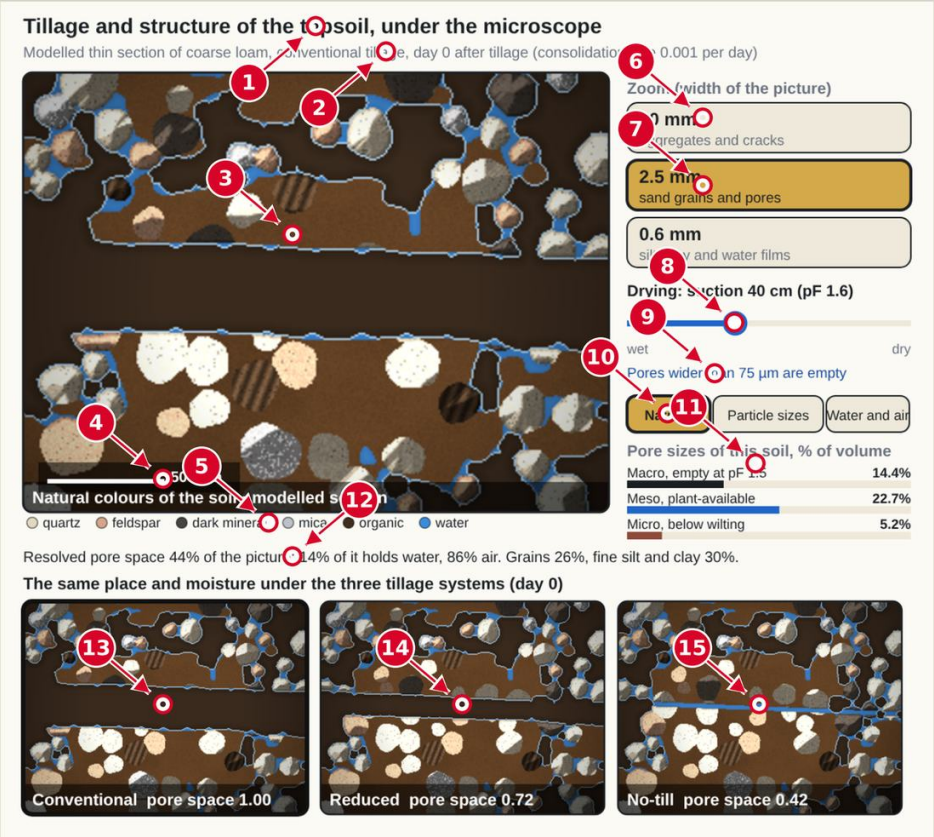


Figure 8.1: Tillage and structure layer: thin section, tillage comparison and the drainage consequences. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

8.2 Soil hydraulic drivers (V7)

8.2.1 What it shows

The question *"What drives change in soil hydraulic properties?"* answered for all ten soils together: for each soil sorted by clay content, a blue bar for the **tillage effect** (conventional minus no-till) and an orange bar for the **consolidation effect** (fastest minus slowest rate) on annual drainage, then scatter plots against clay content, and the 1-m soil profiles of the three tillage systems.

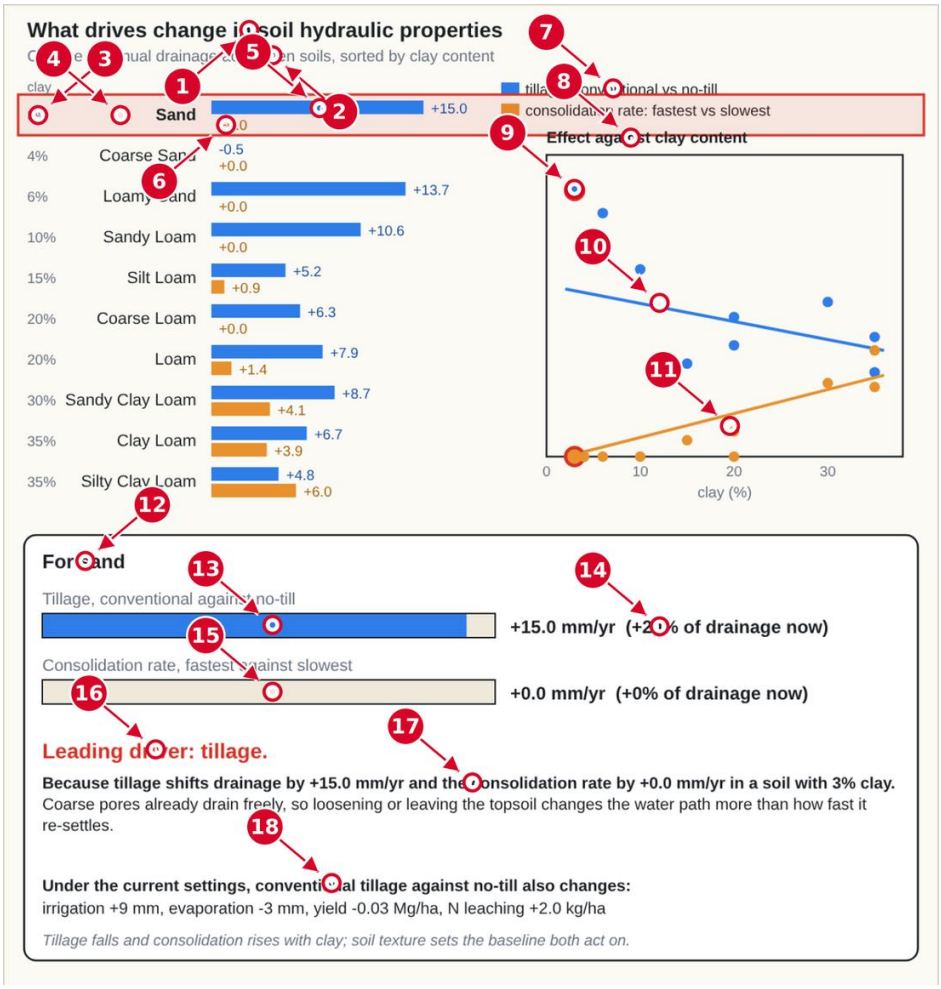


Figure 8.2: Soil hydraulic drivers. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

8.2.2 Generation

For each soil s the Daisy run set E5 gives annual drainage $D_s(\text{tillage}, \tau_c)$. The two bars are

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*), \qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \tag{8.5}$$

Provenance (Eq. 8.5): **[Data-driven]**: differences of Daisy run-set E5 results.

with the current settings as starred values; bar length is proportional up to 16 mm yr^{-1}. The lines in the scatter plots are ordinary least-squares fits of each effect on clay content c,

$$\hat{\Delta} = a + b\,c, \qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2}, \quad a = \bar{\Delta} - b\,\bar{c}. \tag{8.6}$$

Provenance (Eq. 8.6): **[Data-driven]**: ordinary least squares applied to Daisy results.

The headline “Leading driver: tillage” or “consolidation rate” compares Δ_{till} with Δ_{cons} for the selected soil, and the percentage after each bar is $100\,\Delta / D_{\text{now}}$, the share of the present drainage. The last line adds, for conventional against no-till at the current mulch and rain, the changes in irrigation, evaporation, yield and N leaching, obtained by running the meta-model (Eqs. 5.4–5.12) twice (conventional at τ index 2 and no-till) and subtracting.

The lower part of the layer draws, for the three tillage systems, a 48 cm wide, 1 m deep face of the selected soil. The upper 24 cm (the tilled layer) is generated with the pore index p of Eq. (8.1) at day 120 (adjustable), clod and crack density from f_{till}, and the wetness factor $\text{clip}\big((p - 0.30)/0.14;\, 0.3,\, 1.4\big) \cdot \text{clip}(0.75 + 0.012\,\text{clay};\, 0.8,\, 1.4)$; below it the subsoil is untouched by tillage. The drainage figures printed under each column are Daisy results (E5).

8.2.3 Steps

1. Open the layer. Click any soil bar or any scatter point to select that soil.
2. Click a point on the *Drainage against consolidation rate* graph to apply a rate and a tillage system at once.
3. Read the verdict, and compare with the profile drawings.

8.3 Root structure (R1)

8.3.1 What it shows

Rooting depth in all ten soils at the same day and weather: for each soil a column with the deepest roots (20-year mean, brown dashed), and ticks at the depth that holds 50 % and 90 % of the water taken up (green ticks). Below it, a time view of rooting depth through the 20 years and the daily uptake, and a set of heat maps from the repository's **E15** runs: 1,000 Daisy runs on the Coarse Loam, rooting depth 30–180 cm × ten application amounts × ten irrigation methods, mapped to yield, irrigation, drainage, transpiration, N leached and water-stress days.

8.3.2 Generation

Roots and uptake come from Engine 2 (Eqs. 5.18–5.23). For each soil the engine is run in turn with that soil's parameters (and, in own mode, its own trigger). With per-layer uptake $u_i(d)$ on day d the **depth holding a fraction p of the uptake** is

$$z_p = k_p\,\Delta z, \qquad k_p = \min\Big\{k :\ \frac{\sum_{i\le k} U_i}{\sum_i U_i} \ge p\Big\}, \qquad U_i = \sum_{d\in\text{year}} u_i(d), \quad p = 0.5,\ 0.9, \tag{8.7}$$

Provenance (Eq. 8.7): **[Engineered]**: diagnostic defined for the workbench.

with the layer index counted from 1 (z_p is the *bottom* of layer k_p). It is computed for each of the 20 years from that year's uptake, then averaged over the years; the share of uptake in the top 30 cm is $\sum_{z_i\le 30} U_i \big/ \sum_i U_i$. The model stops roots at 1 m, whereas the Daisy runs reach 120 cm, as the layer notes. The statement "*Across the soils the deepest roots differ by …*" is the range of the 20-year-mean $\max_d R_d$ across soils.

What these roots are. The drawing states it: *"These are modelled roots, not a scan of real ones"*; root density is *relative* (Eq. 5.20), not measured root length. The E15 heat maps are **[Data-driven]** and exist for the Coarse Loam only ("they do not change with the soil you picked").

8.3.3 Steps

1. Open the layer. Click a column to make that soil the working soil.
2. Click or drag on the upper graph to move the day; the lower panel shows the uptake on that day.
3. In the E15 panel choose an output (yield, irrigation, …), click a rooting depth and an application amount in the heat map; the table shows the change against the present setting and tells you the best depth, for example *"Roots at 30 cm leave the crop stressed …"*.

Chapter 9

Mulch and evaporation layers

This group follows the water that mulch keeps in the soil: how mulch lowers evaporation (V4, V6), the physics of drying (E1), where the saved water goes (N2) and when evaporation or drainage dominates (N3).

9.1 Mulch against bare soil (V4)

What it shows. Three soil blocks side by side: bare soil, the soil under *your* mulch, and under the strongest mulch (plastic film, $\varphi = 0.1$), with bars for evaporation, irrigation and drainage, the six materials as buttons (click one to apply it), and a verdict.

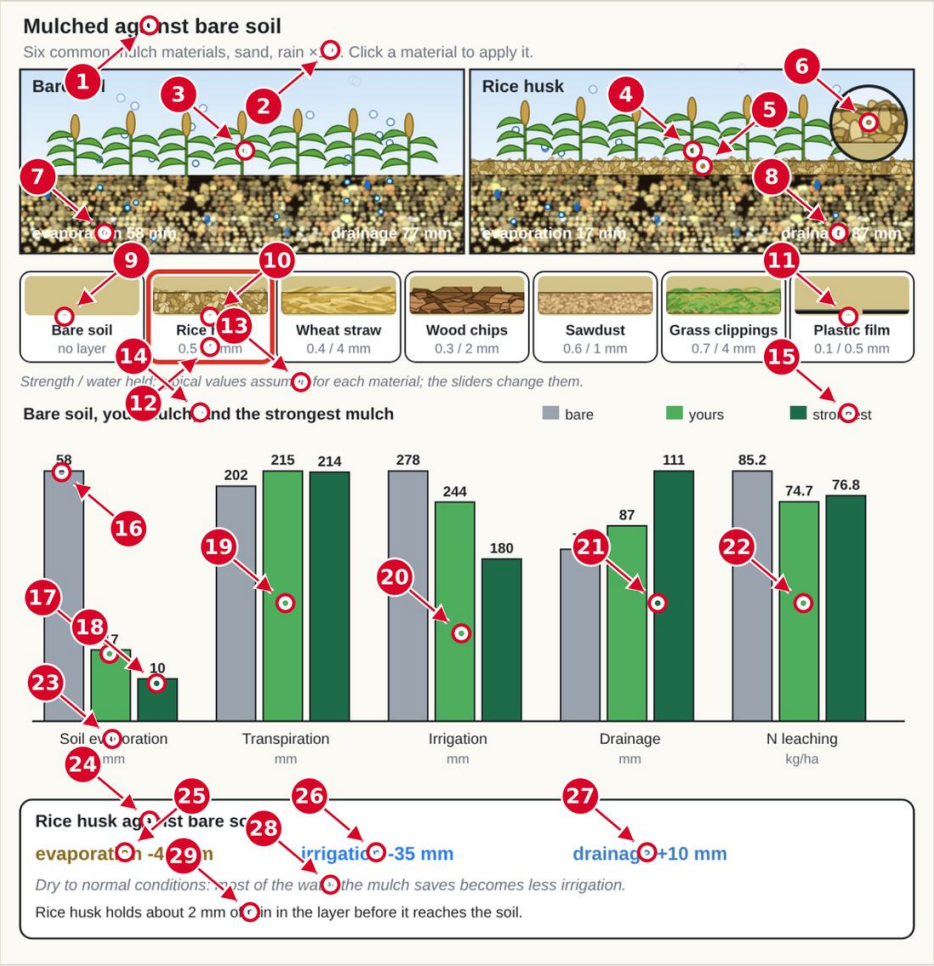


Figure 9.1: Mulch against bare soil. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

How the numbers are generated. All three blocks are Engine 1 evaluated at $\varphi = 1.0$ (bare), at your φ and capacity, and at $\varphi = 0.1$, at the current rainfall. With E_φ, I_φ, D_φ from Eqs. (5.4), (5.7), (5.9),

$$\Delta E = E_{1.0} - E_\varphi, \qquad \Delta I = I_{1.0} - I_\varphi, \qquad \Delta D = D_\varphi - D_{1.0}, \tag{9.1}$$

Provenance (Eq. 9.1): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

are the evaporation saved, the irrigation saved and the extra drainage, in mm yr^{-1}, printed under the blocks. The verdict compares ΔI with ΔD (“*Wet conditions: much of the water the mulch saves now leaves as drainage*” or “*Dry to normal conditions: most of the water the mulch saves becomes less irrigation*”). The block of text for plastic film states that “*plastic film stops vapour almost completely but holds no water, so rain runs to the planting holes*”, which is the capacity of 0.5 mm in Table 3.1.

Steps. (1) Open *Mulch and evaporation* ▸ *Mulch against bare soil*. (2) Click a material. (3) Raise the **Rainfall scale** to \$×\$1.3 and watch ΔD overtake ΔI. (4) Move the **Strength** slider to see the evaporation saving grow convexly.

9.2 Mulch control (V6)

What it shows. Two matrices. The first is **soil evaporation against mulch factor (ten rows, 0.1–1.0) by interception capacity (four columns, 0.5–4 mm)**. The second is **drainage against tillage system (three) by consolidation rate (five)**. Colour darkens with the value; a red outline marks your current setting. The text states what fraction of bare-soil evaporation your mulch removes and whether the soil responds to tillage.

Generation. Each cell of the first matrix is $E(\varphi, c)$ and each cell of the second $D(\text{till}, \tau_c)$, both from Engine 1 at the current soil, rainfall and temperature (Eqs. 5.4 and 5.9). The sentence "*No response here: this soil does not react to tillage or consolidation*" appears when the spread $\max D - \min D$ of the second matrix is below 0.5 mm yr^{-1}. The percentage in the text is $100\,(1 - E_{\varphi,c}/E_{1.0})$.

Steps. Click a cell to apply that mulch factor and capacity or that tillage and rate; the red outline moves; the Dashboard changes at once.

9.3 Evaporation physics (E1)

What it shows. A modelled thin section of the topsoil drying after a wetting, with liquid water in the pores (blue), vapour moving up (white), the **drying front** (orange) and, for comparison, the bare-soil curve (grey). To the right: the evaporation rate $e(t)$ and cumulative loss $Q(t)$ for 24 days, split into **stage 1** (the surface stays wet, loss at the atmospheric rate) and **stage 2** (a dry layer grows, loss is limited by the soil).

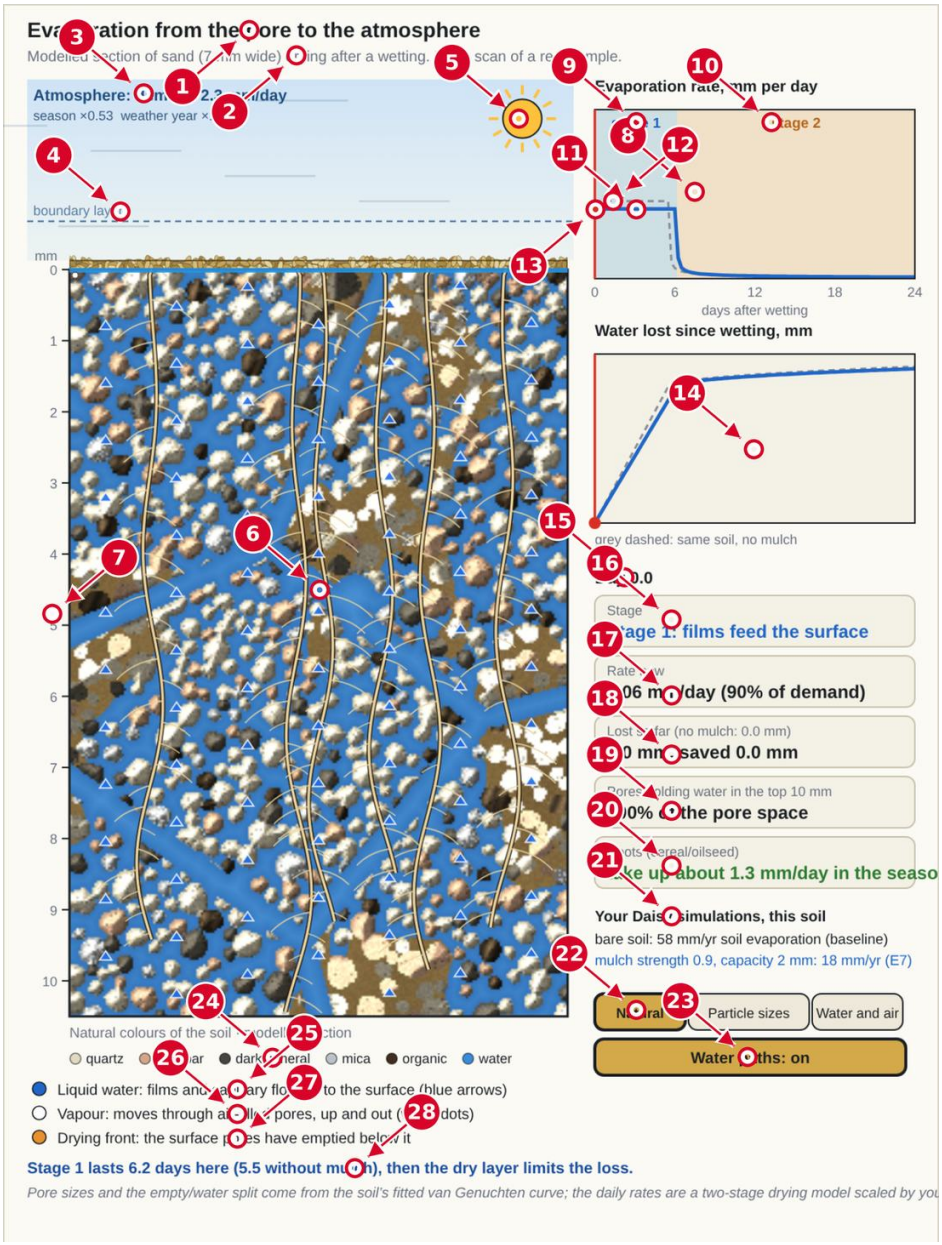


Figure 9.2: Evaporation physics layer. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

Governing relations. The loss is described by the classical two-stage model of soil drying (Philip, 1957; Ritchie, 1972): stage 1 at the atmospheric rate e_0, then stage 2 with a rate falling roughly as $t^{-1/2}$ (cumulative loss $\propto \sqrt{t}$). The implementation in the workbench is

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

Provenance (Eq. 9.2): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

$$Q(t) = q_0 + \begin{cases} e_0\, t, & t \le t_1 \\ (Q_1 - q_0) + 2\, e_0\, \tau\Big(\sqrt{1 + (t - t_1)/\tau} - 1\Big), & t > t_1 \end{cases} \tag{9.3}$$

Provenance (Eq. 9.3): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

(the rate is the time derivative of the cumulative loss, so the two are consistent), with the following parameters:

Symbol	Definition	Value or formula
$e_{0,b}$	bare-soil atmospheric demand	Eq. (5.26) (mm d^{-1})
e_0	demand under mulch and canopy	$e_0 = e_{0,b}\,\varphi\,(1 - 0.4\,c)$ with canopy cover c on the date
Q_1	stage-1 storage (mm), water held in the top 40 mm between the wet and the surface-dry state	$Q_1 = \max\{2,\ [\theta(h_i) - \theta(h_s)]\, L_1\, (0.6 + 0.4\rho)\}$, $L_1 = 40$ mm
h_i, h_s	initial and surface-dry suctions (cm)	$h_i = \dfrac{0.25}{\alpha}(1.5 - 0.5\rho)$; $h_s = 4/\alpha$
τ	stage-2 time scale (d)	$\tau = 0.006 + 0.03\,(\text{clay} + \text{silt})/100$
q_0	loss already experienced when the water was placed below the surface (drip)	$q_0 = (1 - w_f)\,0.95\,Q_1$, $w_f = \text{clip}(1 - z_d/50;\ 0.05,\ 1)$, z_d the drip depth in cm; 0 for surface methods
$\Delta\theta$	water lost across the dry layer	$\Delta\theta = \text{clip}(\theta(h_i) - \theta(10^4);\ 0.05,\ 0.4)$

$\theta(h)$ is Eq. (8.3) and α the van Genuchten parameter of Table 5.3; ρ is the rainfall scale. **[Engineered]** — every constant in the table (0.25, 1.5, 0.5, 4, 40 mm, 0.6, 0.4, 0.006, 0.03, 0.95, 50 cm, 0.05–1) is a presentation parameter chosen so that stage 1 lasts a few days in a loam and stage 2 gives the characteristic $\sqrt{t}$ fall; they are **not** calibrated to a laboratory evaporation experiment and the drawing says so ("*the daily rates are a two-stage drying model scaled by your mulch factor*"). The simulated annual evaporation of the workbench (Eq. 5.4) is *Daisy*, not this curve.

The drying front. The depth of the dry layer after stage 1 is

$$\ell(t) = \min\Big\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\Big\}\ \text{mm} \quad (Q > Q_1), \tag{9.4}$$

Provenance (Eq. 9.4): **[Engineered]**: drying-front geometry chosen for drawing.

and the suction profile with depth z is an exponential approach of $\log_{10} h$ with length $\lambda = L_1/3$:

$$\log_{10} h(z) = \log_{10} h_s + (\log_{10} h_i - \log_{10} h_s)(1 - e^{-(z-\ell)/\lambda}) \quad (z \ge \ell), \qquad h = 10^5 \text{ cm for } z < \ell. \tag{9.5}$$

Provenance (Eq. 9.5): **[Engineered]**: drying-front geometry chosen for drawing.

Pore picture. Which pores are water-filled at each depth is given by the capillary rule of Eq. (8.4) with the local suction of Eq. (9.5); vapour particles are released from the drying front and rise through air-filled pores. The picture is an **illustration of the mechanism**, not a pore-network simulation.

Steps. (1) Open *Evaporation physics* and press **Play** (24 days). (2) Raise the mulch (lower φ): the grey bare-soil curve stays, your curve lengthens stage 1. (3) Change the soil: stage 1 lasts longest in the sands (largest Q_1, about 3 days for the Sand at the season-mean demand) and is shortest in the silty clay loam (about 1 day), because the window between h_i and h_s scales with $1/\alpha$; in stage 2 the fine soils lose water more slowly (larger τ). (4) Set a drip method: $q_0 > 0$

shortens stage 1 because the surface has less water to lose. (5) Hover over a pore to read its radius and the suction at which it empties.

9.4 Fate of saved water (N2)

What it shows. For the soil and mulch you set (a preview of $\varphi = 0.5$ is used if you have not set mulch), four curves against rainfall scale $\times 0.5$–$\times 1.4$: evaporation saved, irrigation saved, extra drainage, extra transpiration. A shaded band marks the rainfall above which drainage exceeds the irrigation saved ("*drainage takes over near* $\times 1.2$"). Below, a bar divides each mm of evaporation saved among less irrigation, more drainage, more transpiration and "storage and other".

Generation. For each rainfall ρ in the ten-value list, with φ^* the mulch factor,

$$\begin{aligned} \Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho), \\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho), \end{aligned} \tag{9.6}$$

Provenance (Eq. 9.6): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

all from Engine 1 at the current soil, tillage and temperature, and the partition at the current rainfall is

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\ f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\ f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\ f_{\text{rest}} = \max\{0,\ 1 - f_I - f_D - f_T\}. \tag{9.7}$$

Provenance (Eq. 9.7): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

The crossover rainfall is the first ρ with $\Delta D \geq \Delta I$ and $\Delta D > 1$ mm. The study reports about 60 % of the saved water going to reduced irrigation in dry years and about 63 % to extra drainage in wet years, with the crossover near $\rho \approx 1.2$ (repository README).

Steps. Choose a material; slide the **Rainfall scale**; the red dashed cursor and the bar move; read the three sentences at the bottom.

9.5 Regime map (N3)

What it shows. A map of **mulch factor (rows) against rainfall scale (columns)** coloured by which loss dominates, together with soil columns in which vapour rises (grey) or droplets fall (blue), and a verdict for your settings.

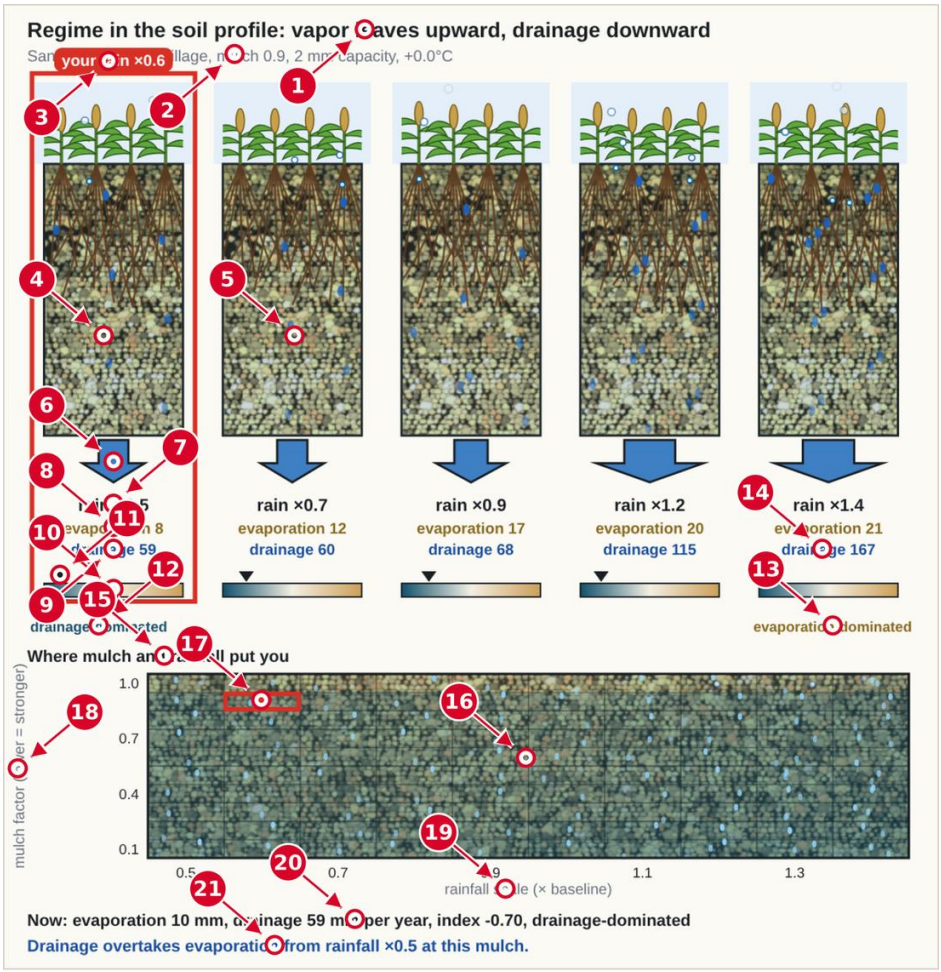


Figure 9.3: Regime map. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

Generation. For each cell the engine takes annual soil evaporation E and drainage D from Engine 1, and forms the **regime index**

$$\mathcal{I} = \frac{E - D}{\max\{1,\ E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{I} > +0.05 \\ \text{drainage-dominated} & \mathcal{I} < -0.05 \\ \text{on the boundary} & |\mathcal{I}| \le 0.05 \end{cases} \tag{9.8}$$

Provenance (Eq. 9.8): **[Engineered]**: index defined for the workbench.

so $\mathcal{I} = +1$ is all evaporation and -1 all drainage. The statement "*Climate and mulch act non-additively ... reaching 15 % of irrigation at combined extremes*" is a result of the study (the interaction of mulch and climate on irrigation); it is not recomputed by the layer.

Steps. Click a cell to apply that mulch factor and rainfall; the verdict says *"the main loss is evaporation (...), so mulch will pay off here"* or *"the main loss is drainage ..., so mulch saves less"*.

Chapter 10

Feedback loops (V5)

10.1 What it shows

One large picture of a field, from sky to groundwater, on which red arrows join the parts that act on each other: temperature and the crop; rain and soil water; mulch and evaporation; the sensor trigger and irrigation; drainage and nitrogen leaching; nitrogen and the roots; tillage structure and drainage. Each arrow is numbered 1–7, and its **width grows with the size of the effect of one step** of the control behind it. Under the picture a panel lists “*What one step of each control does*”; click a row or an arrow to highlight it.

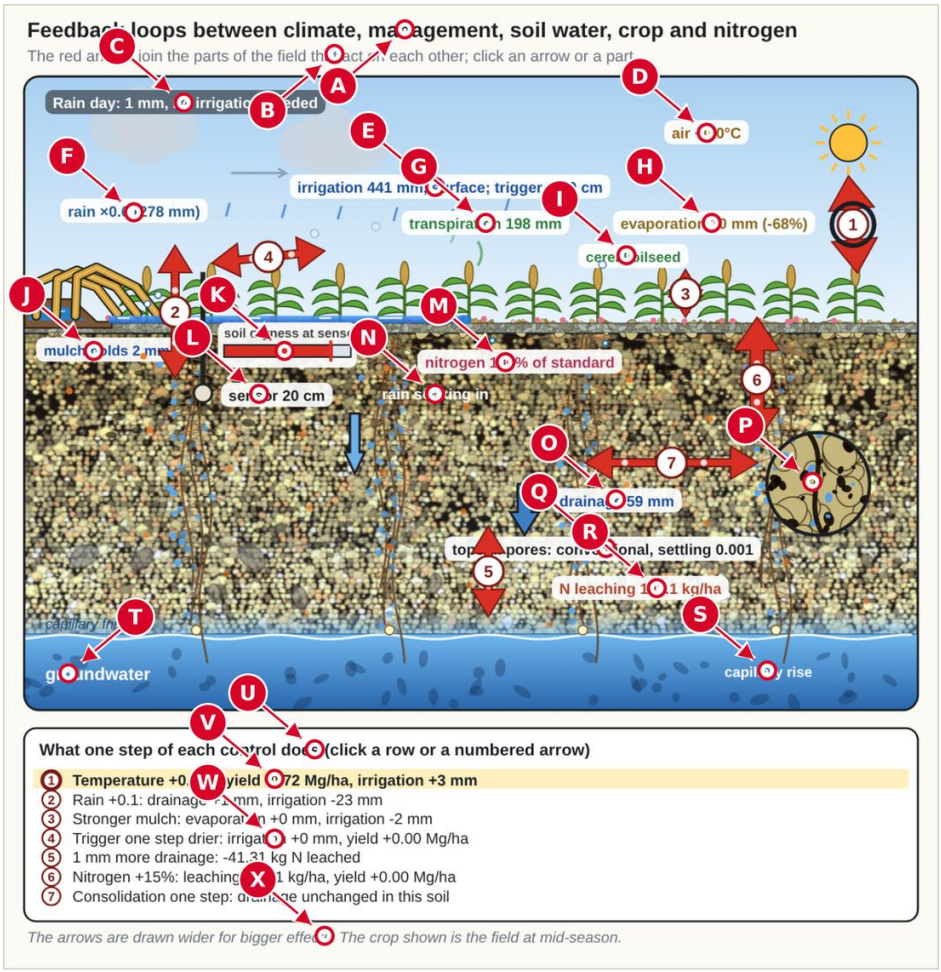


Figure 10.1: Feedback loops. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

The picture is also a set of controls: clicking the clouds cycles the rainfall, the sun the temperature, the mulch layer the material, the sensor the trigger (three grid steps drier each click, wrapping), the soil the tillage, the crop the crop category, the emitter or sprinkler the irrigation method, and the nitrogen band the nitrogen rate.

10.2 How the seven effects are computed

For each control the engine moves **one step** on its list (in the direction that stays inside the list: if the control is at its maximum it steps down) and re-evaluates Engine 1 (Eqs. 5.4–5.12). The change relative to the present result is printed. With m the present result and $m^{(\pm)}$ the result after one step, and f_{crop} the crop factor:

#	Control step	Printed effect	Arrow-width input μ
1	Temperature ± 0.5 °C	$\Delta Y\, f_{\text{crop}}$, ΔI	$8\,\lvert\Delta Y\rvert$
2	Rain ± 0.1 (scale)	ΔD, ΔI	$\lvert\Delta D\rvert/12$
3	Stronger mulch (one step lower φ)	ΔE, ΔI	$\lvert\Delta E\rvert/14$
4	Trigger one step drier or wetter	ΔI, $\Delta Y\, f_{\text{crop}}$	$\lvert\Delta I\rvert/8$

#	Control step	Printed effect	Arrow-width input μ
5	One more mm of drainage	$\Delta N_L/\Delta D$ kg N per mm (from the rain step), or "no change" if $\lvert\Delta D\rvert \le 0.5$	$\lvert\Delta D\rvert/10$
6	Nitrogen ± 15 %	ΔN_L, $\Delta Y f_{\text{crop}}$	$\lvert\Delta N_L\rvert/8$
7	Consolidation one step	ΔD (or "unchanged in this soil" if $\lvert\Delta D\rvert < 0.05$)	$\lvert\Delta D\rvert/2$

and the arrow line width in pixels is

$$w_k = 5 + \text{clip}(\mu_k;\, 0,\, 4). \tag{10.1}$$

Provenance (Eq. 10.1): **[Engineered]**: drawing rule.

The sensitivity printed in row 5, the **nitrate cost of drainage**, is the ratio

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D} \quad [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

Provenance (Eq. 10.2): **[Data-driven]**: one-step differences of Daisy results.

which the study reports as roughly 1.06 kg N ha^{-1} per mm of drainage (README). The numbers are **[Data-driven]** one-step differences, not partial derivatives, so a step at the end of a list can be smaller than the step in the middle.

10.3 The sentence under the picture

A status pill reports the day: *"Irrigating now: 28 mm (soil reached the trigger)"*, *"Rain day: 12 mm, no irrigation needed"* or *"Irrigation stopped: soil wet enough"*. The text for the trigger says whether *"the sensor has reached the trigger (depletion f)"*, to what depth *"the water wets the soil"* (the wetting front of the day's irrigation, Section 5.2) and, when that is shallower than the sensor, *"the sensor may stay beyond the trigger"*. If no irrigation is due, it states how many days are free in the next 14 and that rain is needed to keep it that way.

10.4 Steps

1. Open *Feedback loops*. The crop shown is the field at mid-season.
2. Click an arrow (number 4, for example). The matching row is highlighted: a line of the form *"Trigger one step drier: irrigation −X mm, yield −Y Mg/ha"* (the numbers depend on your settings).
3. Click the picture parts to change controls and watch the arrow widths change.
4. Use **Pin baseline** first to compare the 20-year averages for the whole combination.

Chapter 11

Surface irrigation design (S2)

The surface-irrigation layer designs a **border strip, furrow or basin** for *your* soil: it derives the soil's intake from texture, solves the advance of the water front, chooses a cut-off time, computes where the water goes, and then searches for a better layout. Everything on this page is a closed-form hydraulic model; no Daisy table is read, except the crop need and the trigger, which come from the DSS (Chapter 5).

The layer is a long scrolling drawing. The sub-section bar above it (*Whole design, Layout design, Simulation, Hydrograph, Advance and opportunity, Results, Design table*) jumps to a section, as do the sub-items of the layer in the Layers panel. The twelve sections are, in order:

#	Section (sub-item)	Purpose
1	Inputs and *Whole design* header	Choose the starting point, the system, intake, cut-off, and the sliders
2	Layout in plan: the advance front, shaded by the depth already infiltrated	Plan view of the field with the front
3	Longitudinal section: water running over your soil and soaking in	Side view with Z_0, Z_L, Z_n
4	Hydrograph: inflow, infiltration and runoff through time	Flow partition with time
5	Cross-section at the inlet: the wetted bulb in your soil, and the intake curve	Intake curve $Z(\tau)$ and the soil behind it
6	Advance and opportunity time	$t_x(x)$ and $\tau(x)$
7	Infiltrated depth along the field against the required depth Z_n	$Z(x)$ against Z_n
8	Performance indices and the water balance of one irrigation	E_a, E_s, DP, RO, CU, volumes
9	Design table	Iterations towards $E_a \geq 85\,\%$
10	What the efficiency is most sensitive to	Tornado chart
11	What this means for the season and the farm	Season, sets, hours
12	Irrigation schedule and *Intervals, efficiency and the advice in short*	Dates, gaps, "When to irrigate"

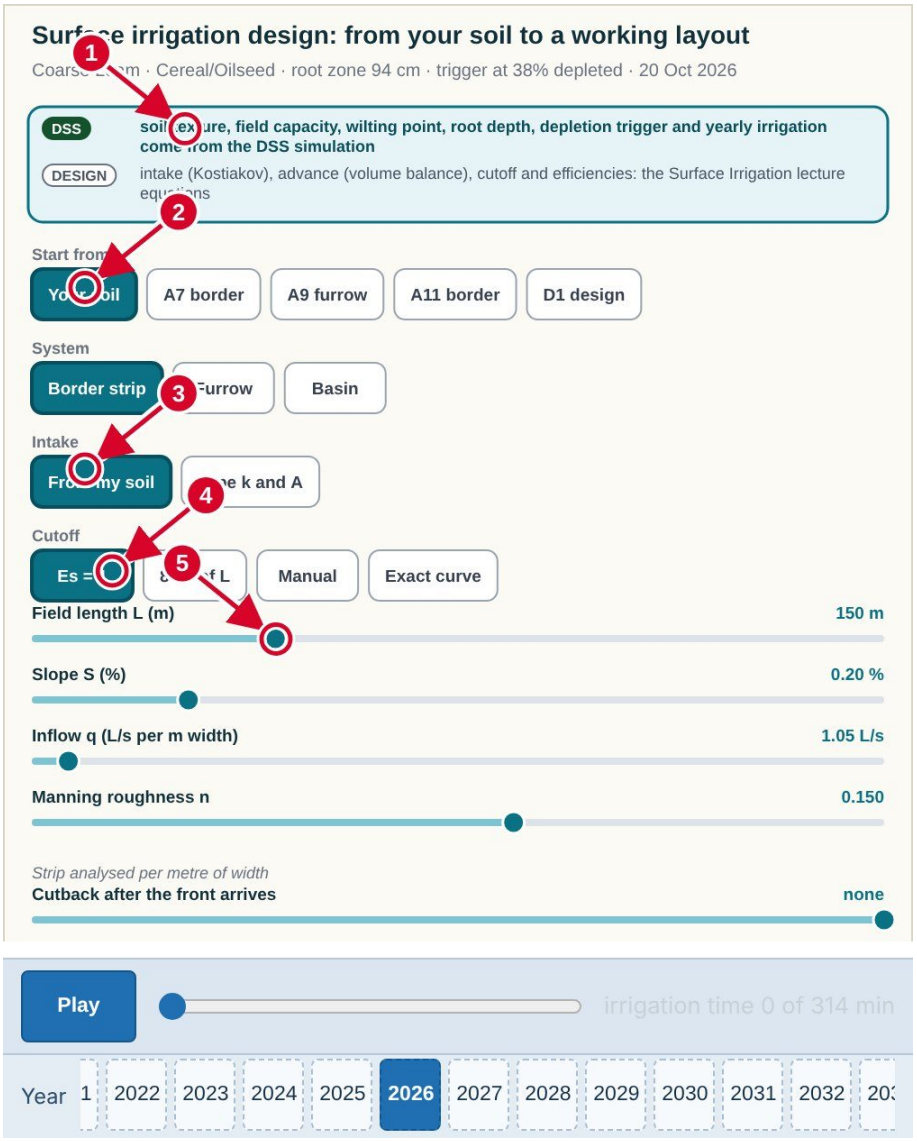


Figure 11.1: Surface irrigation design: the inputs (top) and the layout in plan. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

11.1 Inputs, step by step

1. Open *Irrigation design* ▸ *Surface irrigation design*. The header states the soil, crop, root-zone depth, trigger depletion and date (e.g. *"Coarse Loam · Cereal/Oilseed · root zone 94 cm · trigger at 38 % depleted · 18 Oct 2026"*). The green *DSS* badge lists what is taken from the simulation; the *DESIGN* badge what the design chooses.
2. **Start from** one of five rows:

Chip	Meaning
Your soil	Intake, required depth, cut-off and best inflow are all derived from your soil (default).
A7 border	Lecture worked example: border, $L = 200$ m, $S = 0.20$ %, $q = 2.5$ L s^{-1} m^{-1}, $n = 0.15$, $k = 2$ mm min^{-A}, $A = 0.8$, $Z_n = 90$ mm, given advance $t_L = 120$ min, cut-off 170 min.

Chip	Meaning
A9 furrow	Furrow, $L = 226$ m, $S = 0.30$ %, $q = 2.0$ L s^{-1}, $n = 0.04$, $w = 0.7$ m, $k = 4$, $A = 0.7$, $Z_n = 80$ mm, $\sigma_z = 0.83$.
A11 border	Border, $L = 240$ m, $S = 0.20$ %, $q = 1.667$, $n = 0.15$, $k = 7$, $A = 0.5$, $Z_n = 80$ mm, advance from field data $t_x = x^{1.35} q^{-1.5}/300$ (a power law, Eq. 11.12), cut-off at 85 % of L.
D1 design	Furrow, $L = 240$ m, $S = 0.30$ %, $q = 1.667$, $n = 0.04$, $w = 1$ m, $k = 2$, $A = 0.7$, $Z_n = 80$ mm, approximate advance (Eq. 11.13), $\sigma_z = 0.8154$.

The four worked examples reproduce the course exercises so that the program can be checked against hand calculation; choosing one switches the intake to *Type k and A*.

3. **System**: *Border strip*, *Furrow* or *Basin*. This resets L, S, q, n to the system defaults of Table 11.1.
4. **Intake**: *From my soil* (Green–Ampt from the soil's texture, fitted to Kostiakov, Section 11.2) or *Type k and A* (your own Kostiakov constants).
5. **Cutoff**: *Es = 1* (cut off when the tail has just received Z_n), *85 % of L* (cut off when the front reaches 85 % of the length plus the storage time), *Manual* (set the time), and the toggle *Exact curve* (draw the exact Kostiakov $Z(x)$ instead of the straight-line profile of the lecture notes).
6. Move the sliders (Table 11.1). Everything below redraws live.

Table 11.1. Surface-design sliders: ranges (min, max, step) and system defaults.

Control	Border	Furrow	Basin	Notes
Field length L (m)	50–400, 5 (default 150)	60–400, 5 (180)	20–200, 5 (80)	
Slope S (%)	0.02–1, 0.01 (0.20)	0.05–1.5, 0.01 (0.30)	0.02–0.2, 0.01 (0.05)	
Inflow q (L s^{-1})	0.2–20 per m width, 0.05 (6)	0.2–4 per furrow, 0.05 (1.2)	5–60 per m width, 0.5 (25)	In *Your soil* mode q is set automatically (Section 11.9) until you move it
Manning n	0.02–0.25, 0.005 (0.15)	(0.04)	(0.15)	
Furrow spacing w (m)	—	0.5–1.5, 0.05 (0.75 in *Your soil*)	—	$w = 1$ m is used for border and basin (a strip 1 m wide is analysed)
Cutback after the front arrives	0.2–1.0, 0.05 (1 = none)	same	same	Fraction of q after t_L
Required depth Z_n (mm)	20–150, 1	same	same	Default *from the DSS* (Eq. 5.27); tap the label to reset
Farm supply Q_s (L s^{-1})	5–120, 1 (30)	same	same	Farm-level flow
Field width W_f (m)	30–400, 5 (120)	same	same	
Organic matter (%)	0.5–5, 0.1 (1.5)	same	same	Enters Saxton–Rawls
Kostiakov k (mm min^{-A})	0.2–10, 0.05	same	same	Visible in *Type k and A* mode
Kostiakov A	0.2–0.95, 0.01	same	same	Visible in *Type k and A* mode

11.2 Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)

Step 1: soil water properties (Saxton and Rawls, 2006). With sand fraction S, clay fraction C (both 0–1) and organic matter OM (%), the 1500-kPa and 33-kPa water contents are

$$\begin{aligned}
\theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\mathrm{OM} + 0.005\,S\,\mathrm{OM} - 0.013\,C\,\mathrm{OM} + 0.068\,S\,C + 0.031,\\
\theta_{1500} &= \theta_{1500,t} + (0.14\,\theta_{1500,t} - 0.02),\\
\theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\mathrm{OM} + 0.006\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} + 0.452\,S\,C + 0.299,\\
\theta_{33} &= \theta_{33,t} + (1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015),
\end{aligned} \tag{11.1}$$

Provenance (Eq. 11.1): **[Literature]**: Saxton and Rawls (2006), implemented as written.

and saturation and saturated conductivity (mm h^{-1}) follow from

$$\begin{aligned}
\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078,\\
\theta_{(S-33)} &= \theta_{(S-33),t} + (0.636\,\theta_{(S-33),t} - 0.107), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\
B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.
\end{aligned} \tag{11.2}$$

Provenance (Eq. 11.2): **[Literature]**: Saxton and Rawls (2006), implemented as written.

These are the published pedotransfer equations of Saxton and Rawls (2006), implemented as written (field capacity $\theta_{fc} = \theta_{33}$, wilting point $\theta_{wp} = \theta_{1500}$). They are regressions on a large US soil database and carry its scatter; the layer uses them for design, not for prediction at a particular site.

Step 2: wetting-front suction. The effective suction at the wetting front (mm) is taken from the van Genuchten pair (α, n) of Table 5.3, $m = 1 - 1/n$,

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

This is a rational-polynomial fit of the Rawls-type wetting-front suction as a function of m; it is an **[Engineered]** approximation (the code comment calls it the "Lisle/Rawls form") and should be read as giving the right order of magnitude (tens of mm for loams, e.g. 16 mm for the Coarse Loam), not a measured ψ_f.

Step 3: available storage in the wetting zone. The soil starts at the trigger depletion f_T and is wetted to 95 % of saturation,

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

Provenance (Eq. 11.4): **[Engineered]**: initial and wetted water contents chosen for the workbench.

Step 4: Green–Ampt infiltration (Green and Ampt, 1911). Cumulative infiltration $F(t)$ (mm) solves

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

Provenance (Eq. 11.5): **[Literature]**: Green and Ampt (1911).

by Newton iteration from the start value $\max\{K_s t/60,\ \sqrt{2\psi_f \Delta\theta K_s t/60}\}$.

Step 5: Kostiakov fit (Kostiakov, 1932). The design uses the two-parameter empirical intake law

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

and k, A are found by least squares on the log–log plot of $F(\tau)/1000$ at 25 times spaced geometrically between 10 and 600 min:

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

Provenance (Eq. 11.7): **[Literature]**: ordinary least squares on the log–log plot.

with coefficient of determination $R^2 = \big(\sum (x-\bar{x})(y-\bar{y})\big)^2 \big/ \big(\sum (x-\bar{x})^2 \sum (y-\bar{y})^2\big)$, shown on the layer (e.g. *"Ks 13.1 mm/h · ψf 16 mm · Δθ 0.20 · fit R² 0.997"*). This conversion of a physically based law to a power law is what the lecture method requires; it is an approximation whose quality is reported by R^2.

Step 6: required depth. The net depth to refill the root zone is Eq. (5.27): $Z_n = \mathrm{clip}(f_T\,\mathrm{AWC}\,R_{\max};\ 25,\ 150)$ mm unless you move the slider.

11.3 Inlet flow area (Manning)

The flow cross-section at the inlet, A_0 (m^2), is obtained from Manning's equation (Manning, 1891), in SI units,

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

Provenance (Eq. 11.8): **[Literature]**: Manning (1891).

- **Furrow.** A V-shaped section with side slope $z = 1$ (H:V), depth limit $D = 0.25$ m: $A = z y^2$, $P = 2y\sqrt{1+z^2}$. Eq. (11.8) is solved for the flow depth y by 60 bisection steps for $Q = q$ (m^3 s^{-1}) and the slope S (with $S \geq 10^{-4}$); $A_0 = z y^2$. If $y > D$ the layer warns that the furrow overtops.
- **Border and basin.** A wide sheet: $A \approx y$ per unit width, $R_h \approx y$, so $y_0 = \big(q\,n/\sqrt{S}\big)^{0.6}$ with q in m^2 s^{-1}, and $A_0 = y_0$ (m^2 per m width).
- **D1 preset.** $A_0 = \big(q^2 n^2/792\big)^{0.37313}$ with q in m^3 min^{-1}, the empirical furrow relation of the lecture's worked example. **[Engineered, course-specific]**

11.4 Advance (volume balance)

Definition. For an inflow Q (m^3 min^{-1} per analysed width) the position $x(t)$ of the front is obtained from a volume balance of the water on the surface and in the soil (Walker and Skogerboe, 1987; Elliott and Walker, 1982):

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

The code also accepts an optional steady-intake term, $Z = k\tau^A + f_0\tau$ (Kostiakov–Lewis), in which Eq. (11.9) gains the term $f_0\, t\, w/(1+r)$ in the denominator, with r the iterated advance exponent (release 1.1; release 1.0 used a fixed $r = 0.55$); the interface does not expose f_0 (it is zero), and it exists so that the FAO Irrigation Paper 45 examples, which use it, can be reproduced. Here $\sigma_y = 0.8$ is the surface-storage shape factor, w the width analysed (furrow spacing, or 1 m for borders and basins), and σ_z the subsurface-storage shape factor

$$\sigma_z = \frac{A + r\,(1-A) + 1}{(1+A)(1+r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

where r is the exponent of the power-law advance $x = p\,t^r$, found by iterating (10 times, starting from $\sigma_z = 0.8$) from the times t_L and $t_{L/2}$ at which the front reaches L and $L/2$,

$$r = \frac{\ln 2}{\ln(t_L/t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

with each time found by bisection on Eq. (11.9). Eq. (11.10) is the standard shape factor of the volume-balance method for the Kostiakov intake law (Elliott and Walker, 1982). The advance time to position x is then $t_x(x) = x^{-1}(t)$, solved on a 300-point grid.

Alternatives (set by the worked examples; the first two are Eq. 11.12):

- *Given advance:* $t_x(x) = t_L\,(x/L)^{1/r}$ with $r = 0.55$ unless set (A7).
- *Field advance* (A11): $t_x(x) = c\,q^{-q_e}\,x^{x_e}$ with $c = 1/300$, $q_e = 1.5$, $x_e = 1.35$.

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes. * *Approximate* (D1, lecture formula):

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

with t_V the volume-balance solution of Eq. (11.9). **[Engineered]**: this is the closed-form approximation of the course notes, not a published general law.

11.5 Cut-off, opportunity time and the infiltrated depth

The **time to infiltrate the required depth** follows from inverting Eq. (11.6),

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

The **cut-off time** t_{co} is

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

The **opportunity time** at distance x is the time the water stays on the soil, $\tau(x) = t_{co} - t_x(x)$ for $t_{co} > t_x$, and the **infiltrated depth** is

$$Z(x) = k\left[t_{co} - t_x(x)\right]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

The *straight-line profile* of the lecture notes replaces $Z(x)$ by the line between Z_0 and Z_L (and, for the 85 %-of-L rule, a broken line through $(0.85L,\, Z_n)$); *Exact curve* draws Eq. (11.17). If $t_{co} < t_L$ the front never reaches the end of the field and the layer shows a warning (*the stream does not reach the end*).

11.6 Volumes, efficiencies and uniformity

With w the analysed width, $N = 300$ grid intervals (90 while dragging) and the trapezoidal rule, for $Z_i = Z(x_i)$:

$$\begin{aligned} &V_{\text{app}} = Q\,t_{co} \quad \big(\text{with cutback } c_b < 1:\ Q\,t_L + c_b\,Q\,(t_{co} - t_L)\big), \quad V_{\text{req}} = Z_n\,L\,w, \\ &V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx, \\ &V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx, \\ &V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0). \end{aligned} \tag{11.18}$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

The performance indices (Walker and Skogerboe, 1987; Burt et al., 1997) are

$$E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N} |Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}$$

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

E_a is the **application efficiency**, E_s the **storage (requirement) efficiency**, and CU the uniformity coefficient of Christiansen (1942) evaluated on the infiltrated depths. Because $V_{\text{app}} = V_{\text{inf}} + V_{\text{ro}}$ and $V_{\text{inf}} = V_{\text{st}} + V_{\text{dp}}$, the three fractions satisfy $E_a + \text{DP} + \text{RO} = 1$, which the layer prints as the closed water balance of one irrigation. The deficit V_{df} is the missing part of Z_n. The distance at which Z crosses Z_n, if it does, is $X = L\,(Z_0 - Z_n)/(Z_0 - Z_L)$.

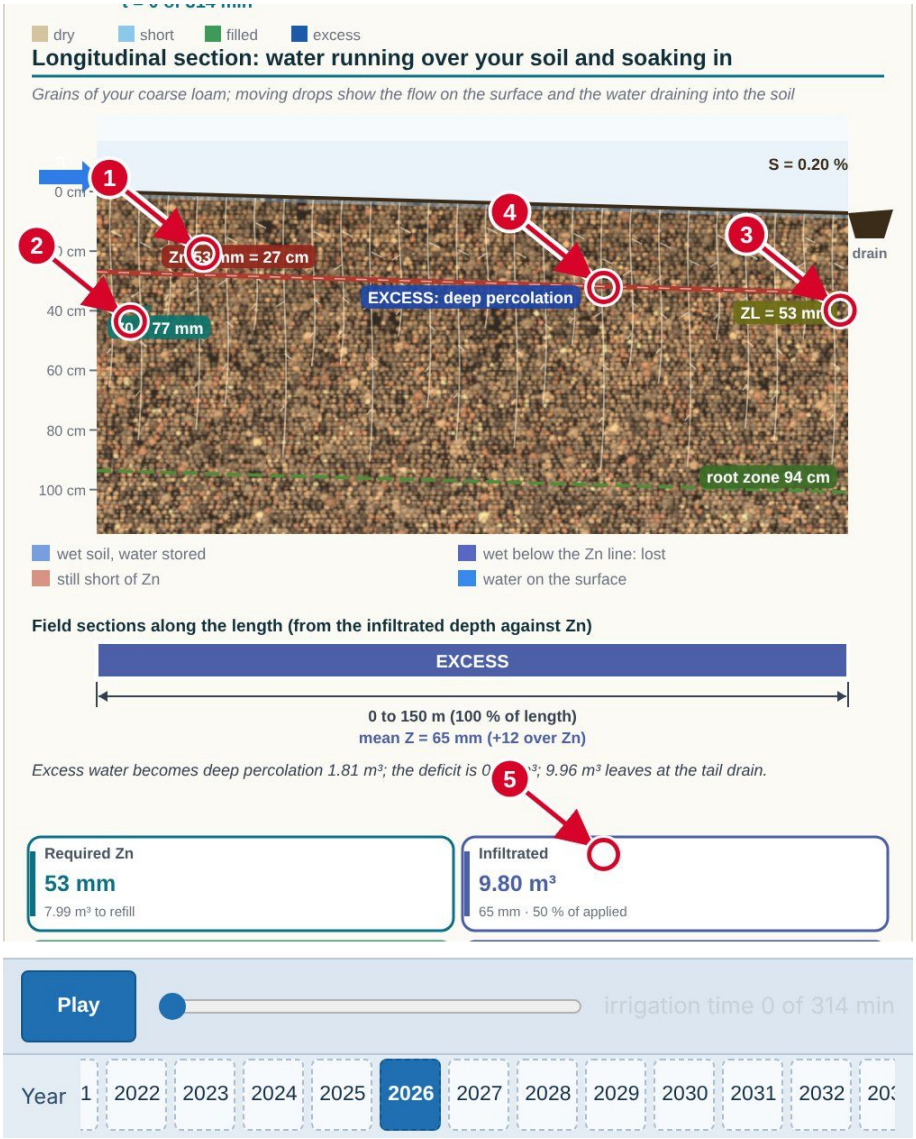


Figure 11.2: Surface design: longitudinal section with Z_0, Z_L, Z_n and the water balance of one irrigation. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

11.7 Hydrograph

The hydrograph (Section 4 of the layer) divides the inflow q into infiltration $I(t)$, runoff $R(t)$ and surface storage. It is built from the same advance and intake:

$$V_c(t) = w \sum_i k \left[\min(t, t_{co}) - \bar{t}_{x,i}\right]_+^A \Delta x \cdot s_c, \qquad s_c = \frac{V_{\inf}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}$$

Provenance (Eq. 11.20): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

where $\bar{t}_{x,i}$ is the advance time at the mid-point of interval i and the factor s_c makes the cumulative infiltration at cut-off equal to $V_{\inf}$ of Eq. (11.18). While $t_L \le t \le t_{co}$ the runoff is $R(t) = \max\{0,\ q - I(t)\}$. After cut-off the surface water *recedes*: with $V_s = \max\{0,\ V_{\text{app}} - V_{\inf} - \int R\,dt\}$ left on the surface, a triangular recession over $t_r = \max\{4,\ \min(0.3\,t_{co},\ 0.25\max(t_L, 10))\}$ min

drains it at the tail,

$$R(t) = \frac{2V_s}{t_r}\Big(1 - \frac{t - t_{co}}{t_r}\Big), \quad t_{co} < t < t_{co} + t_r. \tag{11.21}$$

Provenance (Eq. 11.21): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

The recession shape is **[Engineered]** (the course does not specify it); it conserves the volume V_s. Surface storage at time t is $V_{\text{sto}}(t) = q\,\min(t, t_{co}) - V_c(t) - \int_0^t R\,dt'$.

11.8 Cross-section and intake curve

The *Cross-section at the inlet* section draws, at the inlet, the wetted shape in a soil section generated from the soil's texture, and the intake curve $Z(\tau)$ of Eq. (11.6) beside the Green–Ampt curve $F(\tau)$ of Eq. (11.5), so that the fit can be inspected by eye.

11.9 Best inflow ("Your soil" mode)

In *Your soil* mode, q is chosen automatically from 25 equally spaced values between $q_{\min}$ and $q_{\max}$ of the system as the one that maximises $E_a\,E_s$. For furrows the upper limit is the non-erosive stream

$$q_{\max} = \min\Big\{q^{\text{sys}}_{\max},\ \frac{0.04}{S}\cdot\frac{1000}{60}\Big\}\ \text{L s}^{-1} \quad (S \text{ in \%}), \tag{11.22}$$

Provenance (Eq. 11.22): **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

about $0.67/S$ L s^{-1}, in line with the classical Criddle et al. (1956) limit $Q_{\max} \approx 0.63/S$. The choice is stored in the slider, which becomes manual as soon as you move it.

11.10 The design table

The *Design table* automates one step at a time towards a good design. The **target** is $E_a \geq 85\,\%$ with $E_s \geq 80\,\%$ (the root zone stays at least 80 % refilled). The score of a design is

$$\text{score} = \min(E_a, 1) - 3\,\max(0,\ 0.8 - E_s) - 3\,[c_b < 1]\ \max(0,\ 0.05 - \text{RO}) - 5\,\max(0,\ E_a - 1). \tag{11.23}$$

Provenance (Eq. 11.23): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

The search tries, in this order, ten single-variable moves — **cut-off, inflow, cutback, length, cut-off, cutback, inflow, length, cut-off, cutback** — and, for each move, evaluates a candidate list and keeps the best candidate **only if the score improves by more than 0.003**:

Move	Candidates
Cut-off	$t_{co} \times \{0.5, 0.58, 0.66, 0.74, 0.8, 0.85, 0.9, 0.95, 1, 1.05, 1.1, 1.2\}$
Inflow	12 values spaced geometrically from $q_{\min}$ to $q_{\max}$
Cutback	$c_b \in \{1, 0.85, 0.7, 0.6, 0.5, 0.4, 0.3, 0.25, 0.2\}$
Length	11 values spaced geometrically from $\max(L_{\min}, 30)$ to $\min(L_{\max}, 1.5L)$, rounded to 5 m

For every candidate other than a pure cut-off move, six cut-off times are also tried (0.62 to 1.02 times the "$E_s = 1$" cut-off). The search stops when the target is met or after the ten moves. Each accepted row lists q, L, cutback, cut-off, E_a, E_s, DP and RO; **tap a row (green when it meets the target) to apply it** to the sliders. If the target cannot be reached the layer says so and suggests a shorter field, tailwater reuse, or drip or sprinklers. The same search is run by the toolbox item *Optimise surface layout (best design)*.

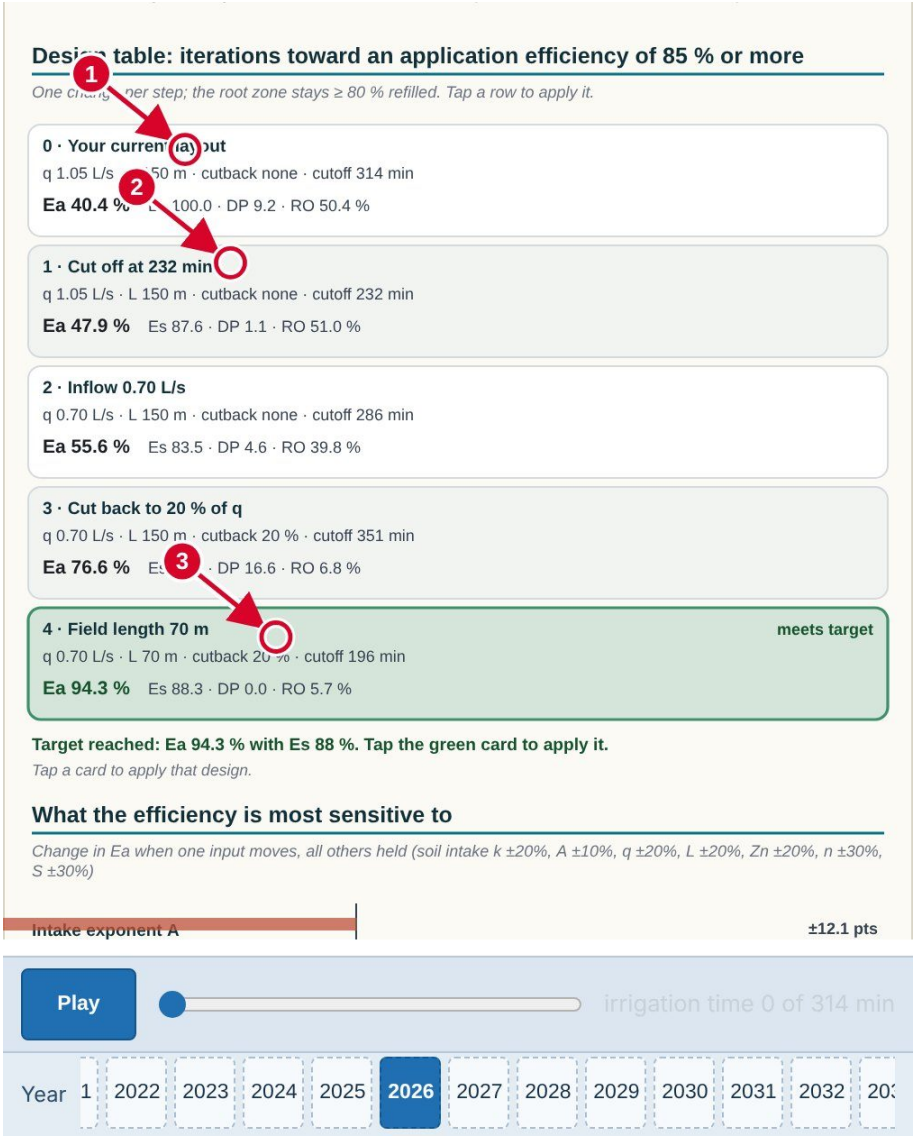


Figure 11.3: Surface design table: iterations toward $E_a \geq 85\,\%$, then the tornado chart. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

11.11 Sensitivity (tornado chart)

Each of seven inputs is moved up and down while the others are held, and the change in E_a is drawn as a bar (green raises E_a, red lowers it):

Input	Intake k	Intake A	Inflow q	Length L	Required depth Z_n	Roughness n	Slope S
Perturbation	±20 %	±10 %	±20 %	±20 %	±20 %	±30 %	±30 %

The bars are sorted by the larger absolute change, and the printed value is $\max(|\Delta E_a^-|, |\Delta E_a^+|)$ in percentage points (*pts*). The test is local: it is the change around the *present* layout, not a global importance.

11.12 Season and farm

Using the irrigation record of Engine 2 for the year of the working date, with $\sum I$ the net irrigation of that year (mm):

$$n_{\text{ev}} = \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g,$$
$$N_{0M} = \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}), \tag{11.24}$$

Provenance (Eq. 11.24): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

and the seasonal pumping time is $n_{\text{ev}} T_{\text{tot}}$ hours. A depth of Z_{season} mm equals $10\, Z_{\text{season}}$ m^3 per hectare. N_{0M} is the number of furrows or strips that can run at once from the farm supply Q_s.

11.13 Irrigation schedule and “When to irrigate”

The last two sections are common to all three design layers, so they are described once, here.

Schedule table. The next eight irrigation dates after the working date are listed from the DSS (Engine 2, Section 5.2): the sensor in layer j_s reaching the trigger depletion f_T. For each date the table gives

Column	Definition
#	order, soonest first
Date	calendar date of the analogue year (Chapter 4)
In	*today*, *tomorrow*, or *n days*
Net mm	the DSS irrigation depth that day, I_k
Pumped mm	gross depth I_k/E_a with the design’s application efficiency (min. $E_a = 0.05$)
Run time	time of one set or one round of the design
Gap to next	days to the following irrigation

Intervals. Over the year of the working date, with irrigation days $d_1 < d_2 < \ldots$ (irrigations above 0.5 mm), the gaps $g_j = d_{j+1} - d_j$ give the mean, the shortest and the longest interval.

Advice message. The message *When to irrigate* is generated from these numbers: the *next irrigation date* and its net and pumped depth; *“Then irrigate about every $\bar{g}$ days (between $g_{\min}$ and $g_{\max}$ days, shorter in hot weather and longer after rain)”*; the efficiencies of the design (E_a, E_s) and the time of one round; and the rule *“If a rain above 10 mm is forecast, delay the irrigation by a day or two”*. If no irrigation is triggered, it says to *keep the sensor at z_s cm and irrigate when it reaches depletion f_T*.

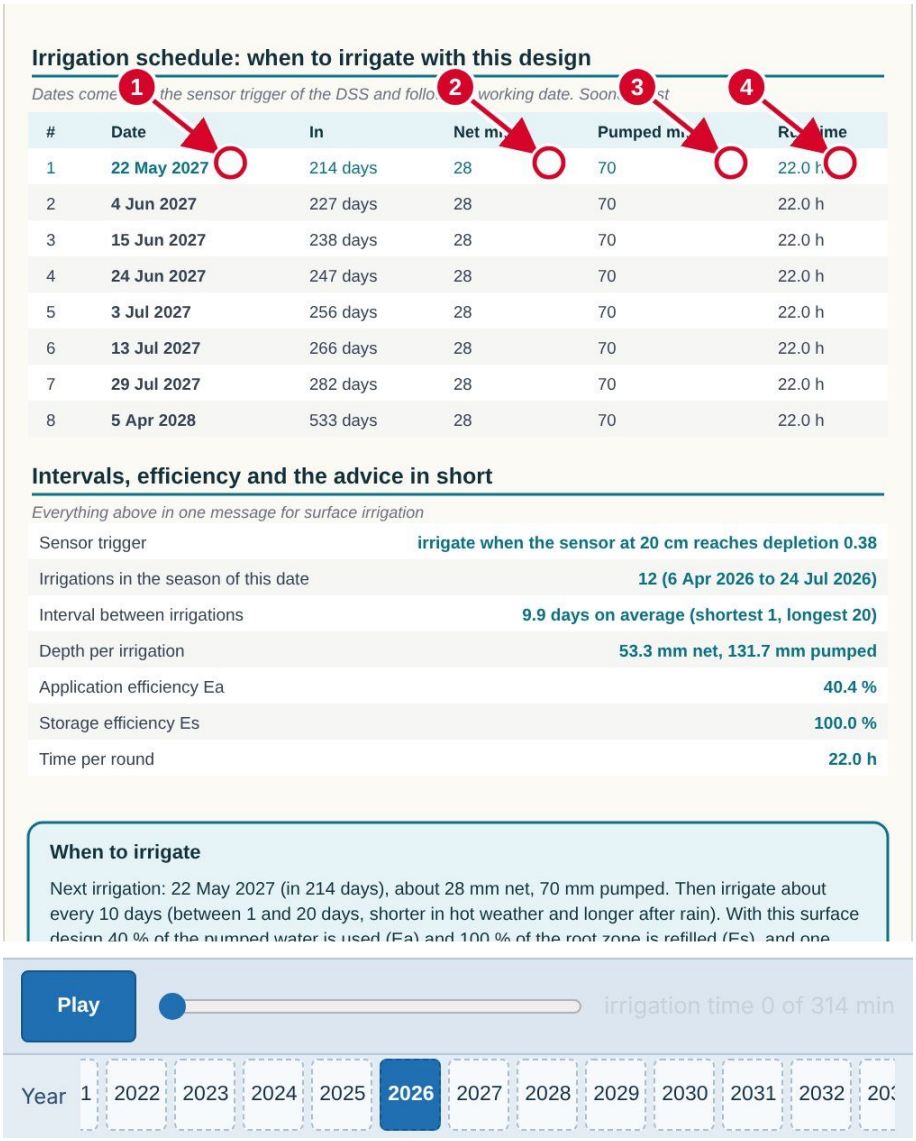


Figure 11.4: The irrigation schedule and the “When to irrigate” message at the end of the surface design. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

Reading it. The first row is today’s decision. If the *Gap to next* is short (2–4 days) the trigger is wet or the soil store is small; if it is long (10+ days) the soil store is large or the weather cool. Move the **Date** to see how the schedule changes with the season, and move the **Trigger** or **Sensor depth** to see the dates shift.

Chapter 12

Drip irrigation design (S3)

The drip layer designs the **emitter, its placement, the lateral and the pipe network** for the soil, crop and trigger set in the rest of the program. It answers four questions in order: how big a wet bulb does one emitter make, how much of the water it delivers stays in the root zone, how uneven is the delivery along the lateral, and what pump and pipes does the farm need. Everything is a closed-form hydraulic or geometric model, recomputed at every slider move.

The layer’s sections, in order: control bar and inputs; soil moisture around the driest, average and wettest emitter (Play animates the wetting); where the water goes; uniformity along the lateral; growth of the wetted bulb with time; result tiles; layout and infrastructure; irrigation schedule; the design table; the sensitivity chart; season and farm; and the “When to irrigate” advice. The last two are shared with the other design layers and are described at the end of Chapter 11.

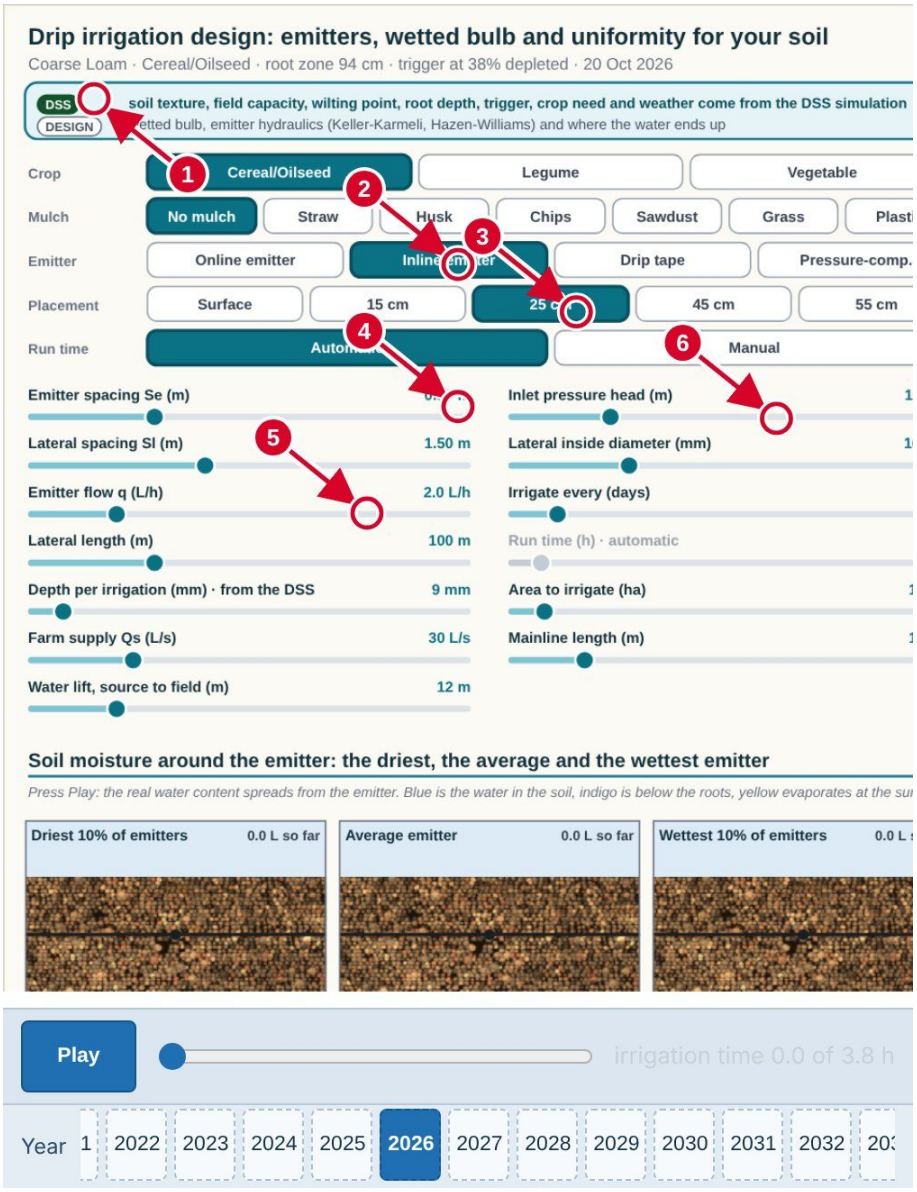


Figure 12.1: Drip design: inputs and the soil moisture around the driest, average and wettest emitter. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

12.1 Inputs, step by step

1. Open *Irrigation design* ▸ *Drip irrigation design*. The crop, soil, root zone, trigger and date come from the DSS (the green badge); the choices below are the design (the amber badge).
2. **Emitter** (a row of four chips). Choosing one sets the nominal flow, spacing, lateral diameter and inlet head of Table 12.1.
3. **Placement**: *Surface* (design only), or a buried line at 15, 25, 45 or 55 cm. A buried choice also sets the explorer's irrigation method to the matching Daisy run (M04, M06, M09 or M10), so that the rest of the program shows the same depth.
4. **Run time**: *Automatic* (time to deliver the depth at the driest emitters, Eq. 12.17) or *Manual*.
5. Move the sliders of Table 12.2. The crop, mulch and irrigation method rows at the top are the same controls as the Properties panel.

Table 12.1. Emitter types (defaults are set when the chip is chosen; c_v manufacturer coefficient

of variation, x flow exponent, n_e emitters per plant or point, $H_{\min}$ compensation threshold).

Emitter	q (L h^{-1})	S_e (m)	D (mm)	H_0 (m)	c_v	x	n_e	$H_{\min}$ (m)
Online emitter	4	0.75	16	10	0.07	0.5	1	—
Inline emitter	2	0.50	16	10	0.04	0.5	1	—
Drip tape	1	0.20	16	10	0.08	0.6	2	—
Pressure-compensating	2	0.60	16	15	0.03	0.05	1	8

Table 12.2. Drip sliders (range, step, default).

Control	Range, step	Default	Meaning
Emitter spacing S_e (m)	0.1–1.5, 0.05	by emitter	Along the lateral
Lateral spacing S_l (m)	0.5–3, 0.05	1.5 (1.0 for cereal, legume; 1.5 for vegetable, reset when the crop changes)	Between laterals
Emitter flow q (L h^{-1})	0.5–8, 0.1	by emitter	Nominal flow at H_0
Lateral length L_L (m)	20–300, 5	100	
Inlet pressure head H_0 (m)	4–30, 0.5	10 (15 for PC)	
Lateral inside diameter D (mm)	10–32, 1	16	
Irrigate every (days)	1–10, 1	2	Sets the default depth
Run time (h)	0.25–48, 0.25	automatic	Greyed while *Automatic*
Depth per irrigation Z_n (mm)	3–80, 1	from the DSS	Tap the label to reset
Area to irrigate A_f (ha)	0.2–10, 0.1	1	
Farm supply Q_s (L s^{-1})	2–120, 1	30	
Mainline length L_m (m)	20–600, 10	120	
Water lift, source to field (m)	0–60, 1	12	

Depth per irrigation. By default the depth is the crop need over the interval, capped by what the root zone can hold at the trigger,

$$Z_n = \min\{ Z_n^{\mathrm{DSS}},\ \max(4,\ \mathrm{ET}_{\mathrm{pk}}\,\Delta t)\}\ \ \mathrm{mm}, \tag{12.1}$$

Provenance (Eq. 12.1): **[Engineered]**: depth and demand rules chosen for the workbench.

where Δt is the chosen interval, $Z_n^{\mathrm{DSS}} = \mathrm{clip}(f_T\,\mathrm{AWC}\,R_{\max}; 25, 150)$ is Eq. (5.27), and $\mathrm{ET}_{\mathrm{pk}}$ is the peak five-day moving mean of simulated evaporation plus transpiration over the year, bounded to 2.5–9 mm d^{-1},

$$\mathrm{ET}_{\mathrm{pk}} = \mathrm{clip}\Big\{ \max_d \tfrac{1}{5}\sum_{j=0}^{4}(E_{d+j} + T_{d+j});\ 2.5,\ 9 \Big\}. \tag{12.2}$$

Provenance (Eq. 12.2): **[Engineered]**: depth and demand rules chosen for the workbench.

The mean evaporative demand used for surface drying, E_0, is the mean of the simulated daily evaporation on days 141–250 of the year (when positive), clipped to 1–5 mm d^{-1}.

12.2 Lateral hydraulics: head loss and flow along the lateral

An emitter lateral of length L_L carries $N = \mathrm{round}(L_L/S_e)$ intervals and $N+1$ emitters, indexed $j = 0 \ldots N$ from the inlet. The friction head loss in the segment feeding emitter j is the Hazen–

Williams equation (Hazen and Williams, 1905), in SI form with a roughness coefficient $C = 140$ for plastic pipe,

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

The head at emitter j is $h_j = h_{j-1} - h_{f,j-1}$ with $h_0 = H_0$ (floored at 0.5 m). Emitter discharge depends on the local head through the emitter equation (Keller and Karmeli, 1974),

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating};\ h_j/H_0 \geq 0.01), \tag{12.4}$$

Provenance (Eq. 12.4): **[Literature]**: Keller and Karmeli (1974) emitter law.

with $x = 0.5$ for a fully turbulent orifice or long-path emitter and $x = 0.6$ for the drip tape of Table 12.1. The pressure-compensating (PC) emitter is modelled as

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \geq H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

Provenance (Eq. 12.5): **[Engineered]**: pressure-compensating emitter model with chosen constants.

i.e. a nearly flat discharge above $H_{\min} = 8$ m (a 4 % change in flow per 100 % change in head) and the orifice law below it. The flows depend on the heads and the heads on the flows, so the pair (12.3)–(12.4) is **iterated to convergence** from the uniform start $q_j = q$, $h_j = H_0$ (the first three passes are plain substitution, then the flows are averaged with the previous pass; the iteration stops when the largest relative change in any emitter flow is below 10^{-7}, at most 500 passes). Version 1 used four passes, which is sufficient only when the head loss is small; the converged solver agrees with an independent solution to 3×10^{-7} for head losses up to 60 % of the inlet head. When the friction loss exceeds 20 % of the inlet head, or the iteration fails to converge, the layer prints a red warning, because uniformity then falls and the design is outside the usual rule of thumb. The mean and minimum discharge are then

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600} \;\; (\mathrm{L\,s^{-1}}), \tag{12.6}$$

Provenance (Eq. 12.6): **[Literature]**: definitions of mean and minimum discharge.

and the loss along the whole lateral is $h_f = H_0 - h_N$, with $H_{\text{end}} = h_N$ the head at the far end (Section “Uniformity along the lateral”).

Emission uniformity. The design emission uniformity follows Keller and Karmeli (1974),

$$E_u = \text{clip}\Big\{\Big(1 - \frac{1.27\, c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

Provenance (Eq. 12.7): **[Literature]**: Keller and Karmeli (1974).

The first factor is the manufacturing variation; the second is the hydraulic variation along the lateral (here $q_{\min}$ replaces the classical quarter-lowest mean, a conservative choice). The *combined* relative standard deviation of the delivery, which drives the spread of bulbs below, is

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

Provenance (Eq. 12.8): **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Reading the lateral plot: the solid line is q_j/q in per cent against distance, the dashed line is the head as a percentage of H_0. A lateral that is too long or too narrow shows a steep fall of both. For a non-compensating emitter a 10 % loss of head costs about 5 % of flow ($x = 0.5$); the PC emitter stays flat until the head drops below $H_{\min}$.

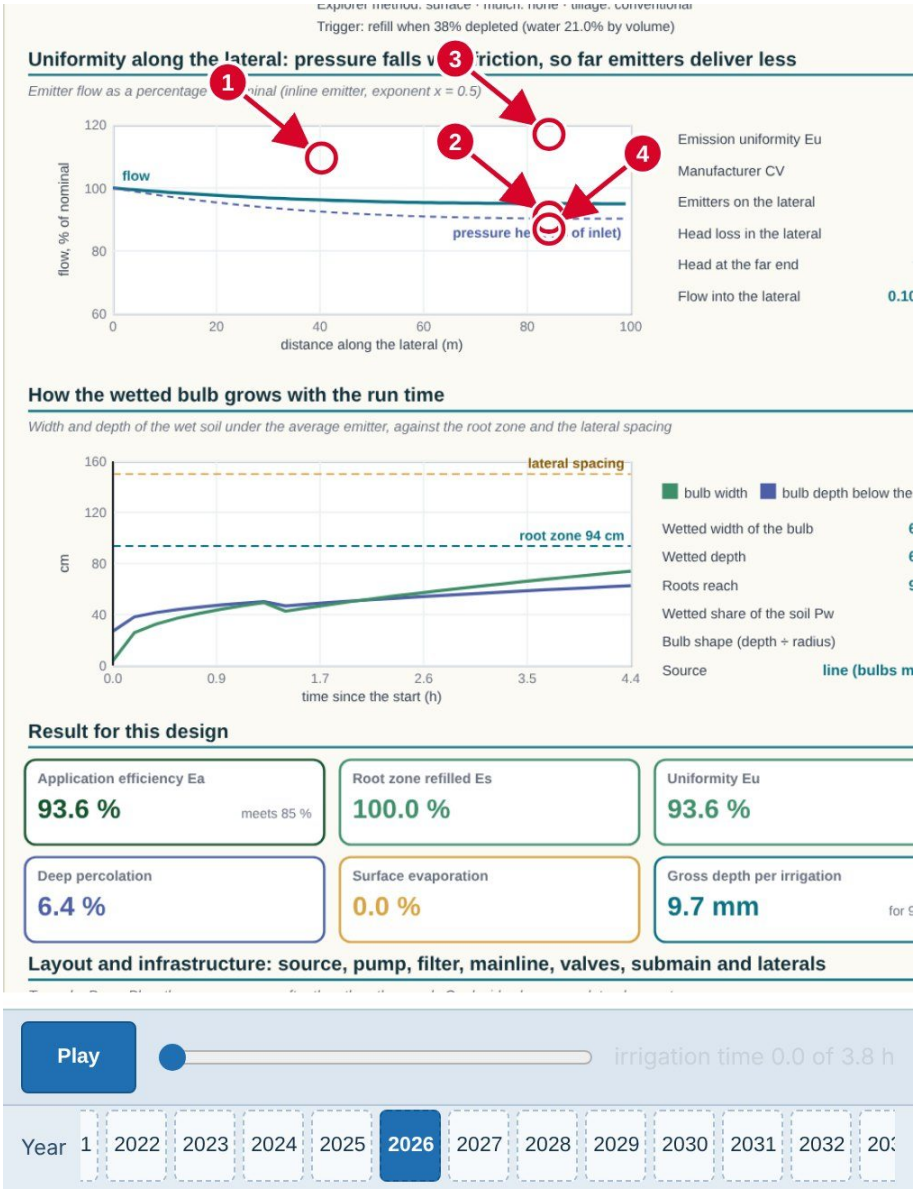


Figure 12.2: Drip design: uniformity along the lateral and the growth of the wetted bulb with run time. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

12.3 The wetted bulb

Water from an emitter spreads into soil that is initially at the trigger water content and is refilled up to field capacity. The layer models the wetted volume around a *buried* point source as a **half-ellipsoid** below the emitter and a short one above it (an **[Engineered]** simplification of the wetted-volume shapes used in drip design; the aspect ratio is derived from the soil, not read from a table).

Water to be stored per unit soil volume.

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Shape. With emitter depth z_d (mm), horizontal radius r (mm) at the emitter level, lower semi-axis $a\,r$ and upper semi-axis $u\,r$ with $u = 0.6$, the wetted radius at depth z is

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Aspect ratio. The shape factor a (depth over radius) increases when gravity is strong relative to capillarity, i.e. for a coarse soil and a low flow. It is evaluated from the ratio of the saturated conductivity to the radial capillary spreading rate $q/(2\pi\lambda^2)$ (the form is a scaling argument and the constants are **[Engineered]**),

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

with K_s in mm h^{-1}, q in L h^{-1} and ψ_f the wetting-front suction of Eq. (11.3) in mm. The result is a bulb that is round ($a = 1$) for a fine soil and elongated downwards (up to $a = 1.9$) for a sand.

Radius. For an emitter that delivers V litres, the wetted volume is $V_w = V\cdot 10^6/\Delta\theta$ mm^3 (1 L $= 10^6$ mm^3). The radius r is found by bisection (38 steps between 5 and 3500 mm) so that the volume of revolution of Eq. (12.10), integrated in 56 layers,

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

equals V_w. When the volume is *line-like*, the cross-section instead of the volume is conserved: for drip tape ($S_e < 0.34$ m) or when neighbouring bulbs overlap ($2r > S_e$ with $S_e < 1.2$ m), the water is spread along the line, $A_w = V\cdot 10^6/(S_e\cdot 1000)/\Delta\theta$ mm^2 per unit length of lateral and the profile area is $2\int \rho\,dz$. The panel is then labelled "line (bulbs merge)" rather than "point". The wetted share of the soil surface is

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Where the water goes. Of the V litres, the part that falls inside the bulb below the root-zone depth $R_{\max}$ is lost to deep percolation, a fraction B_{below}/V_w of V:

$$V_{\text{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

and the part evaporated from the wet surface patch is limited to half the delivery,

$$V_{\text{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\} \quad (\text{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

(for a line source the area is $2\rho(0)S_e \cdot 10^{-6}$ m^2 per emitter). The factor 2 corresponds to about two days of evaporation from a wet surface patch of mean demand E_0; this is an **[Engineered]** simplification of the two-stage evaporation of Chapter 9. For a surface line ($z_d = 0$) the bulb is half-spherical and the evaporating patch is largest, which is why surface drip stores less than buried drip in the table. The remainder is available to the roots, $V_{\text{root}} = V - V_{\text{dp}}^{(1)} - V_{\text{ev}}$.

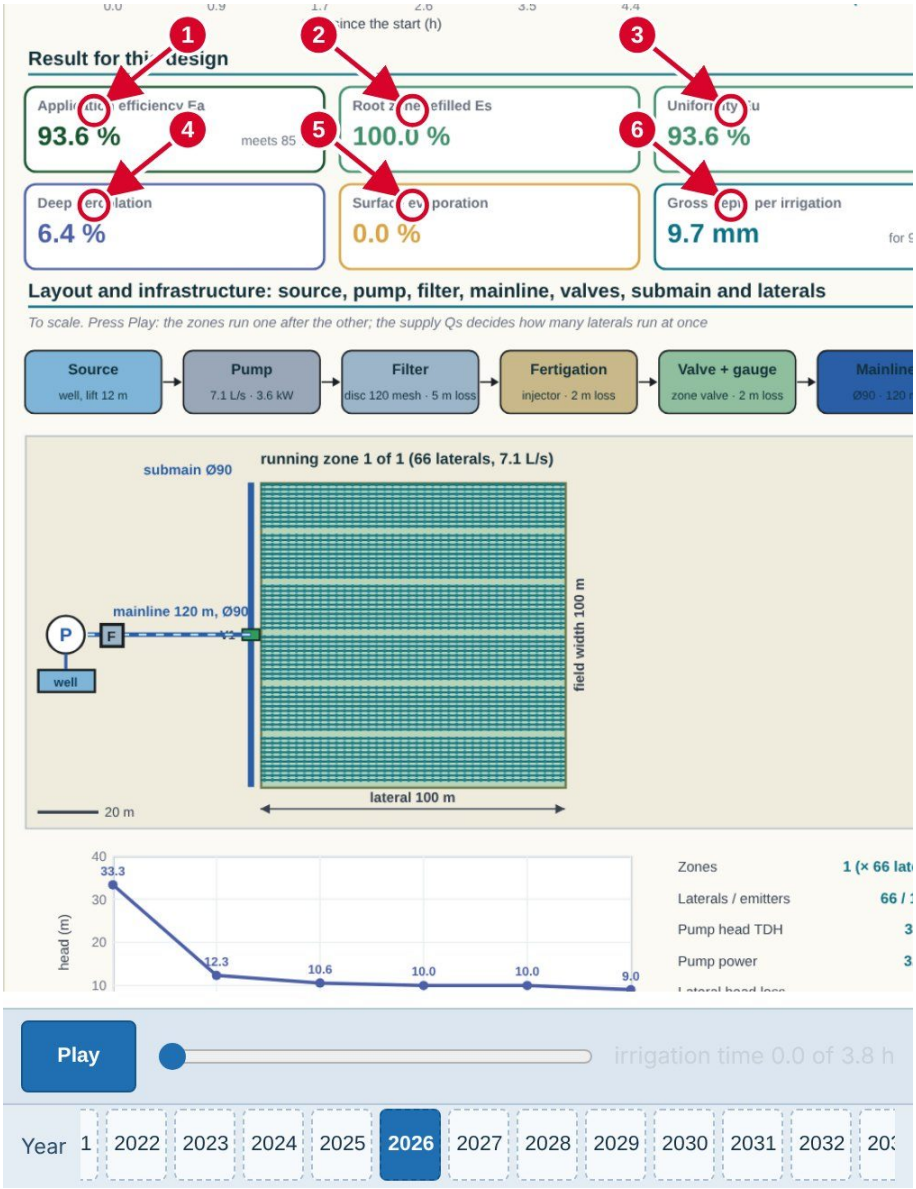


Figure 12.3: Drip design: result tiles and the water partition of one irrigation. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

12.4 Spread between emitters and the efficiencies

Emitters do not all deliver the same volume. The layer evaluates five representative emitters, at standard-normal quantiles $z \in \{-1.28, -0.52, 0, 0.52, 1.28\}$ (the 10th, 30th, 50th, 70th and 90th percentiles, equal weights), each delivering

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1 + \sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

where t is the run time (h). For each, the bulb, deep percolation and evaporation are computed with Eqs. (12.10)–(12.15). The water that the root zone can accept is limited by the need,

$$V_{\text{req}} = Z_n\,S_e\,S_l\ \ (\text{L}), \qquad V_{\text{st},i} = \min\{V_{\text{root},i},\ V_{\text{req}}\}, \qquad V_{\text{ex},i} = V_{\text{root},i} - V_{\text{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Because 1 mm over 1 m^2 is 1 L, V_{req} is the depth needed over the area $S_e S_l$ served by one emitter. **Automatic run time** is the time at which the *driest* emitters deliver the need,

$$t = \text{clip}\Big\{\frac{V_{\text{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \text{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

The efficiencies are means over the five quantiles ($\langle\cdot\rangle$):

$$E_a = \frac{\langle V_{\text{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\text{st}}\rangle}{V_{\text{req}}}, \quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)} + V_{\text{ex}}\rangle}{\langle V\rangle}, \quad \text{EV} = \frac{\langle V_{\text{ev}}\rangle}{\langle V\rangle}, \quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

so that $E_a + \text{DP} + \text{EV} = 1$. Reading them: E_a is how much of the pumped water ends up stored in the root zone; E_s is how well the root zone is refilled (a value below 85 % turns the "Root zone refilled" line red and says *short*); E_u is the lateral uniformity; Z_{gross} is the depth to pump per irrigation. The layer colours $E_a \geq 85$ % green, 65–85 % amber and below 65 % red; $E_u \geq 90$ % is green and 80–90 % amber.

12.5 Layout and infrastructure

The farm side of the design uses the area A_f and the supply Q_s.

Laterals and zones.

$$W_f = \frac{10^4 A_f}{L_L}, \quad N_L = \Big\lfloor \frac{W_f}{S_l} \Big\rfloor, \quad n_{\text{zone}} = \Big\lceil \frac{N_L\,Q_{\text{lat}}}{Q_s} \Big\rceil, \quad n_l = \Big\lceil \frac{N_L}{n_{\text{zone}}} \Big\rceil, \quad Q_z = n_l\,Q_{\text{lat}}, \tag{12.20}$$

Provenance (Eq. 12.20): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

where W_f is the width of the field perpendicular to the laterals, n_{zone} the number of zones that must run in turn so that Q_s is not exceeded, and Q_z (L s^{-1}) the flow of one zone. The total number of emitters is $N_L(\text{round}(L_L/S_e) + 1)$.

Pipe sizing. Submain length $L_{sm} = n_l S_l$, mainline length L_m (slider). The pipe diameter is the smallest of the commercial series

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\} \text{ mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

for which the mean velocity $v = 4Q_z/(\pi D^2) \le 1.5$ m s^{-1} (a common design limit that keeps friction and water-hammer moderate; see Keller and Bliesner, 1990, for pipe design practice). The submain loss is 0.4 of the full-length Hazen–Williams loss, Eq. (12.3) with $Q_j = Q_z$ (the Christiansen (1942) reduction factor F for a pipe with many equally spaced outlets is $F \to 1/2.852 = 0.35$ for $m = 1.852$, so 0.4 is slightly conservative), and the mainline loss is the full value:

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852} D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852} D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Pump. The total dynamic head and the shaft power are

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\text{ m}, \qquad P = \frac{\rho g\, Q_z\, \text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

The constant 9 m covers the filter, valves and fittings, and $\eta = 0.65$ is the overall pump-and-motor efficiency; both are **[Engineered]** design allowances, not measured values. The laterals' own loss is already inside H_0.

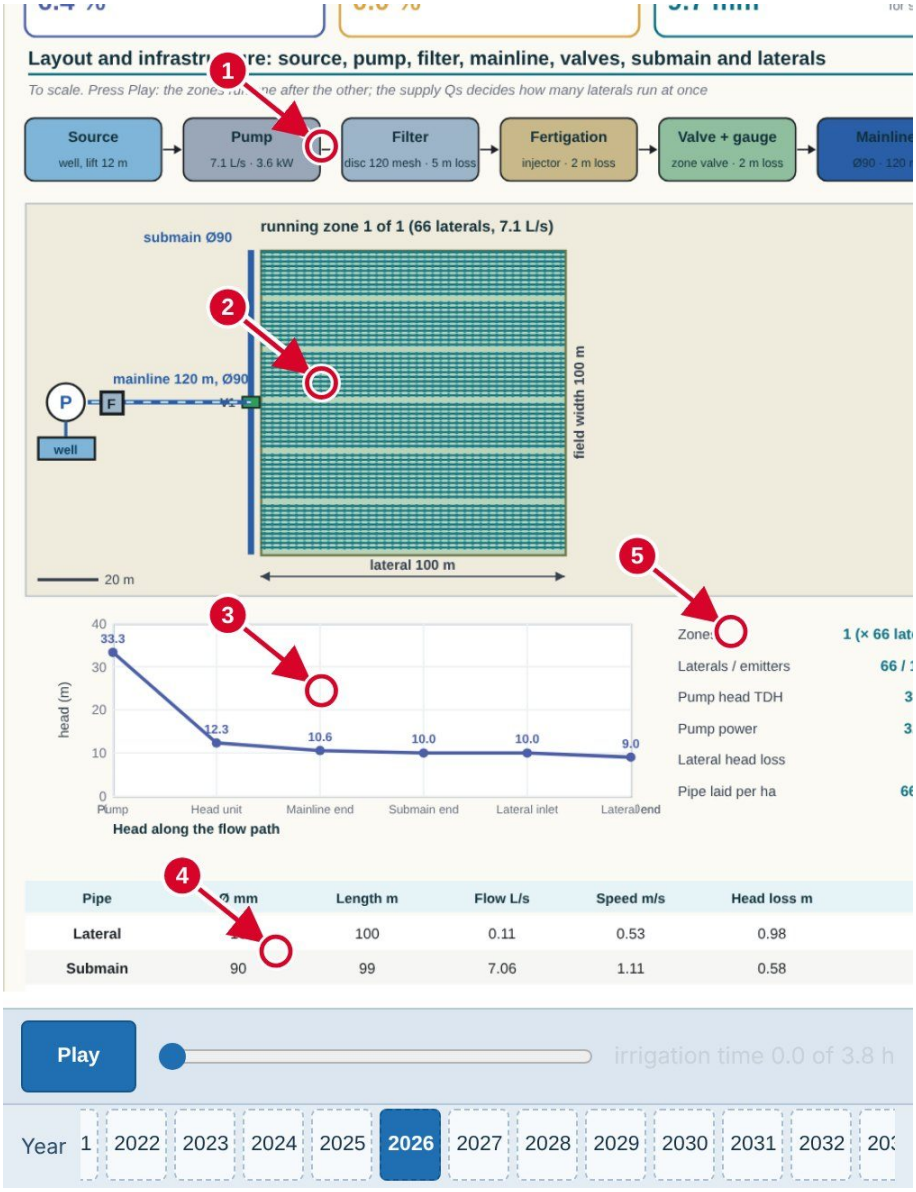


Figure 12.4: Drip design: layout and infrastructure, to scale, with the zones running in turn. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

The drawing shows the source, pump, filter, mainline, valves, submain and laterals to scale; during *Play* the zones run one after the other, and the head along the flow path is drawn underneath. The table beside it gives zones (× laterals per zone), laterals and emitters, TDH, pump power, lateral head loss and pipe laid per hectare, $(N_L L_L + L_{sm})/A_f$.

12.6 Season and farm

With y_I the net seasonal irrigation of the DSS for the year of the working date, E_u and E_a as above,

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

The layer prints these as a table (net irrigation need and number of events, gross water per irrigation and per season in mm and $\text{m}^3\ \text{ha}^{-1}$, peak crop need, flow per hectare, sets for A_f with Q_s, hours of pumping and energy). Then the *Irrigation schedule* and the *When to irrigate* advice, described at the end of Chapter 11, close the layer.

12.7 The design table

The optimiser takes the current design and tries one change at a time, in this order: **placement depth, run time, emitter flow, emitter spacing, placement depth, run time, emitter kind, lateral spacing, emitter flow, run time**. For each move the candidates are:

Move	Candidates
Placement depth	Surface, 15, 25, 45, 55 cm
Run time	$t_0 \times \{0.55, 0.65, 0.75, 0.85, 0.95, 1, 1.1, 1.25\}$, with t_0 the automatic run time of the current design
Emitter flow q	0.6, 1, 1.5, 2, 3, 4, 6, 8 L h^{-1} (run time reset to automatic)
Emitter spacing S_e	0.2, 0.3, 0.4, 0.5, 0.6, 0.75, 1, 1.25 m
Emitter kind	the four emitters of Table 12.1, each with its own default q, S_e
Lateral spacing S_l	0.75, 1, 1.25, 1.5, 1.8, 2.2, 2.6 m

The score is

$$\text{score} = \min(E_a, 1) - 2.5\, \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

so a design is penalised when the root zone is not refilled to 90 %. A move is accepted when its best candidate improves the score by more than 0.002; the search stops when $E_a \geq 85$ % **and** $E_s \geq 85$ %, or after ten moves. One move is performed per screen update, and the search pauses while a slider is being dragged. Each accepted row shows the step, q, S_e, depth, run time, E_a, E_s and DP; a green row meets the target, and **tapping any row applies that design** to the sliders (including the placement depth and the explorer's irrigation method). If the target is not met, the layer reports the best found and suggests a shorter interval or a different emitter.

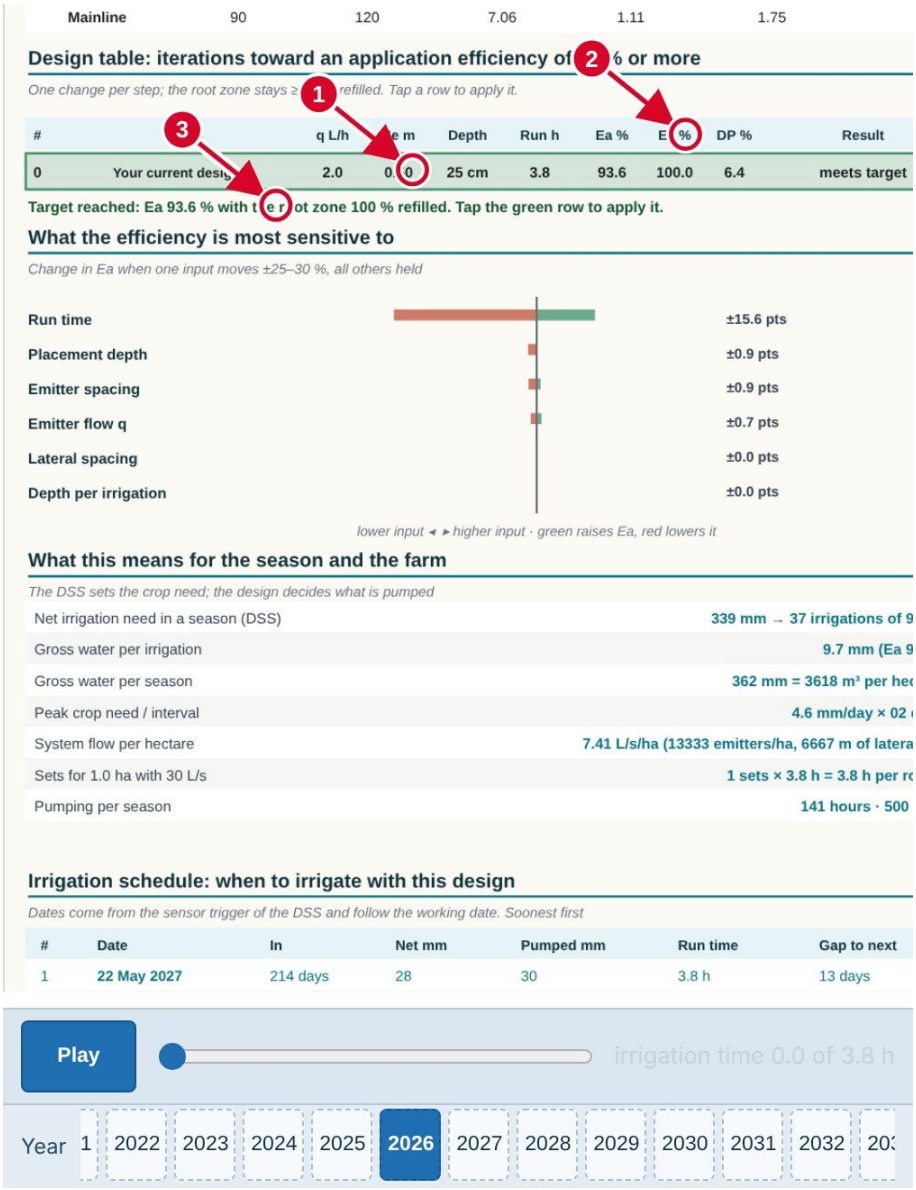


Figure 12.5: Drip design table and the sensitivity chart. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

12.8 Sensitivity

As for the surface layer, each input is moved up and down and the change of E_a is shown as a bar: emitter flow q (± 25 %), emitter spacing (± 25 %), lateral spacing (± 25 %), placement depth (± 30 %, at least 10 cm), depth per irrigation (± 25 %) and run time (± 20 %). The result is a local sensitivity at the present design.

12.9 How to use the layer

- **To choose an emitter for a soil.** Select the soil in the Properties panel, open the layer and compare the four chips with the same run time: the tile of E_a and the line chart of the bulb show which emitter keeps the bulb inside the root zone.

- **To check a lateral length.** Increase L_L until E_u turns amber; the lateral plot shows where the head falls below the pressure at which the emitter holds its flow.
- **To size a pump.** Fix A_f, Q_s, L_m and the lift; read TDH and kW from the layout table.
- **To see the effect of the trigger.** Move *Trigger* in the Properties panel: f_T changes $\Delta\theta$ (Eq. 12.9) and Z_n (Eq. 12.1), and the bulb and the run time adjust.

Chapter 13

Sprinkler irrigation design (S4)

The sprinkler layer designs the **nozzle, pressure, spacing and layout** of a sprinkler system for the soil, crop, slope and wind of the case, and then checks the result against the standard course rules (spacing, application rate, uniformity, pipes and pump). It has two parts that are built on the same run: a *simulation* part (the overlapped spray pattern, the wind drift, the soil intake that causes runoff, and the efficiencies) and a *textbook* part (the ten design checks, the pipe and pump sizing, the kinds of sprinkler system).

The layer reads top to bottom: **A** site, **B** sprinkler and layout, **C** operation and **D** farm and pipes (the inputs); the layout in plan; the profile along the lateral; the intake against the application rate; result tiles and the water partition; the textbook checks and system plans; layout and infrastructure; the irrigation schedule; the design table with the sensitivity chart; the season and the advice.

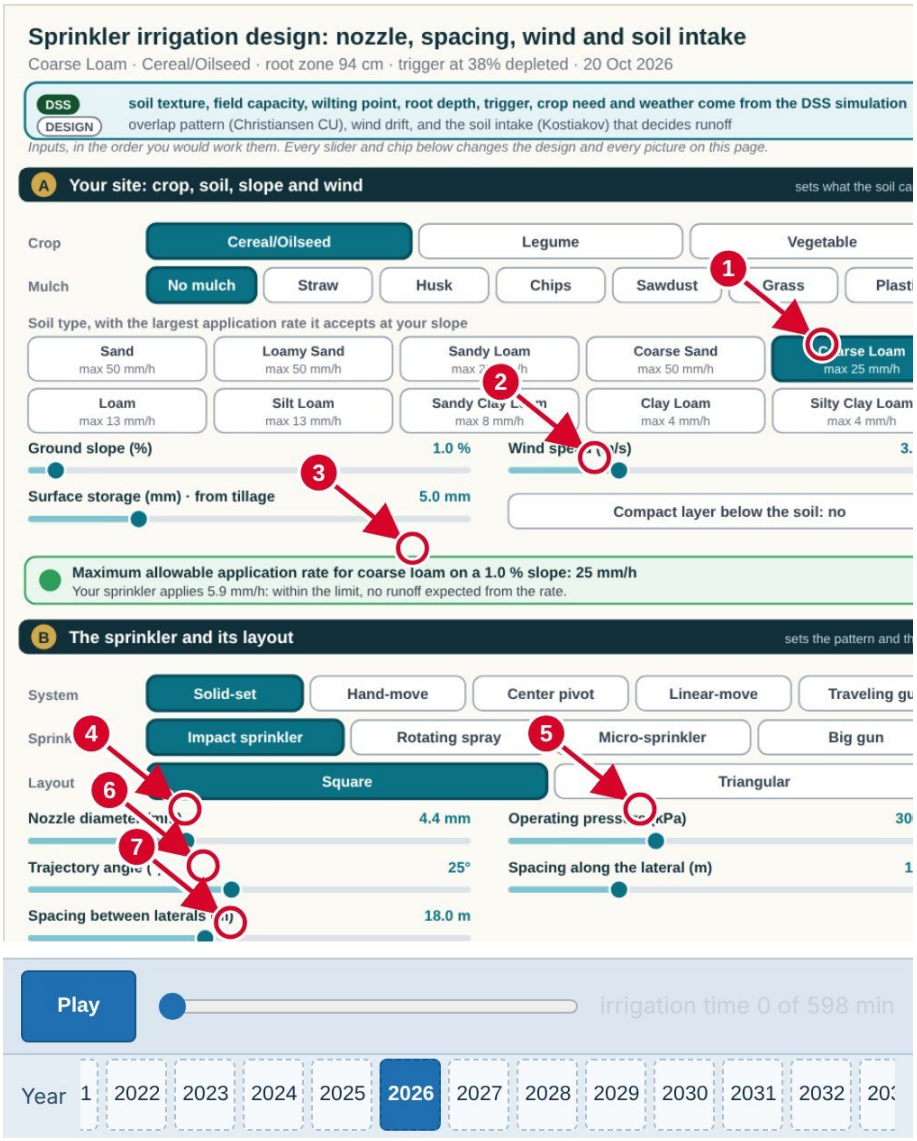


Figure 13.1: Sprinkler design: inputs A to D and the layout in plan with the overlapping spray patterns. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

13.1 Inputs, step by step

A. Your site. Crop, mulch and irrigation method are the same controls as in the Properties panel. The soil row shows each soil with the largest application rate it accepts at the present slope (Table 13.3). Sliders: ground slope S (0–16 %, step 0.5, default 1), wind speed u (0–12 m s^{-1}, step 0.5, default 3), surface storage S_d (0–20 mm, step 0.5; default from tillage, below) and a toggle *Compact layer below the soil* (which lowers the allowed rate, Table 13.3). A banner states the maximum allowable application rate for the soil and slope and whether the present sprinkler is within it.

B. The sprinkler and its layout. *System* (five chips, see *Kinds of sprinkler system* below), *Sprinkler* (four types, Table 13.1), *Layout* (square or triangular), then sliders for the nozzle diameter d, operating pressure P, trajectory angle α (8–45°, default 25°), spacing along the lateral S_s and between laterals S_l. Choosing a type resets d, P, S_s, S_l to its defaults and run time to automatic.

Table 13.1. Sprinkler types: ranges (min–max, step) and defaults; c_R range coefficient, m profile

exponent, F drift factor. The impact value $c_R = 1.35$ is the Keller and Bliesner (1990) coefficient for impact sprinklers (changed from 1.15 in version 1; the earlier value under-predicted the wetted diameter of their worked example by 14 %). The values for the other three types are **[Engineered]** and have not been verified against published data.

Type	d (mm)	P (kPa)	S_s (m)	S_l (m)	c_R	Profile	F
Impact	2.4–8, 0.1 (4.4)	200–500, 10 (300)	6–30, 1 (12)	6–36, 1 (18)	1.35	$1 - x^{1.3}$	1.0
Rotating spray	1.5–5, 0.1 (3.2)	70–250, 5 (170)	3–15, 0.5 (6)	3–15, 0.5 (6)	0.80	ring	1.6
Micro-sprinkler	0.8–2.5, 0.05 (1.3)	100–250, 5 (180)	1.5–8, 0.25 (3)	1.5–8, 0.25 (3)	0.70	ring-2	0.7
Big gun	10–25, 0.5 (16)	350–700, 10 (500)	20–80, 1 (30)	20–90, 1 (36)	0.95	$1 - x$	1.3

C. How you run it. *Run time* automatic or manual; depth per irrigation Z_n (5–100 mm; default Eq. 5.27, tap to reset); run time T (10–1200 min, step 5); operating hours per day (4–24, default 16).

D. Farm, water supply and pipes. Area A_f (0.5–40 ha, step 0.5, default 5), farm supply Q_s (5–200 L s^{-1}, default 40), lateral length L_L (40–500 m, step 10, default 150), mainline length L_m (20–600 m, default 120) and lift (0–60 m, default 12).

Tillage and surface storage. Unless set by hand, the depression storage S_d follows the tillage of the Properties panel: 5 mm conventional, 3.5 mm reduced, 2 mm none. On a slope the effective storage is reduced,

$$S_d^{\text{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

Provenance (Eq. 13.1): **[Engineered]**: depression storage reduction with slope.

This is an **[Engineered]** relation: storage in depressions decreases as the slope rises (compare the slope dependence of depression storage in Hillel, 1998).

13.2 Nozzle discharge and wetted radius

For a nozzle of diameter d (mm) at operating pressure P (kPa) the pressure head is $h = P/9.81$ m. The discharge follows the orifice equation (Torricelli) with a discharge coefficient of 0.95,

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^4 \;\; (\text{L min}^{-1}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2 \;\text{m}^2, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

and the wetted radius is (an empirical range law of the form used in sprinkler manuals, Keller and Bliesner, 1990, with the coefficient c_R of Table 13.1 chosen per type)

$$R = c_R\sqrt{d\,h}\;f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

with d in mm, h in m and α in degrees (3.1α as an angle in degrees). The angle function peaks near $\alpha = 29^\circ$ and equals 1 at the default 25° ($\sin 77.5^\circ = 0.9763$). **[Engineered]**: the exponent 0.45 and the 3.1α term are chosen to give a mild optimum, not fitted to a data set. The reach R is printed in the plan as a percentage of the spacing along the lateral.

13.3 The spray pattern and the overlap

Radial profile. A single sprinkler deposits water at radius r with a shape $\hat{p}(x)$, $x = r/R_e$, taken from Table 13.1:

$$\hat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

These are **[Engineered]** shapes that reproduce the qualitative patterns published for each class: a single sprinkler with a rapidly falling profile (impact, gun), and a pattern that peaks away from the centre (spray and micro types).

Wind. Wind elongates the pattern downwind and shifts it:

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

where θ is the angle from the wind direction and u is the wind speed in m s^{-1}. The pattern is computed on a 45×45 grid over $\pm 1.4R$ and normalised so that its volume equals the discharge, $\iint p\,dA = q$, with p in mm min^{-1} (1 mm over 1 m^2 is 1 L).

Overlap. The application rate at a point of the repeating cell is the sum over neighbouring sprinklers (square grid $S_s \times S_l$; triangular layout: every second lateral shifted by $S_s/2$, repeating cell $S_s \times 2S_l$),

$$a(x, y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\text{mm min}^{-1}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

evaluated on a 14×14 grid of the cell (14×24 for the triangular layout). The mean *gross* application rate is AR $= 60\bar{a}$ mm h^{-1} and the **Christiansen coefficient of uniformity** (Christiansen, 1942) is

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

A CU ≥ 85 % is the usual requirement for a sprinkler system (Burt et al., 1997; Keller and Bliesner, 1990); the tile is green from 84 %. The *Layout in plan* shows the circles of reach, the wind arrow and the depth infiltrated so far (Play animates the time): green is the depth needed, red-brown is short, blue is more than needed, and brown hatching is water running off.

13.4 Wind drift and evaporation loss

The fraction of the water lost between nozzle and soil is

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{\left(1.2 + 0.5u + 0.04u^2\right) F\Big(\frac{P}{250}\Big)^{0.7}\left[1 + 0.025\,(T_a - 20)\right]\Big[0.6 + 0.4\,\frac{\sin\alpha}{0.4226}\Big];\ 0.5,\ 45\Big\}. \tag{13.8}$$

Provenance (Eq. 13.8): **[Engineered]**: loss and ponding rules chosen for the workbench.

where F is the type factor of Table 13.1, P is in kPa, u in m s^{-1} and the air temperature $T_a = 24\,^{\circ}\mathrm{C} + \Delta T$, with ΔT the warming of the Properties panel. The form follows the known trends — loss rises with wind, with pressure (finer droplets), with temperature, and falls with a flatter trajectory — in line with the ranges of 5–15 % usually quoted in textbooks (Keller and Bliesner, 1990); the numerical coefficients are **[Engineered]**. w_d is clipped to 0.5–45 %.

13.5 The soil intake, ponding and runoff

At each cell the water balance is computed with the Kostiakov intake of the soil (Eq. 11.6, from the texture; $K = k \times 1000$ mm min^{-A}). The applied rate after drift is $a_k(1 - w_d)$ and for a run of T min the supply is $\mathrm{sup} = a\,T$. Ponding begins when the application rate exceeds the intake rate, which the layer takes as $a = K t^{A-1}$, i.e. at

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \mathrm{sup}, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \mathrm{sup} - F, \quad \mathrm{RO} = \max(0, R - S_d^{\mathrm{eff}}), \quad I = \mathrm{sup} - \mathrm{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

This is a **simplified point balance** (it uses the cumulative law $F = K\,T^{A}$ after ponding, and it ignores re-infiltration of water that has run to a lower cell, which makes the layer slightly pessimistic about runoff in a field with surface storage and slightly optimistic on a steep slope). A portion of the infiltrated depth is retained on the crop canopy and mulch and evaporates,

$$I_c = \min\{I,\ I_{\mathrm{int}}\}, \qquad I_{\mathrm{int}} = 0.6\,\mathrm{CAP}\,[\mathrm{VFF} < 0.95] + 1.2\,\mathrm{LAI}_{\mathrm{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

where CAP is the mulch storage capacity (mm, Table 3.1), the indicator is 1 when a mulch is present, and $\mathrm{LAI}_{\mathrm{live}}$ is the live leaf-area index of the simulated crop on the working date. The depth that reaches the soil at the cell is $Z_k = I - I_c$.

The intake figure shows the rate the soil accepts, $\dot{I}(t) = 60\,K A\,t^{A-1}$ mm h^{-1}, the mean applied rate $\overline{AR}(1 - w_d)$, the shaded area where the applied rate exceeds the intake (runoff), and the time at which ponding starts. (The plotted curve is the derivative of Kt^{A}; the point balance uses the ponding condition $a = Kt^{A-1}$, so the two differ by the constant factor A in the intake rate, a documented simplification.)

Figure 13.2: Sprinkler design: depth applied and infiltrated along the lateral, and the soil intake against the application rate. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

13.6 Efficiencies and run time

With Z_k the infiltrated depth at cell k and gross $= \bar{a}\,T$ the mean depth discharged:

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{\overline{Z_n}}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

where $\overline{Z}_{25}$ is the mean of the lowest quarter of the Z_k (the distribution uniformity of Merriam and Keller, 1978). The partition is closed, $E_a + \text{DP} + \text{RO} + \text{WD} = 1$. The tiles show E_a (green from 85

%), E_s, CU, runoff, deep percolation and the combined drift, evaporation and interception loss, with a stacked bar of the partition.

Automatic run time. The time T is the smallest value for which the driest quarter of the field receives the required depth,

$$T = \min\{T : \ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\, 2400] \text{ min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

found by 18 bisection steps. If the soil cannot take the water as fast as it is applied, longer runs only raise runoff and the layer shows the run time and low E_a together.

13.7 Maximum allowable application rate

The allowable rate by soil and slope follows the standard table of sprinkler manuals (Keller and Bliesner, 1990, Table of maximum application rates; reproduced here for the textbook check).

Table 13.3. Maximum application rate (mm h^{-1}) by soil type and slope.

Soil group	0–5 %	5–8 %	8–12 %	12–16 %
Deep coarse sandy soils	50	38	25	13
Coarse sandy, compact layer below	38	25	19	10
Deep sandy loam	25	20	15	10
Sandy loam, compact layer below	19	13	10	8
Deep silt loam	13	10	8	5
Silt loam, compact layer below	8	6	4	2.5
Clay or clay loam, strong structure	4	2.5	2	1.5

The ten soils of the program are assigned to rows: Coarse Sand, Sand, Loamy Sand $\rightarrow$ 1; Sandy Loam, Coarse Loam $\rightarrow$ 3; Loam, Silt Loam $\rightarrow$ 5; Sandy Clay Loam $\rightarrow$ 6; Clay Loam and Silty Clay Loam $\rightarrow$ 7. The toggle *Compact layer below the soil* moves an odd row to the next (compact) row. The slope selects the column.

13.8 The ten textbook checks

The “Does your design pass?” panel runs the design through the course rules. Let $h_s = P/9.81$ (m), $q_h = 0.06\,q$ (m^3 h^{-1}), $R_w = R$, the wind in km h^{-1} $W = 3.6\,u$, and the wind factor

$$k = \begin{cases} 1, & W \leq 5.5 \\ 0.9, & 5.5 < W \leq 13 \\ 0.8, & 13 < W \leq 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

Provenance (Eq. 13.13): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

The spacing limits, the field layout and the time budget are

$$S_s \leq k\,R_w, \qquad S_l \leq 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x/S_s \rfloor, \quad N_m = \lfloor L_y/S_l \rfloor, \tag{13.14}$$

Provenance (Eq. 13.14): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}}\ \text{(h)}, \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right)\ \text{days}, \tag{13.15}$$

Provenance (Eq. 13.15): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

where $L_x = L_L$ is the lateral length, h_{op} the operating hours per day and Z_{gross} the gross depth per irrigation. For $n = 1 \dots 4$ laterals running at once, the days needed and the inflow are

$$N_d(n) = \left\lceil \frac{N_m}{n\, n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\, N\, q_h, \tag{13.16}$$

Provenance (Eq. 13.16): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

and the program picks the smallest n with $N_d \le T_{\text{int}}$ and $Q_{\text{in}} \le 3.6\, Q_s$ ($\text{m}^3\ \text{h}^{-1}$); if none satisfies both it picks the smallest n meeting the time, else $n = 4$.

Pipes. Friction losses use the Blasius form (Blasius, 1913) for smooth plastic pipe,

$$J = \frac{0.472\, Q^{1.75}}{D^{4.75}}\ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

Provenance (Eq. 13.17): **[Literature]**: Blasius friction law for smooth pipe.

with the Christiansen reduction factor $F = 1/2.75 + 1/(2N)$ for N equally spaced outlets and a 10 % allowance for local losses. The lateral pipe is the smallest of the standard internal diameters $\{46.4,\ 58,\ 63,\ 65.8,\ 75,\ 84.9,\ 100\}$ mm for which

$$h_{f,\text{lat}} = 1.1\, F\, J(Q_L, D)\, (L_x - S_s/2) \le 0.2\, h_s, \qquad Q_L = 1000\, N\, q_h, \tag{13.18}$$

Provenance (Eq. 13.18): **[Literature]**: 20 % rule of Keller and Bliesner (1990).

the classical 20 %-of-operating-head rule that keeps discharge variation along the lateral under about 10 % (Keller and Bliesner, 1990). The mainline (flow $Q_m = n\, Q_L$, length $L_y - S_l/2$) is the smallest diameter not below the lateral's with $J < 0.05$ m m^{-1}.

Manometric head and pump.

$$H_{\text{man}} = z_{\text{lift}} + J\, L_{\text{conv}} + J\, L_{\text{main}} + h_s + 0.75\, h_{f,\text{lat}}, \tag{13.19}$$

Provenance (Eq. 13.19): **[Engineered]**: head and sets bookkeeping.

$$P_{\text{CV}} = \frac{1000\, (Q_m / 3.6 \times 10^6)\, H_{\text{man}}}{75\, \eta}, \quad \eta = 0.75, \qquad P_{\text{kW}} = 0.7355\, P_{\text{CV}}, \tag{13.20}$$

Provenance (Eq. 13.20): **[Literature]**: hydraulic power definition.

with L_{conv} the conveyance length (the mainline length slider) and $0.75\, h_{f,\text{lat}}$ the head at the lateral inlet above h_s (the textbook inlet-head rule). The ten checks (spacing along the lateral, spacing between laterals, wind below 16 km h^{-1}, rate below the allowed value, rate below K_s, CU ≥ 85 %, lateral loss ≤ 20 % of h_s, mainline gradient, the whole field within the interval, flow within the farm supply) are shown with a tick or an exclamation point and the numbers; "N of 10 checks pass" closes the panel. The textbook sequence in the layer also plots the class of sprinkler (low, medium or high pressure) from h_s: < 20 m, 20–45 m, > 45 m.

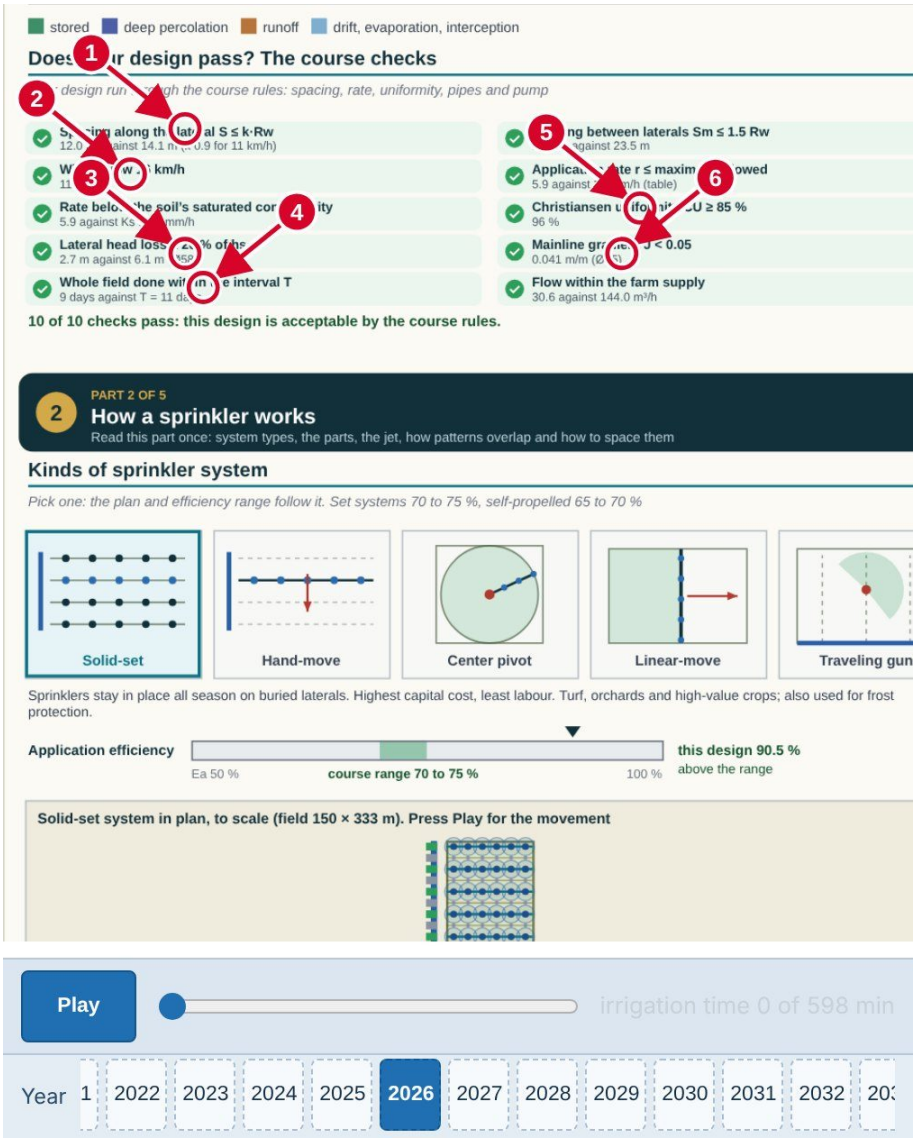


Figure 13.3: Sprinkler design: the ten textbook checks and the kinds of sprinkler system. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

13.9 Kinds of sprinkler system

Five systems are offered, each with its efficiency range (Keller and Bliesner, 1990; Burt et al., 1997 report the range of field values):

System	Principle	Typical E_a
Solid-set	Fixed buried laterals; valves open some at a time	70–75 %
Hand-move	The lateral is moved by hand after every set (1–4 sets a day)	70–75 %
Center pivot	A rotating lateral on towers about 30 m apart; the corners stay dry	65–70 %
Linear-move	A lateral travelling in a straight line	65–70 %
Traveling gun	One gun on a hose reel covers a 220° sector, strip by strip	65–70 %

The plan of the selected system is drawn to scale for the present field; Play moves it (the active lateral, the pivot arm, the travelling lateral or the gun strip). A bar shows where the design's E_a lies against the range for the system.

13.10 Layout and infrastructure

The *Layout and infrastructure* block of this layer uses the same hydraulic tools as the drip layer, with these differences. With $N_s = \lfloor L_L / S_s \rfloor$ sprinklers per lateral and $Q_{\text{lat}} = N_s q / 60$ (L s^{-1}), the number of laterals that can run together is $\lfloor Q_s / Q_{\text{lat}} \rfloor$ and the sets

$$N_L = \Big\lfloor \frac{10^4 A_f / L_L}{S_l} \Big\rfloor, \quad n_l = \min\Big(N_L,\ \Big\lfloor \frac{Q_s}{Q_{\text{lat}}} \Big\rfloor\Big), \quad n_{\text{set}} = \Big\lceil \frac{N_L}{n_l} \Big\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

Provenance (Eq. 13.21): **[Engineered]**: head and sets bookkeeping.

The lateral loss uses the Hazen–Williams equation (Eq. 12.3, $C = 140$) with the Christiansen factor for N_s outlets and the exponent $m = 1.852$,

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m \, h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

Provenance (Eq. 13.22): **[Literature]**: Christiansen reduction factor with Hazen–Williams exponent.

and the lateral pipe is the smallest of the series (12.21) with $h_{f,\text{lat}} \le 0.2\,h$ and $v \le 2.0$ m s^{-1}. The lateral inlet head is $H_0 = h + 1.5 + h_{f,\text{lat}}$ m (1.5 m of riser). The submain, mainline and pump then follow Eqs. (12.22)–(12.23) exactly as for drip, with H_0 from this paragraph. The block's table gives the sets ($\times$ laterals), laterals and sprinklers, TDH, power, lateral loss (≤ 20 %) and pipe laid per hectare.

(Two parallel calculations therefore appear in this layer: the **simulation block** sizes the pipe by Hazen–Williams with $C = 140$, while the **textbook block** uses the Blasius equation and the commercial plastic-pipe sizes of the course. They are separate presentations of the same design question; the numbers will differ by a few per cent, and the textbook block is the one to compare with hand calculations from the course.)

13.11 The design table and sensitivity

The optimiser works like the drip one (Eq. 12.26): score $\min(E_a, 1) - 2.5\max(0, 0.9 - E_s)$, target $E_a \ge 85$ % and $E_s \ge 85$ %, accept a move only if the score rises by more than 0.002. The ten moves are **pressure, spacing along the lateral, run time, nozzle, spacing between laterals, run time, pressure, triangular layout, run time, spacing along the lateral**. For pressure, nozzle and the two spacings the candidates are nine equally spaced values over the range of Table 13.1 (rounded to the step); for the layout the two options; for the run time $T_0 \times \{0.6, 0.7, 0.8, 0.9, 1, 1.1, 1.25\}$ with T_0 the automatic time. Tapping a row applies it. If the target is not met, the layer suggests a lower application rate (a smaller nozzle or wider spacing) or another sprinkler type.

The sensitivity chart moves eight inputs and shows the change of E_a: soil intake k (± 20 %), wind (± 50 %), pressure (± 20 %), nozzle (± 15 %), the two spacings (± 20 %), the depth per irrigation (± 25 %) and the ground slope (± 50 %).

13.12 Season and farm

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60}\ \text{h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

The gross water per irrigation is gross $= \bar{a}\, T$ mm, per season n_{ev} gross mm ($= 10\, n_{\text{ev}}$ gross m^3 ha^{-1}). The table also gives one sprinkler's discharge and reach, the sprinklers for A_f, the sets with Q_s, and the pumping hours and energy of the season. The layer closes with the *Irrigation schedule* and the *When to irrigate* message (end of Chapter 11).

13.13 How to use the layer

- **To pick a sprinkler for a soil.** Choose the soil, set the slope and the wind, then compare the types: the banner tells you the largest allowed rate, the intake plot shows where runoff starts, and the checks name the failing rule.
- **To reduce runoff.** Lower the application rate: a smaller nozzle or lower pressure reduces q; a wider spacing spreads it over a larger area; a triangular layout improves CU at the same rate.
- **To reduce drift.** Reduce the pressure, use a flatter angle, or use a micro-sprinkler ($F = 0.7$); the wind arrow and the w_d share of the partition show the effect.
- **To size the system for the farm.** Set the area, farm supply and operating hours; read the sets, the pipe diameters and the pump power from the layout block, and the textbook block for the hand-calculation version.

Chapter 14

The Dashboard: how every number is produced

The Dashboard is the panel under the drawing. It has four tabs (*Results*, *Design*, *Charts*, *Messages*) and three buttons at the right: **Wider** (enlarges or shrinks the panel), **Pin** (stores the present 20-year averages as a baseline; Ctrl+B) and **Clear** (removes the baseline). It re-reads the engine each time the state changes, so it always agrees with the drawing.

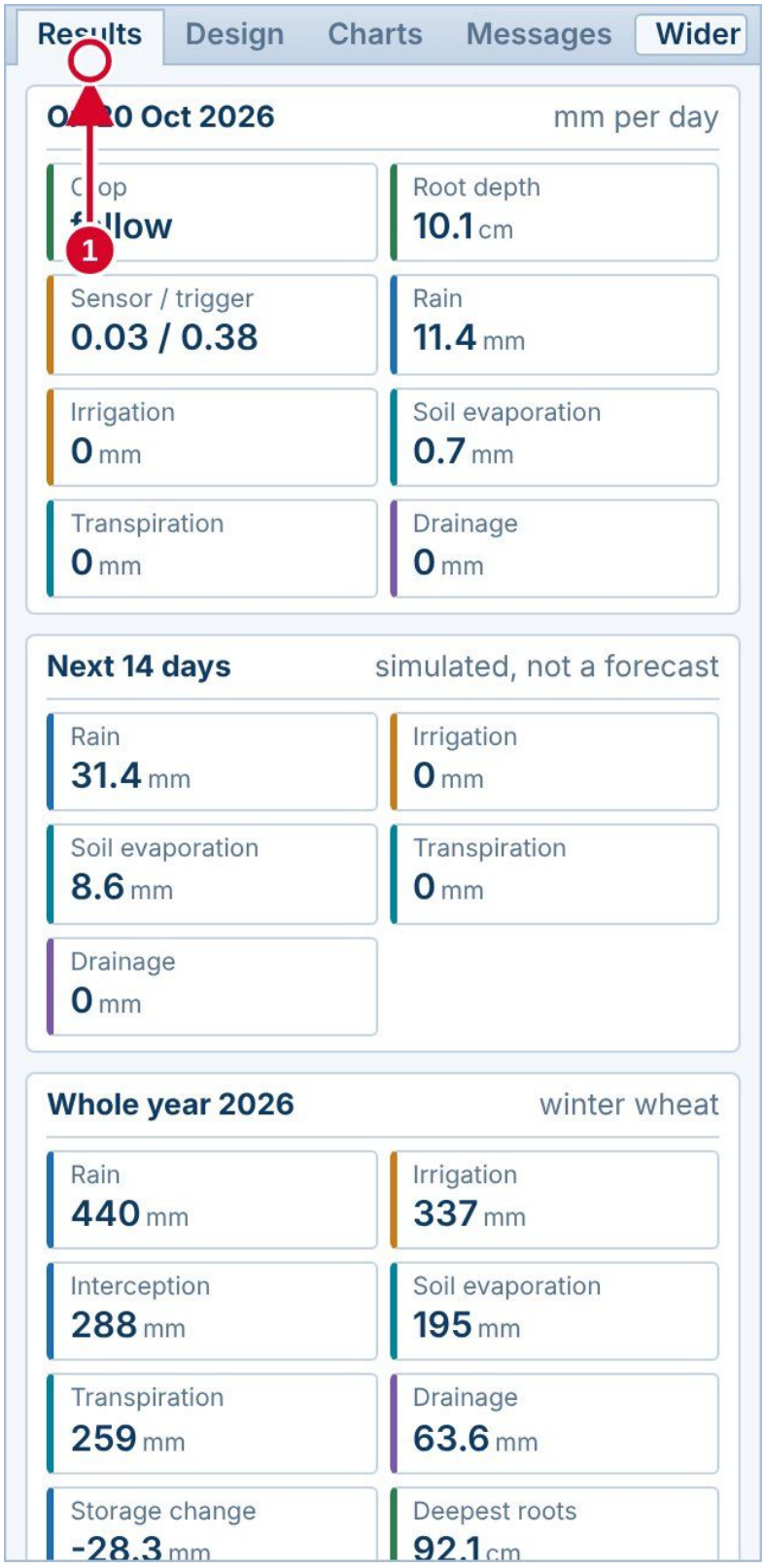


Figure 14.1: The Dashboard, Results tab: the working day, the next days, the whole year and the 20-year averages. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

14.1 Results

The tab has four blocks, each a grid of tiles.

Block 1: On the working date (mm per day unless stated). Crop, root depth (cm), *Sensor / trigger* (the simulated relative depletion f of the sensor layer and the trigger f_T; irrigation fires when $f \geq f_T$, Eq. 6.1), rain, irrigation, soil evaporation, transpiration and drainage on that day. They are the Engine 2 daily series of Chapter 5, read at the day index k of the working date:

$$\text{Rain} = P_k, \quad \text{Irr} = I_k, \quad \text{Evap} = E_k, \quad \text{Transp} = T_k, \quad \text{Drain} = D_k \ \ (\text{mm d}^{-1}). \tag{14.1}$$

Provenance (Eq. 14.1): **[Engineered]**: bookkeeping of Engine 2 series.

Block 2: Next N days ("simulated, not a forecast"). Sums of the same series over the following N days (the look-ahead of Chapter 4, 14 days),

$$\Sigma X = \sum_{j=1}^{N} X_{k+j}, \qquad X \in \{P, I, E, T, D\}. \tag{14.2}$$

Provenance (Eq. 14.2): **[Engineered]**: bookkeeping of Engine 2 series.

Because the weather after today is the analogue year, not a forecast, these numbers describe what the *scenario* implies, not what will happen.

Block 3: Whole year. For the calendar year of the working date: rain, irrigation, interception, soil evaporation, transpiration, drainage, the change in storage and the deepest roots (cm). The water balance of the year closes as

$$\Delta S = P + I - E_{\text{int}} - E - T - D, \tag{14.3}$$

Provenance (Eq. 14.3): **[Literature]**: water-balance identity.

with E_{int} the interception loss (Section 5.2); the tile *Storage change* is ΔS.

Block 4: 20-year averages at these settings. These are the annual-model outputs of Engine 1 (Chapter 5), one tile each: soil evaporation, transpiration, evapotranspiration ET $= E + T$, drainage, irrigation, yield, irrigation efficiency, nitrogen leaching, rainfall, and *Irrigation without mulch* (the same case with no mulch, for the saving). Each tile shows the value as the engine reports it (with its unit).

14.1.1 The pinned baseline

Press **Pin** (or Ctrl+B): the ten tiles of Block 4 are stored. After any change in Properties the tiles show the difference from the pinned values,

$$\Delta_i = x_i - x_i^{\text{pin}}, \tag{14.4}$$

Provenance (Eq. 14.4): **[Engineered]**: bookkeeping of Engine 2 series.

with a coloured arrow: ▲ (increase) or ▼ (decrease), green when the change is in the *better* direction and red when it is worse, and "= no change" when $|\Delta_i| < 10^{-9}$. The "better" direction is fixed per tile: lower is better for soil evaporation, drainage, irrigation and nitrogen leaching; higher is better for yield and irrigation efficiency; transpiration, evapotranspiration, rainfall and *Irrigation without mulch* are neutral (no colour). The values are compared numerically after reading the first number in each tile's text, so the comparison is meaningful only between runs of the same version and units.

Typical use: pin the *Present* case, change one control (for example Mulch → Straw), read the green or red arrows, change it back, and repeat for another control. This is how the effect matrix of Chapter 3 was built.

14.2 Design

The tab lists the live results of the design layer that is open (it refreshes about every 1.2 s). With no design layer open it shows a note. The tiles are:

Layer	Tiles (quantity, unit)
Surface (S2)	E_a, E_s (%); deep percolation, runoff (%); CU (%); cut-off time, time to fill the root zone, advance time (min); applied, required, deep-percolation and runoff volumes
Drip (S3)	E_a, E_s, E_u (%); deep percolation (%); wetted fraction P_w (%); gross application (mm); irrigation time (h); applied and required volume per emitter (L)
Sprinkler (S4)	E_a, E_s, DU, CU, deep percolation (%); runoff; gross and net application (mm); set time (min); sprinkler discharge; application rate

The definitions are Eqs. (11.18)–(11.19), (12.19) and (13.11). The Design tab and the *Design results (CSV)* export read the same object, so the numbers agree digit for digit.

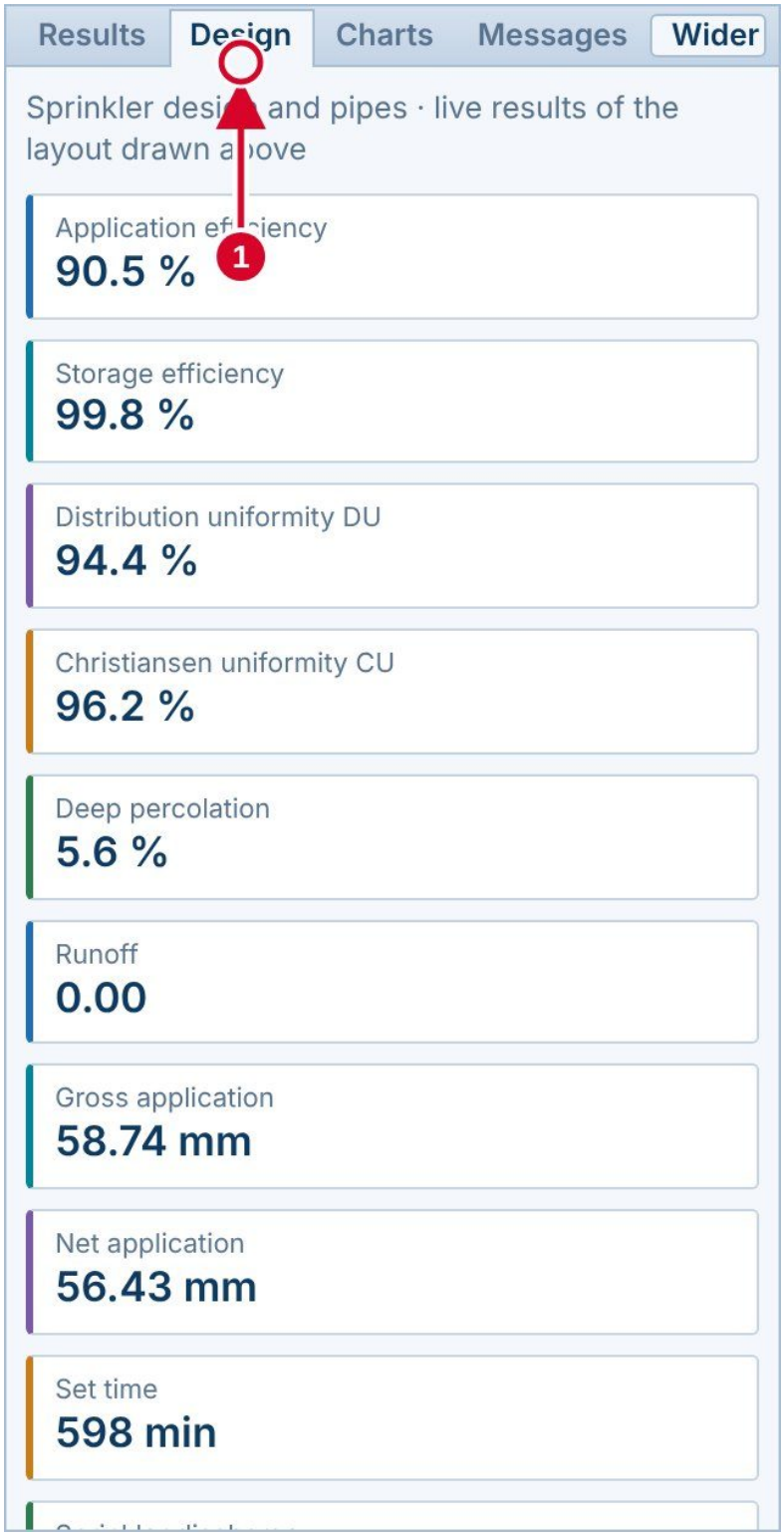


Figure 14.2: The Dashboard, Design tab, with the sprinkler layer open. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

14.3 Charts

Three charts of the 20 years around the working date (calendar years on the horizontal axis, mm per year): *Rainfall and irrigation* (bars), *Drainage below the root zone* (bars) and *Evaporation and transpiration* (lines). They are the annual sums of the Engine 2 daily series,

$$Y^{(y)} = \sum_{d \in y} X_d, \qquad X \in \{P, I, D, E, T\}. \tag{14.5}$$

Provenance (Eq. 14.5): **[Engineered]**: bookkeeping of Engine 2 series.

They show variability between weather years (dry years need more irrigation and drain less) and the trend of the scenario; they are the series saved by *Yearly results (CSV)*.

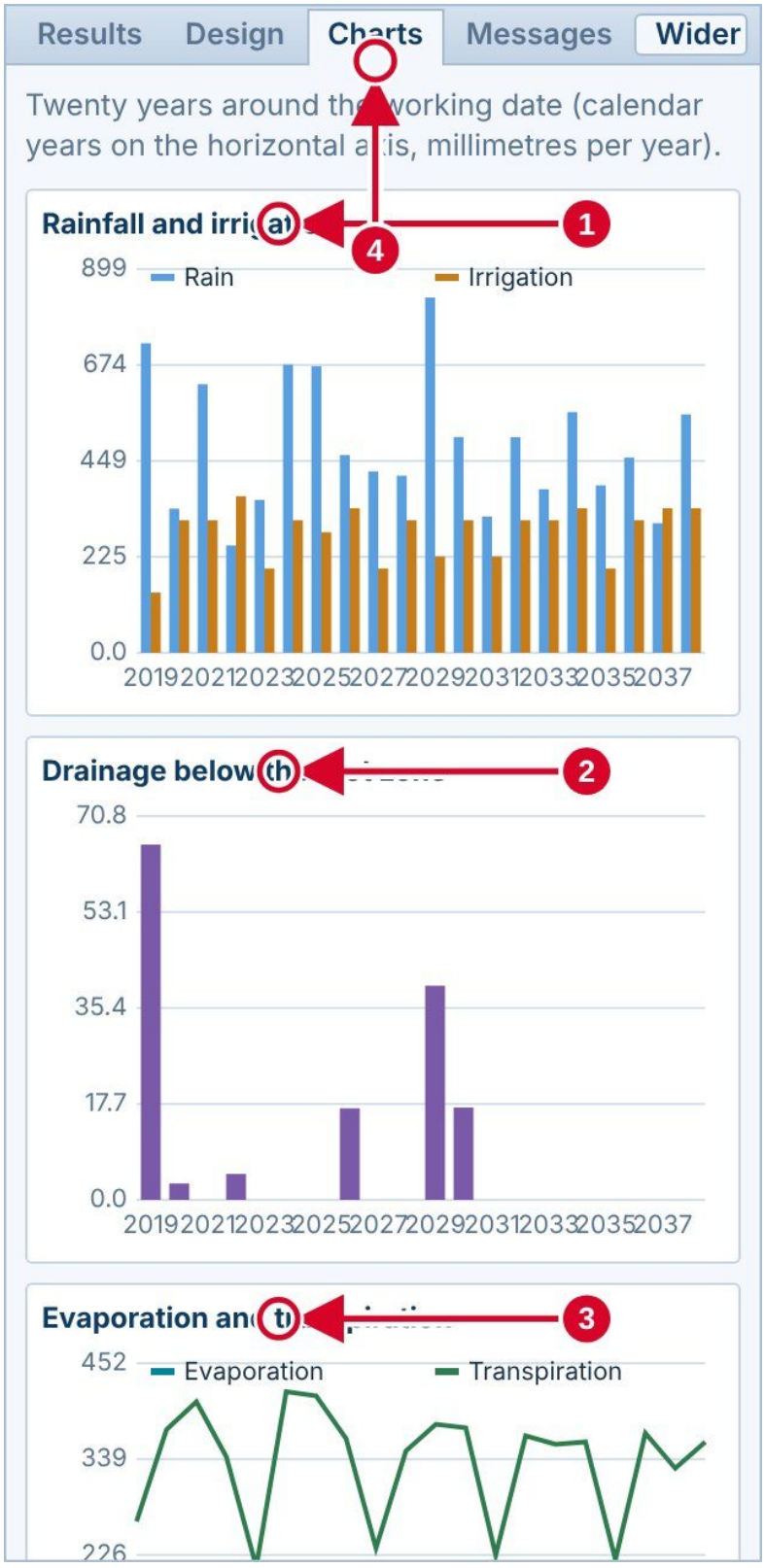


Figure 14.3: The Dashboard, Charts tab. The numbered arrows are explained in the atlas of chapter 18, under the same screen.

14.4 Messages

A time-stamped log of what the program did (project saved, baseline pinned, file saved, engine messages) with warnings and errors coloured. Newest first. Use it to find the path of a saved file.

Chapter 15

Worked workflows

Each workflow is a complete sequence of clicks, the numbers to expect, and the equations behind them.

15.1 Workflow 1: How much does straw mulch save?

1. *Tools* ▸ *Ready-made cases* ▸ *Present* (or set soil Coarse Loam, tillage reduced, crop cereal/oilseed, mulch none, method Surface).
2. Press **Baseline** (Ctrl+B). The ten tiles of the 20-year block are stored.
3. In Properties ▸ Mulch choose **Straw**, then set *Soil cover* (VFF) to 0.8. Read the arrows: soil evaporation falls (green), irrigation falls (green), yield changes slightly.
4. Open *Field balance* and press Run to see the evaporation bar shrink.

Why: straw has CAP $= 4$ mm storage and a thermal-resistance weight 0.4 (Table 3.1); the cover fraction scales the evaporation and the interception of Engine 2 (Section 5.2, "Infiltration, redistribution and drainage" and "Transpiration, evaporation"). The saving reported in the tile *Irrigation without mulch* minus *Irrigation* is the mulch effect on the whole-year balance.

15.2 Workflow 2: When should I irrigate?

1. Set the working date to today and choose the case.
2. Open *Irrigation design* ▸ *Surface irrigation design* (or Drip or Sprinkler) and scroll to the *Irrigation schedule*; the first row is the next irrigation date. The *When to irrigate* message at the end repeats it with the mean interval.
3. Move the **Trigger** slider drier (e.g. $-300 \rightarrow -1000$ cm of suction): the interval lengthens and the number of irrigations falls (Eq. 6.3 shifts f_T).
4. Move the **Sensor depth** to see how the date changes: a shallow sensor dries first and triggers earlier.

15.3 Workflow 3: Design a border strip for a loam

1. Set the soil to Loam; open *Surface irrigation design*; *Start from: Your soil*; *System: Border*.
2. The layer computes K_s, ψ_f and the Kostiakov constants (Eqs. 11.1–11.7); read R^2.
3. Set the length to your field (e.g. 200 m). The inflow is chosen automatically (Eq. 11.22 for furrows, the $E_a E_s$ optimum for borders).
4. Read E_a and E_s in the result section. If $E_a < 85$ % open the design table and tap the green row; the sliders move to it.
5. Read the tornado chart: if A or k dominate, the soil intake is the main uncertainty, so measure it in the field.

15.4 Workflow 4: Choose and size a drip system

1. Choose the crop (vegetable $\rightarrow S_l = 1.5$ m), the soil and the emitter (inline).
2. Lateral length: increase until E_u turns amber (Eq. 12.7).
3. *Placement*: compare *Surface*, *25 cm* and *45 cm*: buried drip loses less to evaporation (Eq. 12.15) but a bulb that is too deep loses water below the roots (Eq. 12.14).
4. Set A_f and Q_s; read zones, TDH and kW from the layout block (Eqs. 12.20–12.23).
5. Tap the best row of the design table to apply it.

15.5 Workflow 5: Check a sprinkler against the course rules

1. Choose the soil and the slope (e.g. Sandy Loam, 3 %); the banner states the allowed rate (Table 13.3).
2. Choose *Impact*; set pressure and spacing; read CU and the application rate.
3. Open the ten checks (Section "The ten textbook checks"); fix the red ones in the order the text names.
4. Compare the system chips (solid-set, hand-move, pivot, linear, gun) and their efficiency ranges.

15.6 Workflow 6: A warmer, drier future

1. Set the date to 2050-07-01; choose *dry* as the weather year; set the temperature offset (+2 °C) and rainfall factor (0.8).
2. Pin the baseline *before* you change the scenario, so the arrows show the climate-change effect.
3. Check the *Charts* tab: irrigation rises and drainage falls.
4. Remember these are scenarios (Chapter 17).

15.7 Workflow 7: Save, share and reproduce

Save the project (*File* ▸ *Save project*) after each case, export the yearly CSV and the design CSV, and note the version number shown in the status bar. Reopen the `.swafl` to restore the state exactly.

Chapter 16

Exports, files and troubleshooting

16.1 What can be saved

Item	Menu / toolbox	File	Contents
Picture of the window	*File* ▸ *Export* ▸ *Picture of the window (PNG)*, Ctrl+E, toolbar *Picture*	`SWAFL_<layer>.png`	The drawing canvas of the open layer
Yearly results	*File* ▸ *Export* ▸ *Yearly results (CSV)*, toolbar *Results*	`SWAFL_yearly_results.csv`	Two comment lines with soil, tillage, crop, mulch and method; then columns `Year, Rain_mm, Irrigation_mm, Drainage_mm, Evaporation_mm, Transpiration_mm`
Design results	*File* ▸ *Export* ▸ *Design results (CSV)*	`SWAFL_design_<id>.csv`	Two columns `Quantity,Value`: every number the open design layer computed
Project	*File* ▸ *Save project*	`SWAFL_project.swafl`	JSON: app name and version, the open layer, the state, the year and the date
Print	*File* ▸ *Print*, Ctrl+P	PDF	The open layer

In the yearly CSV, *Year* is the index 1–20 of the 20-year run starting at the calendar year shown on the Charts tab; the values are the annual sums of Eq. (14.5), rounded to 0.01 mm. In the design CSV the quantity names are the internal names of the model (for example `Ea`, `Es`, `dp`, `ro` for the surface layer; `Eu`, `gross`, `t` for drip; `CU`, `DU`, `AR`, `T` for sprinkler); the efficiencies are fractions (0–1).

16.2 Troubleshooting

Problem	Fix
“Running scripts is disabled” in PowerShell	Use a Command Prompt terminal, or run `Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned`, or type `cmd /c npm install`.
`npm` is not recognised	Install Node.js, then close and reopen VS Code.
`ENOENT … package.json not found`	You are in the wrong folder; the zip makes a folder inside a folder. `cd` into the one that contains `package.json`.
Install or build is slow or fails inside OneDrive	Copy the project to `C:\SWAFL_Desktop` and run it there.
“Windows protected your PC”	Click *More info* ▸ *Run anyway* (the app is not code-signed).

Problem	Fix
Build error about symbolic links or permissions	Turn on Windows Developer Mode or run the terminal as Administrator; repeat `npm run get-app`.
Download of Electron fails	Check internet, VPN and firewall; delete `node_modules` and run `npm install` again.
White or empty window	Wait 10 seconds (the engine loads), then *Refresh* ▸ *Reload simulation engine*. Close any second SWAFL window.
A design layer shows “the stream does not reach the end”	The cut-off time is shorter than the advance time: increase the cut-off or the inflow (Eq. 11.16).
Efficiency E_a is low and runoff is high (sprinkler)	The application rate is above the soil intake: reduce the nozzle or pressure, widen the spacing (Chapter 13, “The soil intake, ponding and runoff”).
The Design tab says to open a design layer	Open one of the three layers under *Irrigation design*.

If a problem remains, open an issue on the project page, and attach the `.swafl` file and the text of the *Messages* tab.

Chapter 17

Validity, limits and honest use

The program is a **teaching and design-exploration tool** built on a research data set. This chapter says what each part can and cannot support.

Engine 1 (the annual meta-model) reproduces, by interpolation, the Daisy 7.1 runs it was fitted to (run sets E2, E3, E4, E5, E7, E11, E13 and E14; the study as a whole has 10,190 runs). Within the ranges of the controls it is an *emulator* of those runs: it is not a new physical model, and it does not extrapolate outside the soils, mulches, climates and irrigation methods that were simulated. It was calibrated to one location's weather record (1980–1999) and its projections to 2100 are **scenarios built from analogue years**, not forecasts.

Engine 2 (the daily bucket model) is a 40-layer, 1-D, daily water balance. It uses closed-form root, evaporation and transpiration functions with a single calibrated coefficient k_t so that its seasonal transpiration matches Engine 1. It does not solve the Richards equation, has no lateral flow, no runoff on slopes and no water table (upward flow is a redistribution between layers only). Its day-to-day irrigation dates are therefore indicative: use them to compare scenarios, not to schedule a real irrigation to the day.

Design layers (S2–S4) are textbook models (Chapters 11–13). They are suitable for teaching, for comparing layouts, and for a first sizing. They are **not a substitute for field evaluation** of intake, advance, pressure and flow, or for a hydraulic design by a qualified engineer. In particular:

- the intake of the soil is derived from texture by pedotransfer functions with their own scatter; measure it in the field for real design;
- the sprinkler pattern is a parametric shape, not a measured catch-can test; the wind-drift loss is a parametric estimate;
- the drip bulb is a geometric simplification of two-dimensional soil water redistribution;
- pump power and pipe sizes use stated allowances (9 m of fittings, 0.65 pump efficiency, $C = 140$) that you should replace with your own;
- the "Engineered" items of the label scheme (Section "About this manual") are modelling choices, not published laws.

Data and reproducibility. The same inputs and version give the same outputs (no random numbers in the design layers; the rainfall generator and all animations of the engine use a deterministic generator seeded by the soil name). Save the `.swafl` file and the version number with any result you report.

Not included. The package does not contain a soil-erosion model, a crop-pest model, an economic optimiser, or a groundwater model.

When you cite results from the program, state the version, the case (soil, crop, mulch, method, trigger, date or weather year) and whether the numbers come from Engine 1, Engine 2 or a design layer.

Chapter 18

Screenshot atlas: every screen with its governing equations

This part of the manual shows every screen of the workbench, one after another, each accompanied by the equations that produce what it displays. The screenshots were taken from the SWAFL workbench itself (the window in `app/index.html`) with the default settings (Coarse Loam, conventional tillage, cereal/oilseed rotation, wheat-straw mulch of strength 1.0, Surface irrigation, typical year). For each screen the first picture is the whole workbench and the second is the drawing area at higher magnification. For the fourteen single layers, whose drawing is longer than the drawing area and is scrolled inside it, the pictures of the whole drawing from top to bottom follow, together with its playback states and with one picture for each control that changes the drawing (every mulch material, every zoom and view of the soil section, every irrigation layout, every day button, and so on). Every picture carries numbered arrows, each ending on the item that its legend entry names. Below the pictures every equation that the screen draws on is printed in full, with its number, the chapter that explains it and its provenance label, so that no number on a screen is without its equation. A reverse index at the end lists, for each of the 134 numbered equations, the screens that use it.

Equation numbers are those of the chapters (for example Eq. 5.4 is the fourth equation of the chapter on the engines). The symbols are defined where the equation first appears; the *Where explained* line gives that place.

18.1 The window and its controls

18.1.1 The welcome dialog

Shown on first start; it carries no equations.

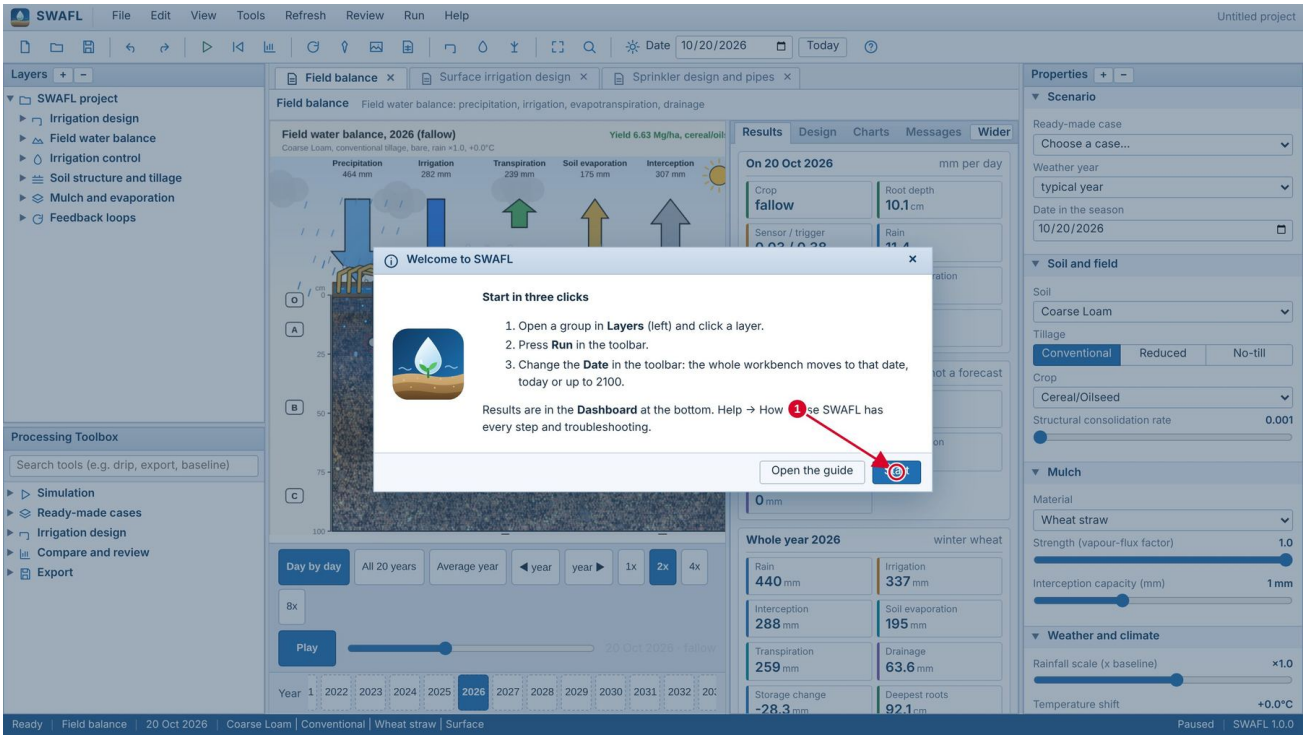


Figure 18.1: The welcome dialog.

What the arrows point to.

- **1** The Start button closes the dialog and opens the workbench; the dialog carries no equation.

18.1.2 The workbench

Layers (left), drawing area (centre), Dashboard (below and right of the drawing) and Properties (right). The date shown in the toolbar sets the whole workbench to that day.

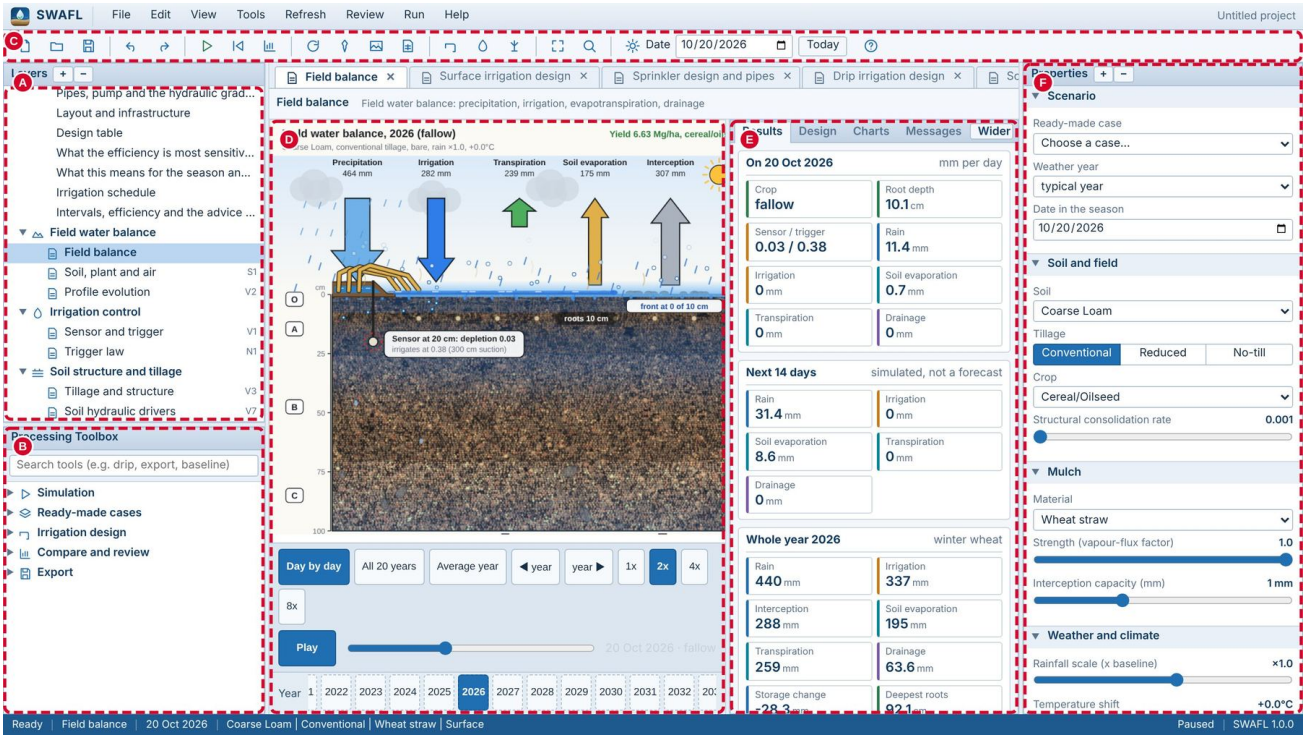


Figure 18.2: The workbench.

What the arrows point to.

- **L** Layers panel: choose a layer or one of its sections. Every choice is one screen of this atlas.
- **X** Processing Toolbox: run, cases, design, compare and export commands.
- **T** Toolbar with the Date control: the date sets the whole workbench to that day (Eqs. 4.1 and 4.2).
- **D** Drawing area: the picture produced by the equations printed under each screen.
- **R** Dashboard: the numerical results of the same equations (Engine 1, Eqs. 5.4 to 5.12; Engine 2, Eqs. 5.13 to 5.25).
- **C** Properties panel: the controls that move the equations (soil, tillage, mulch, weather, irrigation, nutrients).

Equations behind this screen. Eq. 4.1, Eq. 4.2.

Eq. 4.1. Where explained: chapter "Dates", section "The date control".

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \quad (4.1)$$

Provenance (Eq. 4.1): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

Eq. 4.2. Where explained: chapter "Dates", section "The date control".

$$a = \begin{cases} \operatorname{argsort}_y \left(P_{\mathrm{thr},y}\right)_{[10]} & \text{typical (median, 11th smallest of 20)} \\ \arg\min_y P_{\mathrm{thr},y} & \text{dry} \\ \arg\max_y P_{\mathrm{thr},y} & \text{wet} \\ Y_0 - 1980 & 1980 \le Y_0 \le 1999 \end{cases} \tag{4.2}$$

Provenance (Eq. 4.2): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

18.1.3 The File menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

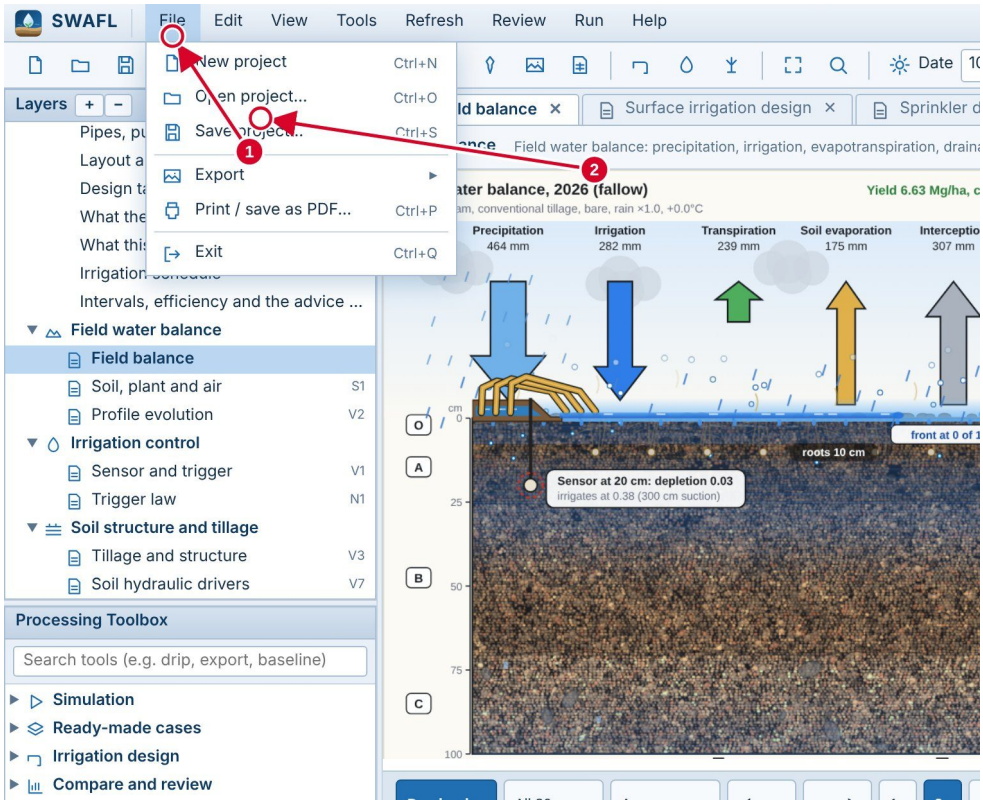


Figure 18.3: The File menu.

What the arrows point to.

- **1** The File menu in the menu bar.
- **2** Its list of commands.

18.1.4 The Edit menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

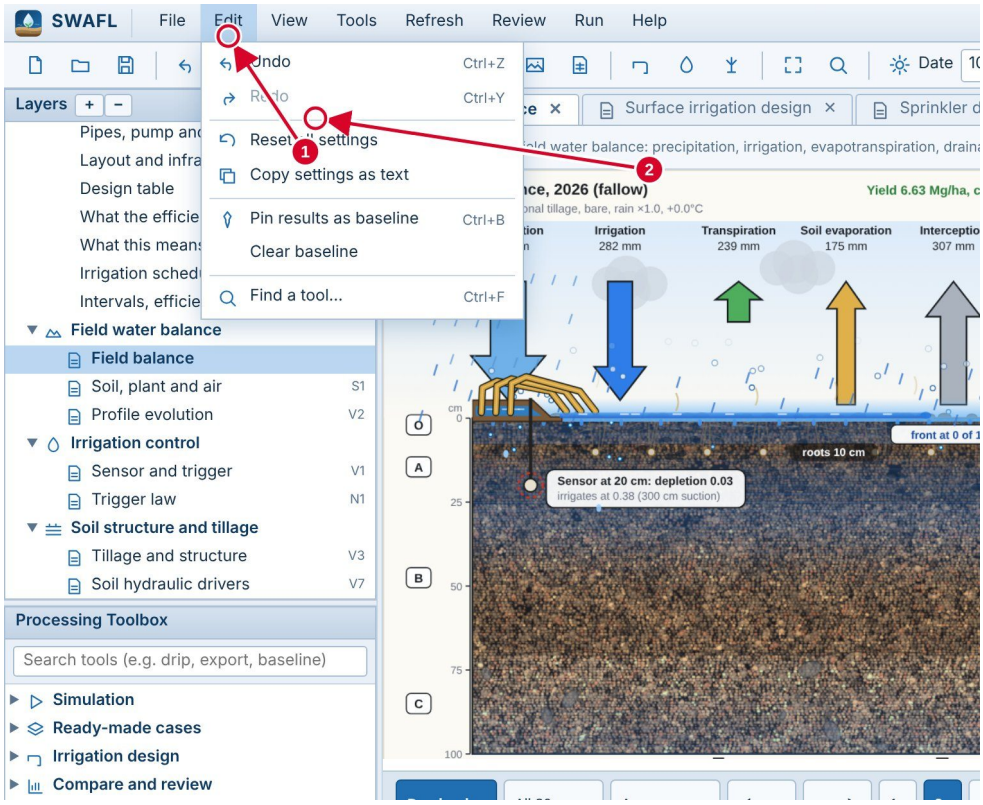


Figure 18.4: The Edit menu.

What the arrows point to.

- **1** The Edit menu in the menu bar.
- **2** Its list of commands.

18.1.5 The View menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

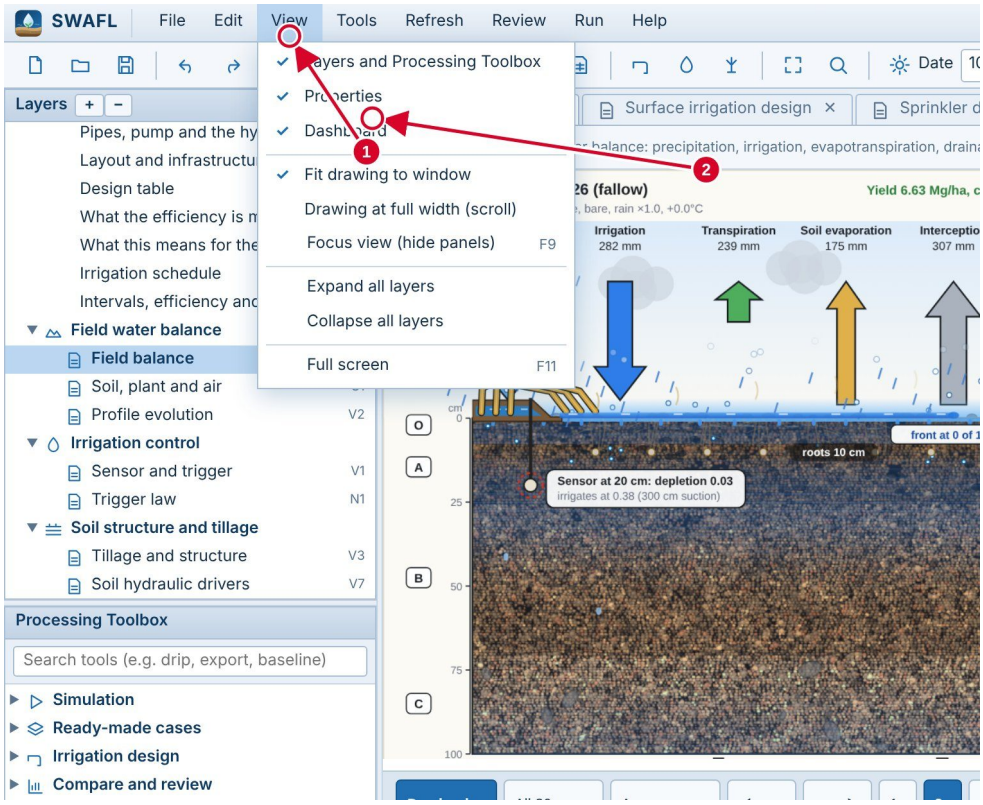


Figure 18.5: The View menu.

What the arrows point to.

- **1** The View menu in the menu bar.
- **2** Its list of commands.

18.1.6 The Tools menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

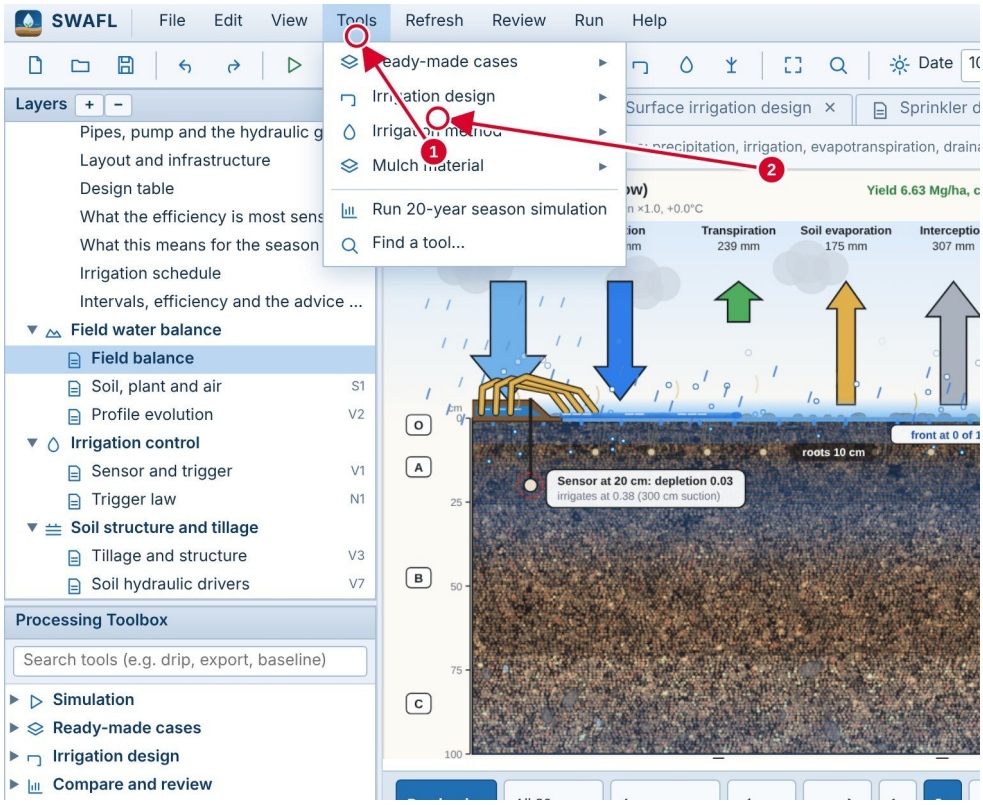


Figure 18.6: The Tools menu.

What the arrows point to.

- **1** The Tools menu in the menu bar.
- **2** Its list of commands.

18.1.7 The Refresh menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

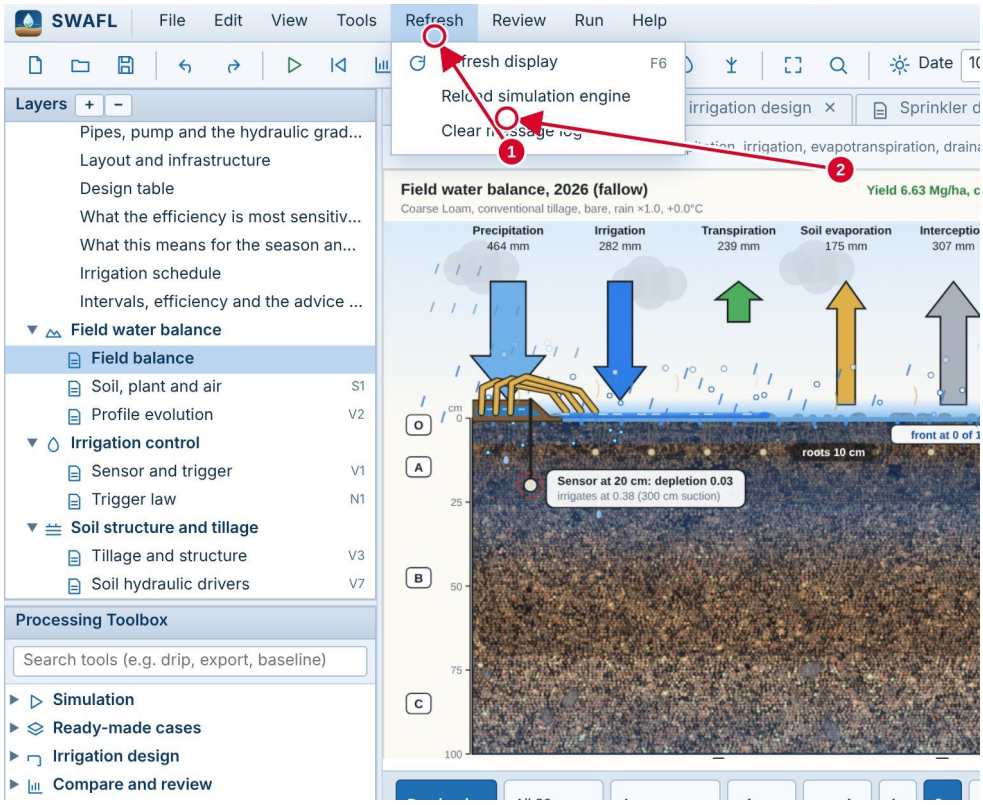


Figure 18.7: The Refresh menu.

What the arrows point to.

- **1** The Refresh menu in the menu bar.
- **2** Its list of commands.

18.1.8 The Review menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

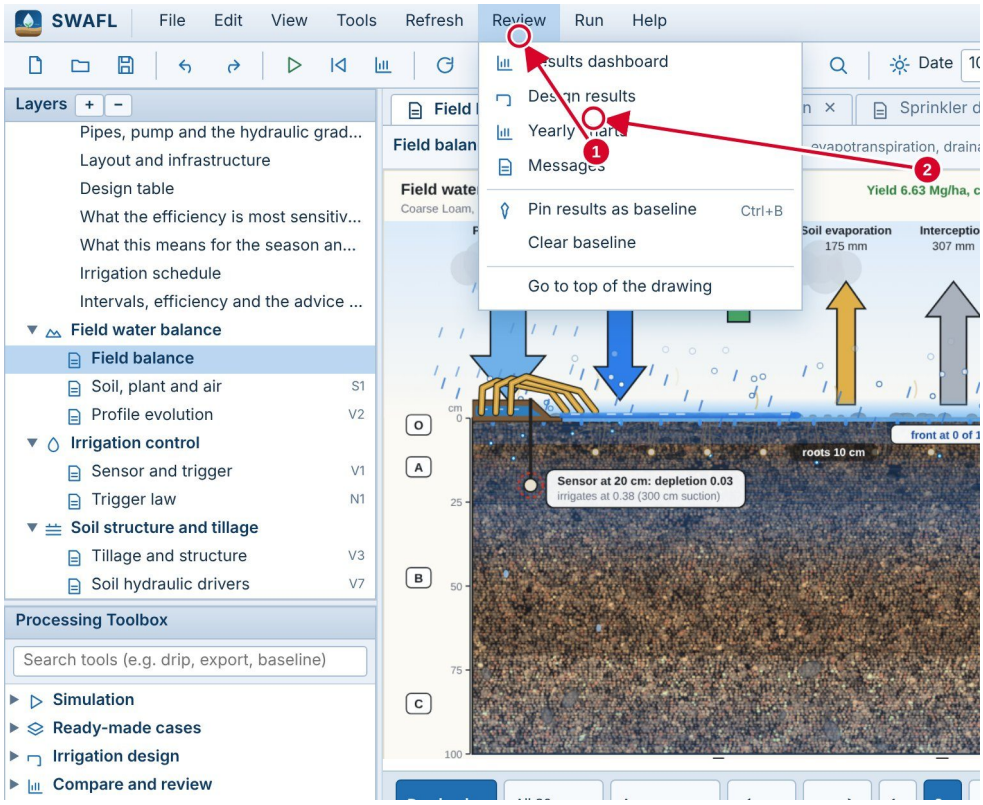


Figure 18.8: The Review menu.

What the arrows point to.

- **1** The Review menu in the menu bar.
- **2** Its list of commands.

18.1.9 The Run menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

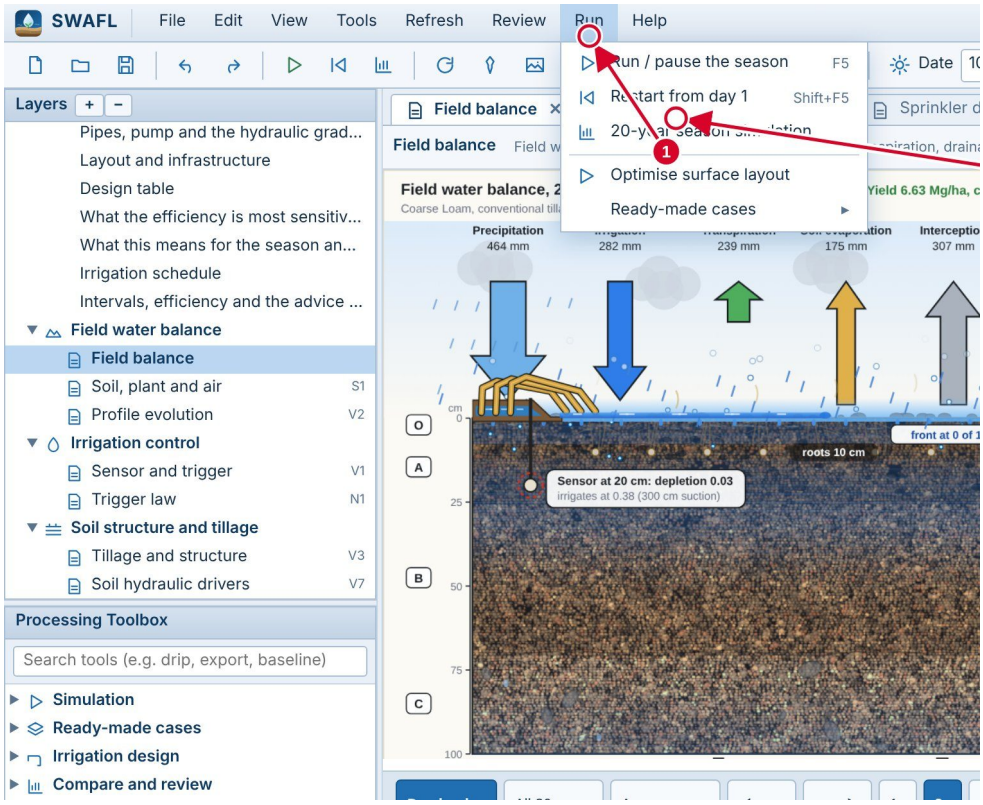


Figure 18.9: The Run menu.

What the arrows point to.

- **1** The Run menu in the menu bar.
- **2** Its list of commands.

18.1.10 The Help menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

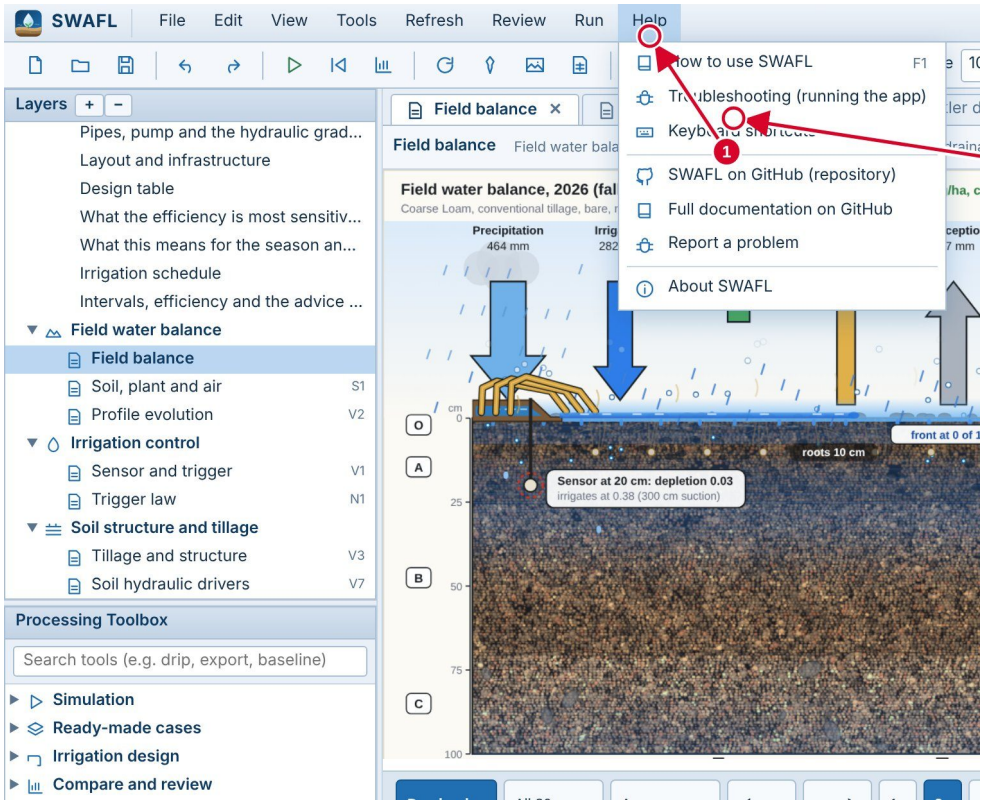


Figure 18.10: The Help menu.

What the arrows point to.

- **1** The Help menu in the menu bar.
- **2** Its list of commands.

18.1.11 The command palette (Ctrl+K)

Searches every layer, control and tool by name.

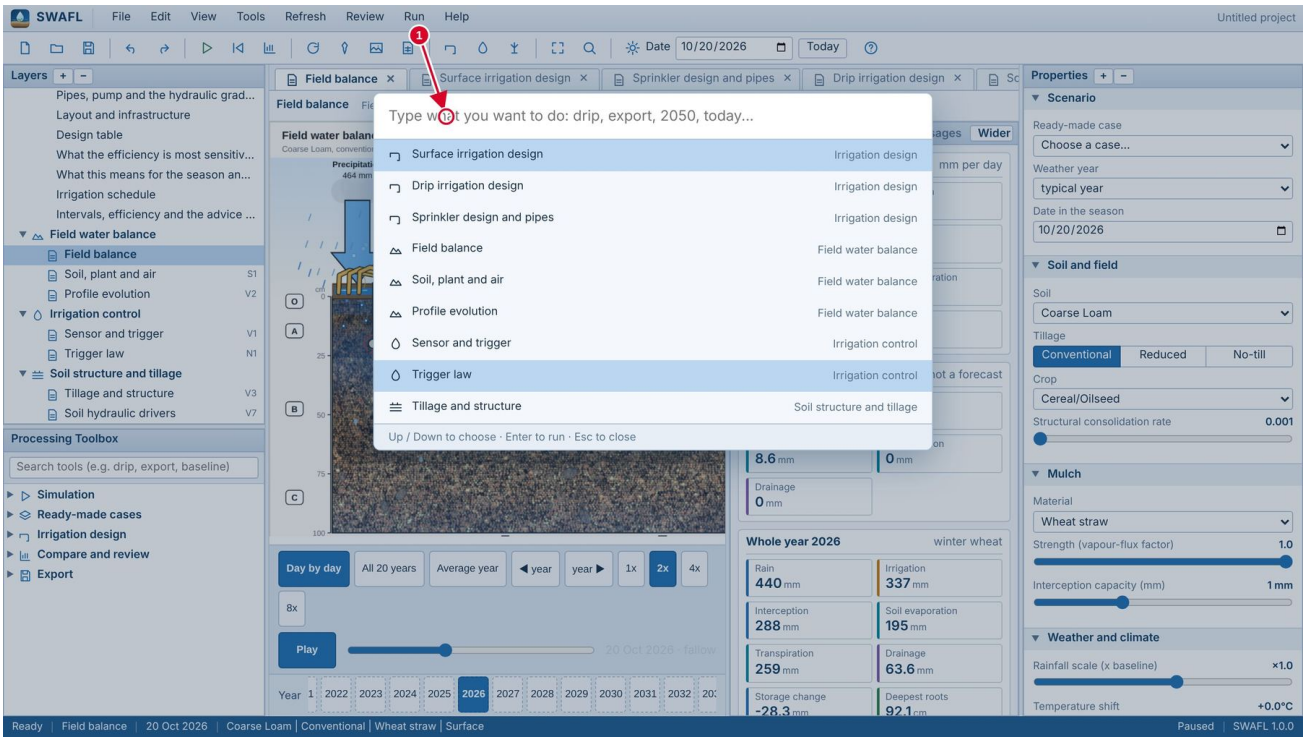


Figure 18.11: The command palette (Ctrl+K).

What the arrows point to.

- **1** Search box of the command palette (Ctrl+K): type a layer, control or tool and press Enter.

18.1.12 The Processing Toolbox

Simulation, ready-made cases, irrigation design, compare and review, export.

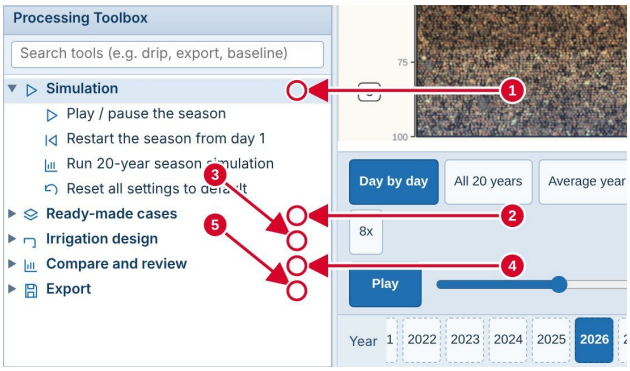


Figure 18.12: The Processing Toolbox.

What the arrows point to.

- **1** Simulation: play or pause the season, restart, 20-year run (Engine 2, Eqs. 5.13 to 5.25).
- **2** Ready-made cases: load a stored combination of soil, tillage, mulch, weather and irrigation.
- **3** Irrigation design: open the surface, drip or sprinkler design layer (Eqs. 11.1 to 13.23).
- **4** Compare and review: pin the current results as the baseline (Eqs. 14.4 and 14.5), clear it, show the dashboard.
- **5** Export: picture, yearly results (CSV), design results (CSV), project, print.

18.1.13 The toolbar

Run, restart, 20-year run, refresh, baseline, picture, focus, search and the Date control (Eqs. 4.1, 4.2).

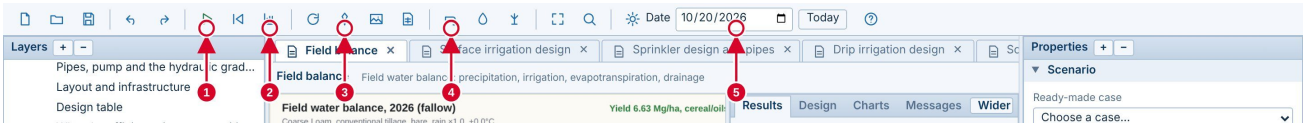


Figure 18.13: The toolbar.

What the arrows point to.

- **1** Run or pause the season (Engine 2).
- **2** Run the 20-year simulation.
- **3** Pin results as baseline (Eq. 14.4).
- **4** Open a design layer: Surface, Drip or Sprinklers.
- **5** Date control: the working date, Eqs. 4.1 and 4.2.

Equations behind this screen. Eq. 4.1, Eq. 4.2.

Eq. 4.1. Where explained: chapter "Dates", section "The date control".

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \tag{4.1}$$

Provenance (Eq. 4.1): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

Eq. 4.2. Where explained: chapter "Dates", section "The date control".

$$a = \begin{cases} \operatorname{argsort}_y \left(P_{\text{thr},y}\right)_{[10]} & \text{typical (median, 11th smallest of 20)} \\ \arg\min_y P_{\text{thr},y} & \text{dry} \\ \arg\max_y P_{\text{thr},y} & \text{wet} \\ Y_0 - 1980 & 1980 \le Y_0 \le 1999 \end{cases} \tag{4.2}$$

Provenance (Eq. 4.2): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

18.1.14 The status bar

The current layer, date, soil, tillage, mulch and irrigation method.

Figure 18.14: The status bar.

What the arrows point to.

- **1** Layer and section on screen.
- **2** Working date.
- **3** Soil, tillage, mulch and irrigation method in force.

18.1.15 The Dashboard, Results tab

Working day, next 14 days, whole year and 20-year averages.

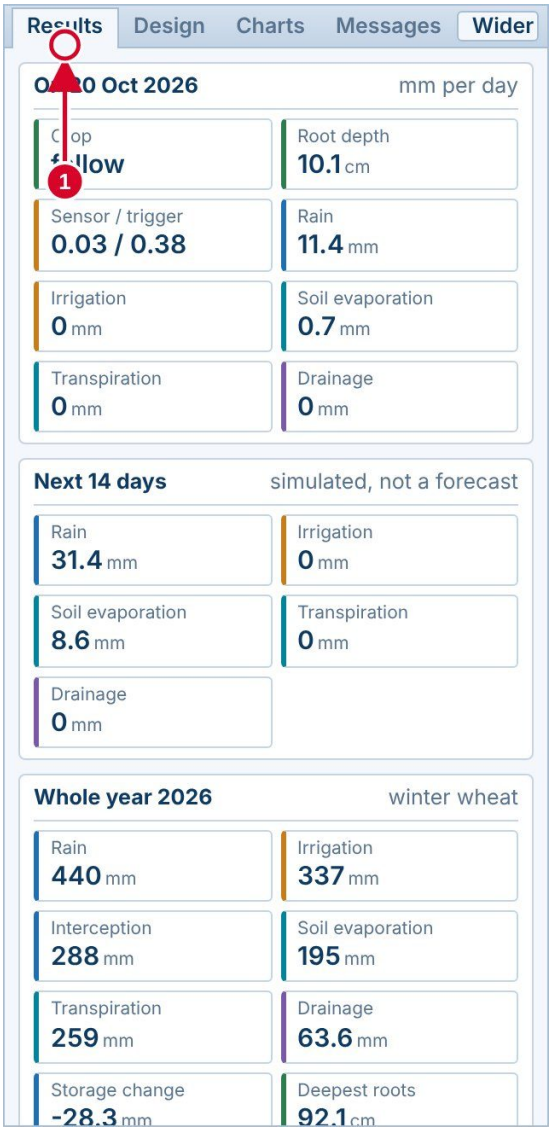


Figure 18.15: The Dashboard, Results tab.

What the arrows point to.

- **1** Working day: crop, root depth, sensor against trigger (Eq. 5.3), rain, irrigation, evaporation, transpiration and drainage of that day, Eqs. 14.1 to 14.3.
- **2** Next 14 days: simulated, not a forecast, from Engine 2.
- **3** Whole year: the annual fluxes of Engine 1, Eqs. 5.4 to 5.12, and the closure of Eq. 5.12.
- **4** Tabs: Results, Design, Charts and Messages.

Equations behind this screen. Eq. 6.1, Eq. 14.1, Eq. 14.2, Eq. 14.3, Eq. 14.4.

Eq. 6.1. Where explained: chapter "Irrigation control layers", section "How the numbers are generated".

$$N_y(h_T) = \#\{\, d \in \text{growing season of year } y : h_d > h_T \,\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

Provenance (Eq. 6.1): **[Data-driven]**: counts over the simulated daily series.

Eq. 14.1. Where explained: chapter “The Dashboard”, section “Results”.

$$\text{Rain} = P_k, \quad \text{Irr} = I_k, \quad \text{Evap} = E_k, \quad \text{Transp} = T_k, \quad \text{Drain} = D_k \ \ (\text{mm d}^{-1}). \tag{14.1}$$

Provenance (Eq. 14.1): **[Engineered]**: bookkeeping of Engine 2 series.
Eq. 14.2. Where explained: chapter “The Dashboard”, section “Results”.

$$\Sigma X = \sum_{j=1}^{N} X_{k+j}, \qquad X \in \{P, I, E, T, D\}. \tag{14.2}$$

Provenance (Eq. 14.2): **[Engineered]**: bookkeeping of Engine 2 series.
Eq. 14.3. Where explained: chapter “The Dashboard”, section “Results”.

$$\Delta S = P + I - E_{\text{int}} - E - T - D, \tag{14.3}$$

Provenance (Eq. 14.3): **[Literature]**: water-balance identity.
Eq. 14.4. Where explained: chapter “The Dashboard”, section “The pinned baseline”.

$$\Delta_i = x_i - x_i^{\text{pin}}, \tag{14.4}$$

Provenance (Eq. 14.4): **[Engineered]**: bookkeeping of Engine 2 series.

18.1.16 The Dashboard, Design tab

Results of the open design layer.

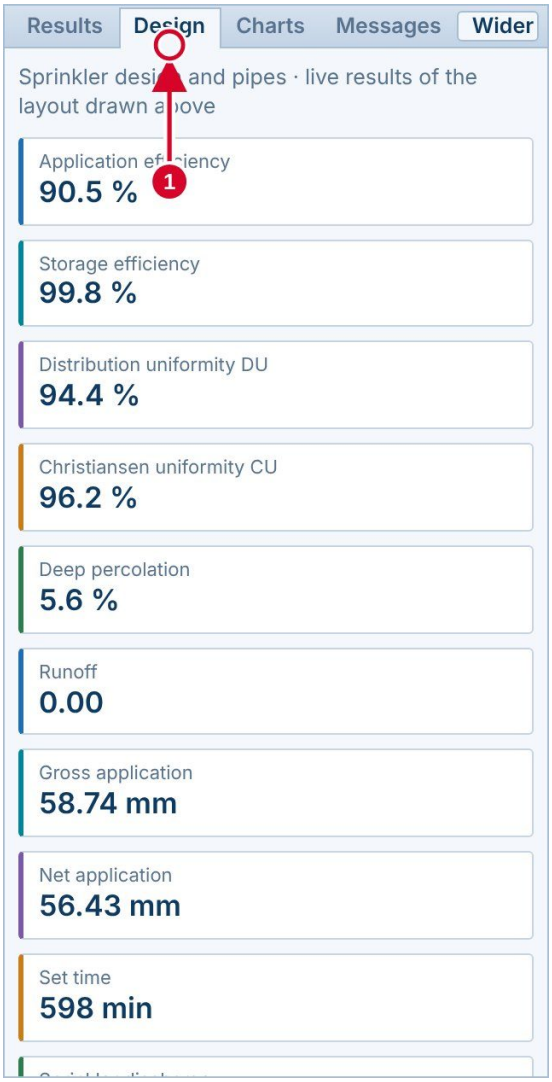


Figure 18.16: The Dashboard, Design tab.

What the arrows point to.

- **1** Design tab with a design layer open: efficiencies, uniformity, times and volumes of the system being designed.
- **2** Tabs.

Equations behind this screen. Eq. 11.18, Eq. 11.19, Eq. 12.19, Eq. 13.11.

Eq. 11.18. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".

$$
\begin{aligned}
&V_{\text{app}} = Q\,t_{co} \quad \big(\text{with cutback } c_b<1:\ Q\,t_L + c_b\,Q\,(t_{co}-t_L)\big), \quad V_{\text{req}} = Z_n\,L\,w,\\
&V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z,Z_n)\,dx,\\
&V_{\text{dp}} = w\int_0^L \max(Z-Z_n,0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n-Z,0)\,dx,\\
&V_{\text{ro}} = \max(V_{\text{app}}-V_{\text{inf}},\ 0).
\end{aligned}
\tag{11.18}
$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.19. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".

$$
E_a = \frac{V_{\text{st}}}{V_{\text{app}}},\quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}},\quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}},\quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}},\quad \text{CU} = 100\Big(1-\frac{\frac{1}{N+1}\sum_{i=0}^{N}|Z_i-\bar{Z}|}{\bar{Z}}\Big)\ \%.
\tag{11.19}
$$

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 12.19. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$
E_a = \frac{\langle V_{\text{st}}\rangle}{\langle V\rangle},\quad E_s = \frac{\langle V_{\text{st}}\rangle}{V_{\text{req}}},\quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)}+V_{\text{ex}}\rangle}{\langle V\rangle},\quad \text{EV} = \frac{\langle V_{\text{ev}}\rangle}{\langle V\rangle},\quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l},
\tag{12.19}
$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 13.11. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".

$$
\begin{aligned}
&E_a = \frac{\overline{\min(Z_k,Z_n)}}{\text{gross}}, && E_s = \frac{\overline{\min(Z_k,Z_n)}}{Z_n},\quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},\\
&\text{DP} = \frac{\overline{\max(Z_k-Z_n,0)}}{\text{gross}},\quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, && \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}},
\end{aligned}
\tag{13.11}
$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

18.1.17 The Dashboard, Charts tab

Yearly charts and the pinned baseline.

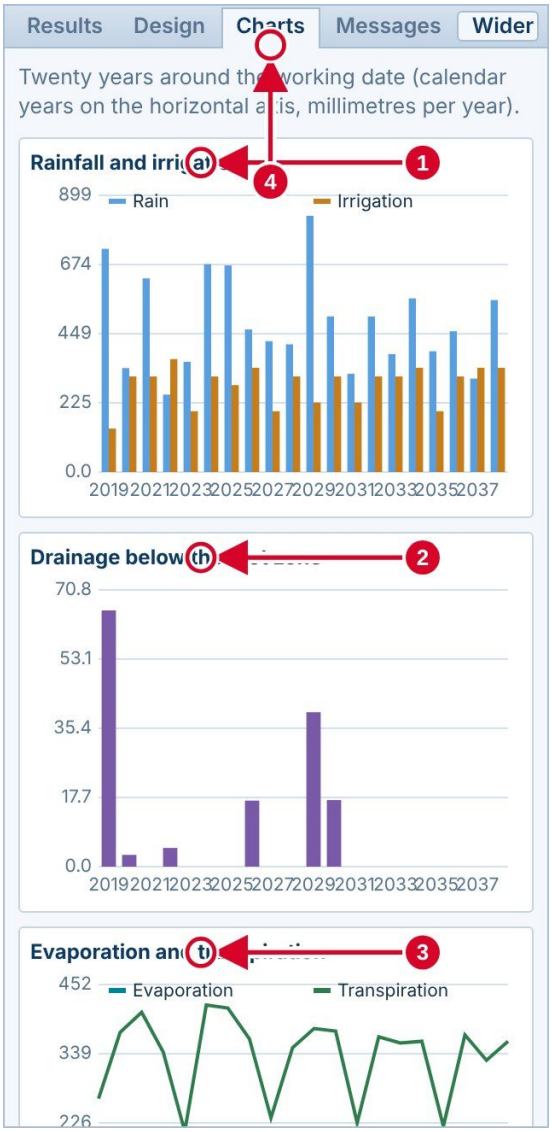


Figure 18.17: The Dashboard, Charts tab.

What the arrows point to.

- **1** Rainfall and irrigation for twenty years (Engine 1 annual values, Eqs. 5.4 to 5.12), with the pinned baseline of Eq. 14.5.
- **2** Drainage below the root zone, D of Eq. 5.9.
- **3** Evaporation and transpiration, E and T of Eqs. 5.4 to 5.12.

Equations behind this screen. Eq. 14.5.

Eq. 14.5. Where explained: chapter "The Dashboard", section "The pinned baseline".

$$Y^{(y)} = \sum_{d \in y} X_d, \qquad X \in \{P, I, D, E, T\}. \tag{14.5}$$

Provenance (Eq. 14.5): **[Engineered]**: bookkeeping of Engine 2 series.

18.1.18 The Dashboard, Messages tab

Plain-language messages.

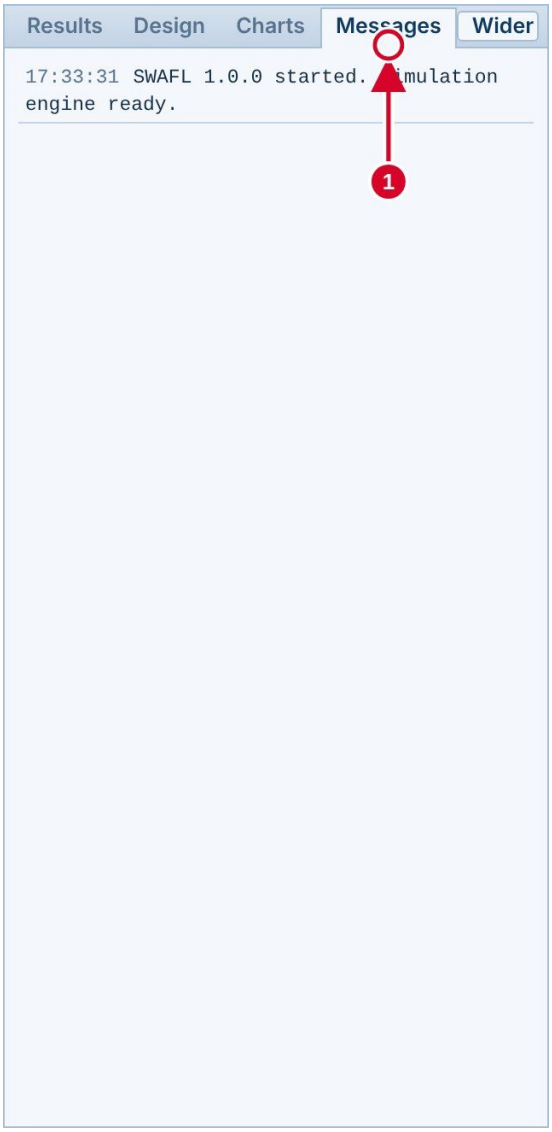


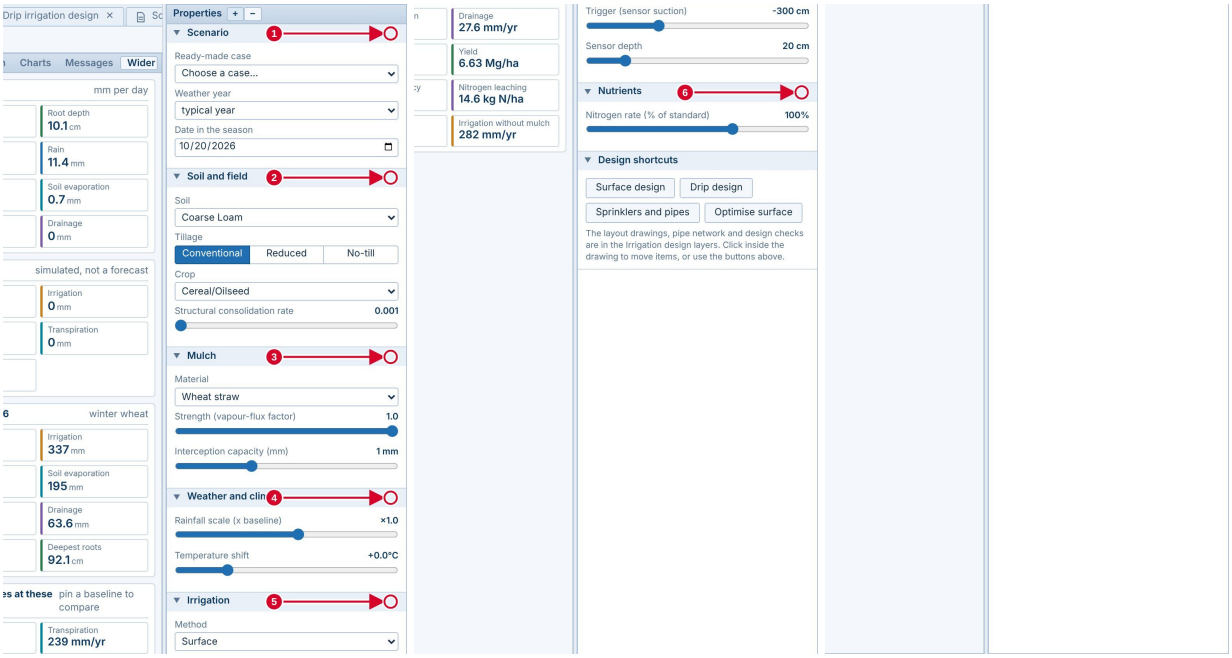
Figure 18.18: The Dashboard, Messages tab.

What the arrows point to.

- **1** Message log: engine start and plain-language notices.

18.1.19 The Properties panel

Every control; the equations are those that the controls move.



What the arrows point to.

- **1** Scenario: ready-made case, weather year and date (Eqs. 4.1 and 4.2).
- **2** Soil and field: soil, tillage, crop and consolidation rate tau_c (Eqs. 5.1, 5.2, 8.1).
- **3** Mulch: material, strength phi and interception capacity c (Eqs. 5.4 to 5.10, 9.1).
- **4** Weather and climate: rainfall scale rho and temperature shift (Eqs. 5.4 to 5.10).
- **5** Irrigation: method, trigger suction h_T and sensor depth (Eqs. 5.3, 5.11, 5.27).
- **6** Nutrients: nitrogen rate (Eqs. 5.10, 5.11).

Equations behind this screen. Eq. 5.18, Eq. 5.26, Eq. 8.1, Eq. 13.9.

Eq. 5.18. Where explained: chapter "The two computing engines", section "Roots".

$$R_{\max} = \mathrm{clip}\Big(R_c\,(0.85{+}0.3\,\mathrm{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \mathrm{cm}, \quad (5.18)$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.26. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$E_{0,\mathrm{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65\cdot\tfrac{1}{2}[1+\sin(2\pi(\mathrm{doy}-100)/365)]}{0.675}\,\mathrm{clip}(1{-}0.12(\rho{-}1);\,0.85,\,1.12)\sqrt{\gamma_y}\ \mathrm{m} \quad (5.26)$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

Eq. 8.1. Where explained: chapter "Soil structure and tillage layers", section "Pore index and consolidation".

$$p(t) = p_c + (p_0 - p_c)\,e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \quad (8.1)$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

Eq. 13.9. Where explained: chapter "Sprinkler design", section "The soil intake, ponding and runoff".

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \text{sup}, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \text{sup} - F, \quad \text{RO} = \max(0, R - S_d^{\text{eff}}), \quad I = \text{sup} - \text{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

18.2 S2 Surface irrigation design

18.2.1 The whole layer: inputs and overview

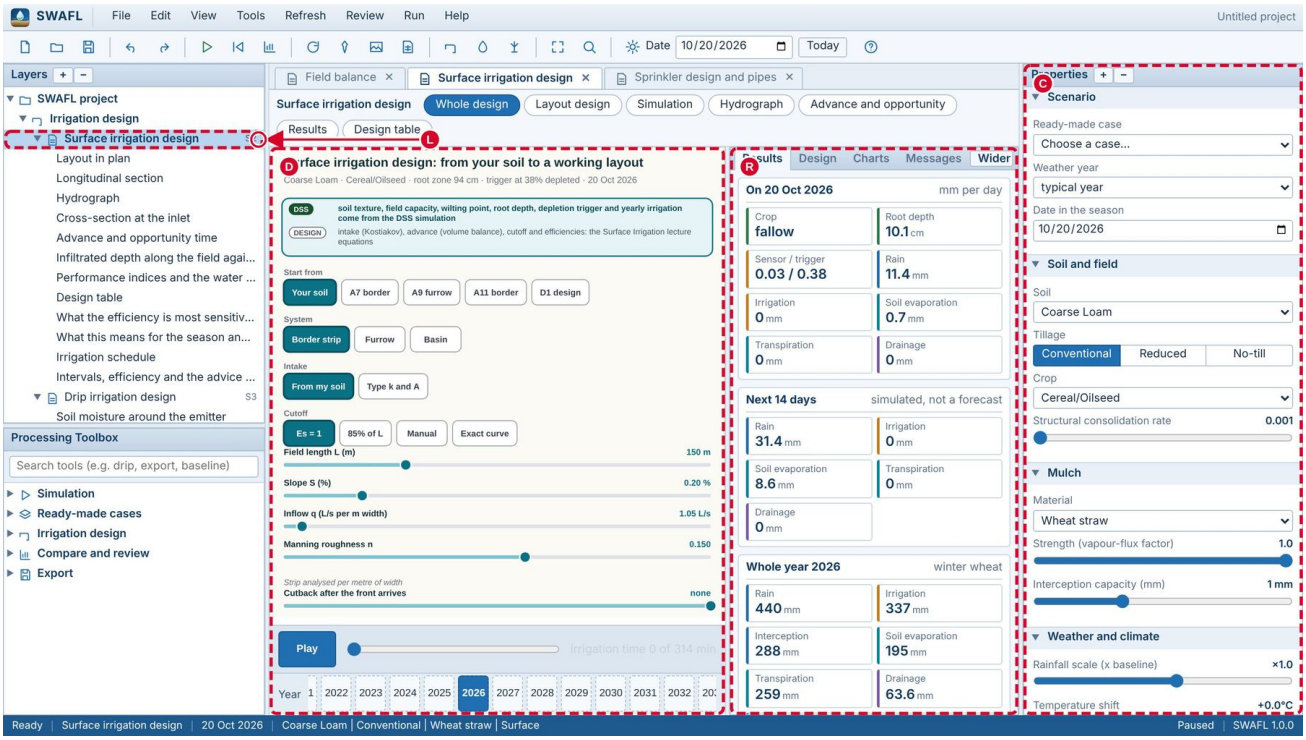


Figure 18.19: S2 Surface irrigation design: the whole workbench.

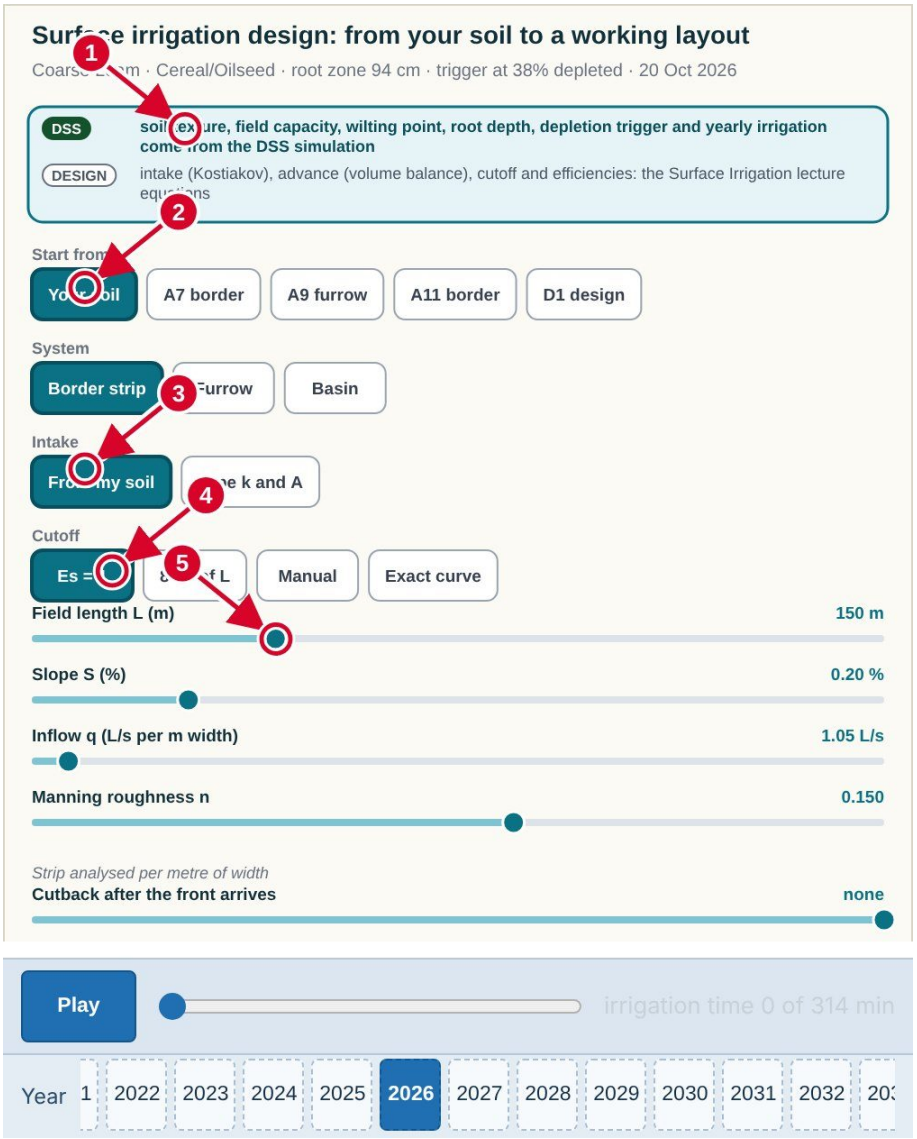


Figure 18.20: S2 Surface irrigation design: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** DSS badge: soil, root zone, trigger and yearly irrigation come from the Engine 1 and 2 results (Eqs. 5.1 to 5.12, 5.27).
- **2** “Start from”: in Your soil mode intake, Z_n and the best inflow are derived from the soil, Eqs. 11.1 to 11.7, 5.27 and 11.22.
- **3** Intake “From my soil”: Saxton-Rawls, Green-Ampt and Kostiakov, Eqs. 11.1 to 11.7.
- **4** Cut-off “Es = 1”: cut-off time t_co, Eq. 11.16.
- **5** Field length L, slope S, inflow q and Manning n: the inputs of Eqs. 11.8 and 11.9.

Equations behind this screen. Eq. 5.27, Eq. 11.1, Eq. 11.2, Eq. 11.3, Eq. 11.4, Eq. 11.5, Eq. 11.6, Eq. 11.7, Eq. 11.8, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13.

Eq. 5.27. Where explained: chapter “The two computing engines”, section “Evaporation under warming (used by the E1 layer)”.

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.1. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\begin{aligned}
\theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\mathrm{OM} + 0.005\,S\,\mathrm{OM} - 0.013\,C\,\mathrm{OM} + 0.068\,S\,C + 0.031,\\
\theta_{1500} &= \theta_{1500,t} + \big(0.14\,\theta_{1500,t} - 0.02\big),\\
\theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\mathrm{OM} + 0.006\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} + 0.452\,S\,C + 0.299,\\
\theta_{33} &= \theta_{33,t} + \big(1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015\big),
\end{aligned} \tag{11.1}$$

Provenance (Eq. 11.1): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.2. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\begin{aligned}
\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078,\\
\theta_{(S-33)} &= \theta_{(S-33),t} + \big(0.636\,\theta_{(S-33),t} - 0.107\big), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\
B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.
\end{aligned} \tag{11.2}$$

Provenance (Eq. 11.2): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 11.4. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

Provenance (Eq. 11.4): **[Engineered]**: initial and wetted water contents chosen for the workbench.

Eq. 11.5. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

Provenance (Eq. 11.5): **[Literature]**: Green and Ampt (1911).

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.7. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

Provenance (Eq. 11.7): **[Literature]**: ordinary least squares on the log–log plot.

Eq. 11.8. Where explained: chapter "Surface irrigation design", section "Inlet flow area (Manning)".

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

Provenance (Eq. 11.8): **[Literature]**: Manning (1891).

Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_{L} = (1 + 0.15\,A)\,\frac{A_{0} L}{Q} + \left(\frac{\sigma_{F}\,k\,w\,L}{Q}\right)^{\frac{1}{1-A}}, \qquad \sigma_{F} = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_{x}(x) = t_{V}(x)\,\frac{t_{L}}{t_{V}(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

18.2.2 Layout in plan: the advance front, shaded by the depth already infiltrated

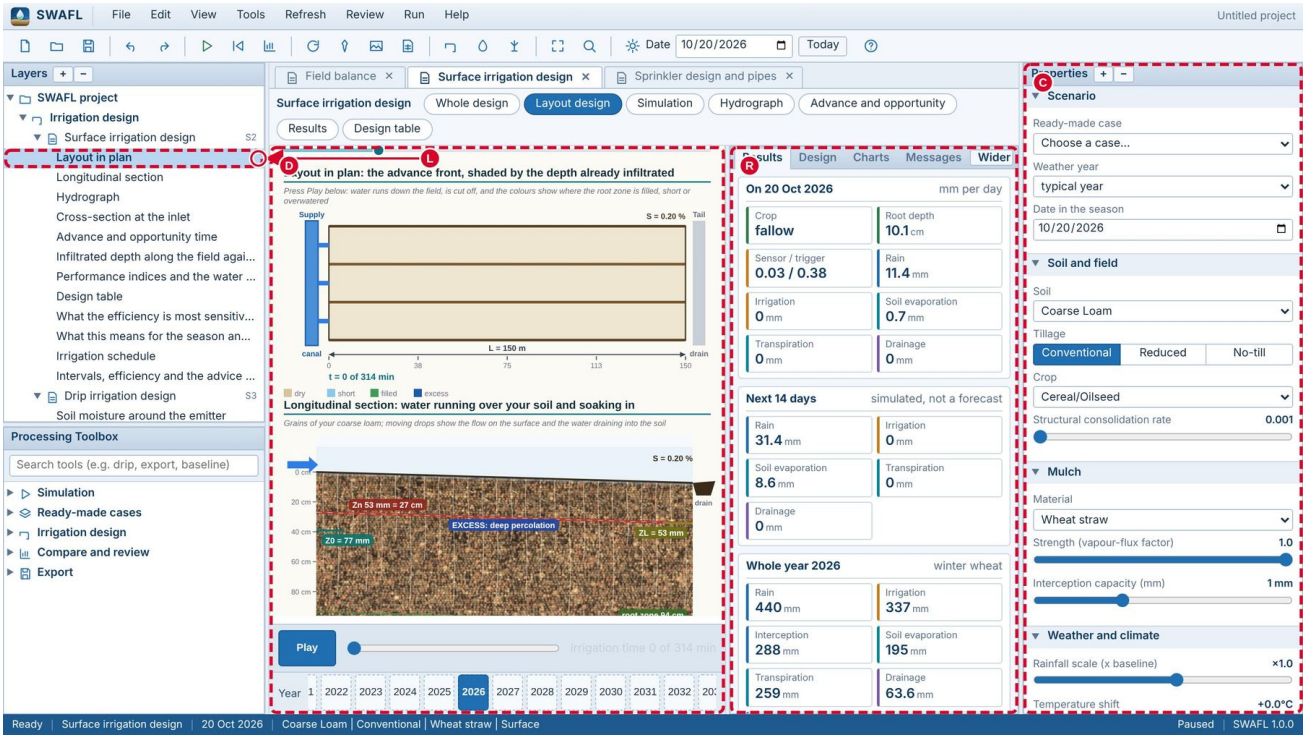


Figure 18.21: S2 Surface irrigation design / Layout in plan: the whole workbench.

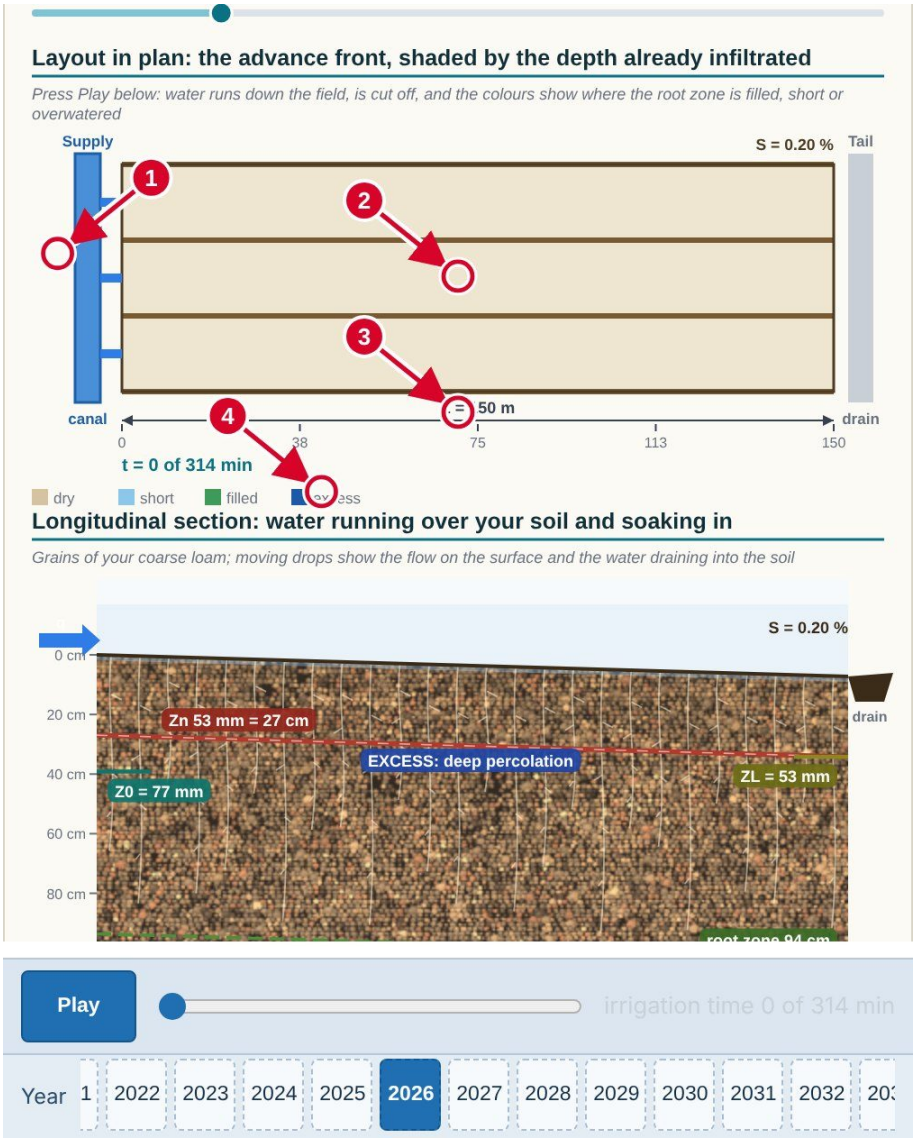


Figure 18.22: S2 Surface irrigation design / Layout in plan: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Supply canal: inflow q enters the strip (flow area A_0 from Manning, Eq. 11.8).
- **2** Strip in plan: front position x(t) from Eq. 11.9; colours show infiltrated depth Z(x) against Z_n, Eq. 11.17.
- **3** Field length L (input).
- **4** Legend dry / short / filled / excess: classes of Z(x) against Z_n, Eq. 11.17.

Equations behind this screen. Eq. 5.27, Eq. 11.8, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13.

Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$Z_n = \mathrm{clip}\left(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\right)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.8. Where explained: chapter "Surface irrigation design", section "Inlet flow area (Manning)".

$$Q = \frac{1}{n}\, A\, R_h^{2/3}\, S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

Provenance (Eq. 11.8): **[Literature]**: Manning (1891).

Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$Q\,t = \sigma_y\, A_0\, x + \sigma_z\, k\, t^{A}\, w\, x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\, k\, t^{A}\, w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_x(x) = t_L \Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\, q^{-q_e}\, x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\, k\, w\, L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

18.2.3 Longitudinal section: water running over your soil and soaking in

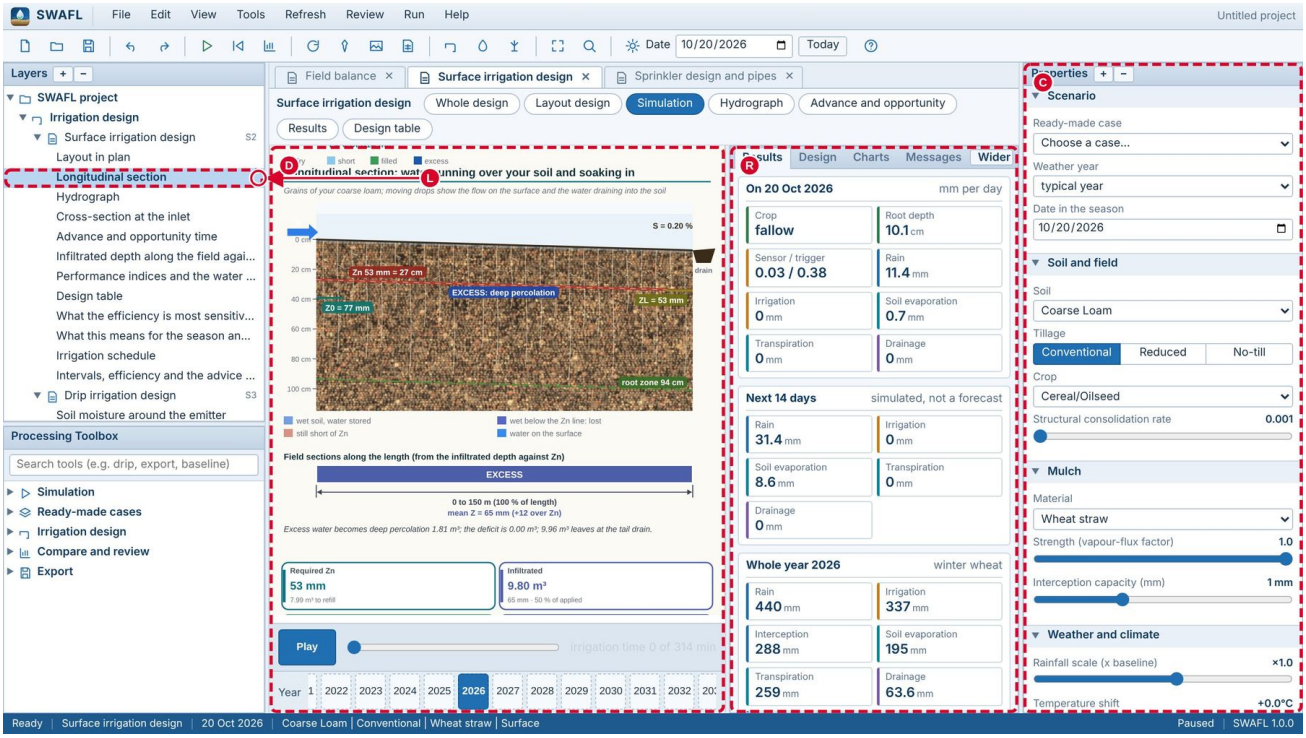


Figure 18.23: S2 Surface irrigation design / Longitudinal section: the whole workbench.

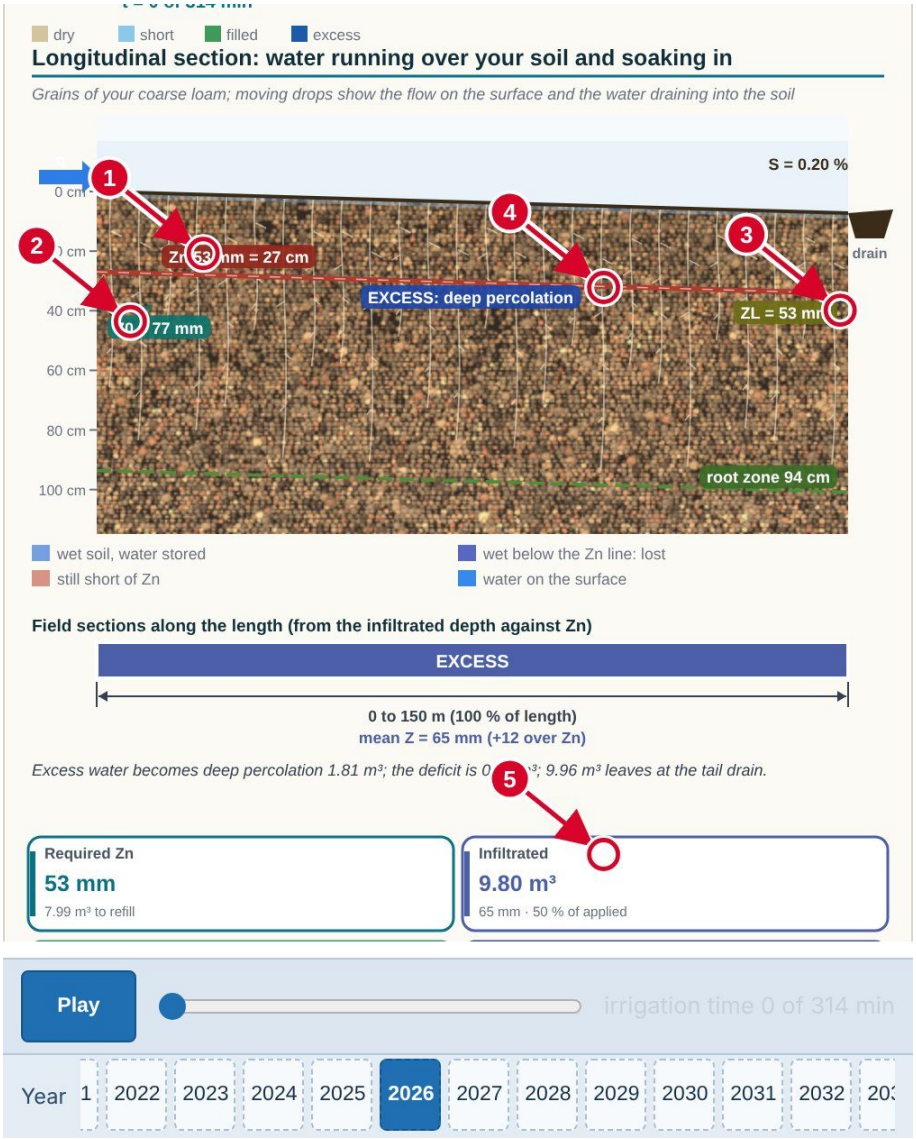


Figure 18.24: S2 Surface irrigation design / Longitudinal section: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Z_n line: required depth from the DSS (Eq. 5.27) drawn as depth of wetted soil.
- **2** Z_0: depth infiltrated at the inlet, Eq. 11.17.
- **3** Z_L: depth infiltrated at the tail, Eq. 11.17.
- **4** EXCESS: water below the Z_n line is lost to deep percolation, V_dp of Eq. 11.18.
- **5** Infiltrated volume V_inf (tile at the foot), Eq. 11.18.

Equations behind this screen. Eq. 11.6, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13, Eq. 11.15, Eq. 11.16, Eq. 11.17.

Eq. 11.6. Where explained: chapter “Surface irrigation design”, section “Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)”.

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \;\Longrightarrow\; x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$\sigma_z = \frac{A + r\,(1-A) + 1}{(1+A)(1+r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.15. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

Eq. 11.16. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{co} = \begin{cases} t_L + t_{st} & \text{"}E_s = 1\text{": the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{"85 \% of L"} \\ t_{co}^{\text{set}} & \text{"Manual"} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

Eq. 11.17. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$Z(x) = k\left[t_{co} - t_x(x)\right]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

18.2.4 Hydrograph: inflow, infiltration and runoff through time

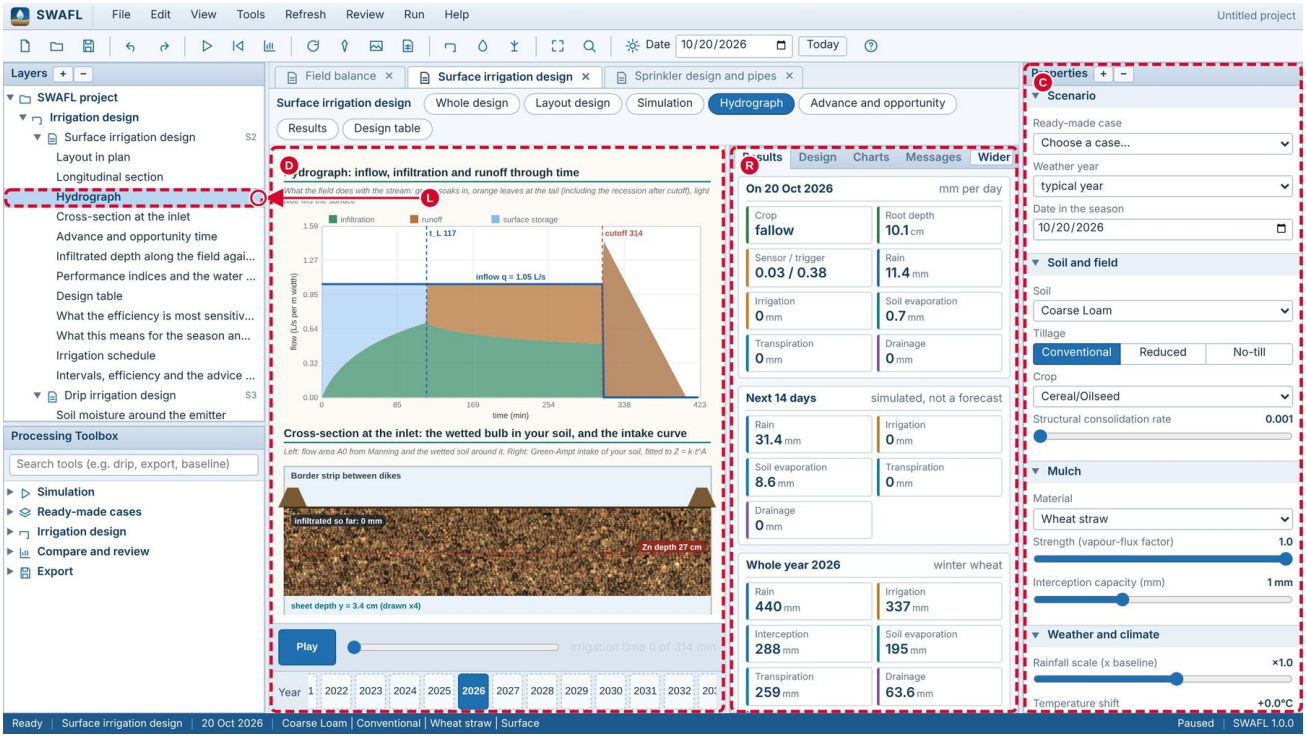


Figure 18.25: S2 Surface irrigation design / Hydrograph: the whole workbench.

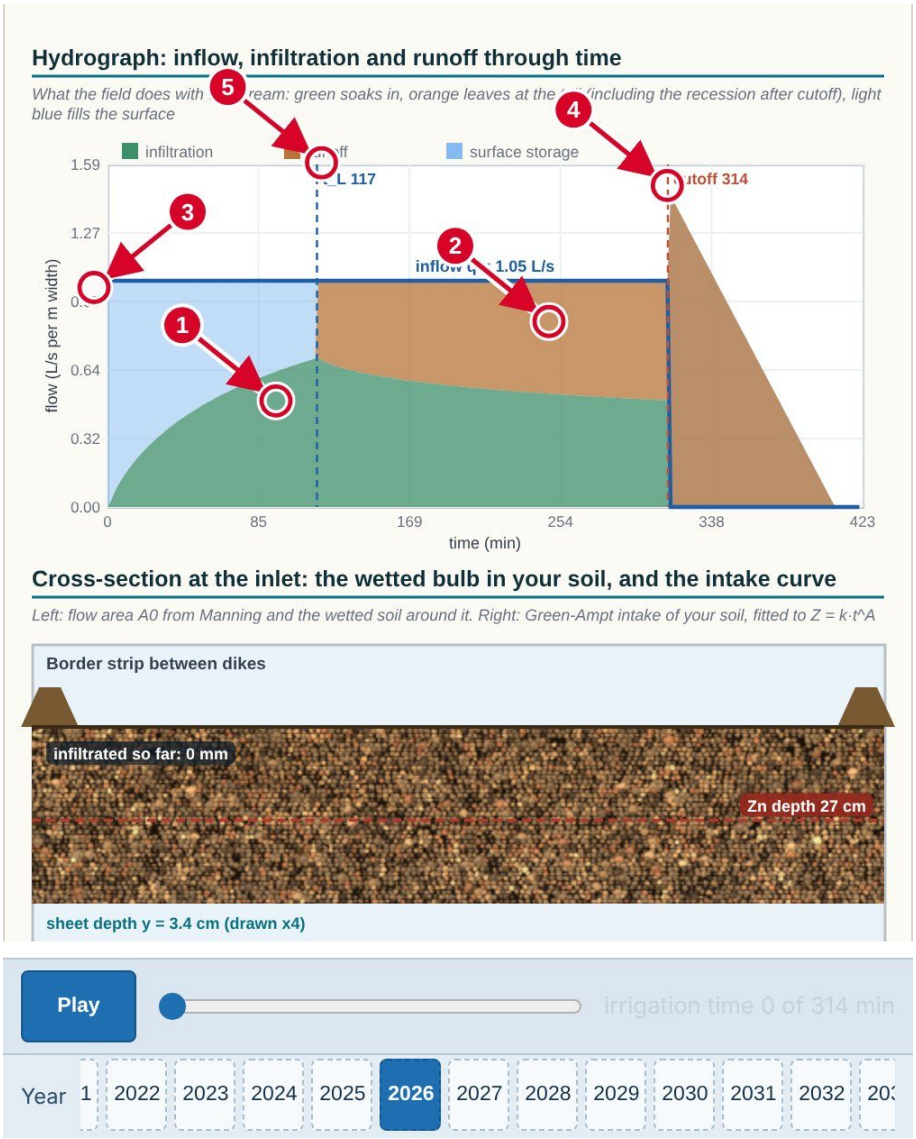


Figure 18.26: S2 Surface irrigation design / Hydrograph: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Green: infiltration I(t): Eq. 11.20.
- **2** Orange: runoff R(t) with its recession after cut-off: Eqs. 11.20 and 11.21.
- **3** Blue: water stored on the surface: Eq. 11.21.
- **4** Cut-off time t_co: Eq. 11.16.
- **5** t_L: advance time to the tail, Eq. 11.9.

Equations behind this screen. Eq. 11.18, Eq. 11.20, Eq. 11.21.

Eq. 11.18. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".

$$V_{\text{app}} = Q\,t_{co} \quad \big(\text{with cutback } c_b < 1:\ Q\,t_L + c_b\,Q\,(t_{co} - t_L)\big), \quad V_{\text{req}} = Z_n\,L\,w,$$
$$V_{\text{inf}} = w\int_0^L Z\,dx, \qquad V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx,$$
$$V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, \qquad V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx,$$
$$V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0). \tag{11.18}$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.20. Where explained: chapter "Surface irrigation design", section "Hydrograph".

$$V_c(t) = w\sum_i k\left[\min(t, t_{co}) - \bar{t}_{x,i}\right]_+^A \Delta x\ \cdot s_c, \qquad s_c = \frac{V_{\text{inf}}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}$$

Provenance (Eq. 11.20): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

Eq. 11.21. Where explained: chapter "Surface irrigation design", section "Hydrograph".

$$R(t) = \frac{2V_s}{t_r}\Big(1 - \frac{t - t_{co}}{t_r}\Big), \quad t_{co} < t < t_{co} + t_r. \tag{11.21}$$

Provenance (Eq. 11.21): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

18.2.5 Cross-section at the inlet: the wetted bulb in your soil, and the intake curve

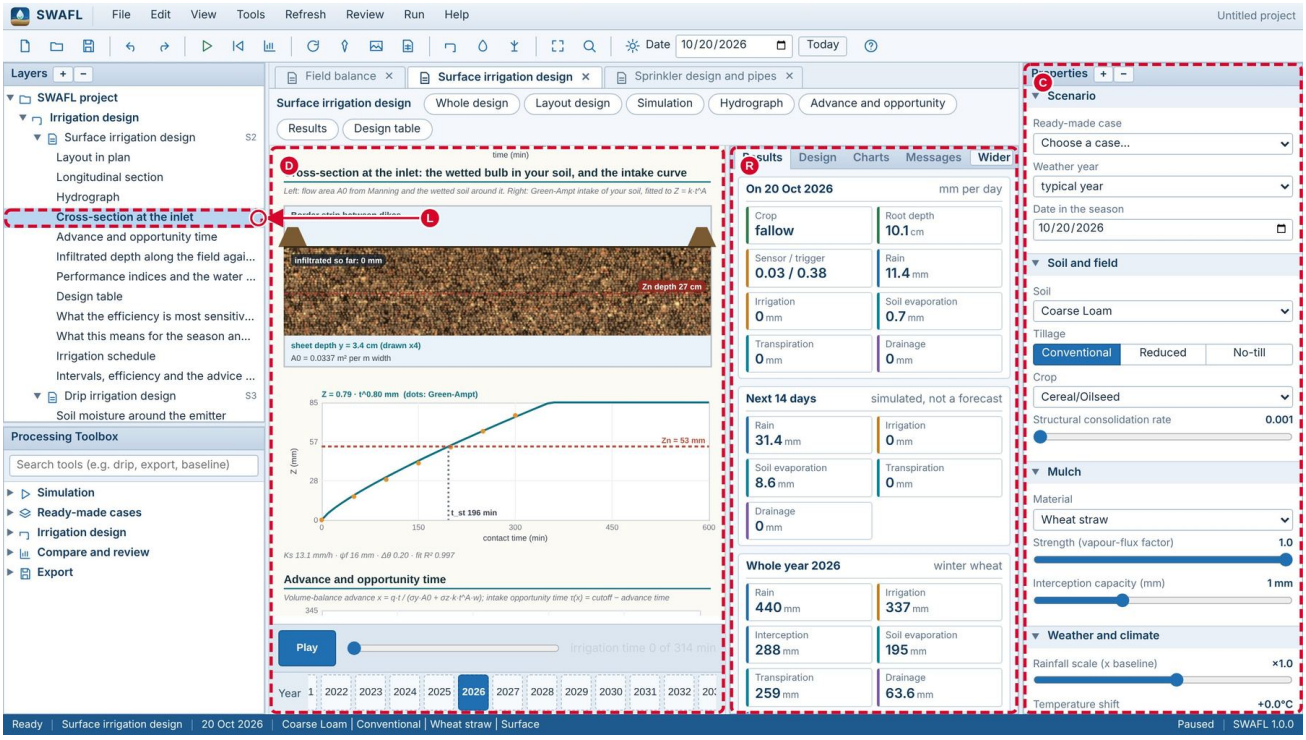


Figure 18.27: S2 Surface irrigation design / Cross-section at the inlet: the whole workbench.

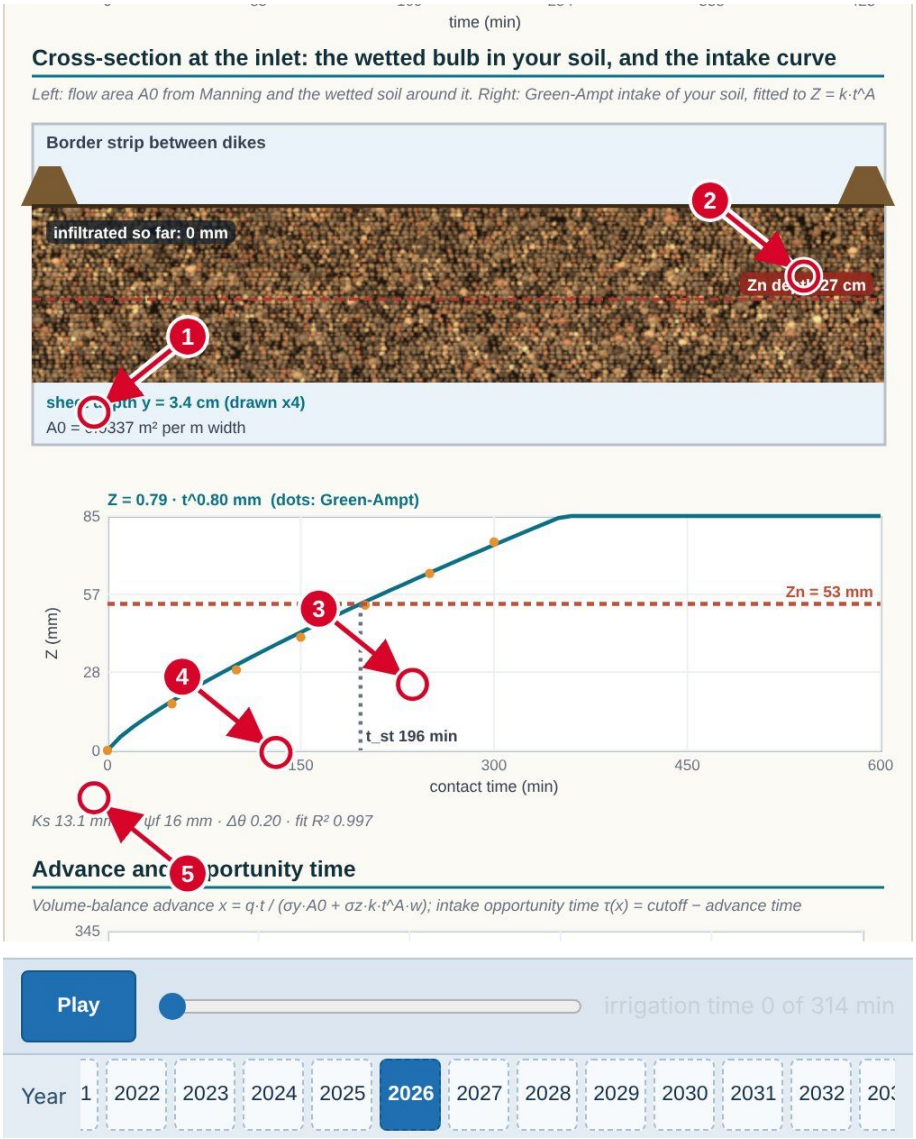


Figure 18.28: S2 Surface irrigation design / Cross-section at the inlet: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sheet depth and flow area A_0 from Manning: Eq. 11.8.
- **2** Z_n depth: the required depth Z_n as a depth of wetted soil (storage of Eq. 11.4).
- **3** Intake curve Z = k tau^A fitted to Green-Ampt: Eqs. 11.5 to 11.7.
- **4** t_st: time to infiltrate the required depth, Eq. 11.15.
- **5** Ks, psi_f and Delta theta of the soil: Eqs. 11.2 to 11.4.

Equations behind this screen. Eq. 5.27, Eq. 11.1, Eq. 11.2, Eq. 11.3, Eq. 11.4, Eq. 11.5, Eq. 11.6, Eq. 11.7.

Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.1. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\begin{aligned}
\theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\mathrm{OM} + 0.005\,S\,\mathrm{OM} - 0.013\,C\,\mathrm{OM} + 0.068\,S\,C + 0.031,\\
\theta_{1500} &= \theta_{1500,t} + (0.14\,\theta_{1500,t} - 0.02),\\
\theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\mathrm{OM} + 0.006\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} + 0.452\,S\,C + 0.299,\\
\theta_{33} &= \theta_{33,t} + (1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015),
\end{aligned} \tag{11.1}$$

Provenance (Eq. 11.1): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.2. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\begin{aligned}
\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078,\\
\theta_{(S-33)} &= \theta_{(S-33),t} + (0.636\,\theta_{(S-33),t} - 0.107), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\
B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.
\end{aligned} \tag{11.2}$$

Provenance (Eq. 11.2): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 11.4. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

Provenance (Eq. 11.4): **[Engineered]**: initial and wetted water contents chosen for the workbench.

Eq. 11.5. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

Provenance (Eq. 11.5): **[Literature]**: Green and Ampt (1911).

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.7. Where explained: chapter “Surface irrigation design”, section “Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)”.

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\; y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

Provenance (Eq. 11.7): **[Literature]**: ordinary least squares on the log–log plot.

18.2.6 Advance and opportunity time

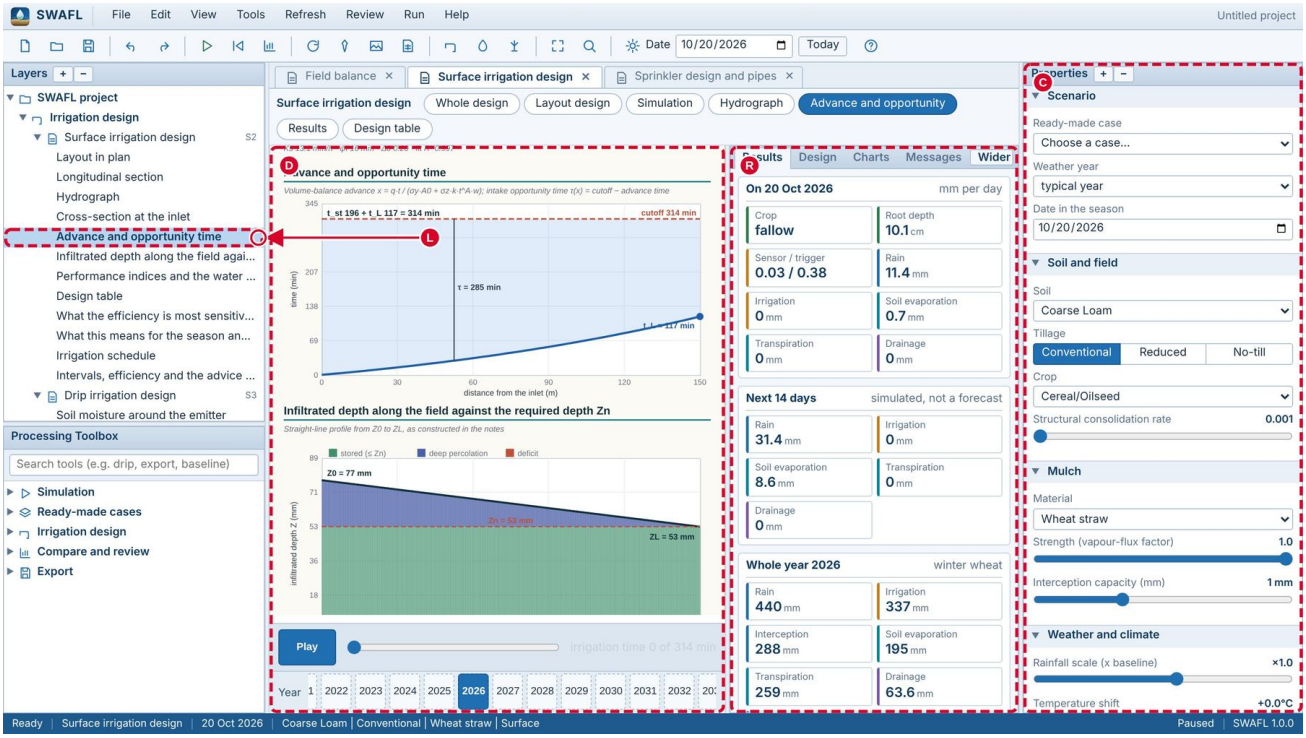


Figure 18.29: S2 Surface irrigation design / Advance and opportunity time: the whole workbench.

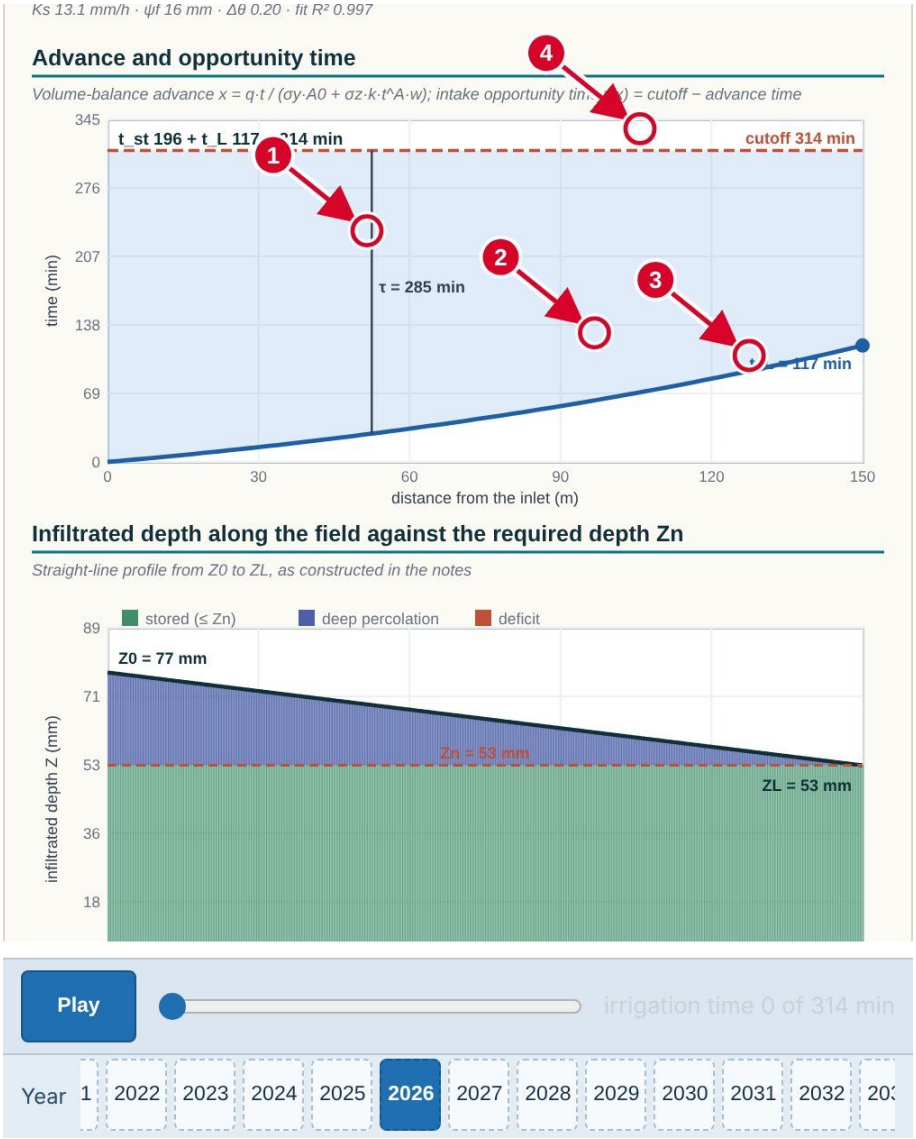


Figure 18.30: S2 Surface irrigation design / Advance and opportunity time: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Opportunity time tau = t_co - t_x: Eq. 11.17.
- **2** Advance curve x(t) from the volume balance: Eq. 11.9.
- **3** t_L: time for the front to reach the tail.
- **4** Cut-off 314 min = t_st + t_L: Eqs. 11.15 and 11.16.

Equations behind this screen. Eq. 11.6, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13, Eq. 11.15, Eq. 11.16, Eq. 11.17.

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume bal-

ance)".

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \;\Longrightarrow\; x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.15. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

Eq. 11.16. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{co} = \begin{cases} t_L + t_{st} & \text{"}E_s = 1\text{": the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{"85 \% of L"} \\ t_{co}^{\text{set}} & \text{"Manual"} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

Eq. 11.17. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$Z(x) = k\left[t_{co} - t_x(x)\right]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

18.2.7 Infiltrated depth along the field against the required depth Zn

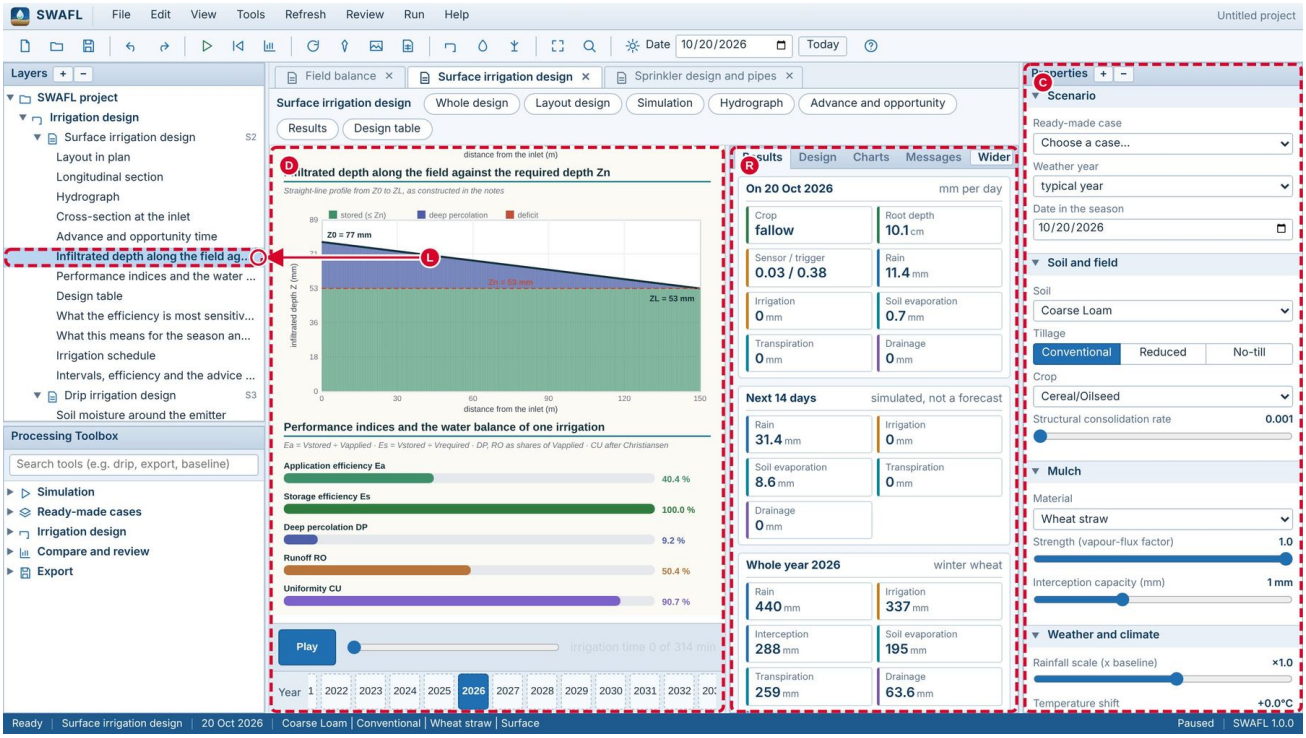


Figure 18.31: S2 Surface irrigation design / Infiltrated depth along the field against the required depth Zn: the whole workbench.

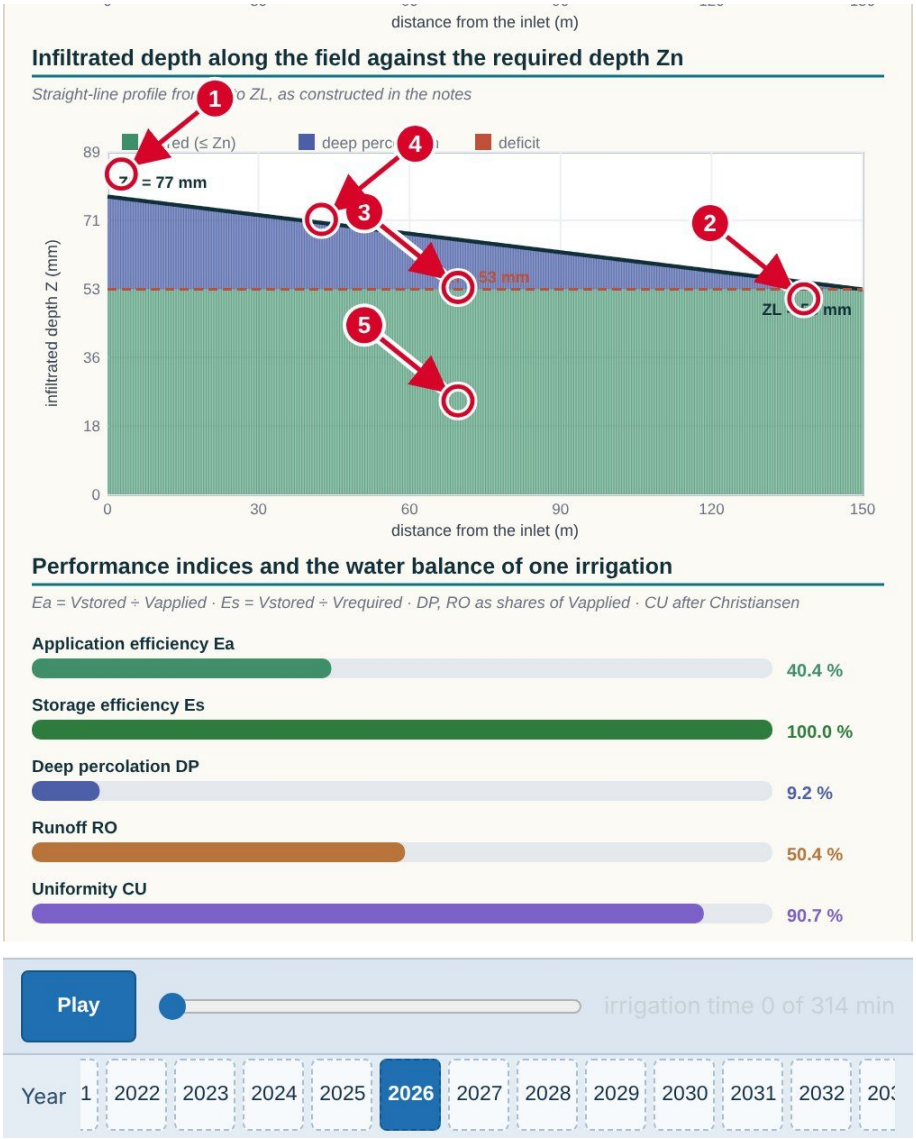


Figure 18.32: S2 Surface irrigation design / Infiltrated depth along the field against the required depth Zn: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Z_0: depth at the inlet, Eq. 11.17.
- **2** Z_L: depth at the tail, Eq. 11.17.
- **3** Z_n: required depth, Eq. 5.27.
- **4** Indigo: depth above Z_n, lost to deep percolation, V_dp of Eq. 11.18.
- **5** Green: water stored in the root zone, V_st of Eq. 11.18.

Equations behind this screen. Eq. 11.6, Eq. 11.15, Eq. 11.16, Eq. 11.17.

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.15. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{st} = \left(\frac{Z_n}{k}\right)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

Eq. 11.16. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

Eq. 11.17. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$Z(x) = k\left[t_{co} - t_x(x)\right]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

18.2.8 Performance indices and the water balance of one irrigation

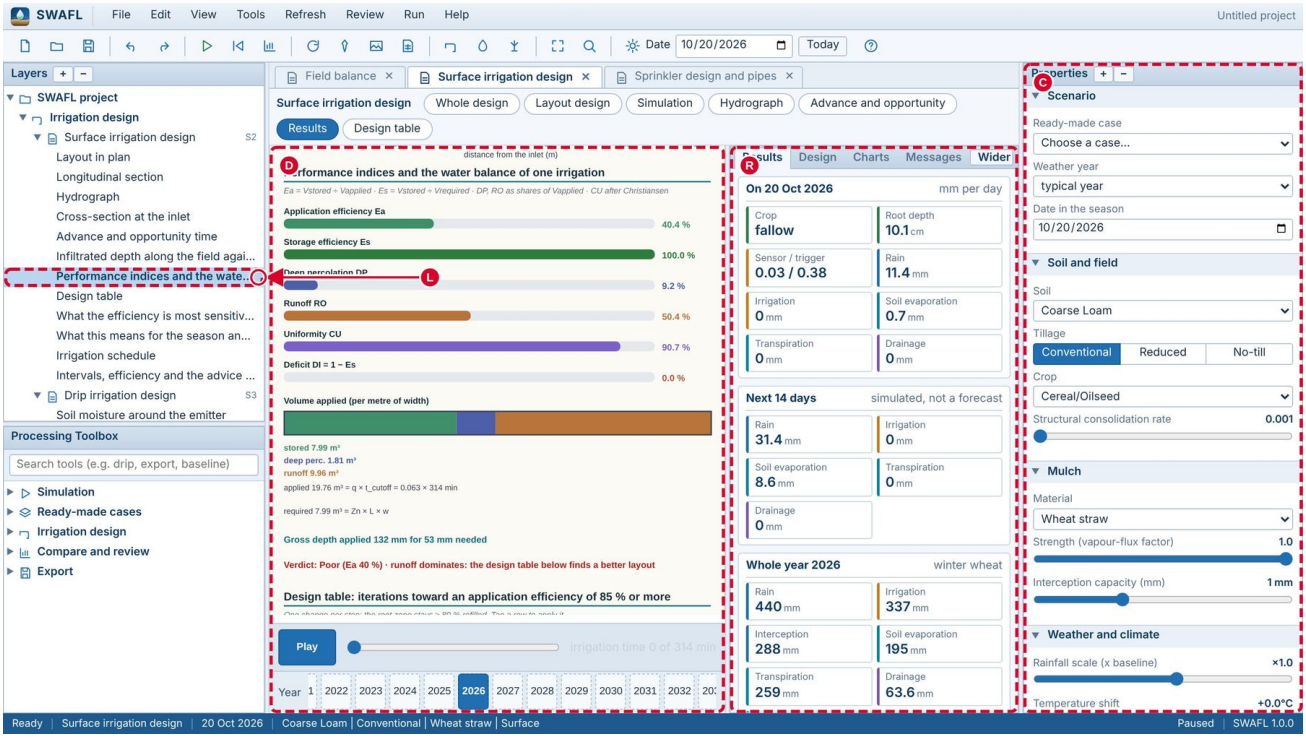


Figure 18.33: S2 Surface irrigation design / Performance indices and the water balance of one irrigation: the whole workbench.

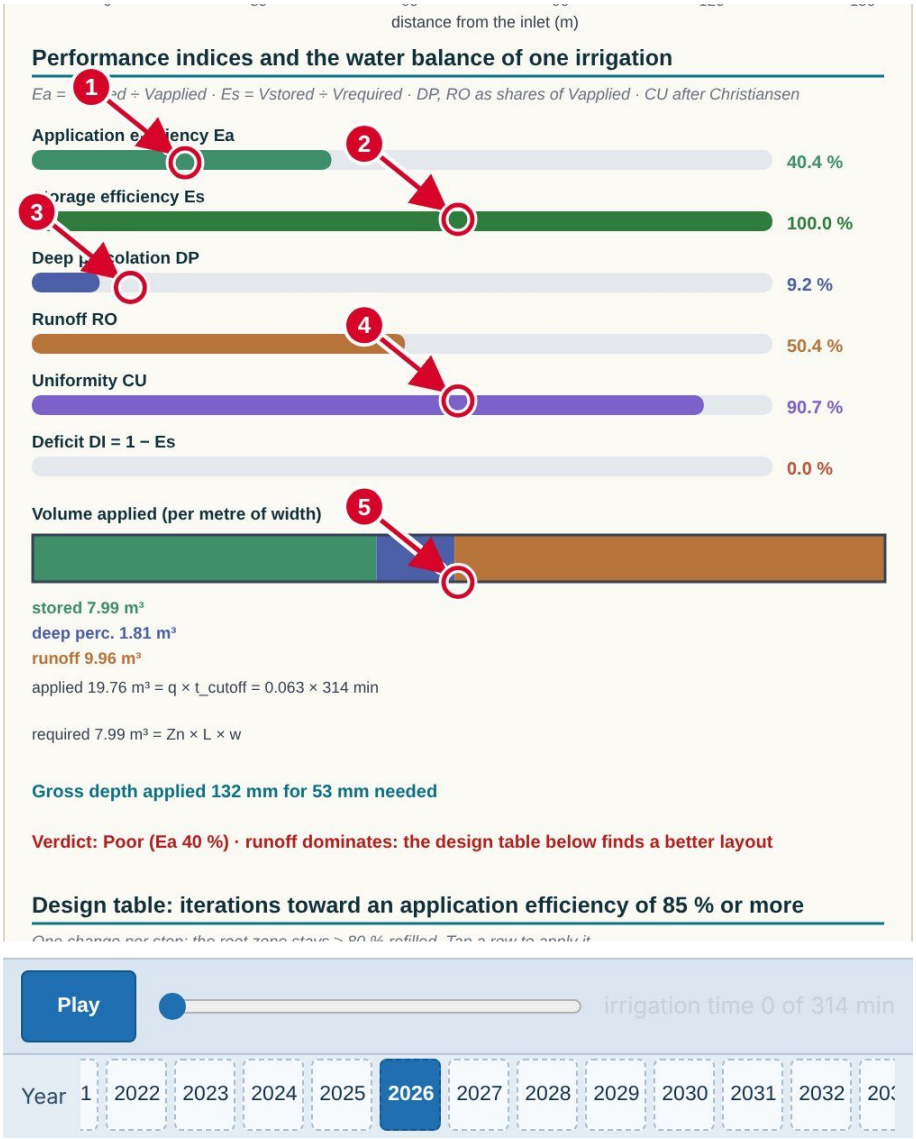


Figure 18.34: S2 Surface irrigation design / Performance indices and the water balance of one irrigation: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Application efficiency E_a = V_st / V_app: Eq. 11.19.
- **2** Storage efficiency E_s = V_st / V_req: Eq. 11.19.
- **3** Deep percolation DP and runoff RO as shares of the applied volume: Eq. 11.19.
- **4** Uniformity CU after Christiansen: Eq. 11.19.
- **5** Volumes applied, stored, percolated and run off: Eq. 11.18.

Equations behind this screen. Eq. 11.18, Eq. 11.19.

Eq. 11.18. Where explained: chapter “Surface irrigation design”, section “Volumes, efficiencies and uniformity”.

$$
\begin{aligned}
&V_{\text{app}} = Q\,t_{co} \quad \big(\text{with cutback } c_b<1:\ Q\,t_L + c_b\,Q\,(t_{co}-t_L)\big), \quad V_{\text{req}} = Z_n\,L\,w,\\
&V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx,\\
&V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx,\\
&V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0).
\end{aligned}
\tag{11.18}
$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.19. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".

$$
E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N}|Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%.
\tag{11.19}
$$

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

18.2.9 Design table: iterations toward an application efficiency of 85 % or more

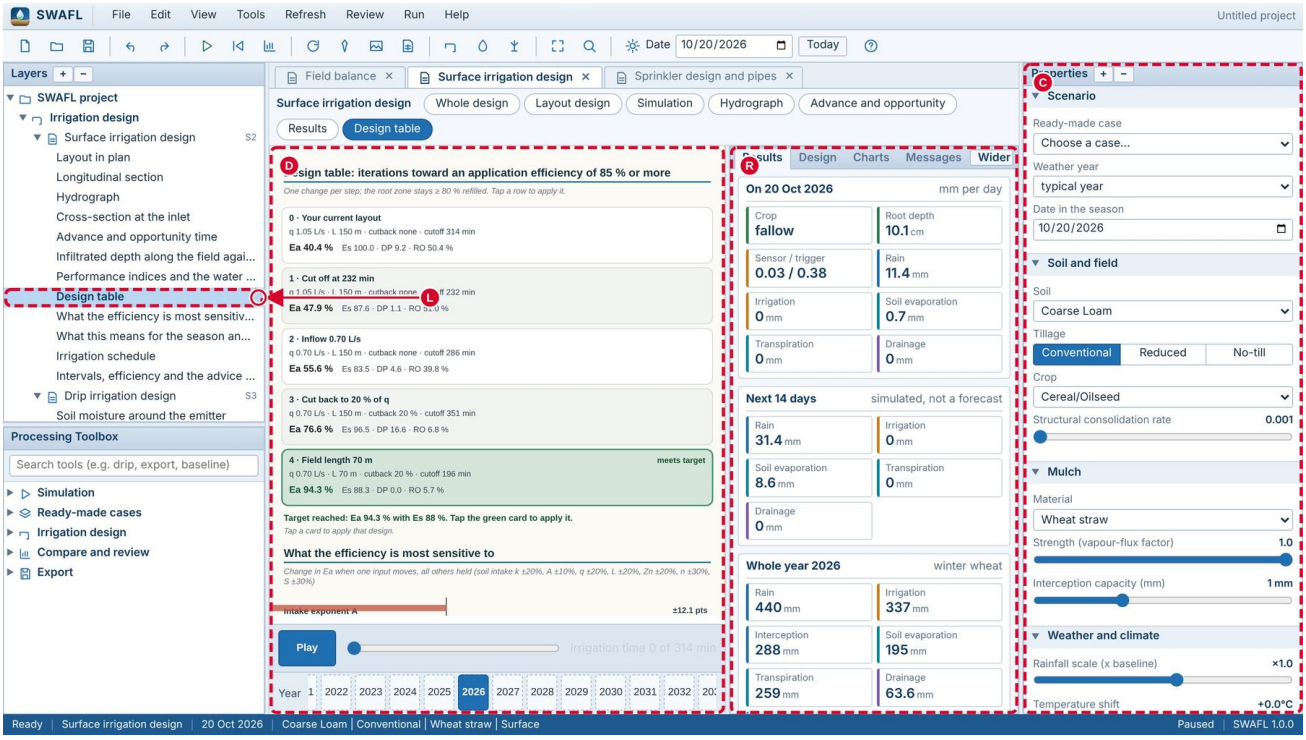


Figure 18.35: S2 Surface irrigation design / Design table: the whole workbench.

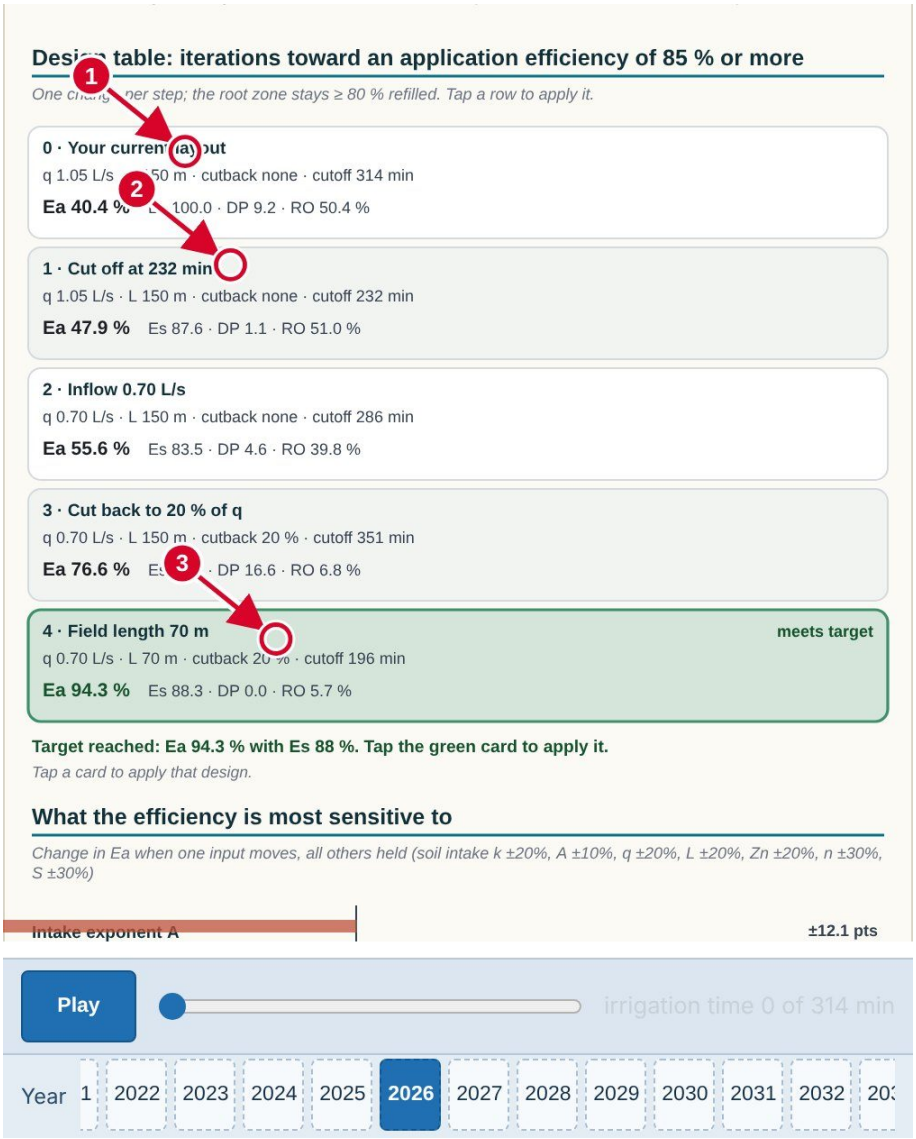


Figure 18.36: S2 Surface irrigation design / Design table: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Row 0: your current layout; its score is Eq. 11.23.
- **2** Candidate moves in cut-off time, Eq. 11.16, then inflow (Eq. 11.22), cutback and length.
- **3** Winning design: E_a of at least 85 % with E_s of at least 80 %, chosen by the score of Eq. 11.23.

Equations behind this screen. Eq. 11.22, Eq. 11.23.

Eq. 11.22. Where explained: chapter "Surface irrigation design", section "Best inflow ("Your soil" mode)".

$$q_{\max} = \min\left\{ q_{\max}^{\mathrm{sys}},\ \frac{0.04}{S} \cdot \frac{1000}{60} \right\} \mathrm{L\,s^{-1}} \quad (S \text{ in } \%), \tag{11.22}$$

Provenance (Eq. 11.22): **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

Eq. 11.23. Where explained: chapter “Surface irrigation design”, section “The design table”.

$$\text{score} = \min(E_a, 1) - 3\max(0,\, 0.8 - E_s) - 3\,[c_b < 1]\,\max(0,\, 0.05 - \text{RO}) - 5\max(0,\, E_a - 1). \tag{11.23}$$

Provenance (Eq. 11.23): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

18.2.10 What the efficiency is most sensitive to

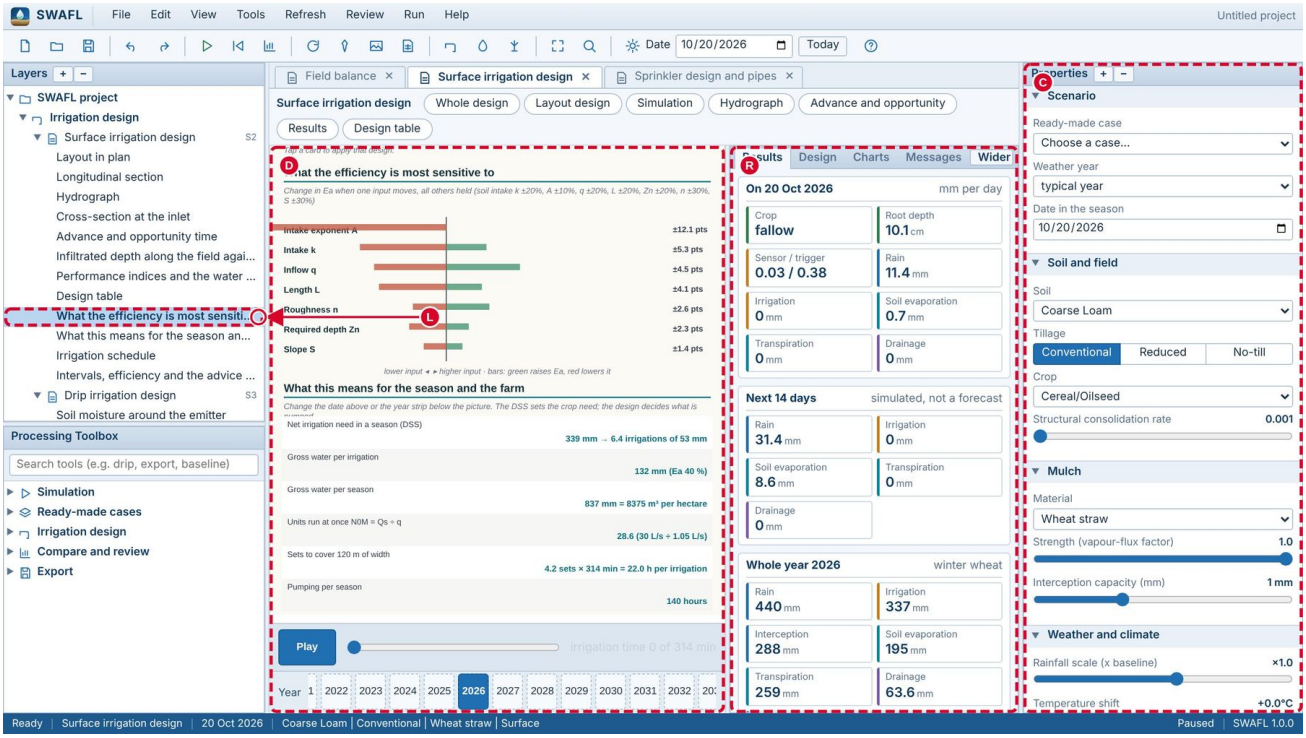


Figure 18.37: S2 Surface irrigation design / What the efficiency is most sensitive to: the whole workbench.

Figure 18.38: S2 Surface irrigation design / What the efficiency is most sensitive to: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Tornado bar: change in E_a (Eq. 11.19) when this input moves by 10 to 30 %.
- **2** Size of the change in percentage points.
- **3** Green raises E_a, red lowers it.

Equations behind this screen. none (the screen shows stored results).

18.2.11 What this means for the season and the farm

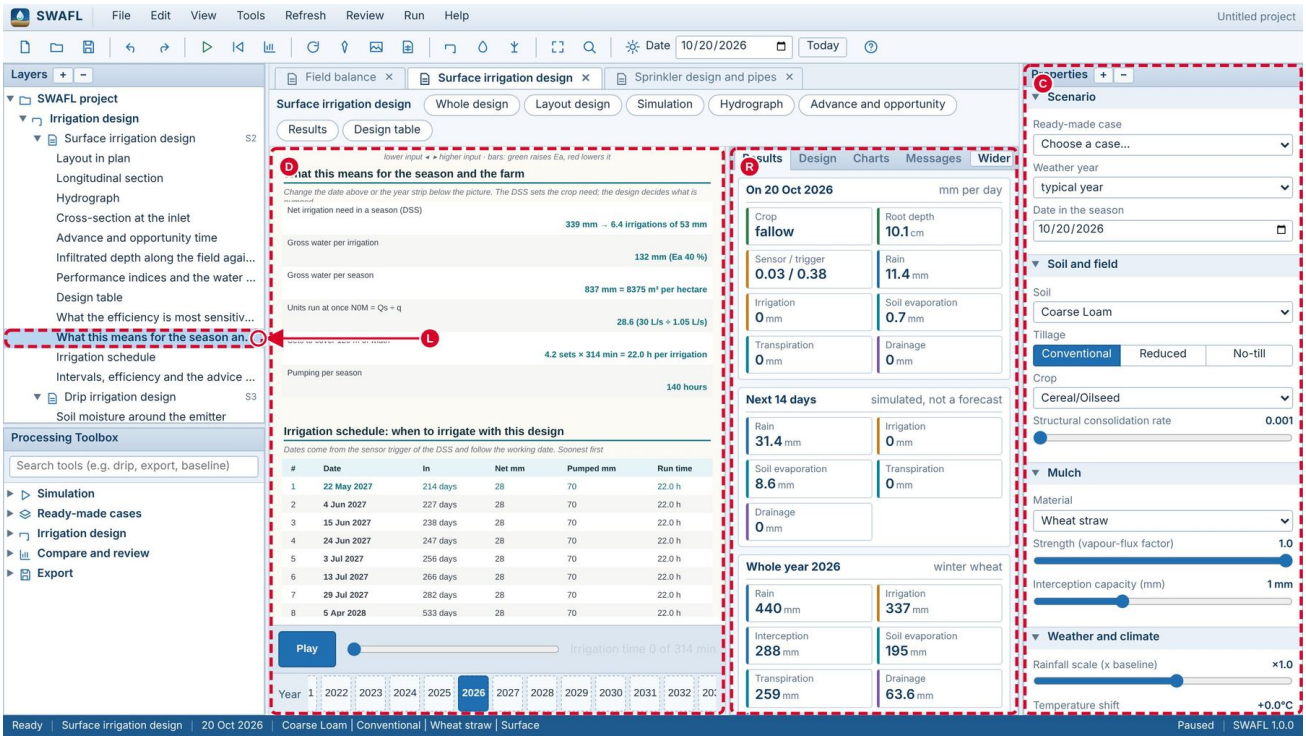


Figure 18.39: S2 Surface irrigation design / What this means for the season and the farm: the whole workbench.

Figure 18.40: S2 Surface irrigation design / What this means for the season and the farm: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Irrigations in a season, n_ev = sum I / Z_n: Eq. 11.24.
- **2** Gross depth per irrigation, Z_g = Z_n / E_a: Eq. 11.24.
- **3** Units run at once, Q_s / q: Eq. 11.24.
- **4** Sets and hours per irrigation: Eq. 11.24.

Equations behind this screen. Eq. 11.24.

Eq. 11.24. Where explained: chapter “Surface irrigation design”, section “Season and farm”.

$$n_{\text{ev}} = \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g,$$
$$N_{0M} = \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\left(1, \frac{N_{\text{units}}}{N_{0M}}\right), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}), \tag{11.24}$$

Provenance (Eq. 11.24): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

18.2.12 Irrigation schedule: when to irrigate with this design

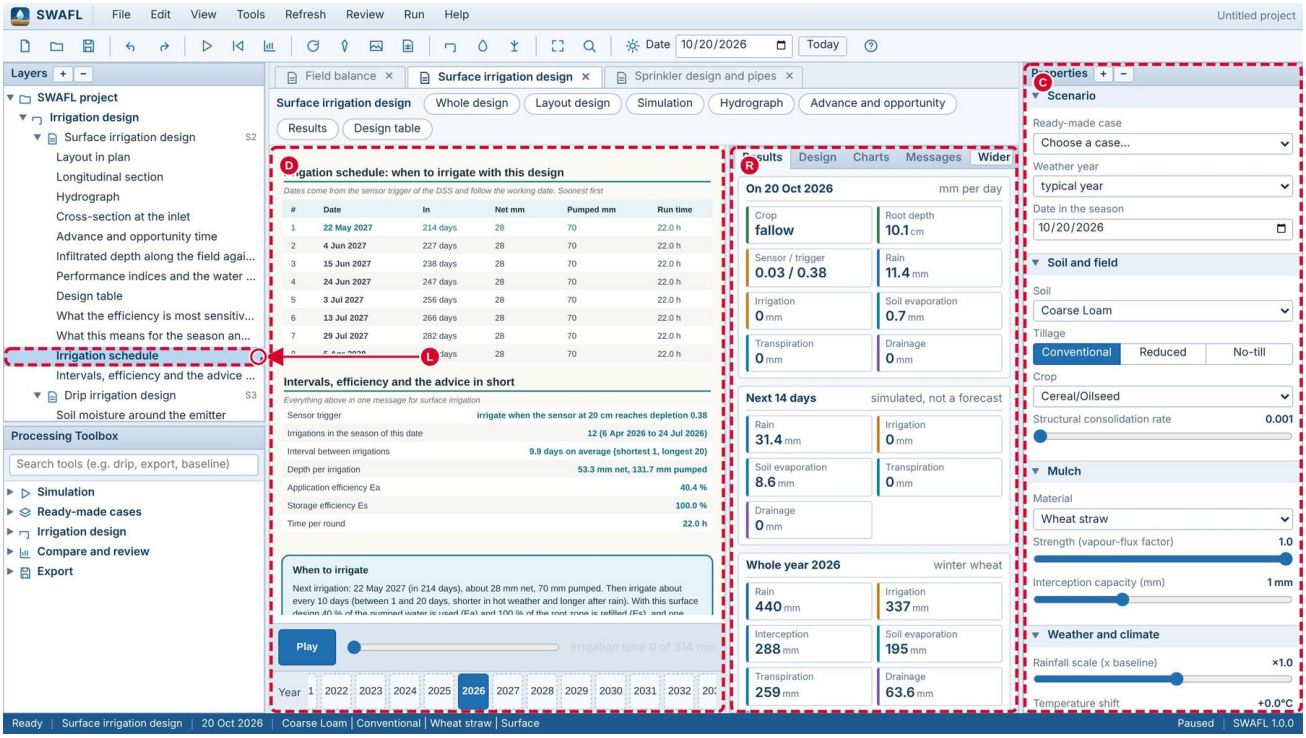


Figure 18.41: S2 Surface irrigation design / Irrigation schedule: the whole workbench.

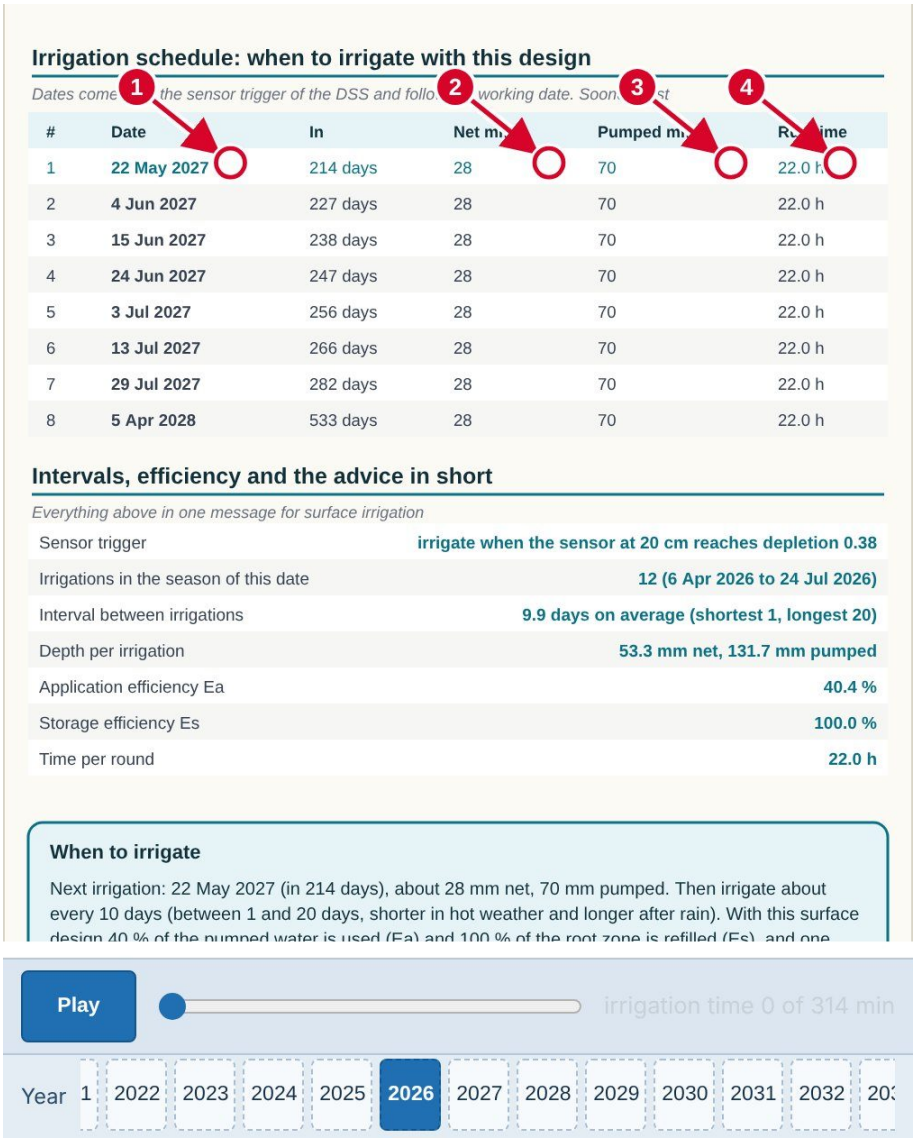


Figure 18.42: S2 Surface irrigation design / Irrigation schedule: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Dates: from the Engine 2 irrigation record of the analogue year (Eqs. 4.1 and 4.2).
- **2** Net mm, I_k: the DSS irrigation depth that day.
- **3** Pumped mm = I_k / E_a: Eq. 11.24.
- **4** Run time of one round: Eq. 11.24.

Equations behind this screen. none (the screen shows stored results).

18.2.13 Intervals, efficiency and the advice in short

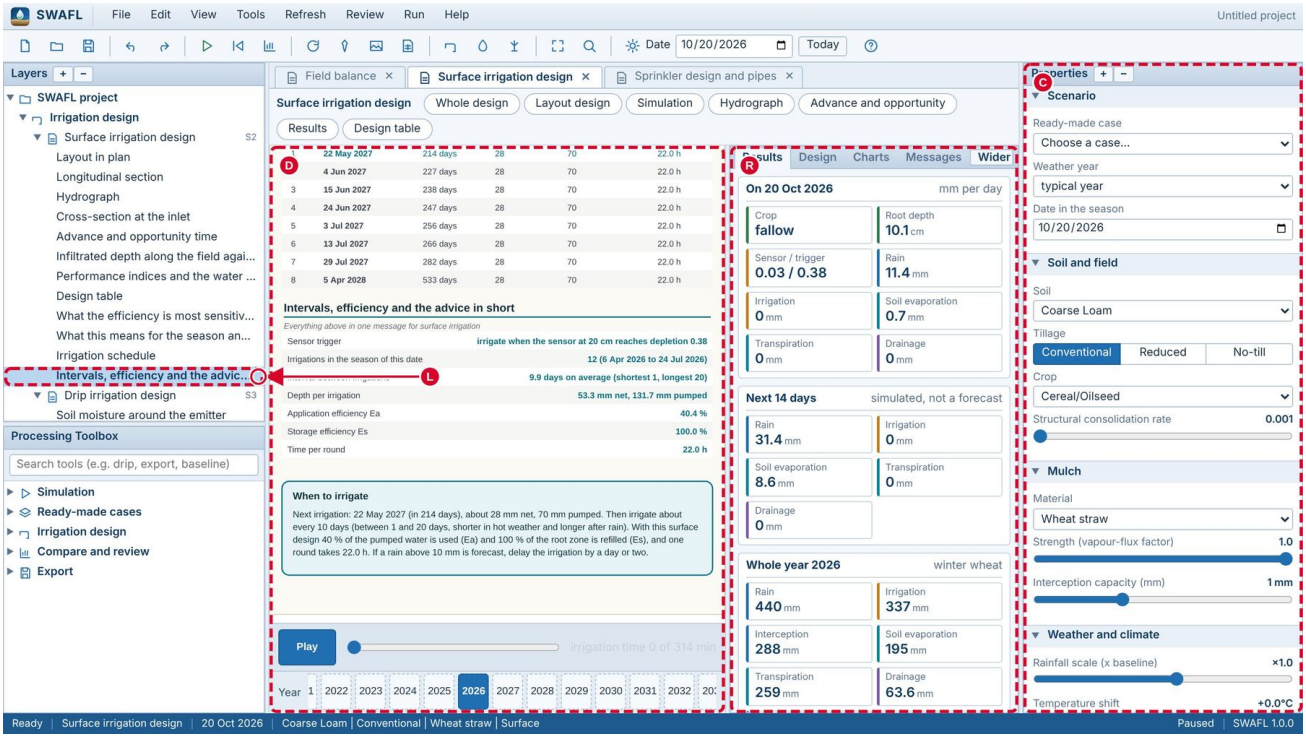


Figure 18.43: S2 Surface irrigation design / Intervals, efficiency and the advice in short: the whole workbench.

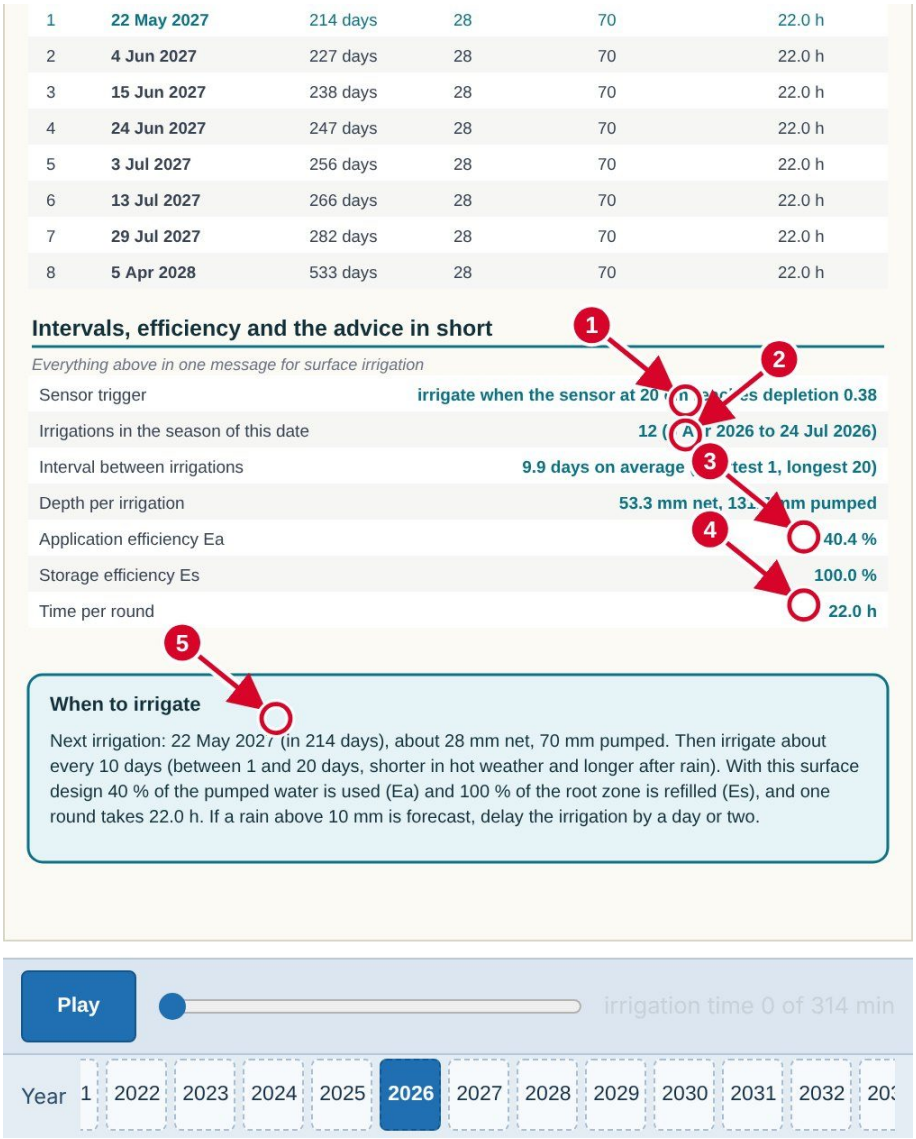


Figure 18.44: S2 Surface irrigation design / Intervals, efficiency and the advice in short: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sensor trigger: depletion f_T, Eqs. 5.3 and 5.27.
- **2** Irrigations in the season: n_ev, Eq. 11.24.
- **3** Application efficiency E_a: Eq. 11.19.
- **4** Time per round: Eq. 11.24.
- **5** “When to irrigate”: a message assembled from the schedule and from E_a and E_s (Eqs. 11.19 and 11.24).

Equations behind this screen. Eq. 11.24.

Eq. 11.24. Where explained: chapter “Surface irrigation design”, section “Season and farm”.

$$
\begin{aligned}
n_{\text{ev}} &= \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g, \\
N_{0M} &= \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}),
\end{aligned}
\tag{11.24}
$$

Provenance (Eq. 11.24): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

18.3 S3 Drip irrigation design

18.3.1 The whole layer: inputs and overview

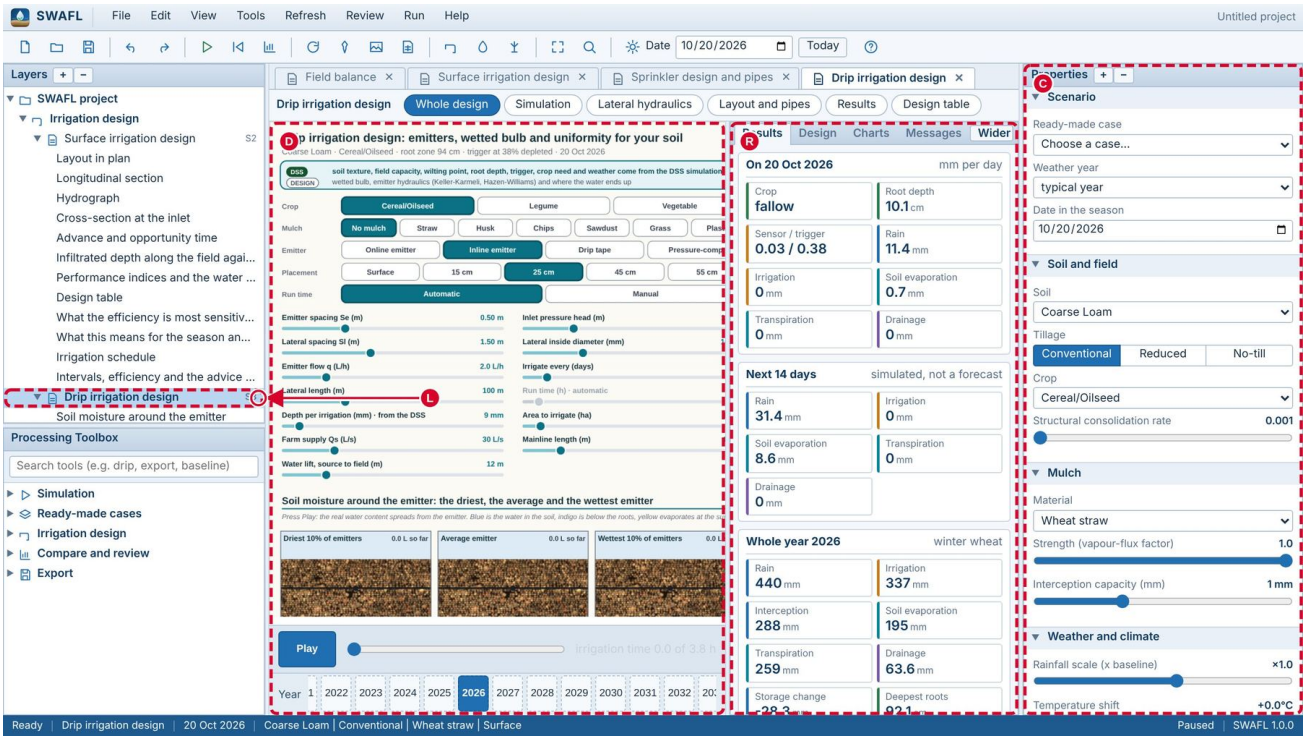


Figure 18.45: S3 Drip irrigation design: the whole workbench.

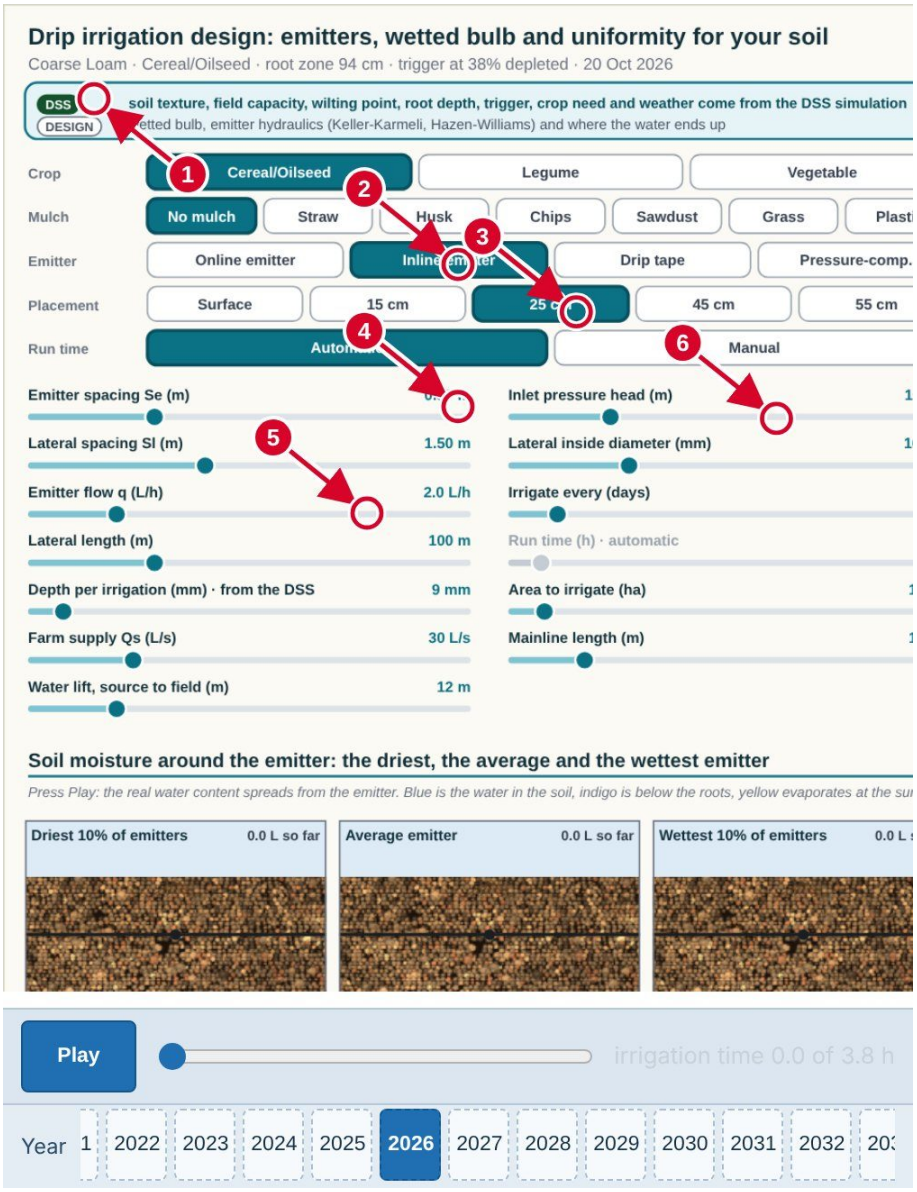


Figure 18.46: S3 Drip irrigation design: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** DSS badge: soil, root depth, trigger and crop need come from Engines 1 and 2.
- **2** Emitter type sets the discharge exponent x: Eqs. 12.4 and 12.5.
- **3** Emitter depth z_d sets the bulb geometry: Eq. 12.10.
- **4** Emitter and lateral spacing, S_e and S_l: area served by one emitter, Eq. 12.17.
- **5** Emitter flow q at the inlet head H_0: Eq. 12.4.
- **6** Inlet pressure head H_0: Eqs. 12.3 and 12.4.

Equations behind this screen. Eq. 5.27, Eq. 11.3, Eq. 12.1, Eq. 12.2, Eq. 12.3, Eq. 12.4, Eq. 12.5, Eq. 12.6, Eq. 12.7, Eq. 12.8, Eq. 12.9, Eq. 12.10, Eq. 12.11, Eq. 12.12, Eq. 12.13, Eq. 12.14, Eq. 12.15, Eq. 12.16, Eq. 12.17, Eq. 12.18, Eq. 12.19.

Eq. 5.27. Where explained: chapter “The two computing engines”, section “Evaporation under warming (used by the E1 layer)”.

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 12.1. Where explained: chapter "Drip irrigation design", section "Inputs, step by step".

$$Z_n = \min\big\{Z_n^{\mathrm{DSS}},\ \max(4,\ \mathrm{ET}_{\mathrm{pk}}\,\Delta t)\big\}\ \ \mathrm{mm}, \tag{12.1}$$

Provenance (Eq. 12.1): **[Engineered]**: depth and demand rules chosen for the workbench.

Eq. 12.2. Where explained: chapter "Drip irrigation design", section "Inputs, step by step".

$$\mathrm{ET}_{\mathrm{pk}} = \mathrm{clip}\Big\{\max_d \tfrac{1}{5}\sum_{j=0}^{4}(E_{d+j} + T_{d+j});\ 2.5,\ 9\Big\}. \tag{12.2}$$

Provenance (Eq. 12.2): **[Engineered]**: depth and demand rules chosen for the workbench.

Eq. 12.3. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$h_{f,j} = \frac{10.67\,S_e\,Q_j^{1.852}}{C^{1.852}\,D^{4.871}}, \qquad Q_j = \sum_{i \ge j+1} q_i\ \ (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.4. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad \text{(non-compensating; } h_j/H_0 \ge 0.01), \tag{12.4}$$

Provenance (Eq. 12.4): **[Literature]**: Keller and Karmeli (1974) emitter law.

Eq. 12.5. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \ge H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

Provenance (Eq. 12.5): **[Engineered]**: pressure-compensating emitter model with chosen constants.

Eq. 12.6. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600}\ \ (\mathrm{L\,s^{-1}}), \tag{12.6}$$

Provenance (Eq. 12.6): **[Literature]**: definitions of mean and minimum discharge.

Eq. 12.7. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$E_u = \mathrm{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

Provenance (Eq. 12.7): **[Literature]**: Keller and Karmeli (1974).

Eq. 12.8. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

Provenance (Eq. 12.8): **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Eq. 12.9. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases}(z - z_d)/(a\,r), & z \ge z_d,\\ (z_d - z)/(u\,r), & z < z_d,\end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.11. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.12. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.13. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.14. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2\cdot 10^{-6}\cdot 2E_0\}\quad (\mathrm{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.16. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_i = \max\{0.02\,\bar V,\ \bar V\,(1+\sigma_q z_i)\},\qquad \bar V = \bar q\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.17. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_{\mathrm{req}} = Z_n\,S_e\,S_l\ \ (\mathrm{L}),\qquad V_{\mathrm{st},i} = \min\{V_{\mathrm{root},i},\ V_{\mathrm{req}}\},\qquad V_{\mathrm{ex},i} = V_{\mathrm{root},i} - V_{\mathrm{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Eq. 12.18. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$t = \mathrm{clip}\Big\{\frac{V_{\mathrm{req}}}{\bar q\,E_u};\ 0.05,\ 96\Big\}\ \ \mathrm{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.19. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$E_a = \frac{\langle V_{\mathrm{st}}\rangle}{\langle V\rangle},\quad E_s = \frac{\langle V_{\mathrm{st}}\rangle}{V_{\mathrm{req}}},\quad \mathrm{DP} = \frac{\langle V_{\mathrm{dp}}^{(1)} + V_{\mathrm{ex}}\rangle}{\langle V\rangle},\quad \mathrm{EV} = \frac{\langle V_{\mathrm{ev}}\rangle}{\langle V\rangle},\quad Z_{\mathrm{gross}} = \frac{\bar V}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

18.3.2 Soil moisture around the emitter: the driest, the average and the wettest emitter

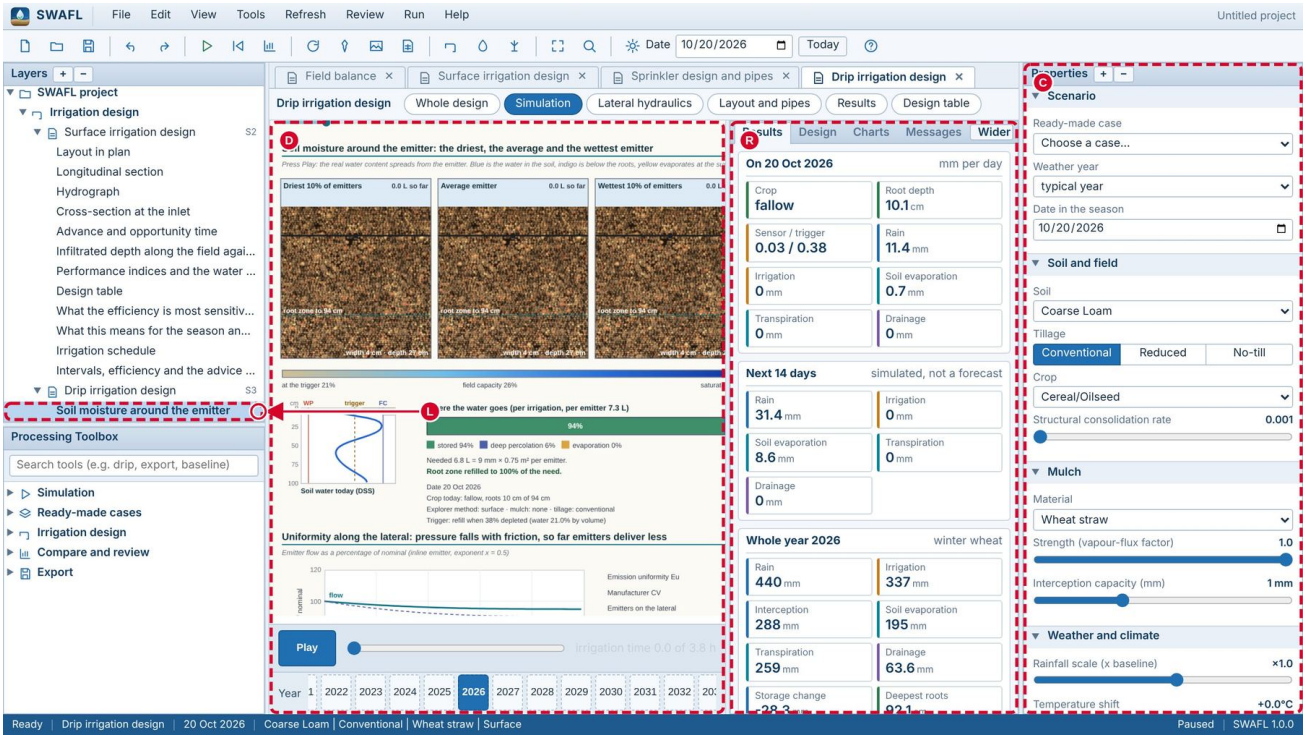


Figure 18.47: S3 Drip irrigation design / Soil moisture around the emitter: the whole workbench.

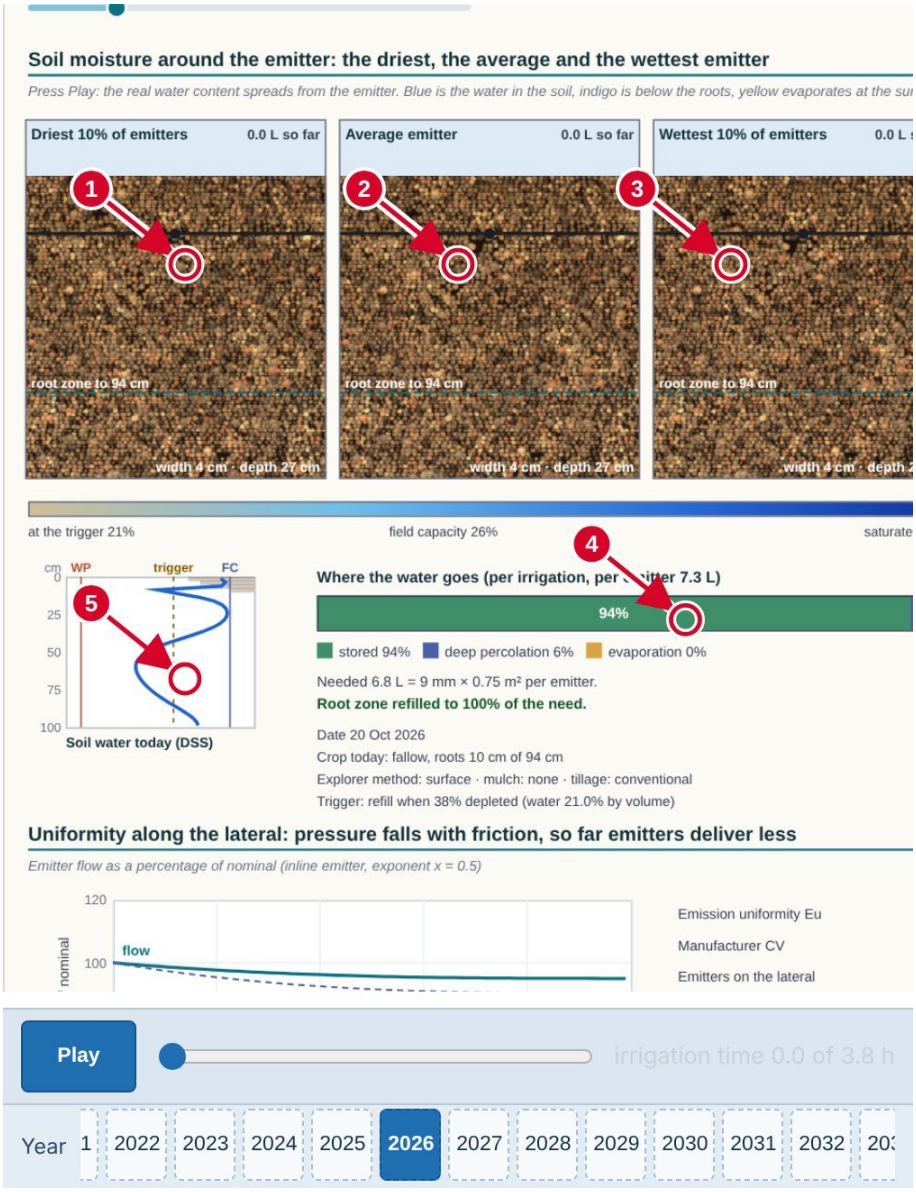


Figure 18.48: S3 Drip irrigation design / Soil moisture around the emitter: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Bulb under the driest 10 % of emitters (low quantile of Eq. 12.16).
- **2** Bulb under the average emitter: Eqs. 12.10 to 12.12.
- **3** Bulb under the wettest 10 % of emitters (high quantile of Eq. 12.16).
- **4** Where the water goes: stored, deep percolation (Eq. 12.14) and evaporation (Eq. 12.15).
- **5** Soil water today from the DSS profile.

Equations behind this screen. Eq. 11.3, Eq. 12.9, Eq. 12.10, Eq. 12.11, Eq. 12.12, Eq. 12.13, Eq. 12.14, Eq. 12.15.

Eq. 11.3. Where explained: chapter “Surface irrigation design”, section “Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)”.

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 12.9. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.11. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.12. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.13. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.14. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2\cdot 10^{-6}\cdot 2E_0\} \quad (\mathrm{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

18.3.3 Uniformity along the lateral: pressure falls with friction, so far emitters deliver less

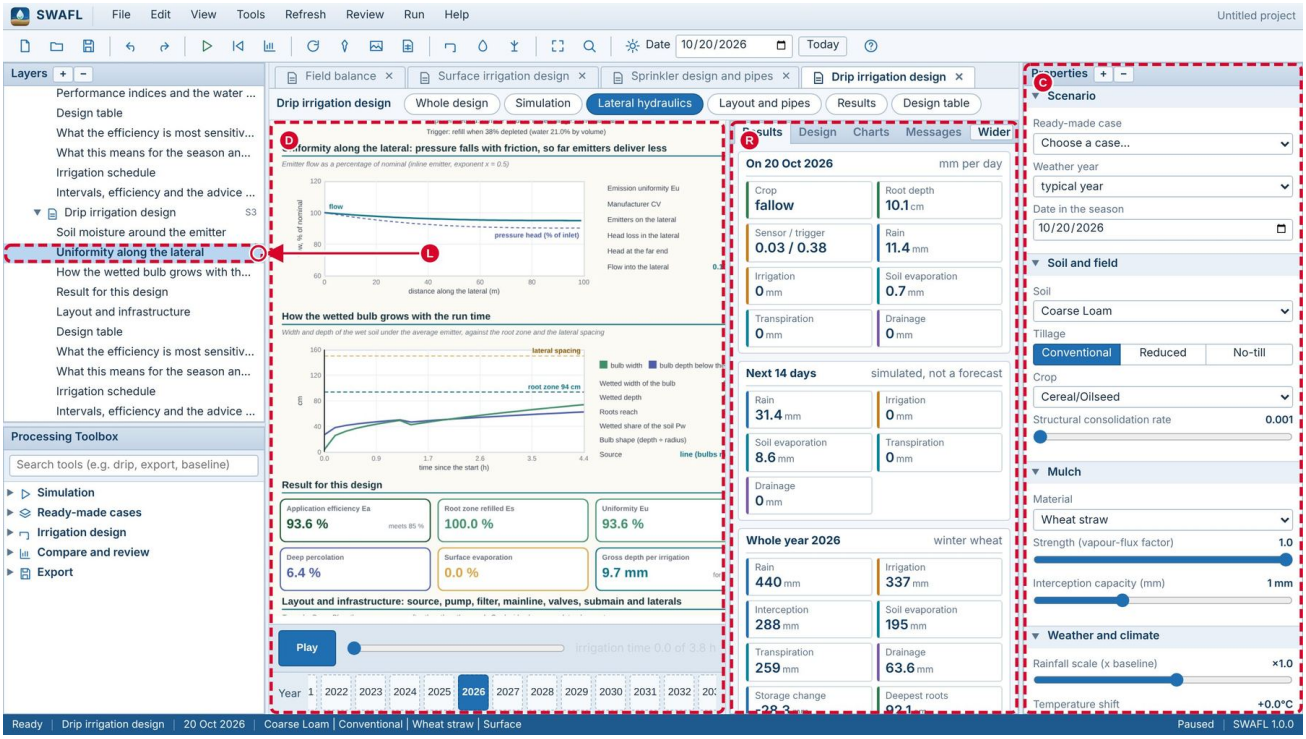


Figure 18.49: S3 Drip irrigation design / Uniformity along the lateral: the whole workbench.

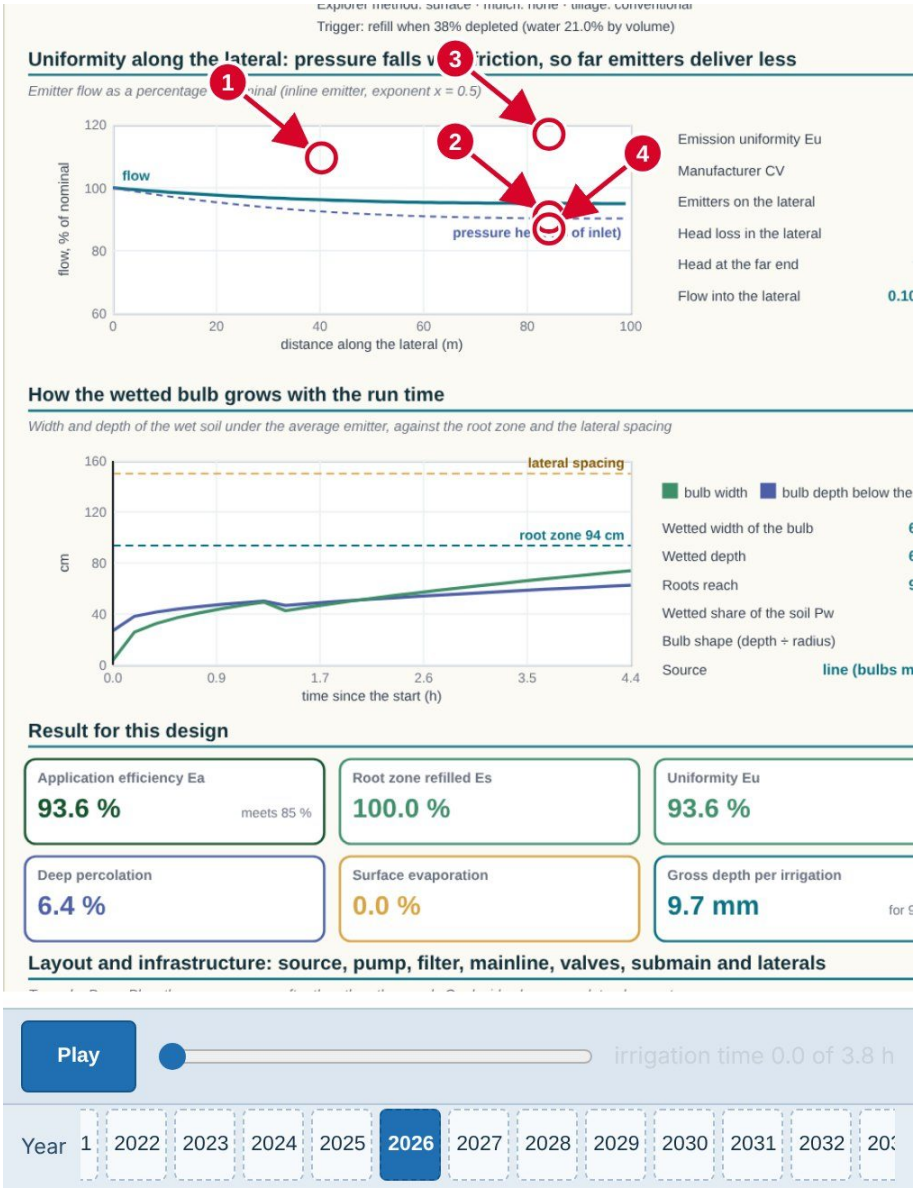


Figure 18.50: S3 Drip irrigation design / Uniformity along the lateral: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Emitter flow q_j along the lateral: Eq. 12.4.
- **2** Pressure head h_j along the lateral, falling with friction: Eq. 12.3.
- **3** Emission uniformity E_u: Eq. 12.7.
- **4** Head loss in the lateral, sum of h_f: Eq. 12.3.

Equations behind this screen. Eq. 12.3, Eq. 12.4, Eq. 12.5, Eq. 12.6, Eq. 12.7, Eq. 12.8, Eq. 12.10, Eq. 12.15, Eq. 12.16, Eq. 12.17, Eq. 12.18, Eq. 12.19.

Eq. 12.3. Where explained: chapter “Drip irrigation design”, section “Lateral hydraulics: head loss and flow along the lateral”.

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.4. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating};\ h_j/H_0 \geq 0.01), \tag{12.4}$$

Provenance (Eq. 12.4): **[Literature]**: Keller and Karmeli (1974) emitter law.

Eq. 12.5. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \geq H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

Provenance (Eq. 12.5): **[Engineered]**: pressure-compensating emitter model with chosen constants.

Eq. 12.6. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600} \;\; (\mathrm{L\,s^{-1}}), \tag{12.6}$$

Provenance (Eq. 12.6): **[Literature]**: definitions of mean and minimum discharge.

Eq. 12.7. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$E_u = \mathrm{clip}\Big\{\Big(1 - \frac{1.27\, c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

Provenance (Eq. 12.7): **[Literature]**: Keller and Karmeli (1974).

Eq. 12.8. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

Provenance (Eq. 12.8): **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\rho(z) = r\,\sqrt{1 - \zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \geq z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \geq 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\} \quad (\mathrm{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.16. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1 + \sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.17. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_{\mathrm{req}} = Z_n\,S_e\,S_l\ \ (\mathrm{L}), \qquad V_{\mathrm{st},i} = \min\{V_{\mathrm{root},i},\ V_{\mathrm{req}}\}, \qquad V_{\mathrm{ex},i} = V_{\mathrm{root},i} - V_{\mathrm{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Eq. 12.18. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$t = \mathrm{clip}\Big\{\frac{V_{\mathrm{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \ \mathrm{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.19. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$E_a = \frac{\langle V_{\mathrm{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\mathrm{st}}\rangle}{V_{\mathrm{req}}}, \quad \mathrm{DP} = \frac{\langle V_{\mathrm{dp}}^{(1)} + V_{\mathrm{ex}}\rangle}{\langle V\rangle}, \quad \mathrm{EV} = \frac{\langle V_{\mathrm{ev}}\rangle}{\langle V\rangle}, \quad Z_{\mathrm{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

18.3.4 How the wetted bulb grows with the run time

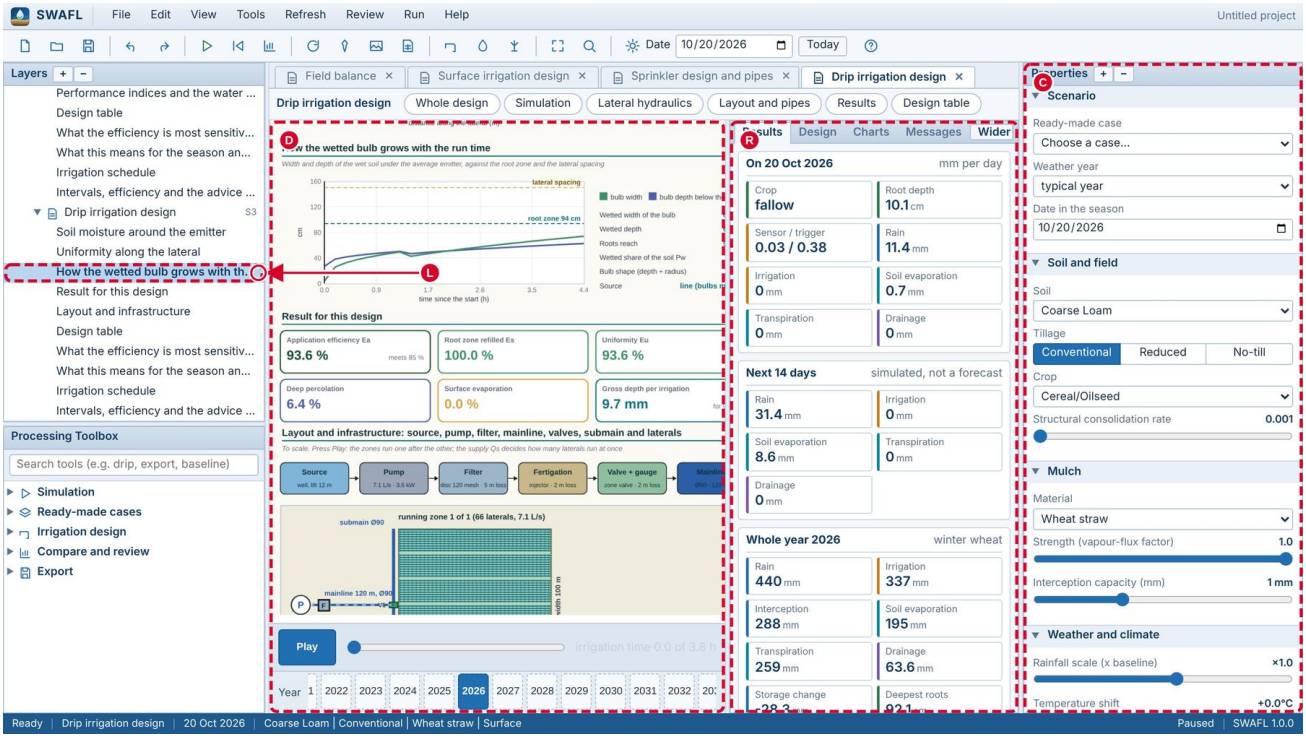


Figure 18.51: S3 Drip irrigation design / How the wetted bulb grows with the run time: the whole workbench.

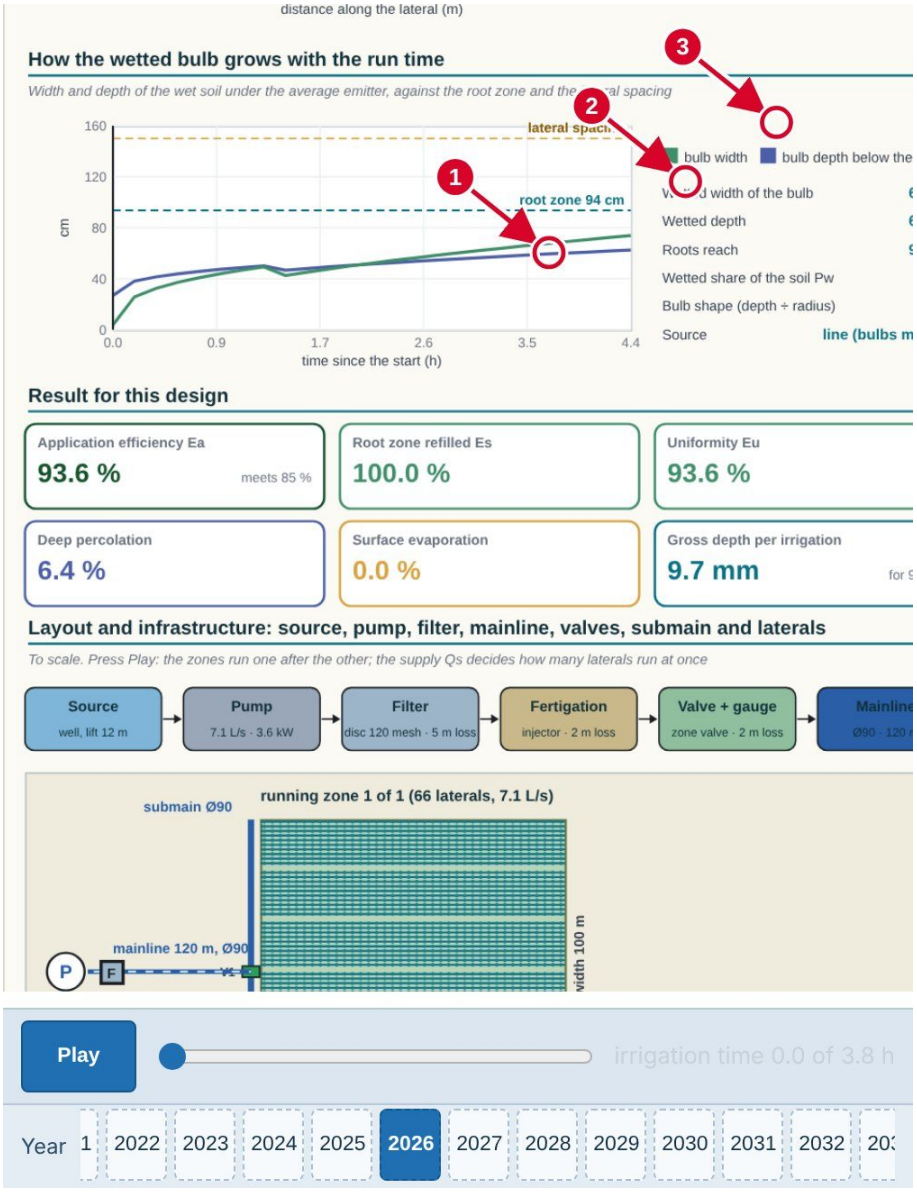


Figure 18.52: S3 Drip irrigation design / How the wetted bulb grows with the run time: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Bulb width (green) and depth (indigo) against time: Eqs. 12.10 to 12.12.
- **2** Root-zone depth R_max, against which the bulb depth gives deep percolation: Eq. 12.14.
- **3** Lateral spacing S_l: the bulbs merge when the width reaches it, P_w of Eq. 12.13.

Equations behind this screen. Eq. 11.3, Eq. 12.9, Eq. 12.10, Eq. 12.11, Eq. 12.12, Eq. 12.13, Eq. 12.14, Eq. 12.15.

Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 12.9. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.11. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^{6}/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.12. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.13. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.14. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2\cdot 10^{-6}\cdot 2E_0\} \quad (\mathrm{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

18.3.5 Result for this design

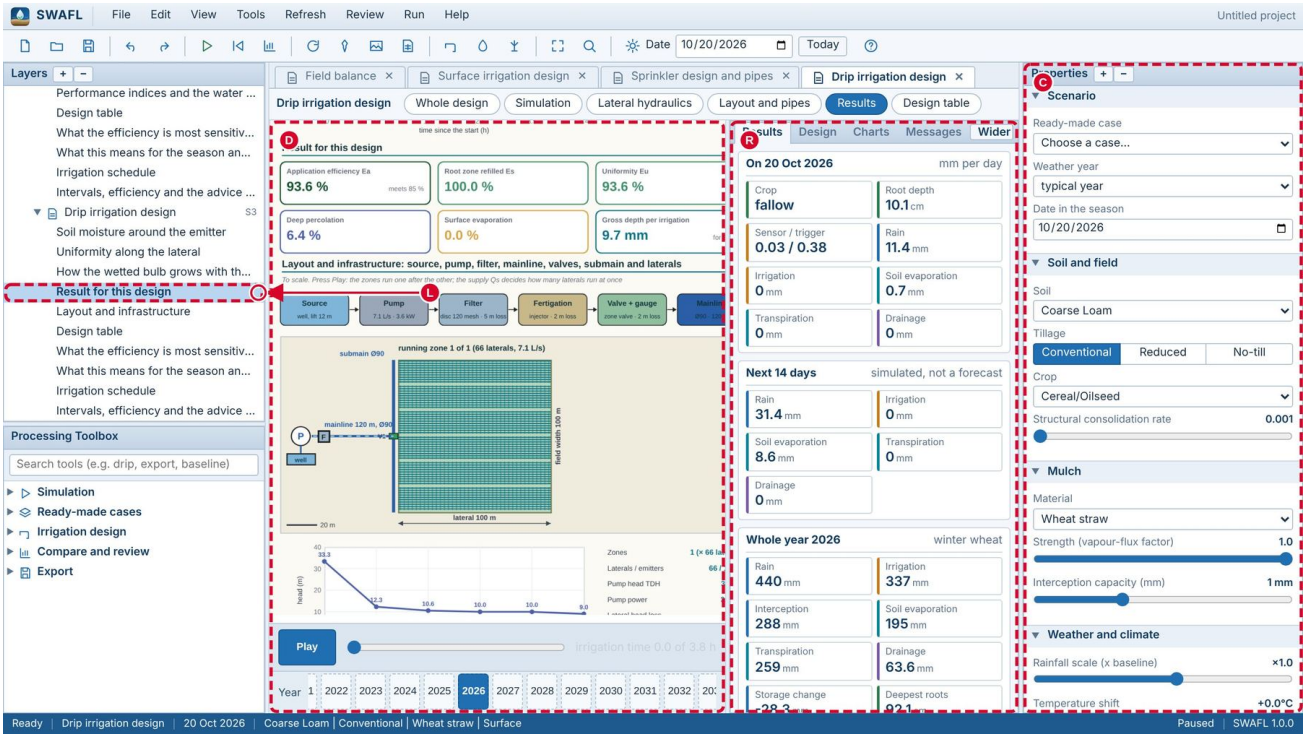


Figure 18.53: S3 Drip irrigation design / Result for this design: the whole workbench.

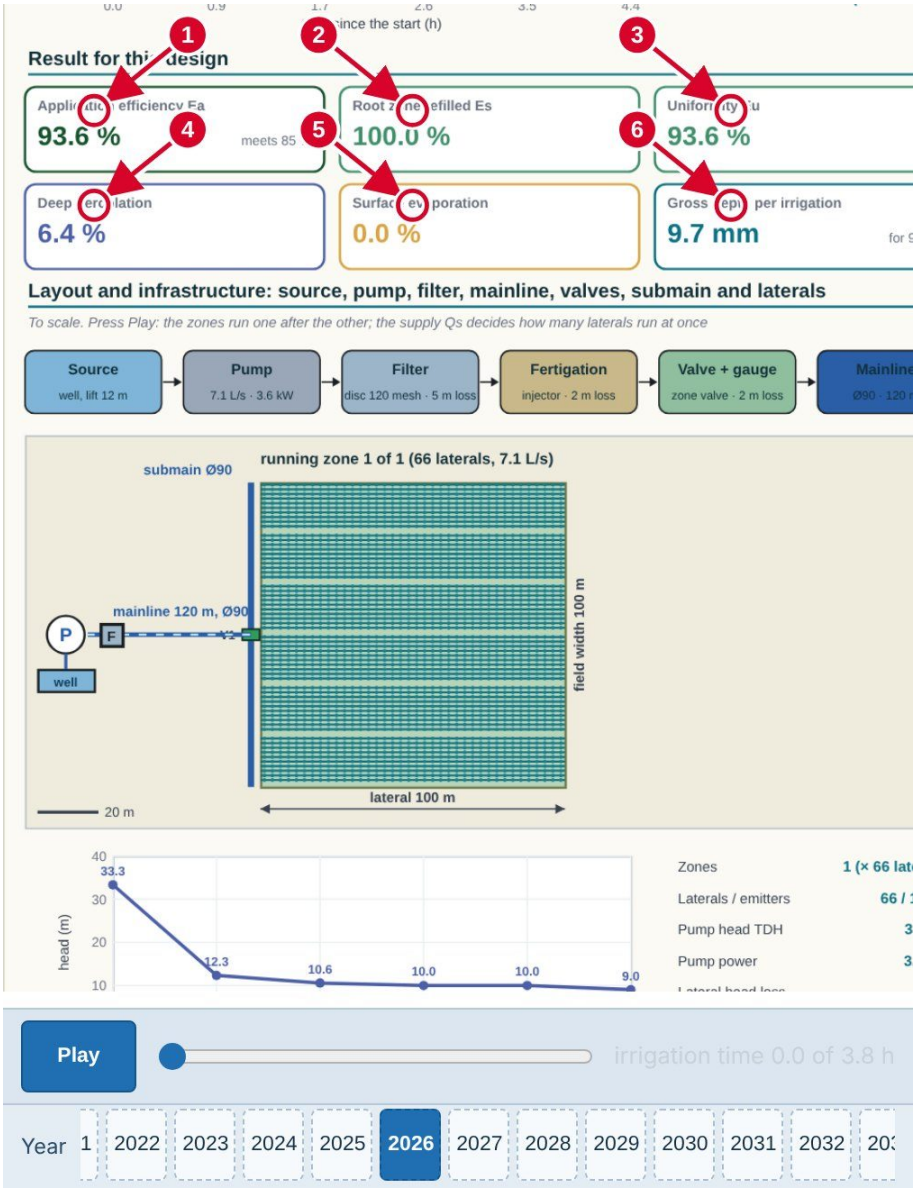


Figure 18.54: S3 Drip irrigation design / Result for this design: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Application efficiency E_a: Eq. 12.19.
- **2** Root-zone refilled E_s: Eq. 12.19.
- **3** Emission uniformity E_u: Eq. 12.7.
- **4** Deep percolation: Eq. 12.14.
- **5** Surface evaporation: Eq. 12.15.
- **6** Gross depth per irrigation: Eqs. 12.17 and 12.18.

Equations behind this screen. Eq. 12.10, Eq. 12.15, Eq. 12.16, Eq. 12.17, Eq. 12.18, Eq. 12.19.

Eq. 12.10. Where explained: chapter “Drip irrigation design”, section “The wetted bulb”.

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\text{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\} \quad (\text{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.16. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1 + \sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.17. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_{\text{req}} = Z_n\,S_e\,S_l \ \ (\text{L}), \qquad V_{\text{st},i} = \min\{V_{\text{root},i},\ V_{\text{req}}\}, \qquad V_{\text{ex},i} = V_{\text{root},i} - V_{\text{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Eq. 12.18. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$t = \text{clip}\Big\{\frac{V_{\text{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\} \ \text{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.19. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$E_a = \frac{\langle V_{\text{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\text{st}}\rangle}{V_{\text{req}}}, \quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)} + V_{\text{ex}}\rangle}{\langle V\rangle}, \quad \text{EV} = \frac{\langle V_{\text{ev}}\rangle}{\langle V\rangle}, \quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

18.3.6 Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals

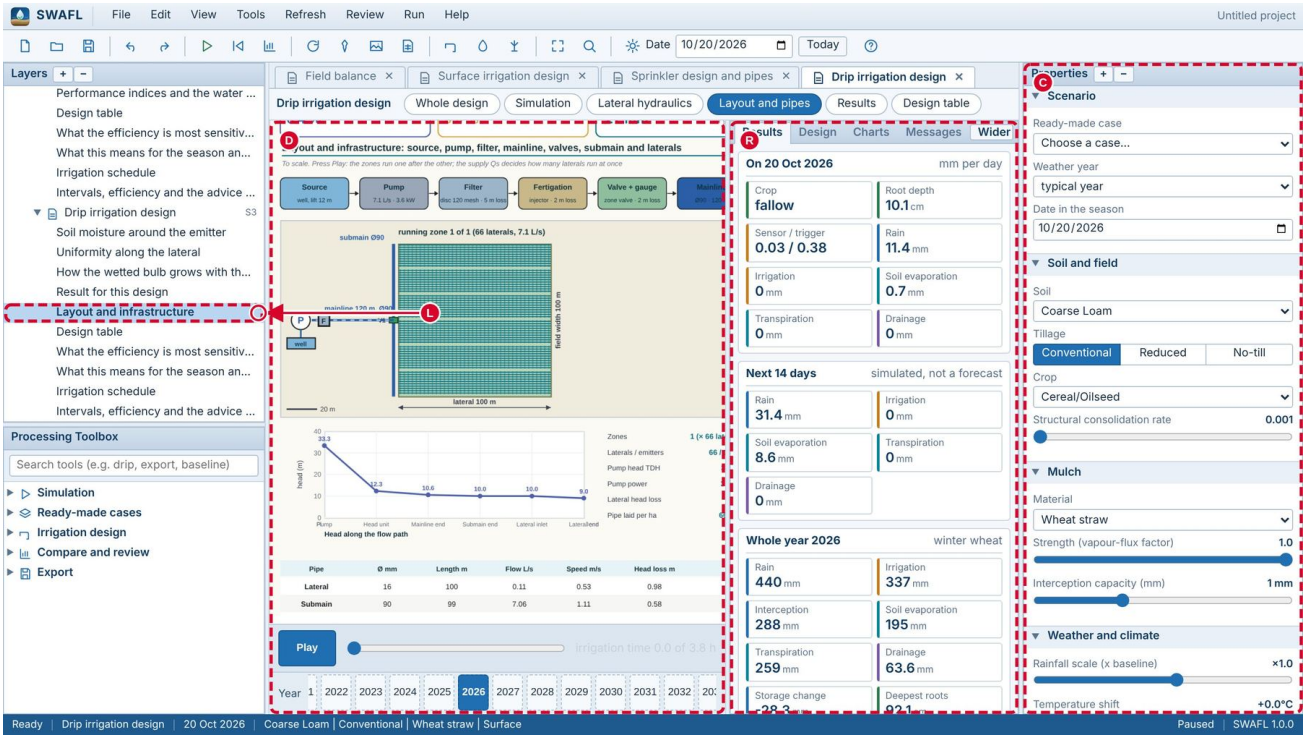


Figure 18.55: S3 Drip irrigation design / Layout and infrastructure: the whole workbench.

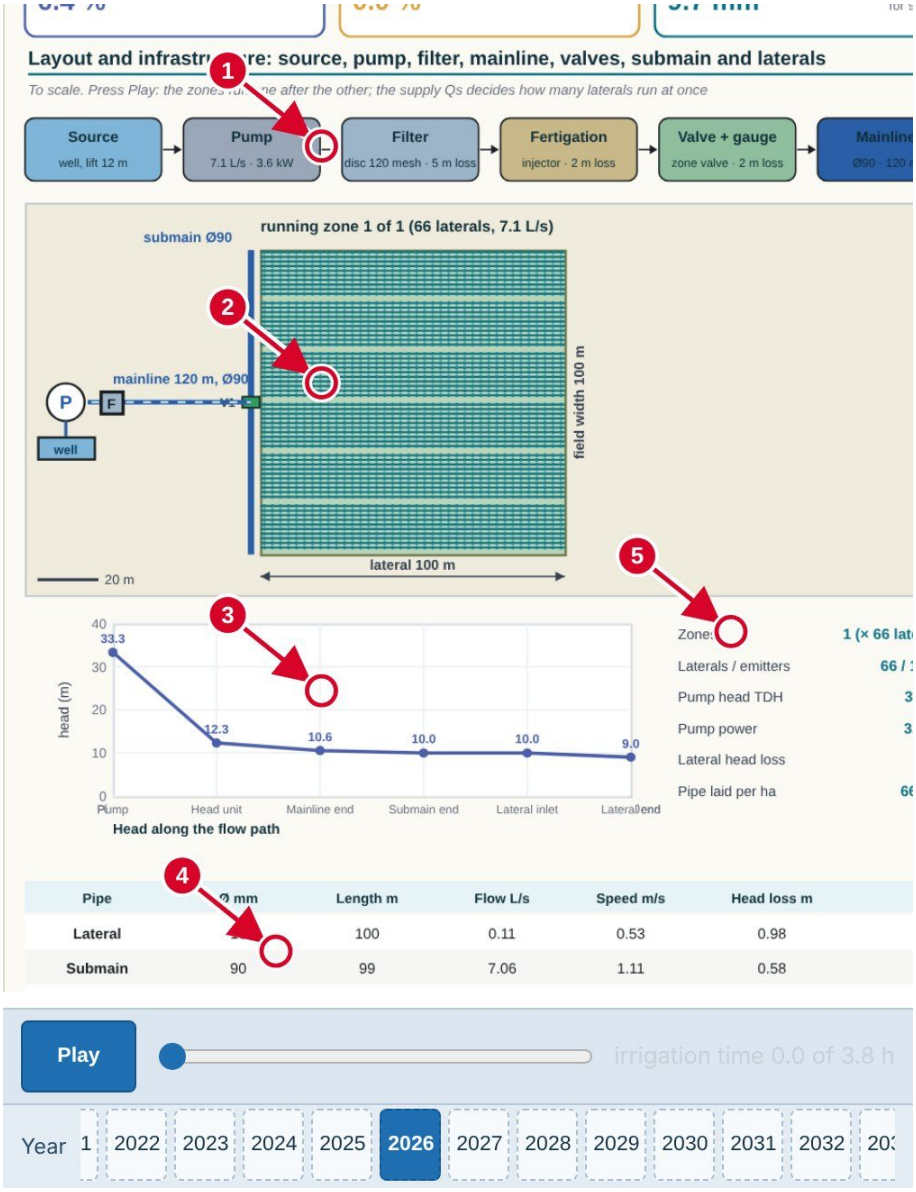


Figure 18.56: S3 Drip irrigation design / Layout and infrastructure: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Source, pump, filter and valves: pump power from the head of Eq. 12.23.
- **2** Laterals across the field, N_L = field width / S_l: Eq. 12.20.
- **3** Head along the flow path: TDH of Eq. 12.23.
- **4** Pipe table: smallest commercial diameter, losses by Eq. 12.22 (Hazen-Williams, Eq. 12.3).
- **5** Zones, laterals and emitters: Eq. 12.20.

Equations behind this screen. Eq. 12.3, Eq. 12.20, Eq. 12.21, Eq. 12.22, Eq. 12.23.

Eq. 12.3. Where explained: chapter “Drip irrigation design”, section “Lateral hydraulics: head loss and flow along the lateral”.

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.20. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".

$$W_f = \frac{10^4 A_f}{L_L}, \quad N_L = \left\lfloor \frac{W_f}{S_l} \right\rfloor, \quad n_{\text{zone}} = \left\lceil \frac{N_L\, Q_{\text{lat}}}{Q_s} \right\rceil, \quad n_l = \left\lceil \frac{N_L}{n_{\text{zone}}} \right\rceil, \quad Q_z = n_l\, Q_{\text{lat}}, \tag{12.20}$$

Provenance (Eq. 12.20): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.21. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\} \text{ mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.22. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".

$$h_{f,sm} = 0.4\, \frac{10.67\, L_{sm}\, Q_z^{1.852}}{C^{1.852} D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\, L_m\, Q_z^{1.852}}{C^{1.852} D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Eq. 12.23. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9 \text{ m}, \qquad P = \frac{\rho g\, Q_z\, \text{TDH}}{\eta} = \frac{9.81\, (Q_z/1000)\, \text{TDH}}{0.65} \text{ kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

18.3.7 Design table: iterations toward an application efficiency of 85 % or more

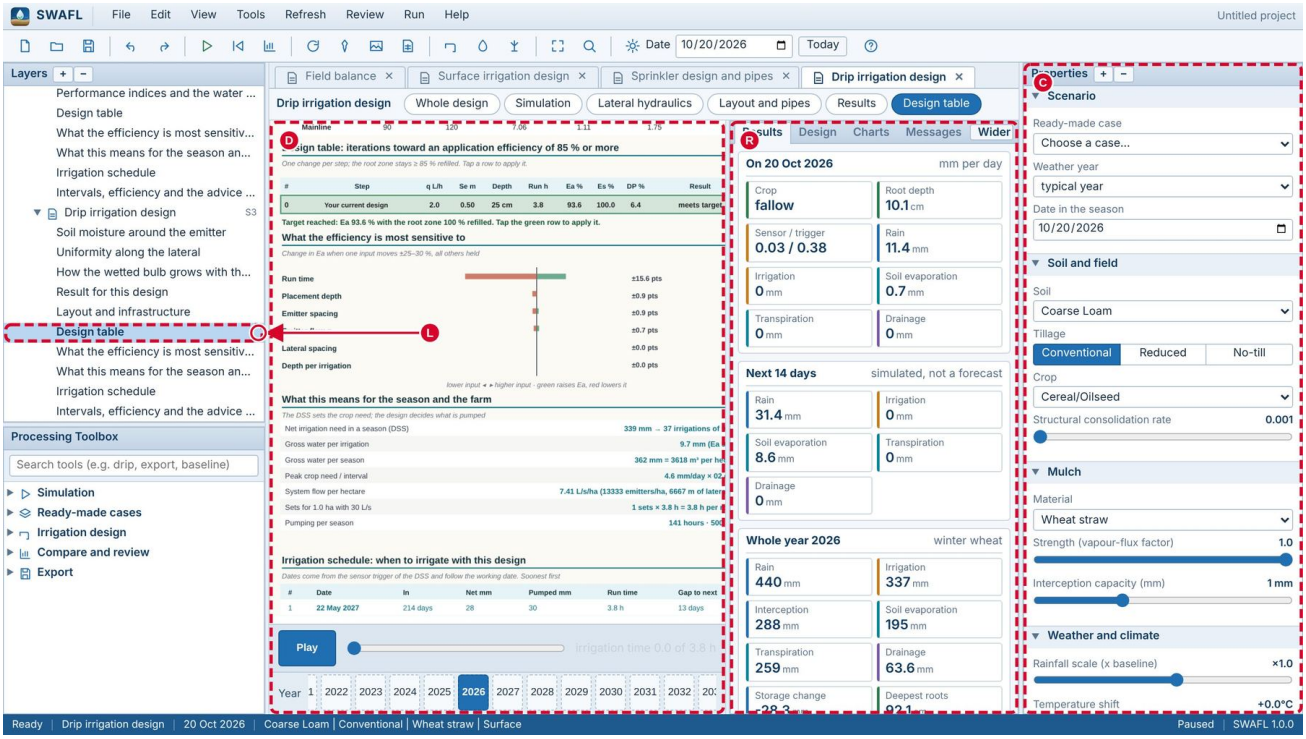


Figure 18.57: S3 Drip irrigation design / Design table: the whole workbench.

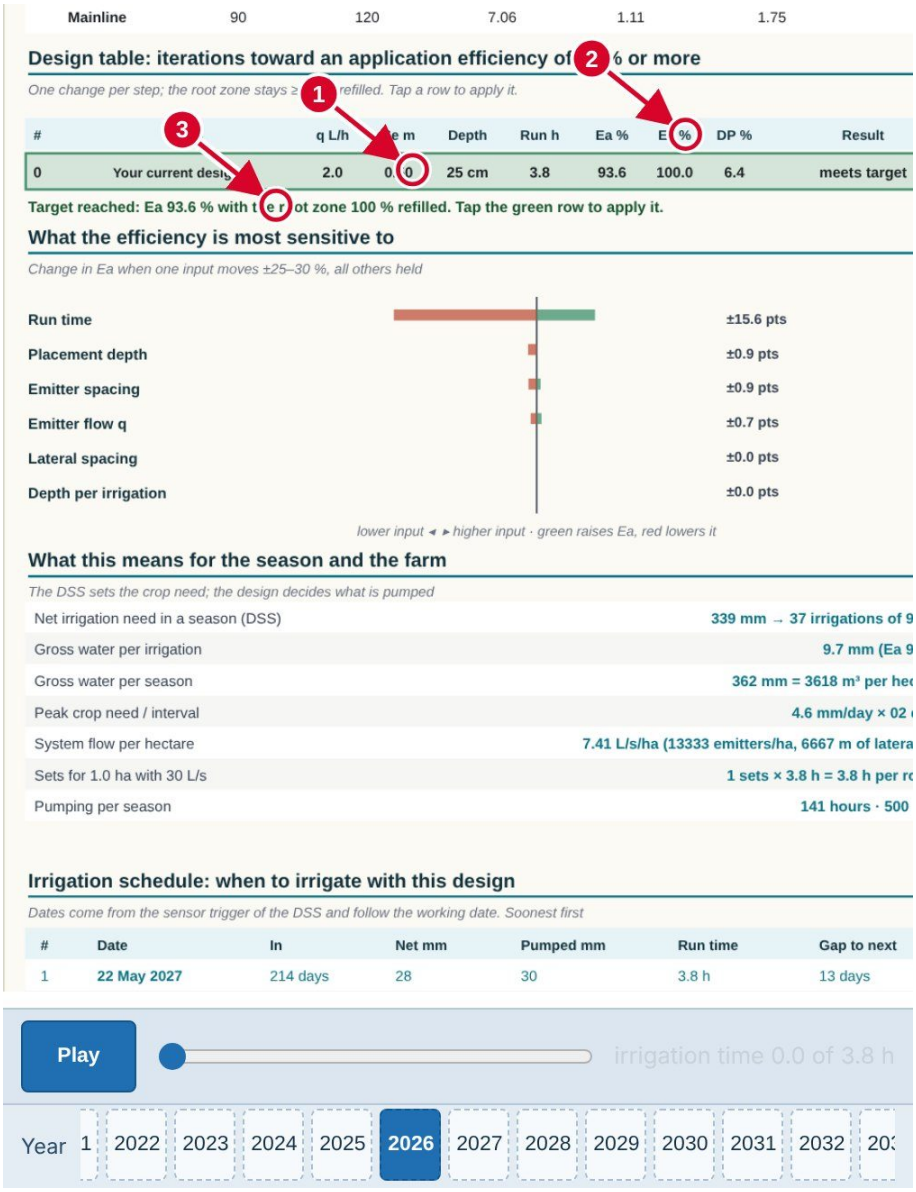


Figure 18.58: S3 Drip irrigation design / Design table: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Row 0: your current design; its score is Eq. 12.26.
- **2** Column E_a: Eq. 12.19.
- **3** Target reached: score = min(E_a, 1) - 2.5 max(0, 0.9 - E_s), Eq. 12.26.

Equations behind this screen. Eq. 12.26.

Eq. 12.26. Where explained: chapter “Drip irrigation design”, section “The design table”.

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

18.3.8 What the efficiency is most sensitive to

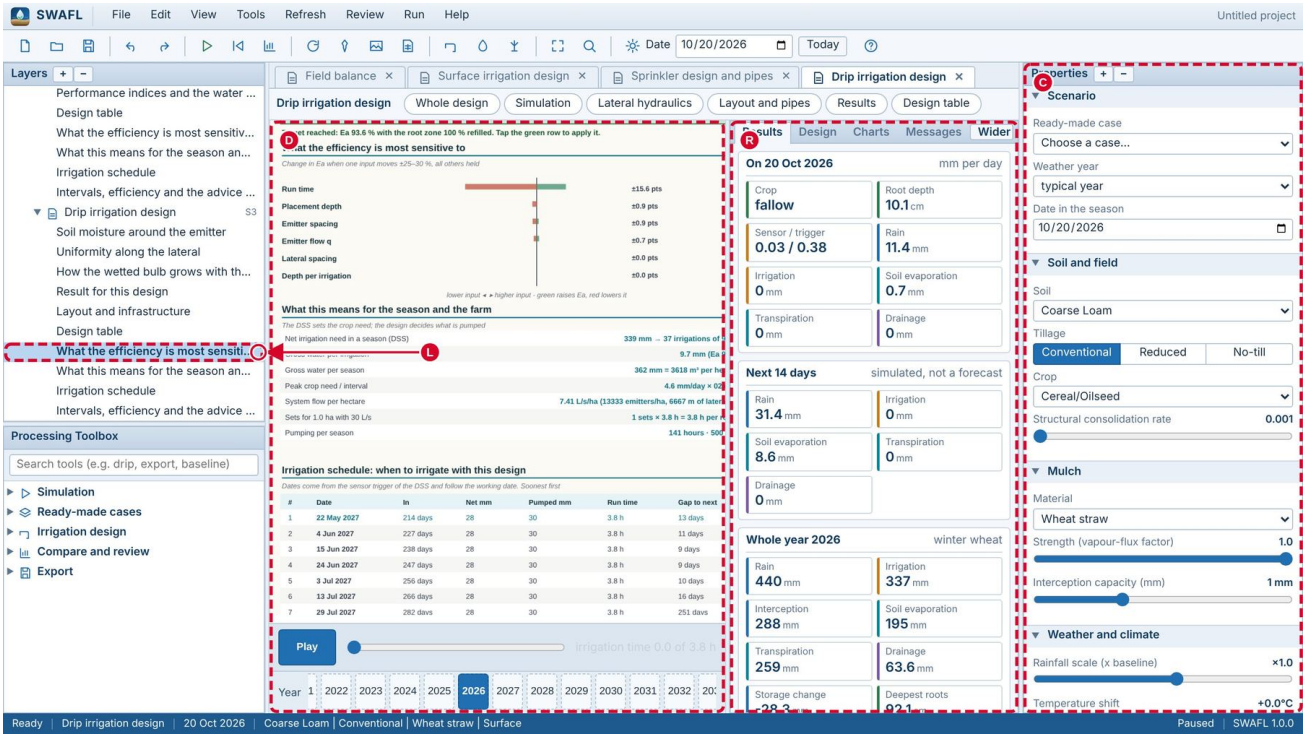


Figure 18.59: S3 Drip irrigation design / What the efficiency is most sensitive to: the whole workbench.

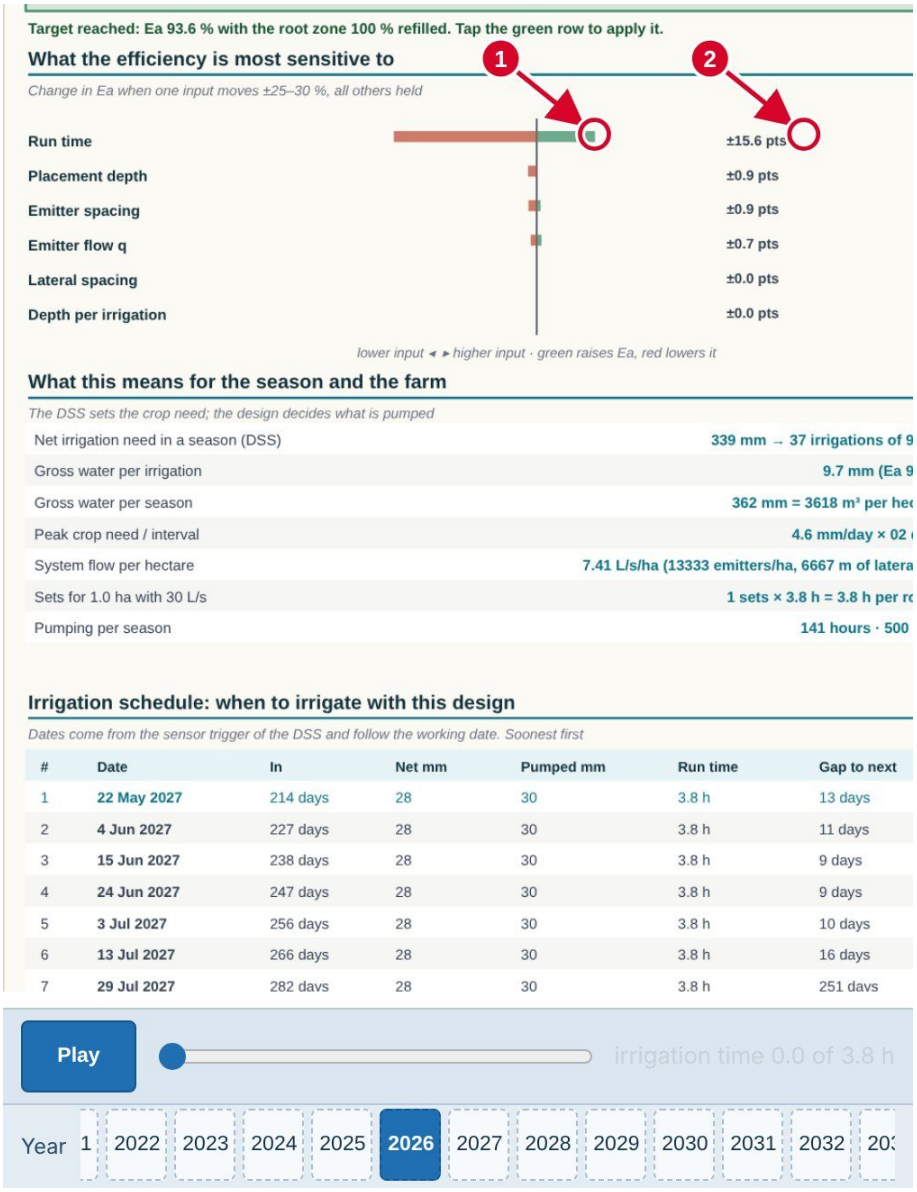


Figure 18.60: S3 Drip irrigation design / What the efficiency is most sensitive to: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Tornado bar: change in E_a (Eq. 12.19) when the run time moves by 25 to 30 %.
- **2** Size of the change in percentage points.

Equations behind this screen. none (the screen shows stored results).

18.3.9 What this means for the season and the farm

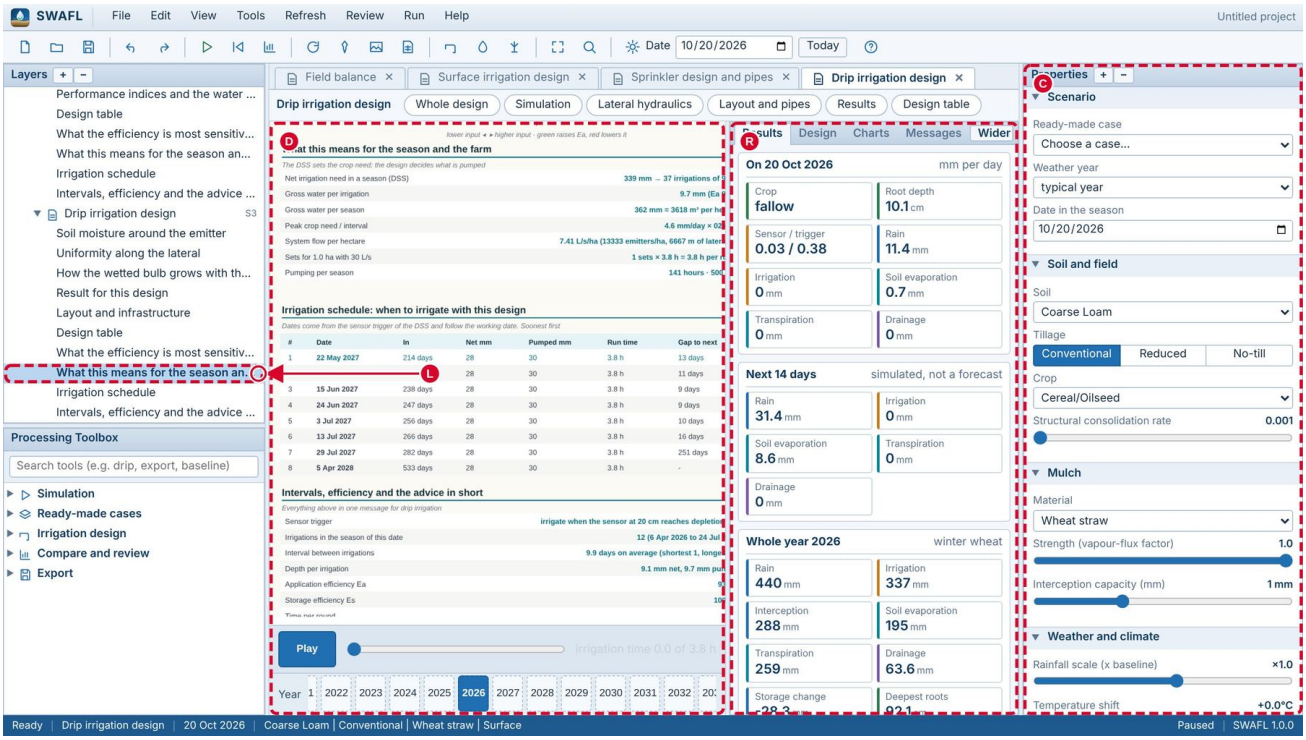


Figure 18.61: S3 Drip irrigation design / What this means for the season and the farm: the whole workbench.

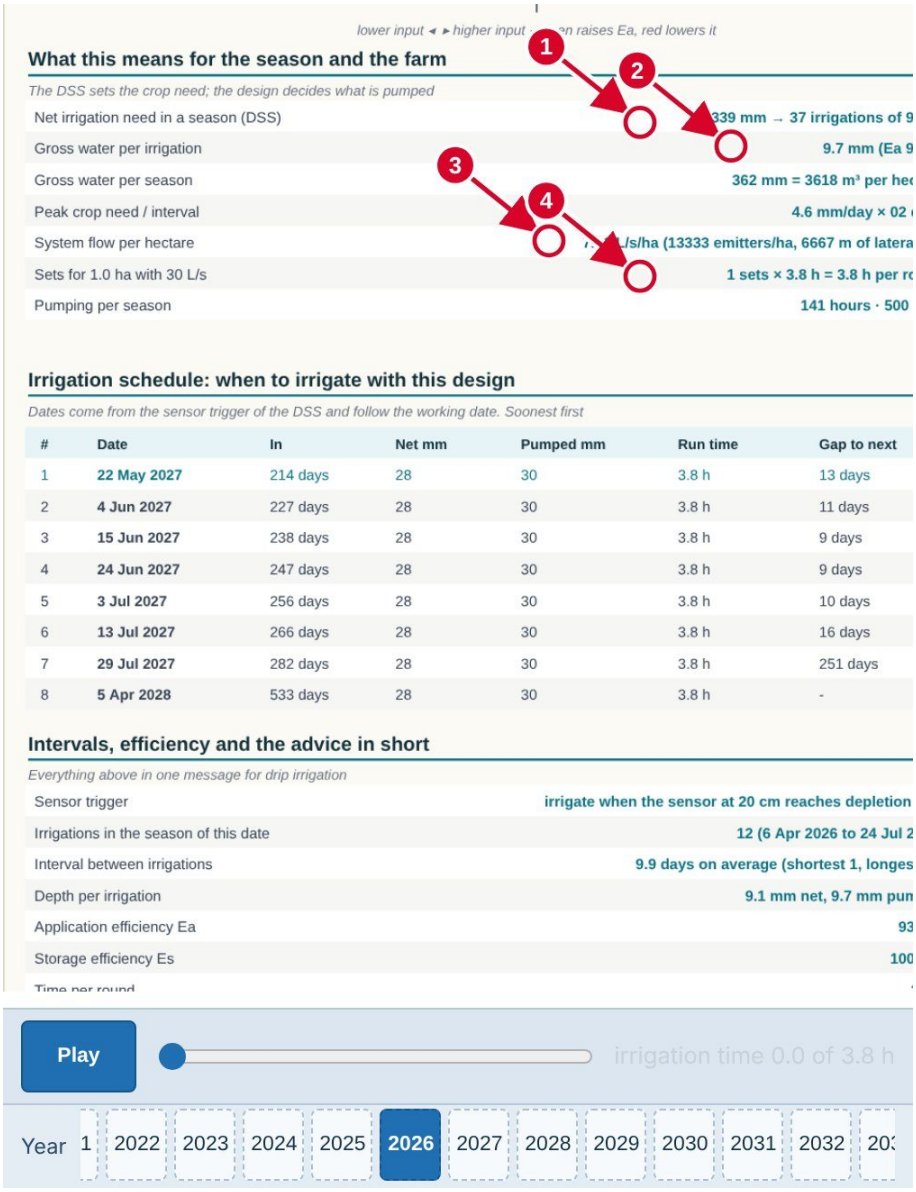


Figure 18.62: S3 Drip irrigation design / What this means for the season and the farm: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Irrigations in a season, n_ev = y_I / Z_n: Eq. 12.24.
- **2** Gross depth per irrigation: Eq. 12.24.
- **3** System flow per hectare: Eq. 12.24.
- **4** Sets and hours: Eq. 12.25.

Equations behind this screen. Eq. 12.24, Eq. 12.25.

Eq. 12.24. Where explained: chapter "Drip irrigation design", section "Season and farm".

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.25. Where explained: chapter “Drip irrigation design”, section “Season and farm”.

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

18.3.10 Irrigation schedule: when to irrigate with this design

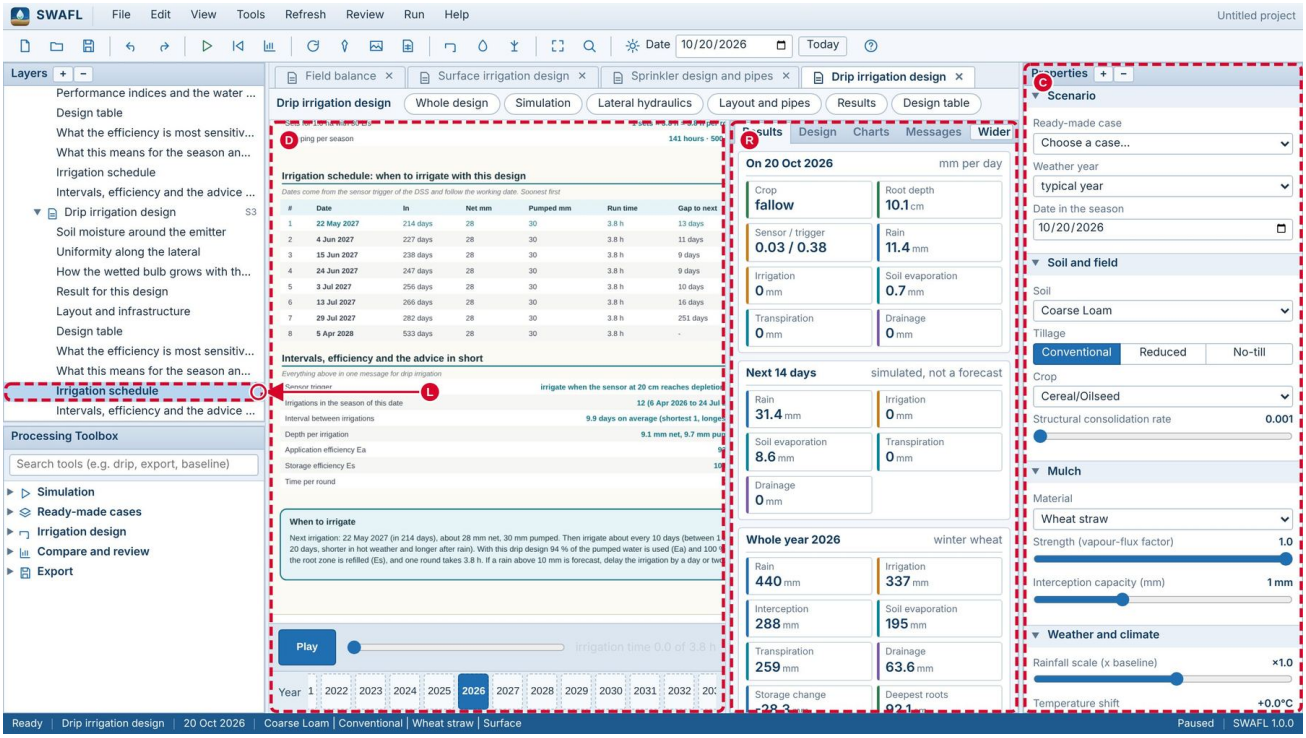


Figure 18.63: S3 Drip irrigation design / Irrigation schedule: the whole workbench.

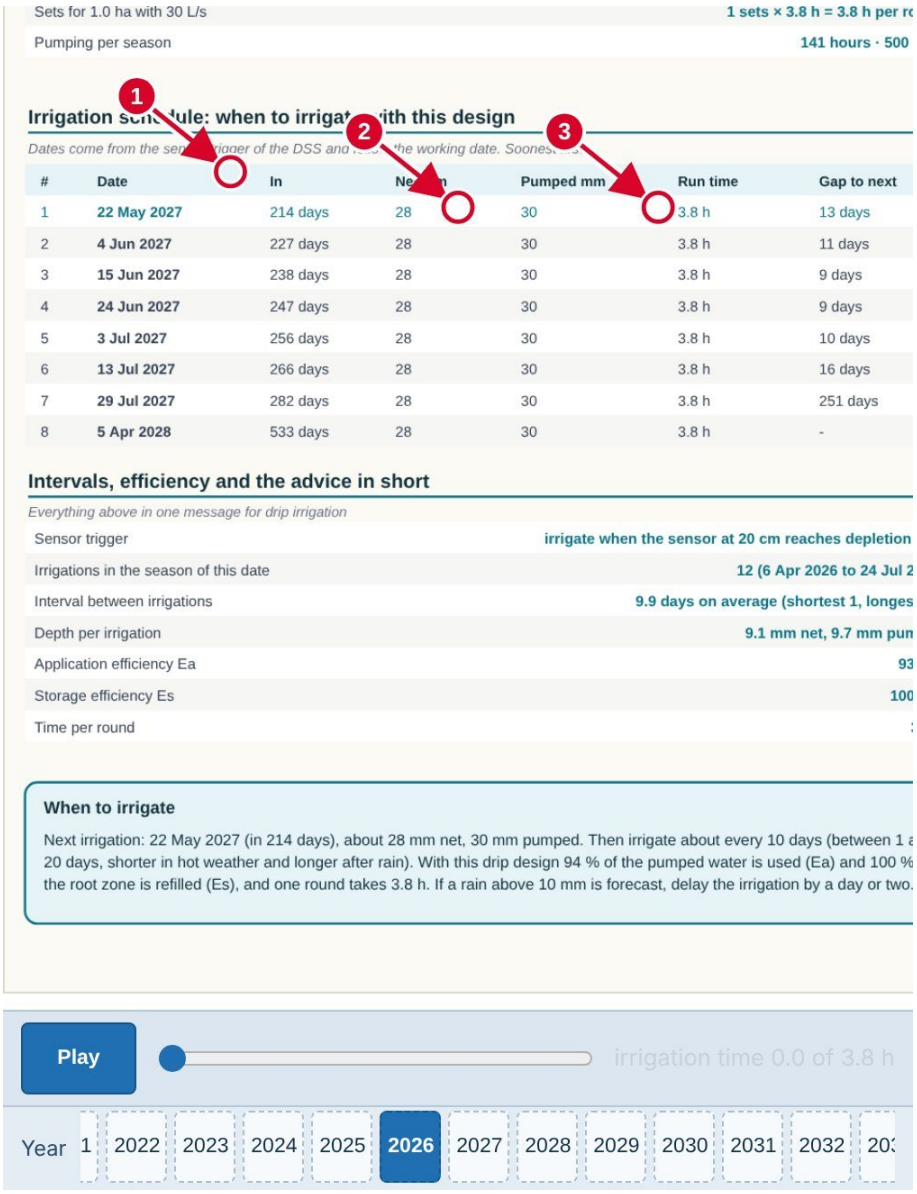


Figure 18.64: S3 Drip irrigation design / Irrigation schedule: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Dates from the Engine 2 irrigation record.
- **2** Pumped mm = I_k / E_a: Eq. 12.24.
- **3** Run time: Eq. 12.18 (automatic run time).

Equations behind this screen. Eq. 12.24, Eq. 12.25.

Eq. 12.24. Where explained: chapter "Drip irrigation design", section "Season and farm".

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.25. Where explained: chapter “Drip irrigation design”, section “Season and farm”.

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

18.3.11 Intervals, efficiency and the advice in short

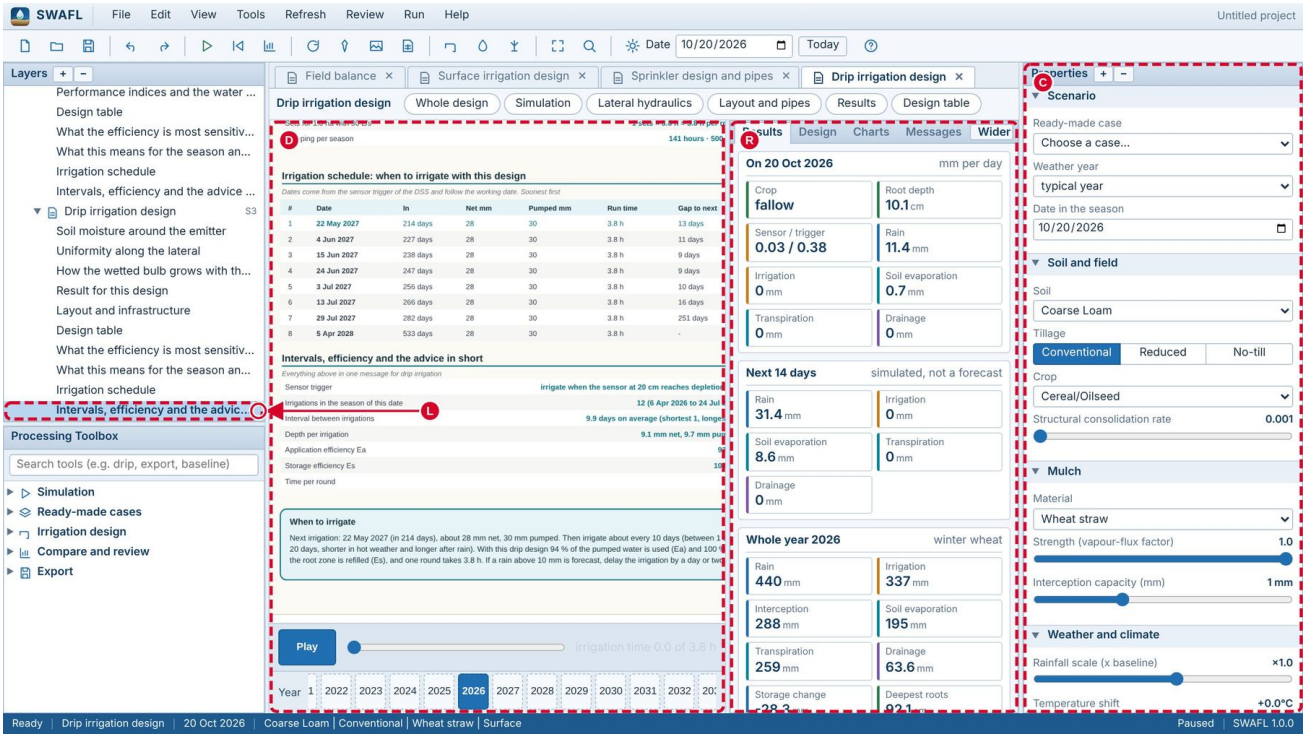


Figure 18.65: S3 Drip irrigation design / Intervals, efficiency and the advice in short: the whole workbench.

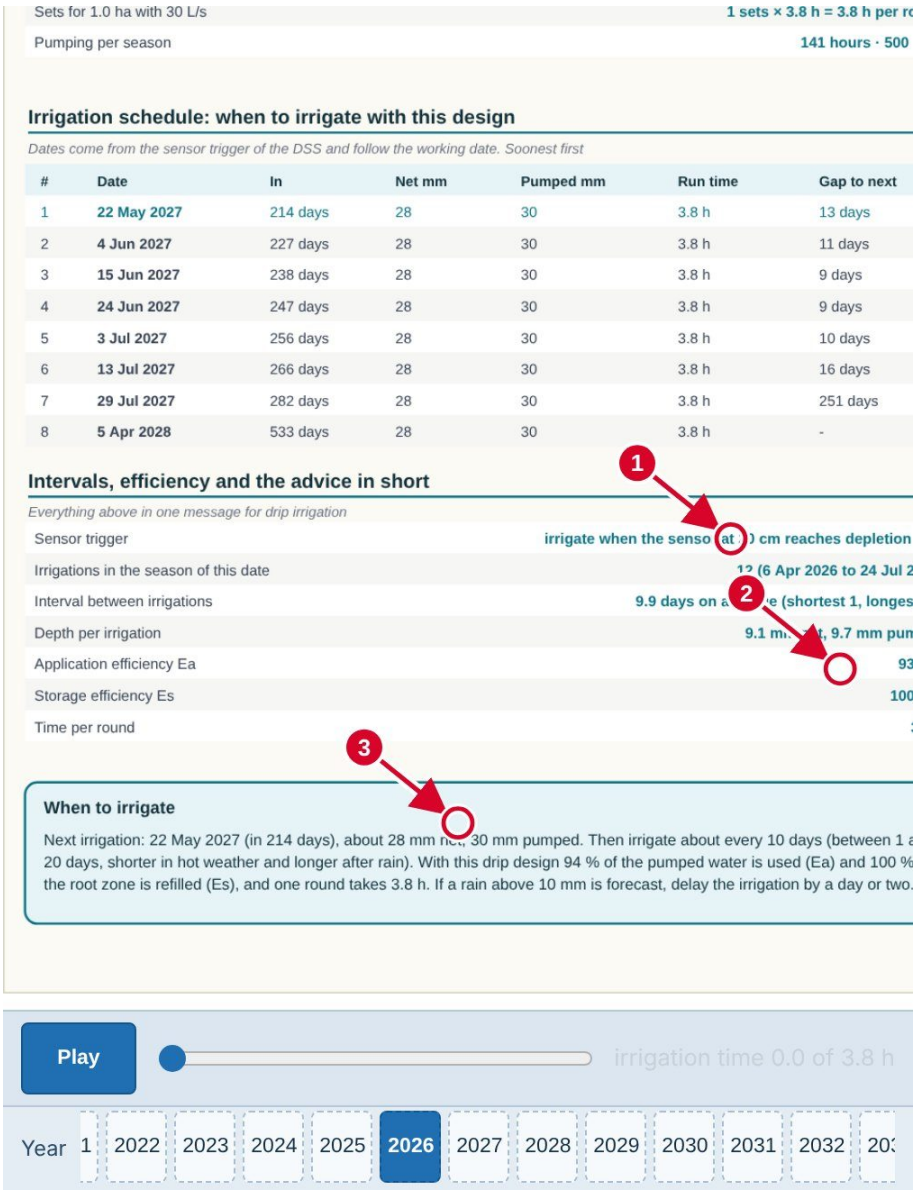


Figure 18.66: S3 Drip irrigation design / Intervals, efficiency and the advice in short: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sensor trigger: Eqs. 5.3 and 5.27.
- **2** Application efficiency E_a: Eq. 12.19.
- **3** "When to irrigate": message built from the schedule and from E_a and E_s.

Equations behind this screen. Eq. 12.24, Eq. 12.25.

Eq. 12.24. Where explained: chapter "Drip irrigation design", section "Season and farm".

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity

limit is a common practice.

Eq. 12.25. Where explained: chapter “Drip irrigation design”, section “Season and farm”.

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

18.4 S4 Sprinkler design and pipes

18.4.1 The whole layer: inputs and overview

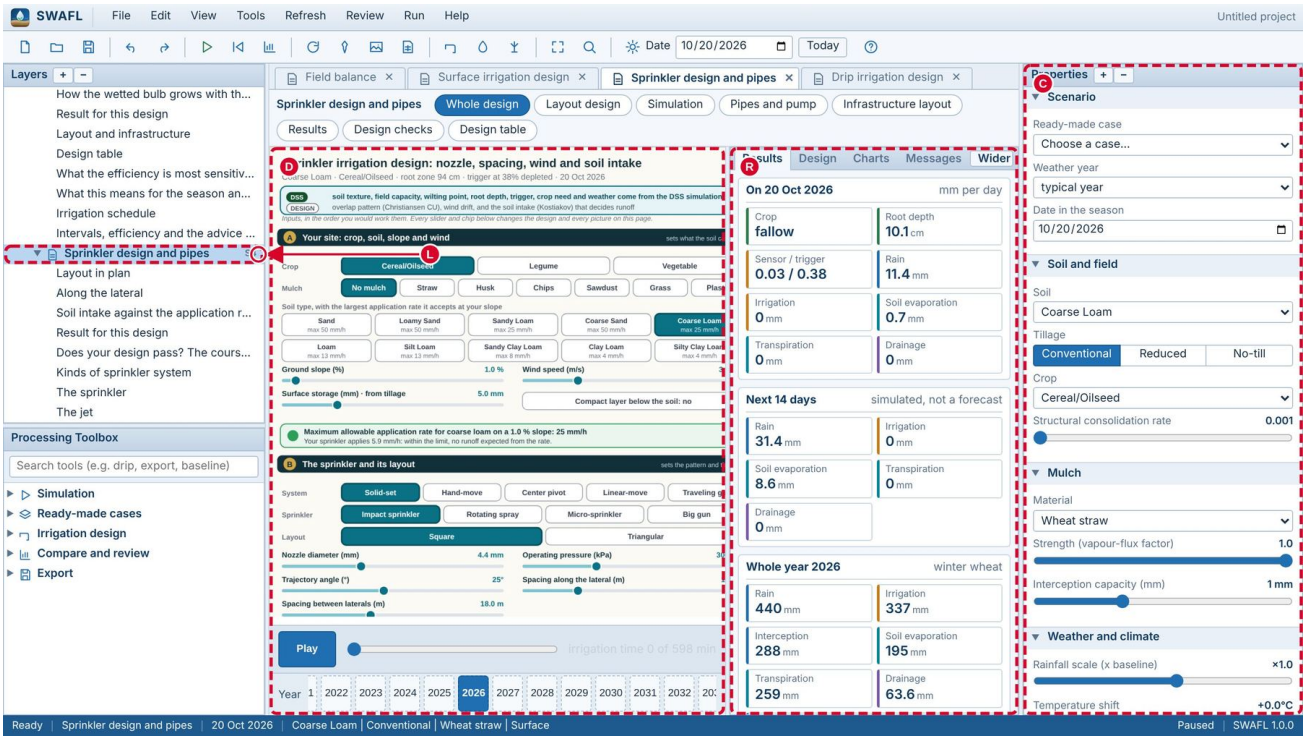


Figure 18.67: S4 Sprinkler design and pipes: the whole workbench.

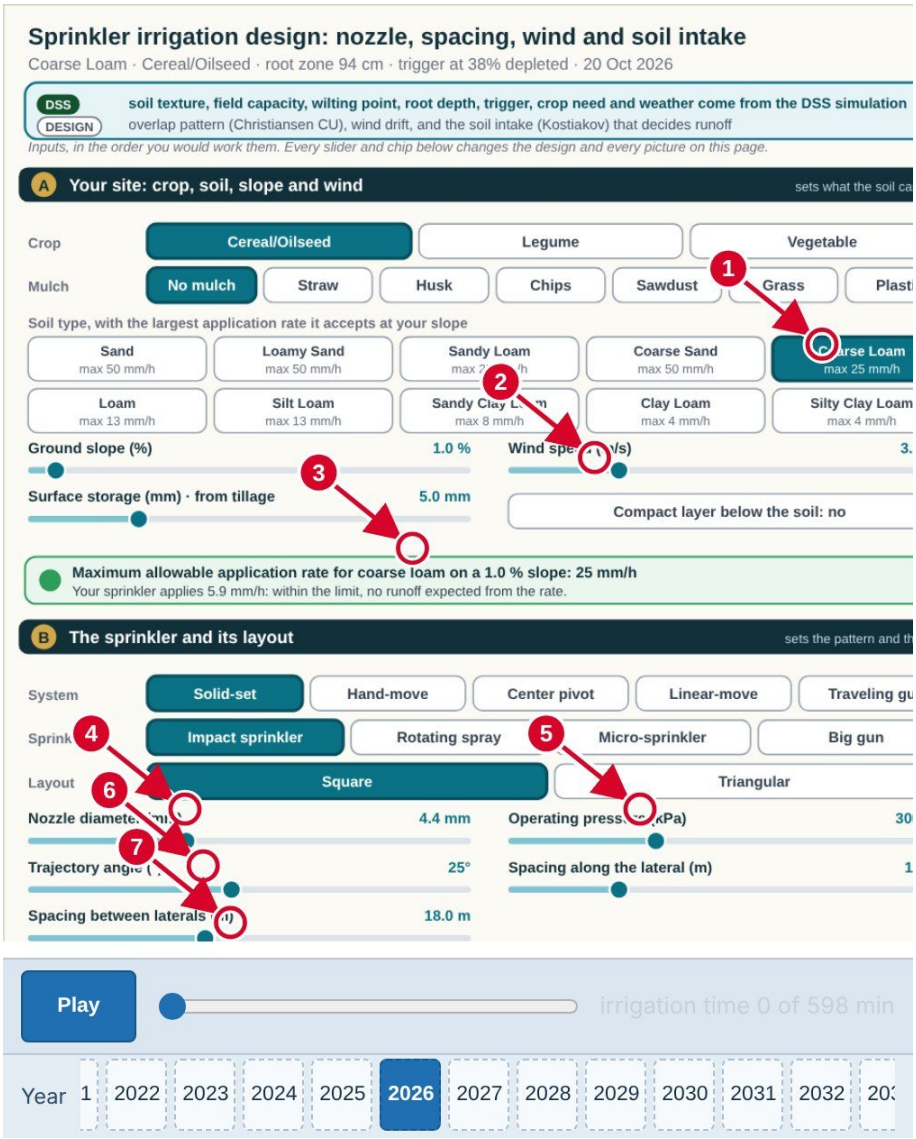


Figure 18.68: S4 Sprinkler design and pipes: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Selected soil: intake K and A and the maximum allowable application rate, Eq. 13.9 and the course table.
- **2** Wind speed u: Eqs. 13.5 and 13.8.
- **3** Banner: application rate against the maximum allowed, Eqs. 13.9 and 13.10.
- **4** Nozzle diameter d: Eqs. 13.2 and 13.3.
- **5** Operating pressure P gives the head h = P / 9.81: Eq. 13.2.
- **6** Trajectory angle alpha: f_alpha in Eq. 13.3.
- **7** Spacing between laterals S_l: Eq. 13.6.

Equations behind this screen. Eq. 5.27, Eq. 11.6, Eq. 13.1, Eq. 13.2, Eq. 13.3, Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7, Eq. 13.9, Eq. 13.10.

Eq. 5.27. Where explained: chapter “The two computing engines”, section “Evaporation under warming (used by the E1 layer)”.

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 13.1. Where explained: chapter "Sprinkler design", section "Inputs, step by step".

$$S_d^{\mathrm{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

Provenance (Eq. 13.1): **[Engineered]**: depression storage reduction with slope.

Eq. 13.2. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^{4}\ \ (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^{2}\ \mathrm{m^2}, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

Eq. 13.3. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".

$$R = c_R\sqrt{d\,h}\ f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

Eq. 13.4. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\widehat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \widehat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

Eq. 13.9. Where explained: chapter "Sprinkler design", section "The soil intake, ponding and runoff".

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \mathrm{sup}, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \mathrm{sup} - F, \quad \mathrm{RO} = \max(0, R - S_d^{\mathrm{eff}}), \quad I = \mathrm{sup} - \mathrm{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.10. Where explained: chapter "Sprinkler design", section "The soil intake, ponding and runoff".

$$I_c = \min\{I,\; I_{\mathrm{int}}\}, \qquad I_{\mathrm{int}} = 0.6\,\mathrm{CAP}\,[\mathrm{VFF} < 0.95] + 1.2\,\mathrm{LAI}_{\mathrm{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

18.4.2 Layout in plan: depth infiltrated so far, from the overlapping spray patterns

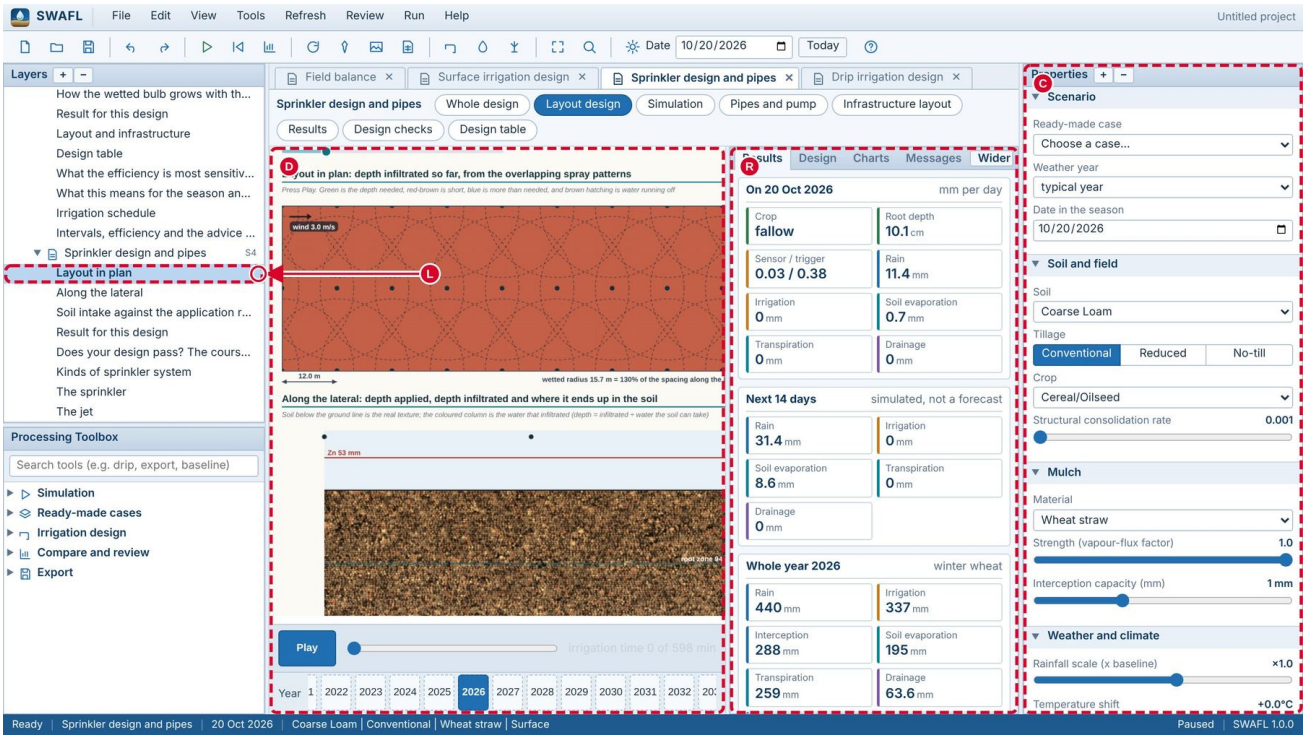


Figure 18.69: S4 Sprinkler design and pipes / Layout in plan: the whole workbench.

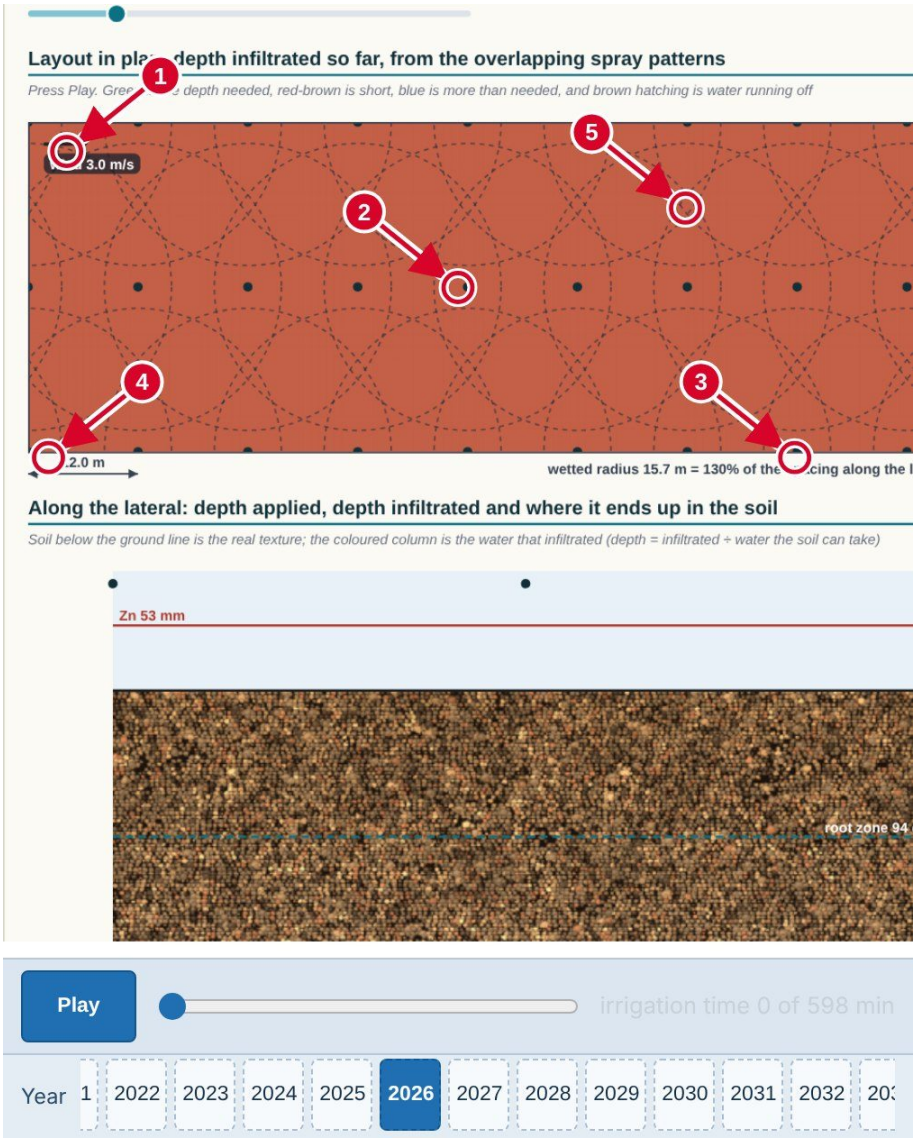


Figure 18.70: S4 Sprinkler design and pipes / Layout in plan: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Wind arrow: wind lengthens and shifts the pattern, Eq. 13.5.
- **2** Overlapping spray patterns: sum over neighbours, Eq. 13.6, with the radial profile of Eq. 13.4.
- **3** Wetted radius R: Eq. 13.3.
- **4** Spacing along the lateral S_s.
- **5** Colour shows infiltrated depth against Z_n: Eq. 13.11.

Equations behind this screen. Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

Eq. 13.4. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$\widehat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \widehat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$a(x, y) = \sum_{i,j} p(x - x_{ij},\ y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

18.4.3 Along the lateral: depth applied, depth infiltrated and where it ends up in the soil

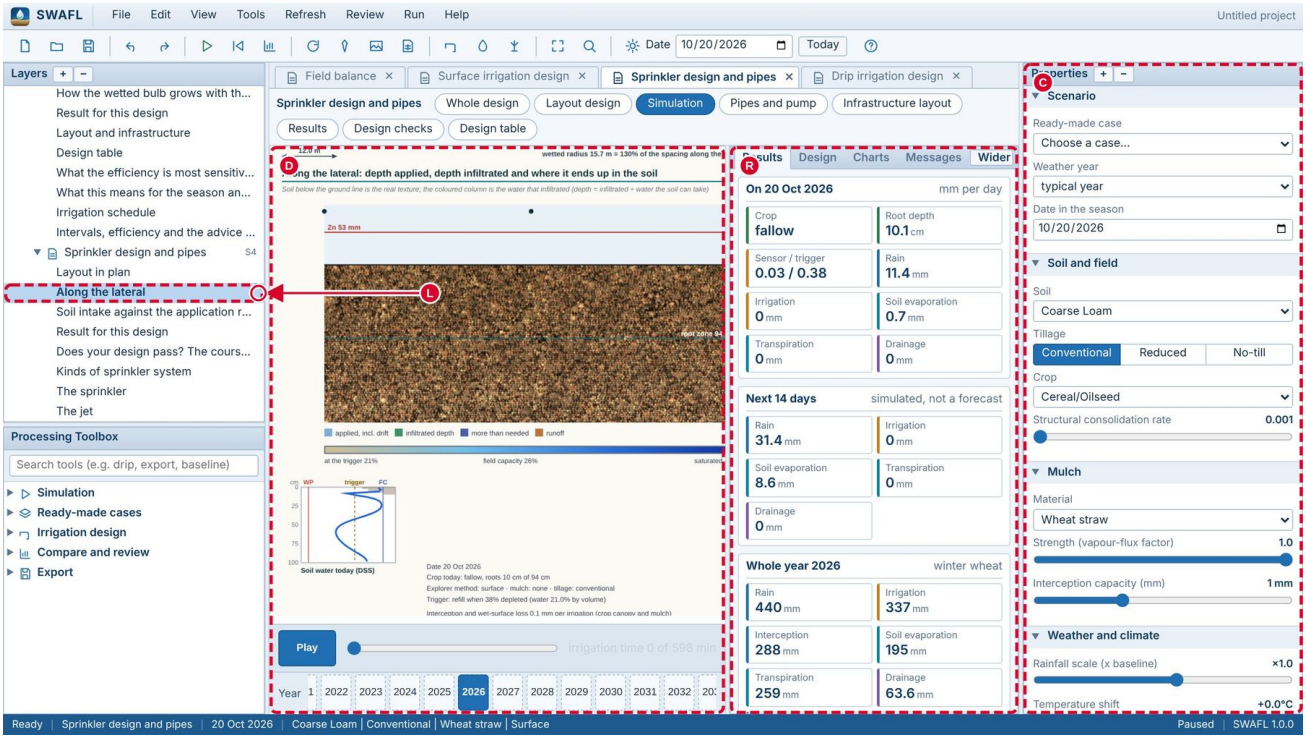


Figure 18.71: S4 Sprinkler design and pipes / Along the lateral: the whole workbench.

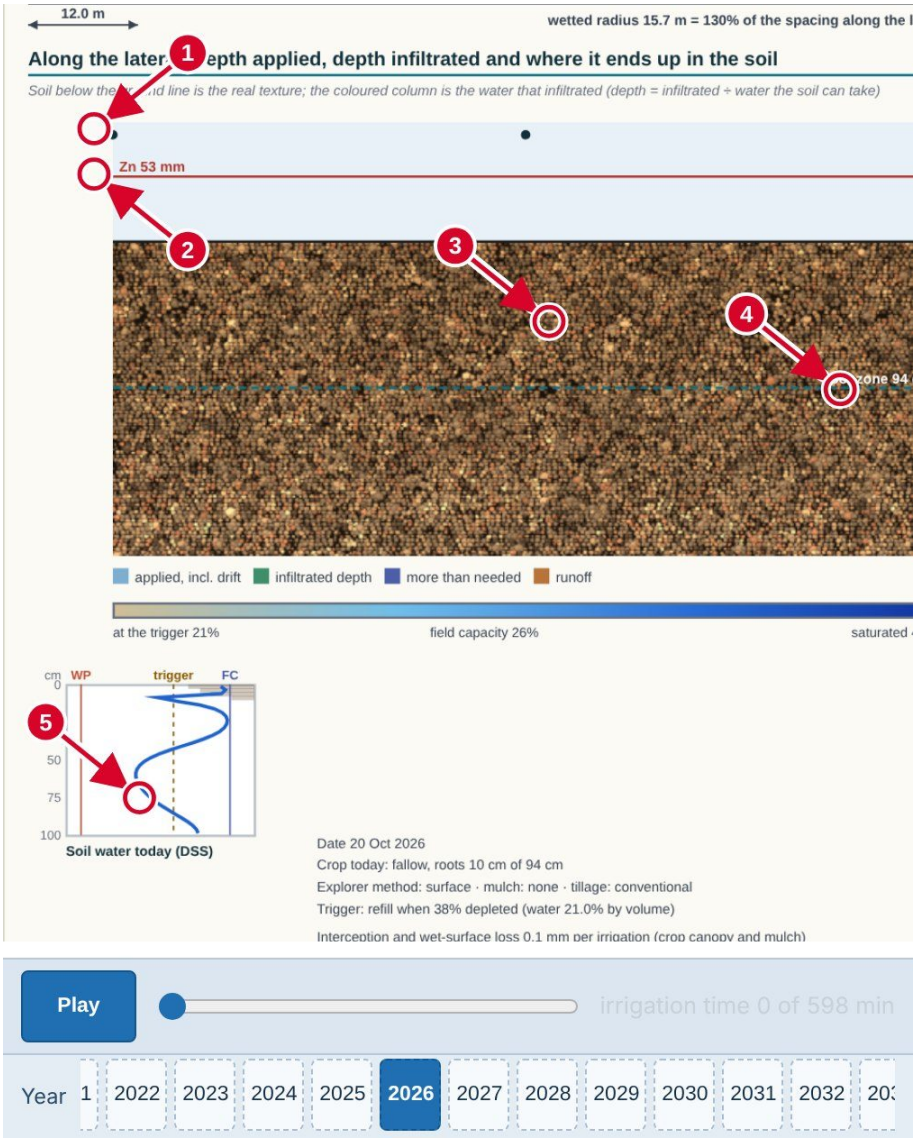


Figure 18.72: S4 Sprinkler design and pipes / Along the lateral: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sprinkler positions.
- **2** Z_n: required depth, Eq. 5.27.
- **3** Infiltrated depth in each cell, Z_k: Eq. 13.11.
- **4** Root-zone depth (94 cm).
- **5** Soil water today from the DSS profile.

Equations behind this screen. Eq. 11.6, Eq. 13.9, Eq. 13.10, Eq. 13.11, Eq. 13.12.

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 13.9. Where explained: chapter "Sprinkler design", section "The soil intake, ponding and

runoff”.

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \text{sup}, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \text{sup} - F, \quad \text{RO} = \max(0, R - S_d^{\text{eff}}), \quad I = \text{sup} - \text{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.10. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.

$$I_c = \min\{I,\ I_{\text{int}}\}, \qquad I_{\text{int}} = 0.6\,\text{CAP}\,[\text{VFF} < 0.95] + 1.2\,\text{LAI}_{\text{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.11. Where explained: chapter “Sprinkler design”, section “Efficiencies and run time”.

$$\begin{aligned} E_a &= \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, & E_s &= \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, & \text{DU} &= \frac{\overline{Z}_{25}}{\overline{Z}}, \\ \text{DP} &= \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, & \text{RO} &= \frac{\overline{\text{RO}_k}}{\text{gross}}, & \text{WD} &= w_d + \frac{\overline{I_c}}{\text{gross}}, \end{aligned} \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

Eq. 13.12. Where explained: chapter “Sprinkler design”, section “Efficiencies and run time”.

$$T = \min\{T :\ \overline{Z}_{25}(T) \ge Z_n\}, \qquad T \in [1,\ 2400]\ \text{min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

18.4.4 Soil intake against the application rate: when the soil cannot keep up, water runs off

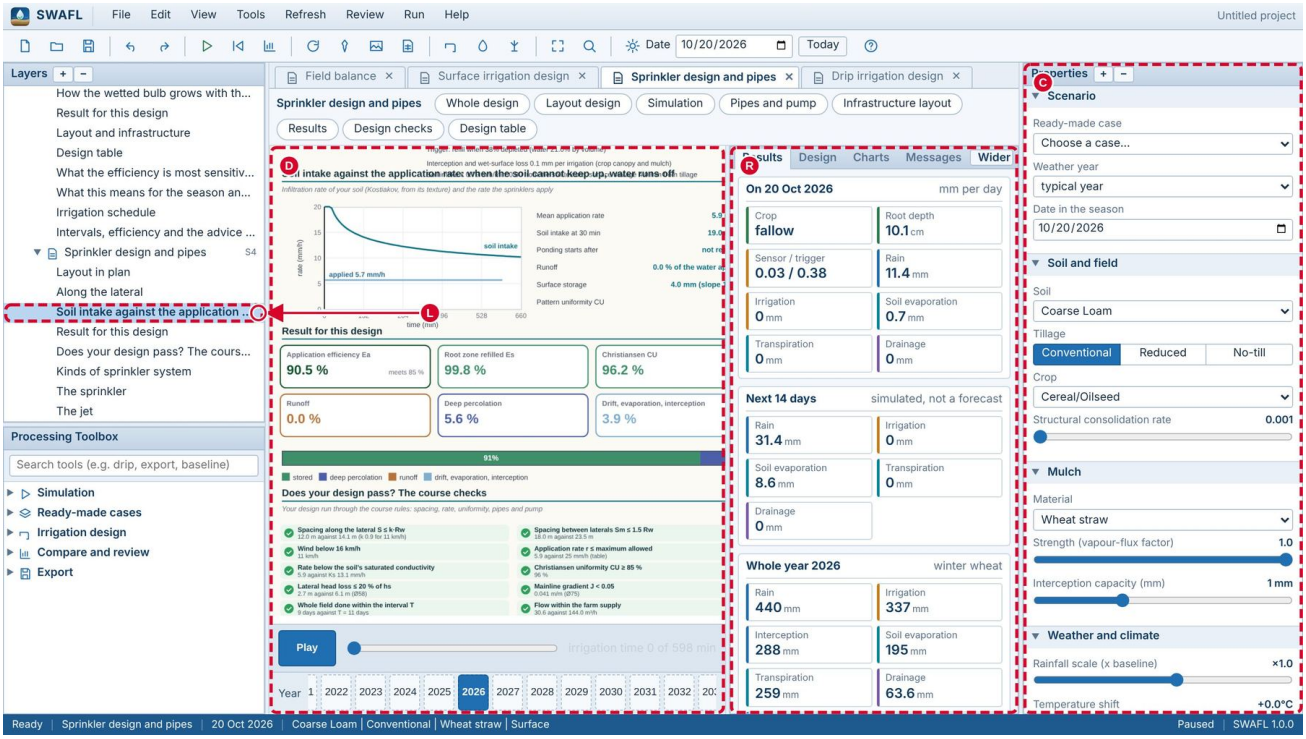


Figure 18.73: S4 Sprinkler design and pipes / Soil intake against the application rate: the whole workbench.

Figure 18.74: S4 Sprinkler design and pipes / Soil intake against the application rate: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Soil intake rate K t^(A-1) (Kostiakov): Eq. 13.9.
- **2** Applied rate a = 60 times the mean depth: Eq. 13.7.
- **3** Ponding starts after t_p: Eq. 13.9.
- **4** Runoff share: Eqs. 13.9 and 13.10.
- **5** Surface storage S_d: Eq. 13.1.

Equations behind this screen. Eq. 11.6, Eq. 13.9, Eq. 13.10.

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 13.9. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \text{sup}, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \text{sup} - F, \quad \text{RO} = \max(0, R - S_d^{\text{eff}}), \quad I = \text{sup} - \text{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.10. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.

$$I_c = \min\{I,\ I_{\text{int}}\}, \qquad I_{\text{int}} = 0.6\,\text{CAP}\,[\text{VFF} < 0.95] + 1.2\,\text{LAI}_{\text{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

18.4.5 Result for this design

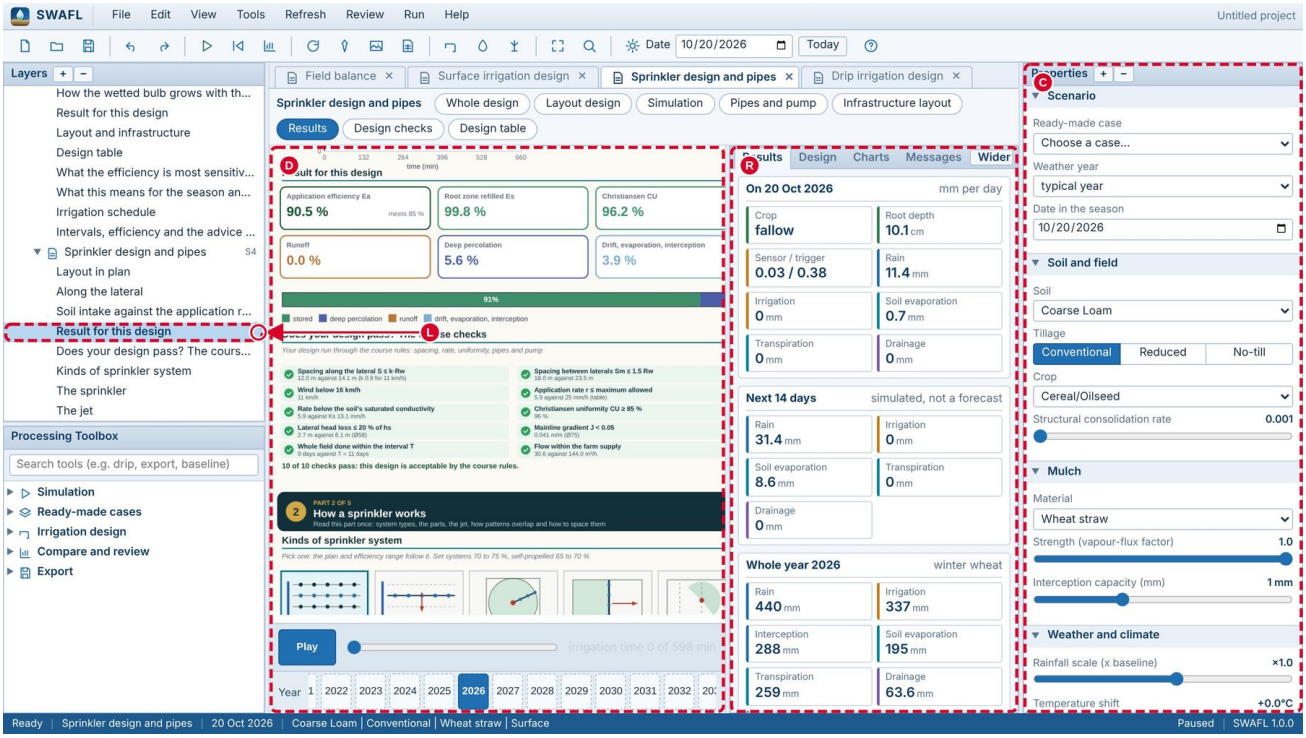


Figure 18.75: S4 Sprinkler design and pipes / Result for this design: the whole workbench.

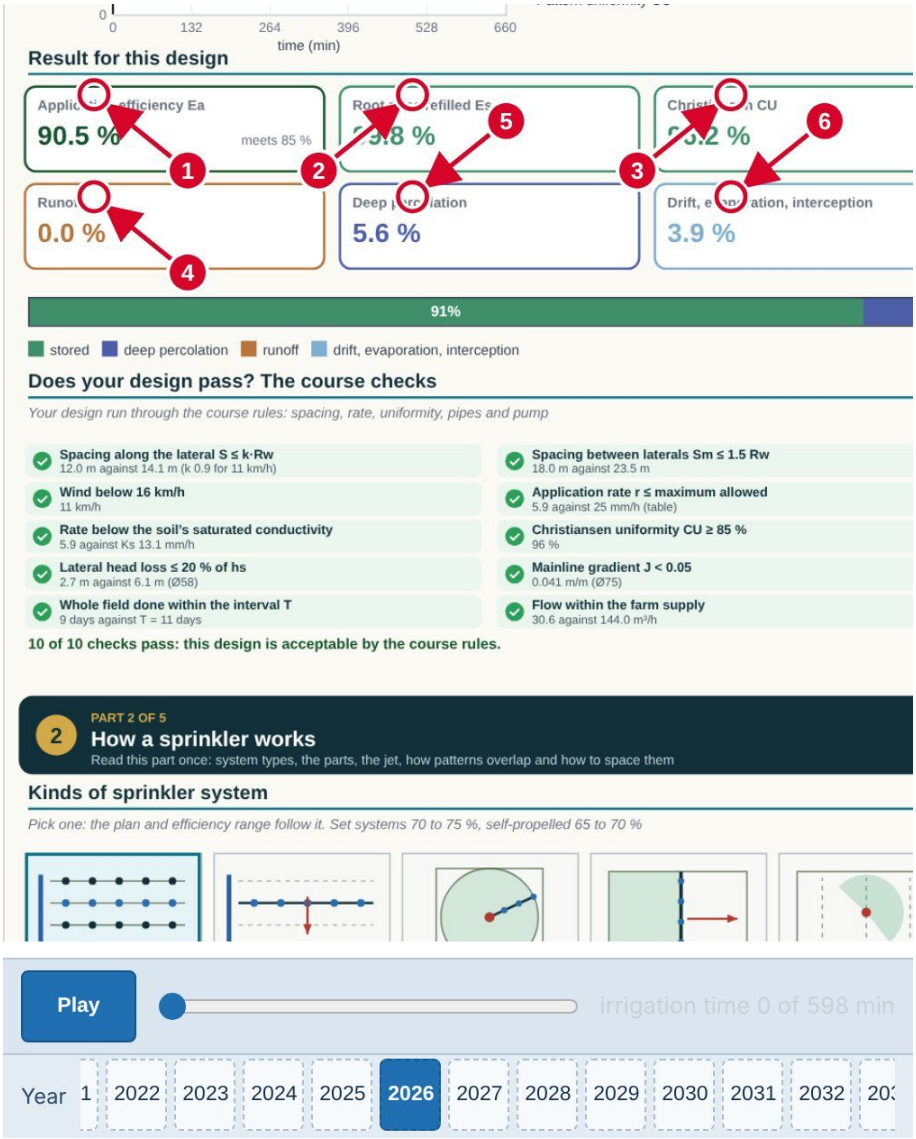


Figure 18.76: S4 Sprinkler design and pipes / Result for this design: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Application efficiency E_a: Eq. 13.11.
- **2** Root-zone refilled E_s: Eq. 13.11.
- **3** Christiansen CU: Eq. 13.7.
- **4** Runoff: Eqs. 13.9 and 13.10.
- **5** Deep percolation: Eq. 13.11.
- **6** Drift, evaporation and interception: Eqs. 13.8 and 13.10.

Equations behind this screen. Eq. 13.11, Eq. 13.12.

Eq. 13.11. Where explained: chapter “Sprinkler design”, section “Efficiencies and run time”.

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

Eq. 13.12. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".

$$T = \min\{T : \ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\, 2400] \text{ min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

18.4.6 Does your design pass? The course checks

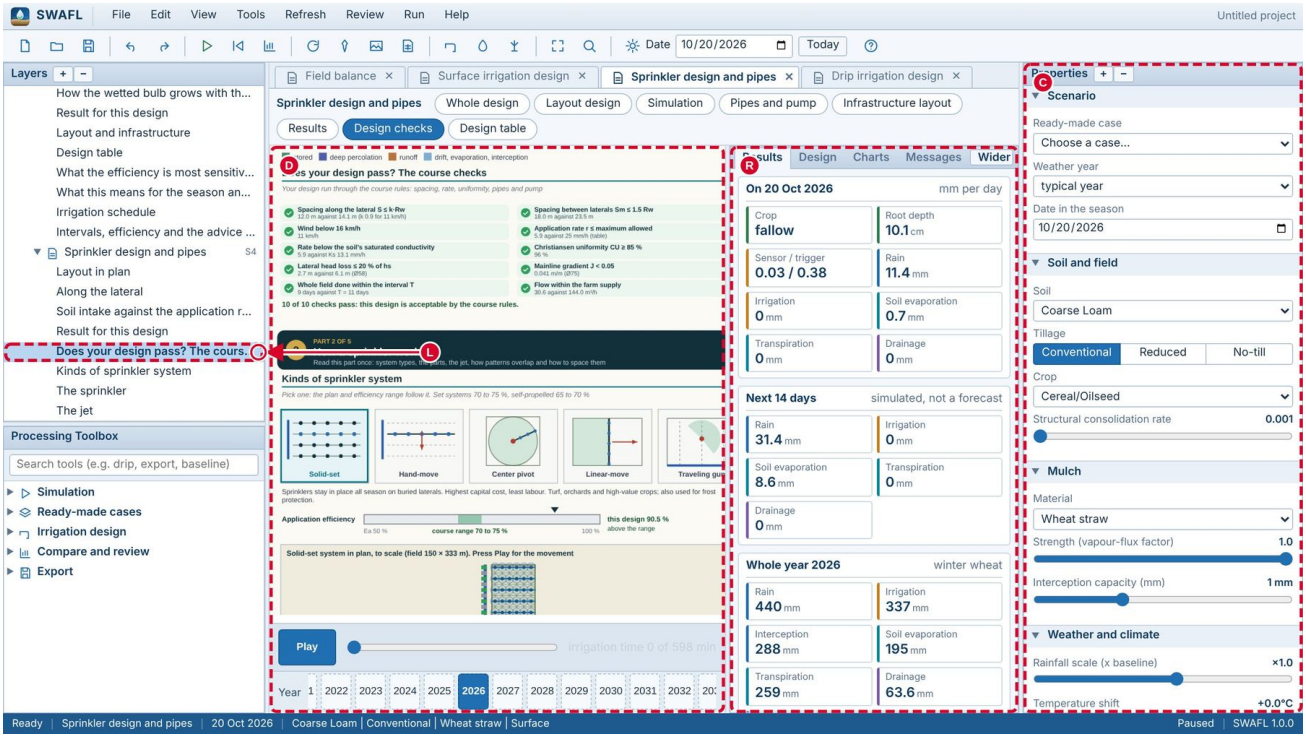


Figure 18.77: S4 Sprinkler design and pipes / Does your design pass? The course checks: the whole workbench.

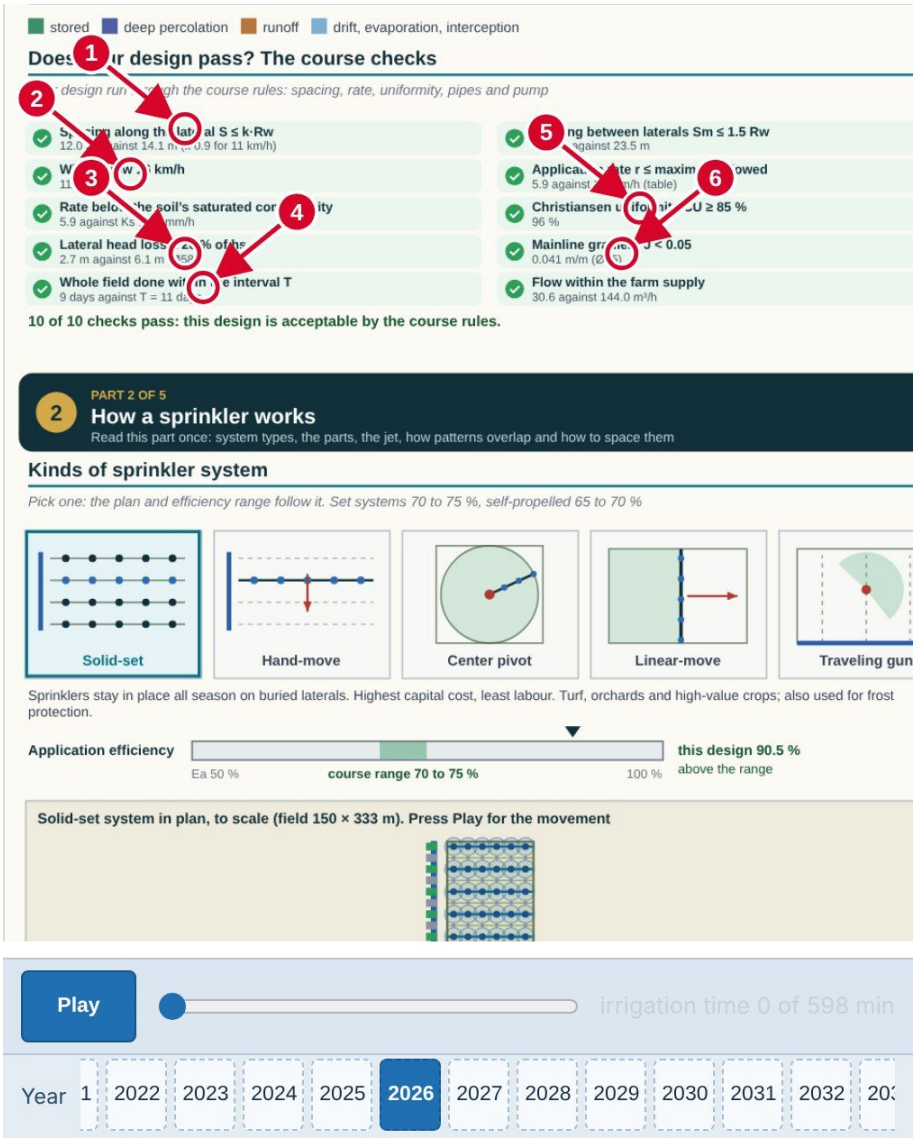


Figure 18.78: S4 Sprinkler design and pipes / Does your design pass? The course checks: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Spacing along the lateral S_s <= k R_w: Eqs. 13.13 and 13.14.
- **2** Wind below 16 km/h: Eq. 13.13.
- **3** Lateral head loss <= 20 % of the head: Eq. 13.18.
- **4** Whole field done within the interval T: Eqs. 13.14 to 13.16.
- **5** Christiansen CU >= 85 %: Eq. 13.7.
- **6** Mainline gradient J < 0.05: Eq. 13.17.

Equations behind this screen. Eq. 13.13, Eq. 13.14, Eq. 13.15, Eq. 13.16, Eq. 13.17, Eq. 13.18, Eq. 13.19, Eq. 13.20.

Eq. 13.13. Where explained: chapter “Sprinkler design”, section “The ten textbook checks”.

$$k = \begin{cases} 1, & W \le 5.5 \\ 0.9, & 5.5 < W \le 13 \\ 0.8, & 13 < W \le 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

Provenance (Eq. 13.13): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.14. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$S_s \le k\,R_w, \qquad S_l \le 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x/S_s \rfloor, \quad N_m = \lfloor L_y/S_l \rfloor, \tag{13.14}$$

Provenance (Eq. 13.14): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.15. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}} \ \text{(h)}, \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right) \text{ days}, \tag{13.15}$$

Provenance (Eq. 13.15): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.16. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$N_d(n) = \left\lceil \frac{N_m}{n\, n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\,N\,q_h, \tag{13.16}$$

Provenance (Eq. 13.16): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.17. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$J = \frac{0.472\,Q^{1.75}}{D^{4.75}} \ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

Provenance (Eq. 13.17): **[Literature]**: Blasius friction law for smooth pipe.

Eq. 13.18. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$h_{f,\text{lat}} = 1.1\,F\,J(Q_L, D)\,(L_x - S_s/2) \le 0.2\,h_s, \qquad Q_L = 1000\,N\,q_h, \tag{13.18}$$

Provenance (Eq. 13.18): **[Literature]**: 20 % rule of Keller and Bliesner (1990).

Eq. 13.19. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$H_{\text{man}} = z_{\text{lift}} + J\,L_{\text{conv}} + J\,L_{\text{main}} + h_s + 0.75\,h_{f,\text{lat}}, \tag{13.19}$$

Provenance (Eq. 13.19): **[Engineered]**: head and sets bookkeeping.

Eq. 13.20. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$P_{\text{CV}} = \frac{1000\,(Q_m/3.6\times10^6)\,H_{\text{man}}}{75\,\eta}, \quad \eta = 0.75, \qquad P_{\text{kW}} = 0.7355\,P_{\text{CV}}, \tag{13.20}$$

Provenance (Eq. 13.20): **[Literature]**: hydraulic power definition.

18.4.7 Kinds of sprinkler system

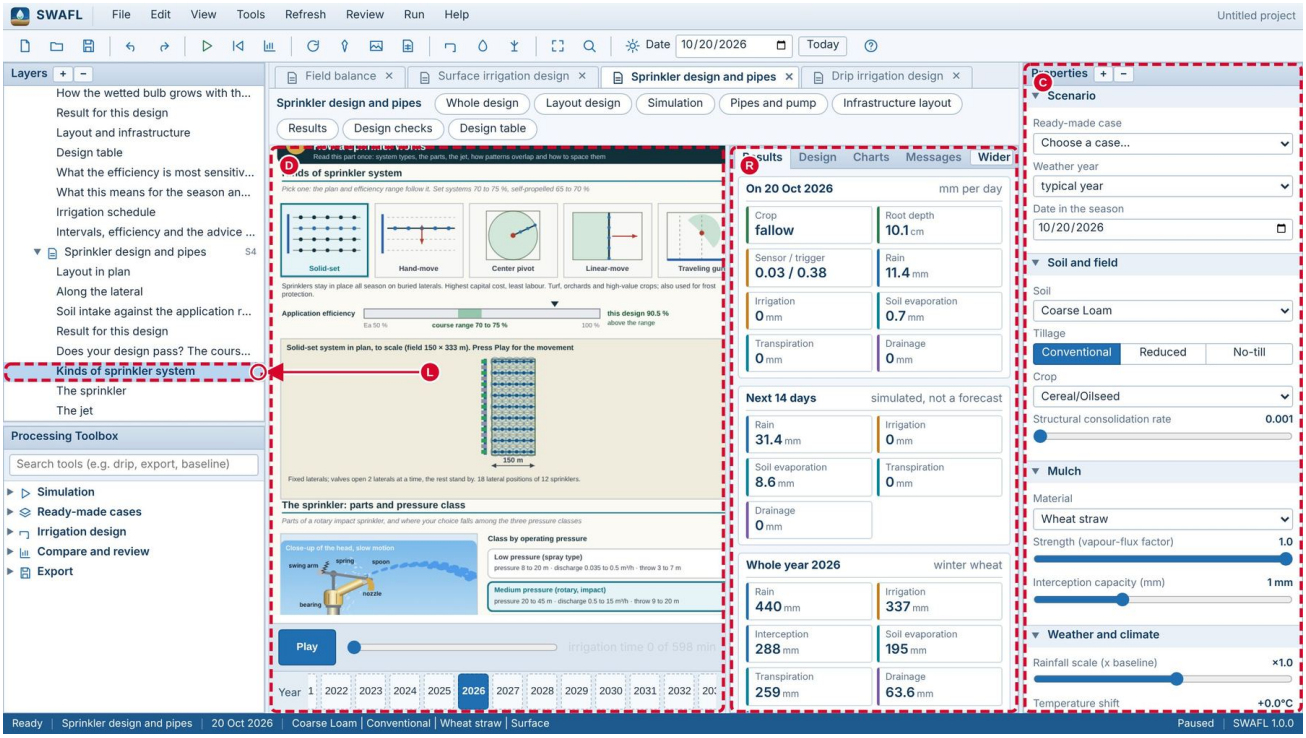


Figure 18.79: S4 Sprinkler design and pipes / Kinds of sprinkler system: the whole workbench.

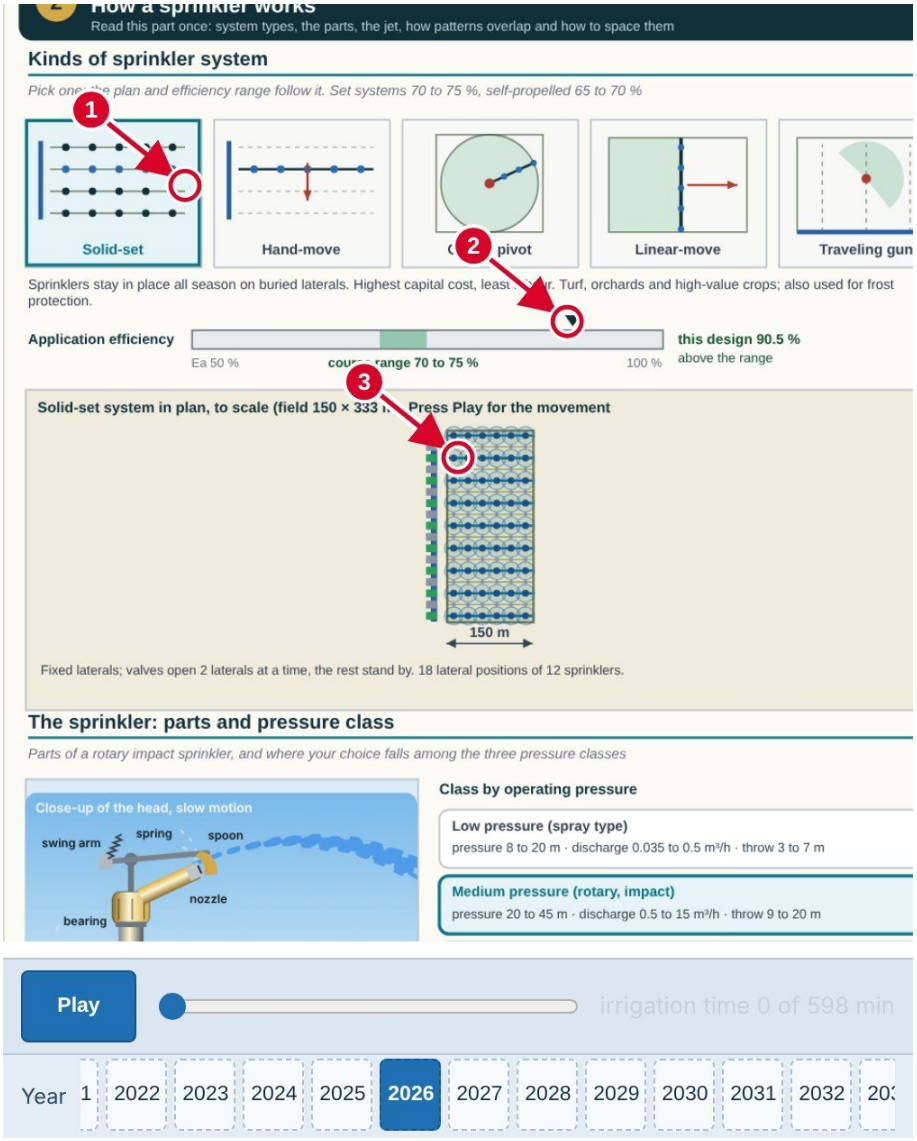


Figure 18.80: S4 Sprinkler design and pipes / Kinds of sprinkler system: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** System type (solid-set selected): sets the efficiency range of the course table.
- **2** This design’s E_a against the course range: Eq. 13.11.
- **3** Field plan to scale: laterals and sets, Eq. 13.21.

Equations behind this screen. none (the screen shows stored results).

18.4.8 The sprinkler: parts and pressure class

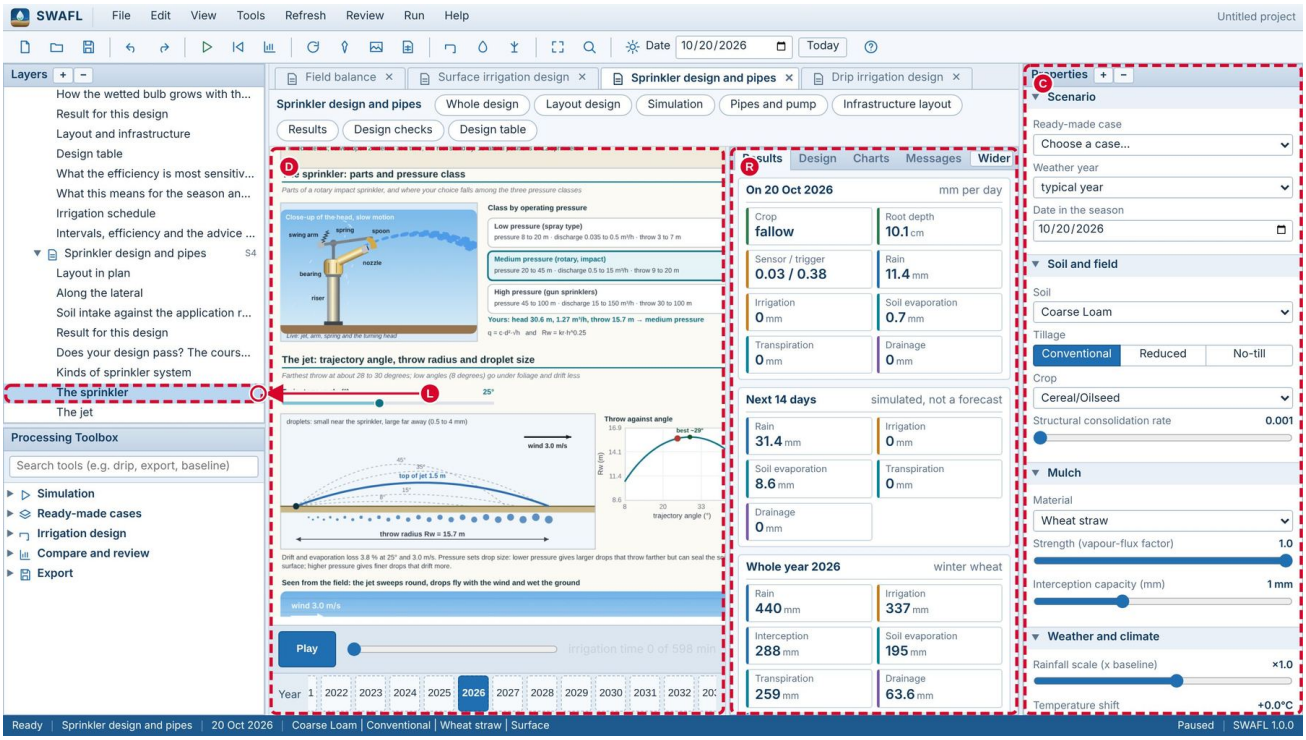


Figure 18.81: S4 Sprinkler design and pipes / The sprinkler: the whole workbench.

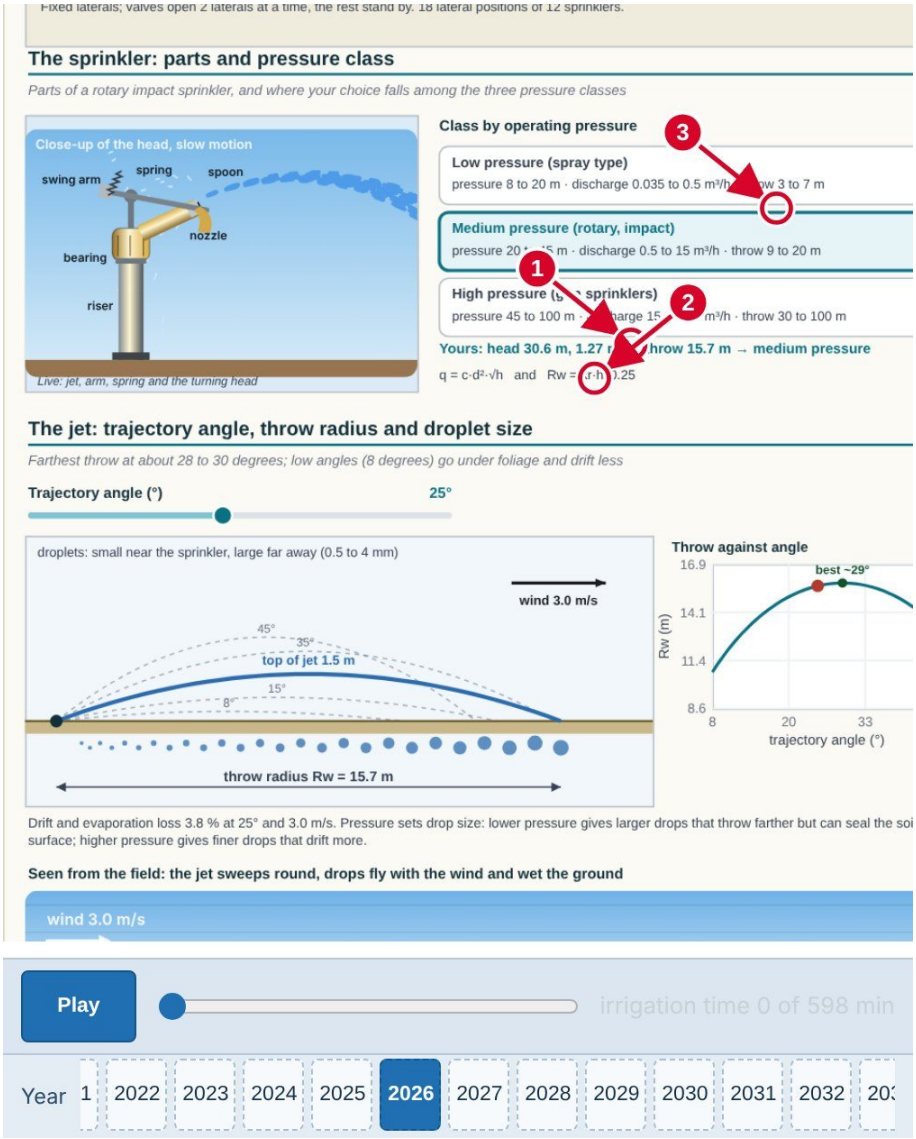


Figure 18.82: S4 Sprinkler design and pipes / The sprinkler: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** “Yours”: head h = P / 9.81 and discharge q (Eq. 13.2), throw R (Eq. 13.3).
- **2** Formulas q = c d^2 sqrt(h) and R = k_r h^0.25: Eqs. 13.2 and 13.3.
- **3** Pressure classes (low, medium, high): table, no equation.

Equations behind this screen. Eq. 13.2, Eq. 13.3.

Eq. 13.2. Where explained: chapter “Sprinkler design”, section “Nozzle discharge and wetted radius”.

$$q = 0.95\, A_n \sqrt{2g\,h} \times 6 \times 10^4 \;\; (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\left(\frac{d}{2000}\right)^2 \mathrm{m^2}, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

Eq. 13.3. Where explained: chapter “Sprinkler design”, section “Nozzle discharge and wetted radius”.

$$R = c_R \sqrt{d\,h}\; f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \left[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\right]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

18.4.9 The jet: trajectory angle, throw radius and droplet size

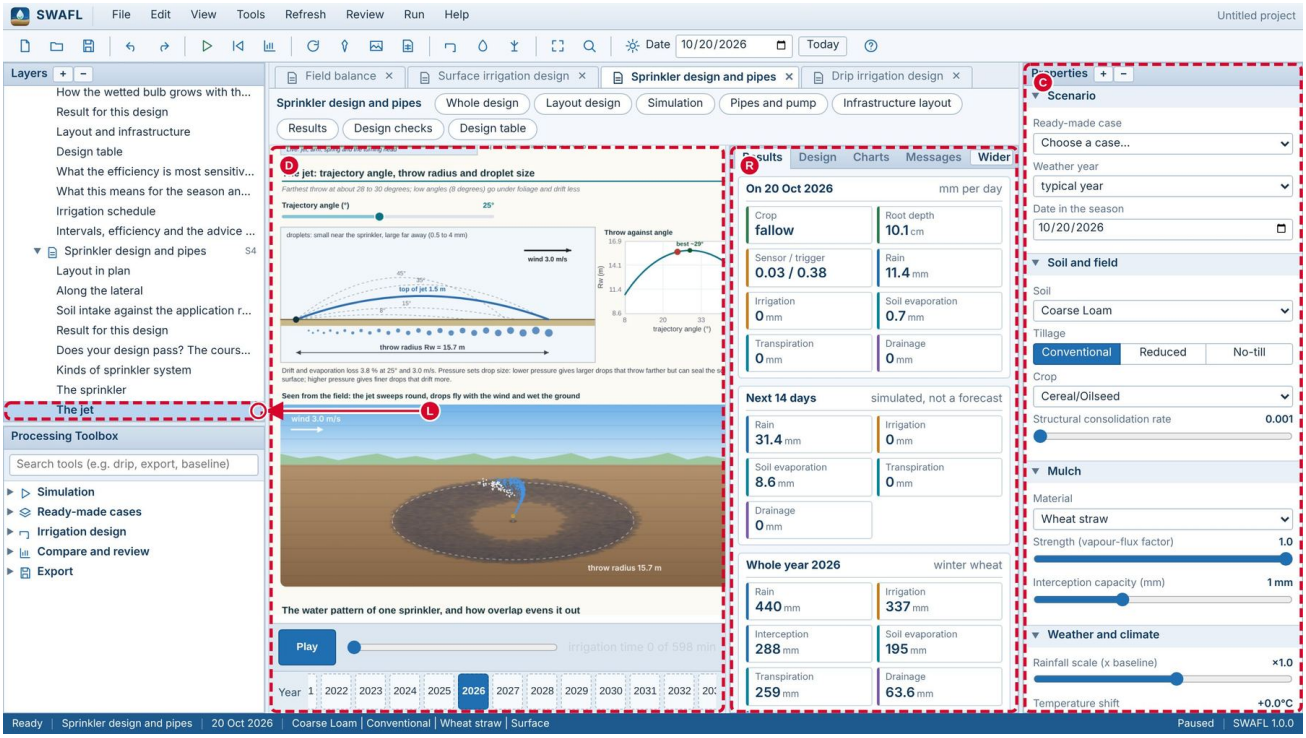


Figure 18.83: S4 Sprinkler design and pipes / The jet: the whole workbench.

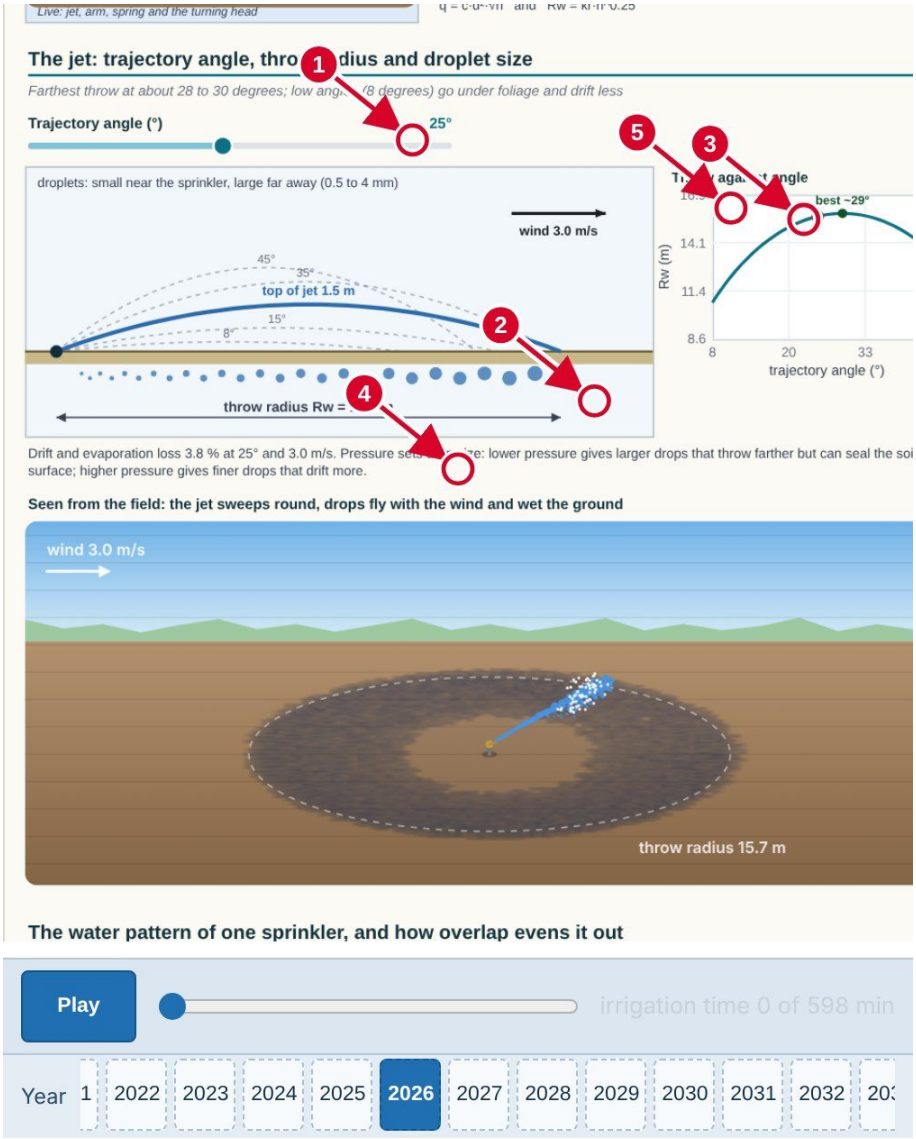


Figure 18.84: S4 Sprinkler design and pipes / The jet: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Trajectory angle alpha: f_alpha(alpha) in Eq. 13.3.
- **2** Throw radius R_w: Eq. 13.3.
- **3** Throw against angle, best near 29 degrees: Eq. 13.3.
- **4** Drift and evaporation loss w_d: Eq. 13.8.
- **5** Wind speed u: Eqs. 13.5 and 13.8.

Equations behind this screen. Eq. 13.2, Eq. 13.3, Eq. 13.8.

Eq. 13.2. Where explained: chapter “Sprinkler design”, section “Nozzle discharge and wetted radius”.

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^4 \;\; (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\left(\frac{d}{2000}\right)^2 \mathrm{m^2}, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

Eq. 13.3. Where explained: chapter “Sprinkler design”, section “Nozzle discharge and wetted radius”.

$$R = c_R \sqrt{d\,h}\; f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \left[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\right]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

Eq. 13.8. Where explained: chapter “Sprinkler design”, section “Wind drift and evaporation loss”.

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{(1.2+0.5u+0.04u^2)\,F\Big(\frac{P}{250}\Big)^{0.7}[1+0.025\,(T_a-20)]\Big[0.6+0.4\,\frac{\sin\alpha}{0.4226}\Big];\ 0.5,\ 45\Big\}. \tag{13.8}$$

Provenance (Eq. 13.8): **[Engineered]**: loss and ponding rules chosen for the workbench.

18.4.10 The water pattern of one sprinkler, and how overlap evens it out

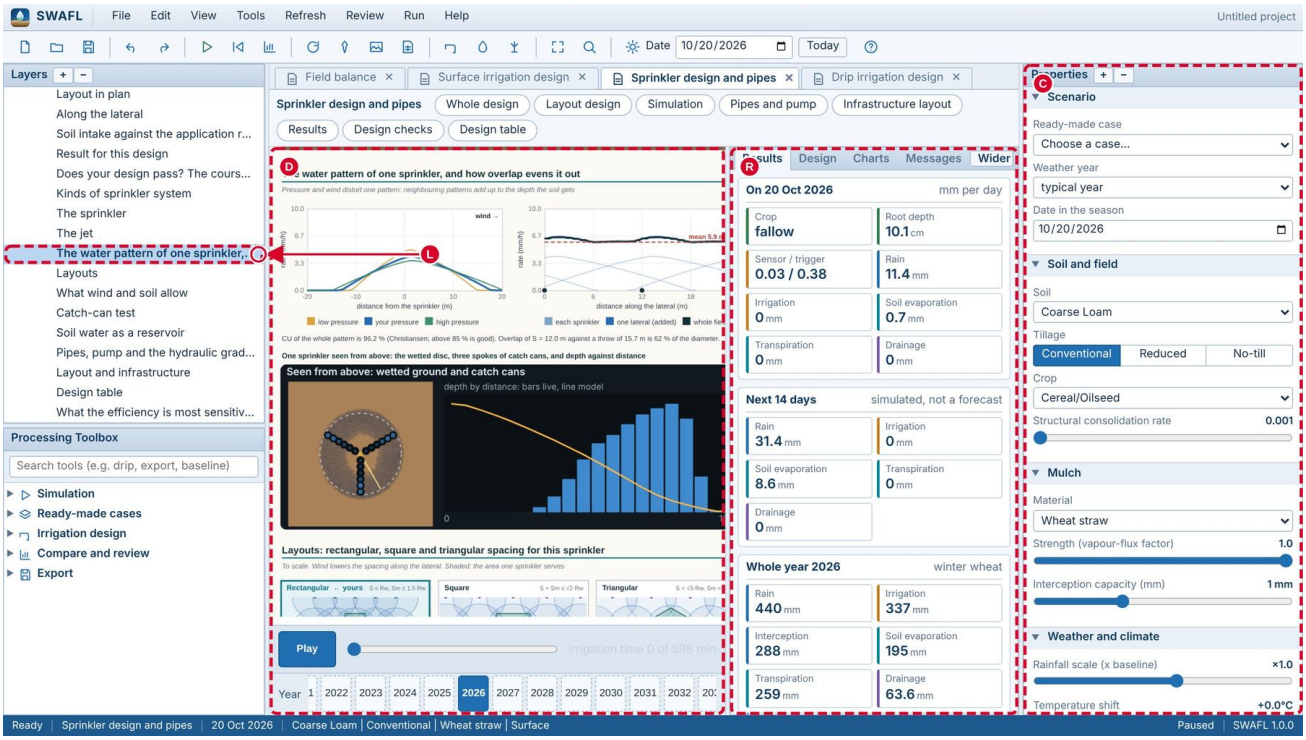


Figure 18.85: S4 Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out: the whole workbench.

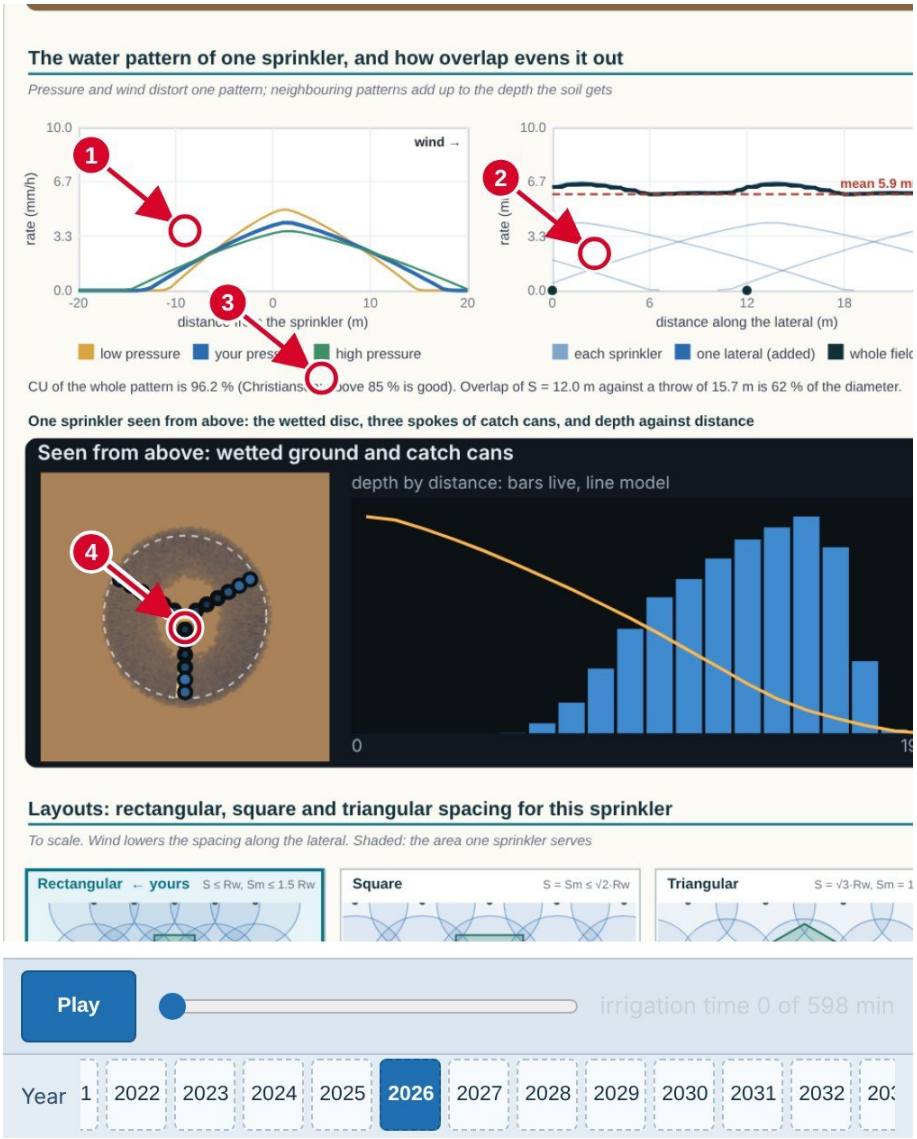


Figure 18.86: S4 Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Radial profile of one sprinkler, p(x): Eq. 13.4, skewed by the wind of Eq. 13.5.
- **2** Sum over neighbouring sprinklers, a(x, y): Eq. 13.6; the dashed line is the mean rate.
- **3** Christiansen coefficient of the whole pattern: Eq. 13.7.
- **4** One sprinkler seen from above with three spokes of catch cans.

Equations behind this screen. Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

Eq. 13.4. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\hat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \quad (13.4)$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

18.4.11 Layouts: rectangular, square and triangular spacing for this sprinkler

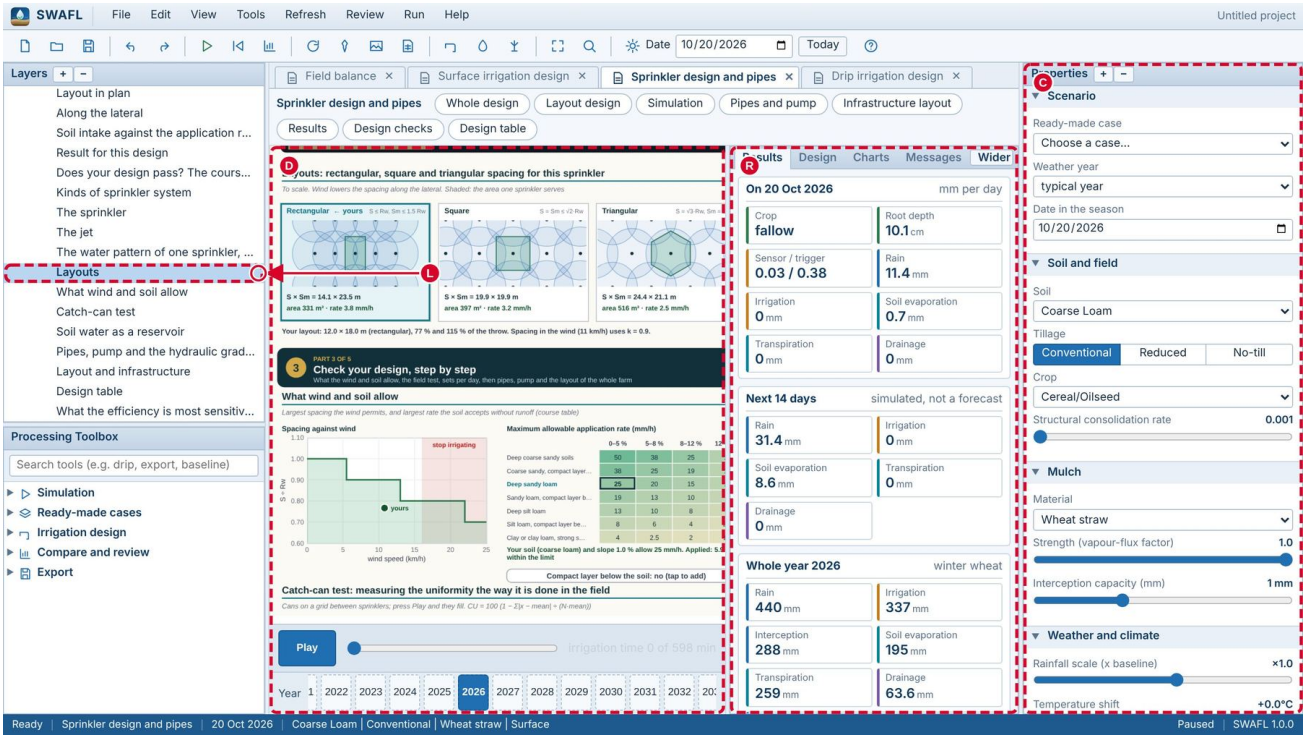


Figure 18.87: S4 Sprinkler design and pipes / Layouts: the whole workbench.

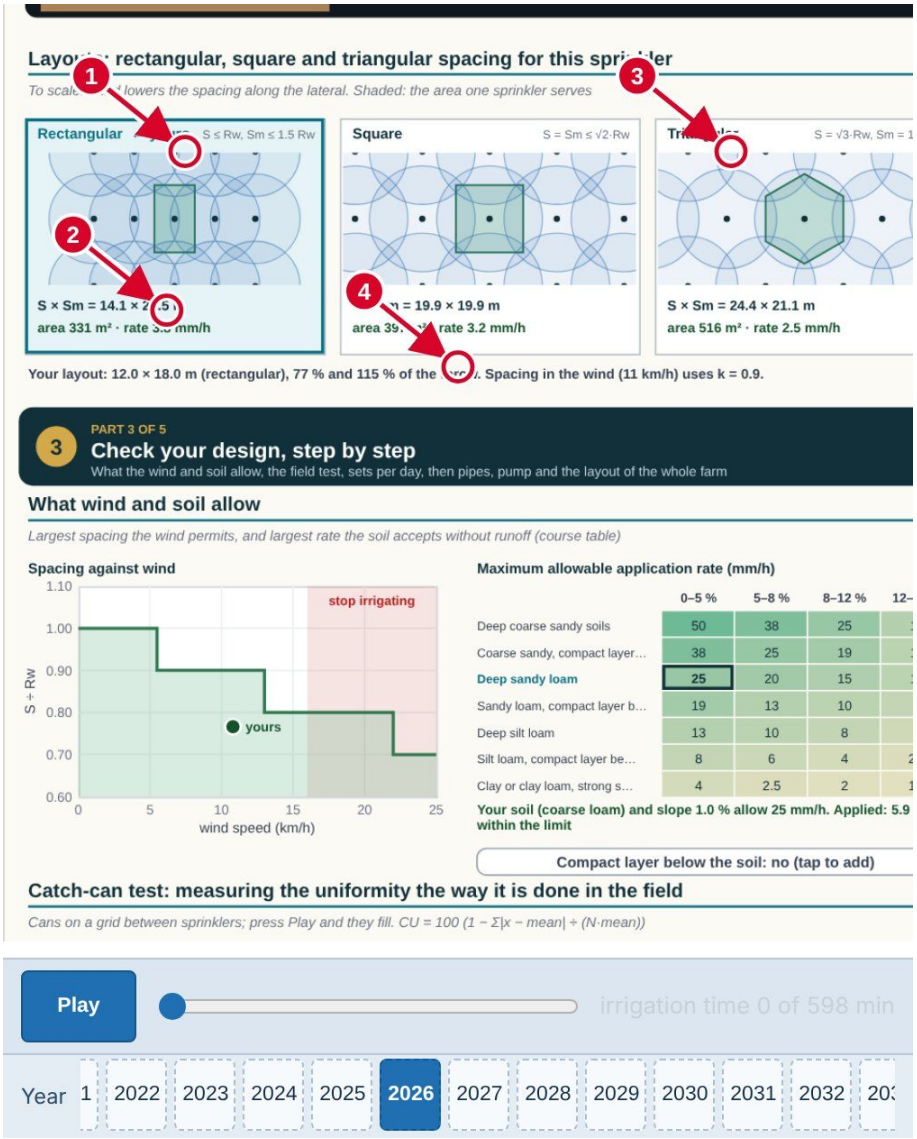


Figure 18.88: S4 Sprinkler design and pipes / Layouts: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Your layout: S_s <= k R_w and S_l <= 1.5 R_w, Eq. 13.14.
- **2** Area per sprinkler and application rate: Eq. 13.7.
- **3** Triangular layout: every second lateral shifted by S_s / 2, Eq. 13.6.
- **4** Wind factor k: Eq. 13.13.

Equations behind this screen. Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

Eq. 13.4. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\hat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \quad (13.4)$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

18.4.12 What wind and soil allow

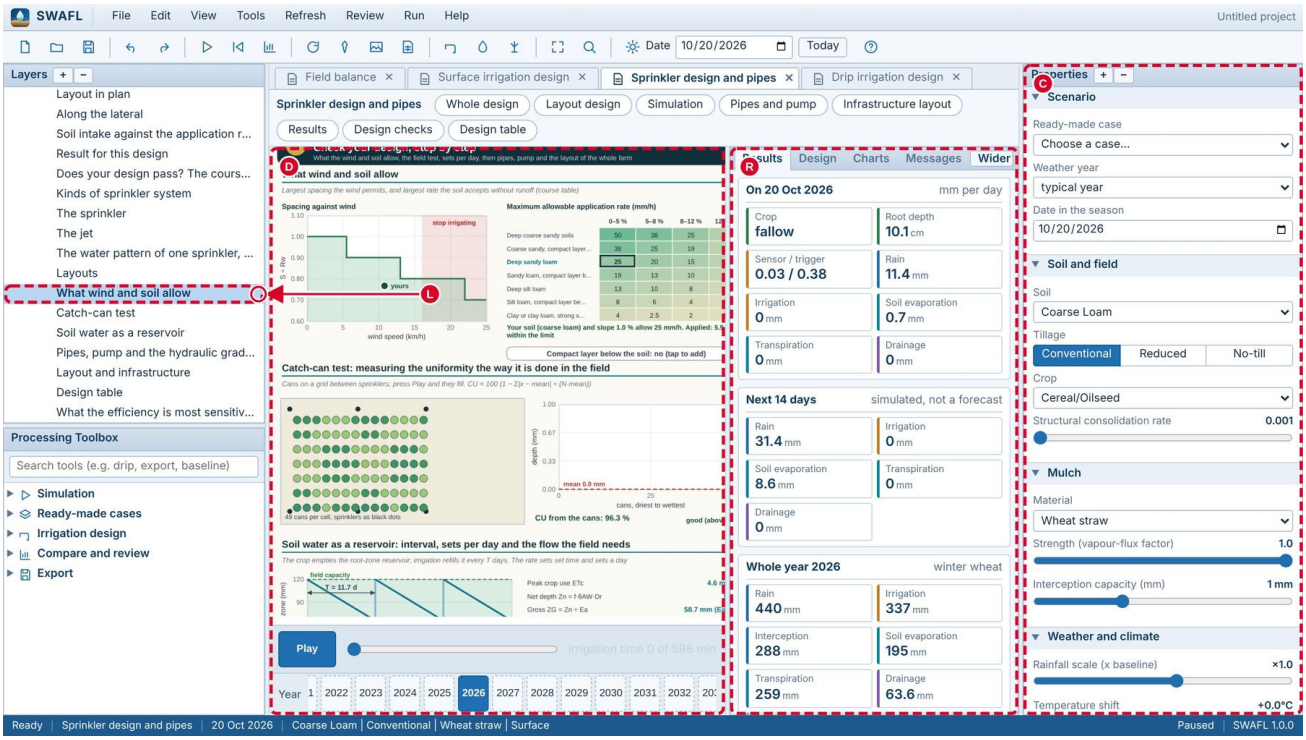


Figure 18.89: S4 Sprinkler design and pipes / What wind and soil allow: the whole workbench.

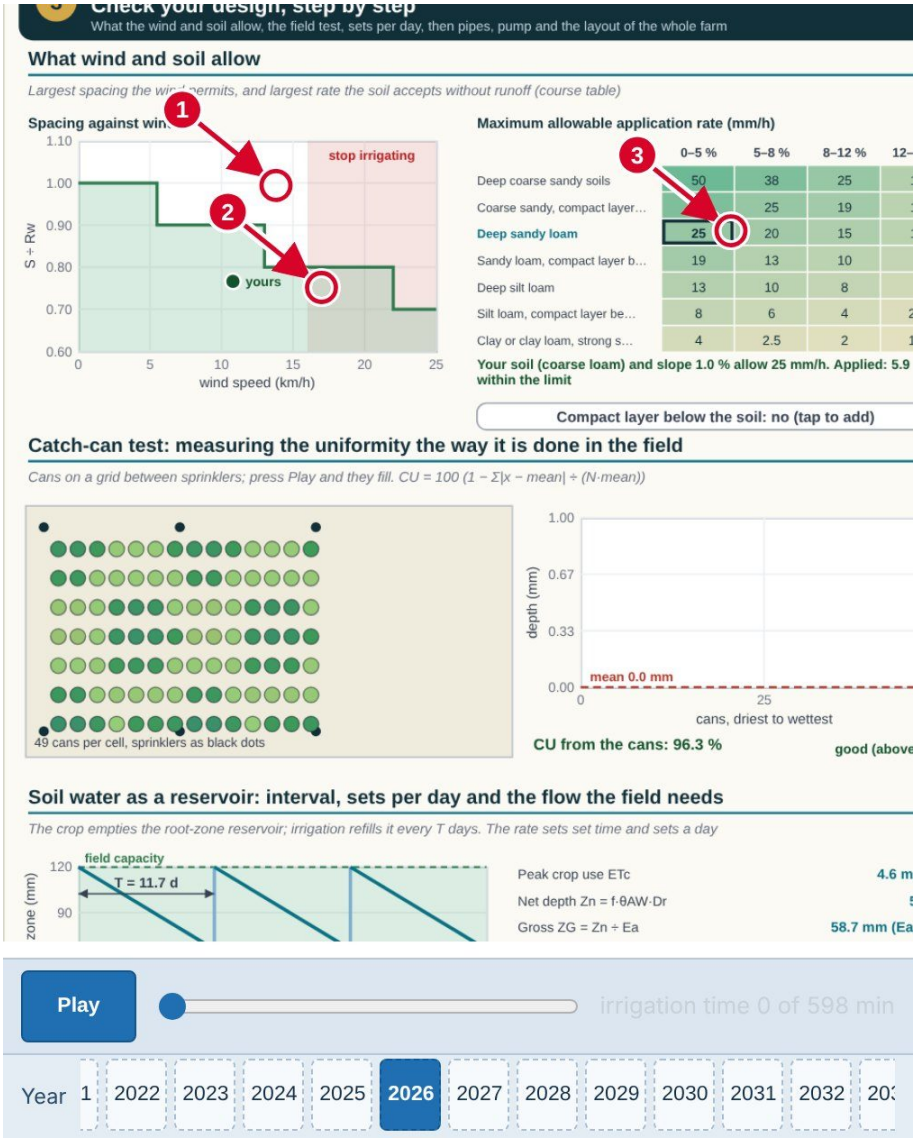


Figure 18.90: S4 Sprinkler design and pipes / What wind and soil allow: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Step line of the wind factor k: Eq. 13.13.
- **2** “Yours”: your wind speed and spacing.
- **3** Maximum allowable application rate by soil and slope (course table), used in Eqs. 13.9 and 13.13.

Equations behind this screen. Eq. 13.8.

Eq. 13.8. Where explained: chapter “Sprinkler design”, section “Wind drift and evaporation loss”.

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{(1.2+0.5u+0.04u^2)\,F\Big(\frac{P}{250}\Big)^{0.7}\,[1+0.025\,(T_a-20)]\,\Big[0.6+0.4\,\frac{\sin\alpha}{0.4226}\Big];\;0.5,\;45\Big\}. \tag{13.8}$$

Provenance (Eq. 13.8): **[Engineered]**: loss and ponding rules chosen for the workbench.

18.4.13 Catch-can test: measuring the uniformity the way it is done in the field

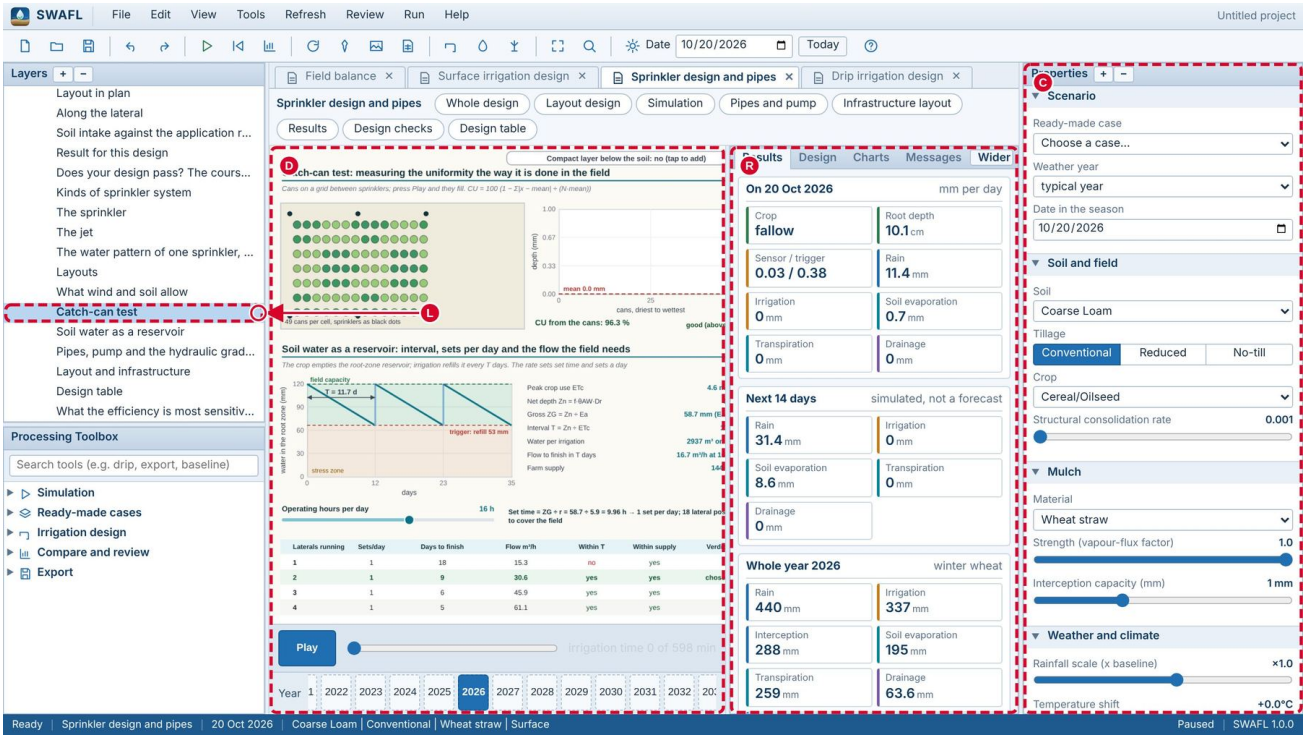


Figure 18.91: S4 Sprinkler design and pipes / Catch-can test: the whole workbench.

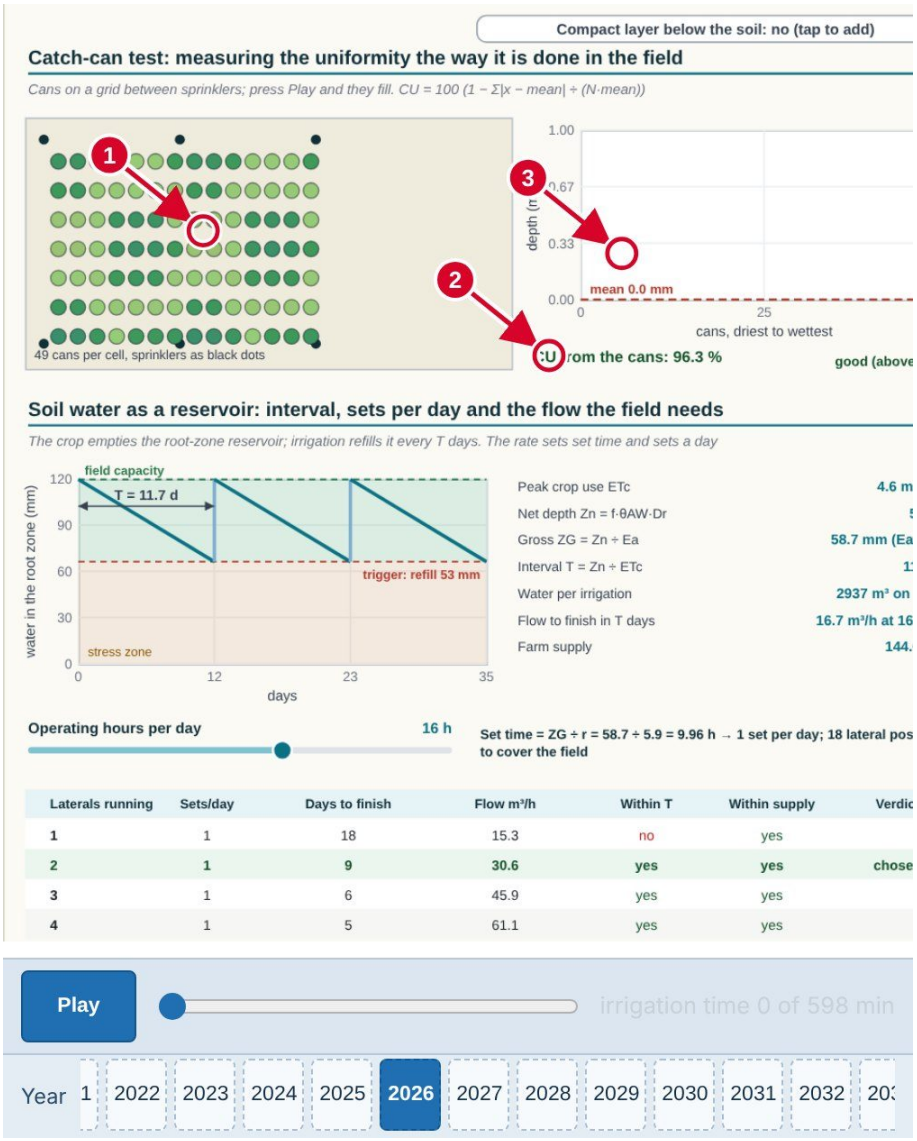


Figure 18.92: S4 Sprinkler design and pipes / Catch-can test: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Catch cans: depth a_k in each can from the pattern of Eq. 13.6.
- **2** CU from the cans = 1 - sum |a_k - mean| / (n mean): Eq. 13.7.
- **3** Cans sorted from driest to wettest, with the mean depth.

Equations behind this screen. Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

Eq. 13.4. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\hat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \quad (13.4)$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published

qualitative patterns.

Eq. 13.5. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

18.4.14 Soil water as a reservoir: interval, sets per day and the flow the field needs

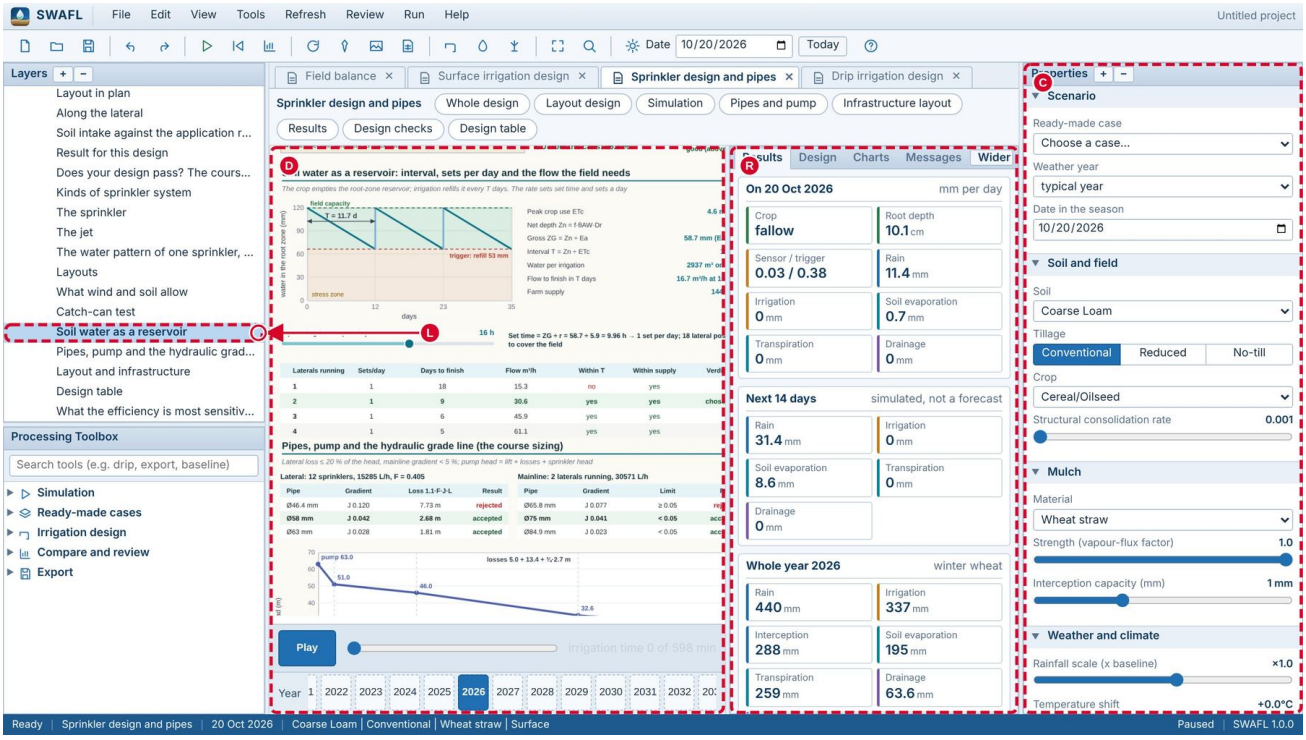


Figure 18.93: S4 Sprinkler design and pipes / Soil water as a reservoir: the whole workbench.

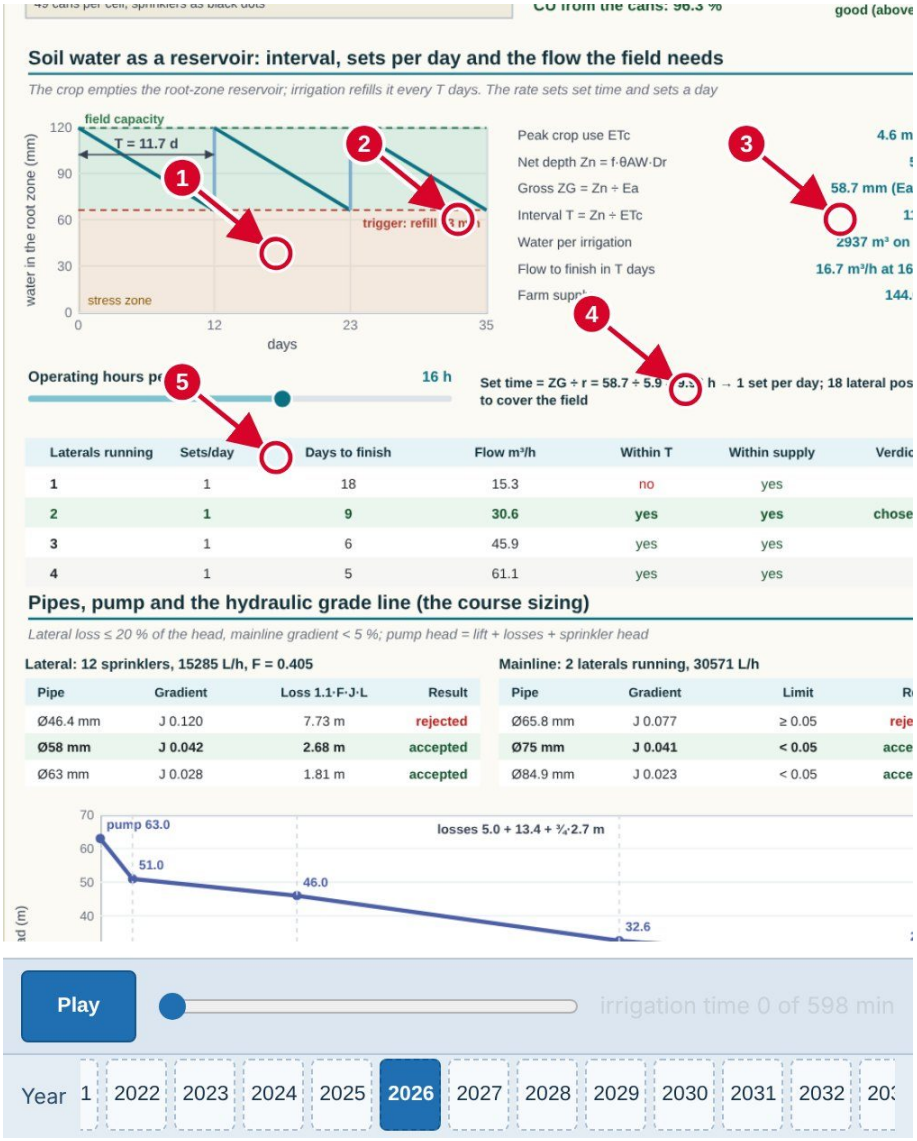


Figure 18.94: S4 Sprinkler design and pipes / Soil water as a reservoir: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Root-zone store drawn down by ET_c and refilled every T days.
- **2** Refill at Z_n from the DSS: Eq. 5.27.
- **3** Gross depth Z_G = Z_n / E_a: Eq. 13.15.
- **4** Set time t = Z_G / r: Eq. 13.15.
- **5** Laterals running together, days to finish and inflow: Eq. 13.16.

Equations behind this screen. Eq. 13.11, Eq. 13.12, Eq. 13.23.

Eq. 13.11. Where explained: chapter “Sprinkler design”, section “Efficiencies and run time”.

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

Eq. 13.12. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".

$$T = \min\{T : \ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\, 2400] \text{ min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.23. Where explained: chapter "Sprinkler design", section "Season and farm".

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60} \text{ h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

18.4.15 Pipes, pump and the hydraulic grade line (the course sizing)

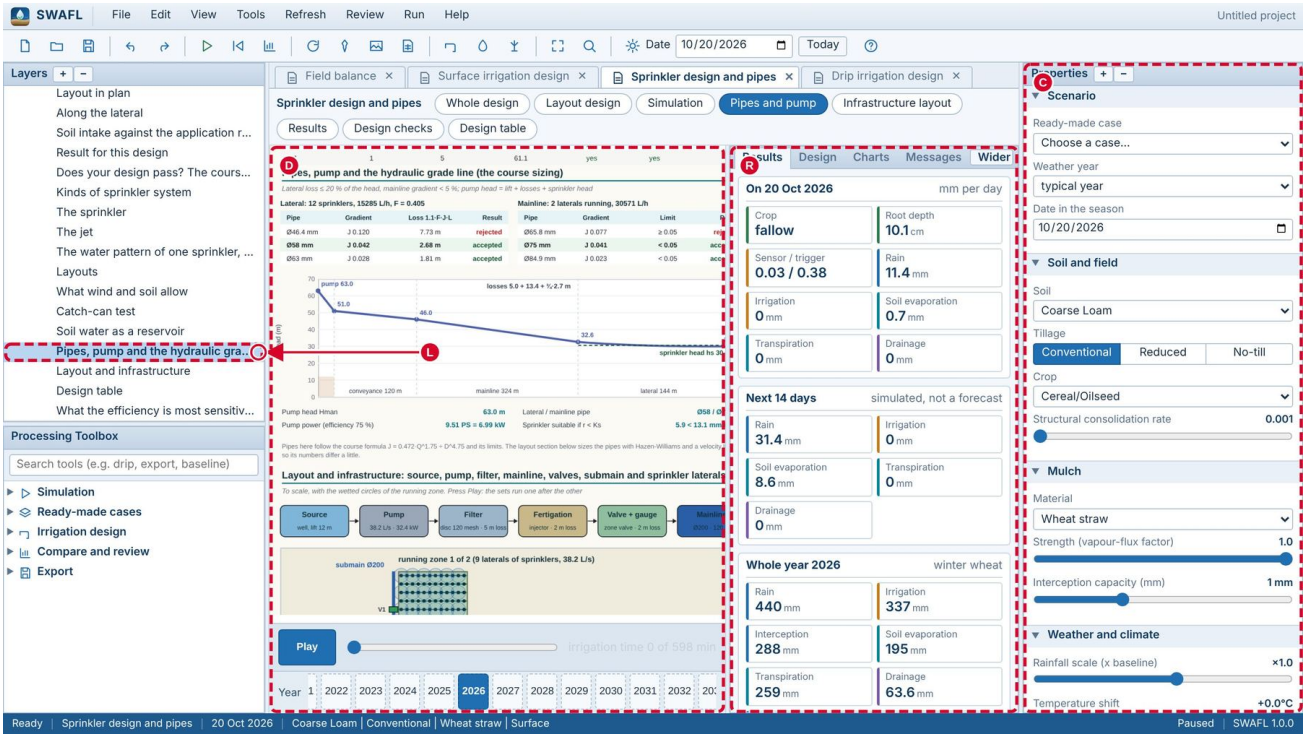


Figure 18.95: S4 Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing): the whole workbench.

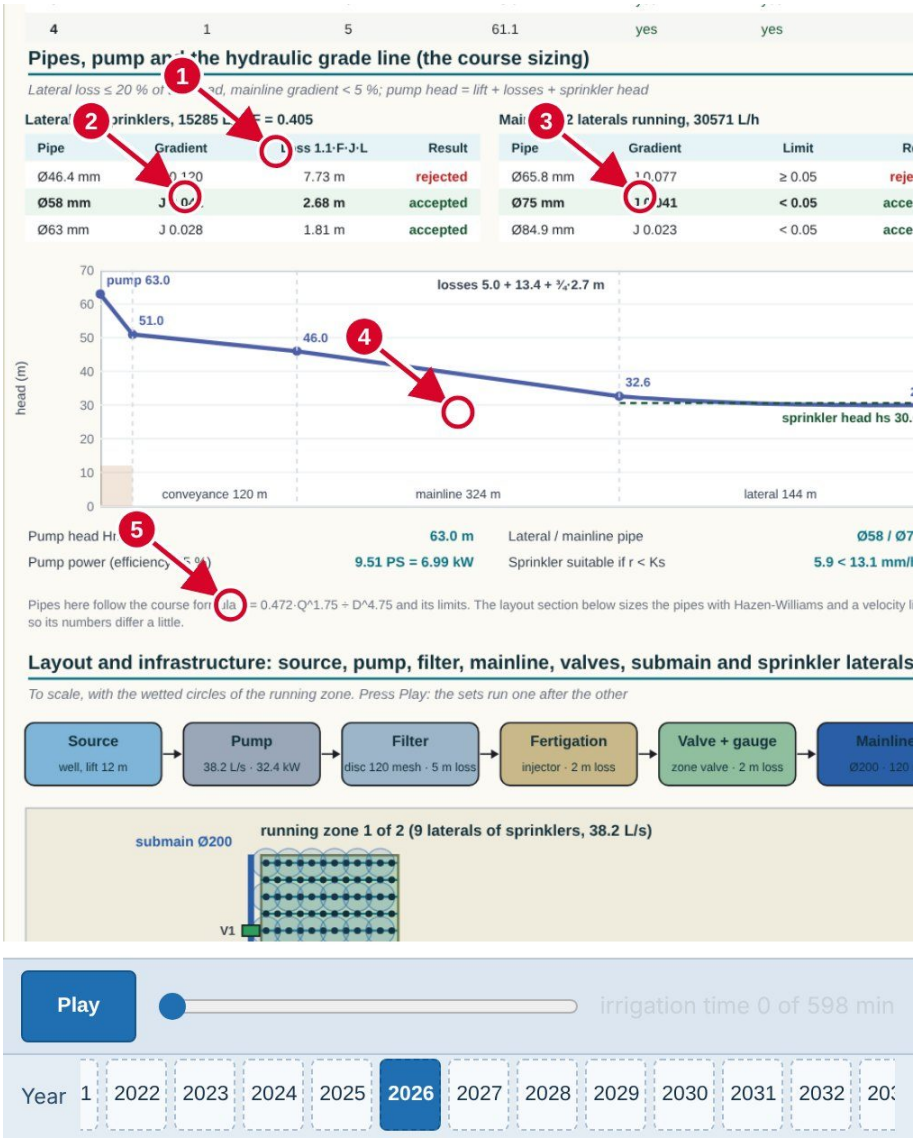


Figure 18.96: S4 Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Gradient J of the pipe (Blasius form): Eq. 13.17.
- **2** Lateral pipe: smallest standard diameter meeting the loss limit, Eq. 13.18.
- **3** Mainline: gradient below 0.05, Eq. 13.17.
- **4** Head along the flow path to the pump, H_man: Eq. 13.19.
- **5** Pump power: Eq. 13.20.

Equations behind this screen. Eq. 12.3, Eq. 12.21, Eq. 12.22, Eq. 12.23, Eq. 13.21, Eq. 13.22.

Eq. 12.3. Where explained: chapter “Drip irrigation design”, section “Lateral hydraulics: head loss and flow along the lateral”.

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.21. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\} \text{ mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.22. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".

$$h_{f,sm} = 0.4\, \frac{10.67\, L_{sm}\, Q_z^{1.852}}{C^{1.852} D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\, L_m\, Q_z^{1.852}}{C^{1.852} D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Eq. 12.23. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".

$$\mathrm{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\,\text{m}, \qquad P = \frac{\rho g\, Q_z\, \mathrm{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\mathrm{TDH}}{0.65} \text{ kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 13.21. Where explained: chapter "Sprinkler design", section "Layout and infrastructure".

$$N_L = \Big\lfloor \frac{10^4 A_f / L_L}{S_l} \Big\rfloor, \quad n_l = \min\Big(N_L,\ \Big\lfloor \frac{Q_s}{Q_{\text{lat}}} \Big\rfloor\Big), \quad n_{\text{set}} = \Big\lceil \frac{N_L}{n_l} \Big\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

Provenance (Eq. 13.21): **[Engineered]**: head and sets bookkeeping.

Eq. 13.22. Where explained: chapter "Sprinkler design", section "Layout and infrastructure".

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m\, h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

Provenance (Eq. 13.22): **[Literature]**: Christiansen reduction factor with Hazen–Williams exponent.

18.4.16 Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

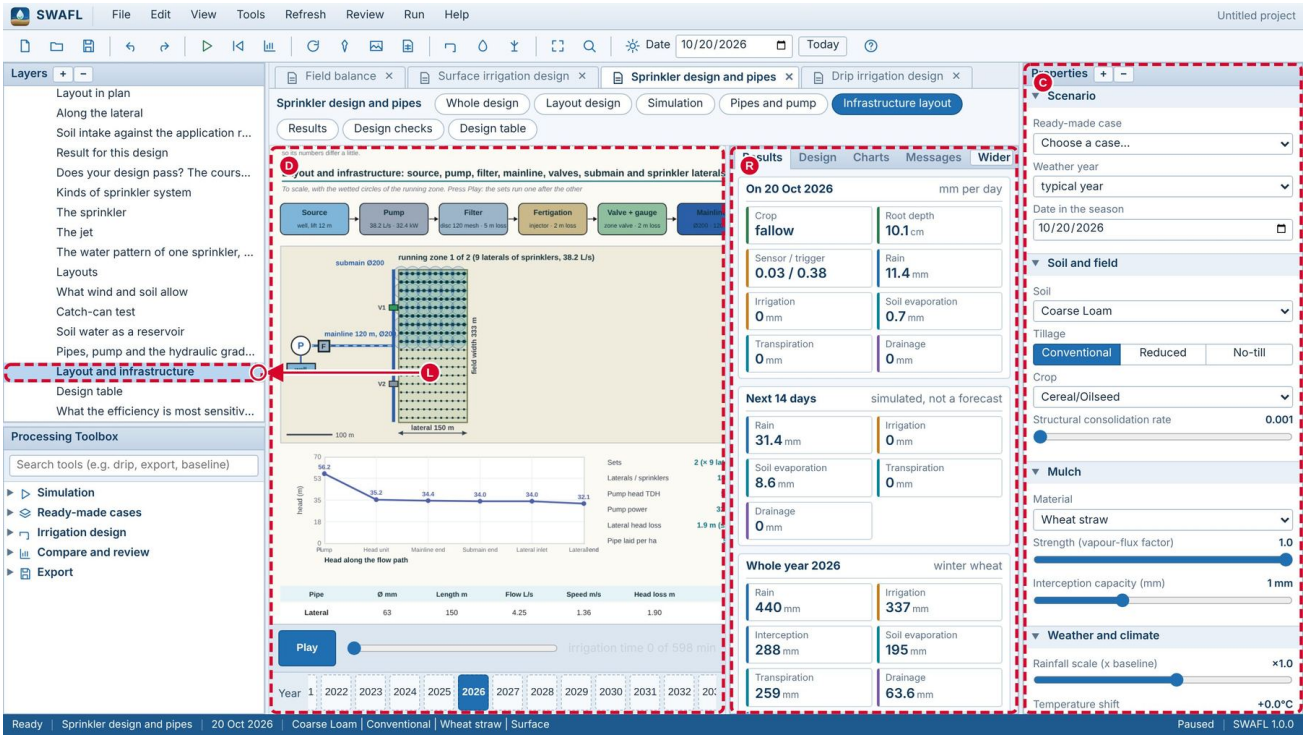


Figure 18.97: S4 Sprinkler design and pipes / Layout and infrastructure: the whole workbench.

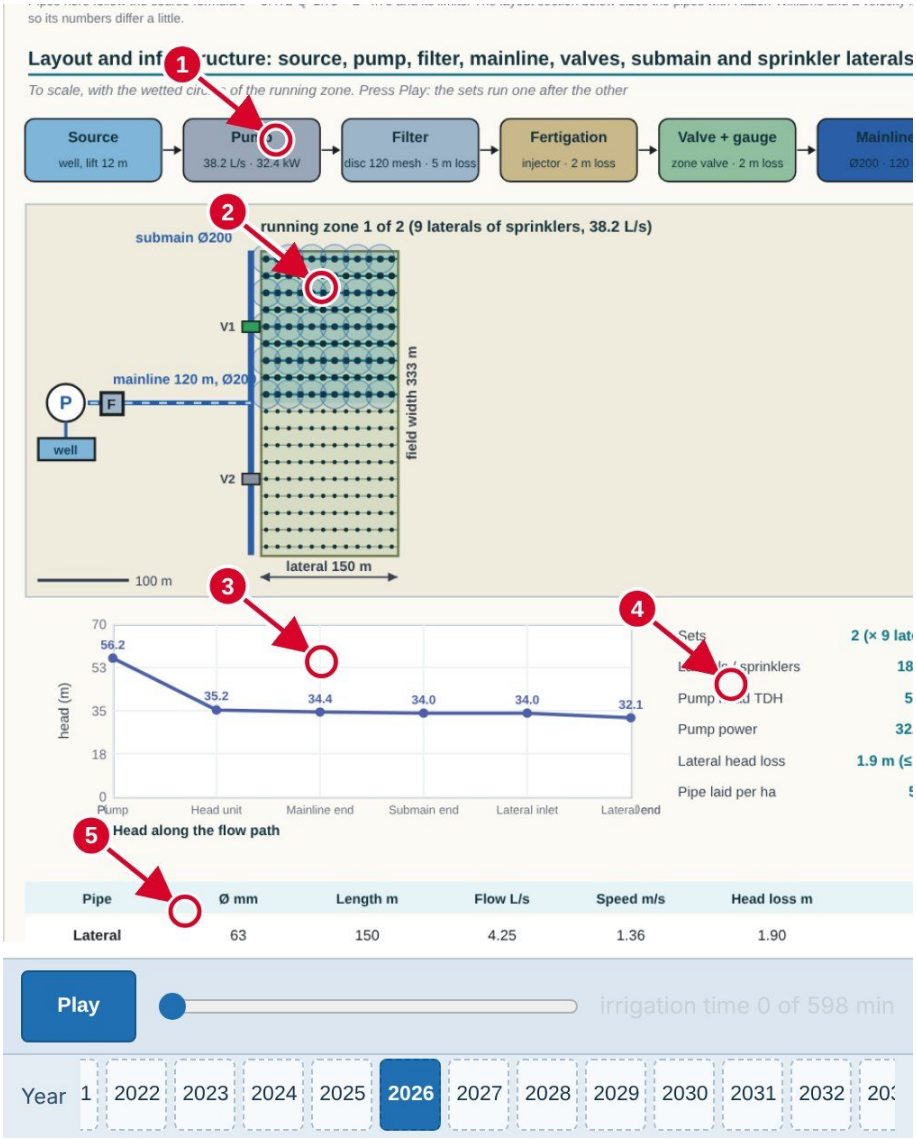


Figure 18.98: S4 Sprinkler design and pipes / Layout and infrastructure: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Source, pump, filter, fertigation and valves in order.
- **2** Laterals, N_L = field length / S_l, and laterals running together: Eq. 13.21.
- **3** Head along the flow path: Eqs. 13.19 and 13.22.
- **4** Lateral head loss with the Christiansen factor F_m: Eq. 13.22.
- **5** Pipe table: Eq. 13.18.

Equations behind this screen. Eq. 12.3, Eq. 12.21, Eq. 12.22, Eq. 12.23, Eq. 13.21, Eq. 13.22.

Eq. 12.3. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m.} \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.21. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \text{mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.22. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852} D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852} D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Eq. 12.23. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\ \text{m}, \qquad P = \frac{\rho g\, Q_z\, \text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 13.21. Where explained: chapter “Sprinkler design”, section “Layout and infrastructure”.

$$N_L = \Big\lfloor \frac{10^4 A_f / L_L}{S_l} \Big\rfloor, \quad n_l = \min\Big(N_L,\ \Big\lfloor \frac{Q_s}{Q_{\text{lat}}} \Big\rfloor\Big), \quad n_{\text{set}} = \Big\lceil \frac{N_L}{n_l} \Big\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

Provenance (Eq. 13.21): **[Engineered]**: head and sets bookkeeping.

Eq. 13.22. Where explained: chapter “Sprinkler design”, section “Layout and infrastructure”.

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m\, h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

Provenance (Eq. 13.22): **[Literature]**: Christiansen reduction factor with Hazen–Williams exponent.

18.4.17 Design table: iterations toward an application efficiency of 85 % or more

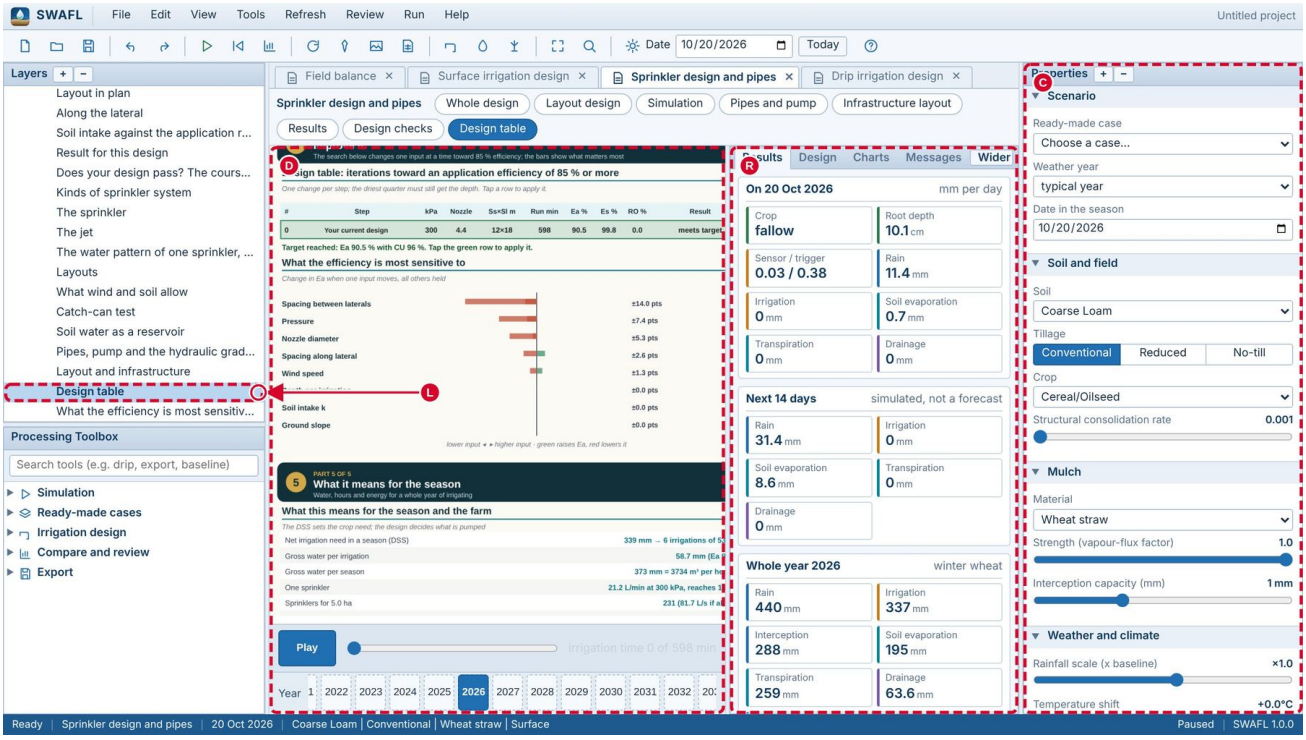


Figure 18.99: S4 Sprinkler design and pipes / Design table: the whole workbench.

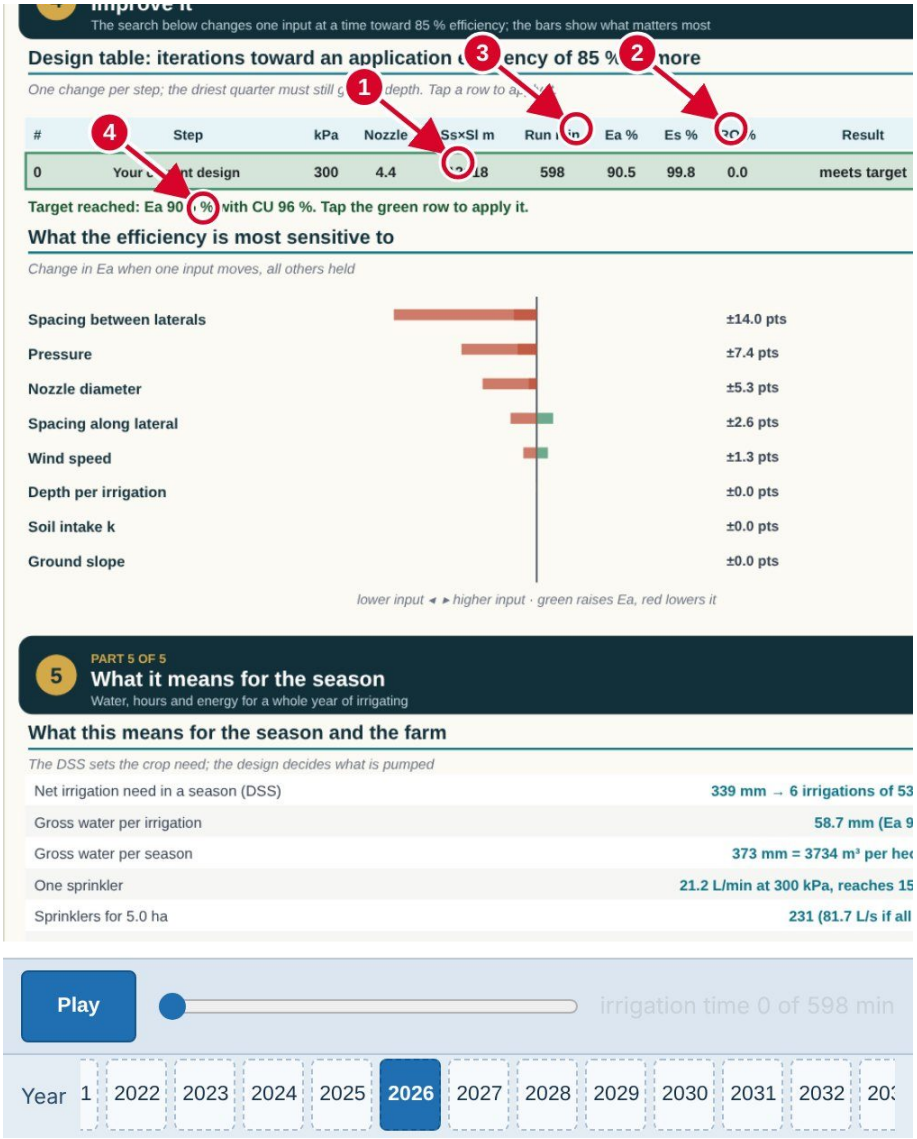


Figure 18.100: S4 Sprinkler design and pipes / Design table: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Row 0: your current design (pressure, nozzle, spacing, run time, E_a, E_s).
- **2** Column E_a: Eq. 13.11.
- **3** Run time T: Eq. 13.12.
- **4** Target reached: E_a of at least 85 % with CU of at least 85 %, Eqs. 13.7 and 13.11.

Equations behind this screen. Eq. 12.26.

Eq. 12.26. Where explained: chapter “Drip irrigation design”, section “The design table”.

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

18.4.18 What the efficiency is most sensitive to

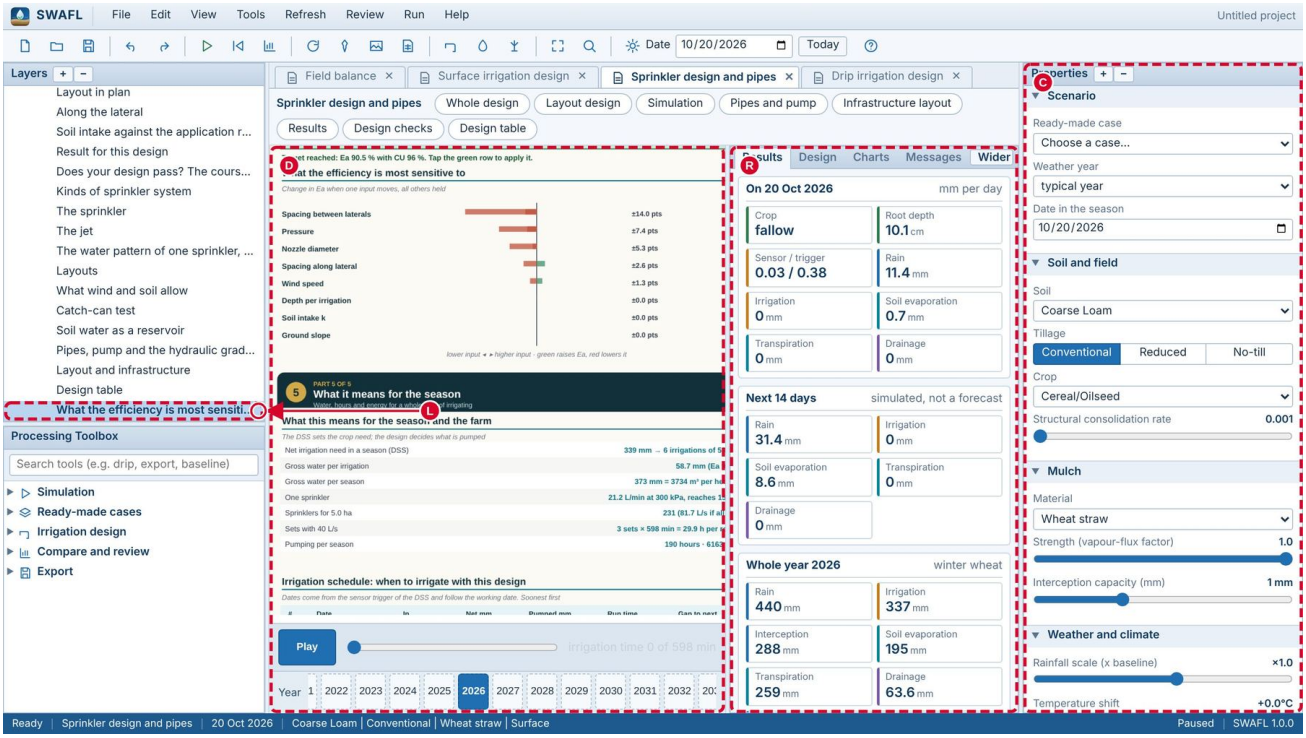


Figure 18.101: S4 Sprinkler design and pipes / What the efficiency is most sensitive to: the whole workbench.

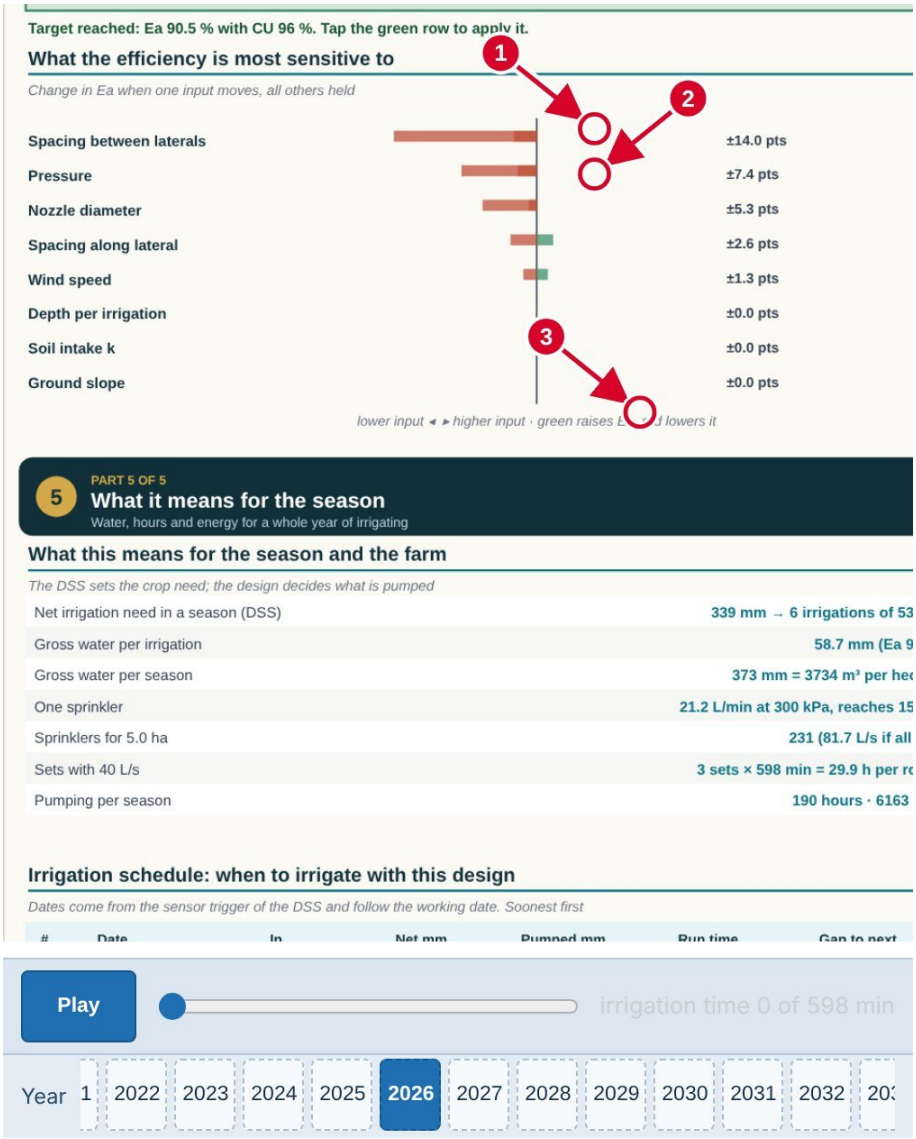


Figure 18.102: S4 Sprinkler design and pipes / What the efficiency is most sensitive to: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Tornado bar: change in E_a (Eq. 13.11) when the input moves.
- **2** Pressure: second most sensitive input.
- **3** Green raises E_a, red lowers it.

Equations behind this screen. Eq. 12.26.

Eq. 12.26. Where explained: chapter "Drip irrigation design", section "The design table".

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

18.4.19 What this means for the season and the farm

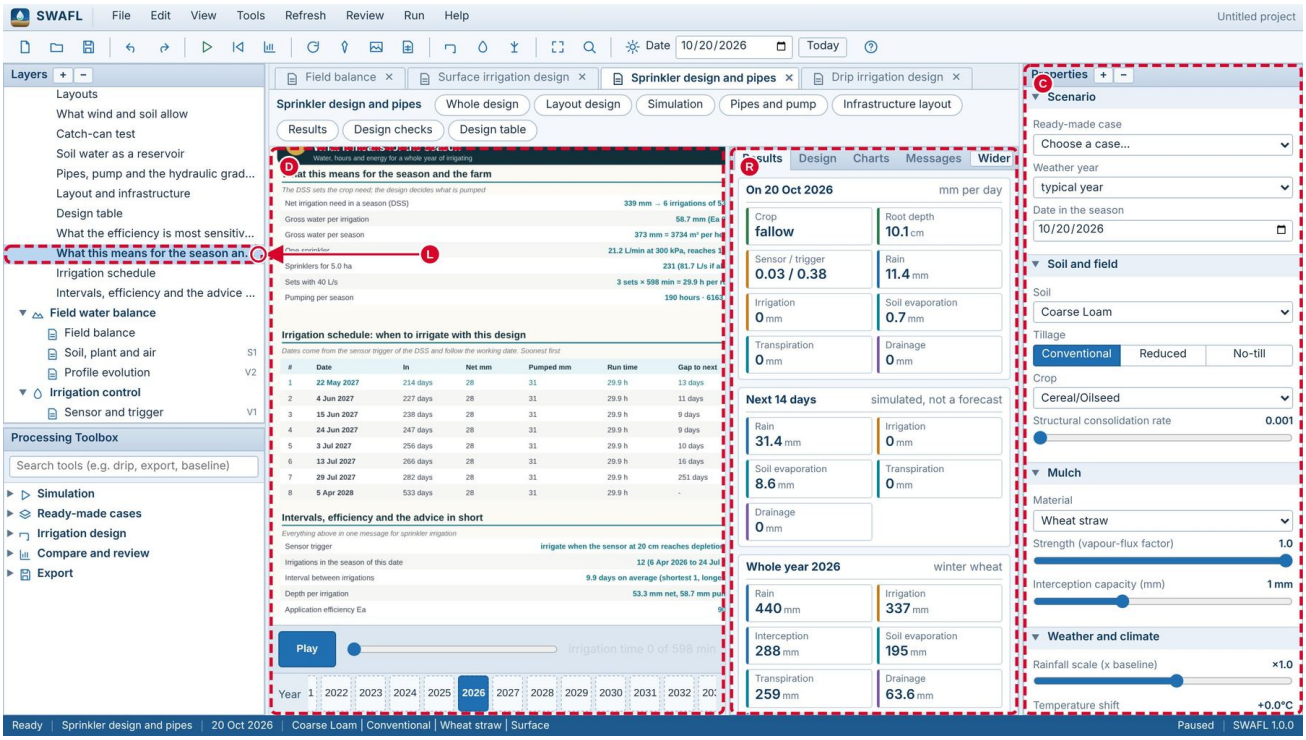


Figure 18.103: S4 Sprinkler design and pipes / What this means for the season and the farm: the whole workbench.

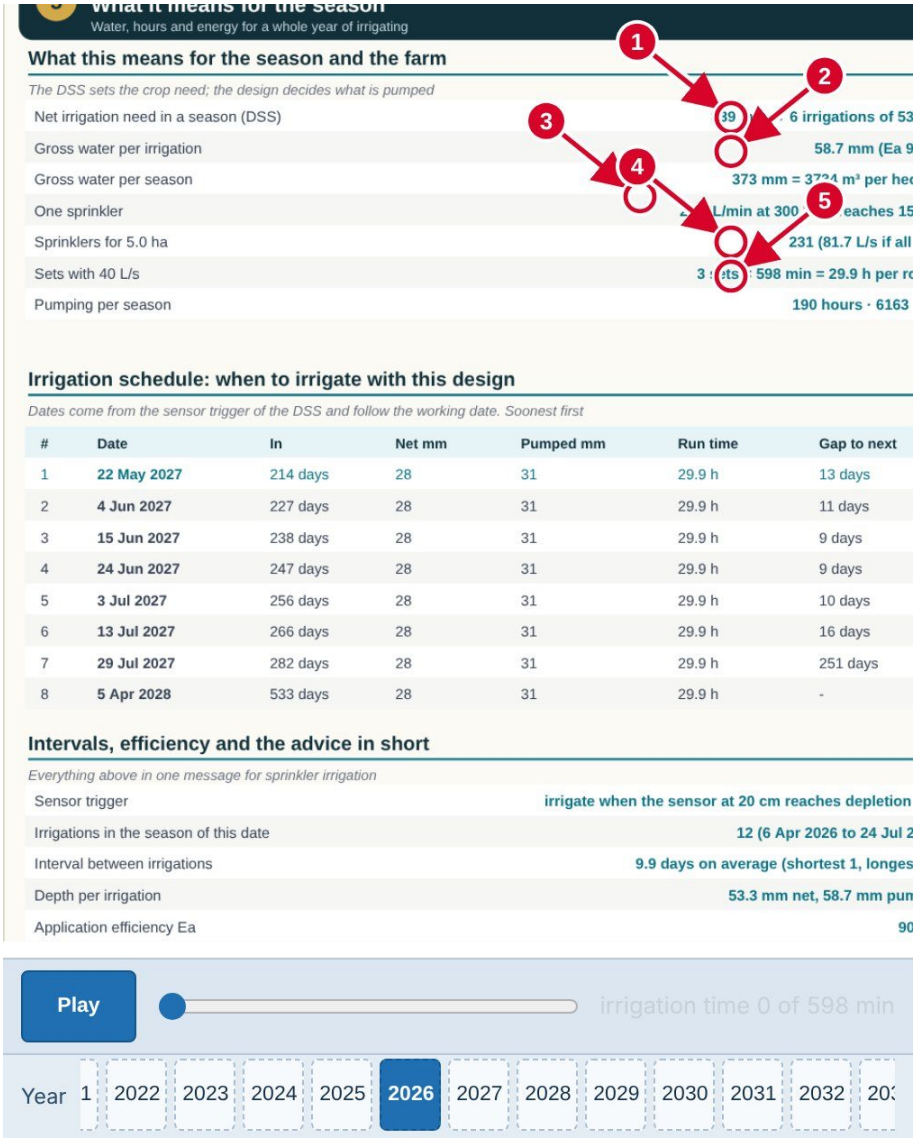


Figure 18.104: S4 Sprinkler design and pipes / What this means for the season and the farm: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Irrigations in a season: Eq. 13.23.
- **2** Gross depth per irrigation: Eq. 13.23.
- **3** One sprinkler discharge q: Eq. 13.2.
- **4** Sprinklers for the farm area, N_s: Eq. 13.21.
- **5** Sets and run time: Eqs. 13.12 and 13.21.

Equations behind this screen. Eq. 13.23.

Eq. 13.23. Where explained: chapter “Sprinkler design”, section “Season and farm”.

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60}\ \text{h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

18.4.20 Irrigation schedule: when to irrigate with this design

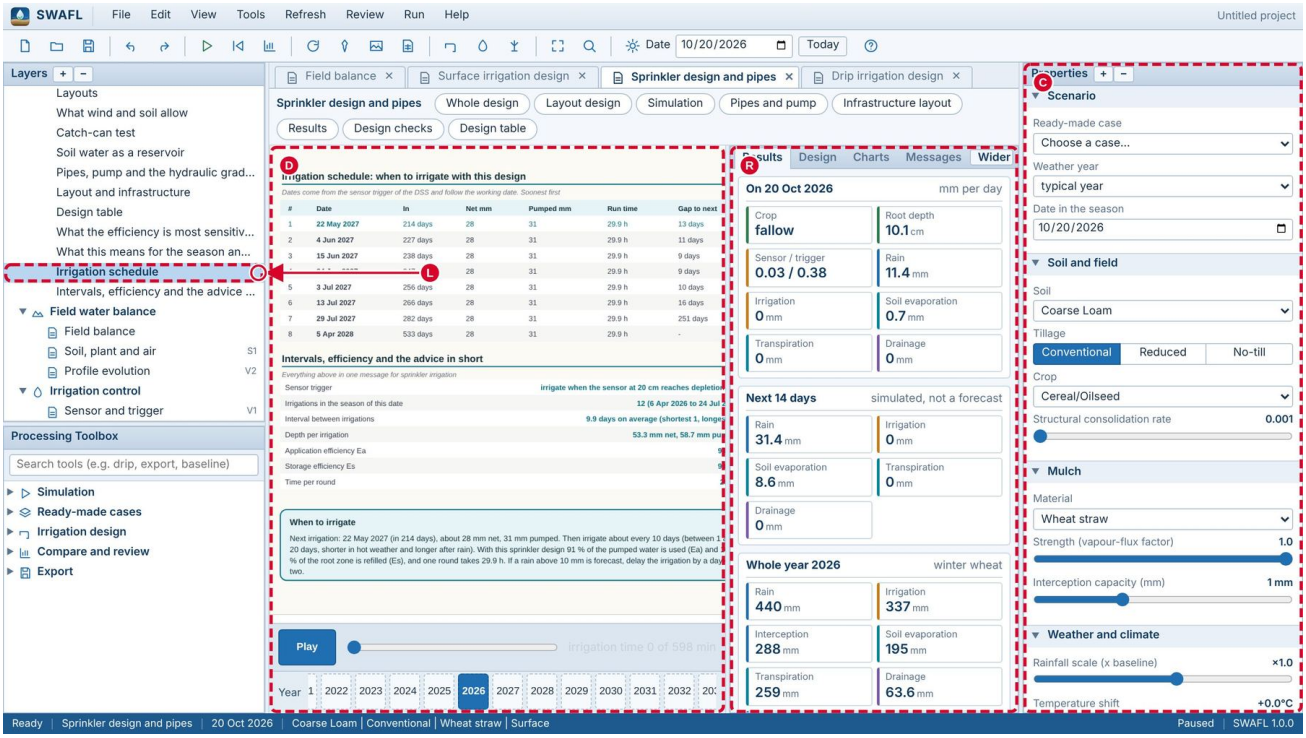


Figure 18.105: S4 Sprinkler design and pipes / Irrigation schedule: the whole workbench.

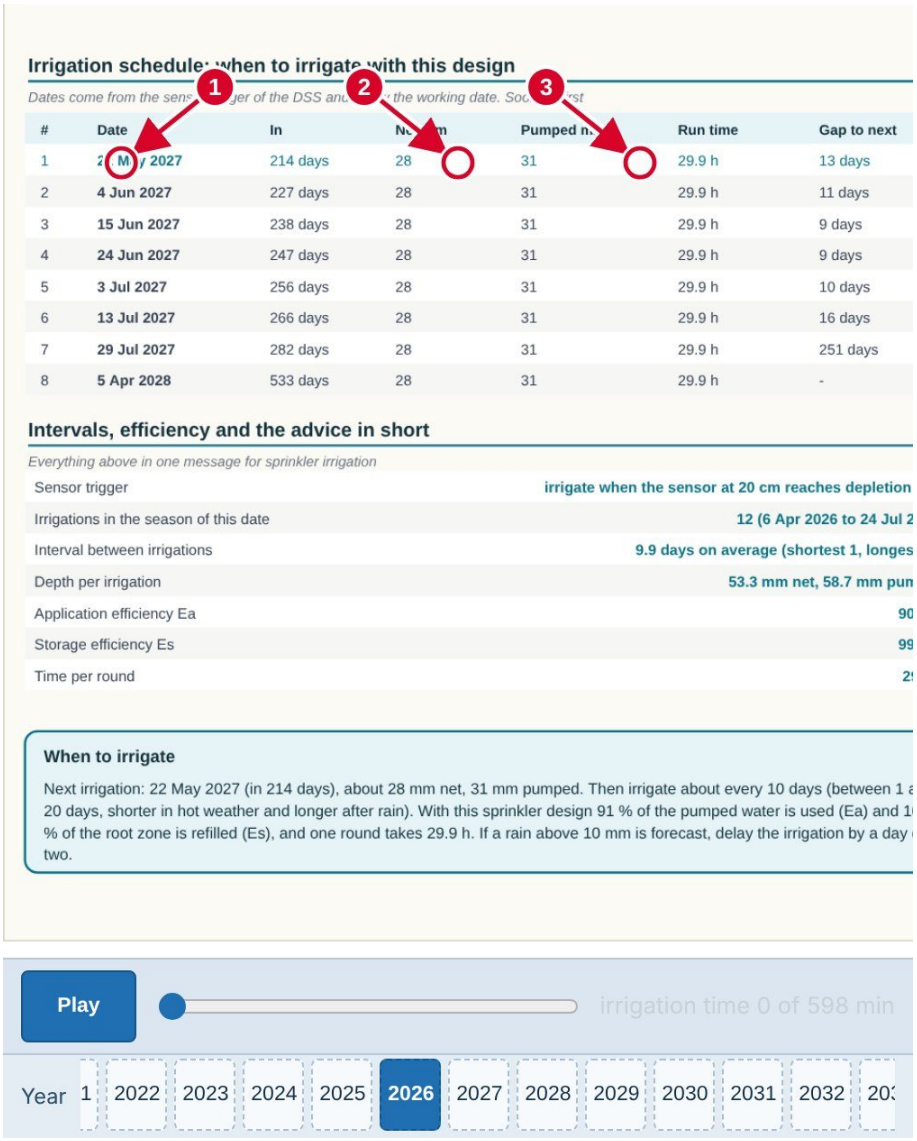


Figure 18.106: S4 Sprinkler design and pipes / Irrigation schedule: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Dates from the Engine 2 irrigation record.
- **2** Pumped mm = I_k / E_a: Eq. 13.23.
- **3** Run time of one round: Eq. 13.12.

Equations behind this screen. Eq. 13.23.

Eq. 13.23. Where explained: chapter “Sprinkler design”, section “Season and farm”.

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \operatorname{round}\!\left(\frac{10^4 A_f}{S_s S_l}\right), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \left\lceil \frac{Q_{\text{tot}}}{Q_s} \right\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60}\ \text{h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

18.4.21 Intervals, efficiency and the advice in short

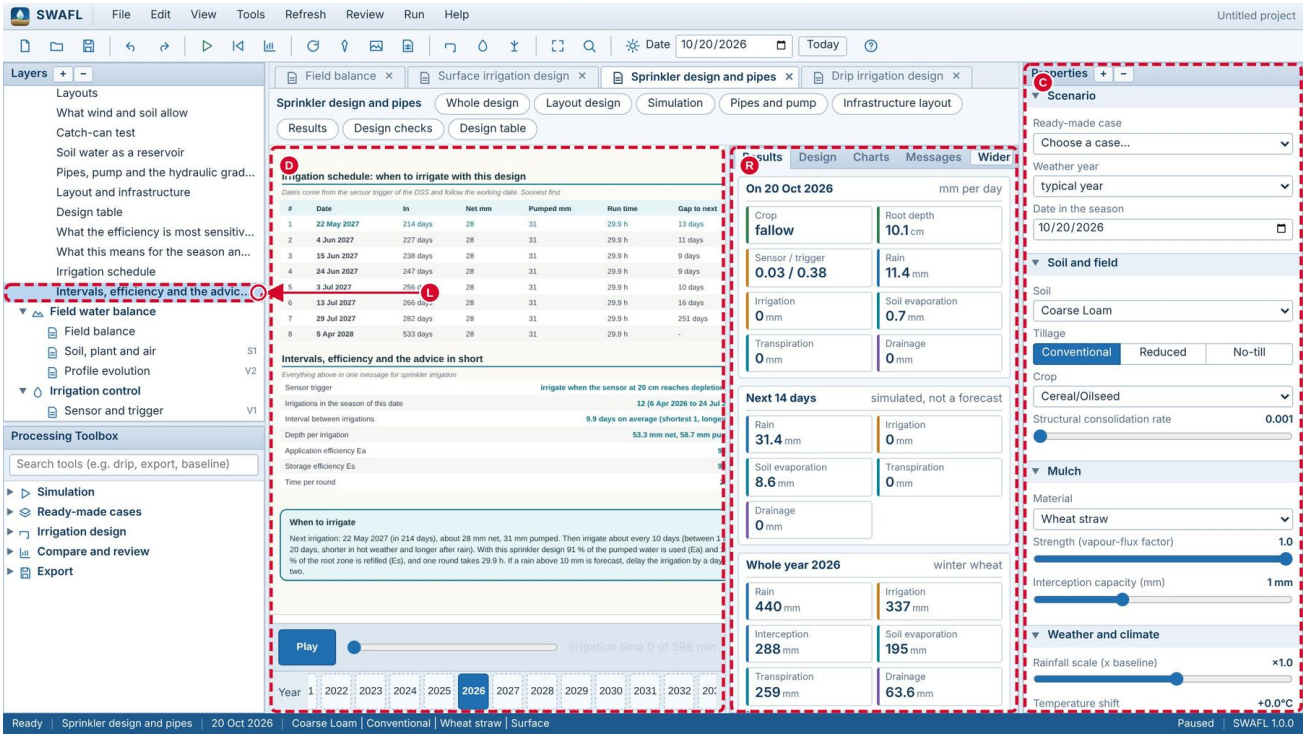


Figure 18.107: S4 Sprinkler design and pipes / Intervals, efficiency and the advice in short: the whole workbench.

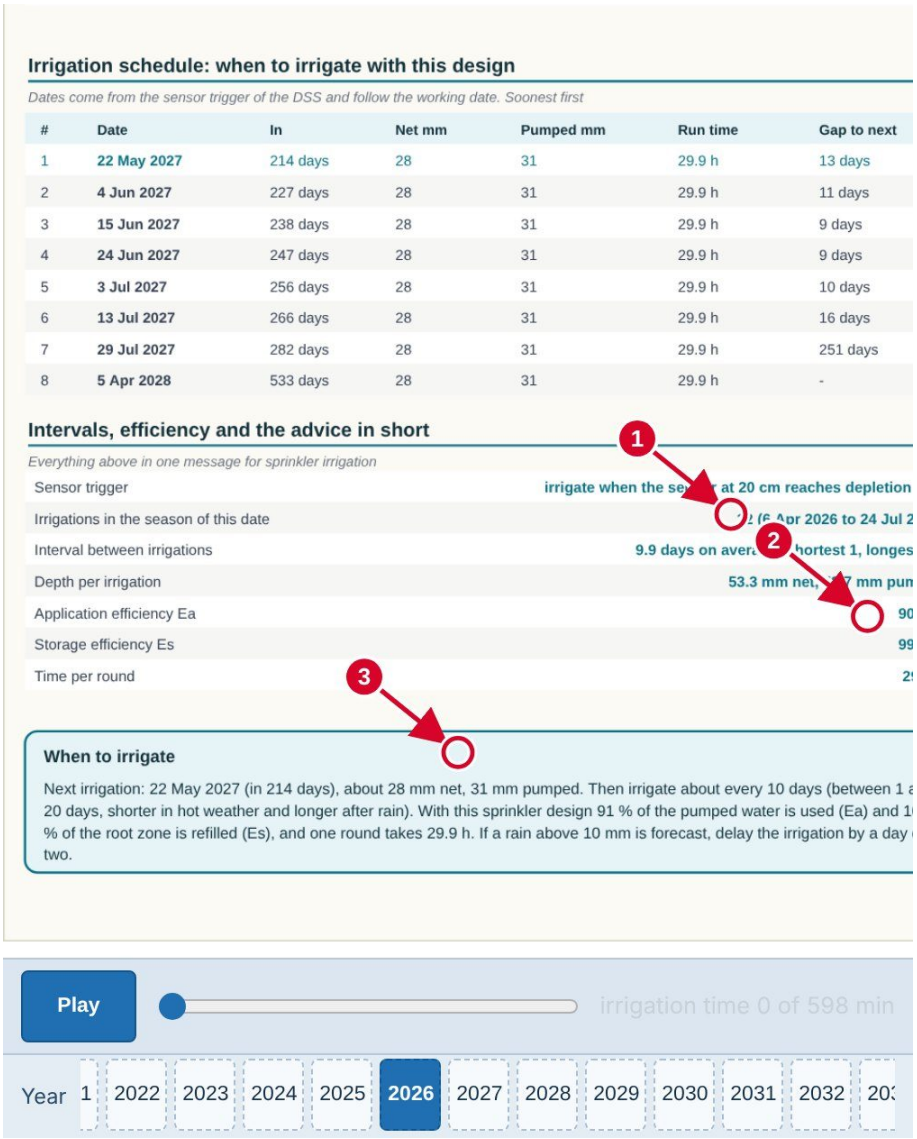


Figure 18.108: S4 Sprinkler design and pipes / Intervals, efficiency and the advice in short: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sensor trigger: Eqs. 5.3 and 5.27.
- **2** Application efficiency E_a: Eq. 13.11.
- **3** “When to irrigate”: message built from the schedule and from E_a and E_s.

Equations behind this screen. Eq. 13.23.

Eq. 13.23. Where explained: chapter “Sprinkler design”, section “Season and farm”.

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}} T}{60}\ \text{h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

18.5 Field balance

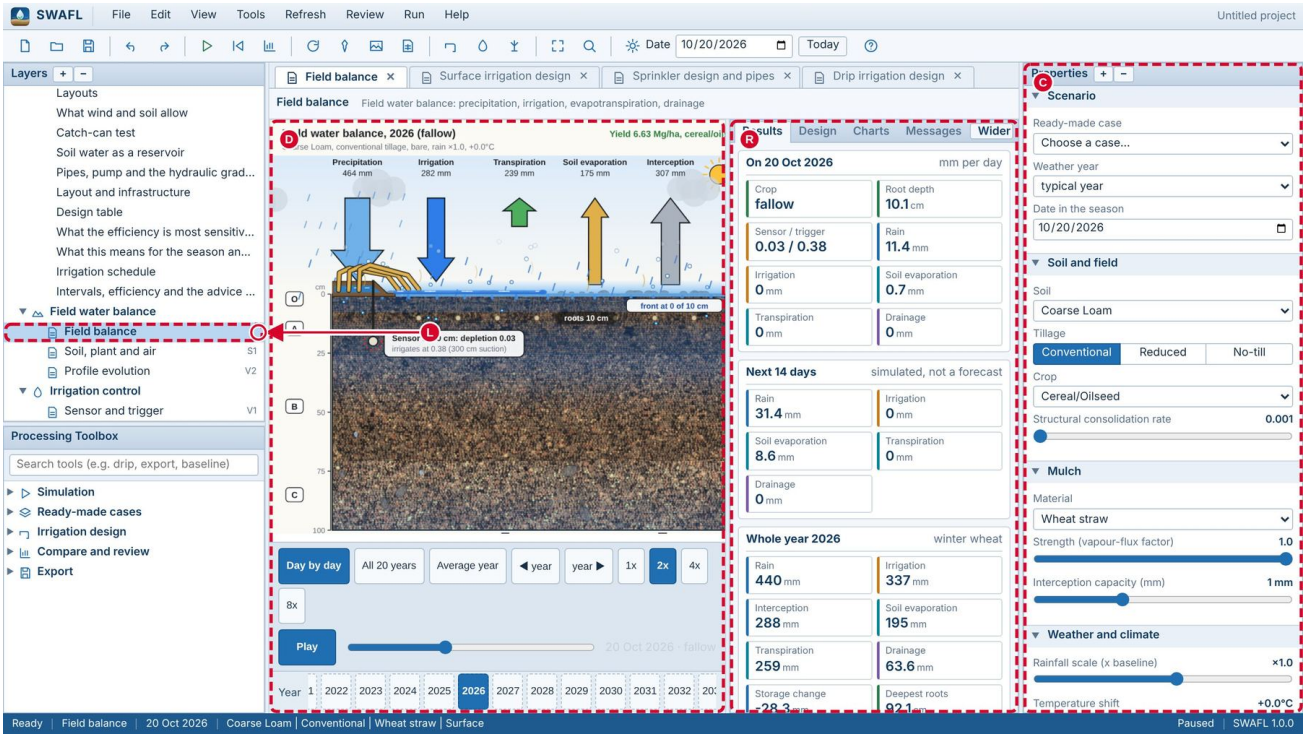


Figure 18.109: Field balance: the whole workbench.

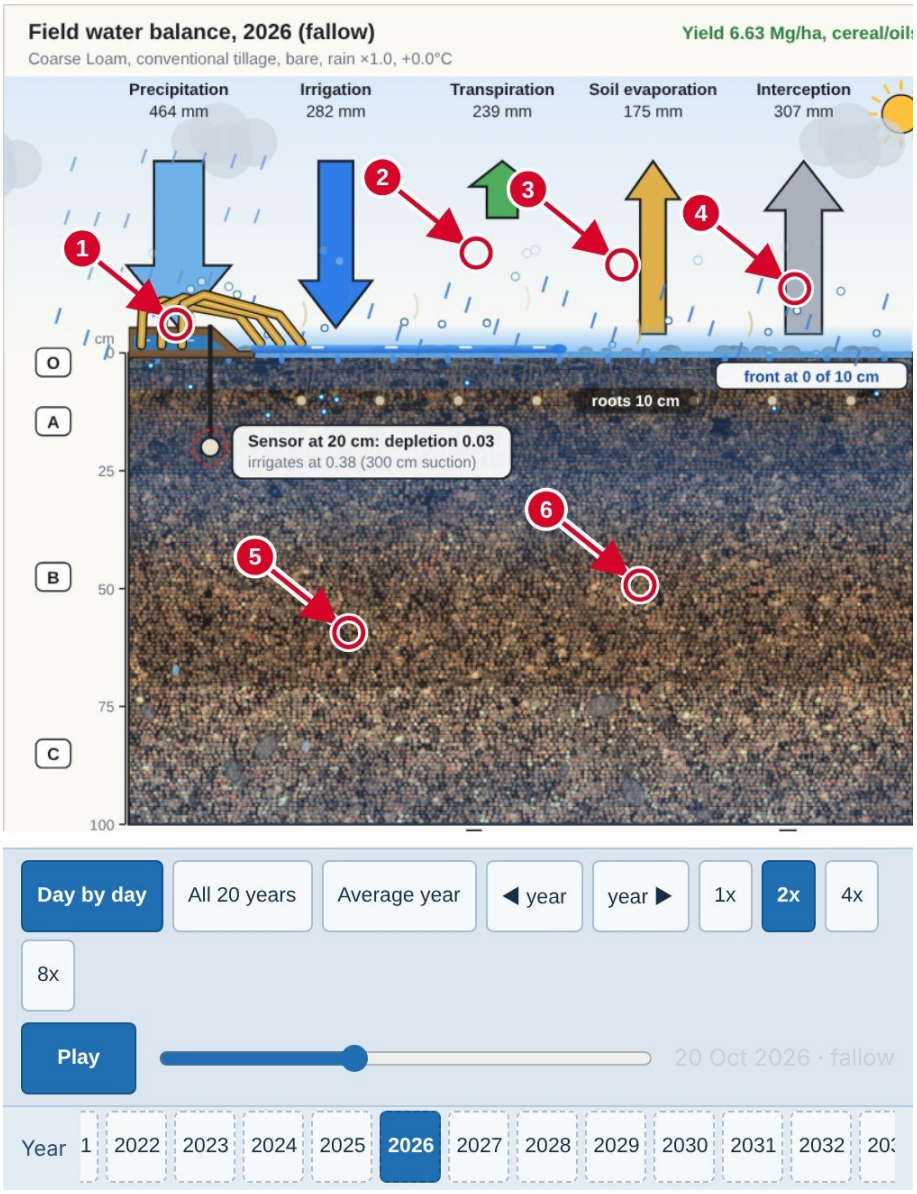


Figure 18.110: Field balance: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Precipitation arrow, P: annual value from Eqs. 5.4 to 5.12 (Average year); daily value from the rainfall generator, Eqs. 5.15 and 5.16 (Day by day).
- **2** Transpiration arrow, T: Eqs. 5.4 to 5.12 (annual) and Eqs. 5.21 to 5.25 (daily).
- **3** Soil evaporation arrow, E: Eqs. 5.4 to 5.12 (annual) and Eqs. 5.21 to 5.26 (daily).
- **4** Interception arrow, E_other: the closure term of Eq. 5.12.
- **5** Sensor tag: depletion f of the sensor layer against the trigger f_T, Eq. 5.3 (Engine 2 layer j_s).
- **6** Roots (depth and density): Eqs. 5.19 and 5.20.

Equations behind this screen. Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 5.13, Eq. 5.14, Eq. 5.15, Eq. 5.16, Eq. 5.17, Eq. 5.18, Eq. 5.19, Eq. 5.20, Eq. 5.21, Eq. 5.22, Eq. 5.23, Eq. 5.24, Eq. 5.25, Eq. 8.1.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".

$$Q \leftarrow Q\,\text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 5.13. Where explained: chapter "The two computing engines", section "Crop calendar and canopy cover".

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp(-(d - d_{\text{mid}})/s)}, \quad s = 13\ (\text{vegetable } 8), \tag{5.13}$$

Provenance (Eq. 5.13): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

Eq. 5.14. Where explained: chapter "The two computing engines", section "Crop calendar and canopy cover".

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12\ \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp(-(d - d_{\text{pk}} + 45)/12)} & \text{otherwise} \end{cases} \tag{5.14}$$

Provenance (Eq. 5.14): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

Eq. 5.15. Where explained: chapter "The two computing engines", section "Rainfall generator".

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0, 1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.16. Where explained: chapter "The two computing engines", section "Rainfall generator".

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}. \tag{5.16}$$

Provenance (Eq. 5.16): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.17. Where explained: chapter "The two computing engines", section "Infiltration, redistribution and drainage".

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

Eq. 5.18. Where explained: chapter "The two computing engines", section "Roots".

$$R_{\max} = \operatorname{clip}\Big(R_c\,(0.85{+}0.3\,\text{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \text{cm}, \tag{5.18}$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.19. Where explained: chapter "The two computing engines", section "Roots".

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\operatorname{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35 + \bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.20. Where explained: chapter "The two computing engines", section "Roots".

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d + 8}\Big)(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.21. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65\cdot\tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.22. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\ \text{yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.23. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".

$$q_i^{T} = T_d \,\frac{\rho_i \, a_i}{\sum_j \rho_j a_j}, \qquad a_i = \mathrm{clip}\Big(\frac{s_i - 0.15}{0.6};\, 0,\, 1\Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.24. Where explained: chapter "The two computing engines", section "Scaling to the annual model".

$$S_P = \frac{P}{\overline{P}_{\mathrm{sim}}}, \quad S_{\mathrm{thr}} = \frac{P - P_{\mathrm{other}}}{\overline{T}_{f,\mathrm{sim}}}, \quad S_E = \frac{E}{\overline{E}_{\mathrm{sim}}}, \quad S_T = \frac{T}{\overline{T}_{\mathrm{sim}}}, \quad S_D = \frac{D}{\overline{D}_{\mathrm{sim}}}, \tag{5.24}$$

Provenance (Eq. 5.24): **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

Eq. 5.25. Where explained: chapter "The two computing engines", section "Scaling to the annual model".

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

Eq. 8.1. Where explained: chapter "Soil structure and tillage layers", section "Pore index and consolidation".

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

18.5.1 The whole layer, part by part

The picture below is the complete drawing of the layer Field balance, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 1 part, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

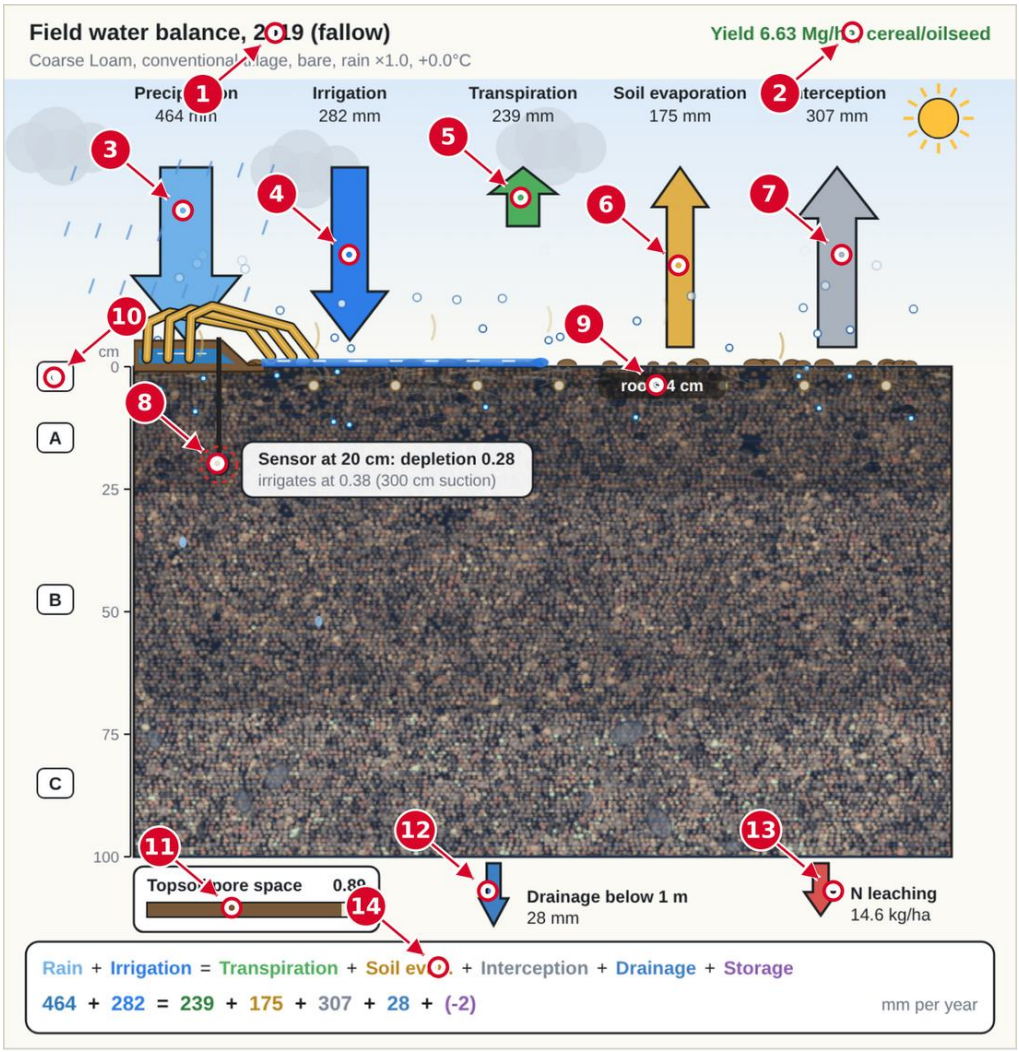


Figure 18.111: Field balance, part 1 of 1.

What the arrows point to.

- **1** The title gives the year, the crop state and the scenario in force (soil, tillage, mulch, rain scaling and warming).
- **2** The yield of the year, with the crop that produced it.
- **3** Precipitation, the wet input of the budget, in mm for the year.
- **4** Irrigation, the applied water decided by the sensor trigger, in mm for the year.
- **5** Transpiration, the water taken up through the roots and released by the leaves.
- **6** Soil evaporation, the water lost directly from the soil surface.
- **7** Interception, the rain held on leaves and mulch and evaporated before reaching the soil.
- **8** The soil sensor at 20 cm, with its depletion and the depletion at which irrigation starts.
- **9** The depth of the roots at this moment of the season.
- **10** The horizons O, A, B and C of the soil profile, with the depth scale in cm.
- **11** The topsoil pore space, the fraction of the topsoil volume that is air and water.
- **12** Drainage below 1 m, the water that leaves the root zone downward.
- **13** Nitrogen leaching carried out of the root zone with the drainage.
- **14** The closed budget strip, in which the inputs equal the outputs plus the change in storage.

18.5.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

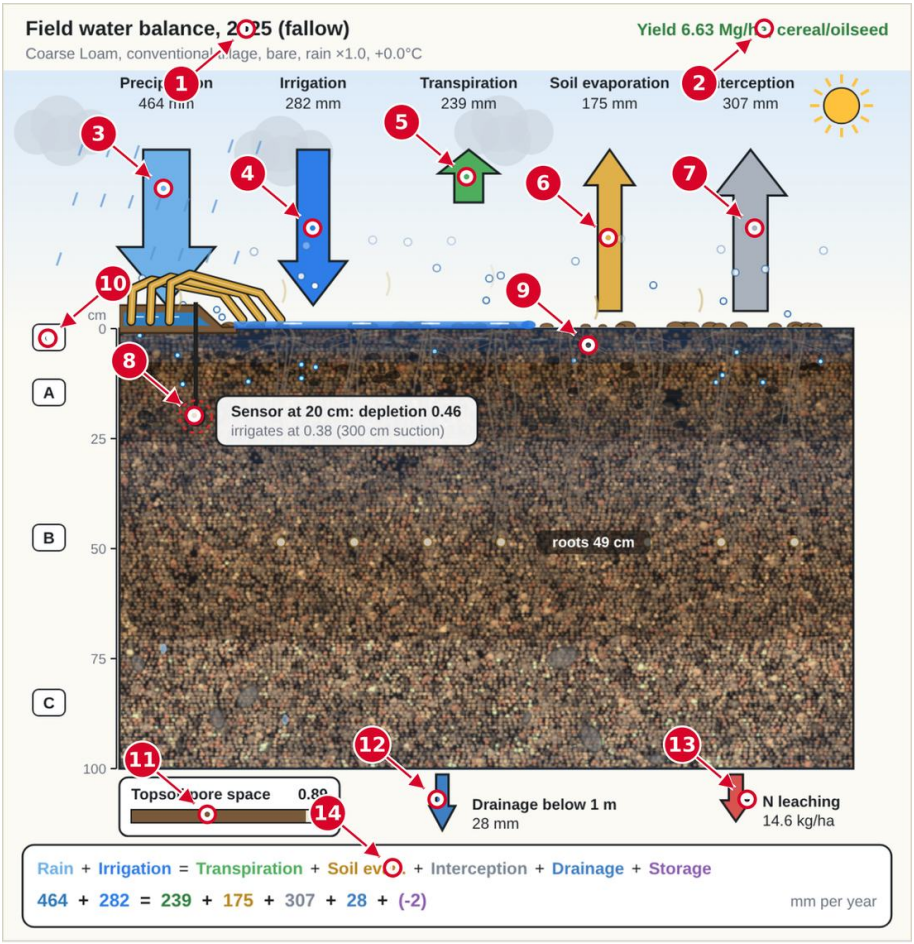


Figure 18.112: Field balance, part 1, playback a third of the way (7 Aug 2025 · fallow).

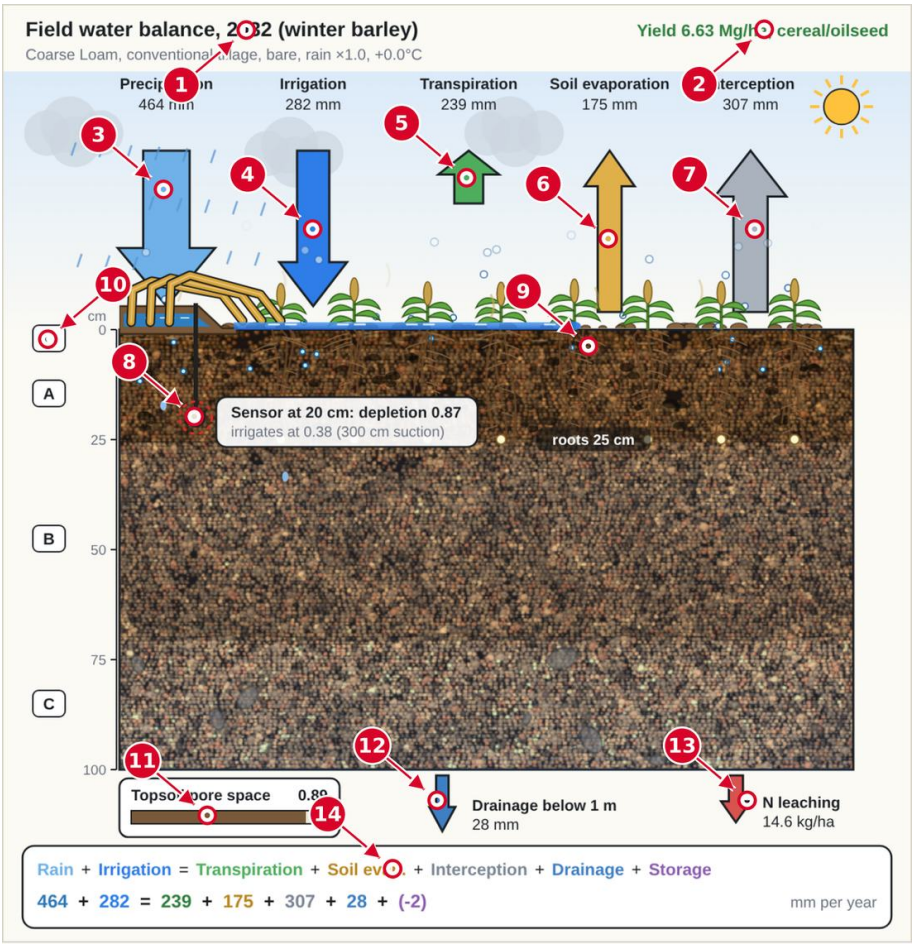


Figure 18.113: Field balance, part 1, playback two thirds of the way (13 Mar 2032 · winter barley).

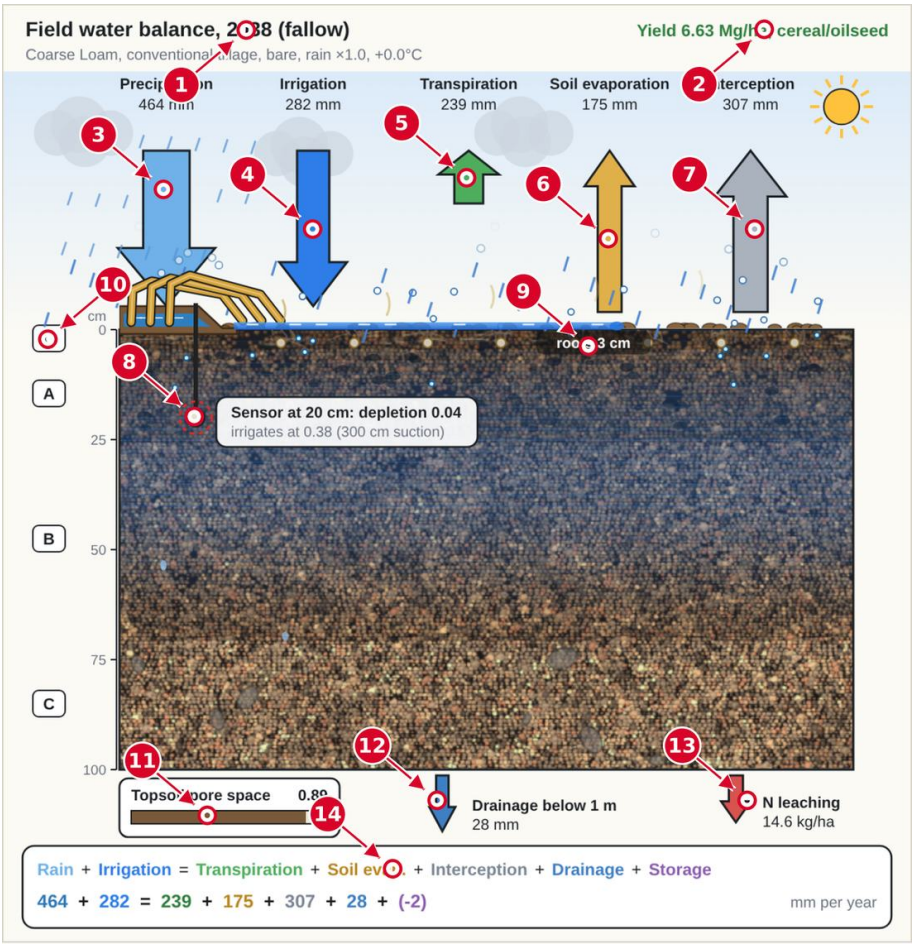


Figure 18.114: Field balance, part 1, playback the end (31 Dec 2038 · fallow).

18.5.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

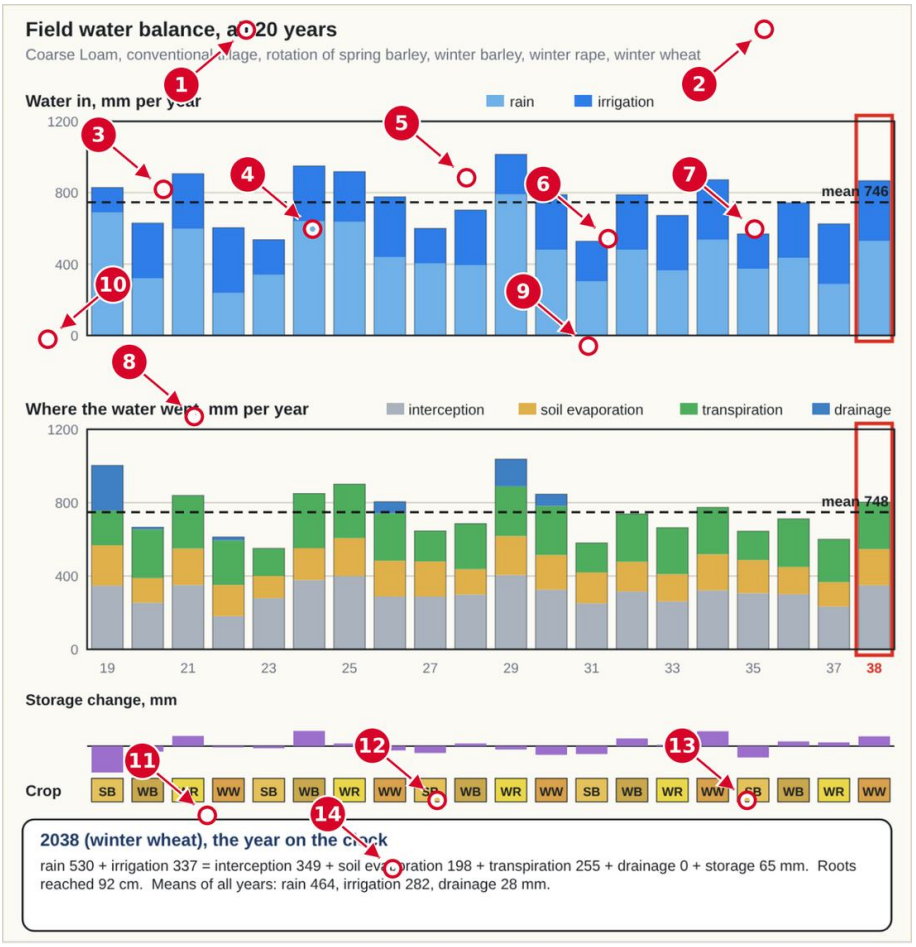


Figure 18.115: Field balance, part 1. The All 20 years button pressed, so the arrows and the budget show the whole twenty-year record in one picture.

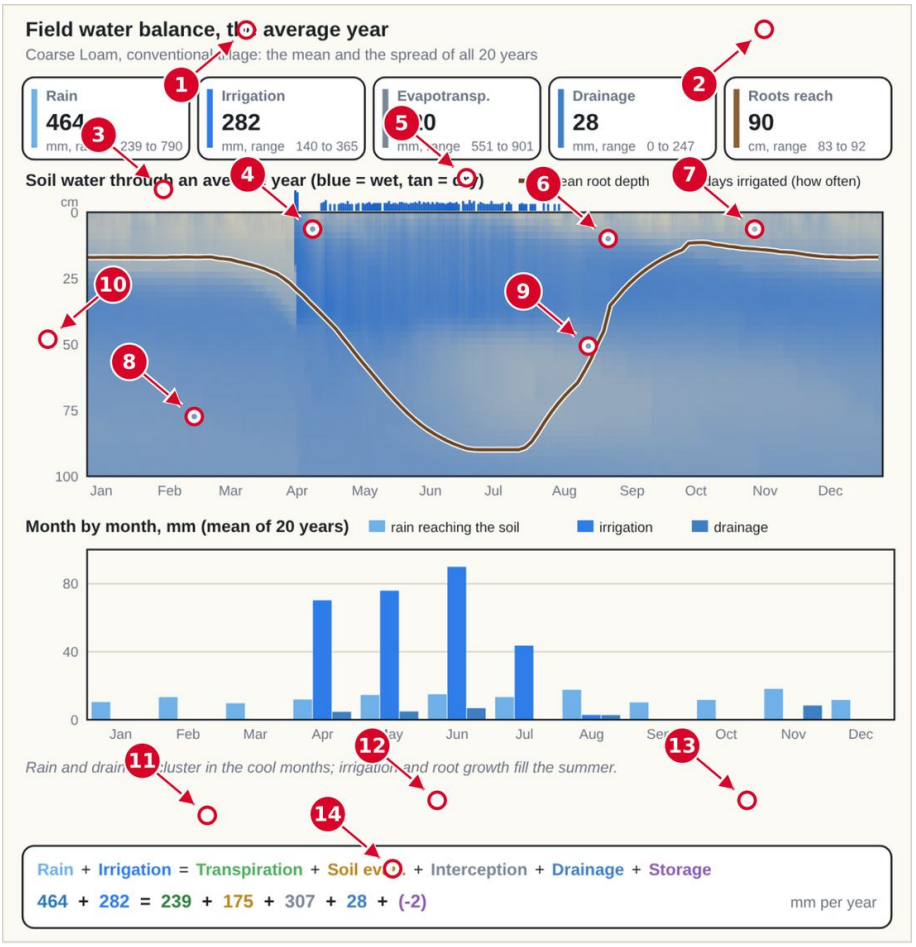


Figure 18.116: Field balance, part 1. The Average year button pressed, so every flux is the mean of the twenty simulated years.

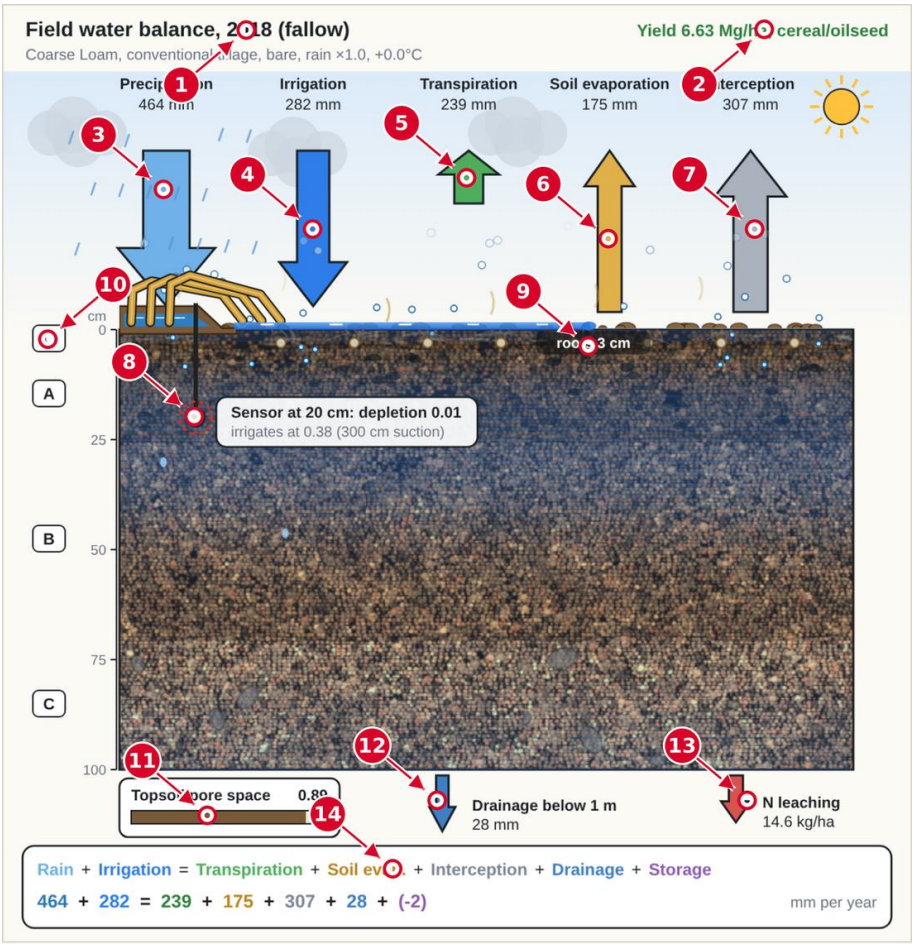


Figure 18.117: Field balance, part 1. The year button 2030 pressed, so the picture shows the budget of that single year.

18.6 Soil, plant and air (S1)

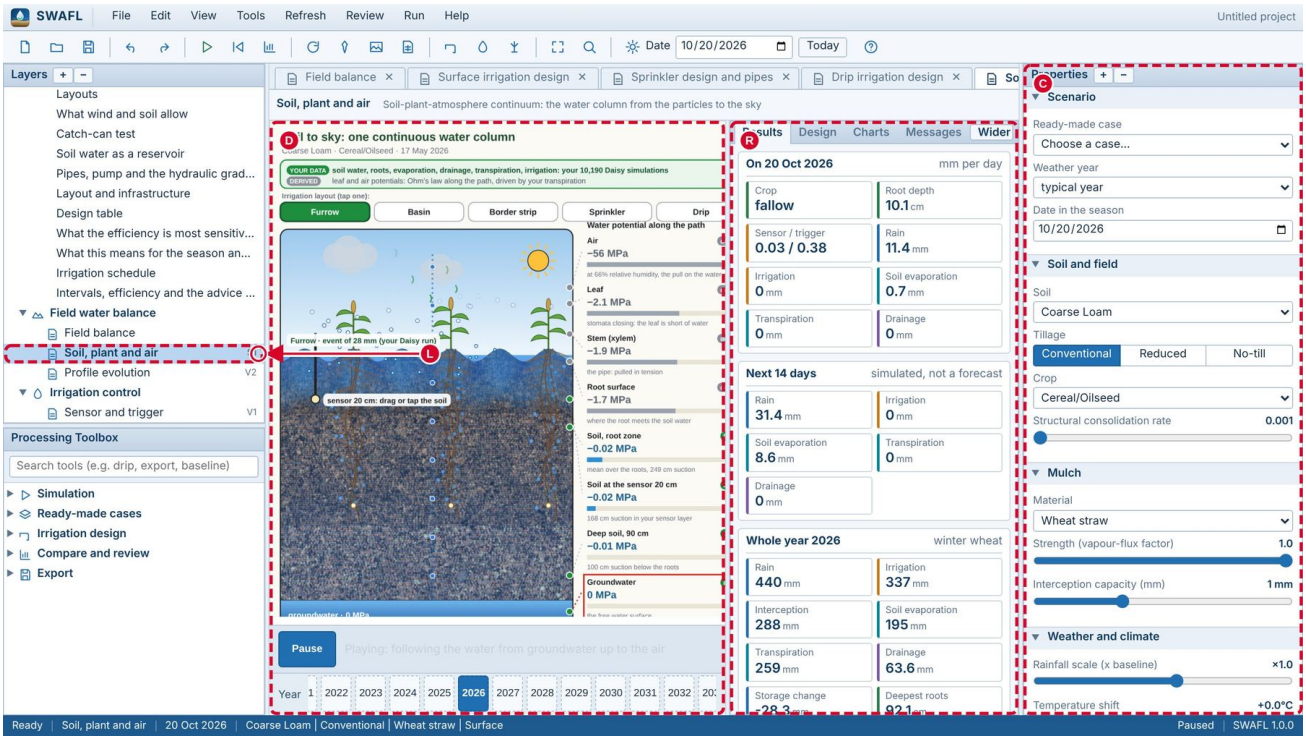


Figure 18.118: Soil, plant and air (S1): the whole workbench.

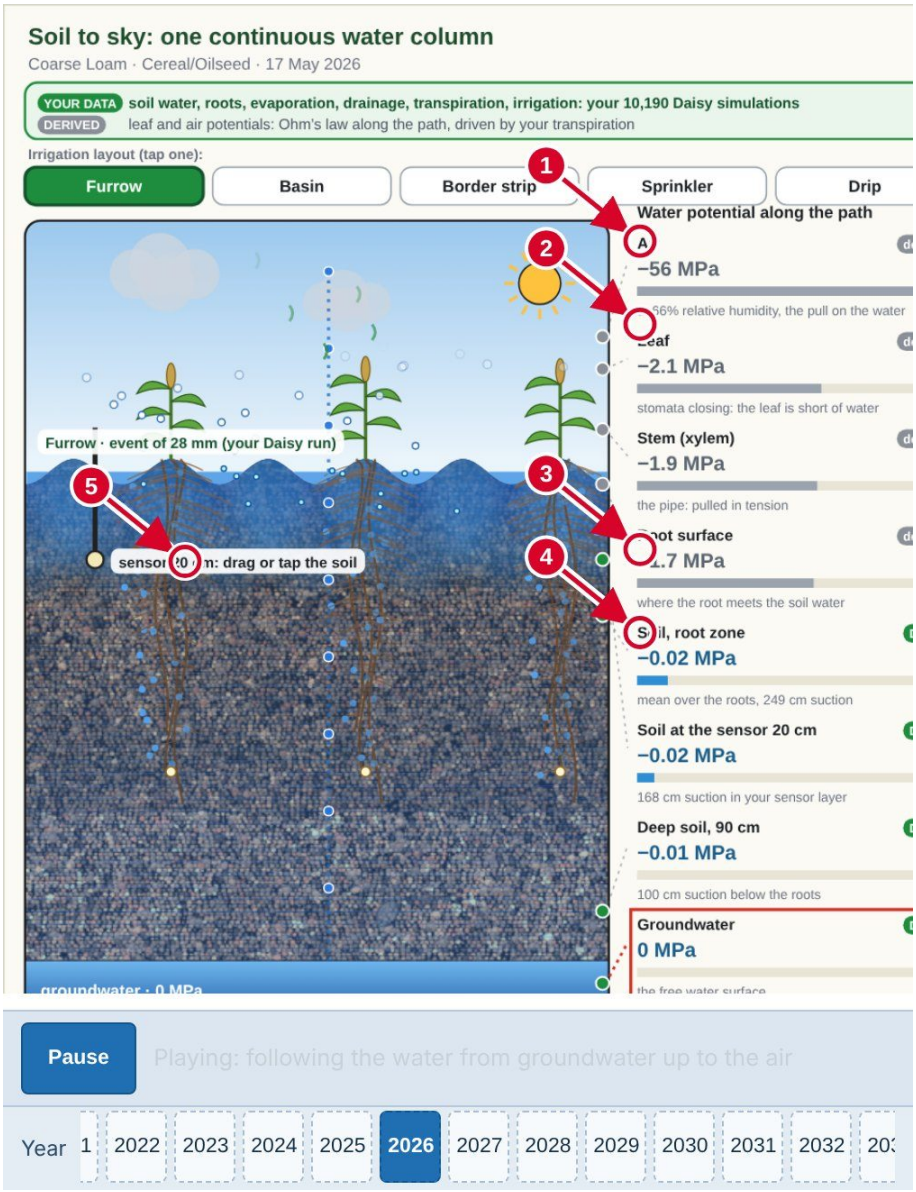


Figure 18.119: Soil, plant and air (S1): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Water potential of the air, psi_air = 135.5 ln RH: Eq. 7.5.
- **2** Leaf potential, psi_leaf: Eq. 7.4.
- **3** Stem and root-surface potentials, psi_stem and psi_root: Eqs. 7.2 and 7.3.
- **4** Root-zone soil potential from the layer suctions: Eq. 7.1 (inverse of Eq. 5.3), with Eqs. 5.1 and 5.2 for the conductivity.
- **5** Sensor at 20 cm: the suction of layer j_s, Eq. 7.1.

Equations behind this screen. Eq. 5.1, Eq. 5.2, Eq. 5.3, Eq. 5.25, Eq. 7.1, Eq. 7.2, Eq. 7.3, Eq. 7.4, Eq. 7.5.

Eq. 5.1. Where explained: chapter "The two computing engines", section "The simulation campaign".

$$S_e(h) = \Big[1 + (\alpha|h|)^{n}\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 5.2. Where explained: chapter "The two computing engines", section "The simulation campaign".

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^{m}\Big]^{2}. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Eq. 5.3. Where explained: chapter "The two computing engines", section "Plant-available water and depletion".

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.25. Where explained: chapter "The two computing engines", section "Scaling to the annual model".

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

Eq. 7.1. Where explained: chapter "Field water balance layers", section "Governing equations".

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \tag{7.1}$$

Provenance (Eq. 7.1): **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

Eq. 7.2. Where explained: chapter "Field water balance layers", section "Governing equations".

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.3. Where explained: chapter "Field water balance layers", section "Governing equations".

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.4. Where explained: chapter "Field water balance layers", section "Governing equations".

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.5. Where explained: chapter "Field water balance layers", section "Governing equations".

$$\psi_{\text{air}} = \frac{RT}{V_w} \ln \text{RH} \;\approx\; 135.5 \ln \text{RH} \;\; \text{MPa} \tag{7.5}$$

Provenance (Eq. 7.5): **[Literature]**: Nobel (2009).

18.6.1 The whole layer, part by part

The picture below is the complete drawing of the layer Soil, plant and air (S1), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 3 parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

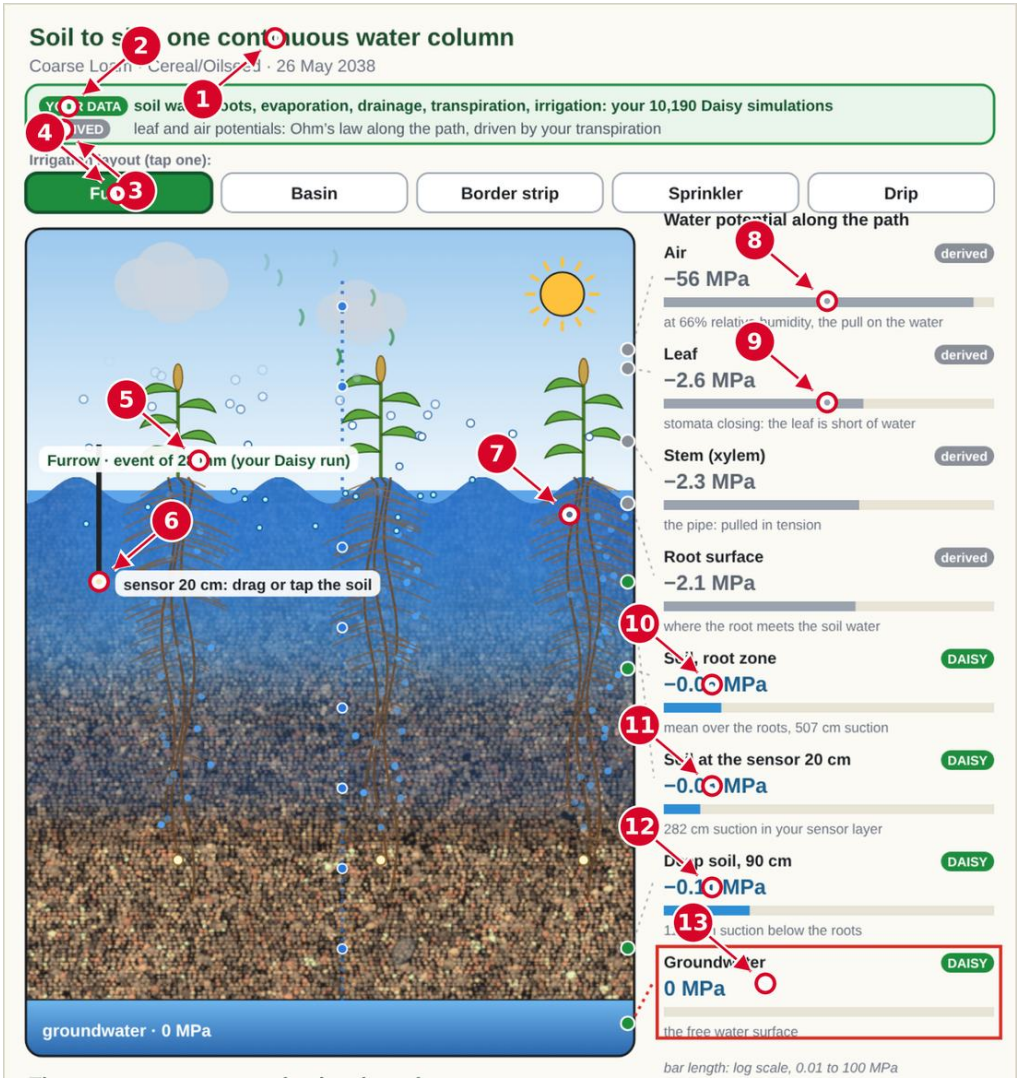


Figure 18.120: Soil, plant and air (S1), part 1 of 3.

What the arrows point to.

- **1** The title and the date of the water column shown, from the groundwater up to the air.
- **2** The YOUR DATA badge marks quantities read from the Daisy simulations of the ten soils.

- **3** The DERIVED badge marks quantities computed along the path from Ohm’s law, the leaf and the air potentials.
- **4** The irrigation layout buttons, here Furrow; the other layouts are Basin, Border strip, Sprinkler and Drip.
- **5** The label of the irrigation event with the depth applied in the run of the Daisy model.
- **6** The soil sensor at 20 cm, which can be dragged or tapped to another depth.
- **7** The roots of the crop, with the depth they have reached.
- **8** The potential of the air at the current relative humidity, the strongest pull on the water.
- **9** The potential of the leaf, set by the stomata and the transpiration.
- **10** The potential of the soil in the root zone, mean over the roots.
- **11** The potential of the soil at the sensor depth.
- **12** The potential of the deep soil below the roots.
- **13** The groundwater, the free water surface at zero potential.

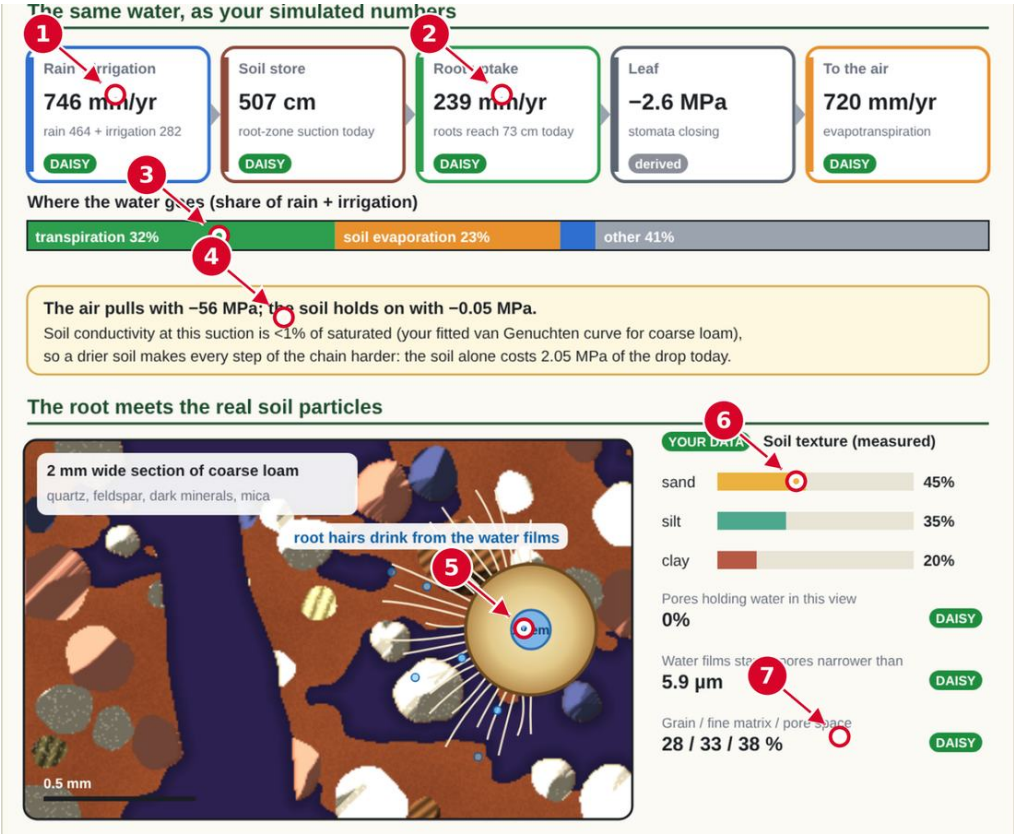


Figure 18.121: Soil, plant and air (S1), part 2 of 3.

What the arrows point to.

- **1** The cards give the same water as numbers, from rain and irrigation to the soil store.
- **2** The root uptake card, the transpiration drawn through the roots in mm per year.
- **3** The partition of rain and irrigation into transpiration, soil evaporation and the other losses.
- **4** The message that compares the pull of the air with the hold of the soil.
- **5** A section of the root surrounded by the real soil particles, with root hairs in the water films.
- **6** The measured texture of the soil, the sand, silt and clay fractions.
- **7** The grain, fine matrix and pore fractions of this section.

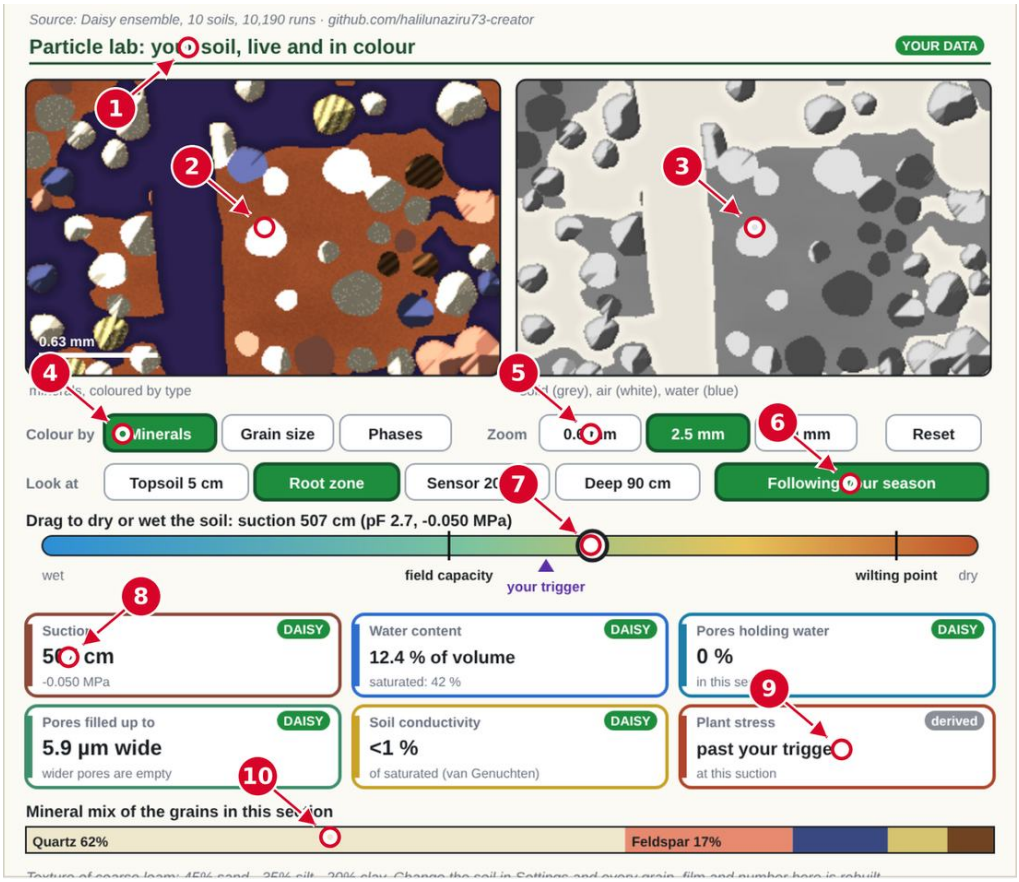


Figure 18.122: Soil, plant and air (S1), part 3 of 3.

What the arrows point to.

- **1** The particle lab, the soil shown live and in colour.
- **2** The left view colours the grains by mineral type.
- **3** The right view shows the solid in grey, the air in white and the water in blue.
- **4** The Colour by buttons choose minerals, grain size or phases.
- **5** The Zoom buttons set the width of the section to 0.6, 2.5 or 10 mm.
- **6** The Look at buttons choose the depth, or follow your season.
- **7** The slider dries or wets the soil, with the field capacity, your trigger and the wilting point marked.
- **8** The suction, the water content and the pores holding water at the chosen state.
- **9** The plant stress flag, past or inside your trigger at this suction.
- **10** The mineral mix of the grains in this section.

18.6.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

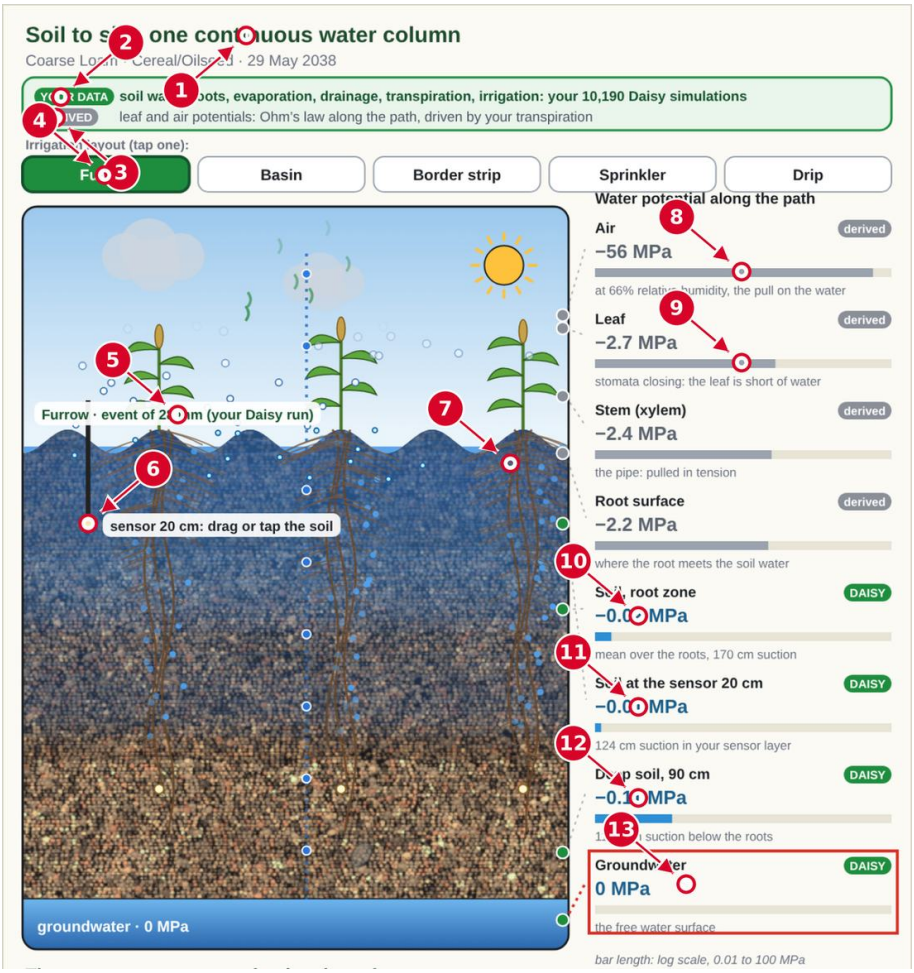


Figure 18.123: Soil, plant and air (S1), part 1, playback a third of the way (Paused: following the water from groundwater up to the air).

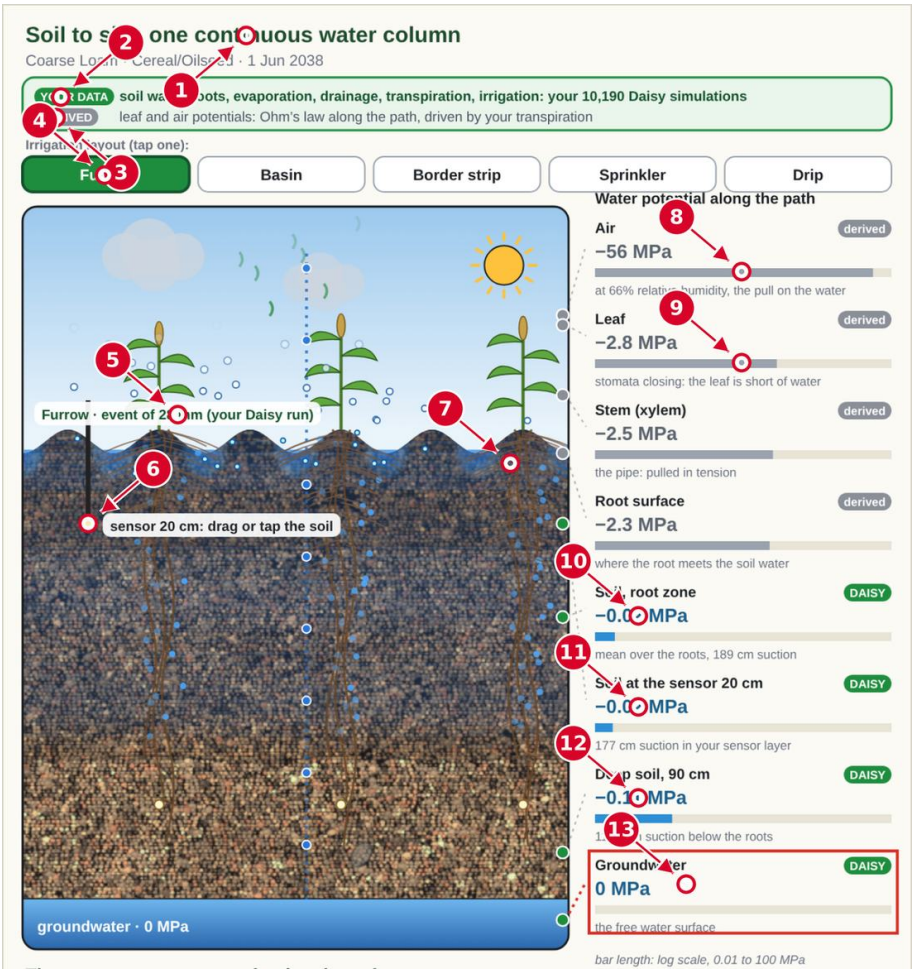


Figure 18.124: Soil, plant and air (S1), part 1, playback two thirds of the way (Paused: following the water from groundwater up to the air).

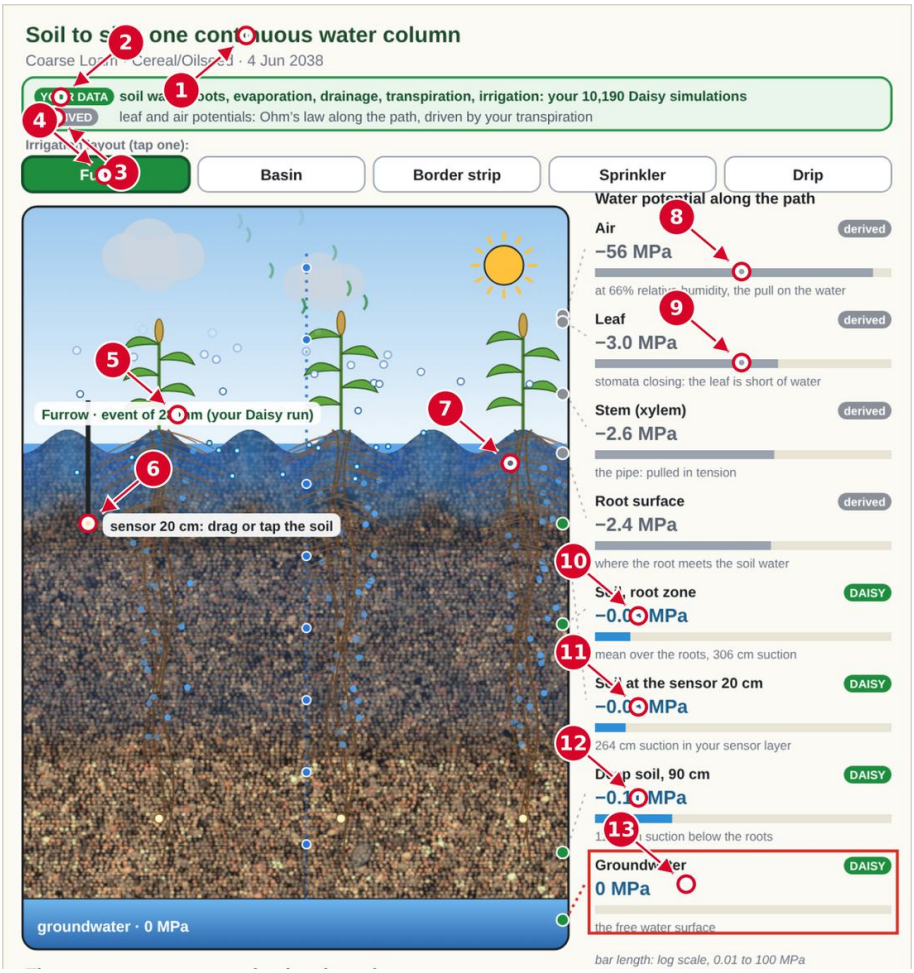


Figure 18.125: Soil, plant and air (S1), part 1, playback the end (Paused: following the water from groundwater up to the air).

18.6.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

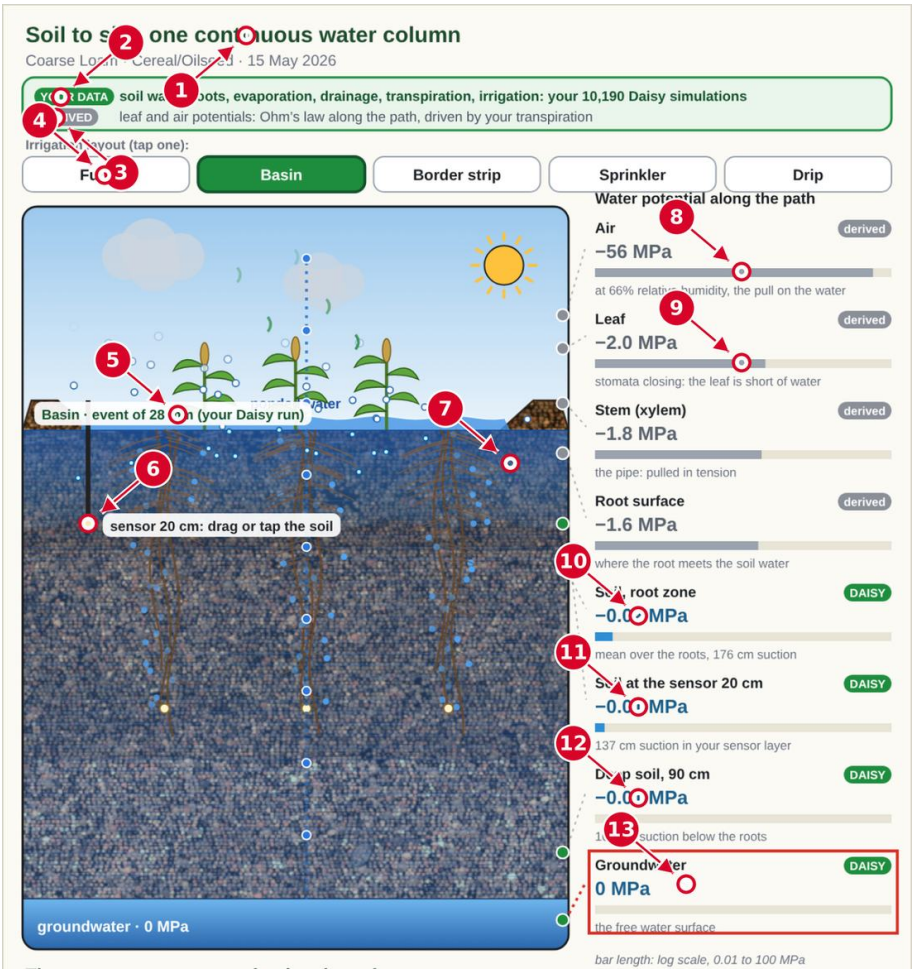


Figure 18.126: Soil, plant and air (S1), part 1. The Basin layout chosen, so the water ponds over the whole level plot.

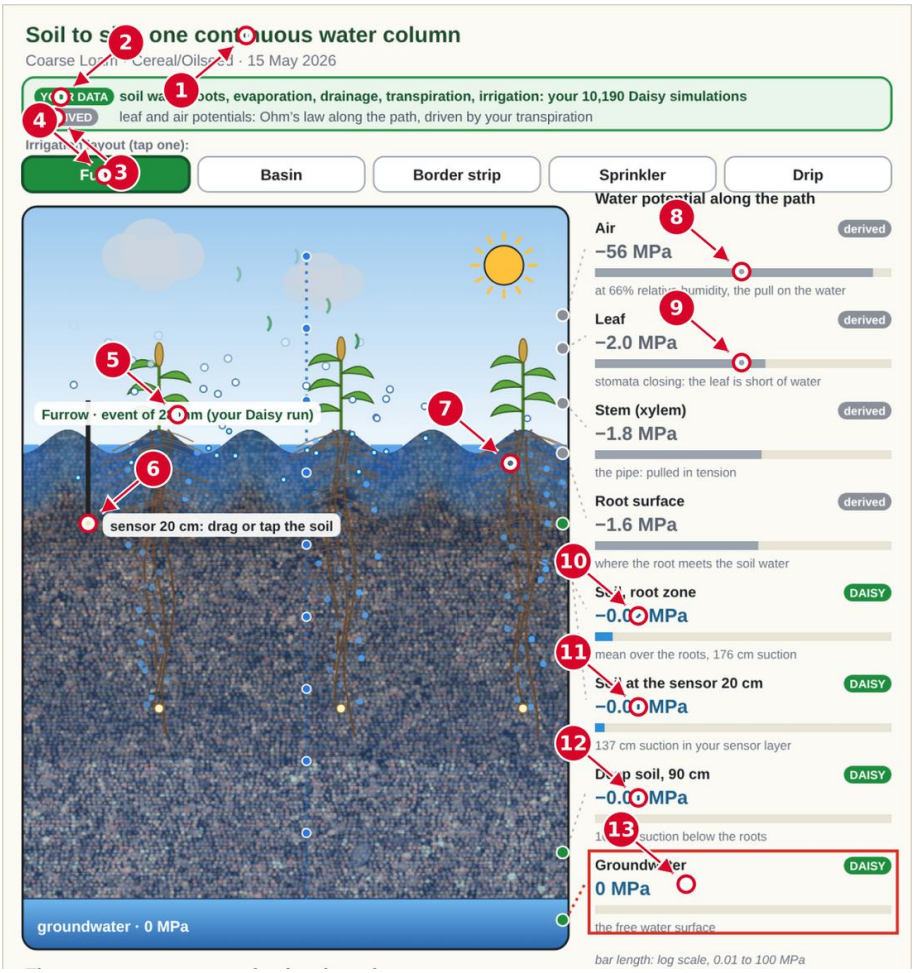


Figure 18.127: Soil, plant and air (S1), part 1. The Border strip layout chosen.

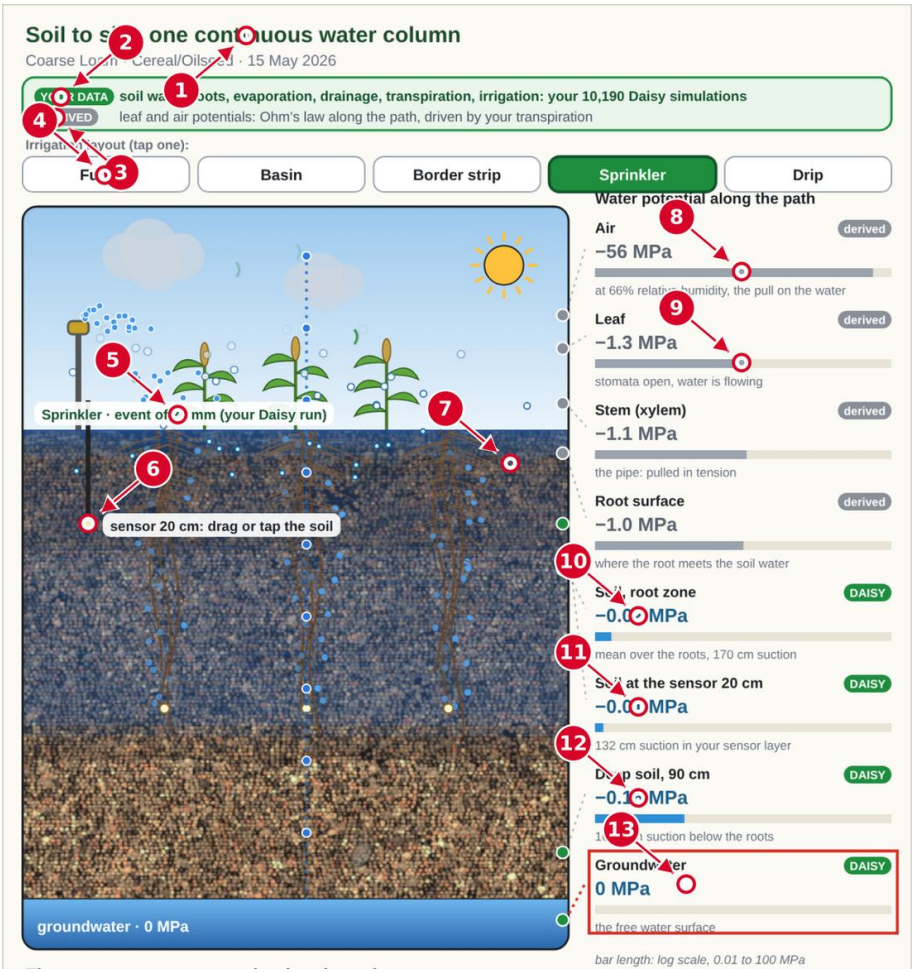


Figure 18.128: Soil, plant and air (S1), part 1. The Sprinkler layout chosen, so the water arrives as spray over the plants.

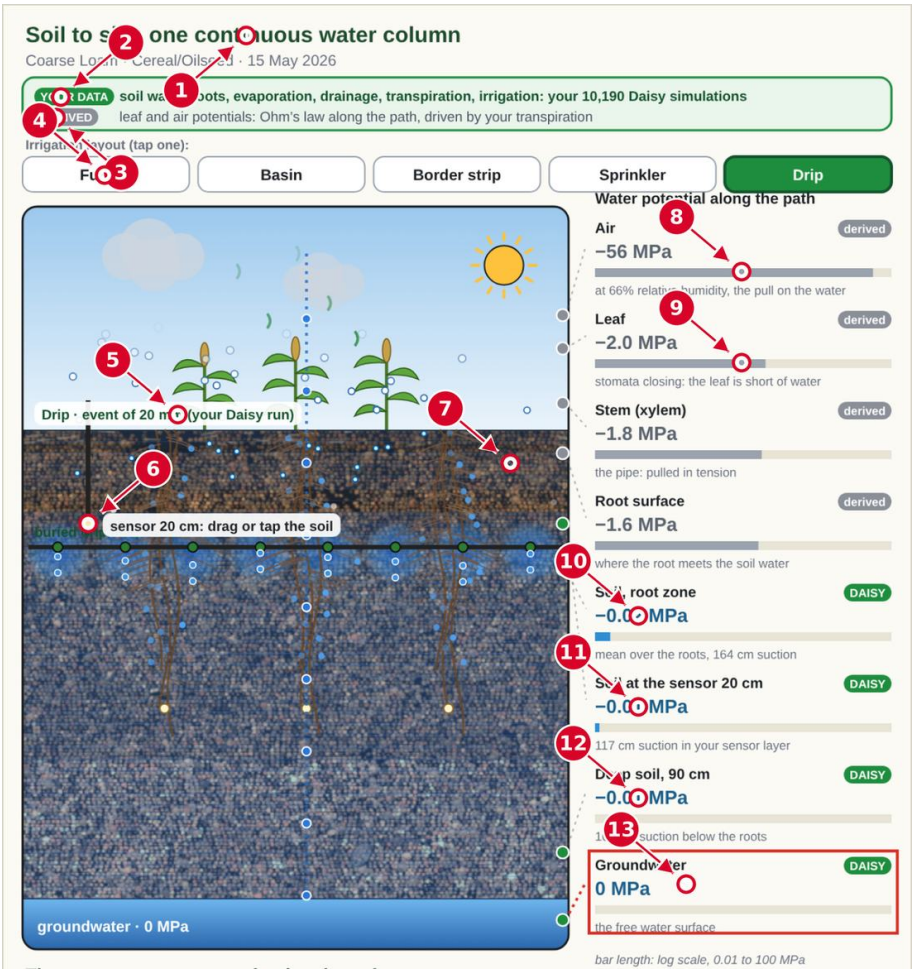


Figure 18.129: Soil, plant and air (S1), part 1. The Drip layout chosen, so the water arrives at the plant row only.

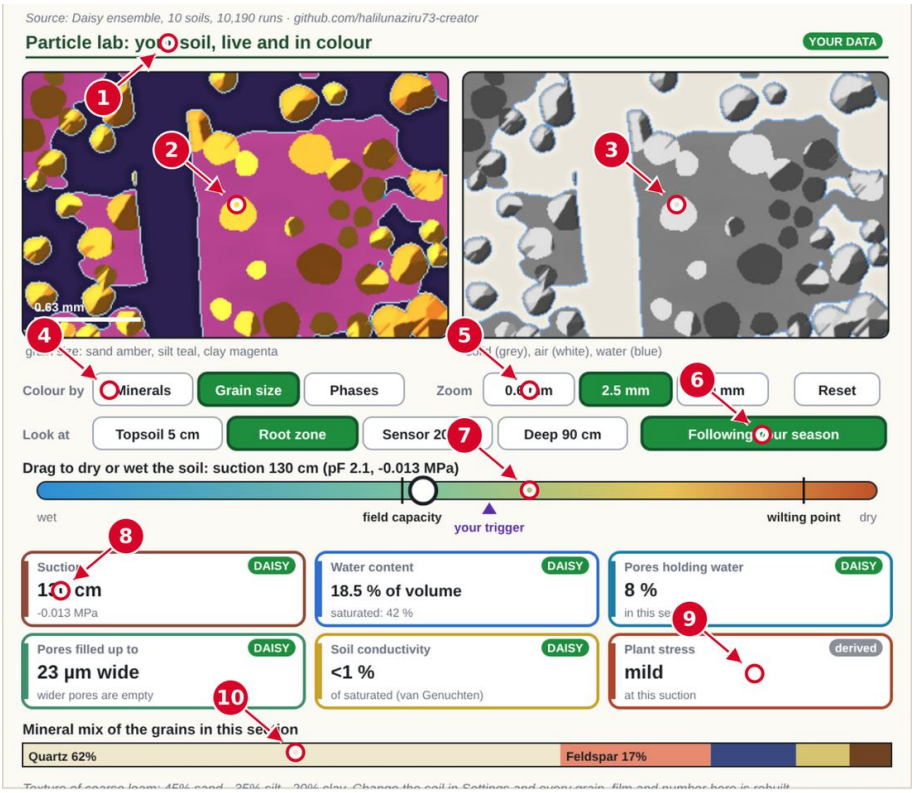


Figure 18.130: Soil, plant and air (S1), part 3. The Colour by Grain size button pressed, so the grains of the section are coloured by their size.

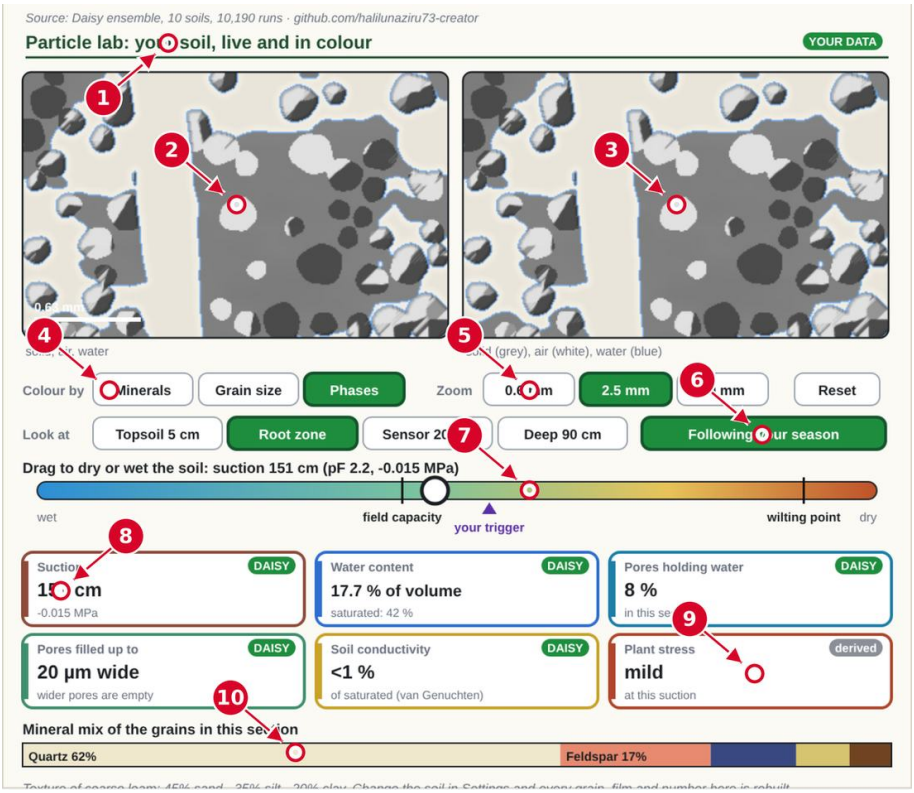


Figure 18.131: Soil, plant and air (S1), part 3. The Colour by Phases button pressed, so the section shows the solid, the air and the water.

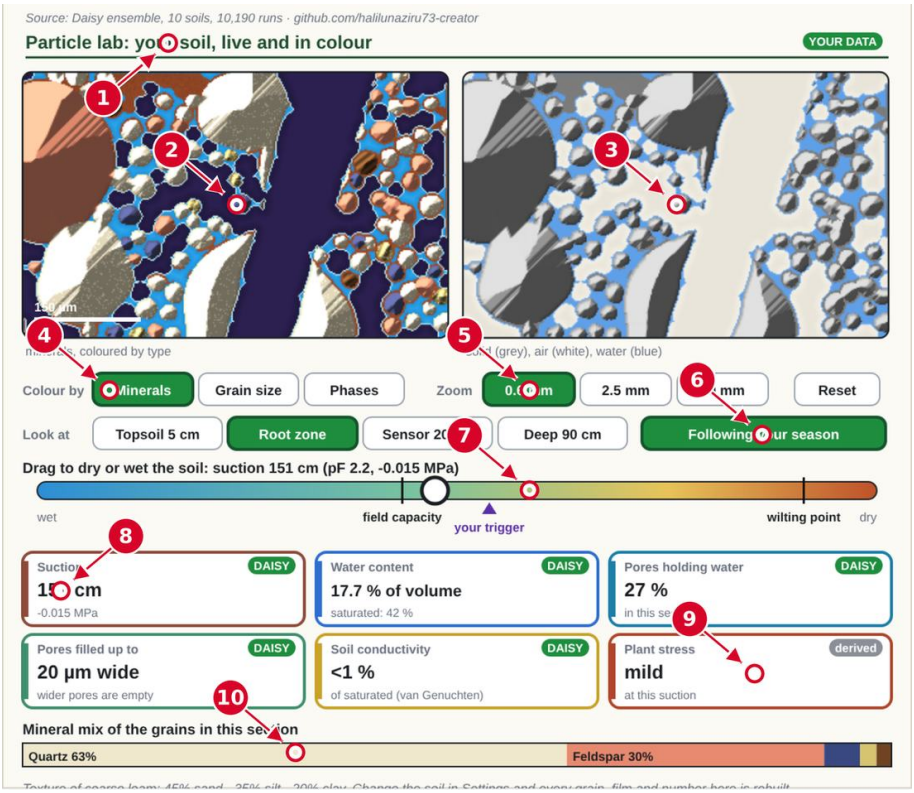


Figure 18.132: Soil, plant and air (S1), part 3. The Zoom 0.6 mm button pressed, so the section shows silt, clay and the water films.

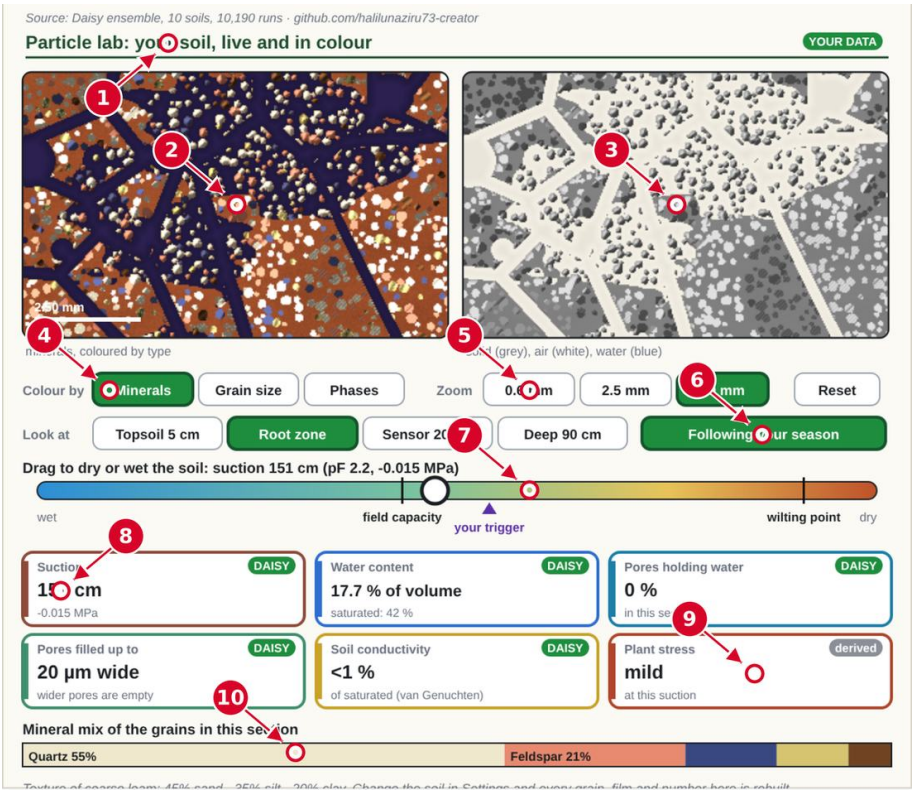


Figure 18.133: Soil, plant and air (S1), part 3. The Zoom 10 mm button pressed, so the section shows aggregates and cracks.

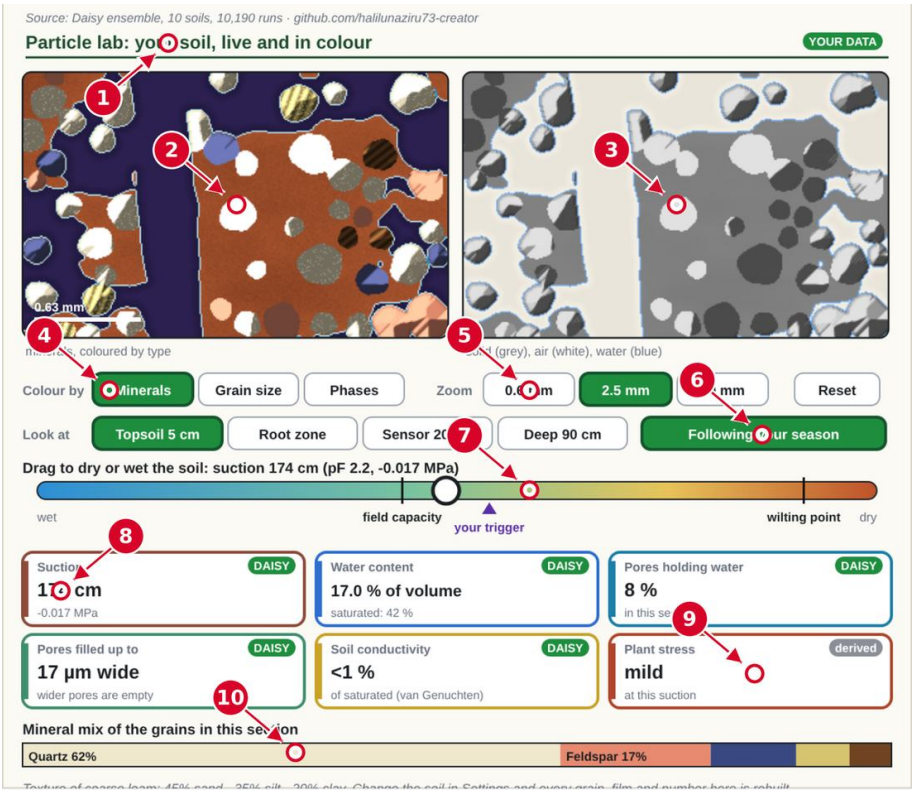


Figure 18.134: Soil, plant and air (S1), part 3. The Look at Topsoil 5 cm button pressed, so the section and the readouts refer to the topsoil.

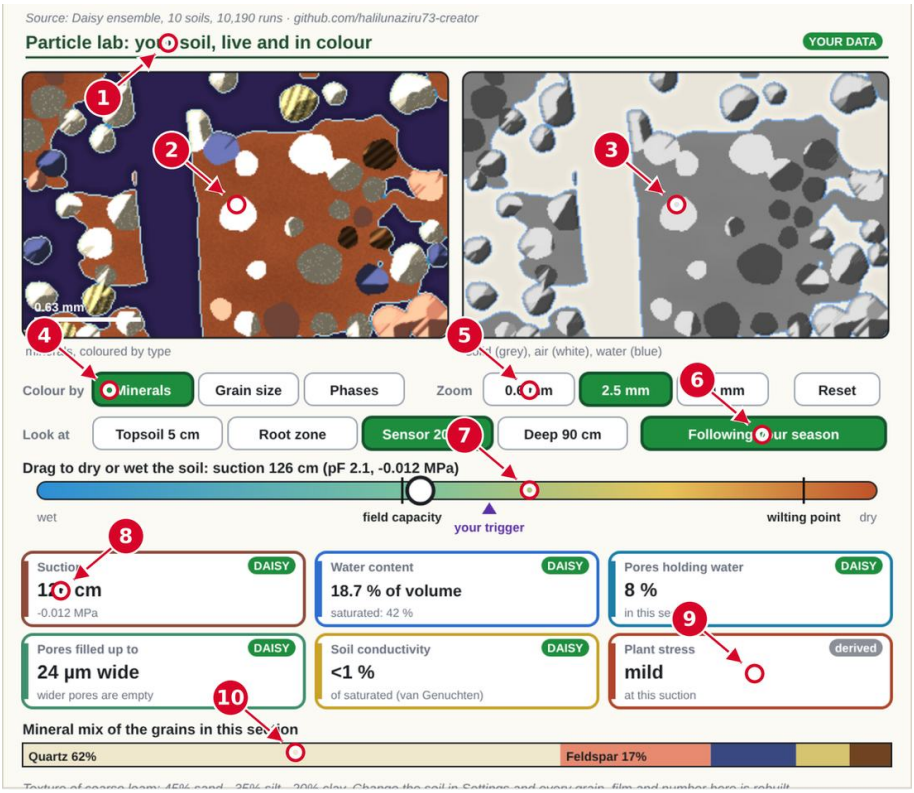


Figure 18.135: Soil, plant and air (S1), part 3. The Look at Sensor 20 cm button pressed.

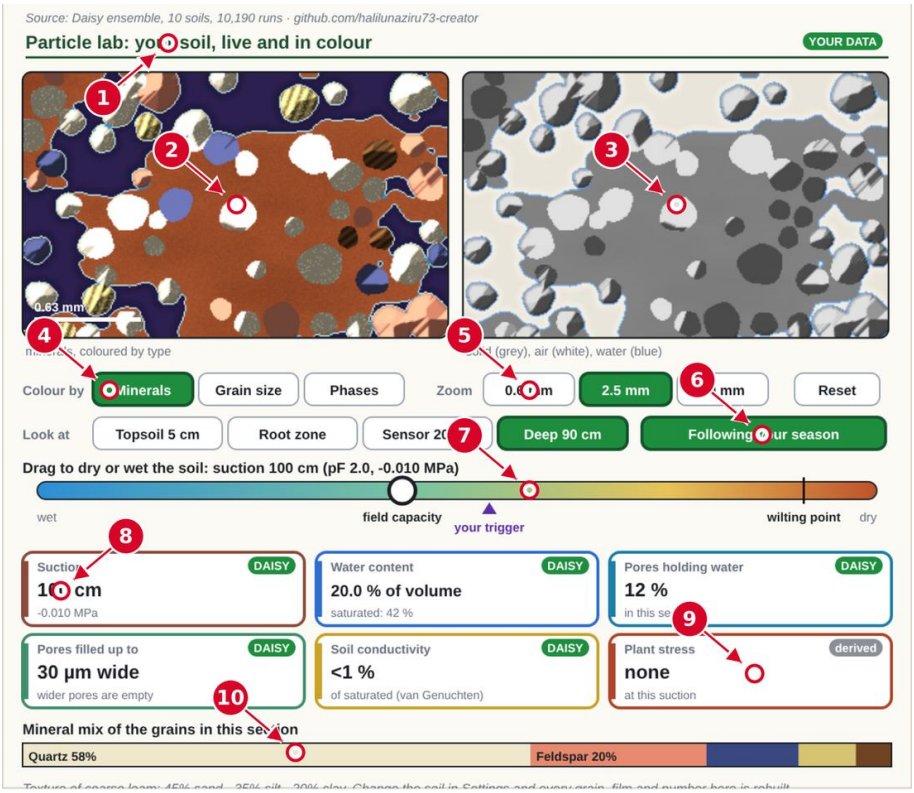


Figure 18.136: Soil, plant and air (S1), part 3. The Look at Deep 90 cm button pressed, so the section and the readouts refer to the deep soil.

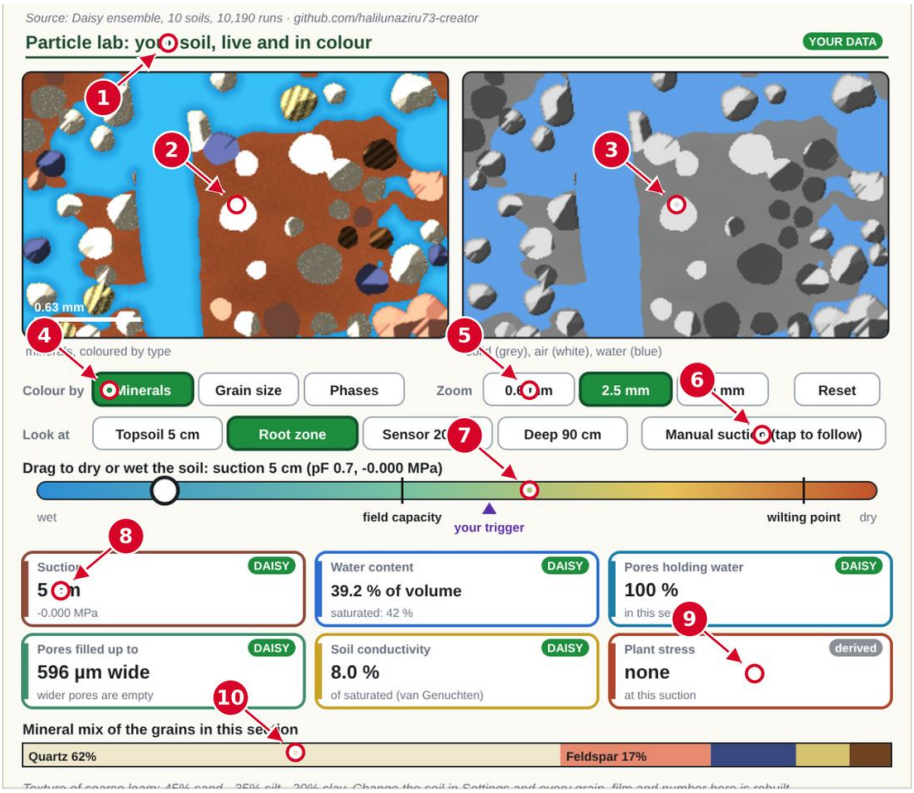


Figure 18.137: Soil, plant and air (S1), part 3. The suction slider dragged to the wet end, so the soil is shown near saturation.

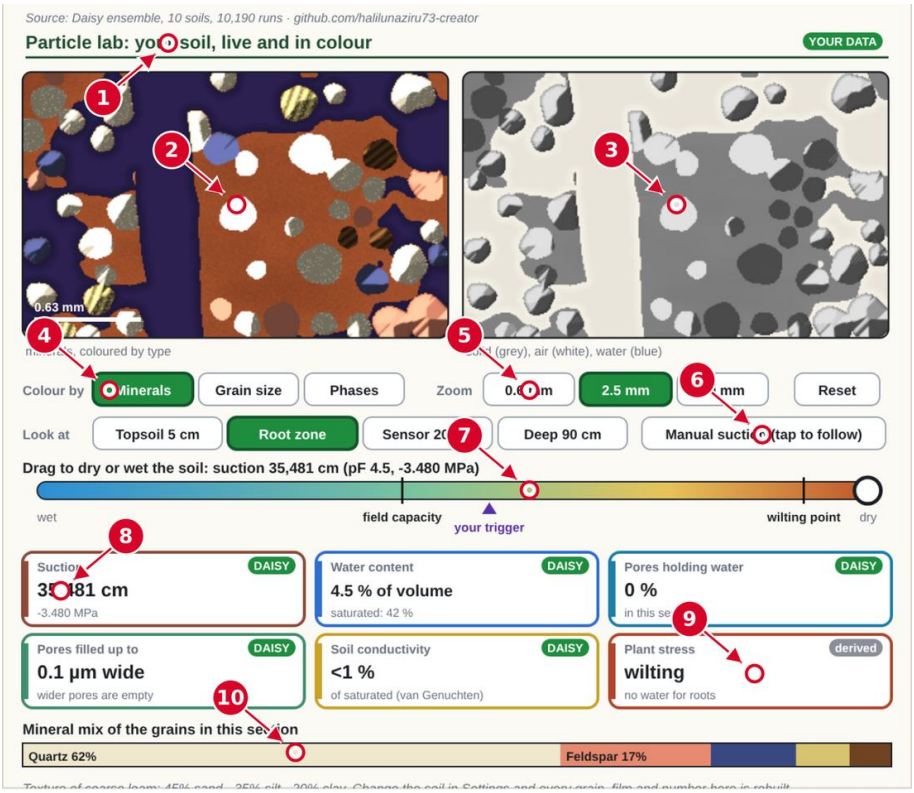


Figure 18.138: Soil, plant and air (S1), part 3. The suction slider dragged to the dry end, so the soil is shown near the wilting point.

18.7 Profile evolution (V2)

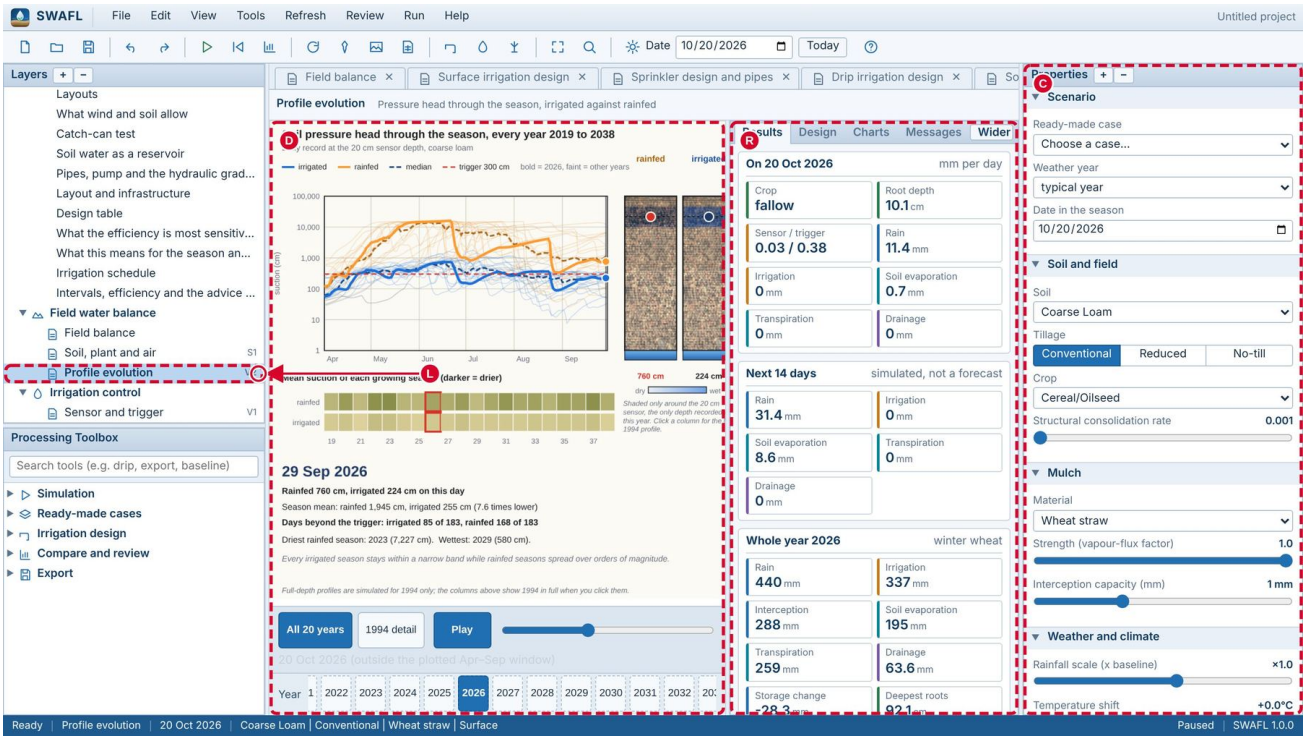


Figure 18.139: Profile evolution (V2): the whole workbench.

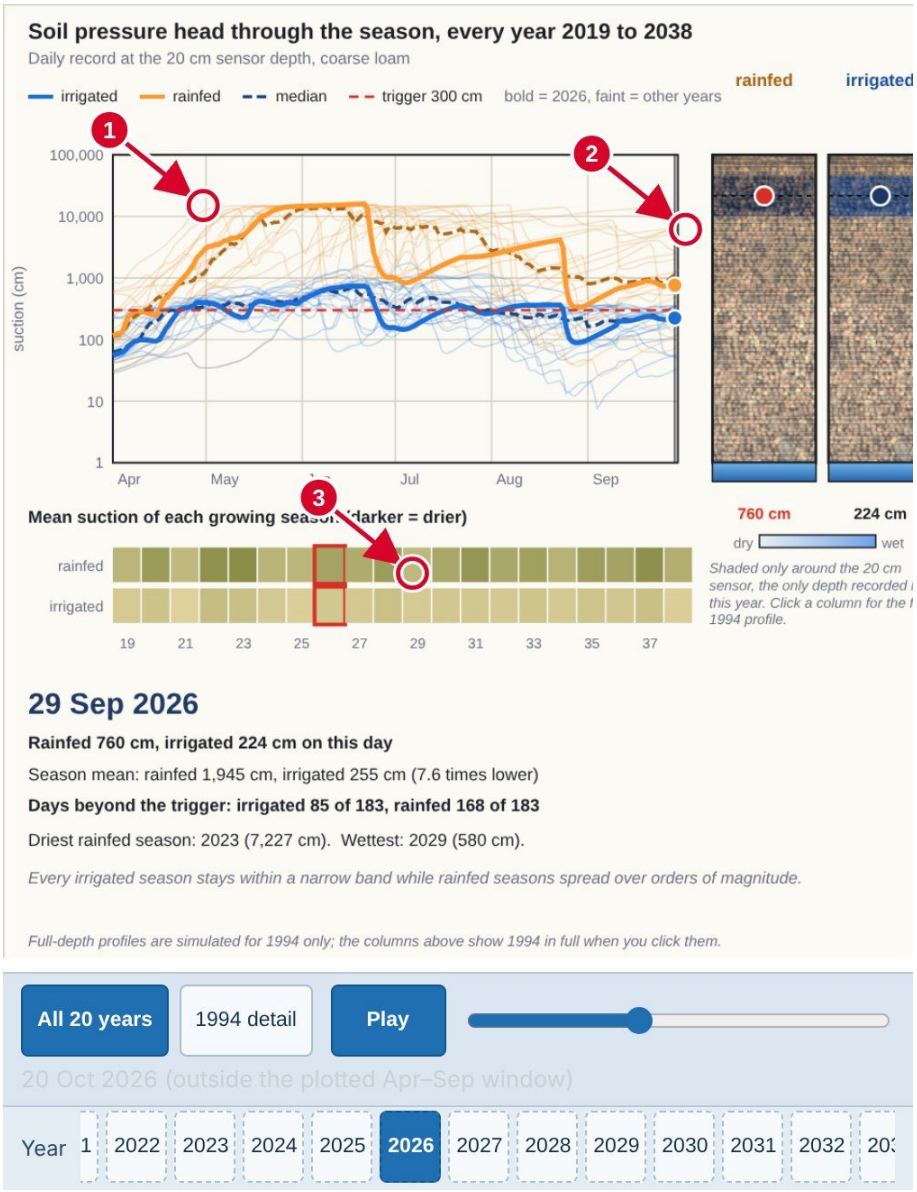


Figure 18.140: Profile evolution (V2): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Daily suction at the 20 cm sensor: stored Daisy series (data-driven, no equation).
- **2** Trigger h_T (dashed red line), set by the slider; the share of the root zone beyond it is Eq. 7.6.
- **3** Heat strip: geometric mean suction of each growing season, text under Eq. 7.6.

Equations behind this screen. Eq. 5.15, Eq. 5.17, Eq. 7.6.

Eq. 5.15. Where explained: chapter "The two computing engines", section "Rainfall generator".

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0,1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.17. Where explained: chapter “The two computing engines”, section “Infiltration, redistribution and drainage”.

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i, \;\; w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

Eq. 7.6. Where explained: chapter “Field water balance layers”, section “How it is generated”.

$$\text{share} = \frac{\sum_{i:\; z_i \le 60\,\text{cm},\; h_i > h_T} \Delta z_i}{\sum_{i:\; z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

Provenance (Eq. 7.6): **[Engineered]**: indicator defined for the workbench.

18.7.1 The whole layer, part by part

The picture below is the complete drawing of the layer Profile evolution (V2), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 1 part, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

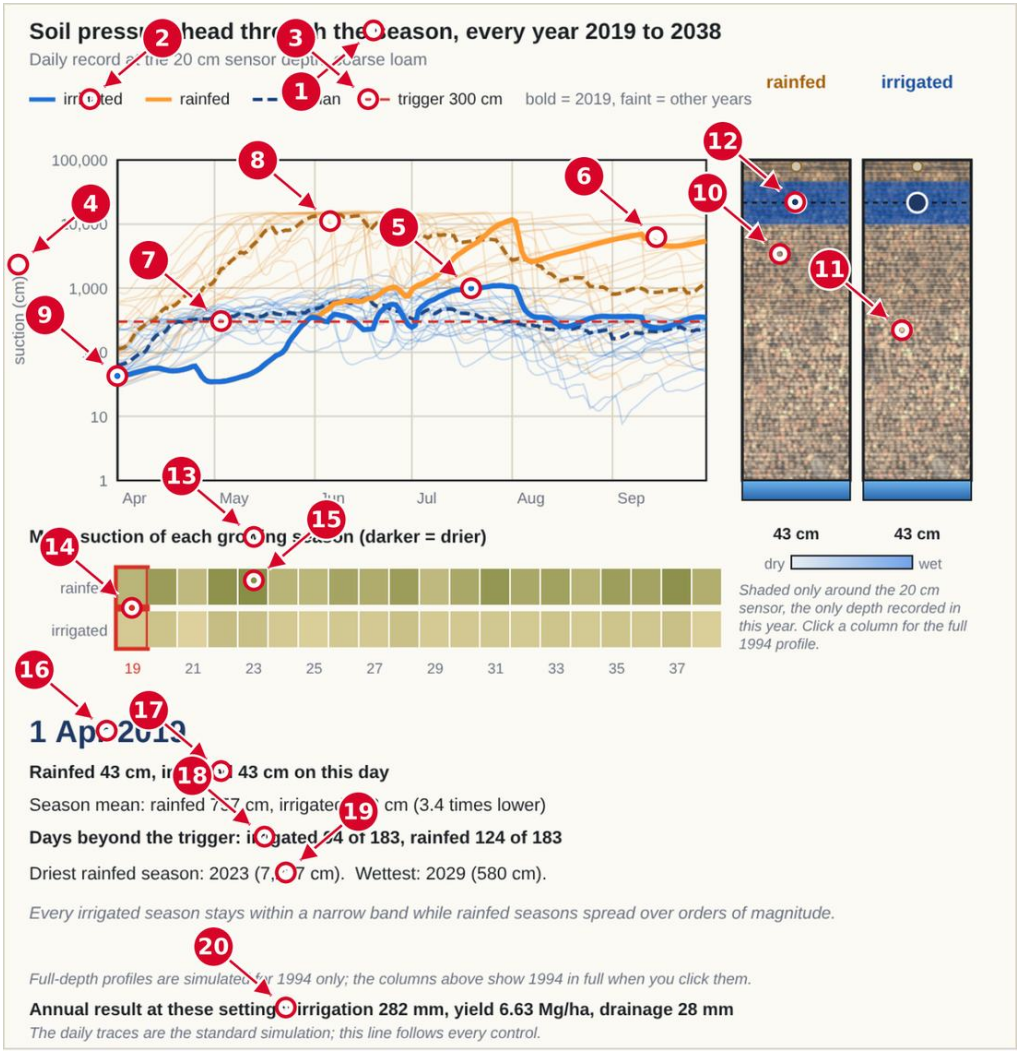


Figure 18.141: Profile evolution (V2), part 1 of 1.

What the arrows point to.

- **1** The title, the daily suction at the 20 cm sensor for every year of the record.
- **2** The legend of the traces, irrigated in blue and rainfed in orange.
- **3** The trigger legend, the dashed red line at the suction that starts an irrigation.
- **4** The suction axis in cm of water on a log scale.
- **5** The bold blue trace, the irrigated season of the year shown.
- **6** The bold orange trace, the rainfed season of the same year.
- **7** The trigger line at 300 cm.
- **8** The faint traces, the other years of the record.
- **9** The marker of the day shown, moving along the year as the animation plays.
- **10** The rainfed soil column, shaded by the suction around the sensor.
- **11** The irrigated soil column.
- **12** The sensor ring at 20 cm, the only depth recorded in this year.
- **13** The strip of the mean suction of every growing season, darker for drier.
- **14** The selected year in the strip; clicking a column selects that year.
- **15** The dark cells are the dry rainfed seasons, far darker than the irrigated row.
- **16** The date of the day shown.
- **17** The suction on this day for the rainfed and the irrigated soil.
- **18** The days beyond the trigger in the growing season, irrigated against rainfed.
- **19** The driest and the wettest rainfed season of the record.
- **20** The annual result at the settings in force, the irrigation, the yield and the drainage.

18.7.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

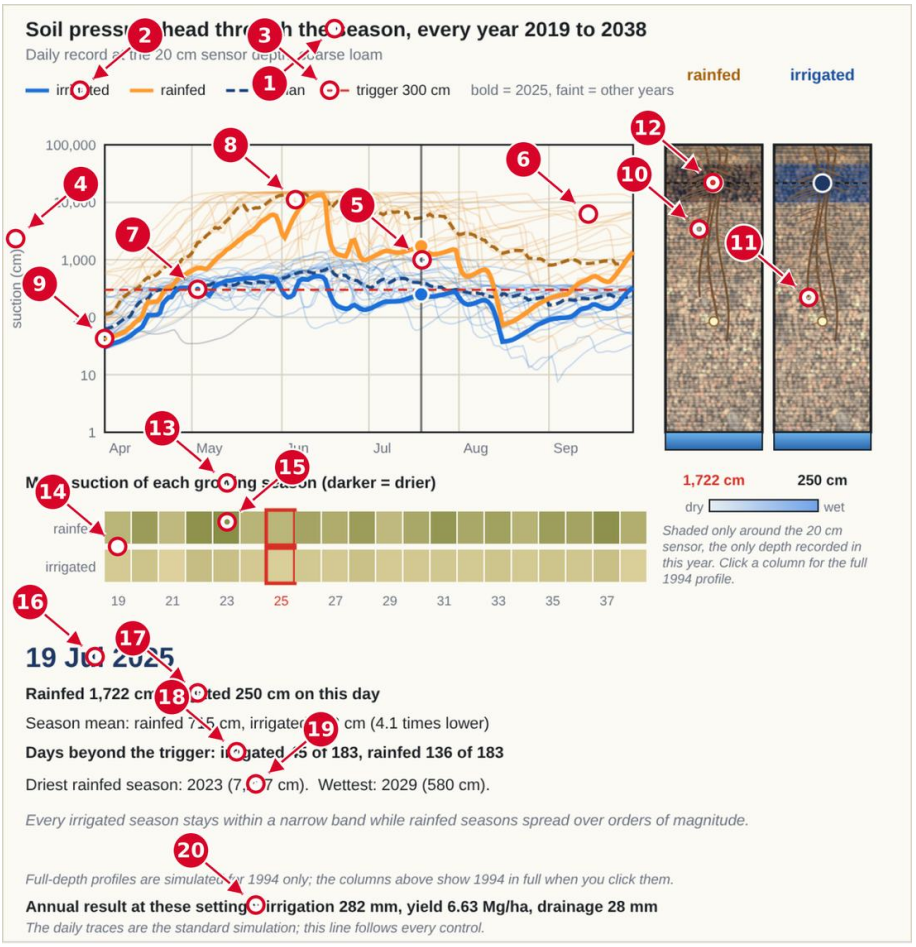


Figure 18.142: Profile evolution (V2), part 1, playback a third of the way (31 Dec 2038 (outside the plotted Apr–Sep window)).

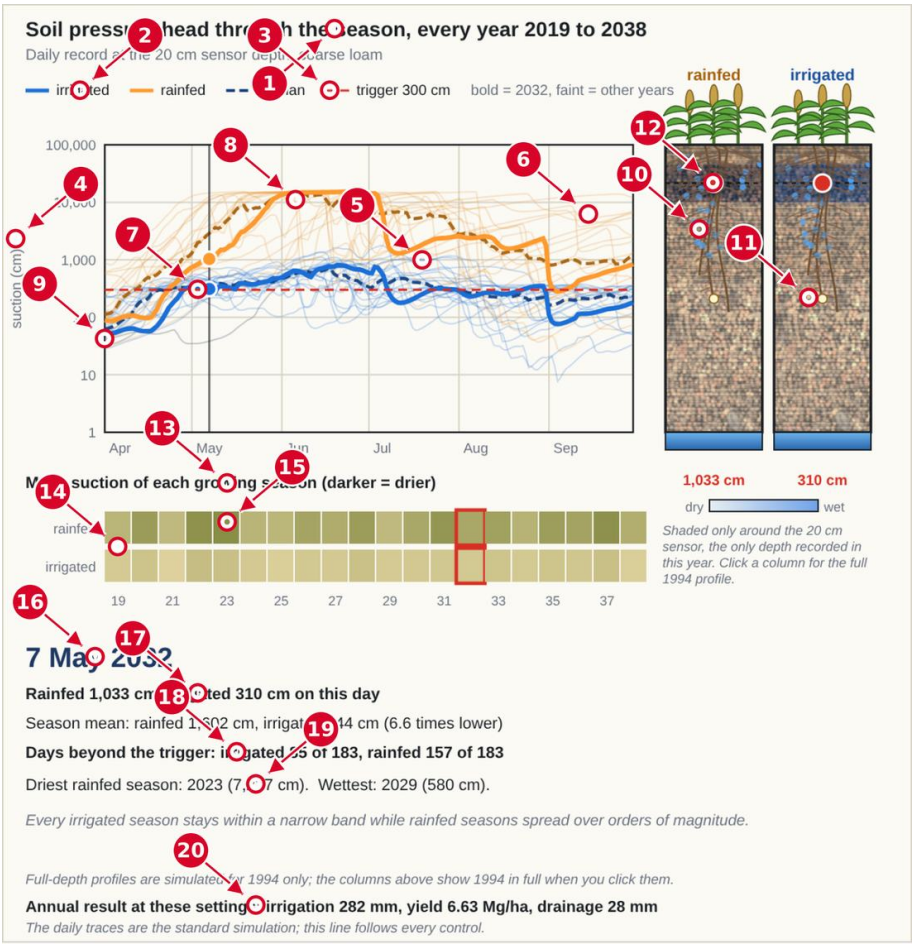


Figure 18.143: Profile evolution (V2), part 1, playback two thirds of the way (31 Dec 2038 (outside the plotted Apr–Sep window)).

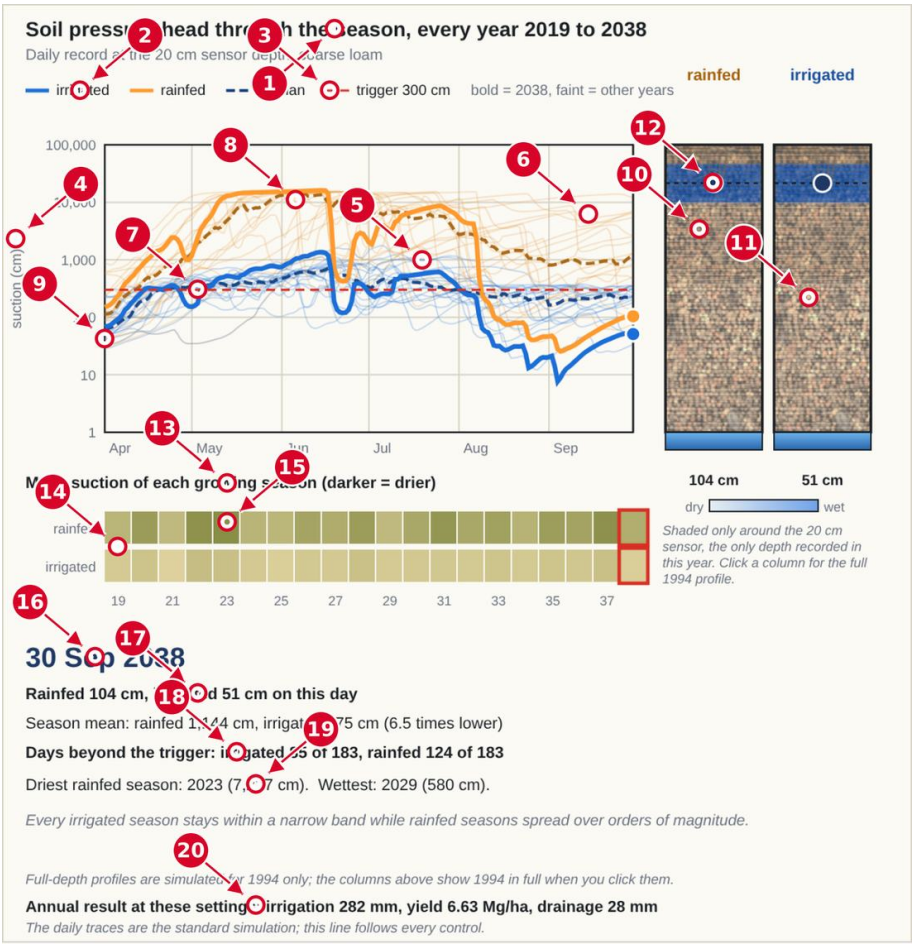


Figure 18.144: Profile evolution (V2), part 1, playback the end (31 Dec 2038 (outside the plotted Apr–Sep window)).

18.7.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

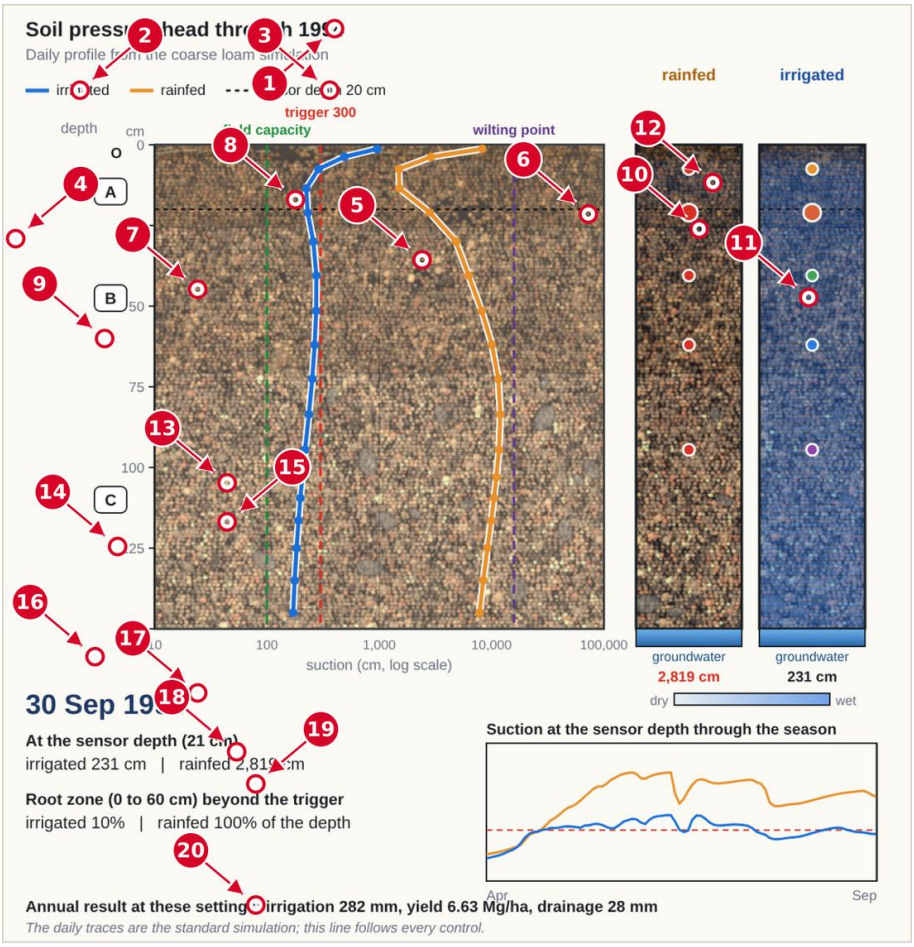


Figure 18.145: Profile evolution (V2), part 1. The 1994 detail button pressed, so the chart shows the daily record of the year 1994 at full depth.

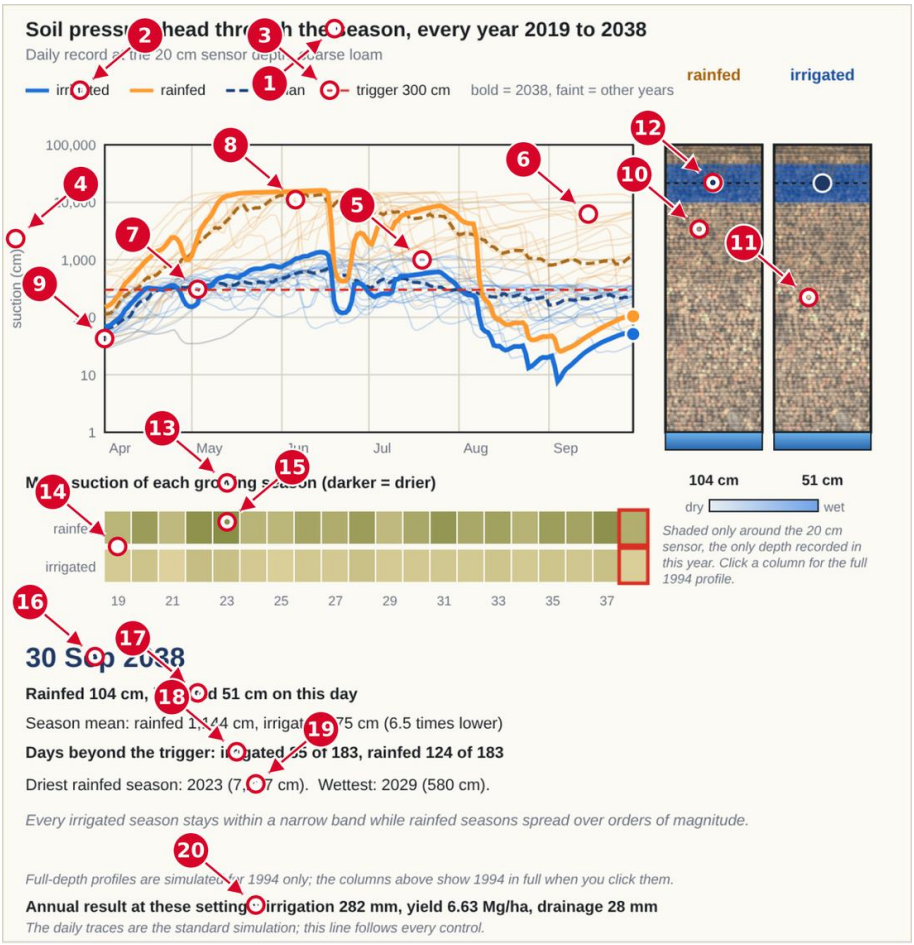


Figure 18.146: Profile evolution (V2), part 1. A column of the season strip clicked, so the chart and the columns show the selected year.

18.8 Sensor and trigger (V1)

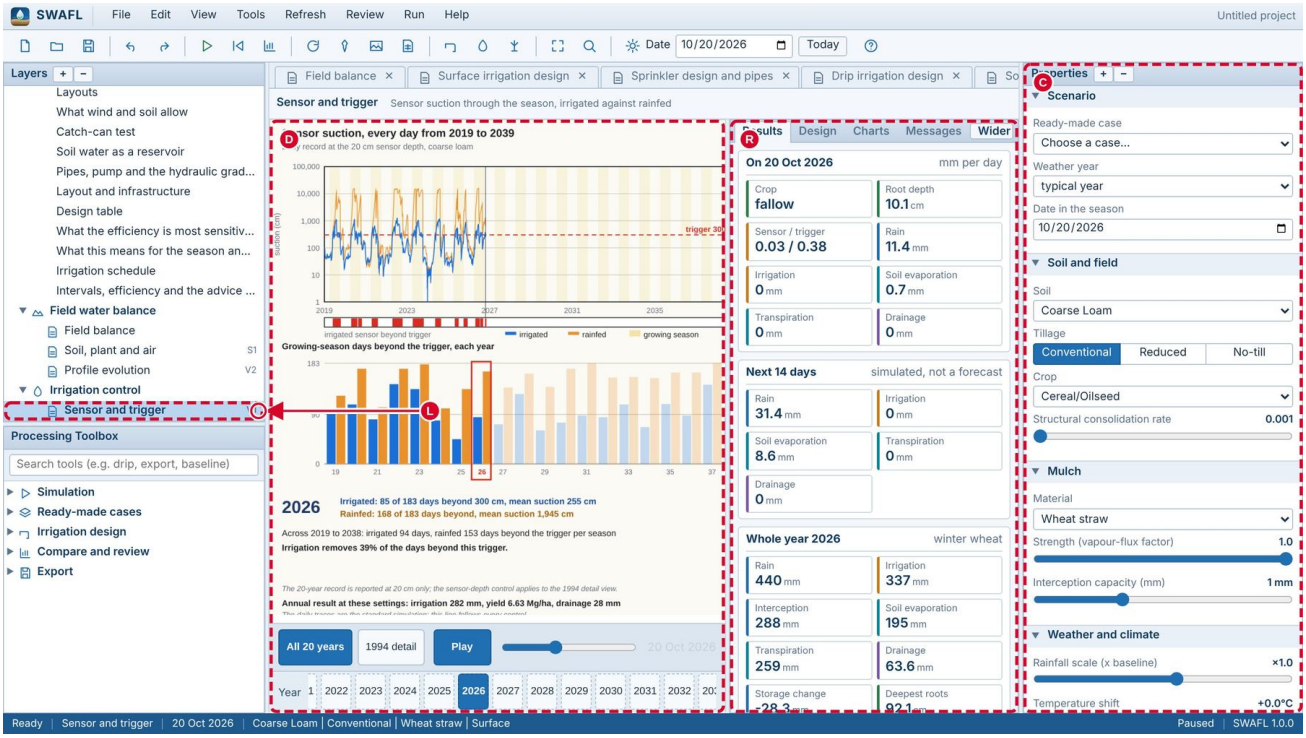


Figure 18.147: Sensor and trigger (V1): the whole workbench.

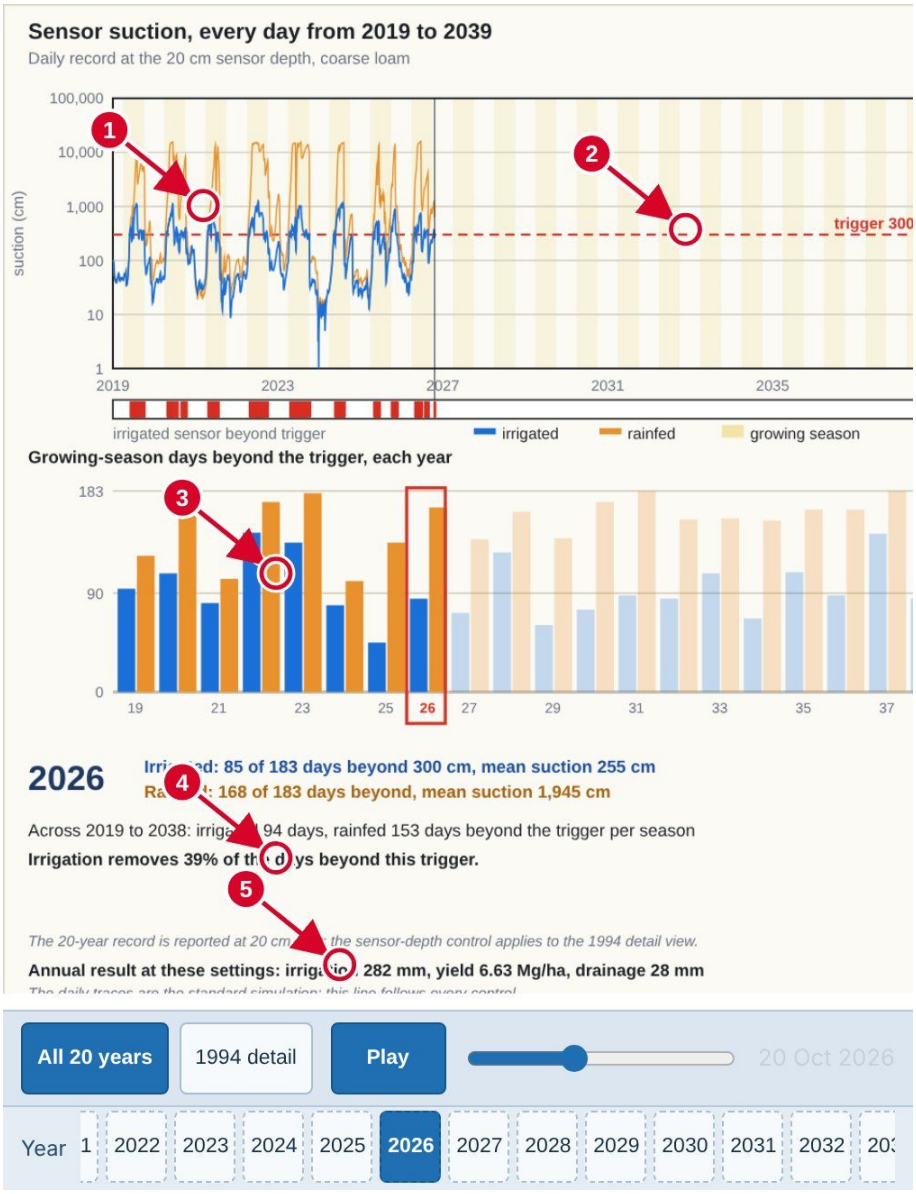


Figure 18.148: Sensor and trigger (V1): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Daily suction at 20 cm, irrigated (blue) and rainfed (orange): stored Daisy series (data-driven).
- **2** Trigger h_T (red dashed line), set in Properties; days beyond it are counted by Eq. 6.1.
- **3** Growing-season days beyond the trigger for each year, N_y(h_T): Eq. 6.1.
- **4** "Irrigation removes 39 % of the days": X = 100(1 - N_irr / N_rainfed), Eq. 6.2.
- **5** Annual result at these settings (irrigation, yield, drainage): Engine 1, Eqs. 5.4 to 5.12.

Equations behind this screen. Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 6.1, Eq. 6.2.

Eq. 5.4. Where explained: chapter "The two computing engines", section "The meta-model".

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".

$$Q \leftarrow Q\,\text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 6.1. Where explained: chapter "Irrigation control layers", section "How the numbers are generated".

$$N_y(h_T) = \#\{\, d \in \text{growing season of year } y : h_d > h_T \,\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

Provenance (Eq. 6.1): **[Data-driven]**: counts over the simulated daily series.

Eq. 6.2. Where explained: chapter "Irrigation control layers", section "How the numbers are generated".

$$X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_y N_y. \tag{6.2}$$

Provenance (Eq. 6.2): **[Data-driven]**: counts over the simulated daily series.

18.8.1 The whole layer, part by part

The picture below is the complete drawing of the layer Sensor and trigger (V1), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 1 part, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

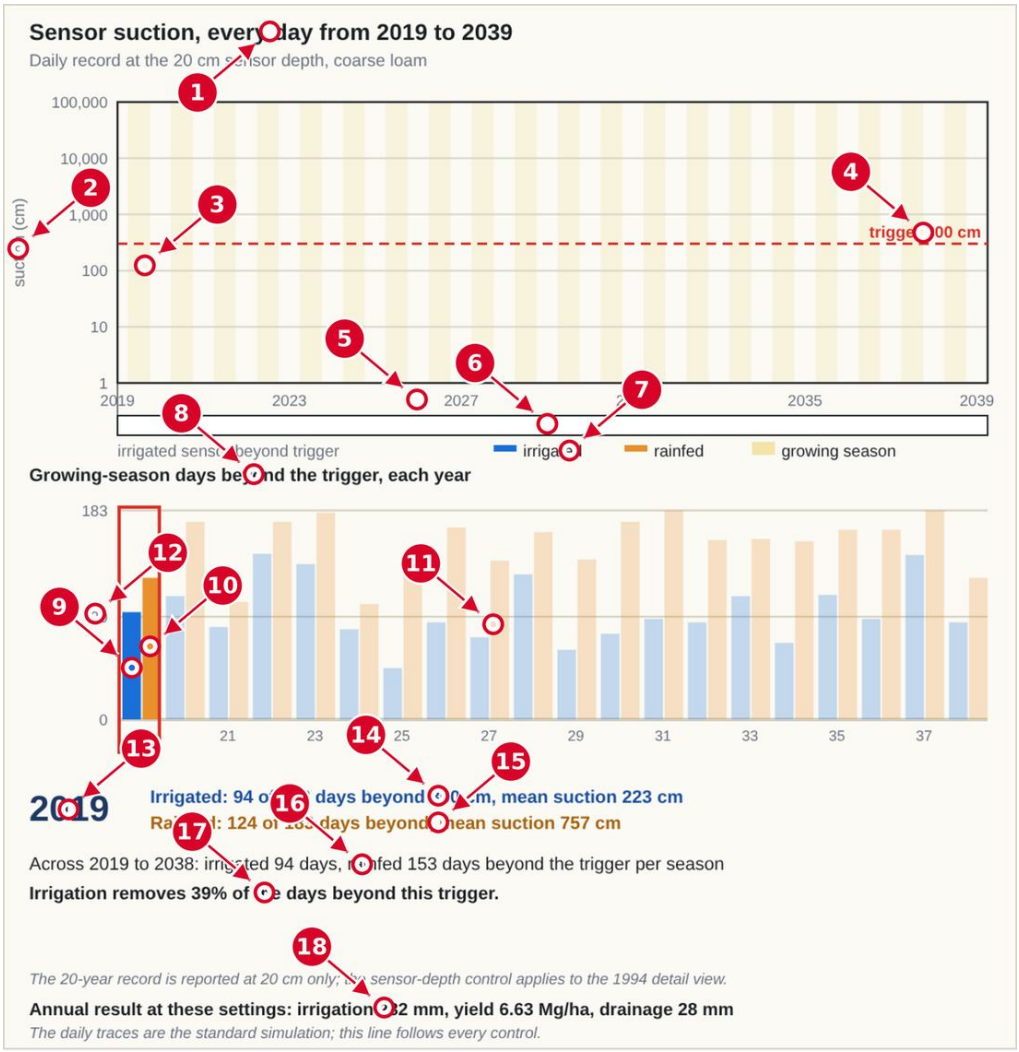


Figure 18.149: Sensor and trigger (V1), part 1 of 1.

What the arrows point to.

- **1** The title, the daily sensor suction for every day of the twenty years.
- **2** The suction axis in cm on a log scale.
- **3** The shaded bands are the growing seasons.
- **4** The trigger line, the suction at which the sensor calls for irrigation.
- **5** The time axis in years.
- **6** The strip that marks the days on which the irrigated sensor is beyond the trigger.
- **7** The legend, irrigated in blue, rainfed in orange and the growing season in tan.
- **8** The chart of the growing-season days beyond the trigger, one pair of bars per year.
- **9** The irrigated bar of the selected year.
- **10** The rainfed bar of the selected year.
- **11** The faint bars are the other years.
- **12** The scale of the days, from 0 to the 183 days of the season.
- **13** The year selected.
- **14** The irrigated days beyond the trigger and the mean suction of the season.
- **15** The rainfed days beyond the trigger and the mean suction of the season.
- **16** The mean over the whole record, irrigated against rainfed.
- **17** The share of the days beyond the trigger that irrigation removes.
- **18** The annual result at the settings in force.

18.8.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

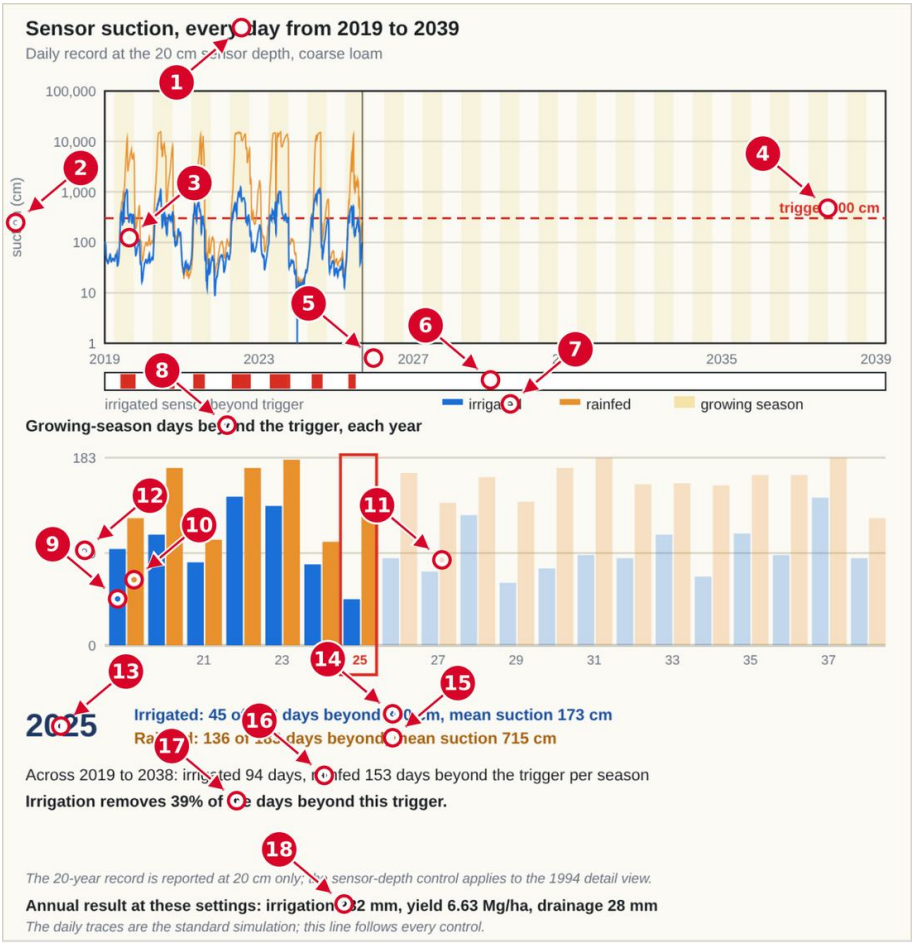


Figure 18.150: Sensor and trigger (V1), part 1, playback a third of the way (7 Sep 2025).

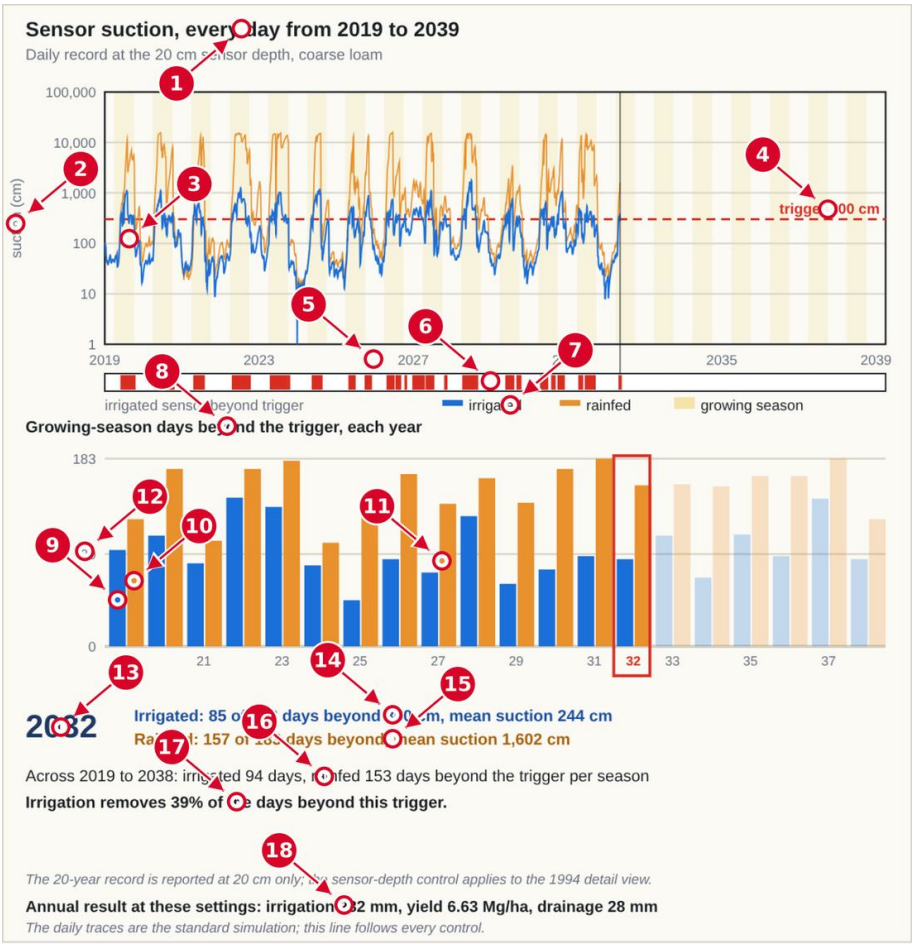


Figure 18.151: Sensor and trigger (V1), part 1, playback two thirds of the way (13 May 2032).

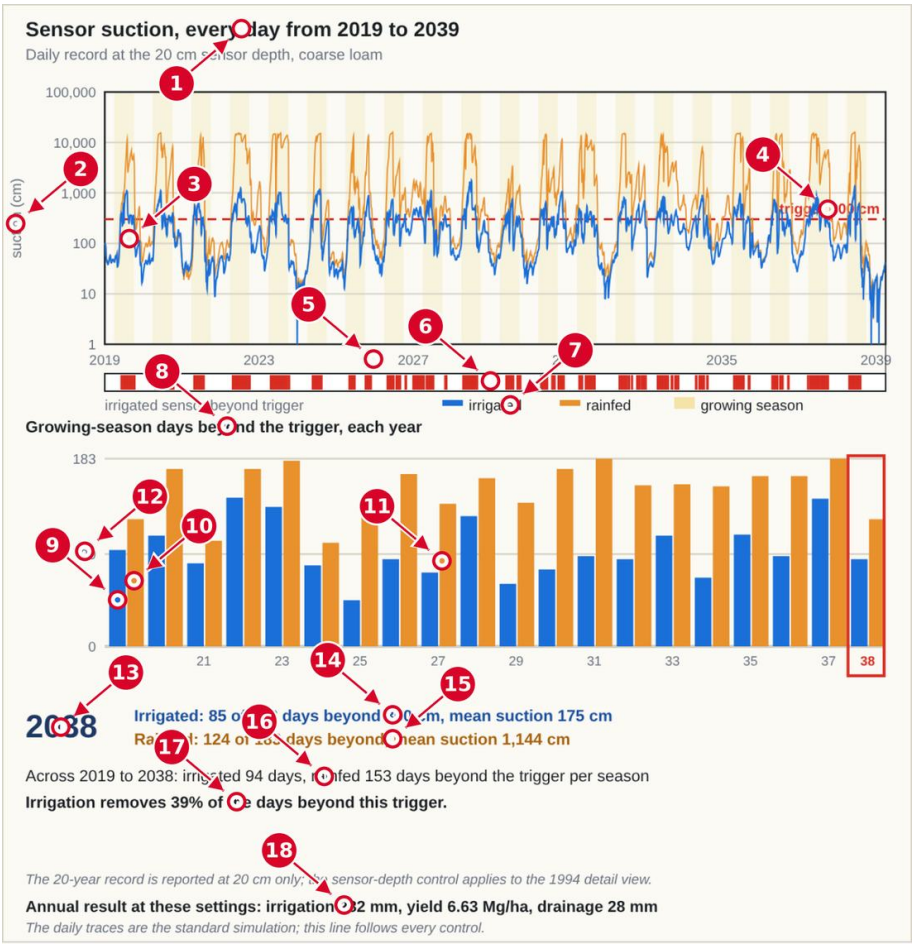


Figure 18.152: Sensor and trigger (V1), part 1, playback the end (1 Apr 2039).

18.8.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

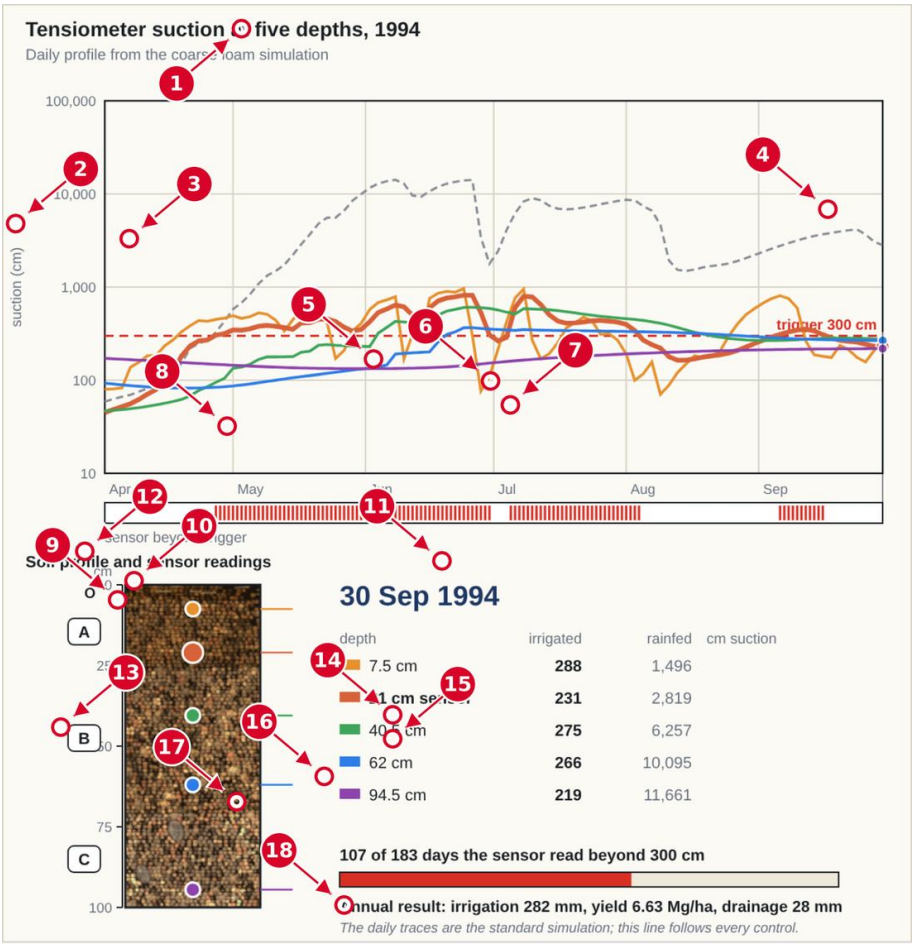


Figure 18.153: Sensor and trigger (V1), part 1. The 1994 detail button pressed, so the chart shows the daily sensor record of the year 1994.

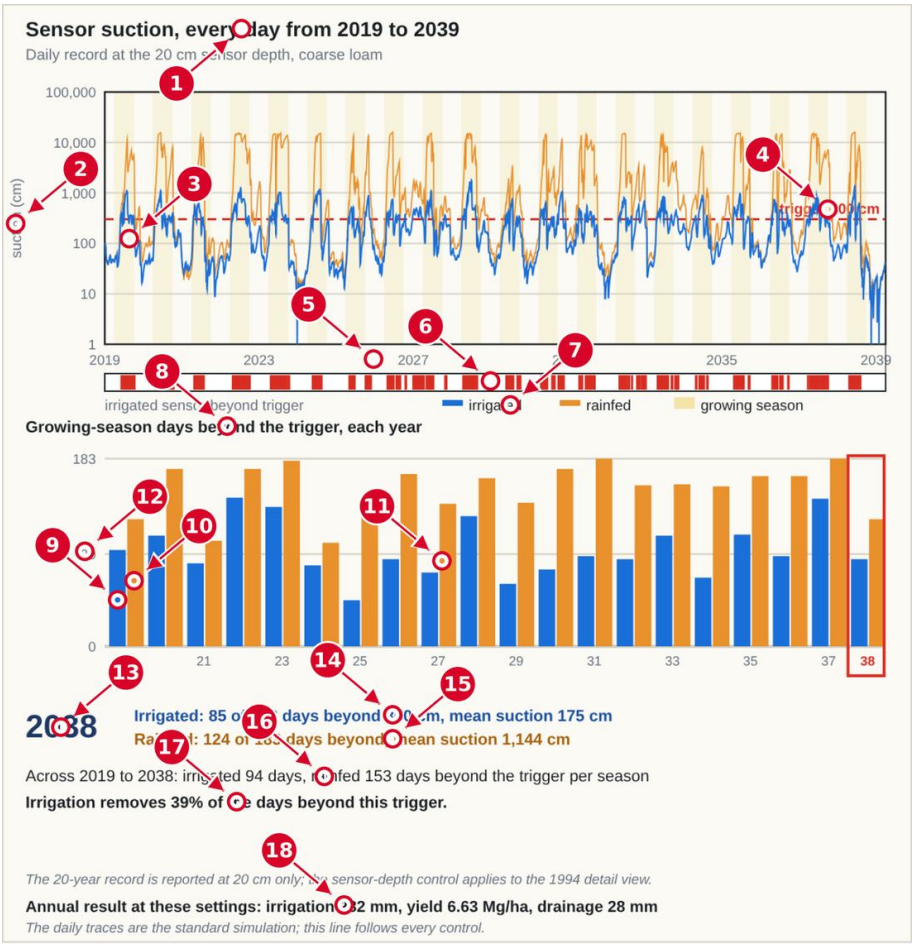


Figure 18.154: Sensor and trigger (V1), part 1. A bar of the lower chart clicked, so the readout below shows the selected year.

18.9 Trigger law (N1)

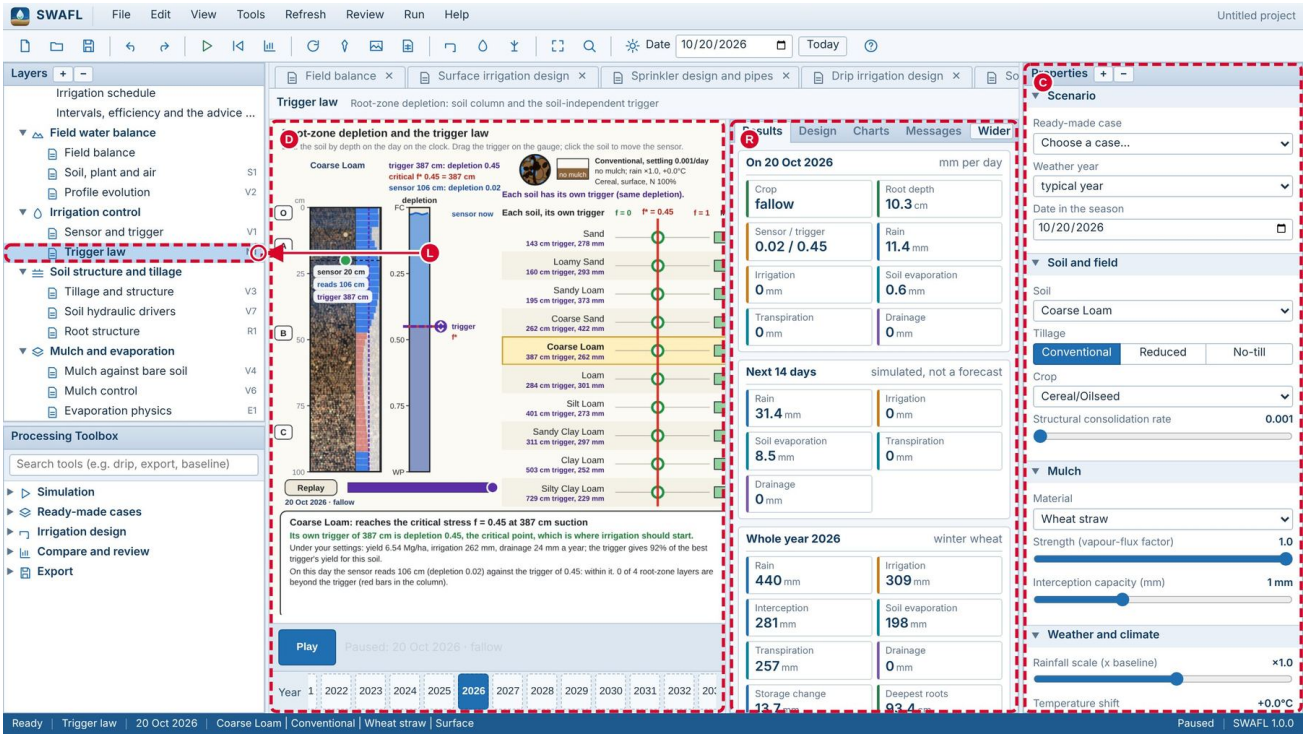


Figure 18.155: Trigger law (N1): the whole workbench.

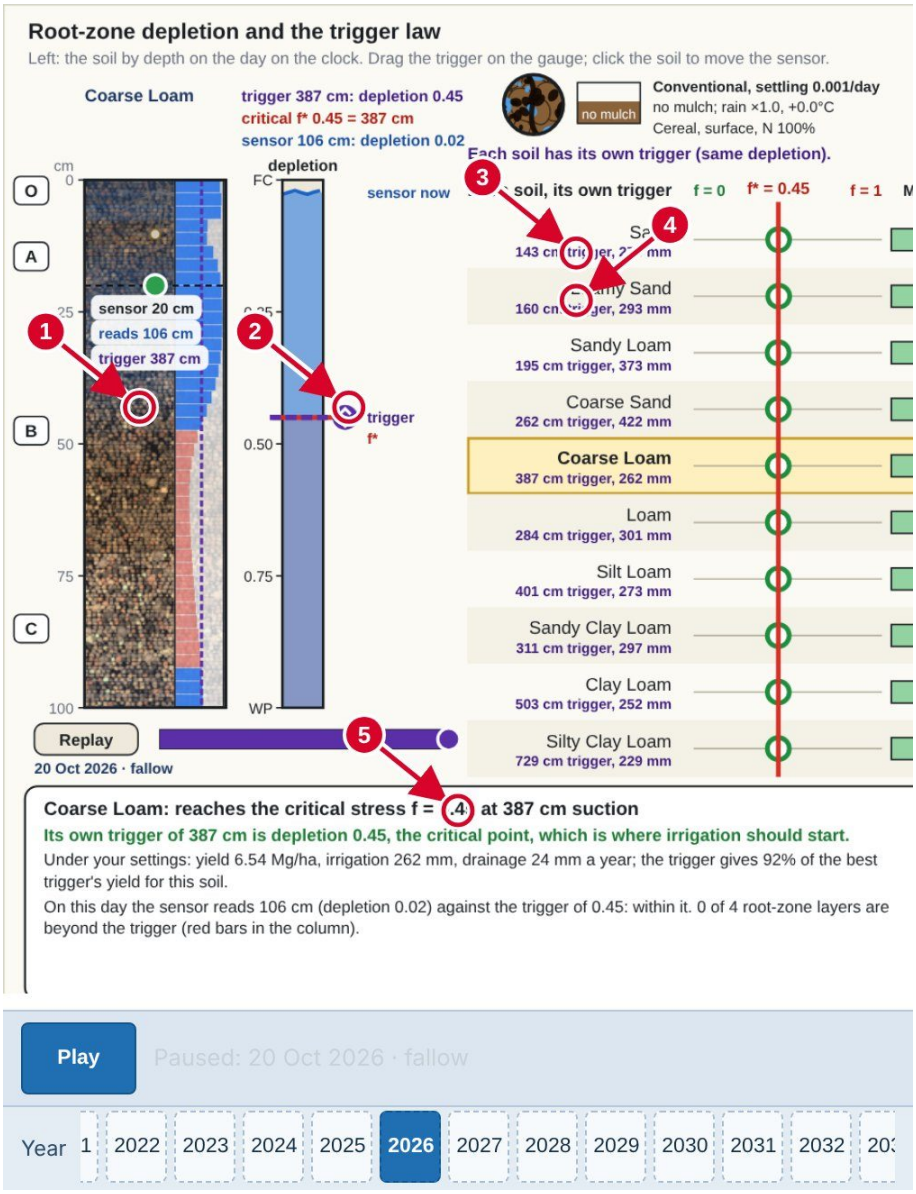


Figure 18.156: Trigger law (N1): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Soil column: each layer coloured by its depletion f, Eq. 5.3 (red = beyond the trigger).
- **2** Critical depletion f* = 0.45, the same for every soil (Eq. 5.3).
- **3** Soil-specific trigger suction h(f*), the inverse of the van Genuchten curve: Eq. 6.3.
- **4** Net irrigation per event Z_n = f AWC R_max: Eq. 6.4.
- **5** Relative yield at the trigger, Y_rel (Eq. 6.5) and the trigger suction (Eq. 6.3) for the selected soil.

Equations behind this screen. Eq. 5.3, Eq. 5.27, Eq. 6.3, Eq. 6.4, Eq. 6.5.

Eq. 5.3. Where explained: chapter “The two computing engines”, section “Plant-available water and depletion”.

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\max};\; 25,\; 150\big)\ \text{mm} \qquad \big(\text{surface; drip and sprinkler use their own limits}\big) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 6.3. Where explained: chapter "Irrigation control layers", section "Governing relations".

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \tag{6.3}$$

Provenance (Eq. 6.3): **[Literature]**: inverse of the van Genuchten curve, solved by bisection.

Eq. 6.4. Where explained: chapter "Irrigation control layers", section "Governing relations".

$$Z_n = f \cdot \text{AWC} \cdot R_{\max} \quad [\text{mm}] \tag{6.4}$$

Provenance (Eq. 6.4): **[Literature]**: same relation as Eq. (5.27) without the clip.

Eq. 6.5. Where explained: chapter "Irrigation control layers", section "Governing relations".

$$Y_{\text{rel}} = \frac{Y(h_T)}{\max_h Y(h)}, \tag{6.5}$$

Provenance (Eq. 6.5): **[Data-driven]**: ratio of Daisy run-set E2 yields.

18.9.1 The whole layer, part by part

The picture below is the complete drawing of the layer Trigger law (N1), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 1 part, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

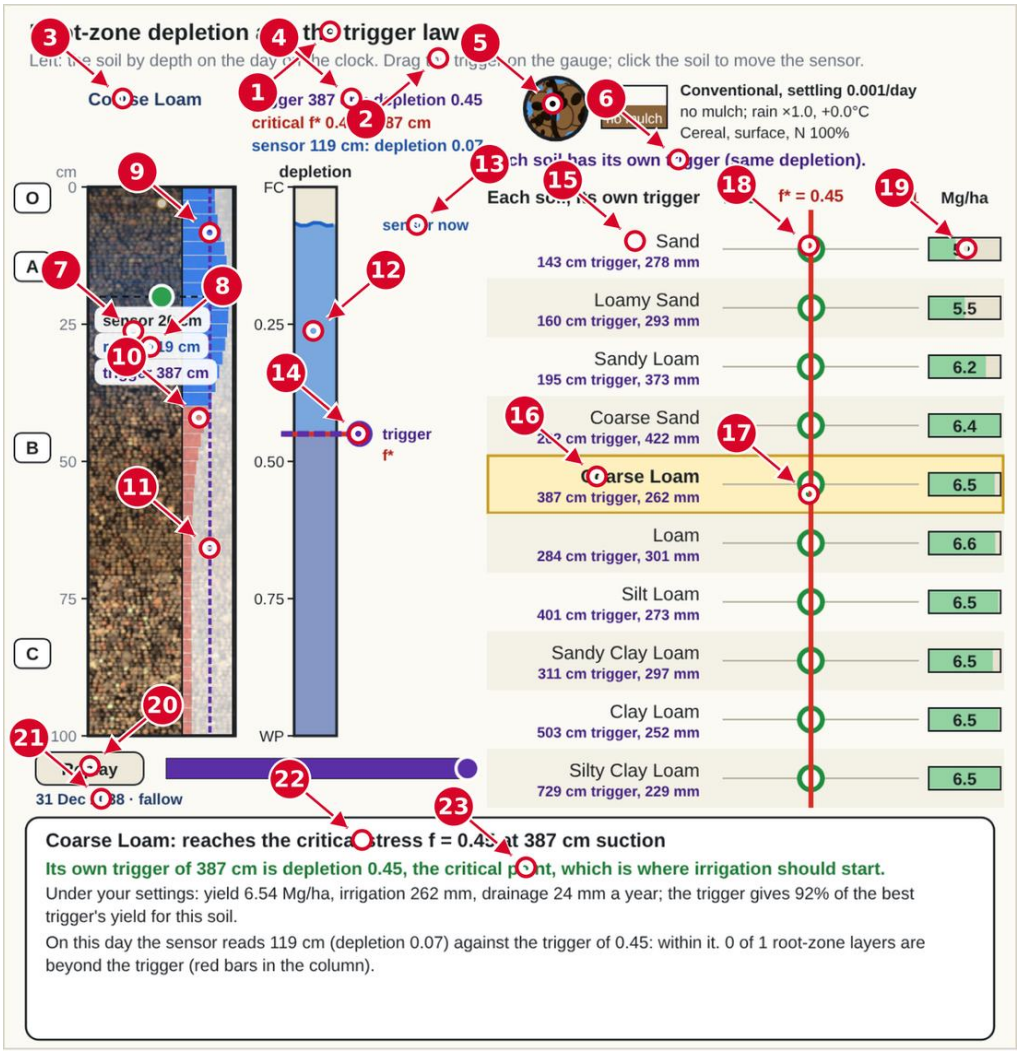


Figure 18.157: Trigger law (N1), part 1 of 1.

What the arrows point to.

- **1** The title, the depletion of the root zone and the trigger law.
- **2** The instruction line, drag the trigger on the gauge and click the soil to move the sensor.
- **3** The soil whose column and gauge are shown.
- **4** The readout of the trigger, the critical depletion f* and the sensor depletion on the day.
- **5** The scenario in force, the tillage, mulch, rain scaling, warming, crop and nitrogen.
- **6** The statement that each soil has its own trigger at the same depletion.
- **7** The soil column by depth with the layers of the root zone.
- **8** The sensor tag, its depth, its reading and the trigger in cm of suction.
- **9** The blue bars are the layers inside the trigger on the day shown.
- **10** The red bars are the layers beyond the trigger.
- **11** The dashed line is the trigger carried down the column.
- **12** The depletion gauge from field capacity (FC) to wilting point (WP).
- **13** The sensor level on the gauge today.
- **14** The handle of the trigger, which is dragged to another depletion.
- **15** The soils ranked in the list, each with its own trigger in cm and its seasonal irrigation.
- **16** The soil selected, highlighted in the list.
- **17** The critical depletion f* as the red line across all ten soils.
- **18** The ring on each soil gives where its own trigger sits between f = 0 and f = 1.

- **19** The simulated yield of each soil at its own trigger, in Mg per ha.
- **20** The Replay button and the time slider that replay the season.
- **21** The date shown on the clock.
- **22** The conclusion, the stress at which the selected soil needs water.
- **23** The sentence that links the soil's own trigger to the critical point.

18.9.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

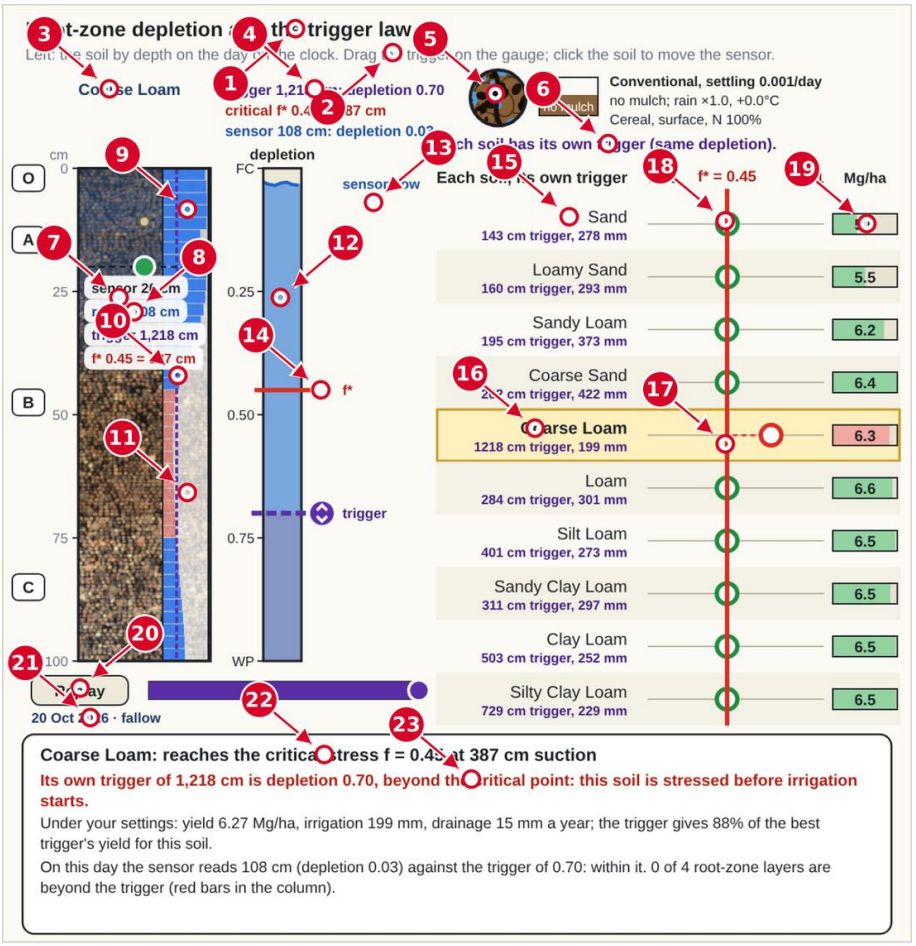


Figure 18.158: Trigger law (N1), part 1. The trigger handle dragged to a deeper depletion on the gauge, so the trigger moves and the number of layers beyond it changes.

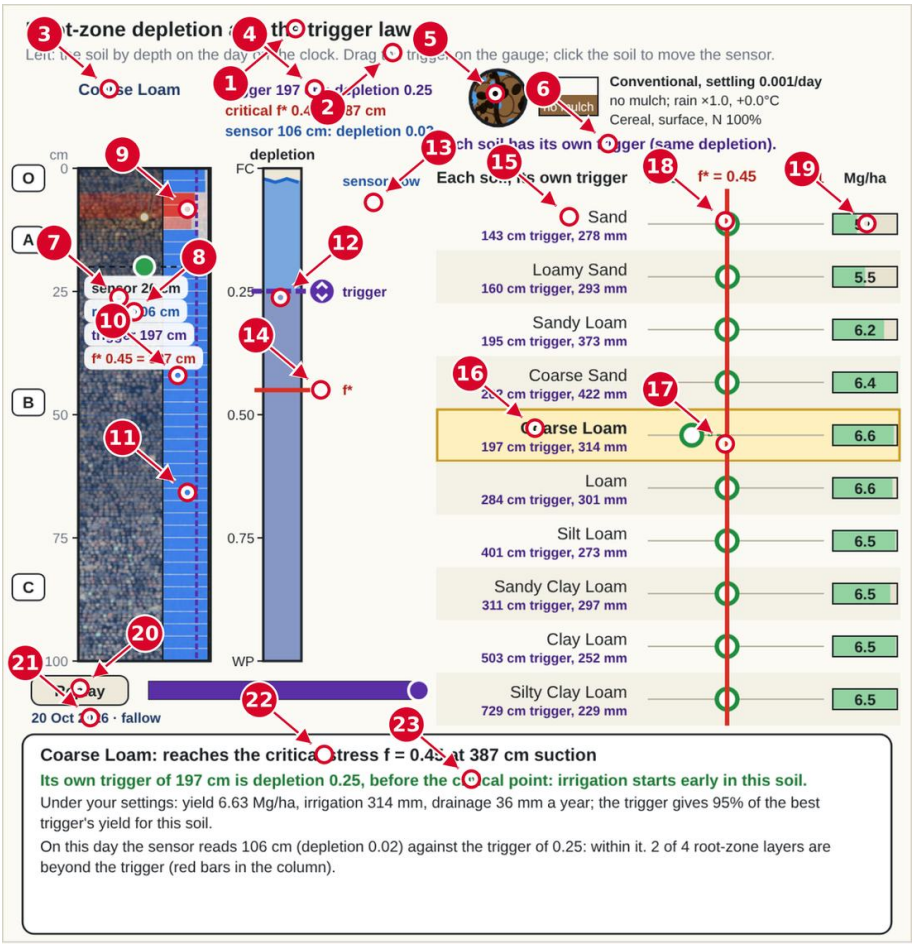


Figure 18.159: Trigger law (N1), part 1. The trigger handle dragged to a shallow depletion on the gauge.

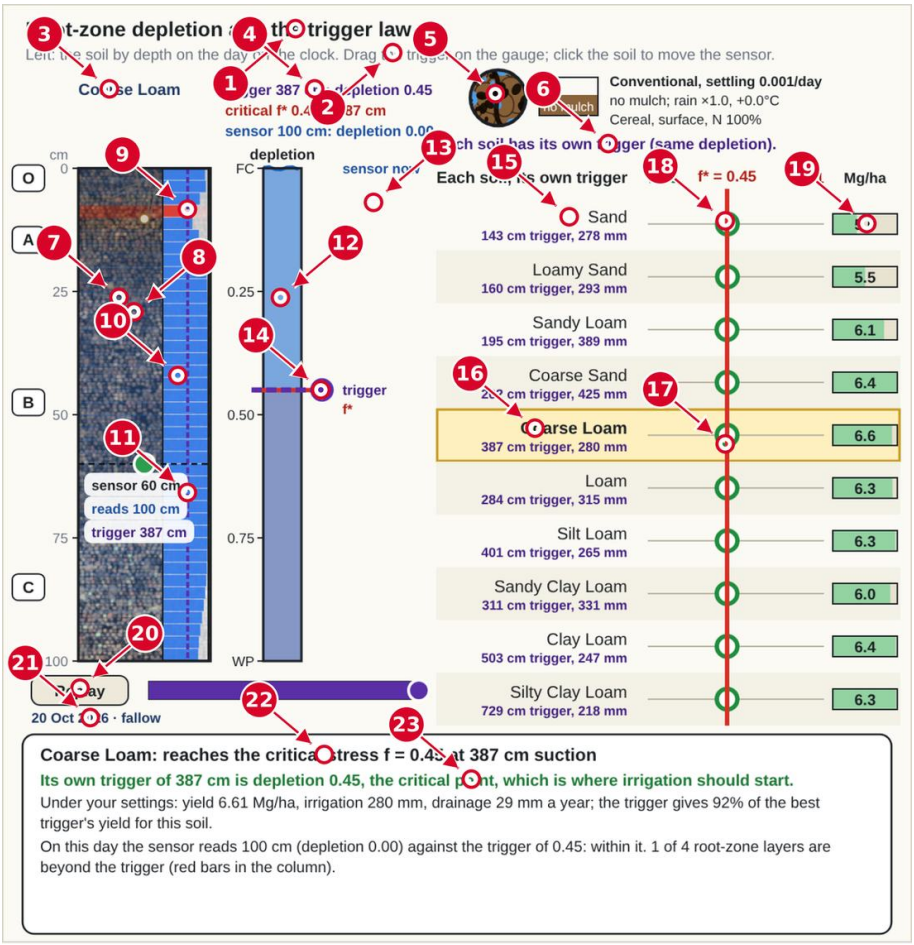


Figure 18.160: Trigger law (N1), part 1. The soil column clicked at 60 cm, so the sensor moves to that depth.

18.10 Tillage and structure (V3)

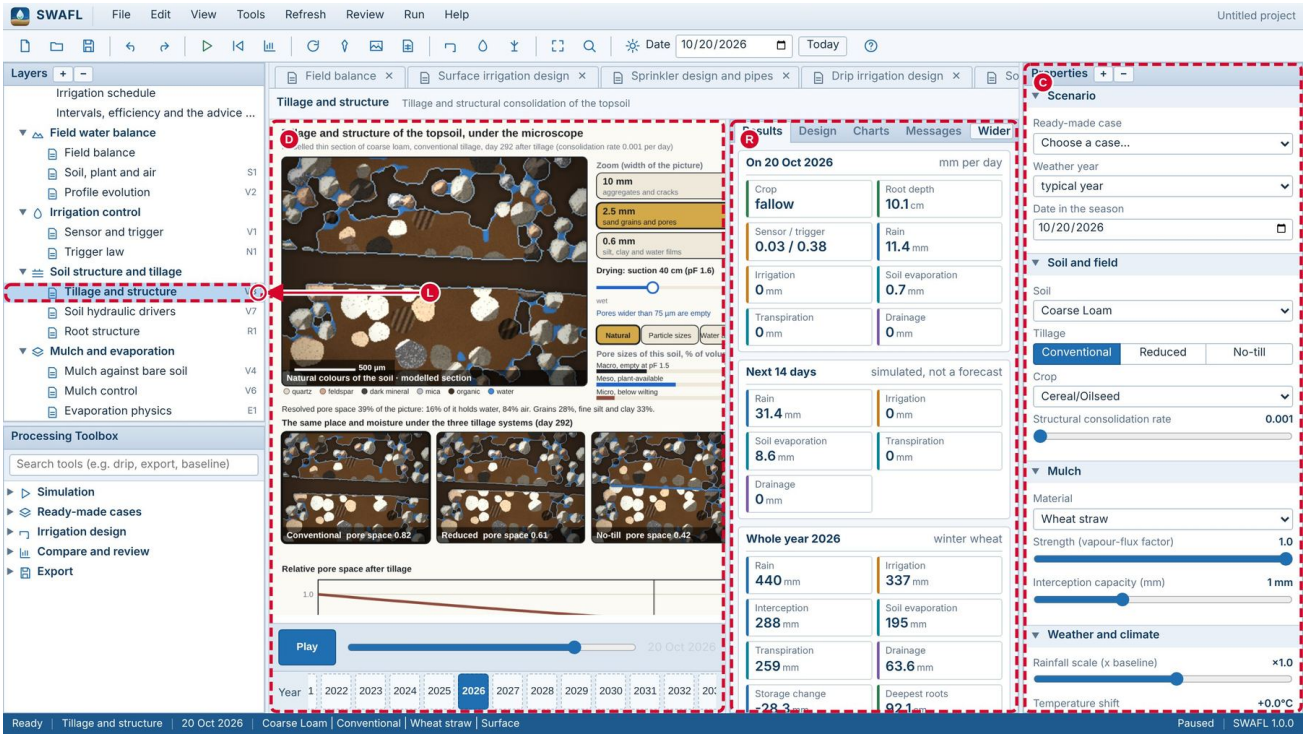


Figure 18.161: Tillage and structure (V3): the whole workbench.

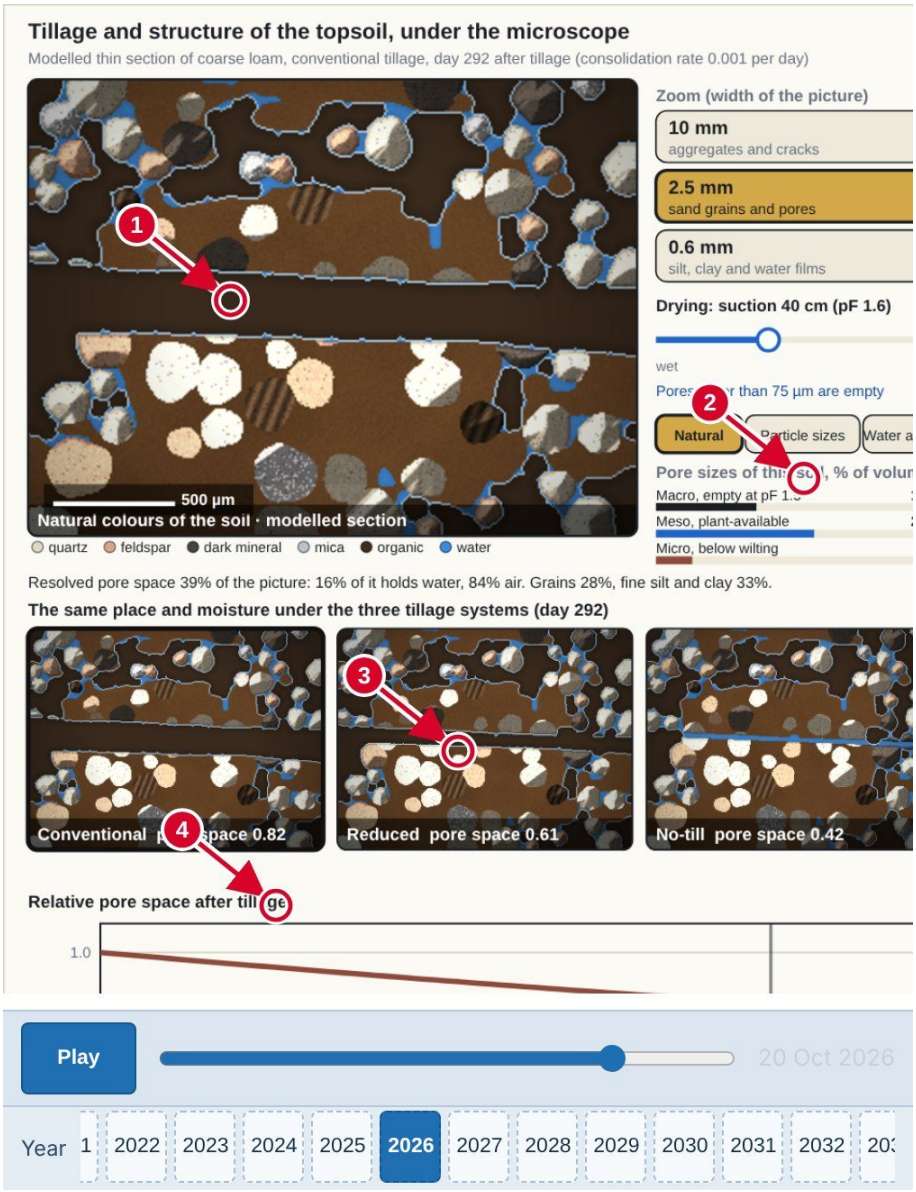


Figure 18.162: Tillage and structure (V3): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Modelled thin section of the topsoil; the share of open pores follows the pore index p(t), Eq. 8.1.
- **2** Pore-size classes from the retention curve, theta(h): Eqs. 8.3 and 8.4 (capillary radius).
- **3** The same place under the three tillage systems; pore space 0.82, 0.61 and 0.42 from Eq. 8.1.
- **4** Relative pore space after tillage, p(t) = p_c + (p_0 - p_c) exp(-tau_c t): Eq. 8.1.

Equations behind this screen. Eq. 5.1, Eq. 8.1, Eq. 8.2, Eq. 8.3, Eq. 8.4.

Eq. 5.1. Where explained: chapter “The two computing engines”, section “The simulation campaign”.

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 8.1. Where explained: chapter "Soil structure and tillage layers", section "Pore index and consolidation".

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

Eq. 8.2. Where explained: chapter "Soil structure and tillage layers", section "How the section is generated".

$$\phi = \text{clip}\big(0.36 + 0.0016\,\text{clay} + 0.0009\,\text{silt};\ 0.34,\ 0.50\big), \qquad \theta_r = 0.01 + 0.0006\,\text{clay}, \tag{8.2}$$

Provenance (Eq. 8.2): **[Engineered]**: porosity relation with chosen coefficients.

Eq. 8.3. Where explained: chapter "Soil structure and tillage layers", section "How the section is generated".

$$\theta(h) = \theta_r + (\phi - \theta_r)\, S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

Eq. 8.4. Where explained: chapter "Soil structure and tillage layers", section "How the section is generated".

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\, g\, h} \;\approx\; \frac{0.149\ \text{cm}^2}{h} \quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

18.10.1 The whole layer, part by part

The picture below is the complete drawing of the layer Tillage and structure (V3), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 2 parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

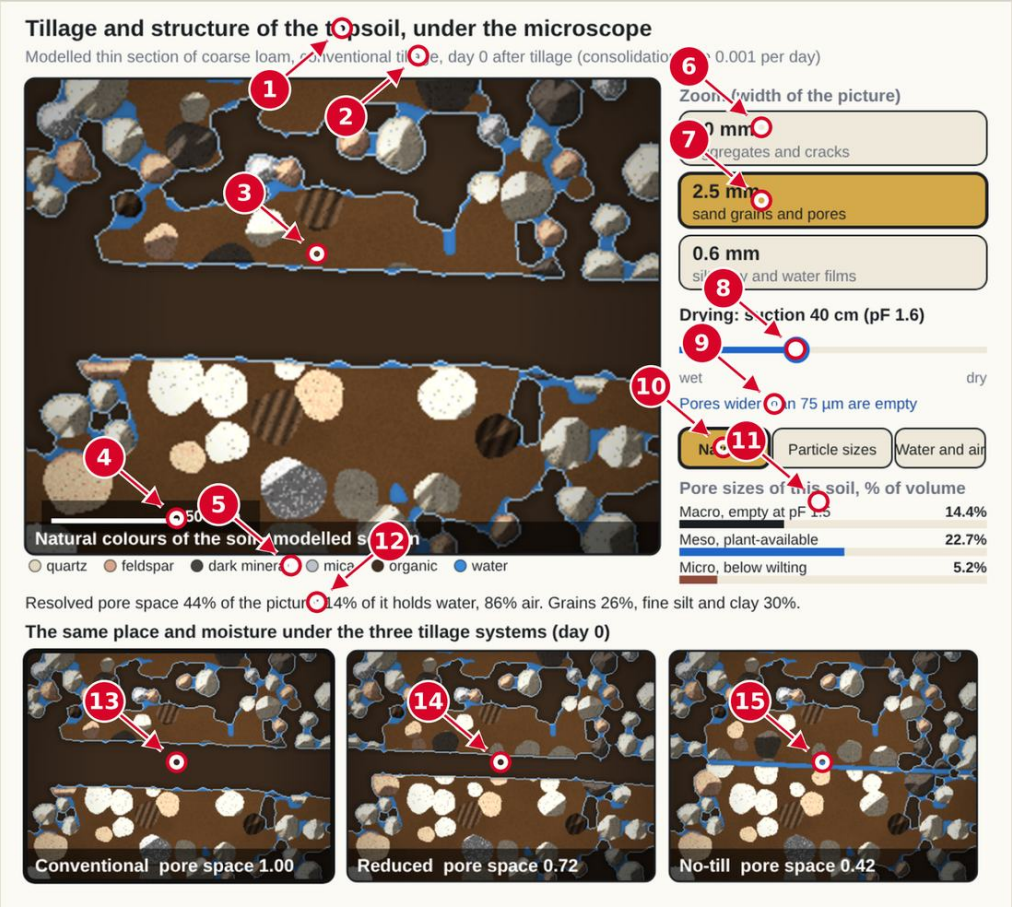


Figure 18.163: Tillage and structure (V3), part 1 of 2.

What the arrows point to.

- **1** The title, tillage and the structure of the topsoil under the microscope.
- **2** The scenario of the section, the soil, the tillage, the day after tillage and the consolidation rate.
- **3** The modelled thin section, with grains, pores and water films drawn at the chosen zoom.
- **4** The scale bar of the section.
- **5** The legend of the colours, quartz, feldspar, dark mineral, mica, organic matter and water.
- **6** The Zoom buttons, 10 mm, 2.5 mm and 0.6 mm wide pictures.
- **7** The zoom selected, here the 2.5 mm picture of sand grains and pores.
- **8** The drying slider, which sets the suction and so which pores hold water.
- **9** The note on the pore width that empties at this suction.
- **10** The view buttons, Natural, Particle sizes and Water and air.
- **11** The pore-size classes of this soil, macro, meso and micro, in percent of the volume.
- **12** The resolved pore space of the picture and the share of it holding water.
- **13** The conventional tillage tile, with its relative pore space of 1.00.
- **14** The reduced tillage tile, with a smaller relative pore space.
- **15** The no-till tile, with the smallest relative pore space.

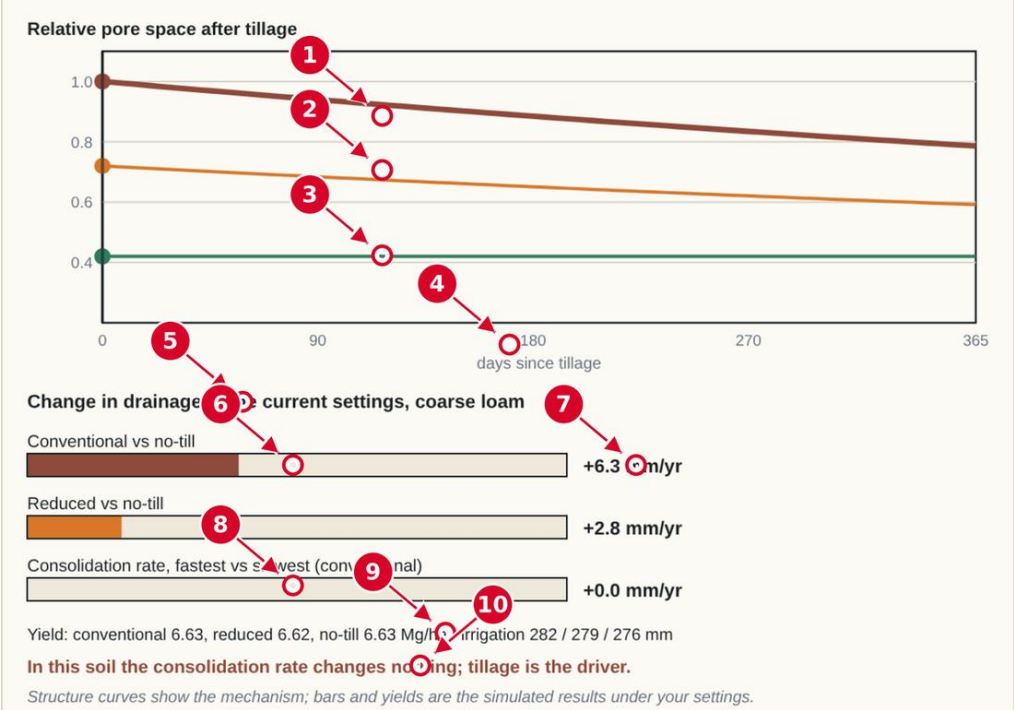


Figure 18.164: Tillage and structure (V3), part 2 of 2.

What the arrows point to.

- **1** The conventional line, which starts at 1.0 and consolidates with time.
- **2** The reduced tillage line.
- **3** The no-till line, flat because the soil is not loosened.
- **4** The time axis in days since tillage.
- **5** The change in drainage at the current settings.
- **6** The bar of drainage for conventional against no-till tillage.
- **7** The value of that difference in mm per year.
- **8** The bar for the consolidation rate, fastest against slowest.
- **9** The yield and the irrigation of the three tillage systems.
- **10** The conclusion, which factor drives the drainage in this soil.

18.10.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

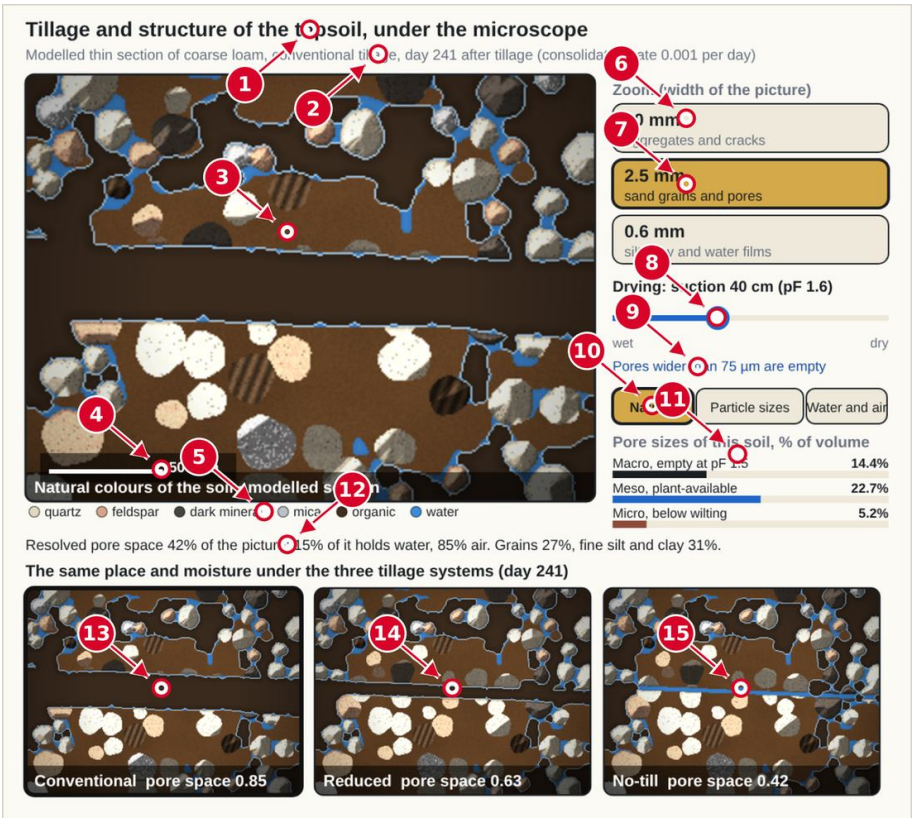


Figure 18.165: Tillage and structure (V3), part 1, playback two thirds of the way (day 241).

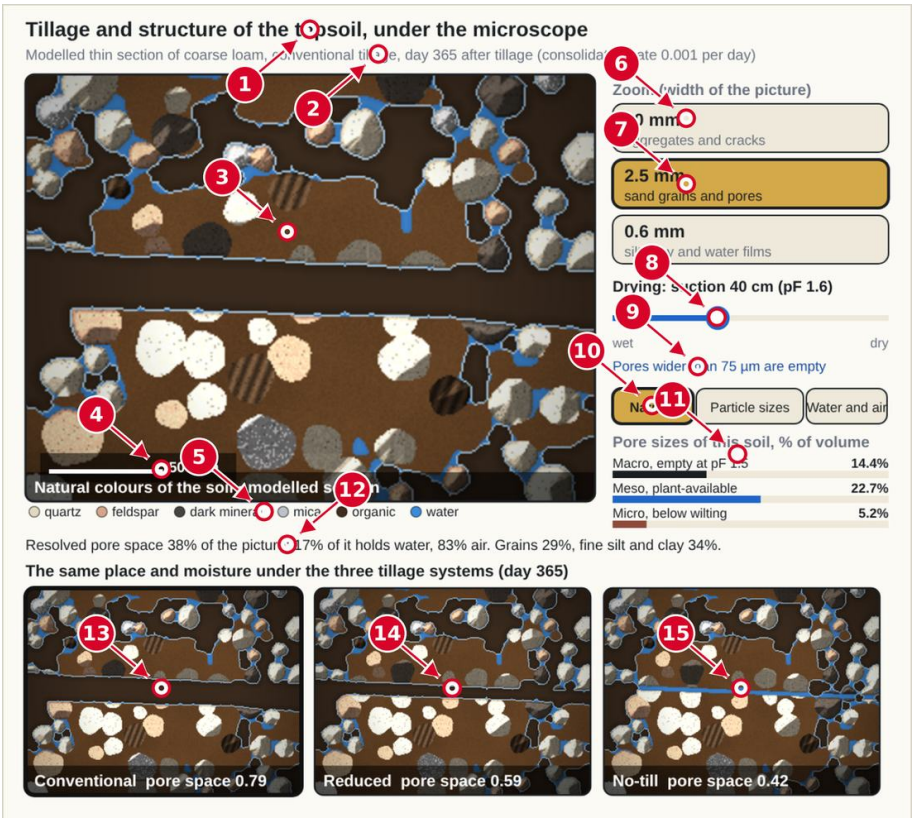


Figure 18.166: Tillage and structure (V3), part 1, playback the end (day 365).

18.10.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

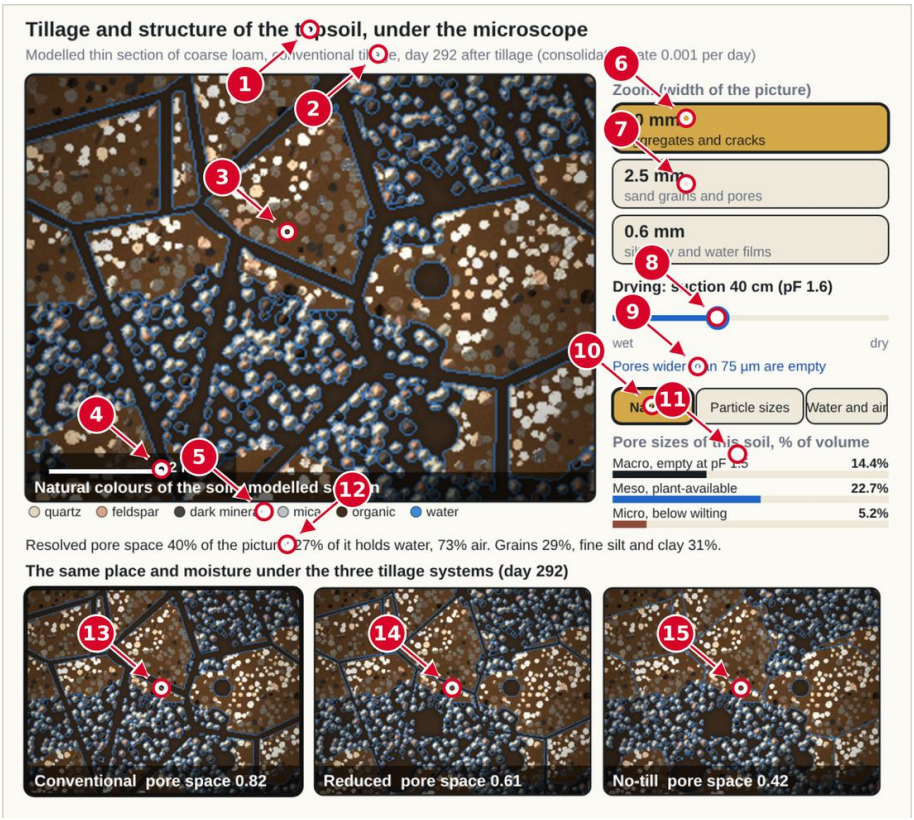


Figure 18.167: Tillage and structure (V3), part 1. The 10 mm zoom chosen, so the picture shows aggregates and cracks.

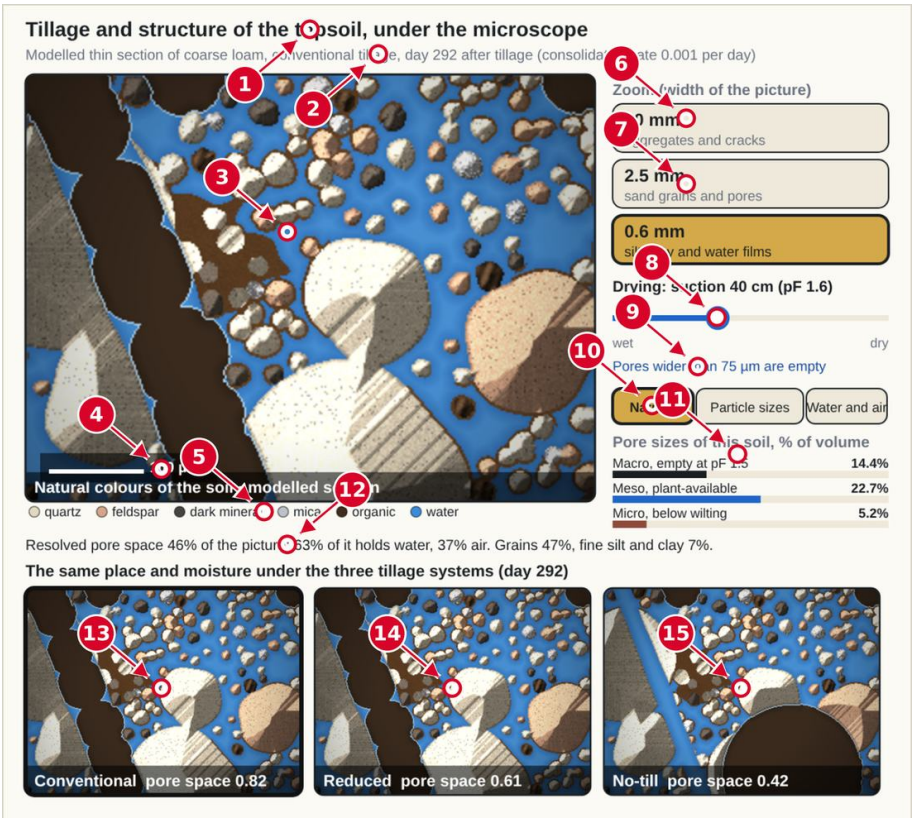


Figure 18.168: Tillage and structure (V3), part 1. The 0.6 mm zoom chosen, so the picture shows silt, clay and the water films.

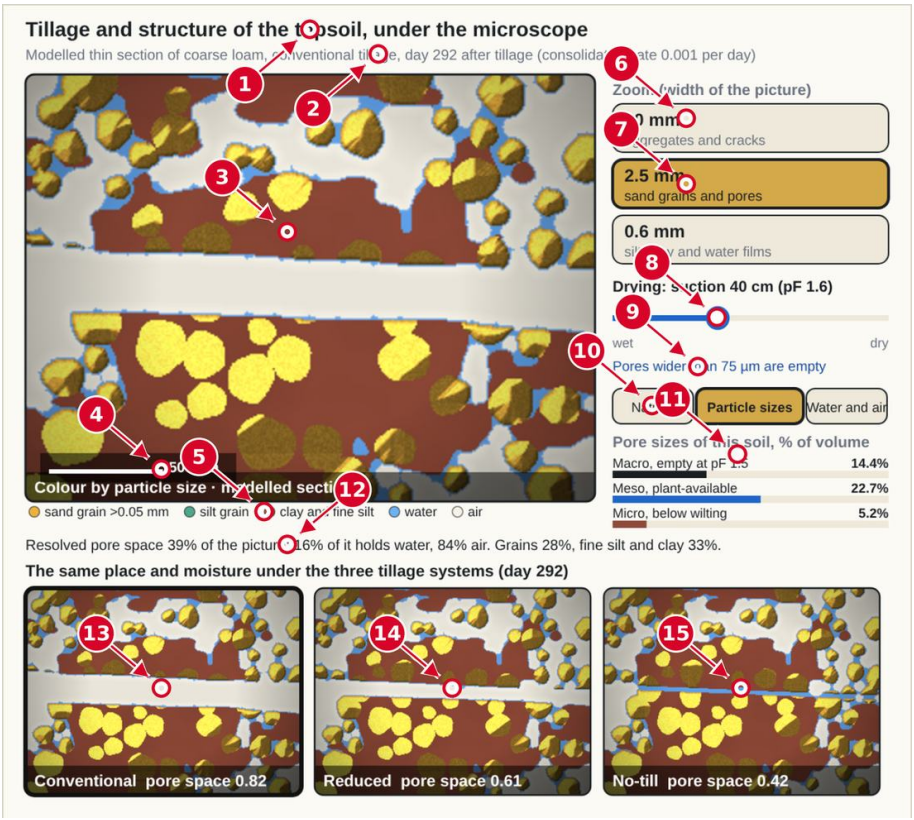


Figure 18.169: Tillage and structure (V3), part 1. The Particle sizes view chosen, so the grains are coloured by size.

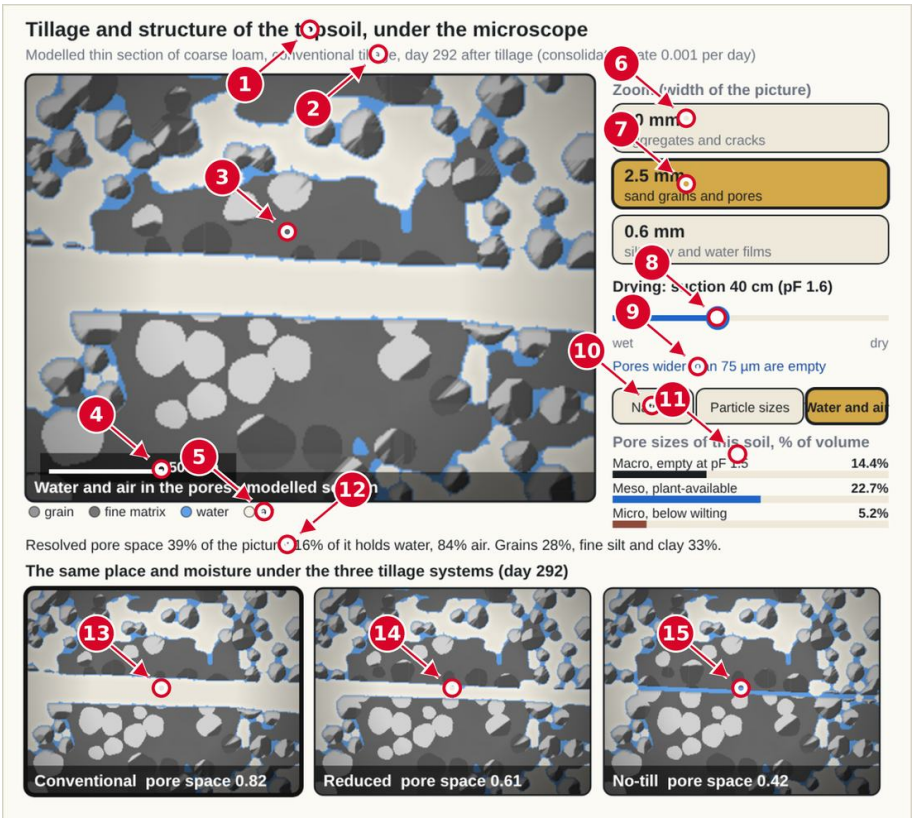


Figure 18.170: Tillage and structure (V3), part 1. The Water and air view chosen, so the pores are coloured by whether they hold water or air.

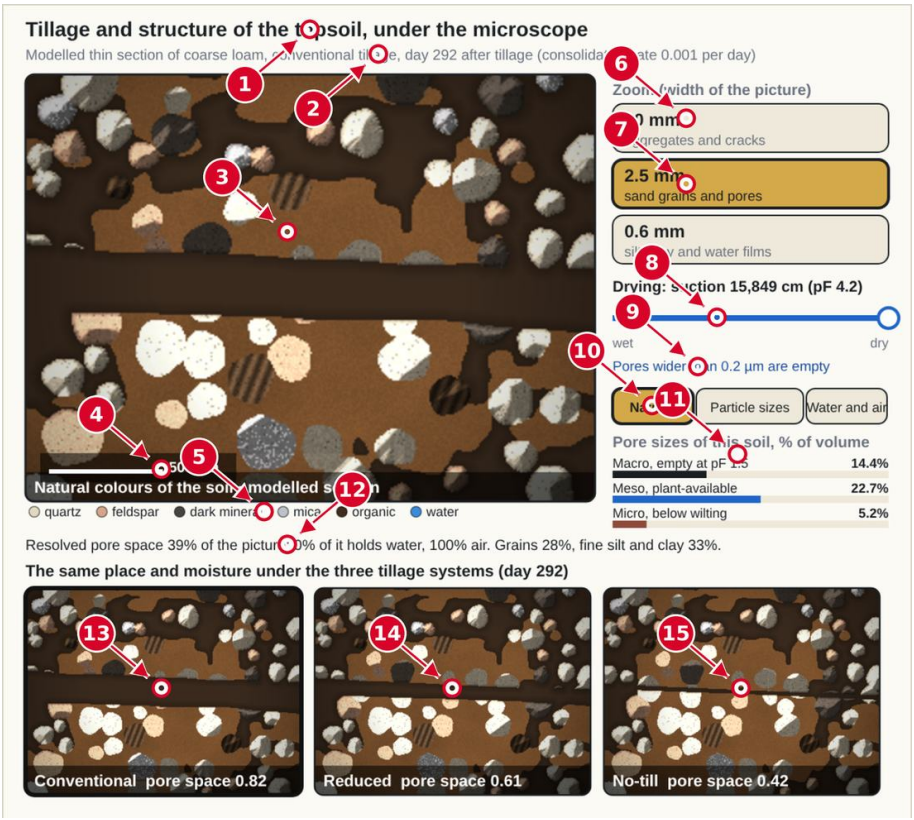


Figure 18.171: Tillage and structure (V3), part 1. The drying slider dragged to the dry end, so most pores are empty.

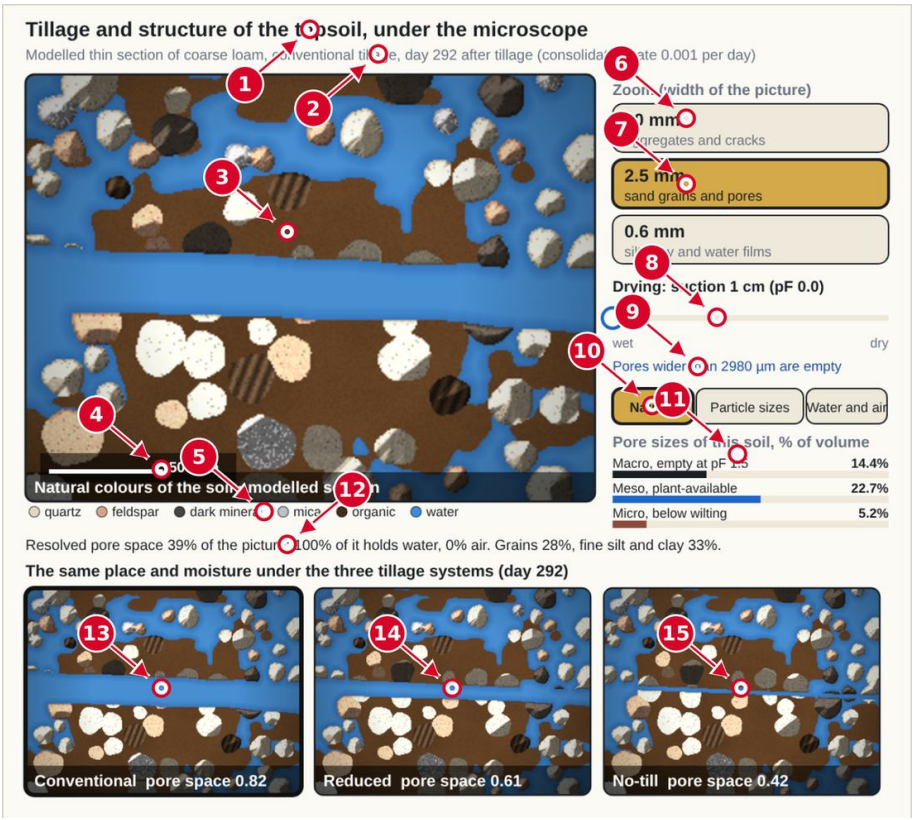


Figure 18.172: Tillage and structure (V3), part 1. The drying slider dragged to the wet end, so most pores hold water.

18.11 Soil hydraulic drivers (V7)

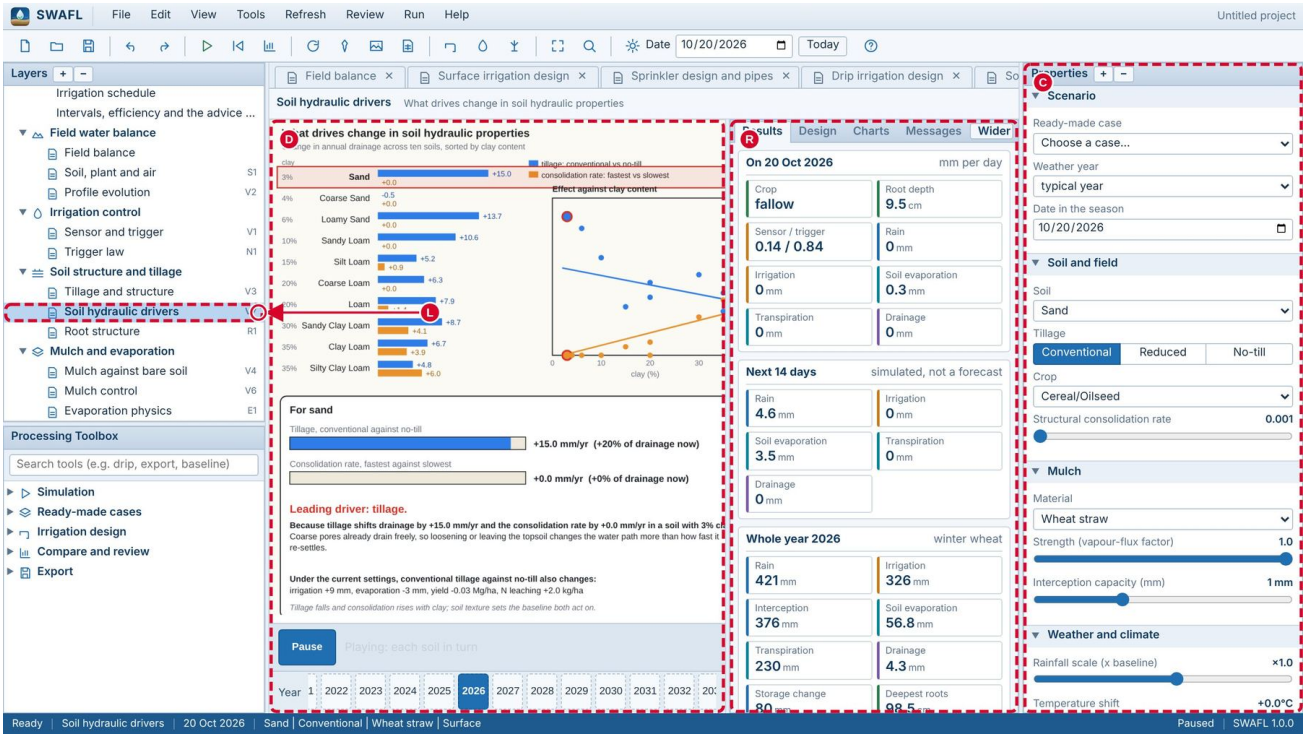


Figure 18.173: Soil hydraulic drivers (V7): the whole workbench.

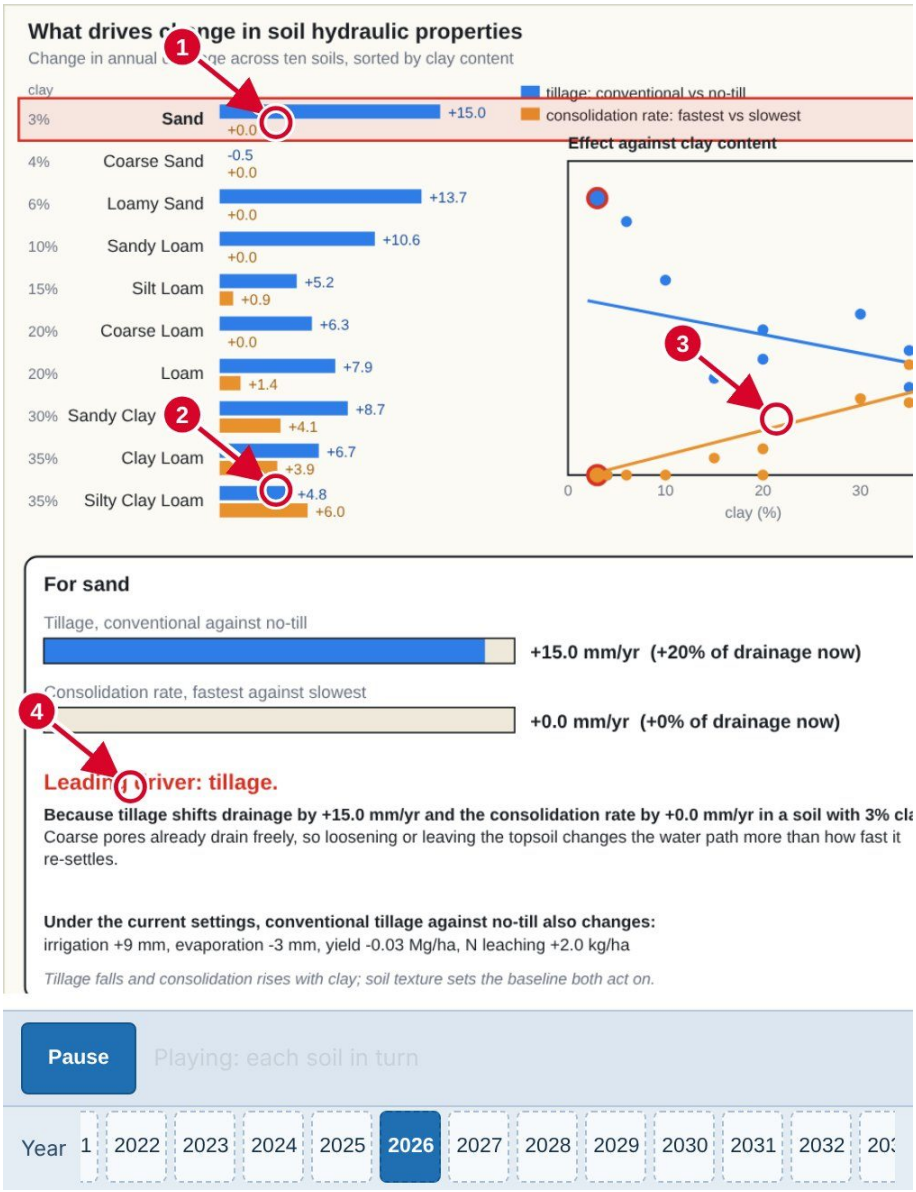


Figure 18.174: Soil hydraulic drivers (V7): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Blue bar: effect of tillage on drainage, Delta_till = D(conventional) - D(no-till): Eq. 8.5.
- **2** Orange bar: effect of the consolidation rate on drainage, Delta_cons: Eq. 8.5.
- **3** Scatter of each effect against clay content with the least-squares fit: Eq. 8.6.
- **4** “Leading driver”: the larger of Delta_till and Delta_cons for the selected soil, Eq. 8.5.

Equations behind this screen. Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 8.1, Eq. 8.5, Eq. 8.6.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".

$$Q \leftarrow Q\,\text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.$$

(5.11)

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 8.1. Where explained: chapter "Soil structure and tillage layers", section "Pore index and consolidation".

$$p(t) = p_c + (p_0 - p_c)\,e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

Eq. 8.5. Where explained: chapter "Soil structure and tillage layers", section "Generation".

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*), \qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \tag{8.5}$$

Provenance (Eq. 8.5): **[Data-driven]**: differences of Daisy run-set E5 results.

Eq. 8.6. Where explained: chapter "Soil structure and tillage layers", section "Generation".

$$\hat{\Delta} = a + b\,c, \qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2}, \quad a = \bar{\Delta} - b\,\bar{c}. \tag{8.6}$$

Provenance (Eq. 8.6): **[Data-driven]**: ordinary least squares applied to Daisy results.

18.11.1 The whole layer, part by part

The picture below is the complete drawing of the layer Soil hydraulic drivers (V7), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 3 parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

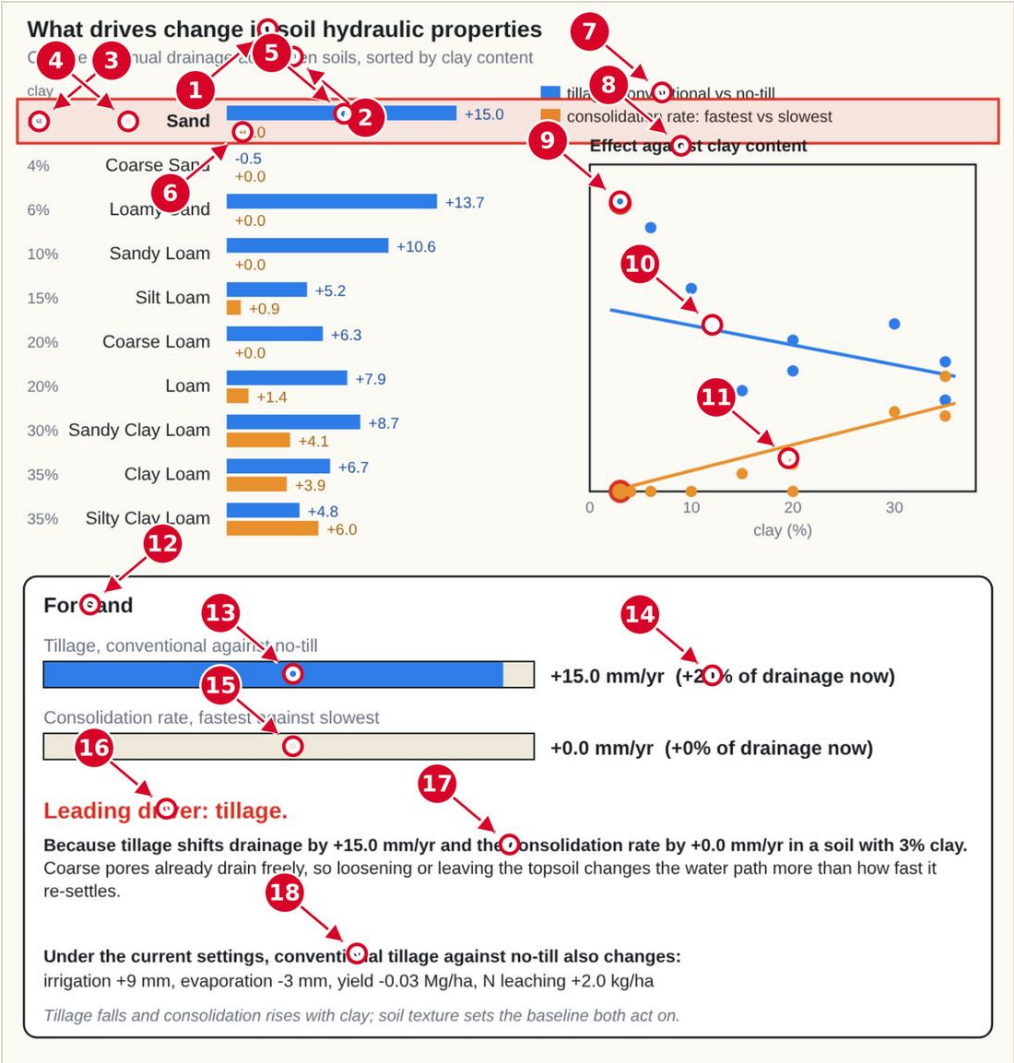


Figure 18.175: Soil hydraulic drivers (V7), part 1 of 3.

What the arrows point to.

- **1** The title, what drives the change in the hydraulic properties of the soil.
- **2** The subtitle, the change in annual drainage across ten soils sorted by clay content.
- **3** The clay content of each soil, the sorting key.
- **4** The soil highlighted in the list, here sand, which every panel below refers to.
- **5** The blue bar, the change in drainage from tillage, conventional against no-till.
- **6** The orange bar, the change in drainage from the consolidation rate.
- **7** The legend of the two bars.
- **8** The scatter chart of the two effects against the clay content.
- **9** The highlighted soil on the scatter chart.
- **10** The blue trend line, the tillage effect falls with clay.
- **11** The orange trend line, the consolidation effect rises with clay.
- **12** The card for the selected soil, with the two effects in mm per year.

- **13** The bar of the tillage effect against its largest value.
- **14** The value of the tillage effect and its share of the drainage now.
- **15** The bar of the consolidation effect.
- **16** The leading driver, tillage or the consolidation rate.
- **17** The reasoning behind the leading driver for this clay content.
- **18** The other changes of the conventional tillage against no-till, irrigation, evaporation, yield and nitrogen leaching.

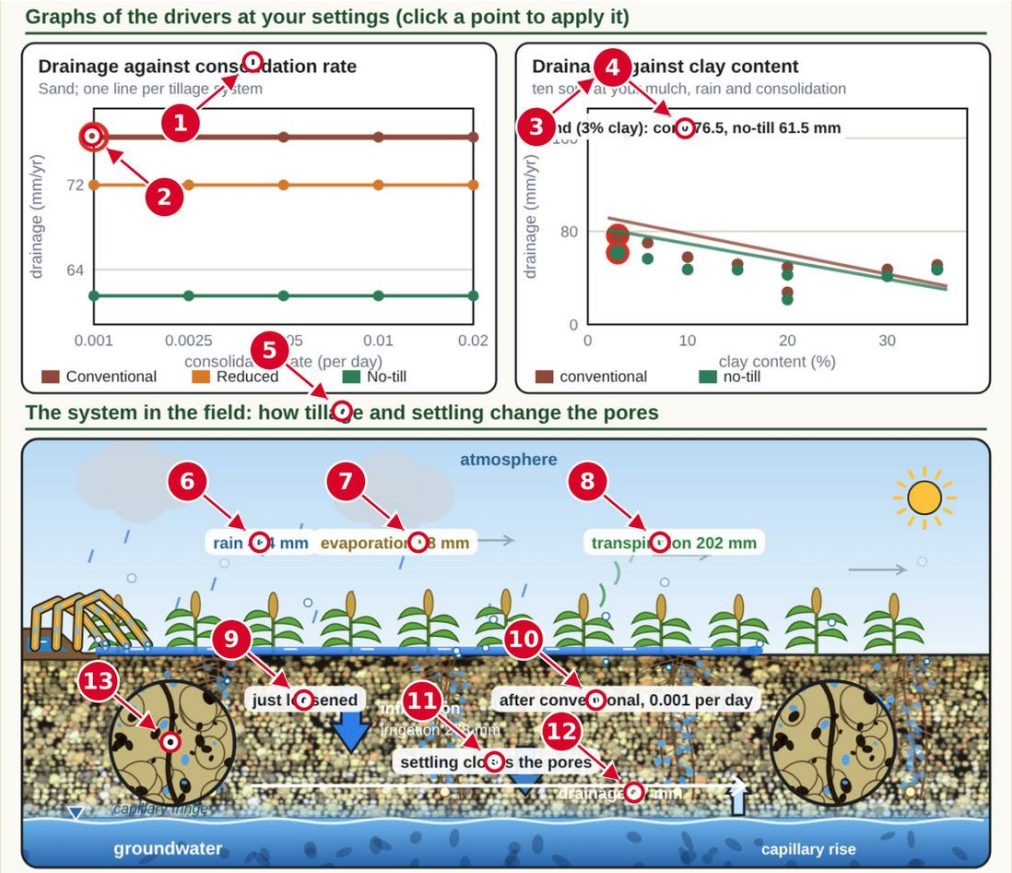


Figure 18.176: Soil hydraulic drivers (V7), part 2 of 3.

What the arrows point to.

- **1** The graph of the drainage against the consolidation rate, one line per tillage system.
- **2** A point of the graph, which applies that value when it is clicked.
- **3** The graph of the drainage against the clay content of the ten soils.
- **4** The readout of the soil selected, drainage of the conventional and no-till systems.
- **5** The system in the field, how tillage and settling change the pores.
- **6** The rain of the year.
- **7** The evaporation of the year.
- **8** The transpiration of the year.
- **9** The loosened layer just after the tillage.
- **10** The state of the layer after tillage at the consolidation rate in force.
- **11** The settling that closes the pores with time.
- **12** The drainage through the profile, from the Daisy simulation.
- **13** The aggregate drawn inside the tilled layer.

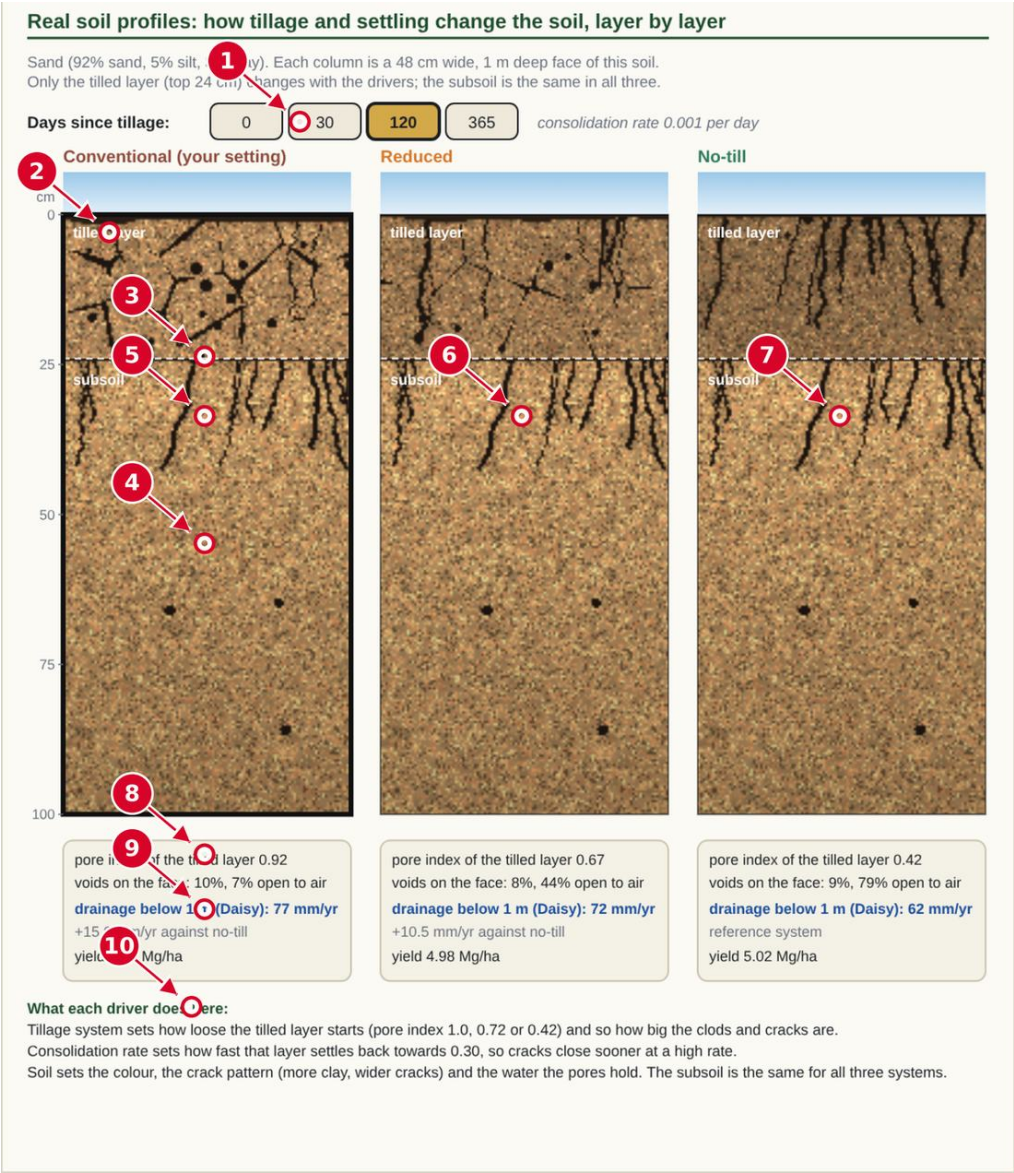


Figure 18.177: Soil hydraulic drivers (V7), part 3 of 3.

What the arrows point to.

- **1** The buttons of the days since tillage, 0, 30, 120 and 365.
- **2** The tilled layer, the top 24 cm, the only layer that changes.
- **3** The boundary of the tilled layer and the subsoil at 25 cm.
- **4** The subsoil, the same in all three systems.
- **5** The conventional column, with the largest clods and cracks.
- **6** The reduced tillage column.
- **7** The no-till column, with fine cracks near the surface.
- **8** The pore index, the voids on the face and the share open to air.
- **9** The drainage below 1 m from the Daisy simulation, against no-till.
- **10** What each driver does, the tillage, the consolidation rate and the soil.

18.11.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

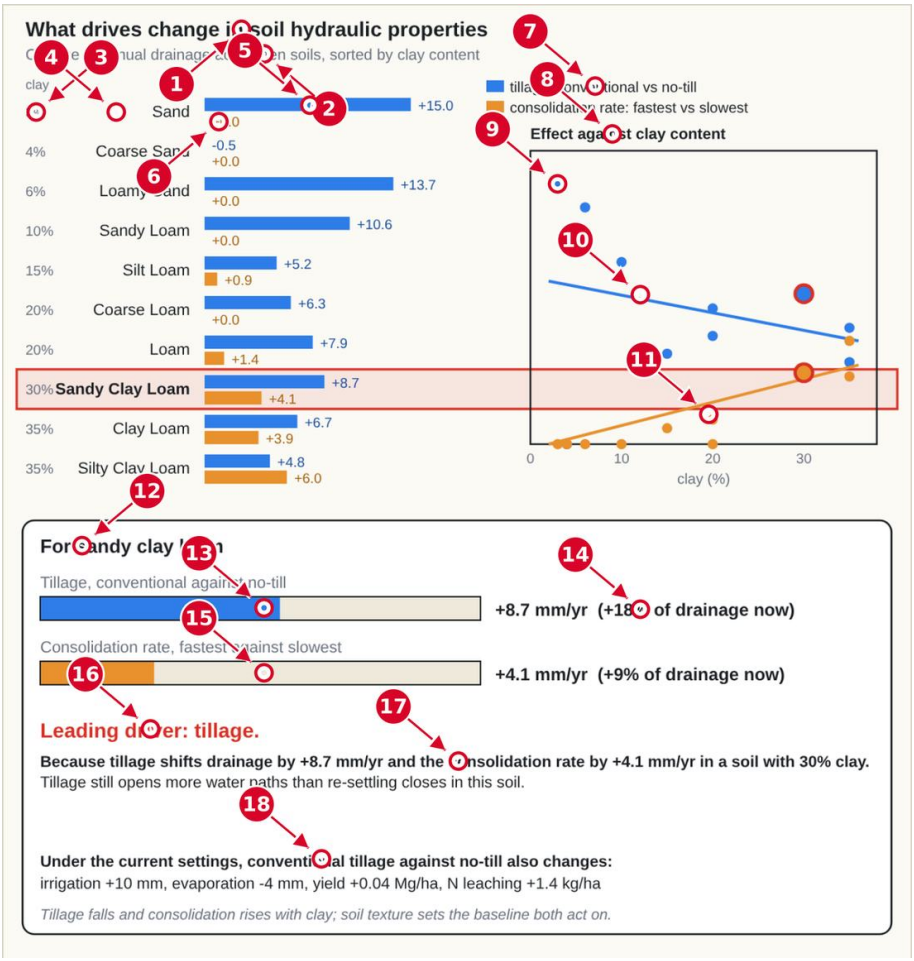


Figure 18.178: Soil hydraulic drivers (V7), part 1. A different soil clicked in the list, so every panel refers to that soil.

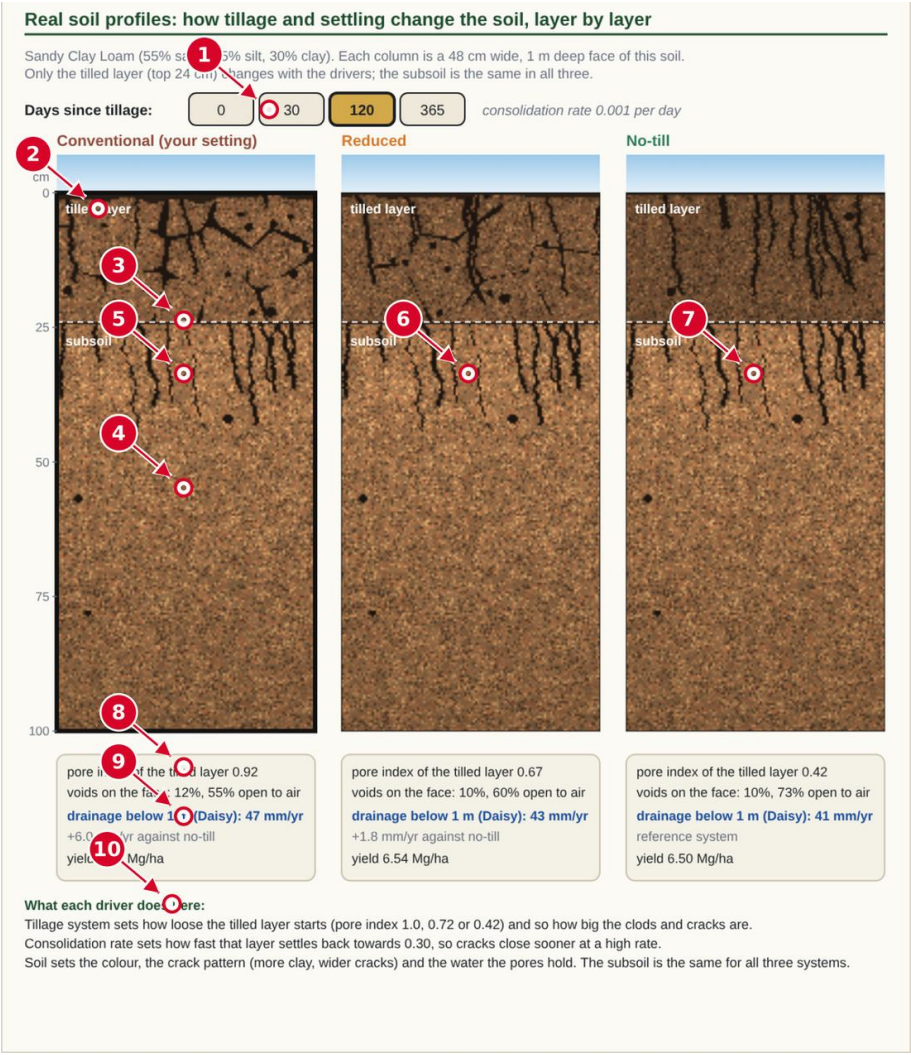


Figure 18.179: Soil hydraulic drivers (V7), part 3. A different soil clicked in the list, so every panel refers to that soil.

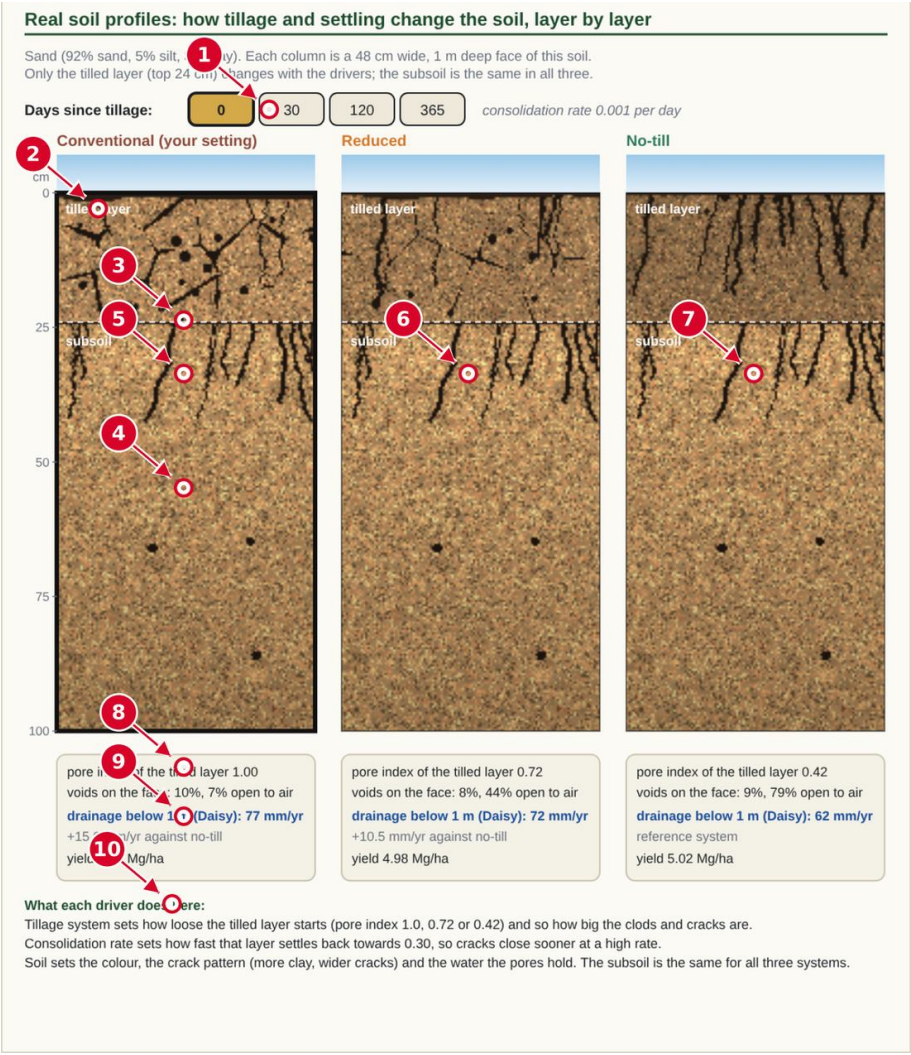


Figure 18.180: Soil hydraulic drivers (V7), part 3. The 0 days since tillage button pressed, so the tilled layer is shown just after tillage.

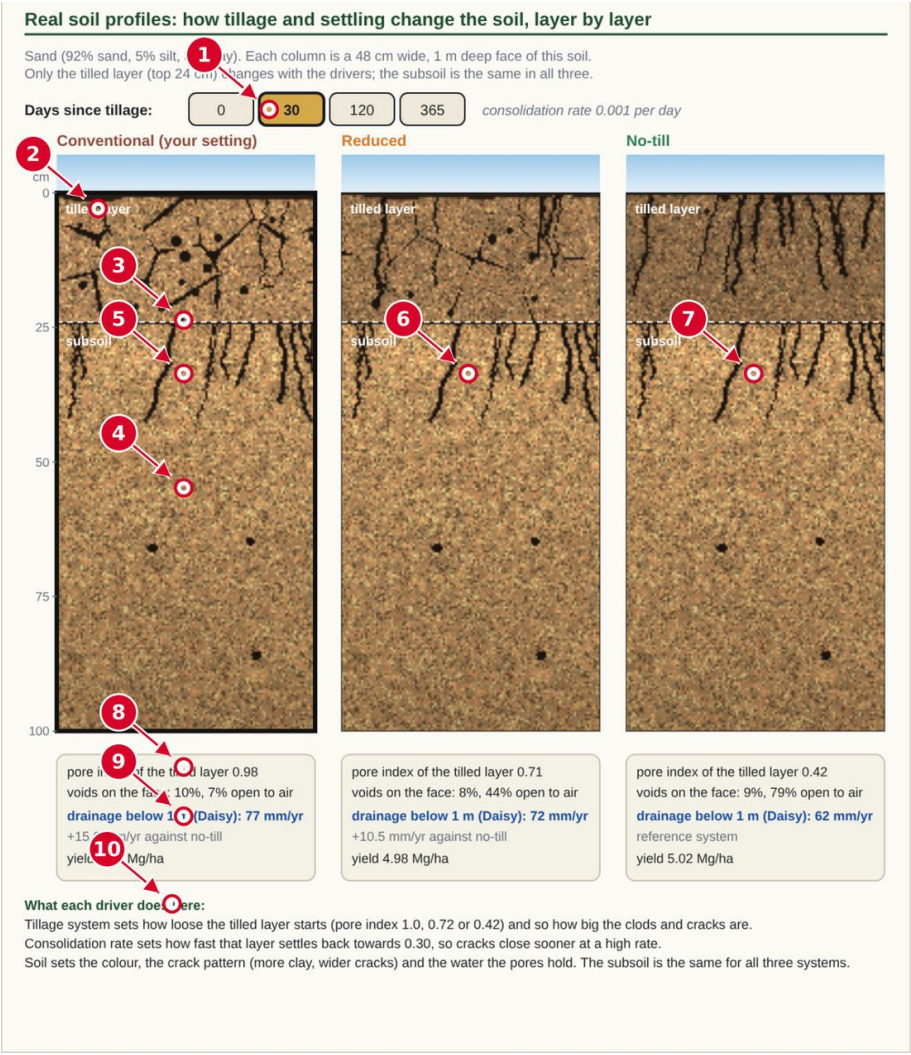


Figure 18.181: Soil hydraulic drivers (V7), part 3. The 30 days since tillage button pressed.

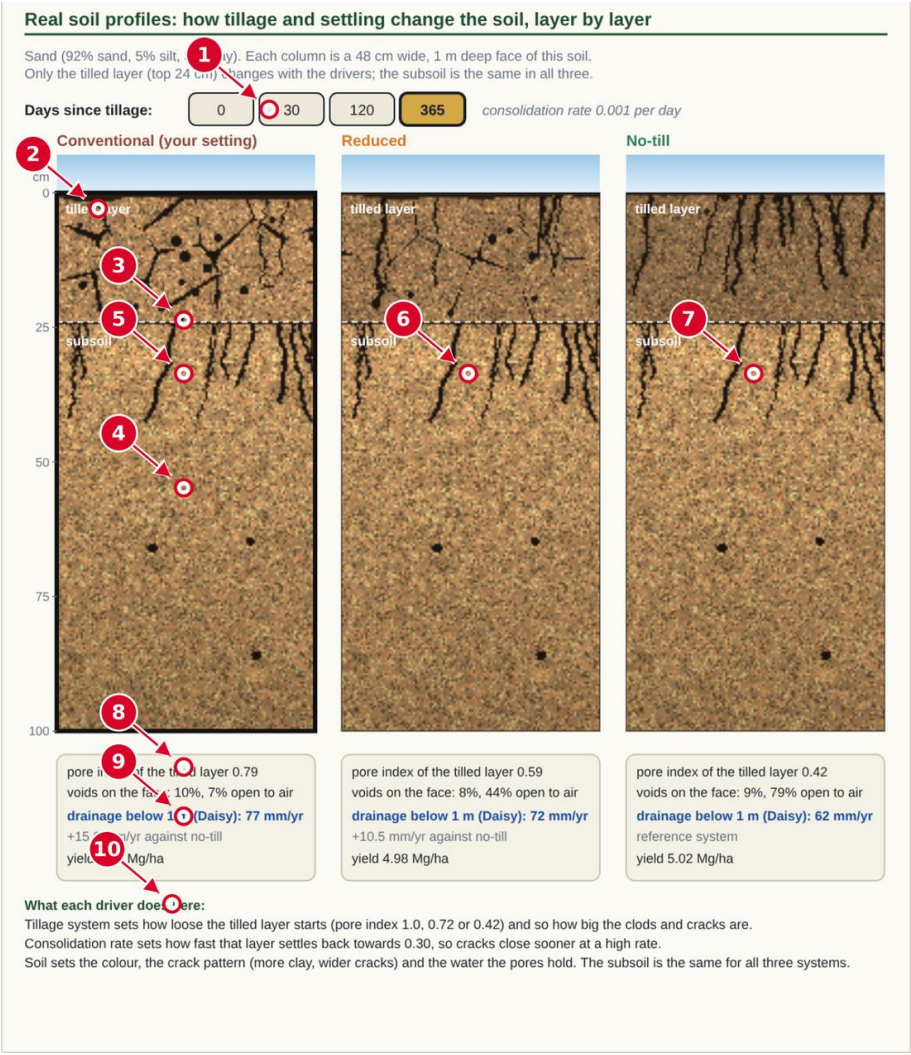


Figure 18.182: Soil hydraulic drivers (V7), part 3. The 365 days since tillage button pressed, so the tilled layer has settled for a year.

18.12 Root structure (R1)

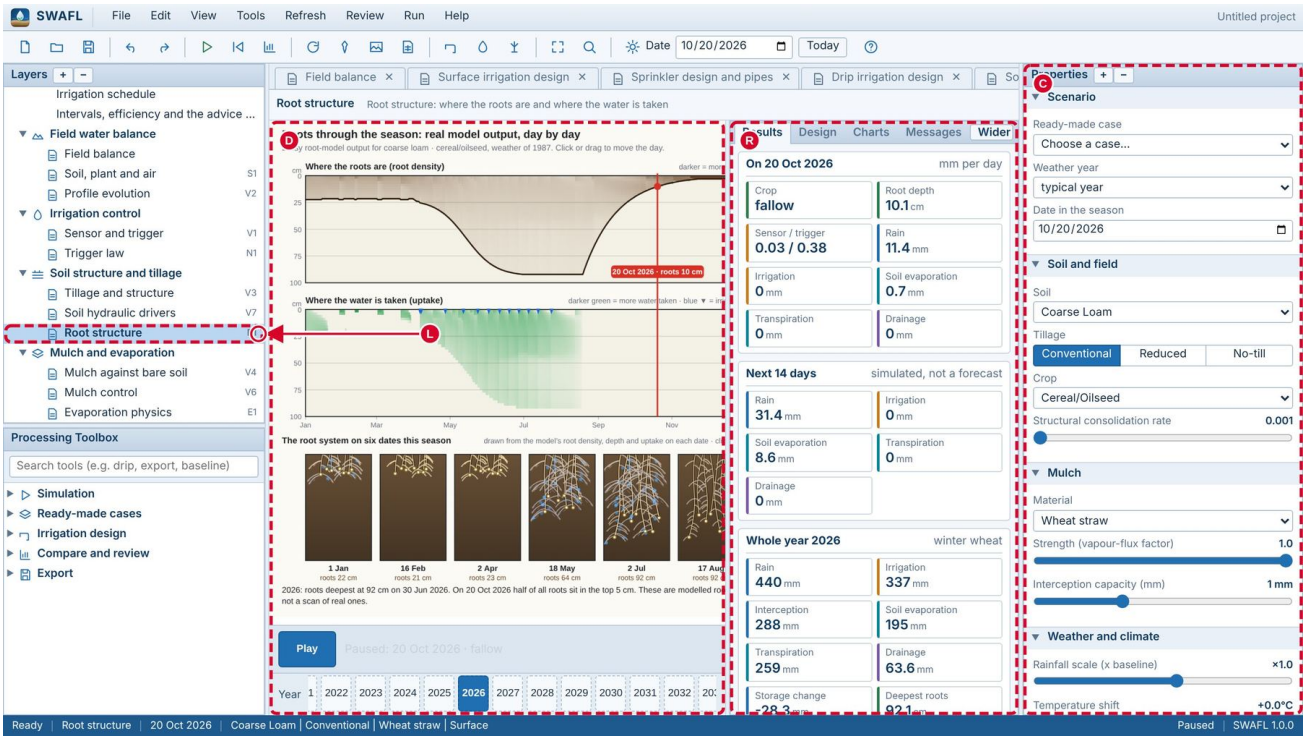


Figure 18.183: Root structure (R1): the whole workbench.

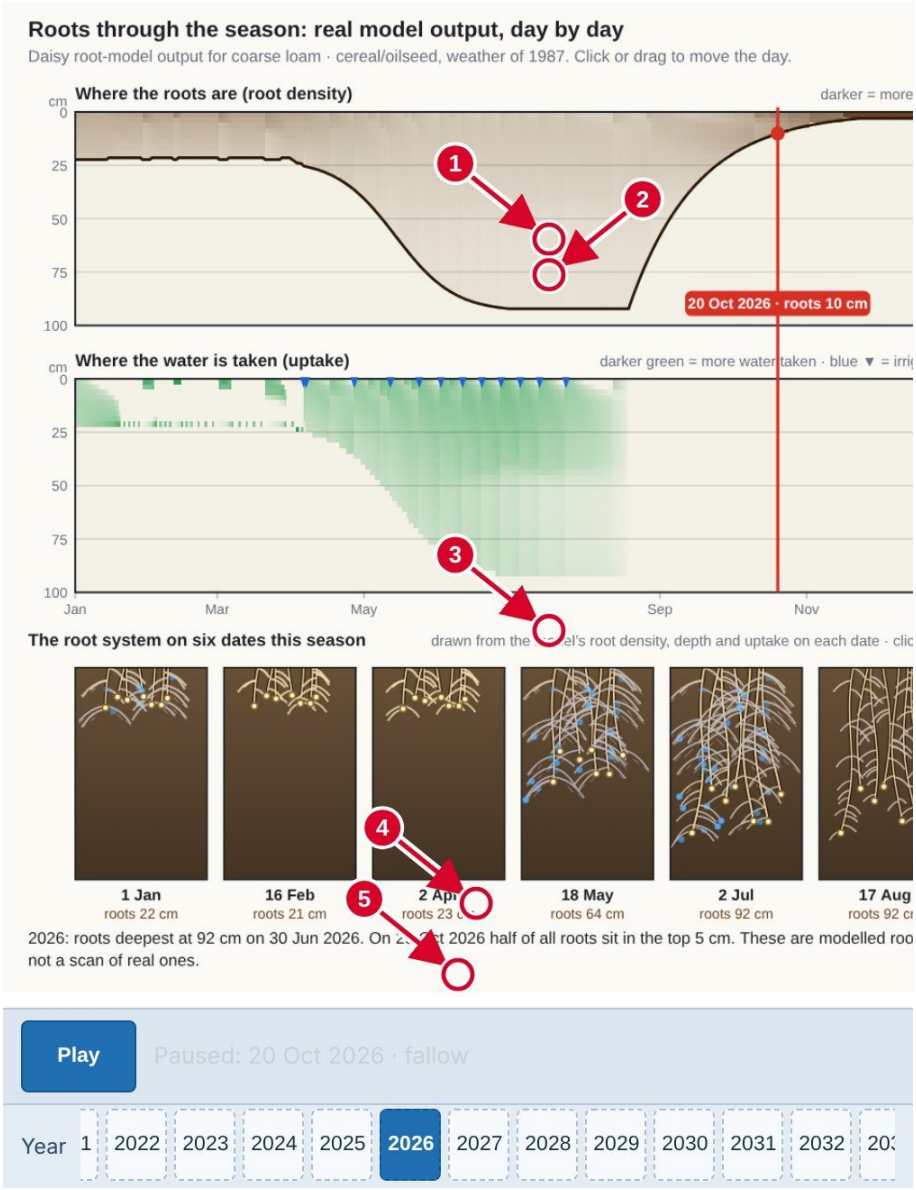


Figure 18.184: Root structure (R1): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Heat map of relative root density by depth and day: Eq. 5.20.
- **2** Rooting depth through the season (black line): Eq. 5.18.
- **3** Heat map of water uptake per layer, u_i(d): Eqs. 5.21 to 5.23.
- **4** Root systems drawn on six dates from the same density: Eqs. 5.19 and 5.20.
- **5** Depth holding half of the uptake, z_p: Eq. 8.7.

Equations behind this screen. Eq. 5.18, Eq. 5.19, Eq. 5.20, Eq. 5.21, Eq. 5.22, Eq. 5.23, Eq. 8.7.

Eq. 5.18. Where explained: chapter "The two computing engines", section "Roots".

$$R_{\max} = \mathrm{clip}\Big(R_c\,(0.85+0.3\,\mathrm{sand})\big[1+0.12\log_{10}(|h_T|/300)\big]\big[1-0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \mathrm{cm}, \quad (5.18)$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.19. Where explained: chapter "The two computing engines", section "Roots".

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\,\mathrm{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d+1.7\,(0.35+\bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.20. Where explained: chapter "The two computing engines", section "Roots".

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d+8}\Big)(0.45+0.9\,s_i)\Big[1+3\,e^{-((z_i-z_d)/12)^2}\Big]_{\text{drip only}},\quad z_i\le R_d,\qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.21. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1-0.3\,c),\qquad w(\text{doy}) = 0.35+0.65\cdot\tfrac{1}{2}\Big[1+\sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.22. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}},\qquad \bar{C} = \tfrac{1}{20}\sum_{20\text{ yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.23. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".

$$q_i^{T} = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j},\qquad a_i = \mathrm{clip}\Big(\frac{s_i-0.15}{0.6};\,0,\,1\Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 8.7. Where explained: chapter "Soil structure and tillage layers", section "Generation".

$$z_p = k_p\,\Delta z,\qquad k_p = \min\Big\{k:\ \frac{\sum_{i\le k} U_i}{\sum_i U_i}\ge p\Big\},\qquad U_i = \sum_{d\in\text{year}} u_i(d),\quad p=0.5,\ 0.9, \tag{8.7}$$

Provenance (Eq. 8.7): **[Engineered]**: diagnostic defined for the workbench.

18.12.1 The whole layer, part by part

The picture below is the complete drawing of the layer Root structure (R1), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 4 parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

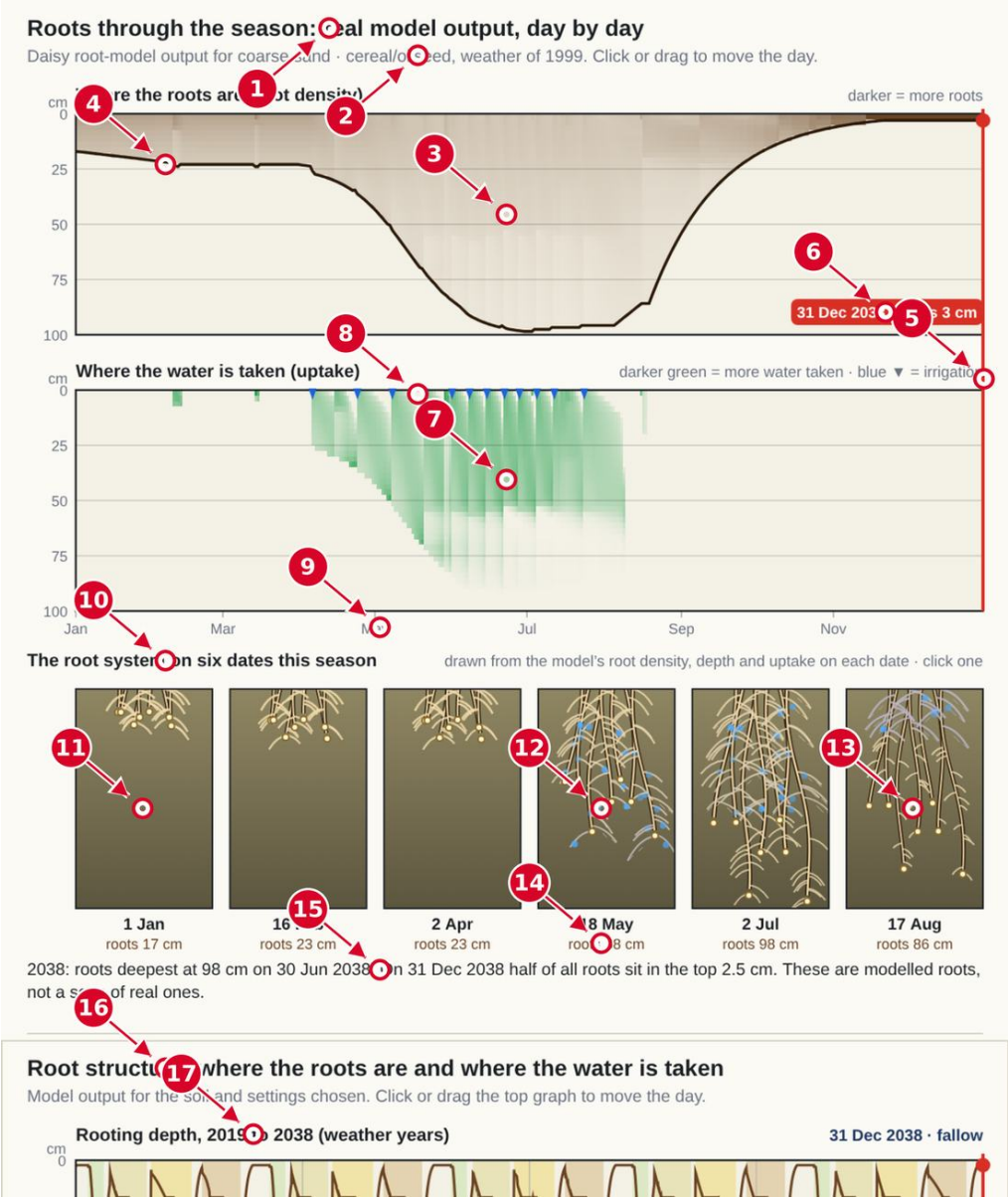


Figure 18.185: Root structure (R1), part 1 of 4.

What the arrows point to.

- **1** The title, the roots through the season as real model output, day by day.
- **2** The instruction, the day is moved by clicking or dragging on the graphs.
- **3** The root density by depth and day, darker where there are more roots.
- **4** The black line is the rooting depth, which falls and grows with the season.
- **5** The red day cursor, the day of the clock.
- **6** The label of the cursor, the date and the rooting depth on that day.
- **7** The uptake of water by depth and day, darker green for more water taken.
- **8** The blue triangles mark the irrigations.
- **9** The month axis of the season.
- **10** The six dates on which the root system is drawn, each clickable.
- **11** The root system on the first date, still small.
- **12** The root system in the middle of the season, deep and branched.
- **13** The root system late in the season with the uptake in blue.

- **14** The date and the rooting depth under each drawing.
- **15** The sentence that reports the deepest roots and where half of all the roots sit.
- **16** The second section, the root structure where the roots are and where the water is taken.
- **17** The rooting depth for every year of the record, with the season colours.

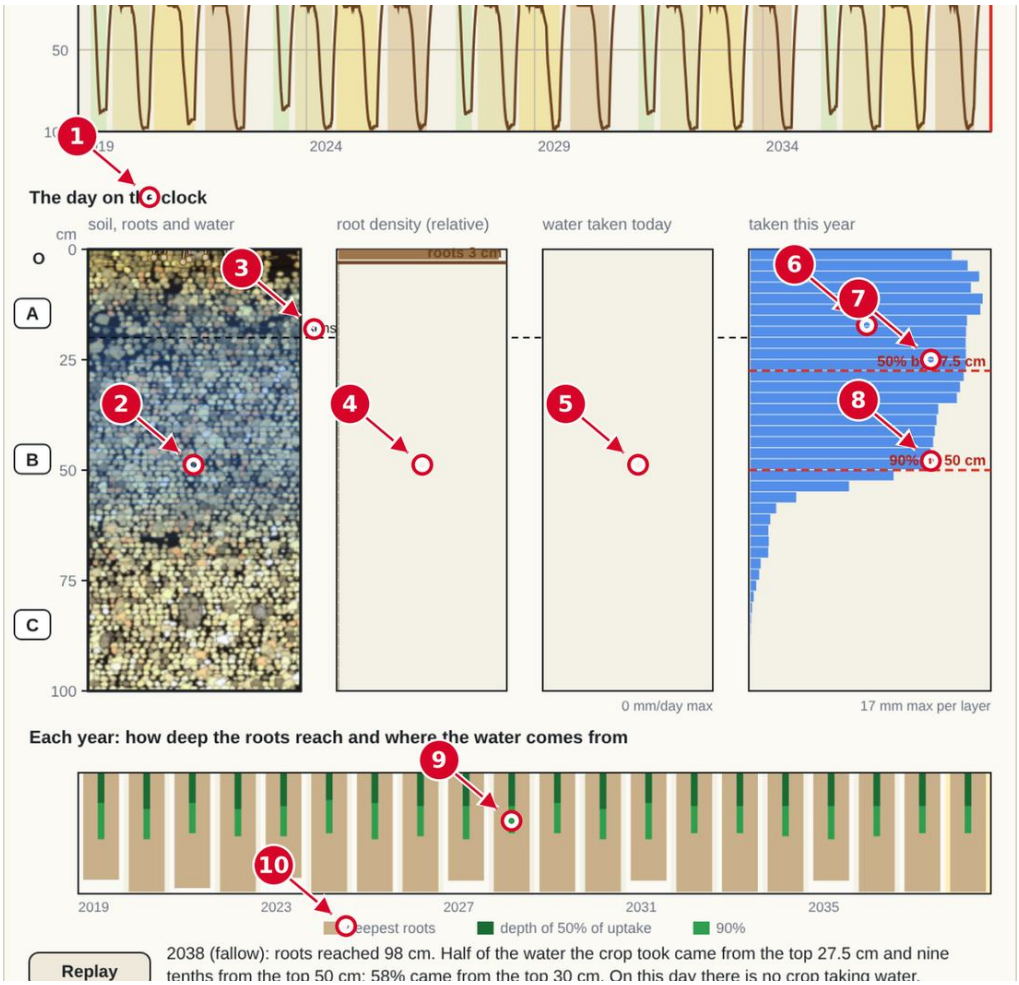


Figure 18.186: Root structure (R1), part 2 of 4.

What the arrows point to.

- **1** The day on the clock, four panels for the selected day.
- **2** The soil, roots and water of the day.
- **3** The dashed line carries the sensor depth across the panels.
- **4** The relative root density by depth on the day.
- **5** The water taken on the day by depth.
- **6** The water taken in the year by depth.
- **7** The depth by which half of the water was taken.
- **8** The depth by which nine tenths of the water was taken.
- **9** The deepest roots of each year and the depth of half and nine tenths of the uptake.
- **10** The legend of the deepest roots and of the two uptake depths.

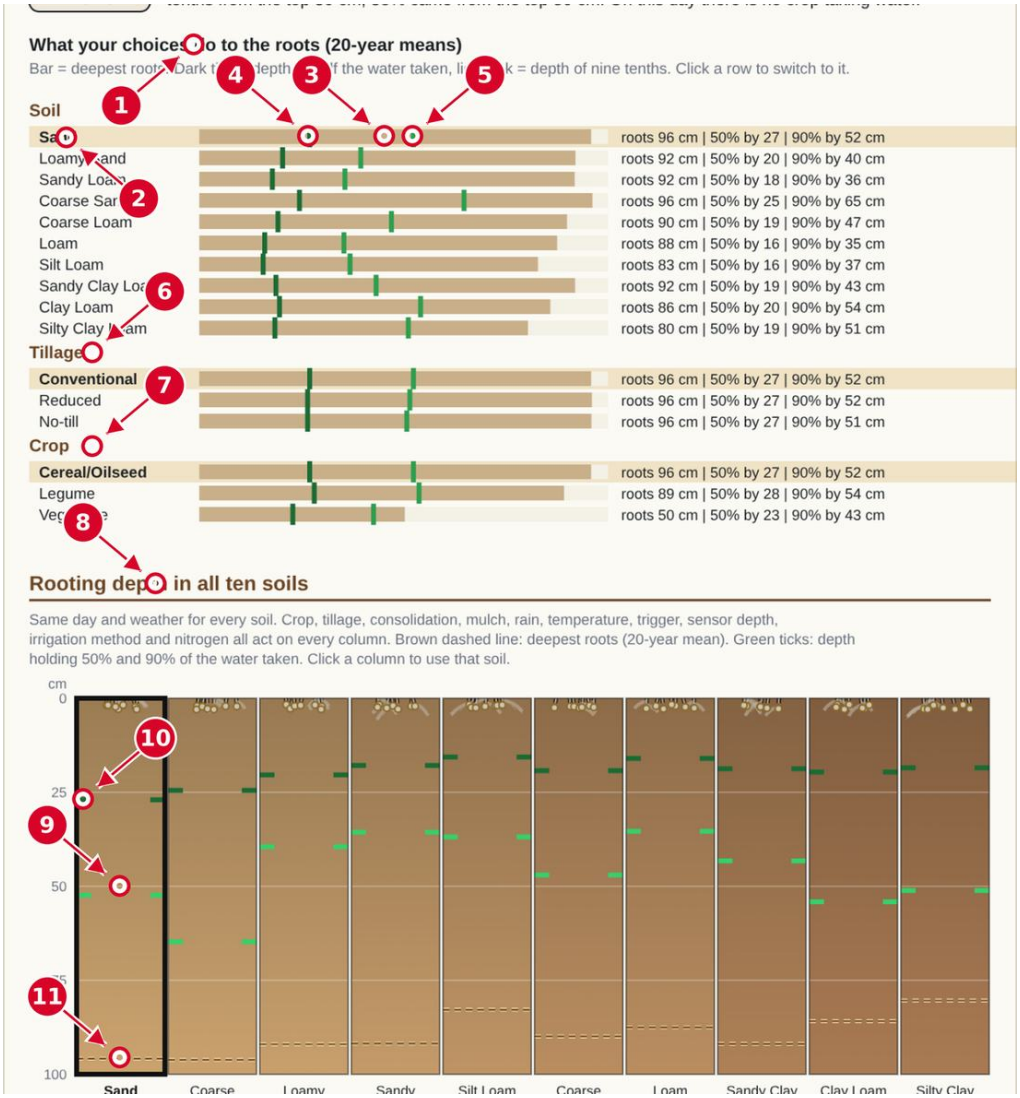


Figure 18.187: Root structure (R1), part 3 of 4.

What the arrows point to.

- **1** What your choices do to the roots, as 20-year means; a row is clicked to switch to it.
- **2** The soil selected, highlighted in the list.
- **3** The bar of the deepest roots of the soil.
- **4** The dark tick, the depth of half of the water taken.
- **5** The light tick, the depth of nine tenths of the water taken.
- **6** The tillage systems compared, conventional, reduced and no-till.
- **7** The crops compared, cereal, legume and vegetable.
- **8** The rooting depth in all ten soils, on the same day and weather.
- **9** The soil column of the selected soil.
- **10** The green ticks, the depths of half and nine tenths of the uptake.
- **11** The dashed line, the deepest roots of the 20-year mean.

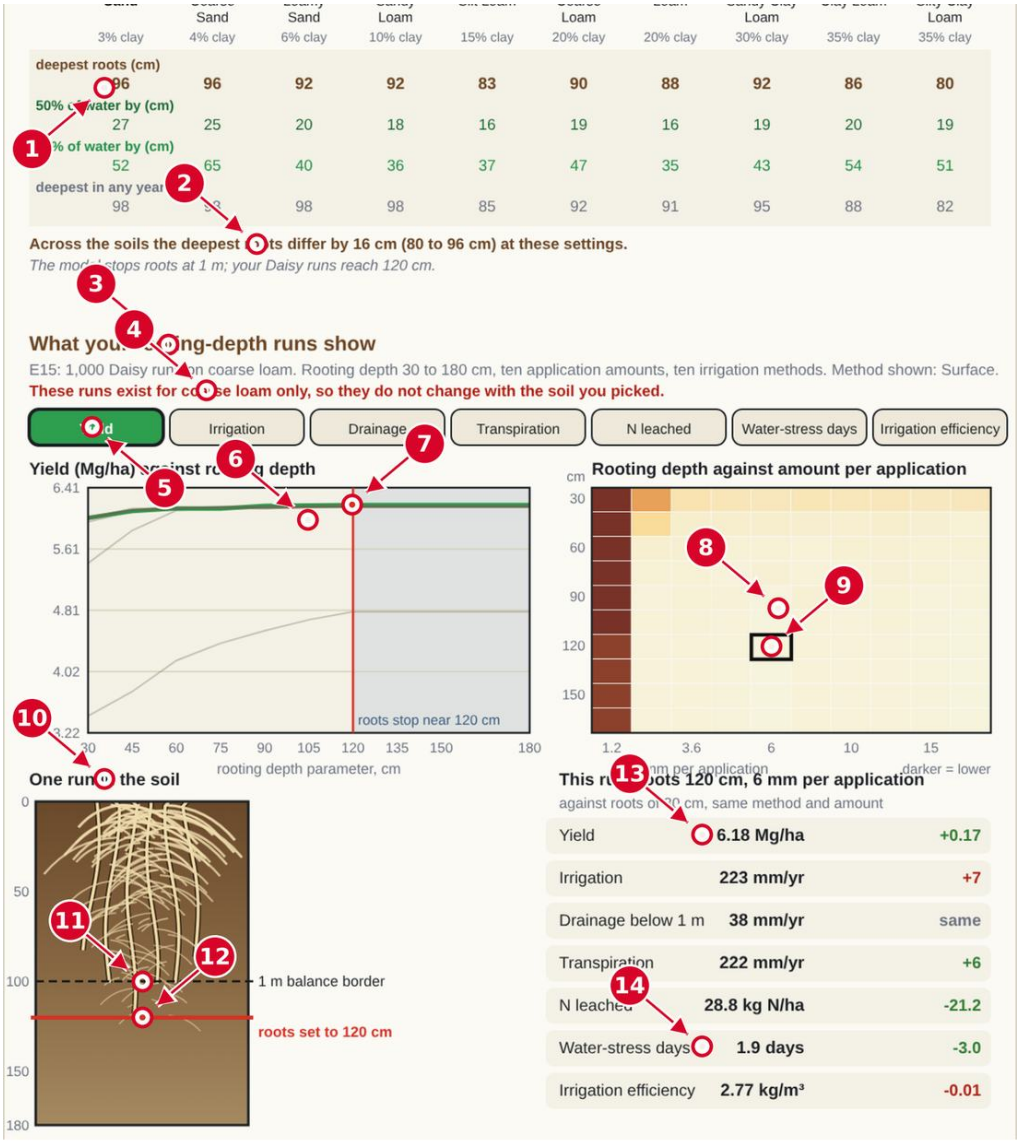


Figure 18.188: Root structure (R1), part 4 of 4.

What the arrows point to.

- **1** The table of the deepest roots and of the uptake depths of the ten soils.
- **2** The range of the deepest roots across the soils at these settings.
- **3** The rooting-depth runs, 1,000 Daisy runs on coarse loam.
- **4** The note that the runs exist for coarse loam only.
- **5** The metric buttons, Yield, Irrigation, Drainage, Transpiration, N leached, Water-stress days and Irrigation efficiency.
- **6** The yield against the rooting depth for ten application amounts.
- **7** The red line, where the roots stop near 120 cm.
- **8** The map of the rooting depth against the amount per application.
- **9** The black square, the run selected.
- **10** One run in the soil, with the roots drawn to scale.
- **11** The dashed line at 1 m, the border of the balance.
- **12** The red line, the rooting depth set for this run.
- **13** The table of the run against the reference run with roots of 30 cm.
- **14** The water-stress days of the run and their change.

18.12.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

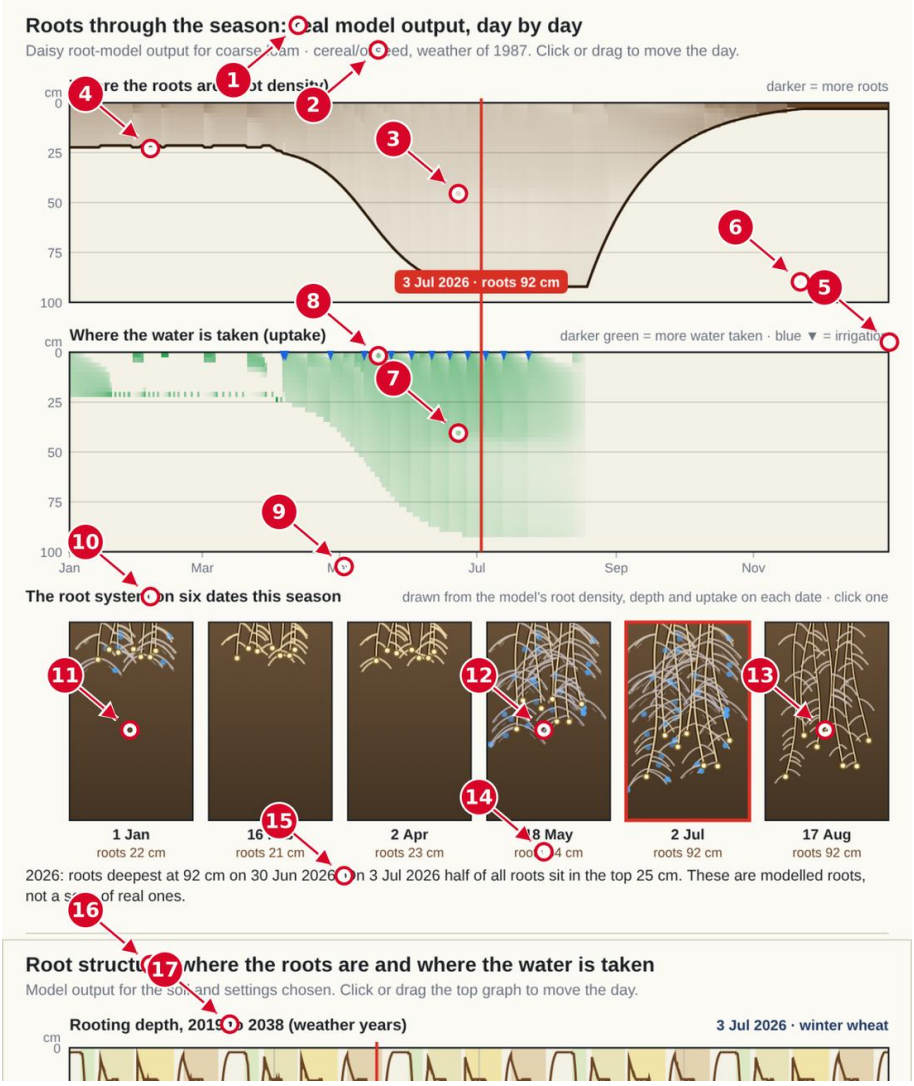


Figure 18.189: Root structure (R1), part 1. The top graph clicked in the middle of the season, so the day on the clock moves there.

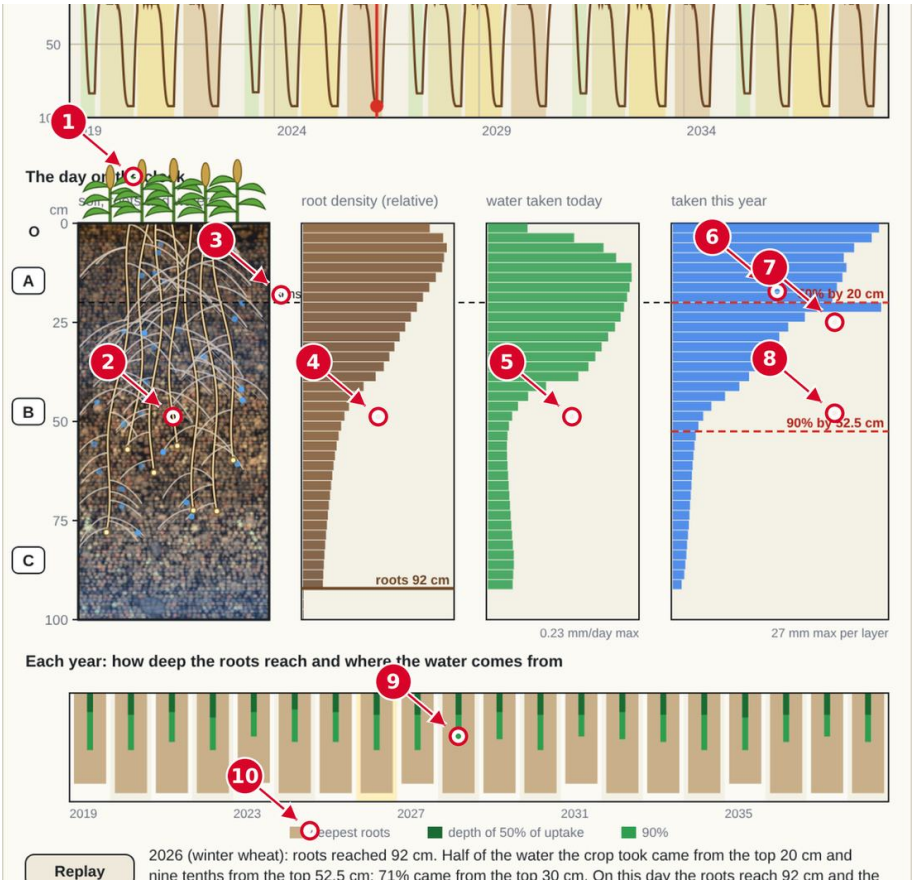


Figure 18.190: Root structure (R1), part 2. The top graph clicked in the middle of the season, so the day on the clock moves there.

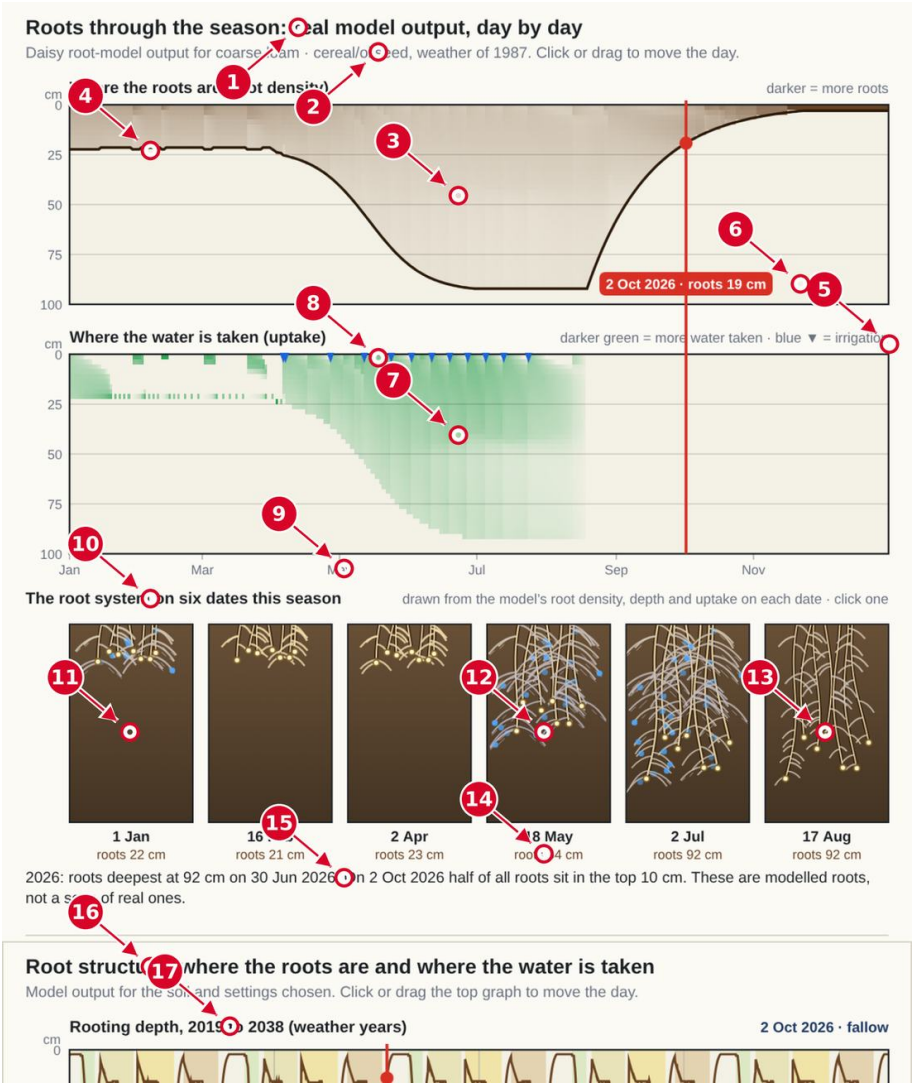


Figure 18.191: Root structure (R1), part 1. The top graph clicked late in the season, so the day on the clock moves there.

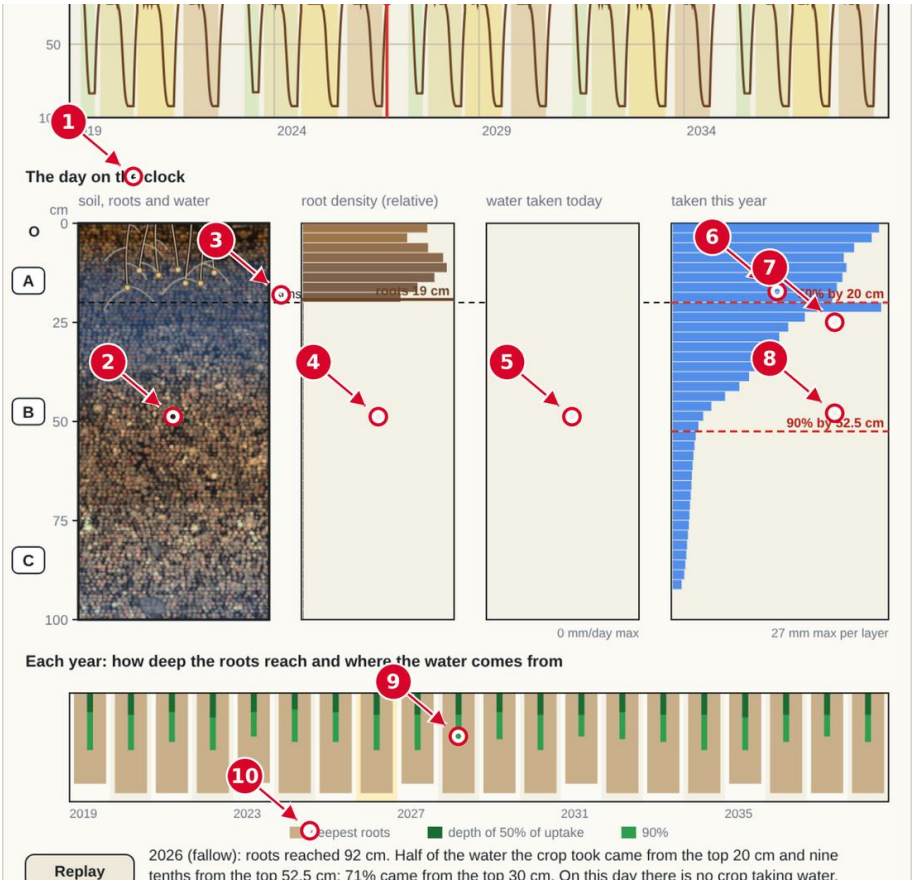


Figure 18.192: Root structure (R1), part 2. The top graph clicked late in the season, so the day on the clock moves there.

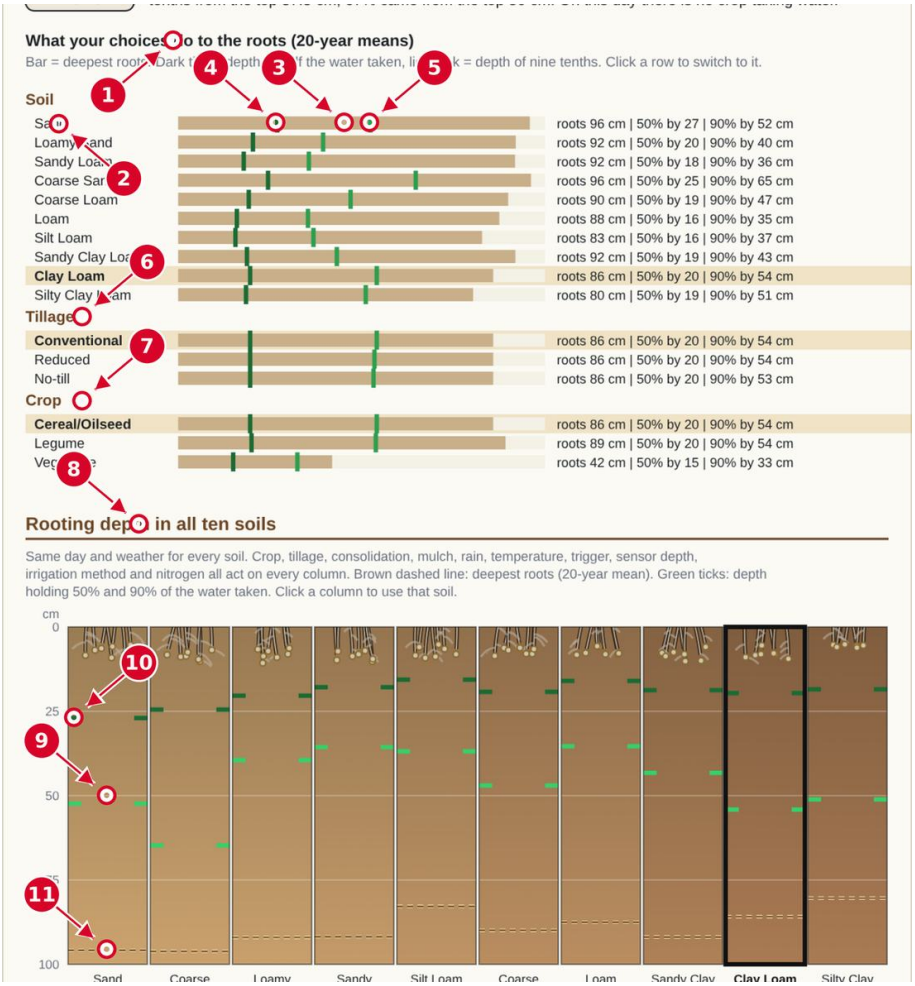


Figure 18.193: Root structure (R1), part 3. A soil row of the list clicked, so the page switches to that soil.

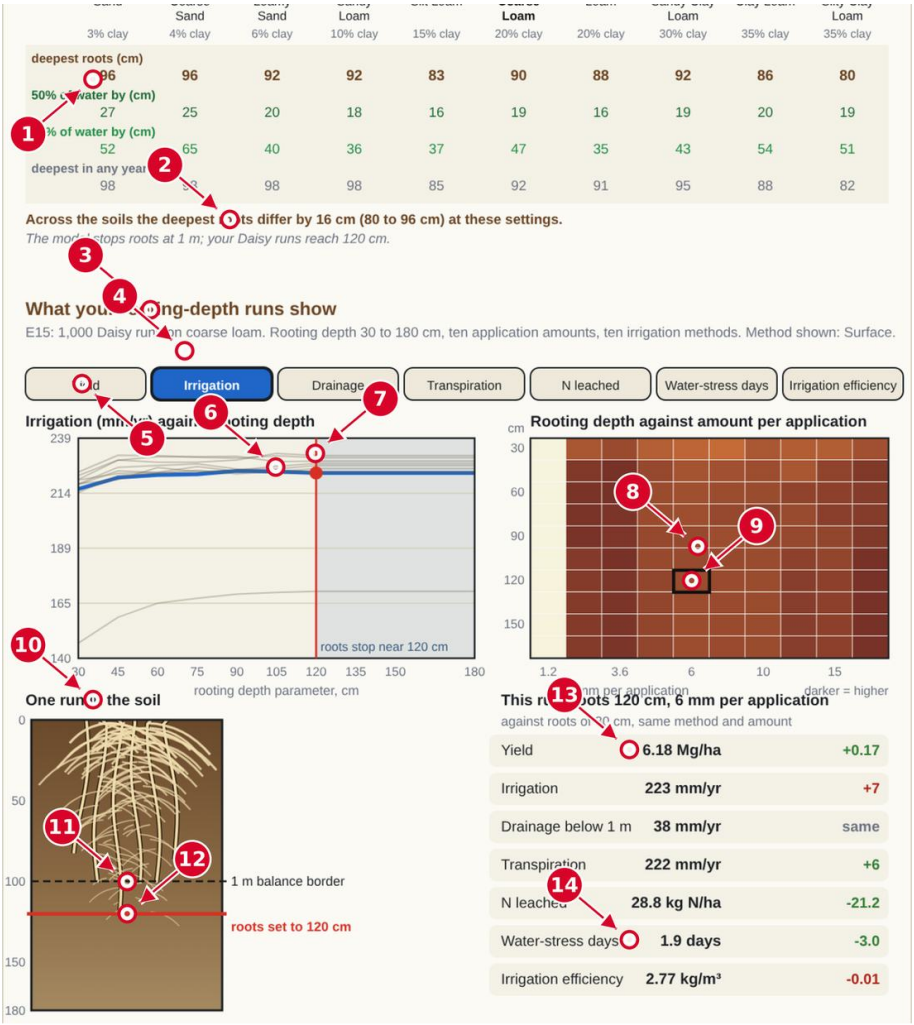


Figure 18.194: Root structure (R1), part 4. The Irrigation button pressed, so the runs show the irrigation against the rooting depth.

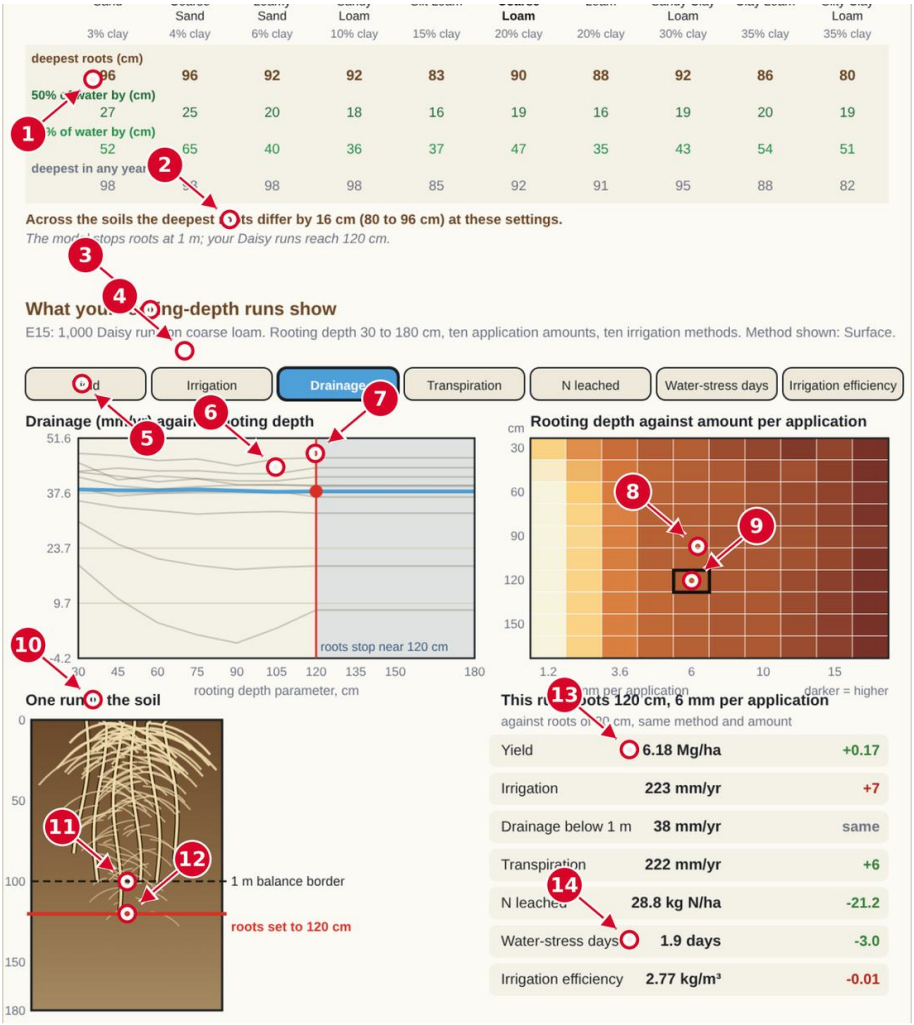


Figure 18.195: Root structure (R1), part 4. The Drainage button pressed, so the runs show the drainage against the rooting depth.

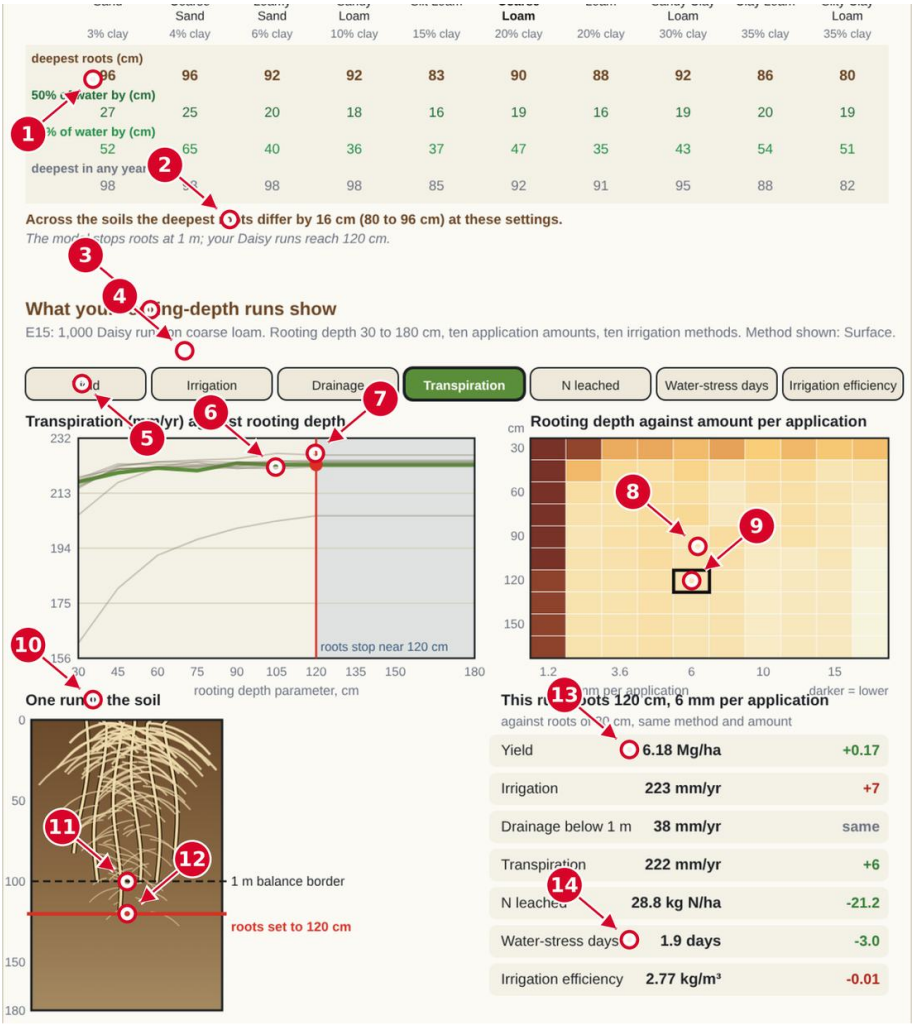


Figure 18.196: Root structure (R1), part 4. The Transpiration button pressed.

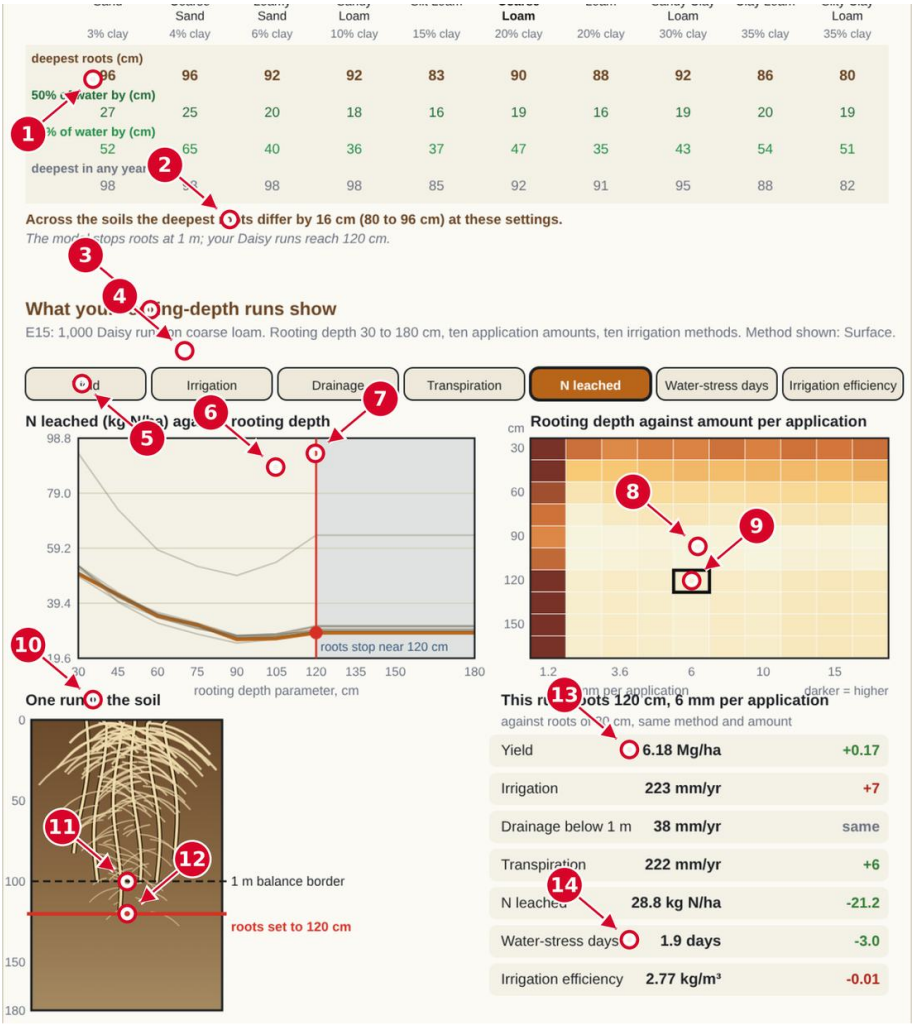


Figure 18.197: Root structure (R1), part 4. The N leached button pressed.

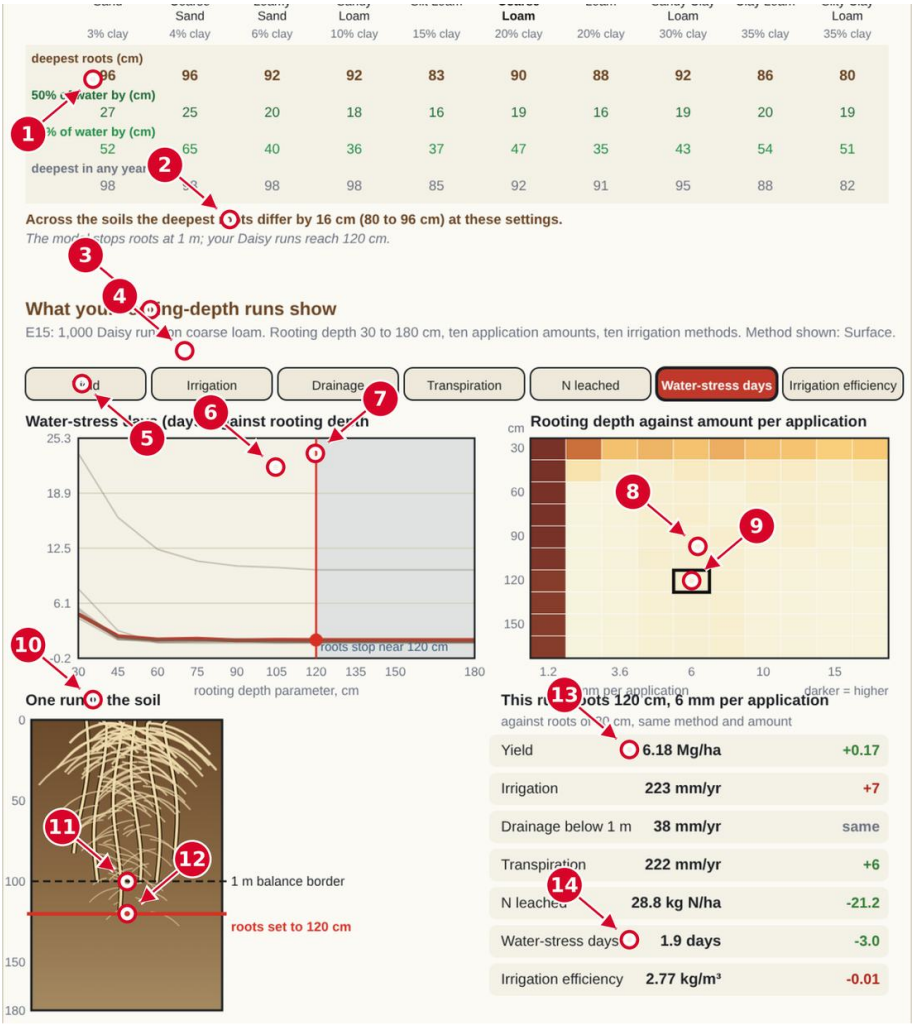


Figure 18.198: Root structure (R1), part 4. The Water-stress days button pressed.

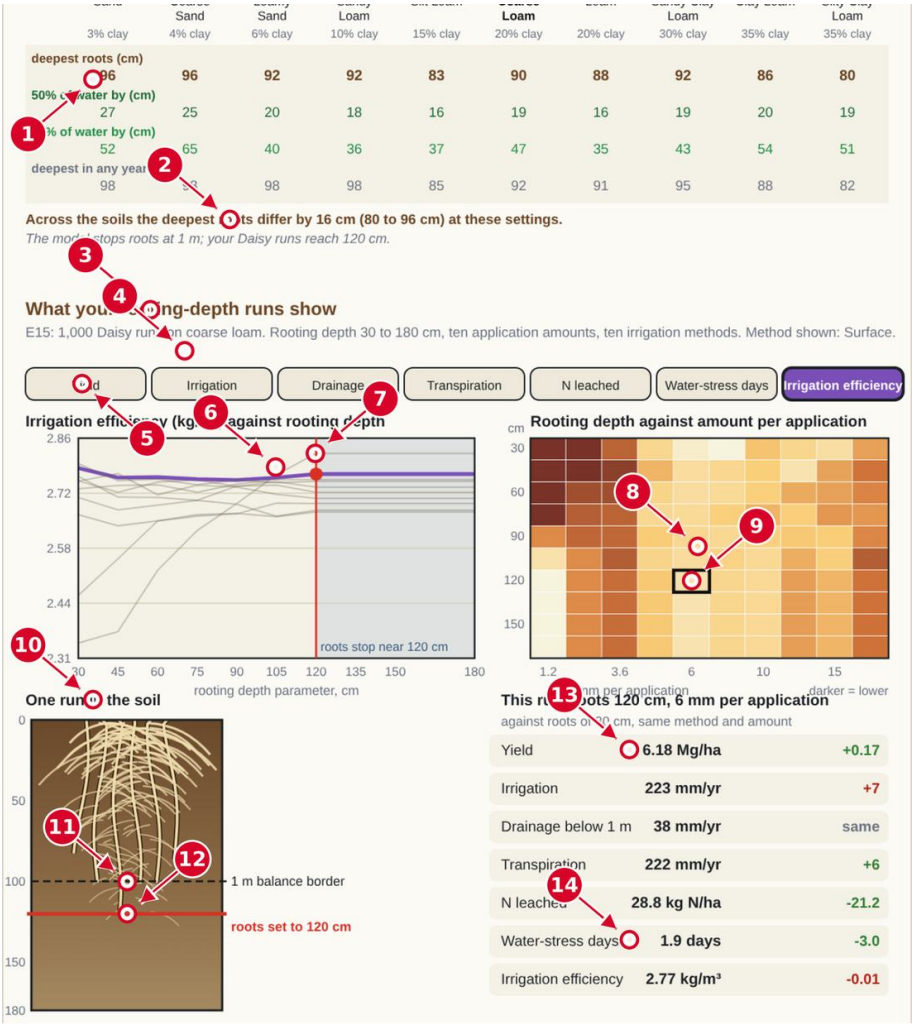


Figure 18.199: Root structure (R1), part 4. The Irrigation efficiency button pressed.

18.13 Mulch against bare soil (V4)

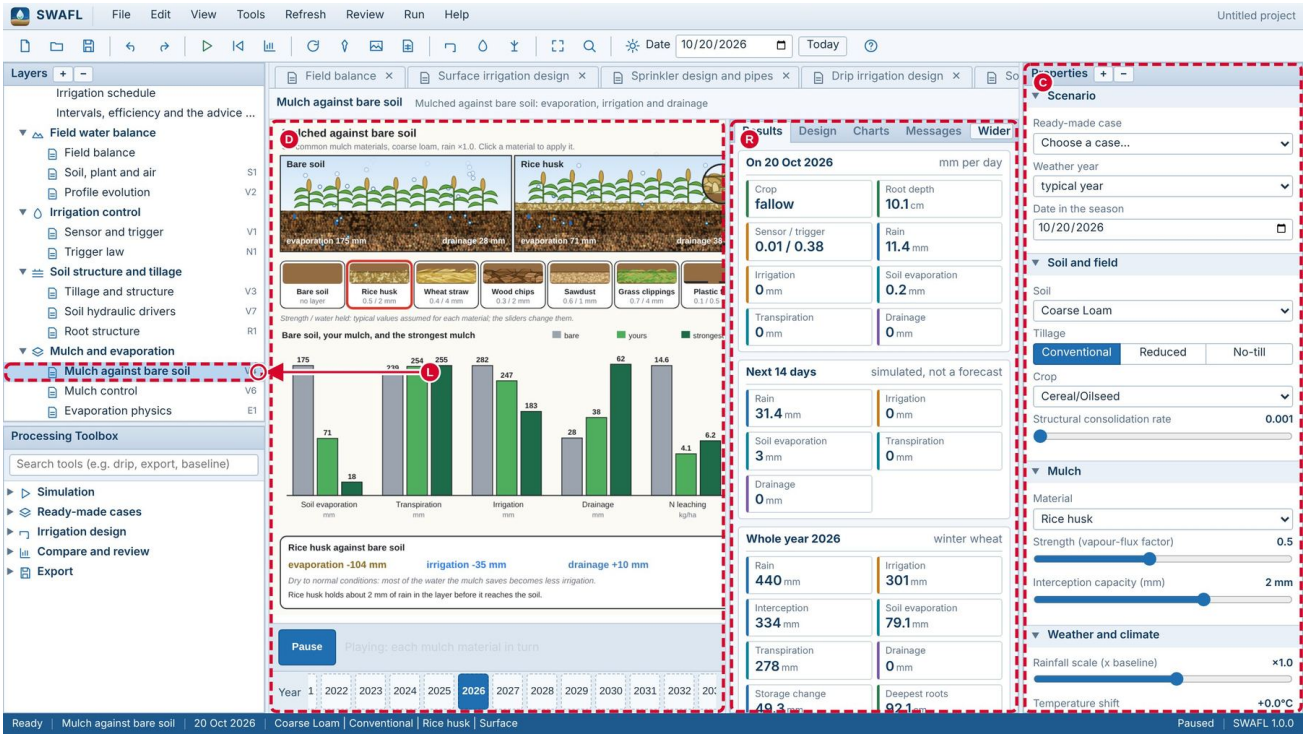


Figure 18.200: Mulch against bare soil (V4): the whole workbench.

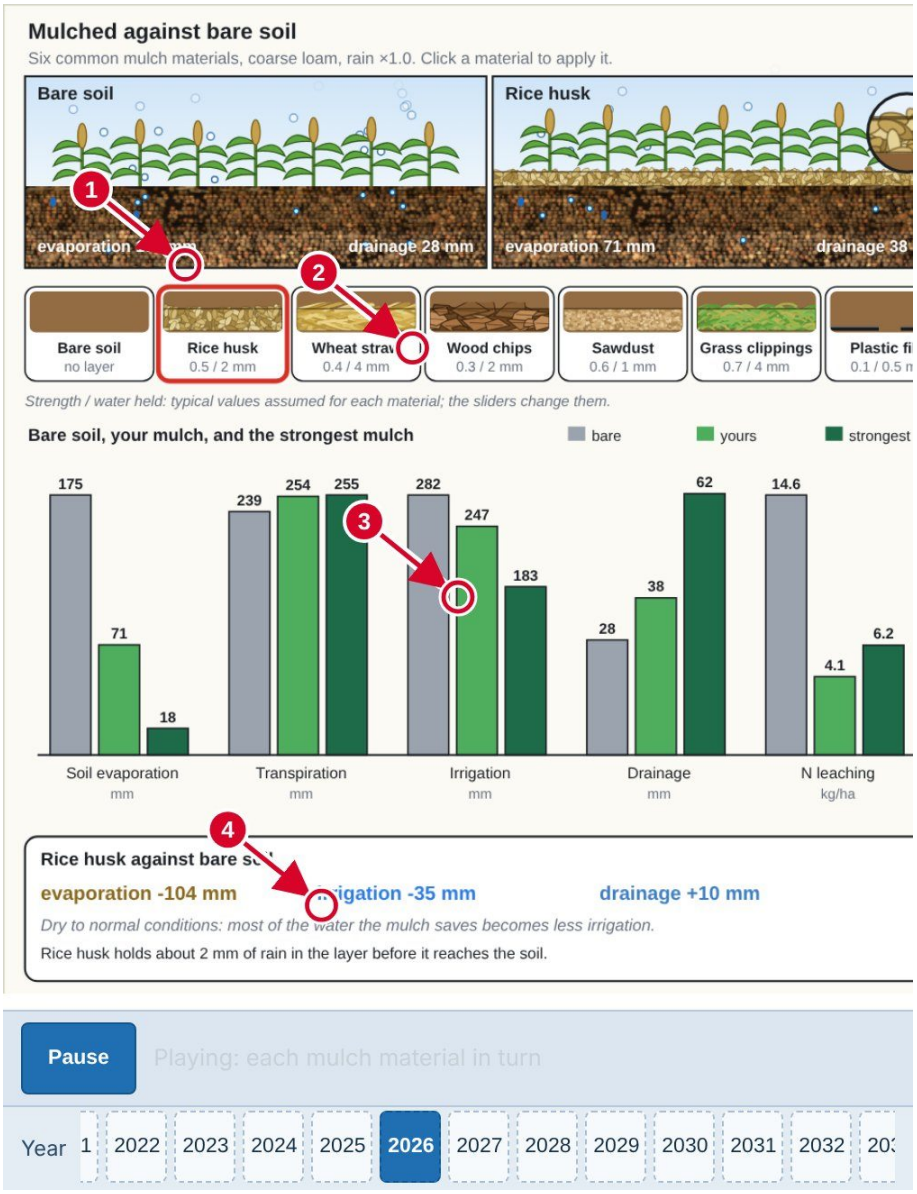


Figure 18.201: Mulch against bare soil (V4): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Bare-soil reference: Engine 1 at mulch factor 1, Eq. 5.4.
- **2** Selected mulch (strength phi, capacity c): the input of Eq. 9.1.
- **3** Evaporation, transpiration, irrigation, drainage and nitrogen leaching of bare soil, your mulch and the strongest mulch: Eqs. 5.4, 5.7, 5.9 and 5.10.
- **4** Differences between bare soil and the selected mulch, Delta E, Delta I, Delta D: Eq. 9.1.

Equations behind this screen. Eq. 5.4, Eq. 5.7, Eq. 5.9, Eq. 9.1.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 9.1. Where explained: chapter “Mulch and evaporation layers”, section “Mulch against bare soil (V4)”.

$$\Delta E = E_{1.0} - E_{\varphi}, \qquad \Delta I = I_{1.0} - I_{\varphi}, \qquad \Delta D = D_{\varphi} - D_{1.0}, \tag{9.1}$$

Provenance (Eq. 9.1): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

18.13.1 The whole layer, part by part

The picture below is the complete drawing of the layer Mulch against bare soil (V4), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 1 part, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

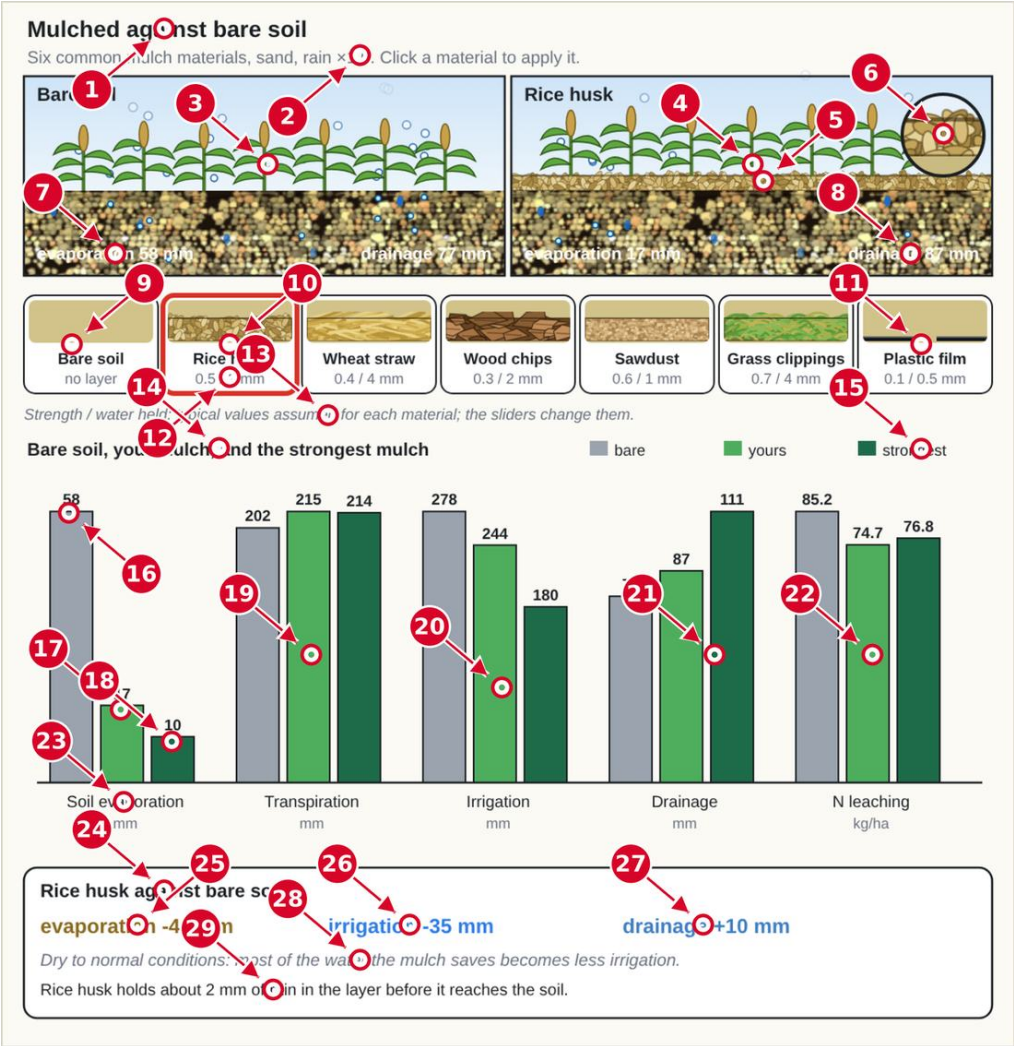


Figure 18.202: Mulch against bare soil (V4), part 1 of 1.

What the arrows point to.

- **1** The title, the mulched soil against the bare soil.
- **2** The instruction, a material is clicked to apply it.
- **3** The bare soil panel, with its evaporation and drainage for the year.
- **4** The panel of the mulch selected, here rice husk.
- **5** The mulch layer drawn on top of the soil.
- **6** The close-up lens that shows the mulch material.

- **7** The evaporation of the bare soil in mm.
- **8** The drainage of the mulched soil in mm.
- **9** The bare soil card, no layer.
- **10** The material selected, highlighted in red.
- **11** The plastic film card, the last of the six materials.
- **12** The strength and the water held by the material, in mm of rain.
- **13** The note that the typical values are assumed and the sliders change them.
- **14** The chart of the bare soil, your mulch and the strongest mulch.
- **15** The legend of the three bars.
- **16** The bar of the bare soil, grey.
- **17** The bar of the mulch selected, light green.
- **18** The bar of the strongest mulch, dark green.
- **19** The transpiration group, nearly the same for the three.
- **20** The irrigation group, lower under mulch.
- **21** The drainage group, higher under mulch.
- **22** The nitrogen leaching group, in kg per ha.
- **23** The unit of each group.
- **24** The result box, the mulch against the bare soil.
- **25** The change of the evaporation.
- **26** The change of the irrigation.
- **27** The change of the drainage.
- **28** The note that explains where the saved water goes.
- **29** The rain held by the layer before it reaches the soil.

18.13.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

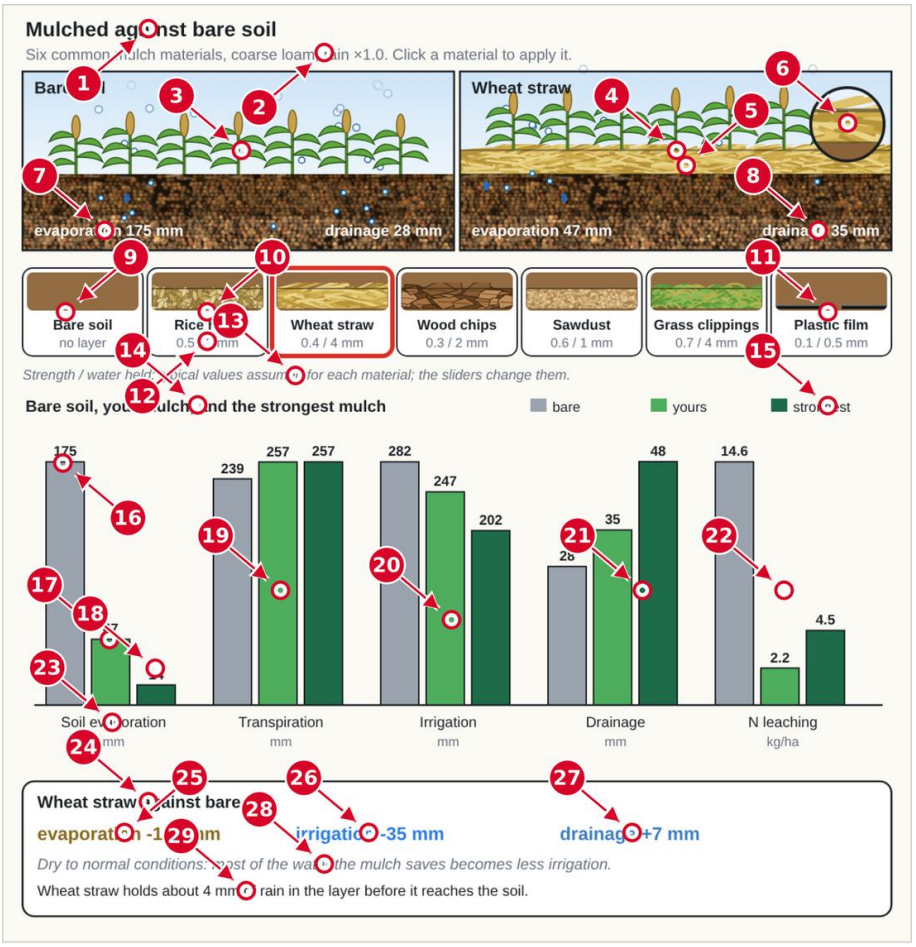


Figure 18.203: Mulch against bare soil (V4), part 1. Wheat straw chosen as the material.

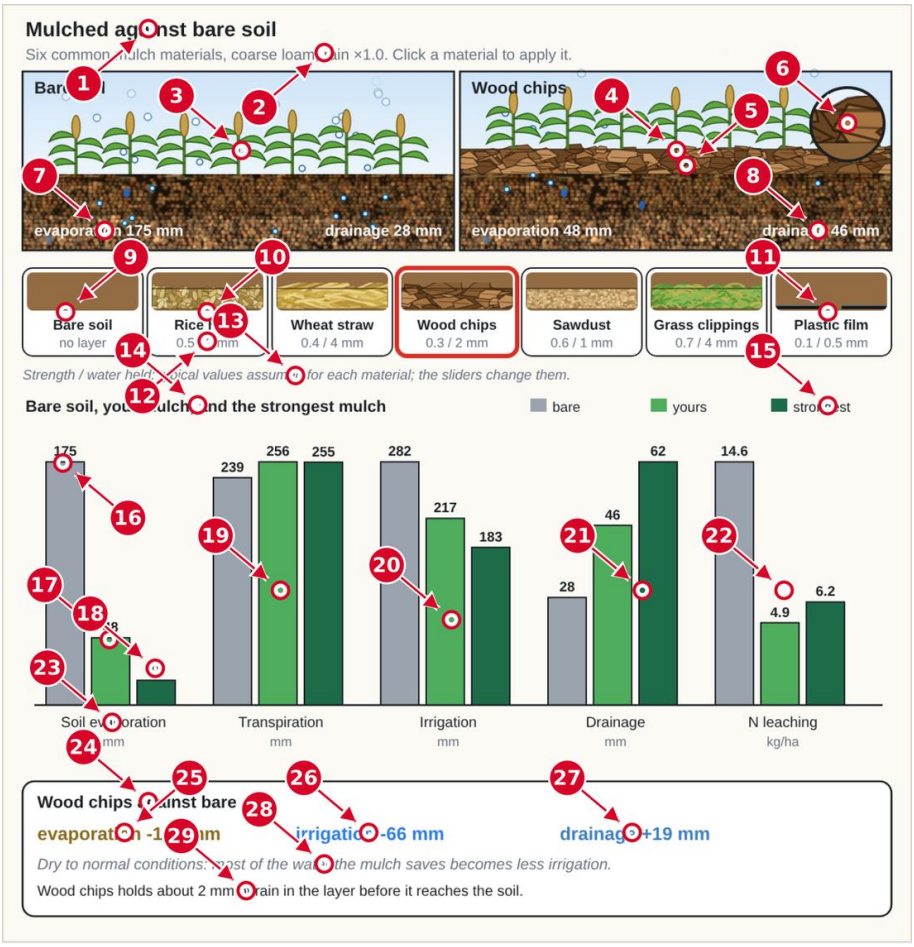


Figure 18.204: Mulch against bare soil (V4), part 1. Wood chips chosen as the material.

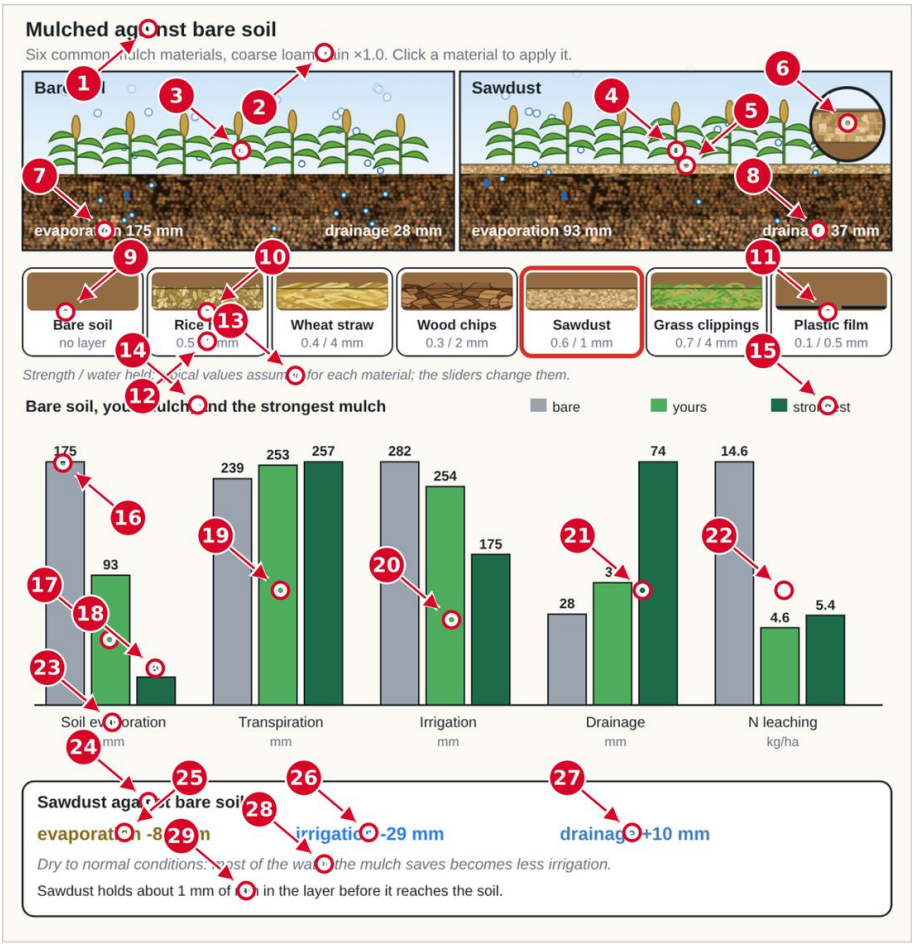


Figure 18.205: Mulch against bare soil (V4), part 1. Sawdust chosen as the material.

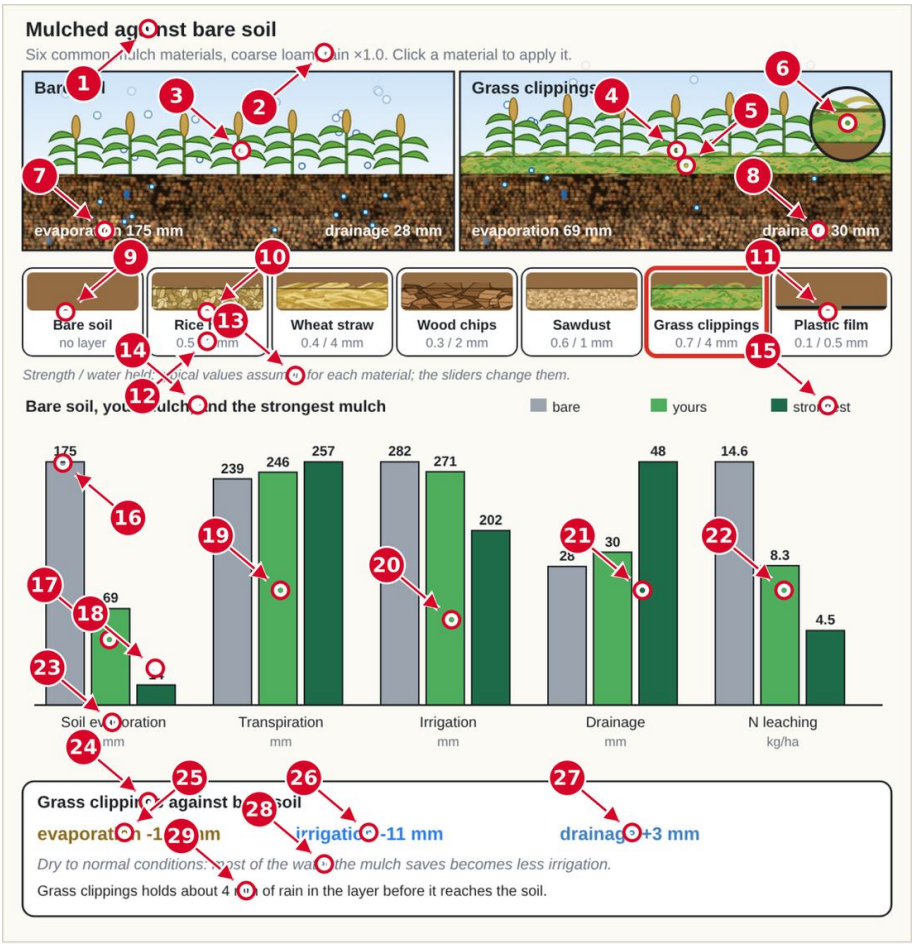


Figure 18.206: Mulch against bare soil (V4), part 1. Grass clippings chosen as the material.

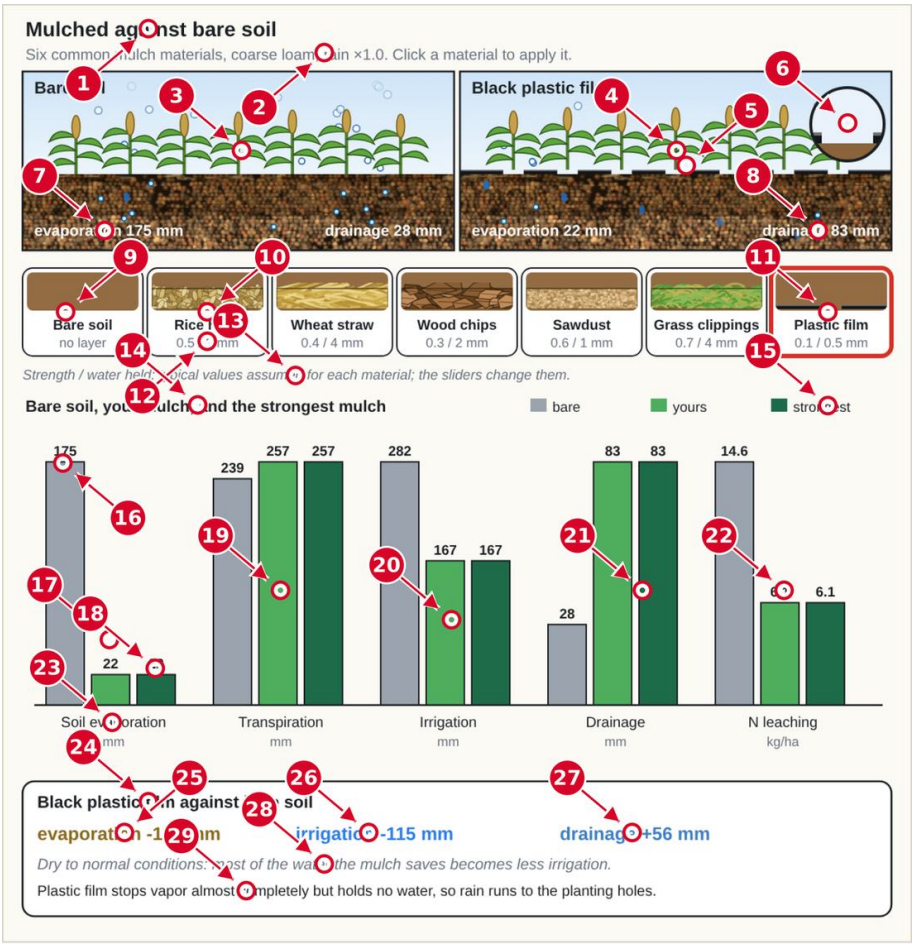


Figure 18.207: Mulch against bare soil (V4), part 1. Plastic film chosen as the material.

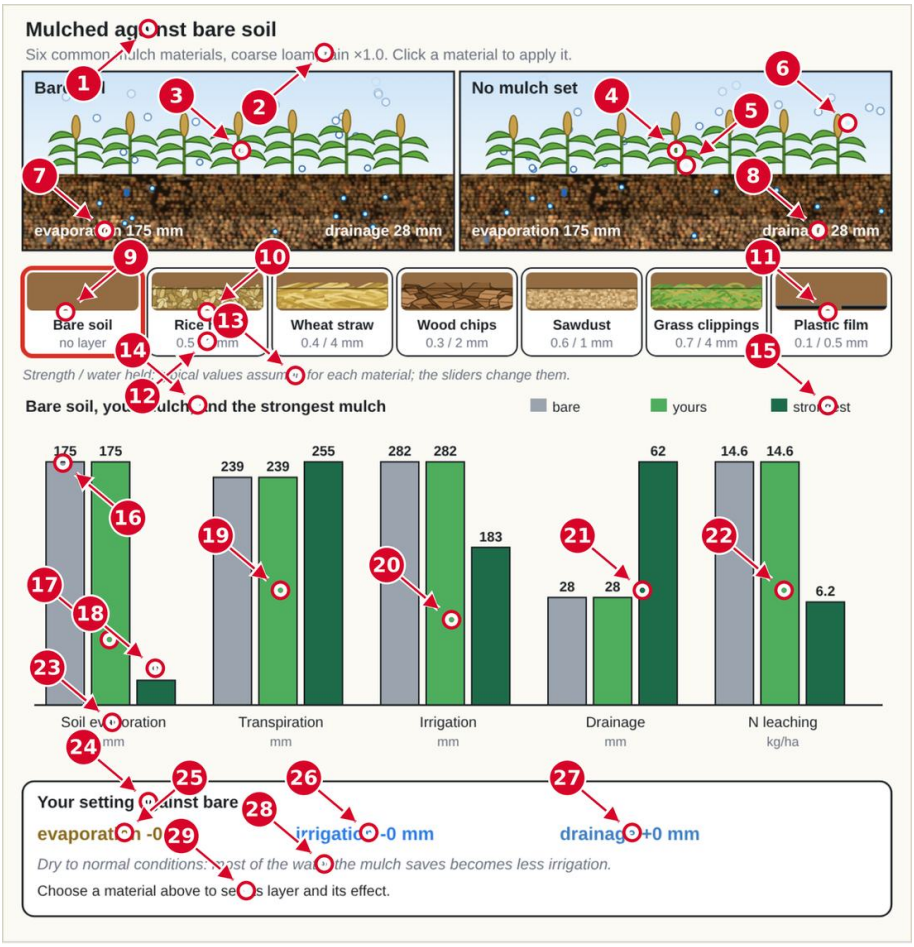


Figure 18.208: Mulch against bare soil (V4), part 1. Bare soil chosen, so the comparison shows no change.

18.14 Mulch control (V6)

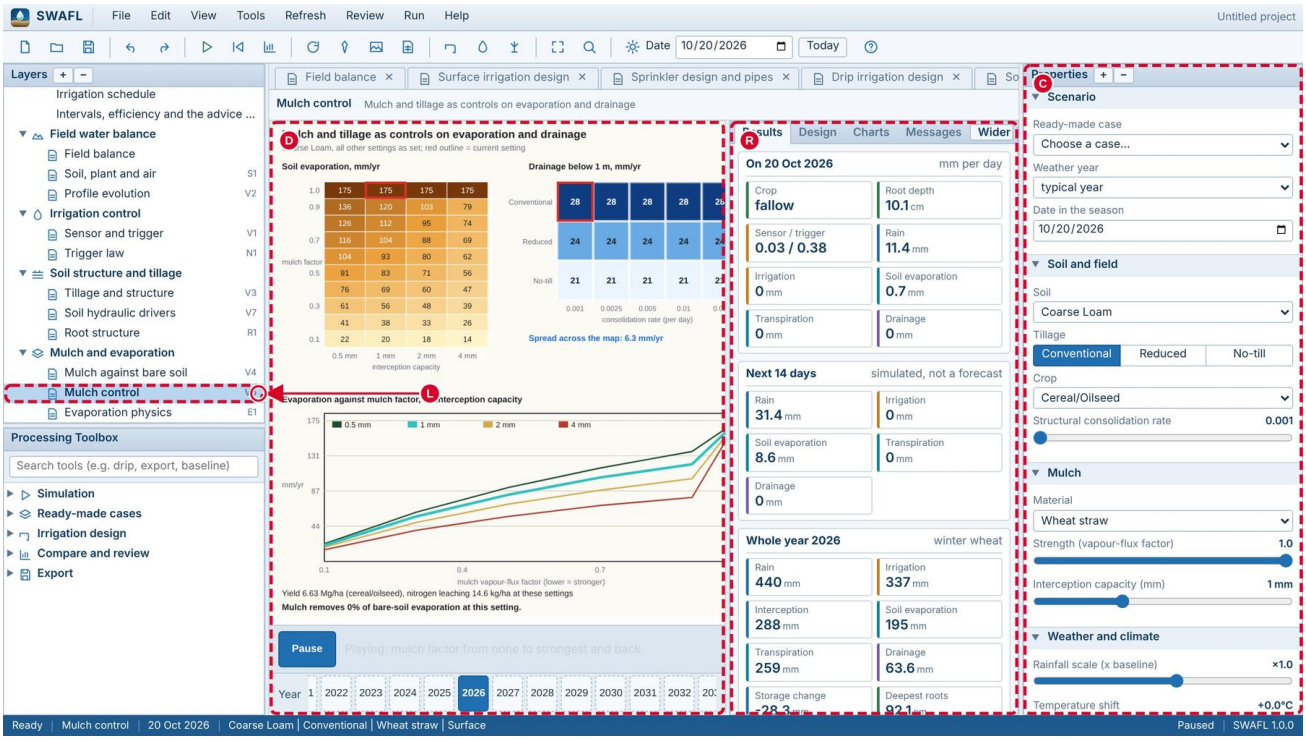


Figure 18.209: Mulch control (V6): the whole workbench.

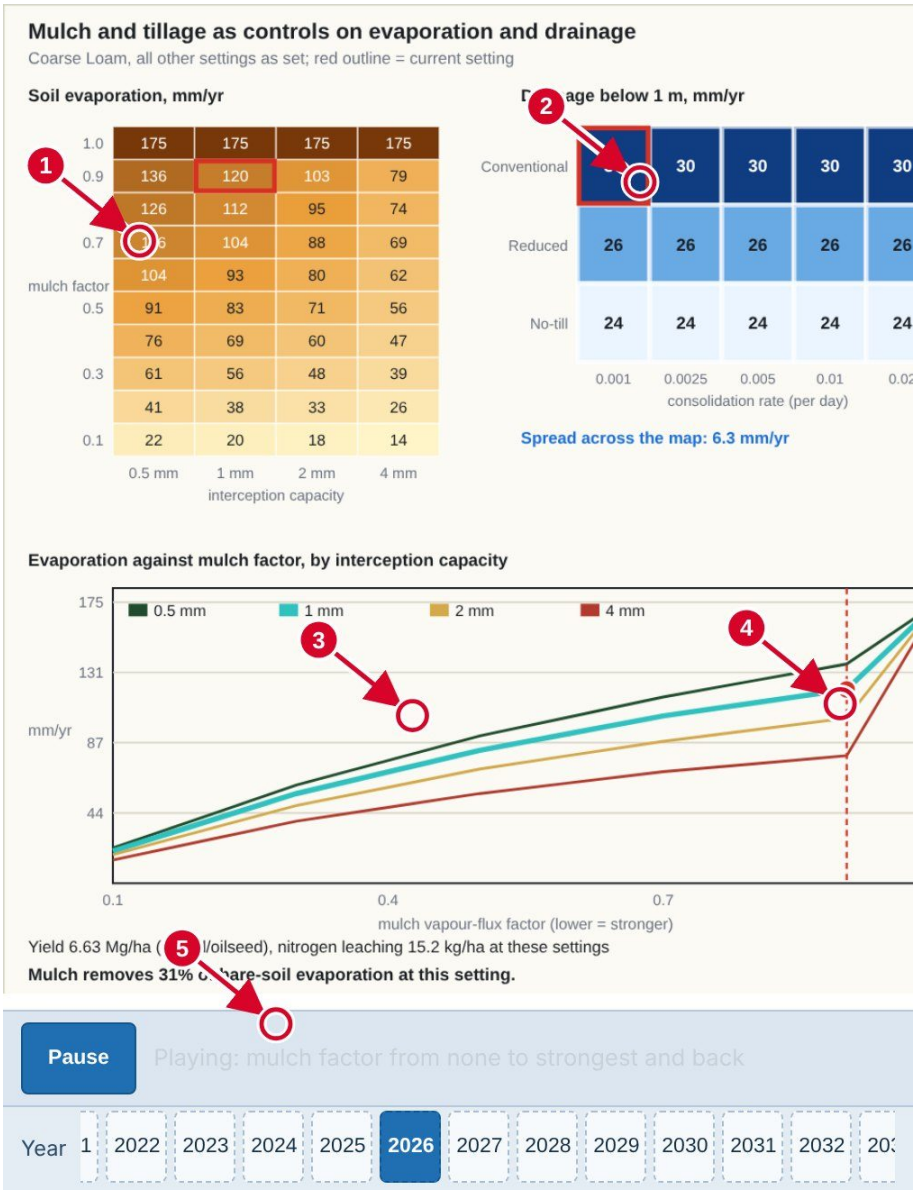


Figure 18.210: Mulch control (V6): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Matrix of soil evaporation E(phi, c) for each mulch factor and interception capacity: Eq. 5.4.
- **2** Matrix of drainage D(tillage, consolidation rate): Eq. 5.9.
- **3** Evaporation against mulch factor for four capacities: Eq. 5.4.
- **4** Red dot: your current setting.
- **5** “Mulch removes 31 % of bare-soil evaporation”: Eq. 9.1.

Equations behind this screen. Eq. 5.4, Eq. 5.9.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

18.14.1 The whole layer, part by part

The picture below is the complete drawing of the layer Mulch control (V6), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 1 part, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

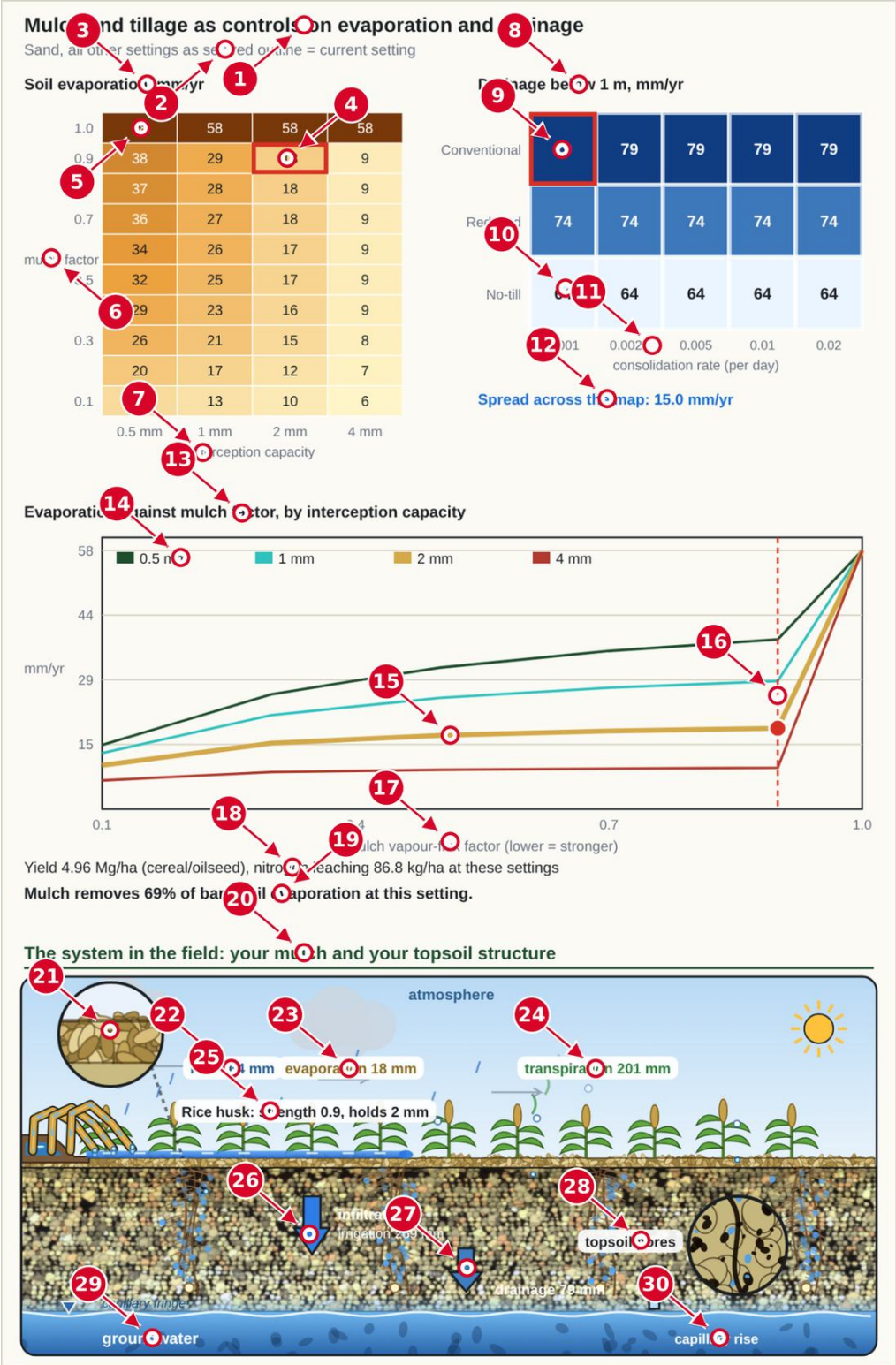


Figure 18.211: Mulch control (V6), part 1 of 1.

What the arrows point to.

- **1** The title, mulch and tillage as controls on evaporation and drainage.
- **2** The note that the red outline marks the current setting.

- **3** The map of the soil evaporation in mm per year.
- **4** The red outlined cell, the evaporation at the current mulch factor and interception capacity.
- **5** The top row, no mulch effect, the bare soil evaporation.
- **6** The mulch factor axis, from 1.0 for no mulch down to 0.1 for the strongest.
- **7** The interception capacity axis, from 0.5 to 4 mm.
- **8** The map of the drainage below 1 m in mm per year.
- **9** The red outlined cell, the drainage under the current tillage and consolidation rate.
- **10** The no-till row, with the smallest drainage.
- **11** The consolidation rate axis, per day.
- **12** The spread of the drainage across the map.
- **13** The curves of the evaporation against the mulch factor.
- **14** The legend of the interception capacities.
- **15** The thick line, the curve of the capacity in force.
- **16** The red dashed line, the current mulch factor.
- **17** The mulch vapour-flux factor axis.
- **18** The yield and the nitrogen leaching at these settings.
- **19** The share of the bare-soil evaporation that the mulch removes.
- **20** The system in the field, your mulch and your topsoil structure.
- **21** The lens that shows the mulch material.
- **22** The rain of the year.
- **23** The evaporation of the year under the mulch.
- **24** The transpiration of the year.
- **25** The mulch tag, its strength and the water held.
- **26** The infiltration arrow with the irrigation applied.
- **27** The drainage arrow.
- **28** The topsoil pores, drawn as an aggregate.
- **29** The groundwater.
- **30** The capillary rise.

18.14.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

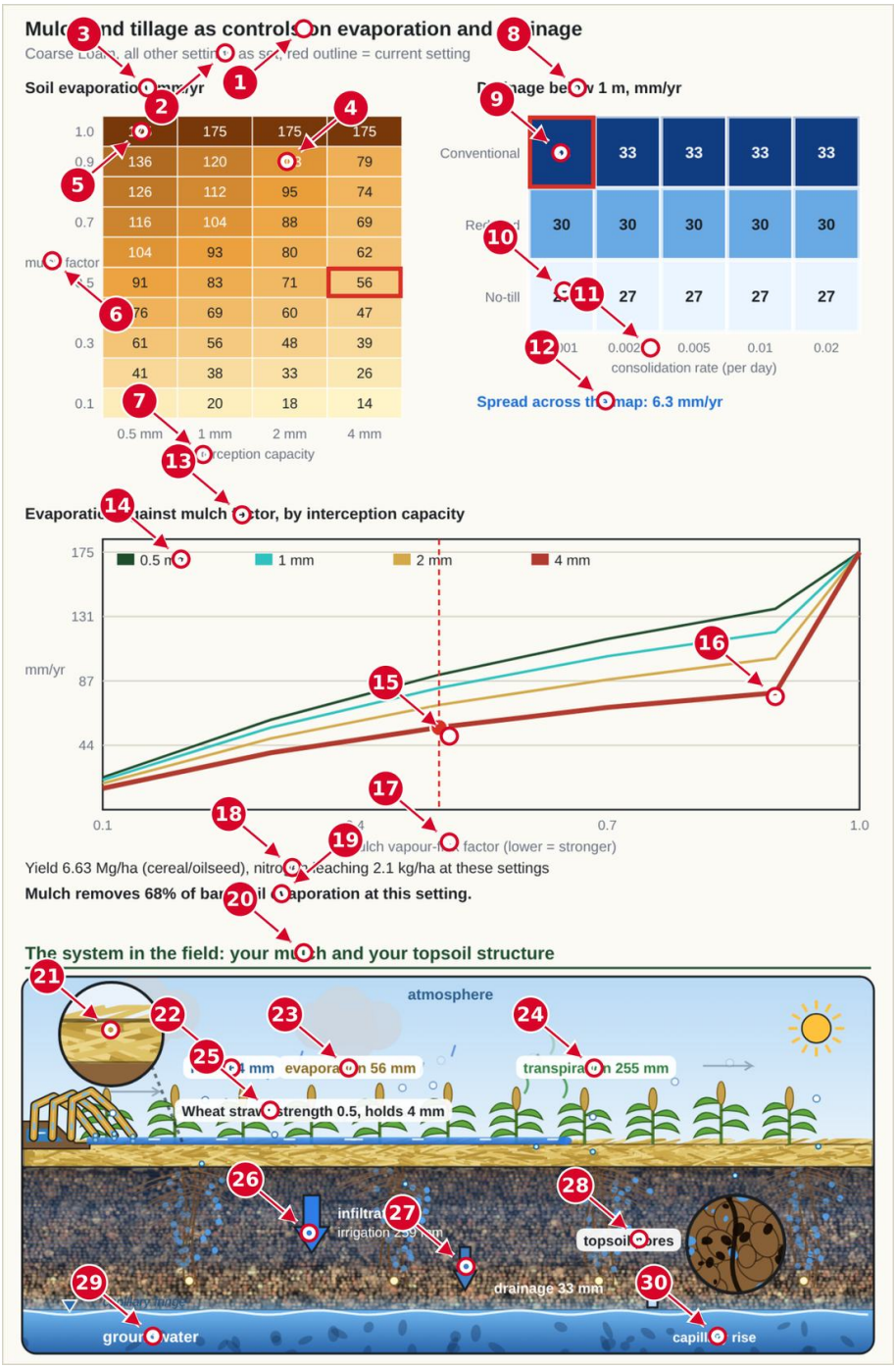


Figure 18.212: Mulch control (V6), part 1. A cell of the evaporation map clicked, so that mulch factor and capacity become the current setting.

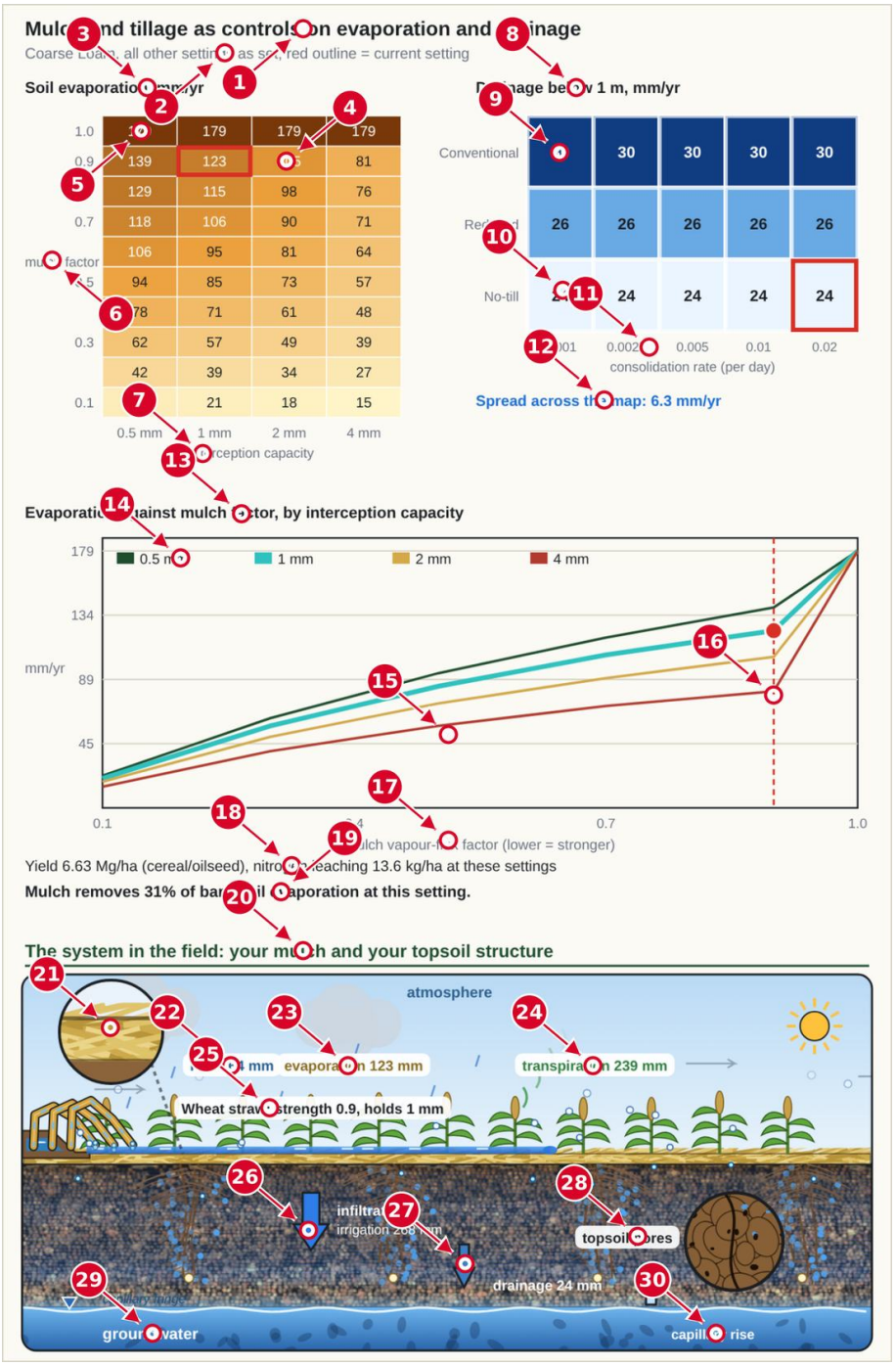


Figure 18.213: Mulch control (V6), part 1. A cell of the drainage map clicked, so that tillage and consolidation rate become the current setting.

18.15 Evaporation physics (E1)

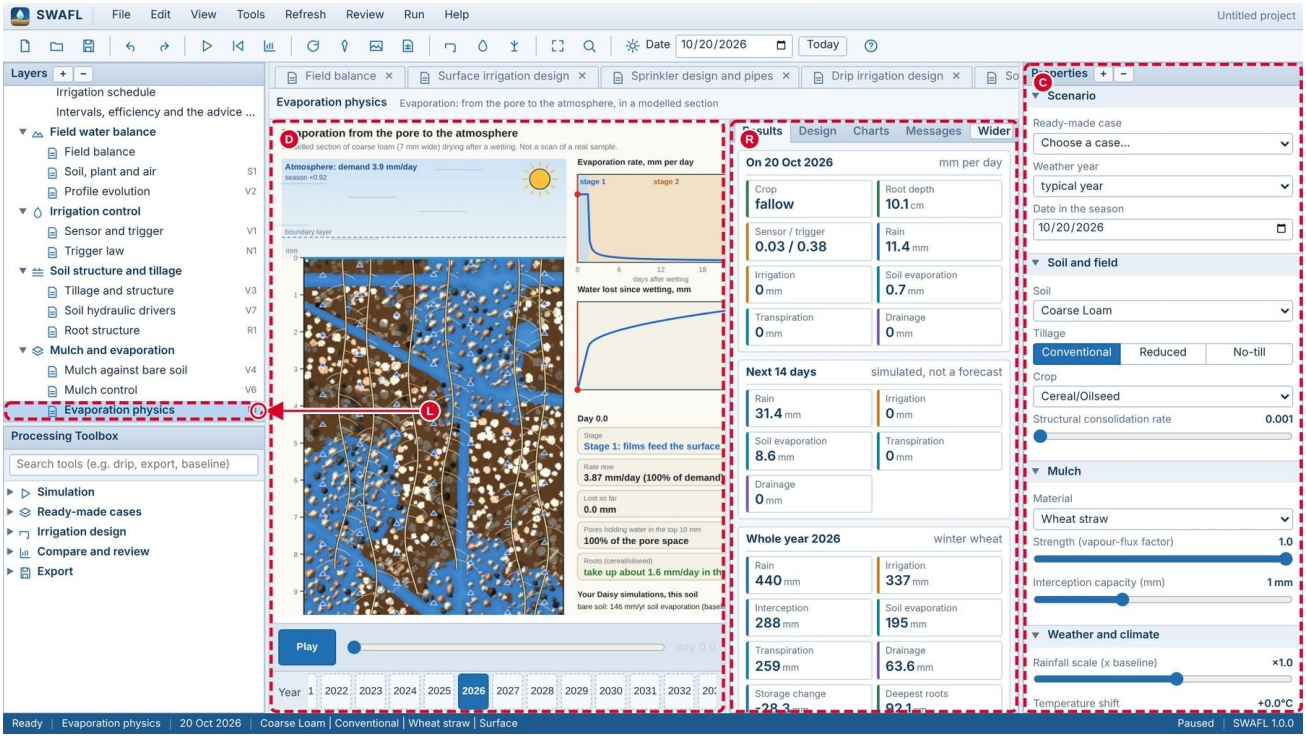


Figure 18.214: Evaporation physics (E1): the whole workbench.

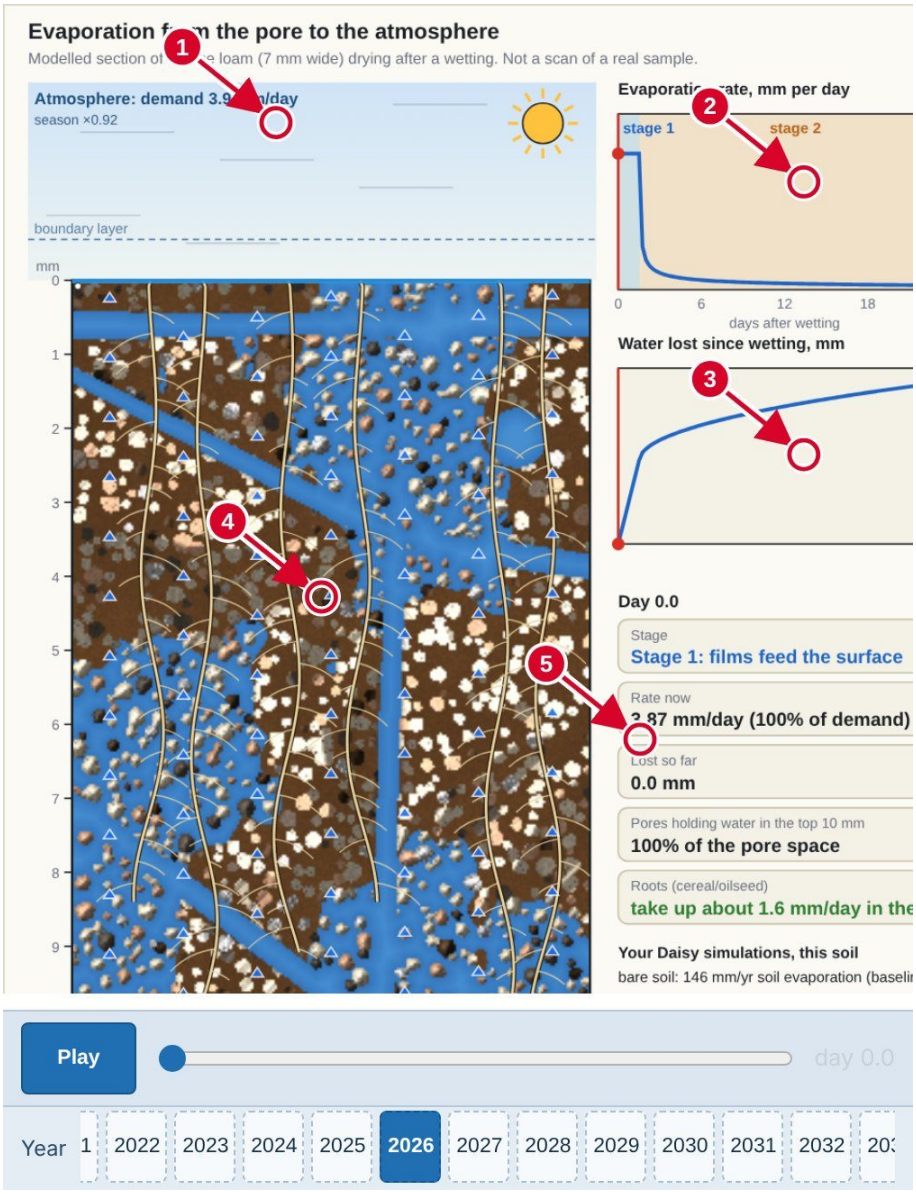


Figure 18.215: Evaporation physics (E1): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Atmospheric demand under mulch and canopy, e_0 = e_0,b phi (1 - 0.4 c): Eq. 5.26.
- **2** Evaporation rate against time since wetting, stage 1 then stage 2: Eq. 9.2.
- **3** Cumulative water lost since wetting, Q(t): Eq. 9.3.
- **4** Section of soil: dry layer depth (Eq. 9.4) and water-filled pores from the capillary rule (Eq. 8.4) with the suction profile of Eq. 9.5.
- **5** Panel with the stage, the rate now and the water lost so far: Eqs. 9.2 and 9.3.

Equations behind this screen. Eq. 5.4, Eq. 5.26, Eq. 8.3, Eq. 8.4, Eq. 9.2, Eq. 9.3, Eq. 9.4, Eq. 9.5.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.26. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$E_{0,\text{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65\cdot\tfrac{1}{2}[1+\sin(2\pi(\text{doy}-100)/365)]}{0.675}\,\text{clip}\big(1{-}0.12(\rho{-}1);\,0.85,\,1.12\big)\sqrt{\gamma_y}\ \text{m} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

Eq. 8.3. Where explained: chapter "Soil structure and tillage layers", section "How the section is generated".

$$\theta(h) = \theta_r + (\phi - \theta_r)\,S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

Eq. 8.4. Where explained: chapter "Soil structure and tillage layers", section "How the section is generated".

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\,g\,h} \;\approx\; \frac{0.149\ \text{cm}^2}{h} \quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

Eq. 9.2. Where explained: chapter "Mulch and evaporation layers", section "Evaporation physics (E1)".

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

Provenance (Eq. 9.2): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

Eq. 9.3. Where explained: chapter "Mulch and evaporation layers", section "Evaporation physics (E1)".

$$Q(t) = q_0 + \begin{cases} e_0\,t, & t \le t_1 \\ (Q_1 - q_0) + 2\,e_0\,\tau\Big(\sqrt{1 + (t - t_1)/\tau} - 1\Big), & t > t_1 \end{cases} \tag{9.3}$$

Provenance (Eq. 9.3): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

Eq. 9.4. Where explained: chapter "Mulch and evaporation layers", section "Evaporation physics (E1)".

$$\ell(t) = \min\!\left\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\right\}\ \text{mm} \quad (Q > Q_1), \tag{9.4}$$

Provenance (Eq. 9.4): **[Engineered]**: drying-front geometry chosen for drawing.

Eq. 9.5. Where explained: chapter "Mulch and evaporation layers", section "Evaporation physics (E1)".

$$\log_{10} h(z) = \log_{10} h_s + (\log_{10} h_i - \log_{10} h_s)(1 - e^{-(z-\ell)/\lambda}) \quad (z \ge \ell), \qquad h = 10^{5}\ \text{cm for } z < \ell. \tag{9.5}$$

Provenance (Eq. 9.5): **[Engineered]**: drying-front geometry chosen for drawing.

18.15.1 The whole layer, part by part

The picture below is the complete drawing of the layer Evaporation physics (E1), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 3 parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

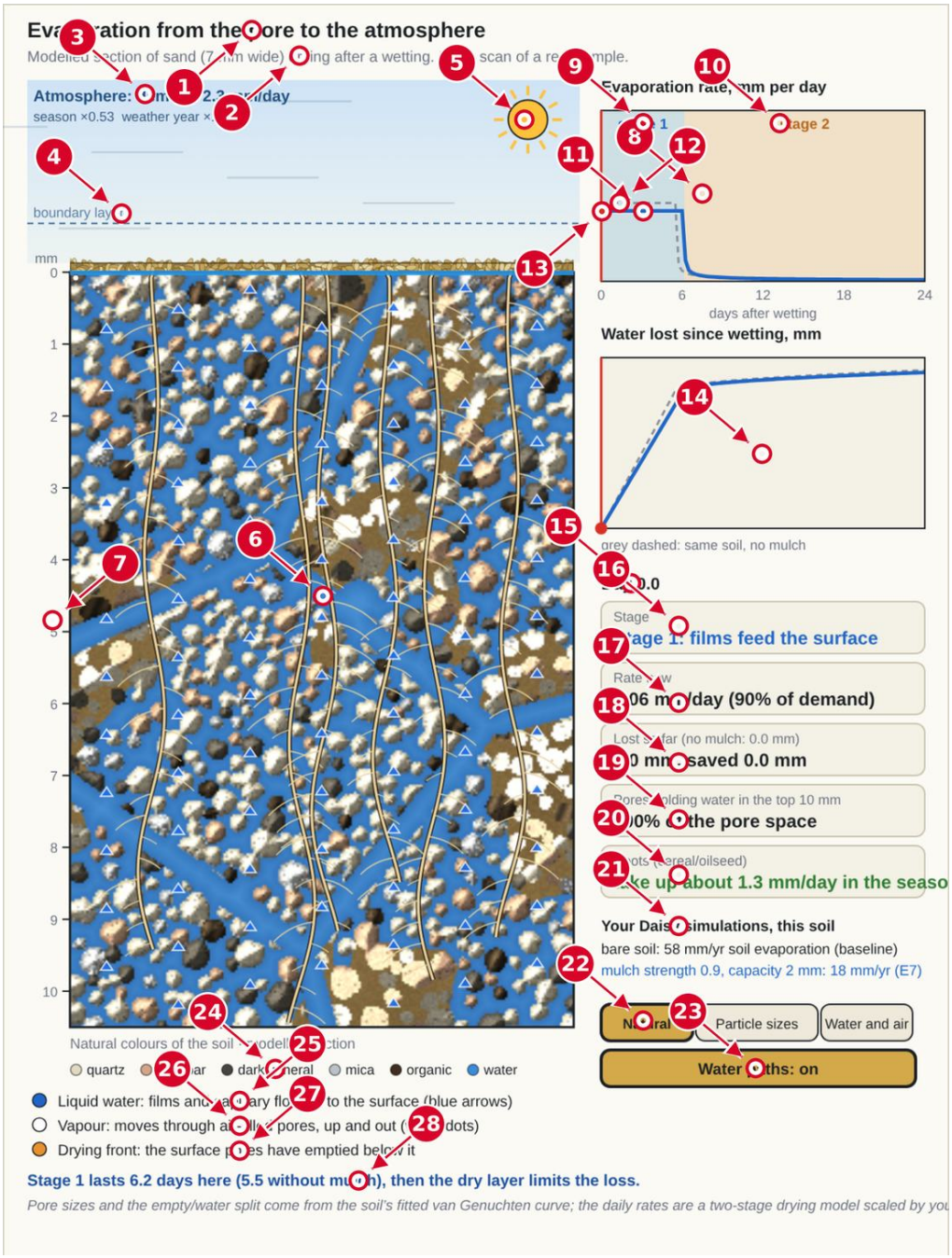


Figure 18.216: Evaporation physics (E1), part 1 of 3.

What the arrows point to.

- **1** The title, the evaporation from the pore to the atmosphere.
- **2** The subtitle, a modelled section of the soil drying after a wetting.
- **3** The atmospheric demand in mm per day, with the season and weather scaling.
- **4** The boundary layer through which the vapour must pass.
- **5** The sun, the energy that drives the demand.
- **6** The modelled section of the soil, 7 mm wide, with grains, films and the water in blue.
- **7** The depth scale of the section in mm.
- **8** The chart of the evaporation rate in mm per day against the days after wetting.
- **9** The stage 1 band, the films feed the surface.

- **10** The stage 2 band, the dry layer limits the loss.
- **11** The blue line, the rate with the mulch.
- **12** The grey dashed line, the same soil without mulch.
- **13** The red dot, the day shown, which moves with the play bar.
- **14** The chart of the water lost since wetting in mm.
- **15** The day shown on the clock.
- **16** The stage card, whether the films feed the surface or the dry layer limits the loss.
- **17** The rate now in mm per day and its share of the demand.
- **18** The water lost so far and the water saved by the mulch.
- **19** The share of the pore space still holding water in the top 10 mm.
- **20** The roots card, the uptake of the crop.
- **21** The yearly figures from your Daisy simulations for this soil.
- **22** The colour buttons of the section, Natural, Particle sizes and Water and air.
- **23** The Water paths button, which draws the flow of the water and the vapour.
- **24** The legend of the minerals and the water.
- **25** The blue arrows, the liquid water moving up by films and capillary flow.
- **26** The white dots, the vapour in the air-filled pores.
- **27** The drying front, below which the surface pores have emptied.
- **28** The duration of stage 1 with the mulch and without it.

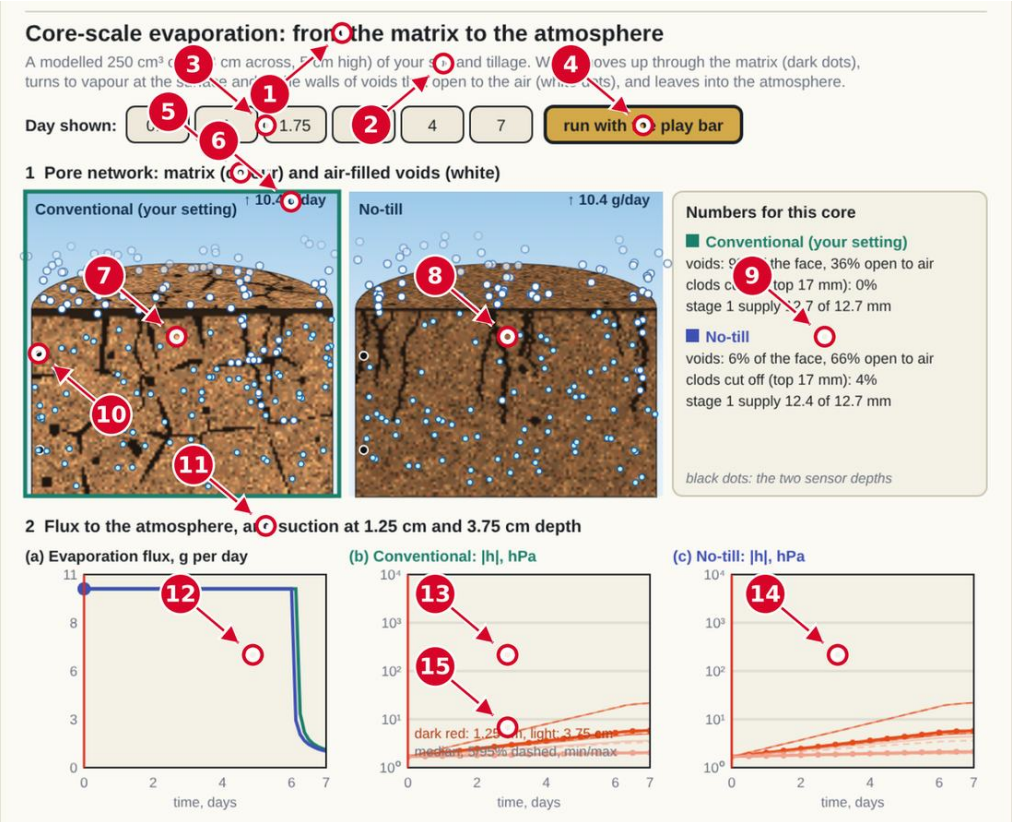


Figure 18.217: Evaporation physics (E1), part 2 of 3.

What the arrows point to.

- **1** The core-scale evaporation, from the matrix to the atmosphere.
- **2** The description of the modelled 250 cm³ core.
- **3** The Day shown buttons, 0.5, 1, 1.75, 2.5, 4 and 7 days.
- **4** The button that runs the core with the play bar.
- **5** The pore network, the matrix in colour and the air-filled voids in white.

- **6** The flux of the core to the atmosphere in g per day.
- **7** The conventional core, with large clods and open cracks.
- **8** The no-till core, with fine open voids.
- **9** The numbers of the two cores, voids on the face, open to air and the stage 1 supply.
- **10** The black dots, the two sensor depths of the core.
- **11** The flux to the atmosphere and the suction at 1.25 and 3.75 cm depth.
- **12** The evaporation flux of the two cores against the days.
- **13** The suction of the conventional core at the two depths.
- **14** The suction of the no-till core at the two depths.
- **15** The dark red line, the median suction at 1.25 cm, with the light line at 3.75 cm.

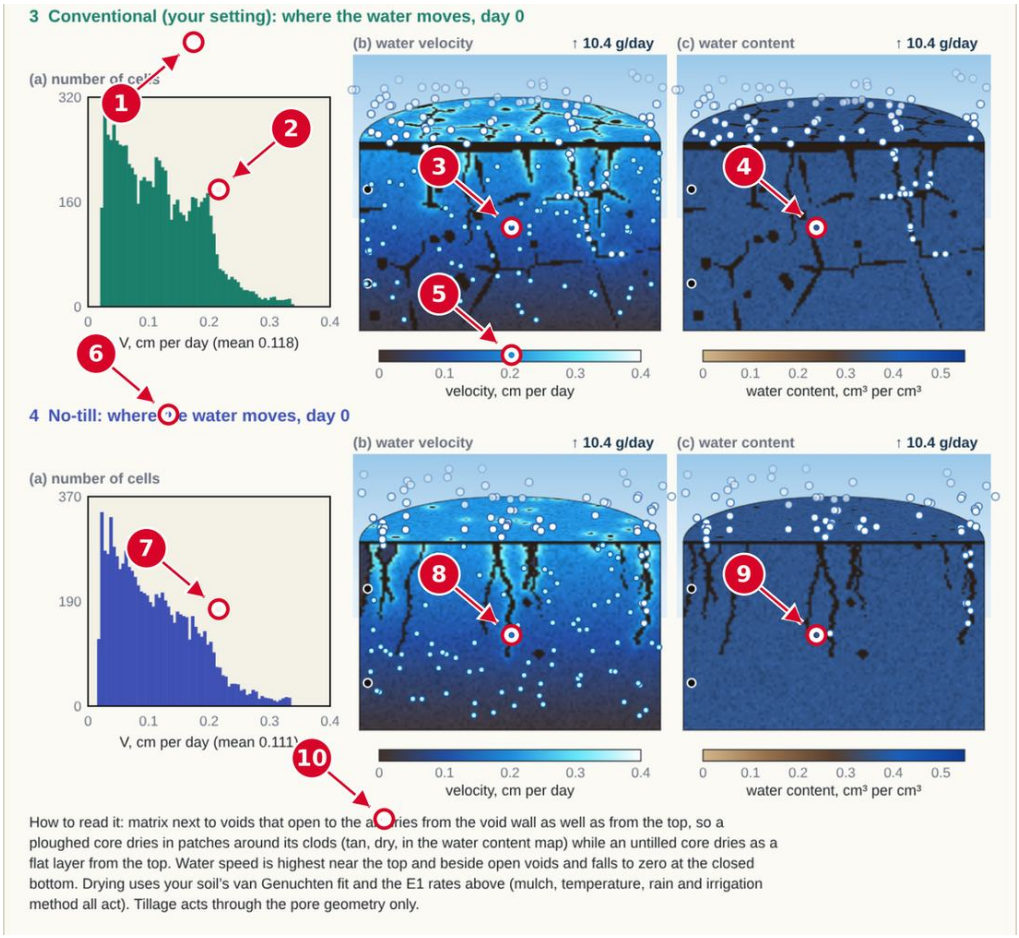


Figure 18.218: Evaporation physics (E1), part 3 of 3.

What the arrows point to.

- **1** The conventional core, where the water moves.
- **2** The histogram of the number of cells by water speed.
- **3** The map of the water velocity.
- **4** The map of the water content.
- **5** The colour scale of the velocity in cm per day.
- **6** The no-till core, where the water moves.
- **7** The histogram of the no-till core.
- **8** The velocity map of the no-till core.
- **9** The water content map of the no-till core.
- **10** How to read the maps, drying from the void wall as well as from the top.

18.15.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

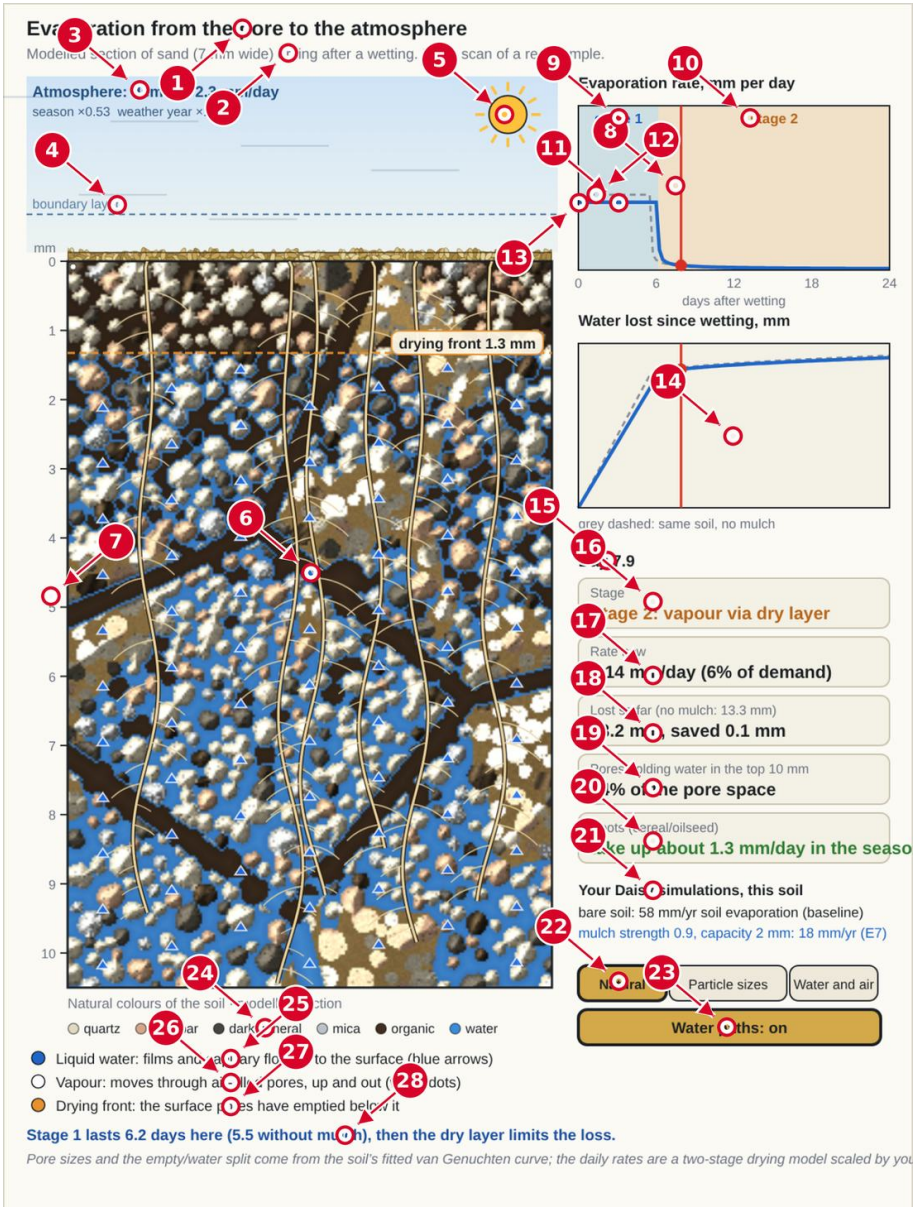


Figure 18.219: Evaporation physics (E1), part 1, playback a third of the way (day 7.9).

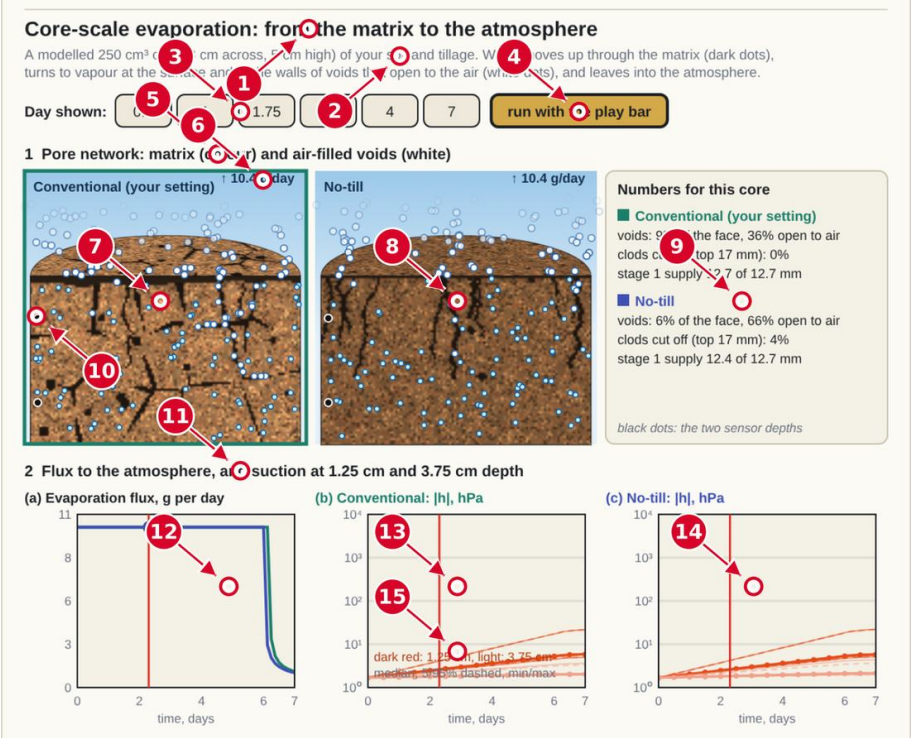


Figure 18.220: Evaporation physics (E1), part 2, playback a third of the way (day 7.9).

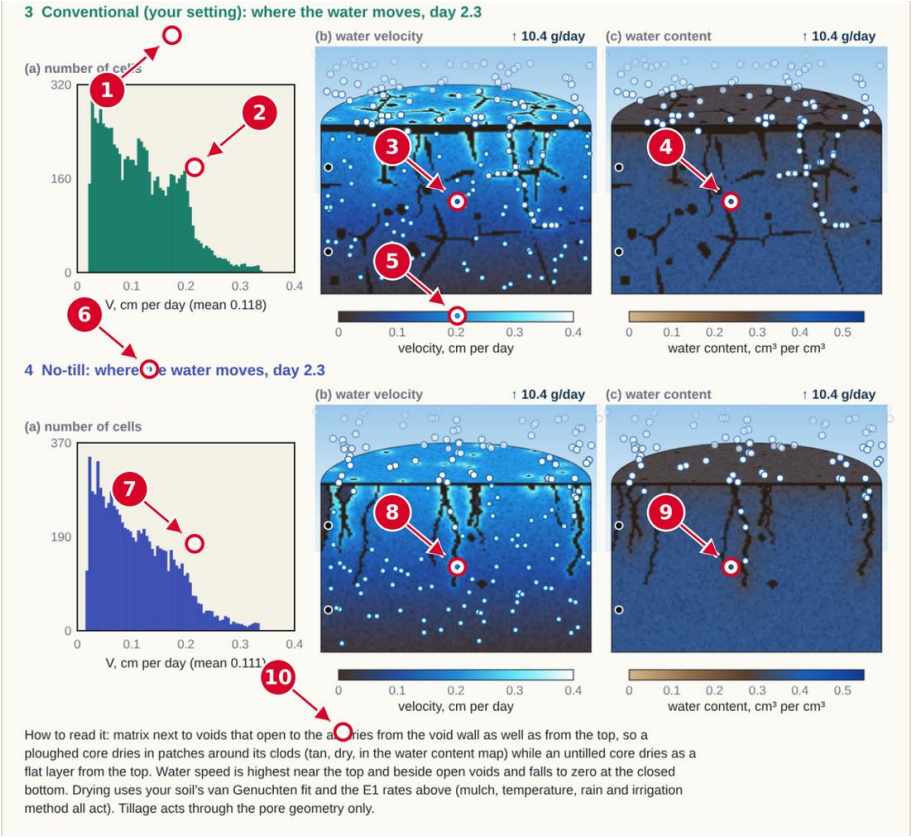


Figure 18.221: Evaporation physics (E1), part 3, playback a third of the way (day 7.9).

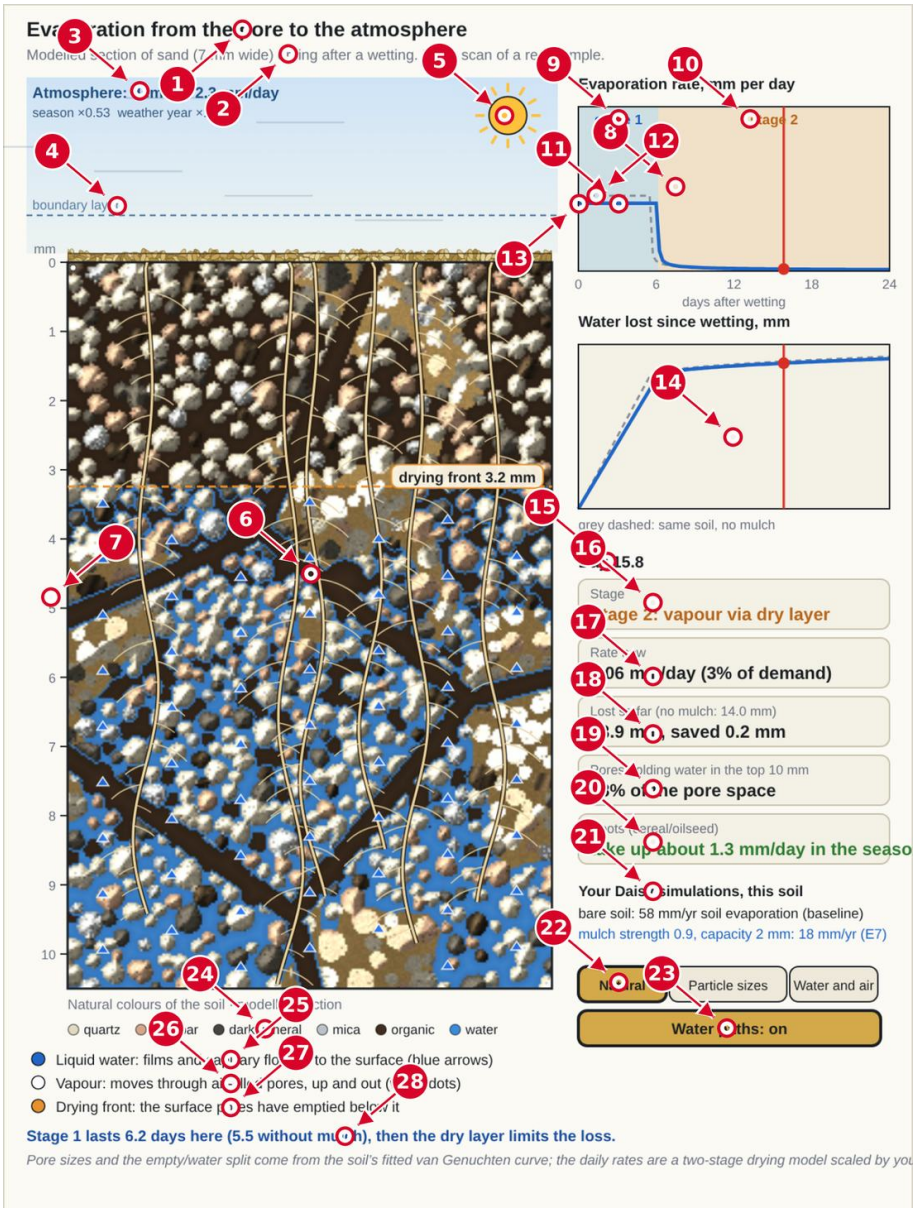


Figure 18.222: Evaporation physics (E1), part 1, playback two thirds of the way (day 15.8).

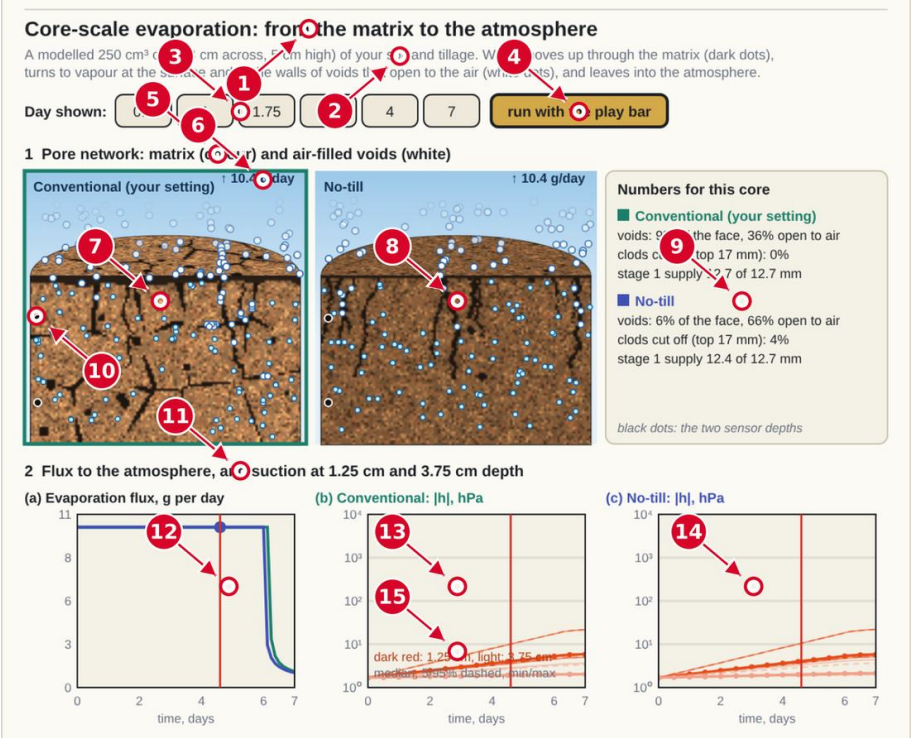


Figure 18.223: Evaporation physics (E1), part 2, playback two thirds of the way (day 15.8).

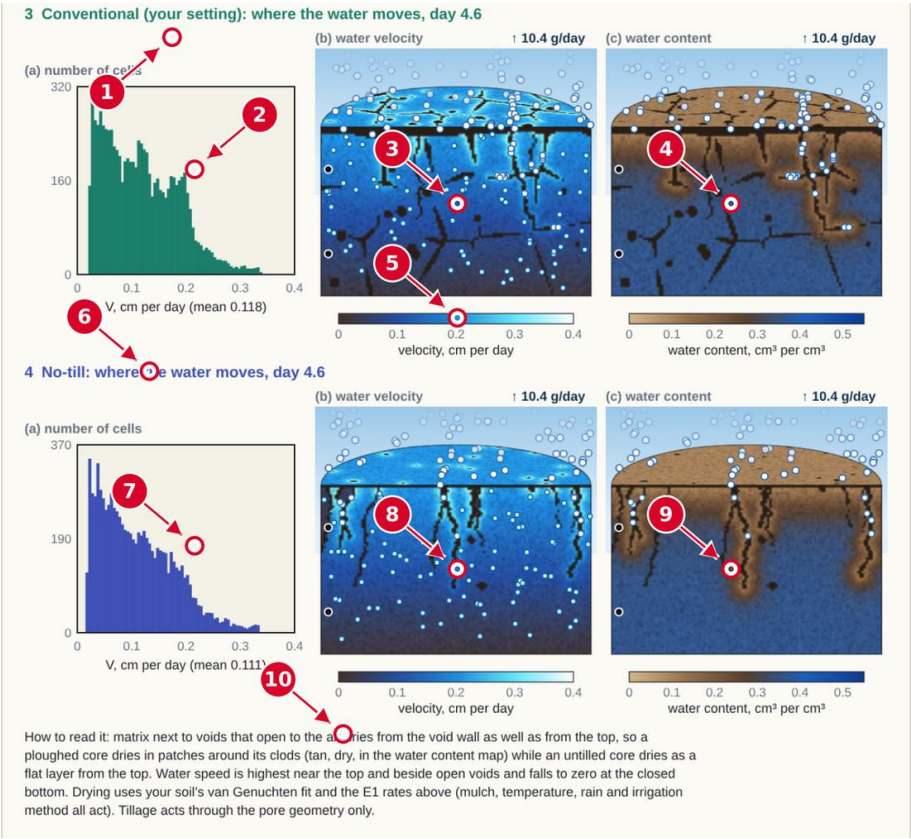


Figure 18.224: Evaporation physics (E1), part 3, playback two thirds of the way (day 15.8).

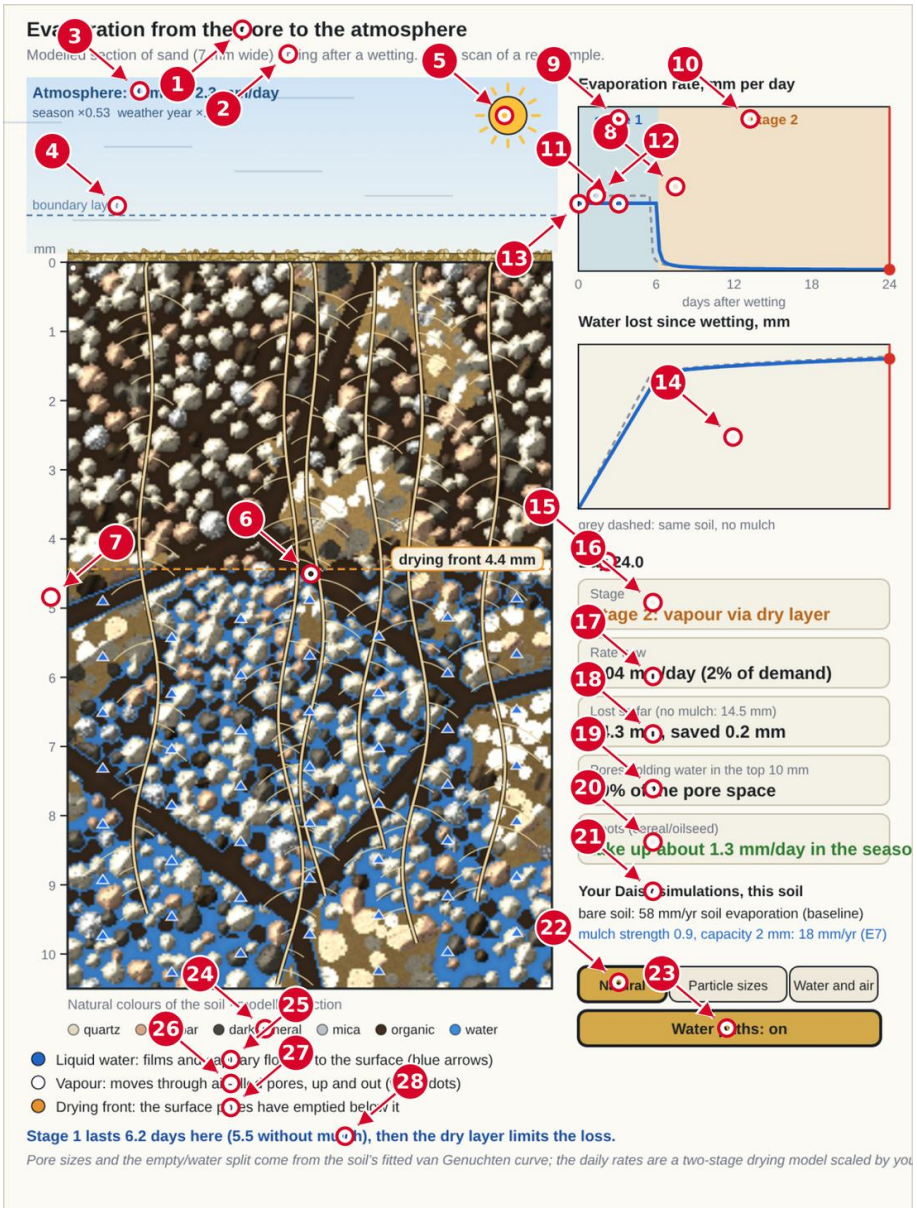


Figure 18.225: Evaporation physics (E1), part 1, playback the end (day 24.0).

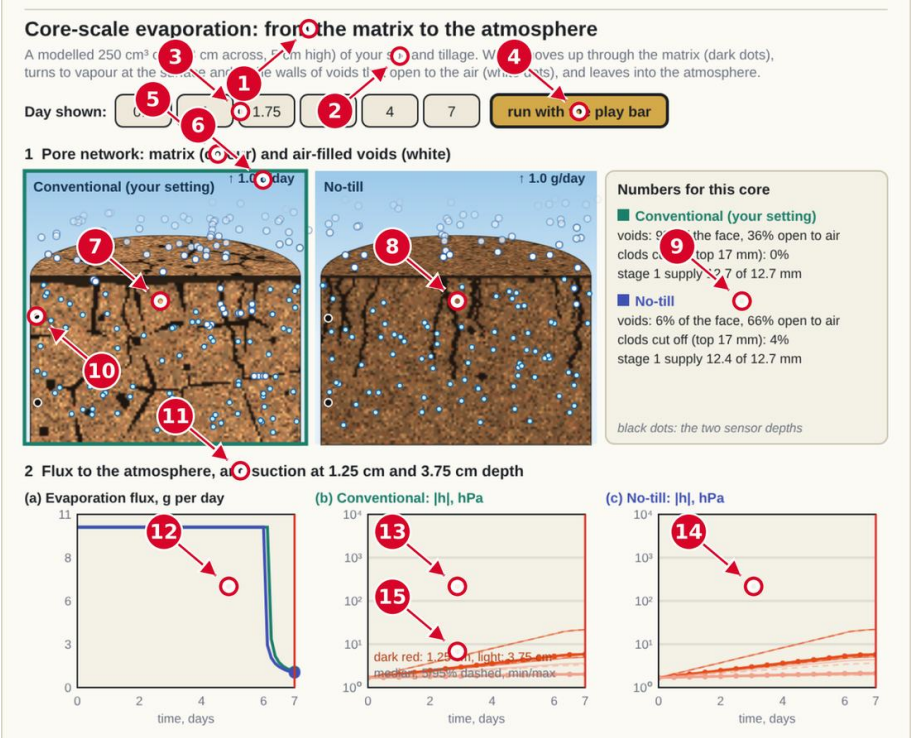


Figure 18.226: Evaporation physics (E1), part 2, playback the end (day 24.0).

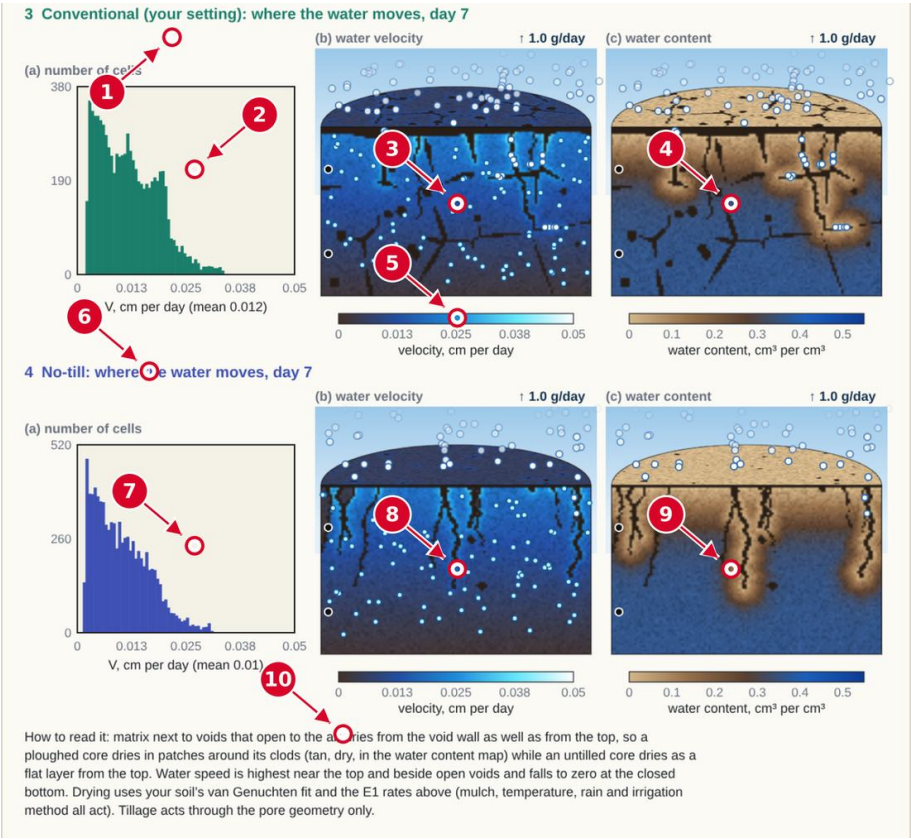


Figure 18.227: Evaporation physics (E1), part 3, playback the end (day 24.0).

18.15.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

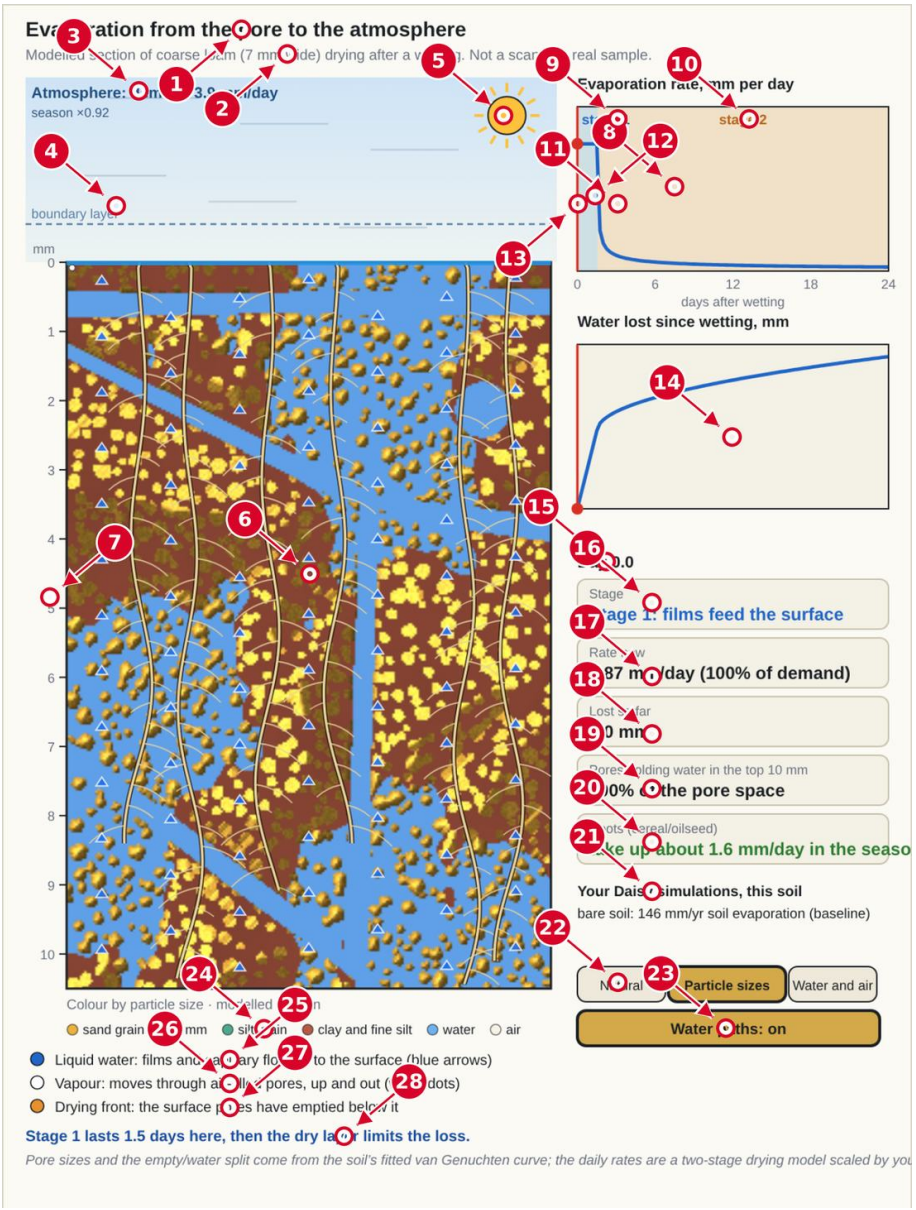


Figure 18.228: Evaporation physics (E1), part 1. The Particle sizes view of the section chosen.

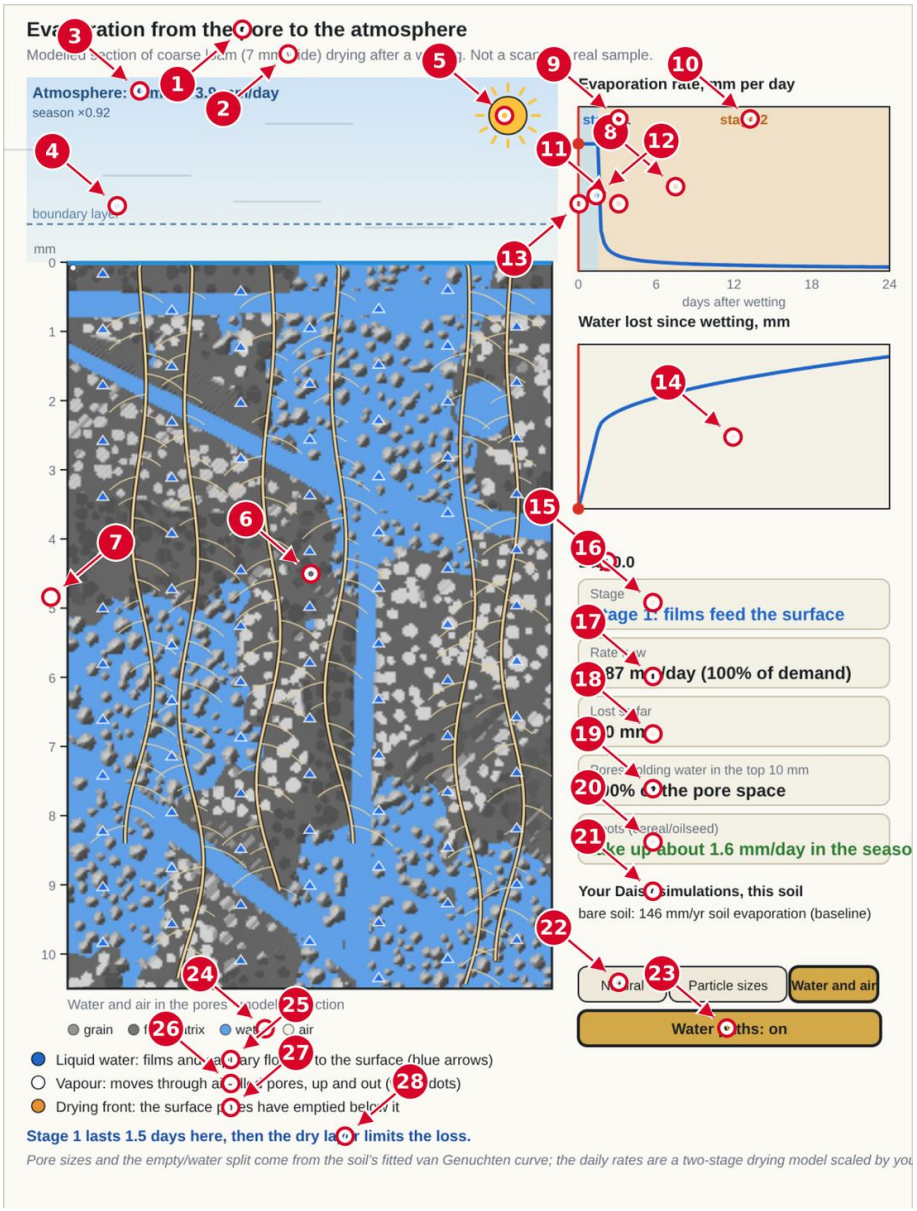


Figure 18.229: Evaporation physics (E1), part 1. The Water and air view of the section chosen.

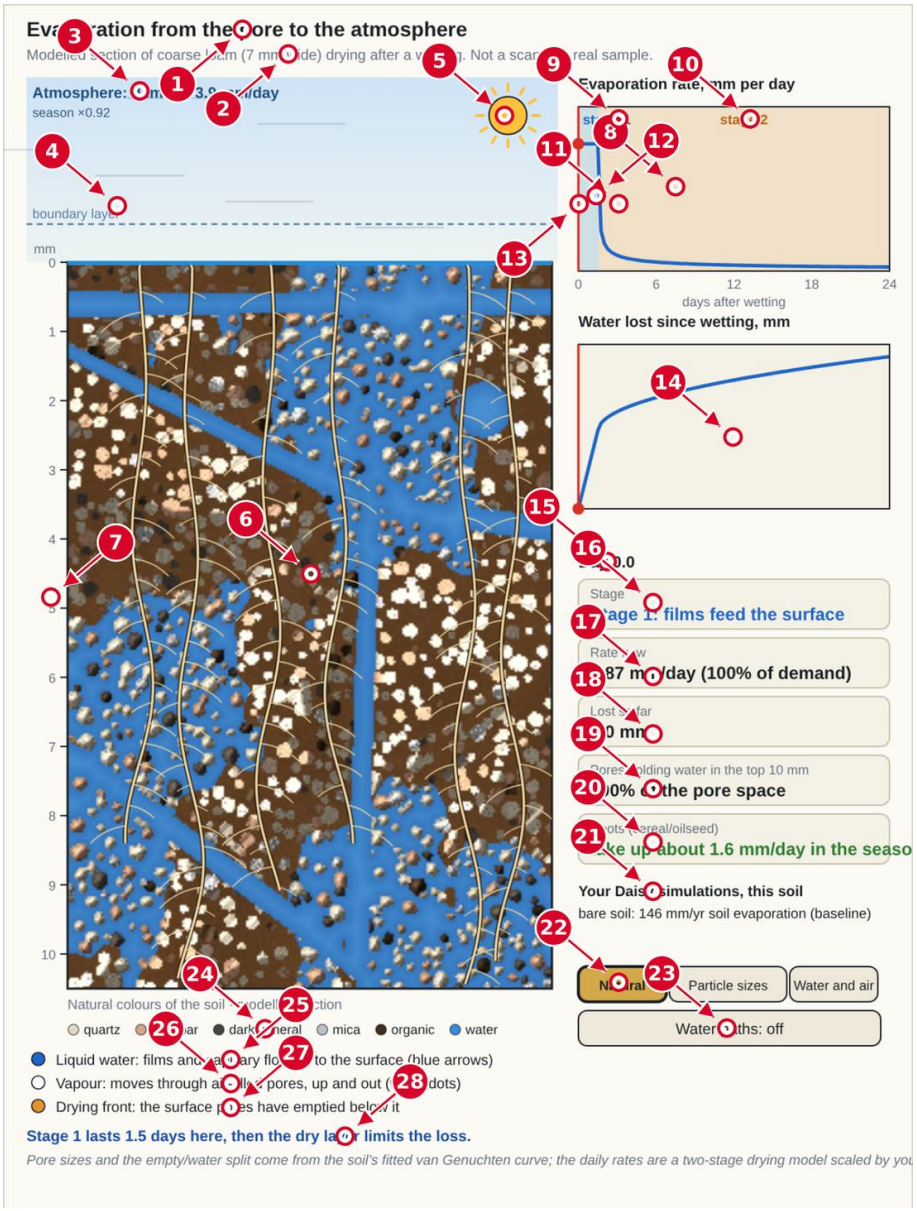


Figure 18.230: Evaporation physics (E1), part 1. The Water paths button switched off, so the arrows and dots of the flow are hidden.

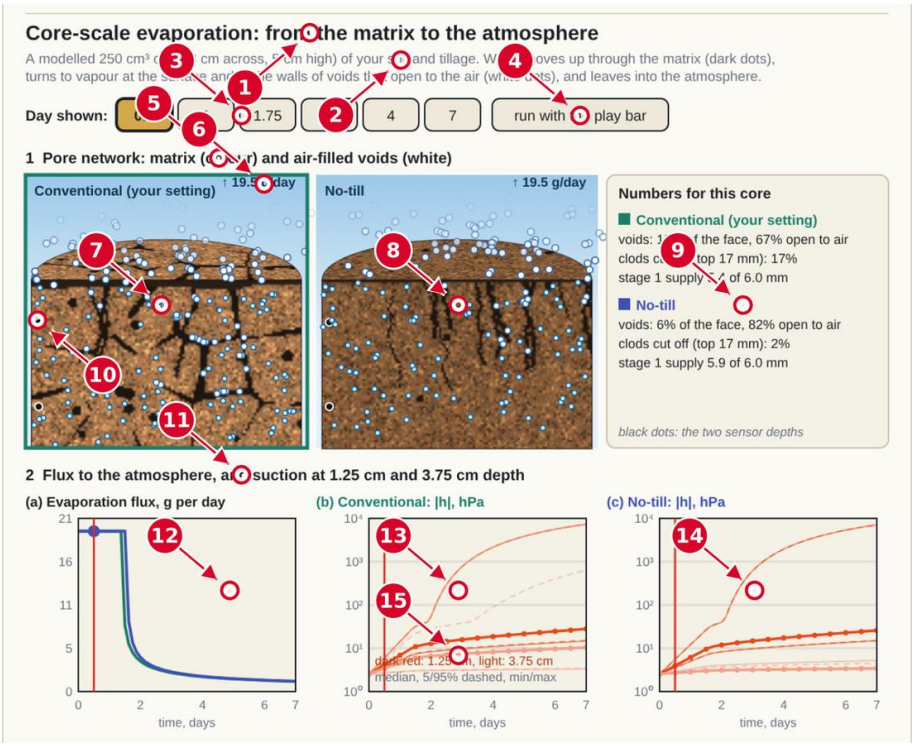


Figure 18.231: Evaporation physics (E1), part 2. The day 0.5 button pressed, so the cores are shown at half a day.

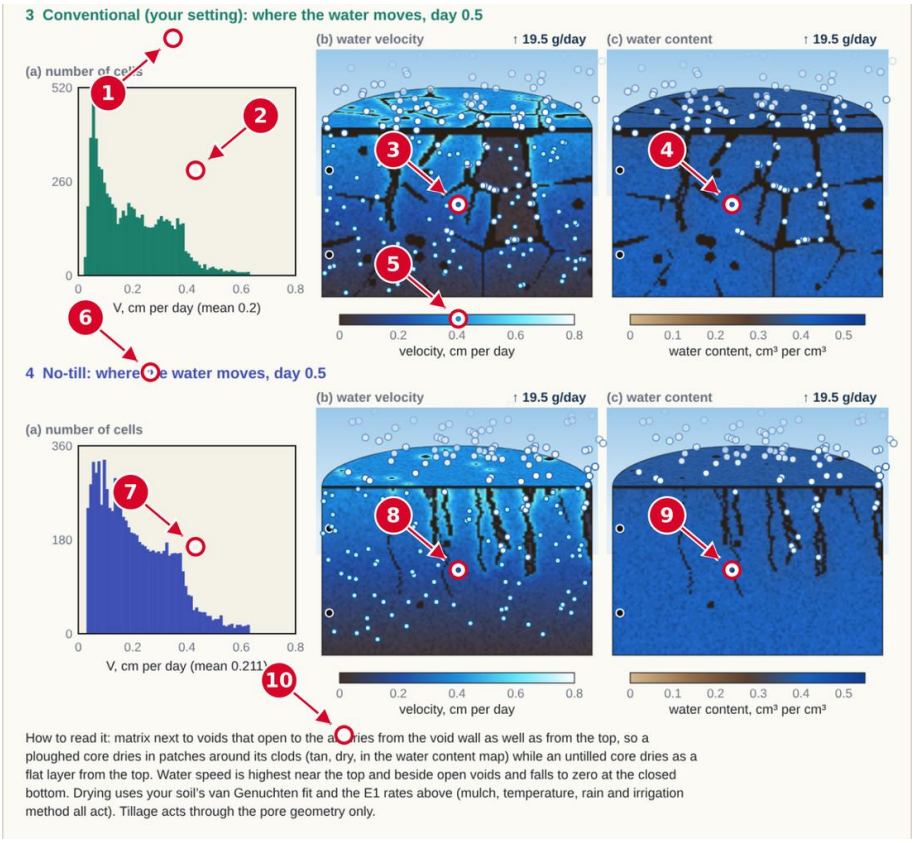


Figure 18.232: Evaporation physics (E1), part 3. The day 0.5 button pressed, so the cores are shown at half a day.

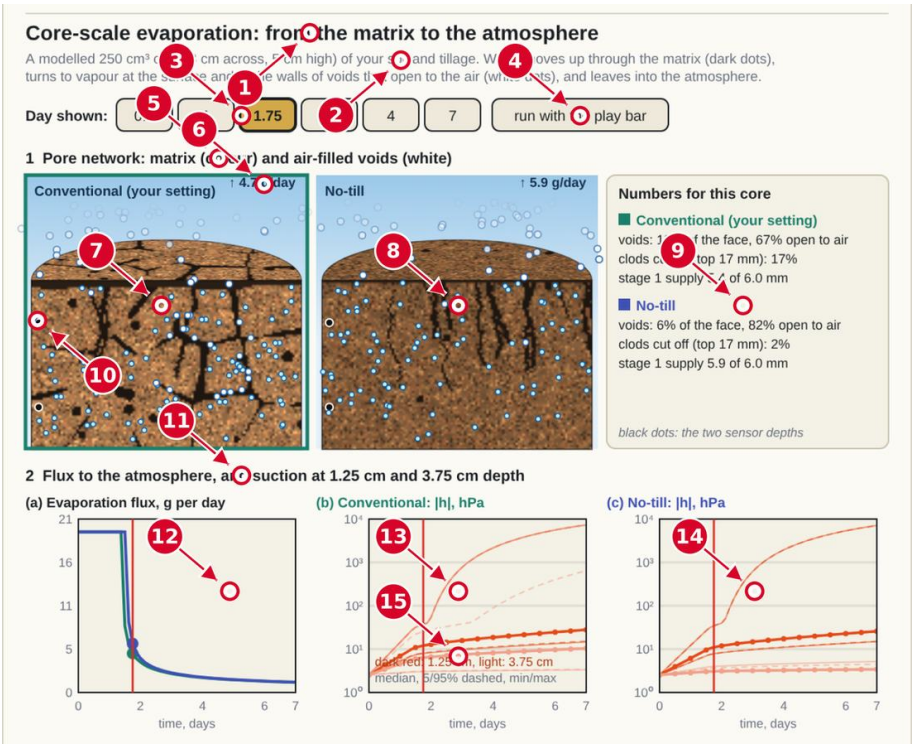


Figure 18.233: Evaporation physics (E1), part 2. The day 1.75 button pressed.

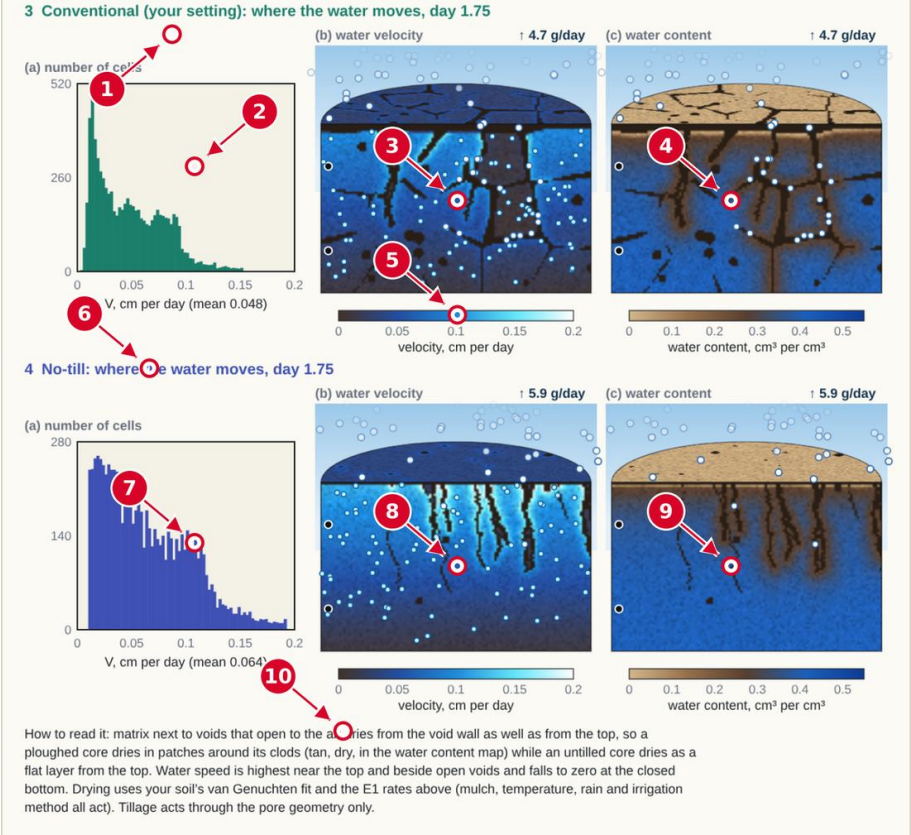


Figure 18.234: Evaporation physics (E1), part 3. The day 1.75 button pressed.

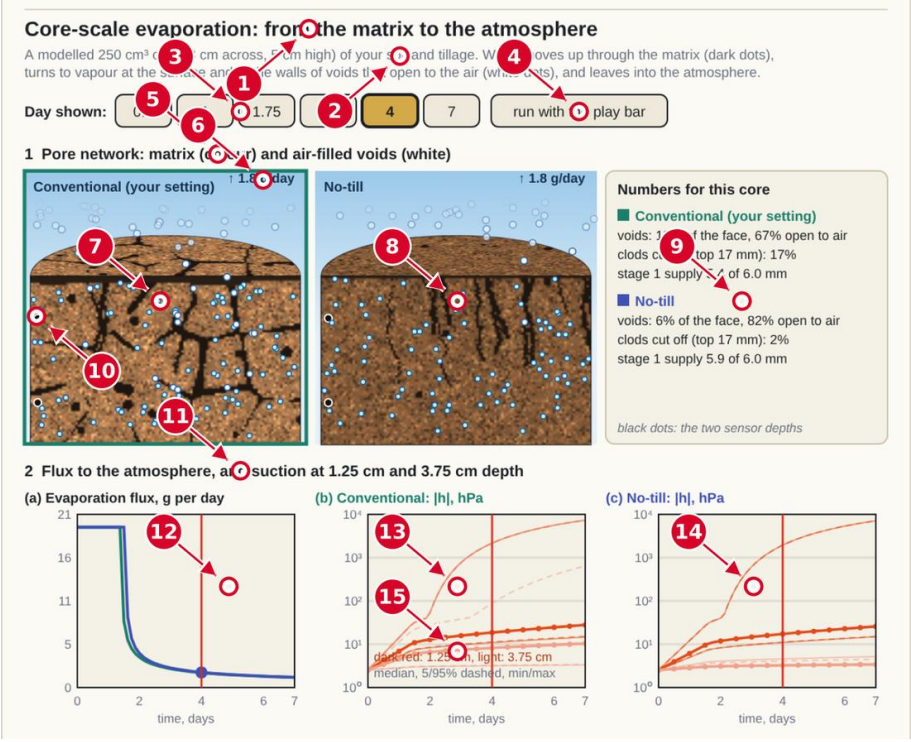


Figure 18.235: Evaporation physics (E1), part 2. The day 4 button pressed.

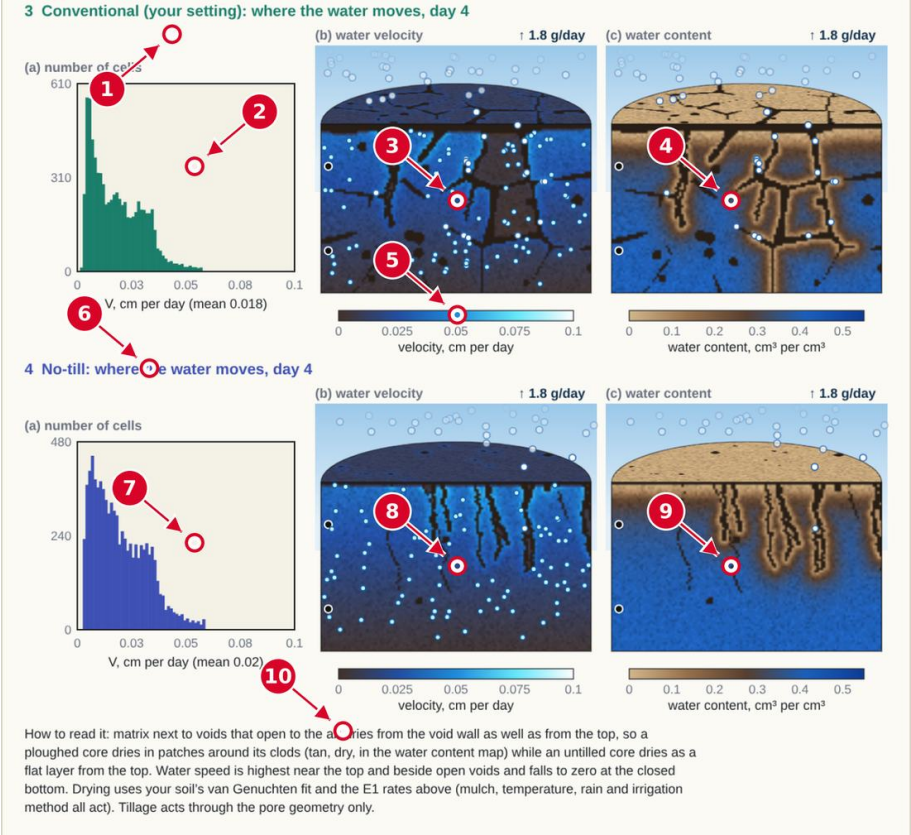


Figure 18.236: Evaporation physics (E1), part 3. The day 4 button pressed.

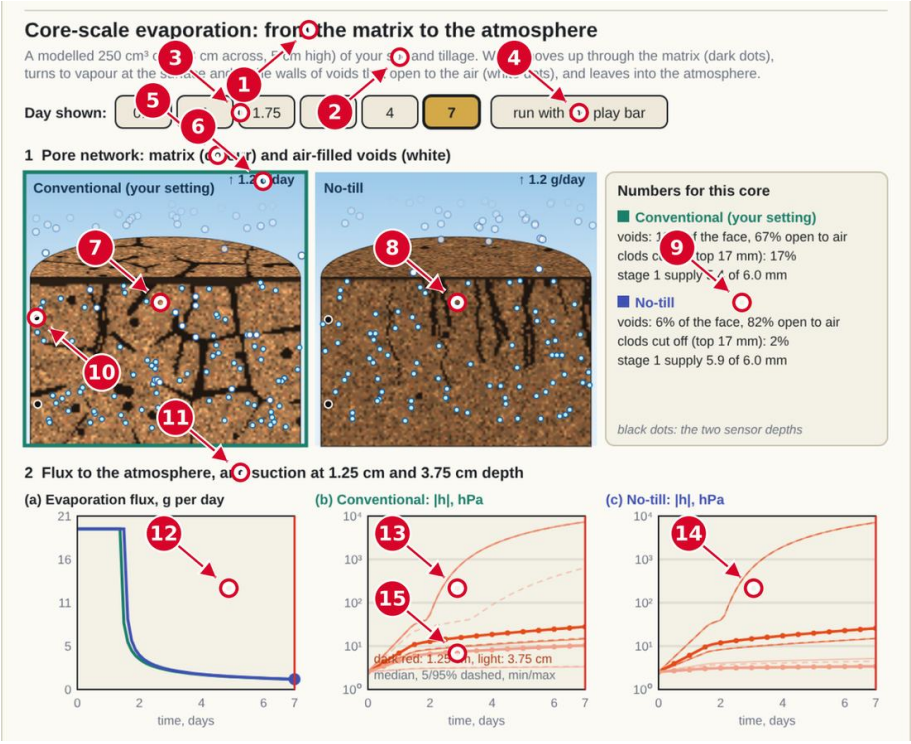


Figure 18.237: Evaporation physics (E1), part 2. The day 7 button pressed, so the cores are shown at the end of stage 1.

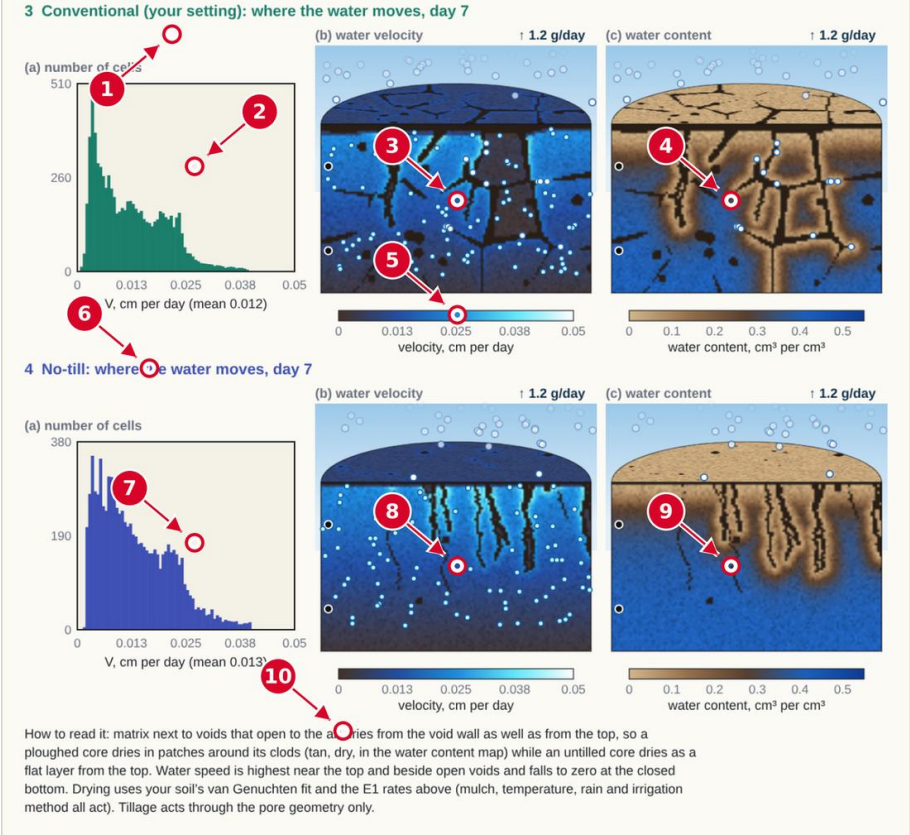


Figure 18.238: Evaporation physics (E1), part 3. The day 7 button pressed, so the cores are shown at the end of stage 1.

18.16 Fate of saved water (N2)

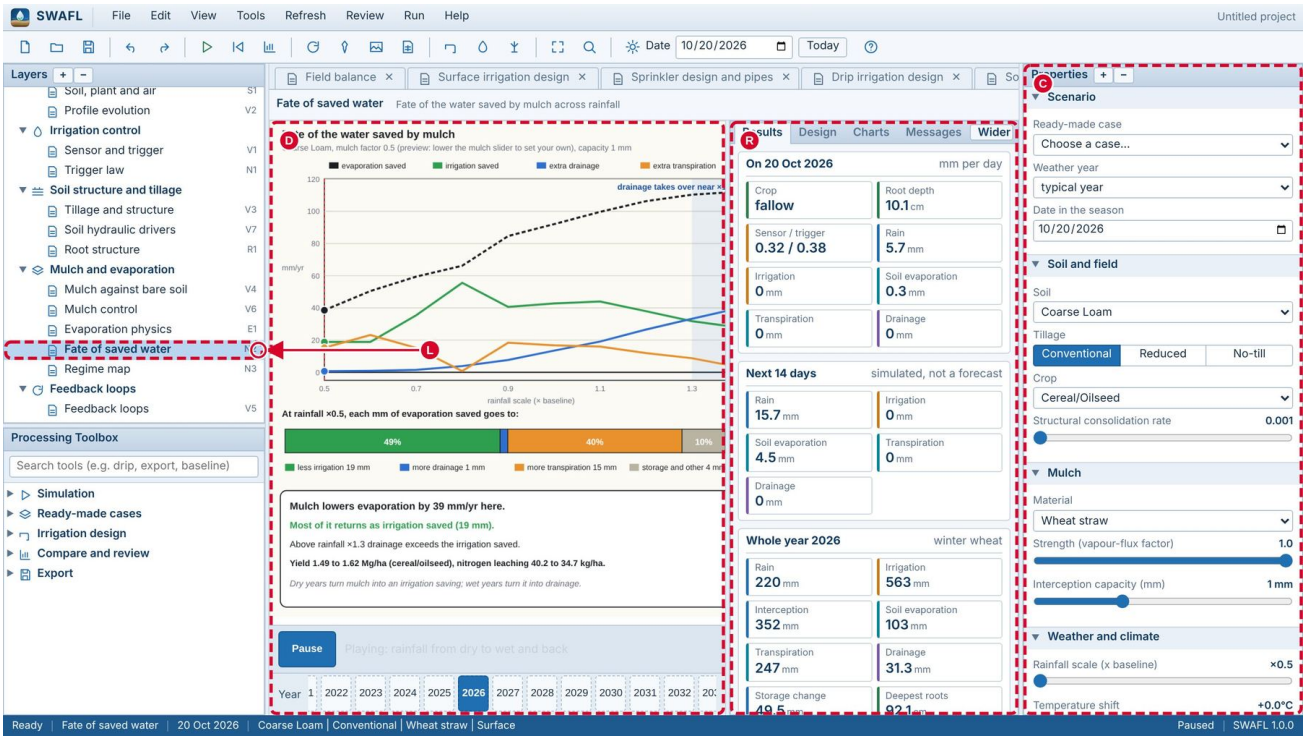


Figure 18.239: Fate of saved water (N2): the whole workbench.

Figure 18.240: Fate of saved water (N2): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Evaporation saved, irrigation saved, extra drainage and extra transpiration against rainfall scale: Eq. 9.6.
- **2** Rainfall at which drainage overtakes the irrigation saved: from the curves of Eq. 9.6.
- **3** Where each mm of evaporation saved goes (irrigation, drainage, transpiration, storage): Eq. 9.7.
- **4** The sentence under the picture is built from the shares of Eq. 9.7.

Equations behind this screen. Eq. 9.6, Eq. 9.7.

Eq. 9.6. Where explained: chapter “Mulch and evaporation layers”, section “Fate of saved water (N2)”.

$$\begin{aligned}\Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho),\\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho),\end{aligned} \tag{9.6}$$

Provenance (Eq. 9.6): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

Eq. 9.7. Where explained: chapter "Mulch and evaporation layers", section "Fate of saved water (N2)".

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\ f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\ f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\ f_{\text{rest}} = \max\{0,\ 1 - f_I - f_D - f_T\}. \tag{9.7}$$

Provenance (Eq. 9.7): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

18.16.1 The whole layer, part by part

The picture below is the complete drawing of the layer Fate of saved water (N2), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 1 part, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

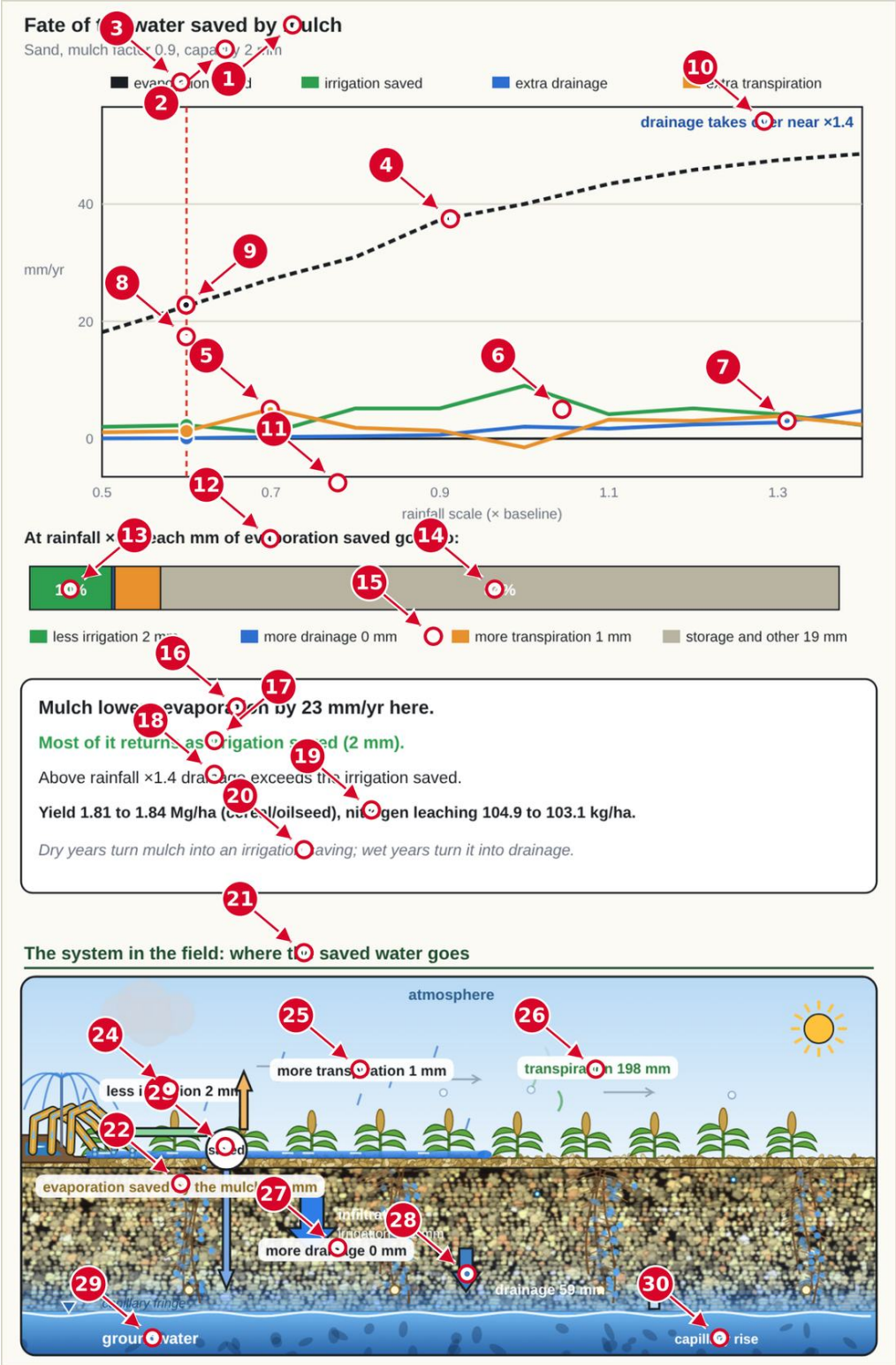


Figure 18.241: Fate of saved water (N2), part 1 of 1.

What the arrows point to.

- **1** The title, the fate of the water saved by mulch.
- **2** The settings of the chart, the soil, the mulch factor and the capacity.

- **3** The legend of the four lines, evaporation saved, irrigation saved, extra drainage and extra transpiration.
- **4** The black dotted line, the evaporation saved by the mulch at each rainfall scale.
- **5** The orange line, the extra transpiration.
- **6** The green line, the irrigation saved.
- **7** The blue line, the extra drainage, which grows with the rainfall.
- **8** The red dashed line, the rainfall scale in force.
- **9** The point of the evaporation saved at that rainfall scale.
- **10** The rainfall scale above which the drainage takes over.
- **11** The rainfall scale axis, as a multiple of the baseline.
- **12** The partition of each mm of evaporation saved at the rainfall in force.
- **13** The share that returns as less irrigation, in green.
- **14** The share that stays as storage and other terms, in grey.
- **15** The legend of the partition with the values in mm.
- **16** The box with the conclusion for these settings.
- **17** The main destination of the saved water.
- **18** The rainfall above which the drainage exceeds the irrigation saved.
- **19** The yield and the nitrogen leaching before and after the mulch.
- **20** The message that dry years turn the mulch into irrigation saving and wet years into drainage.
- **21** The system in the field, where the saved water goes.
- **22** The tag of the evaporation saved by the mulch.
- **23** The saved water leaving the surface, drawn as the circle labelled saved.
- **24** The irrigation saved.
- **25** The extra transpiration.
- **26** The transpiration of the year.
- **27** The extra drainage.
- **28** The drainage arrow.
- **29** The groundwater.
- **30** The capillary rise.

18.17 Regime map (N3)

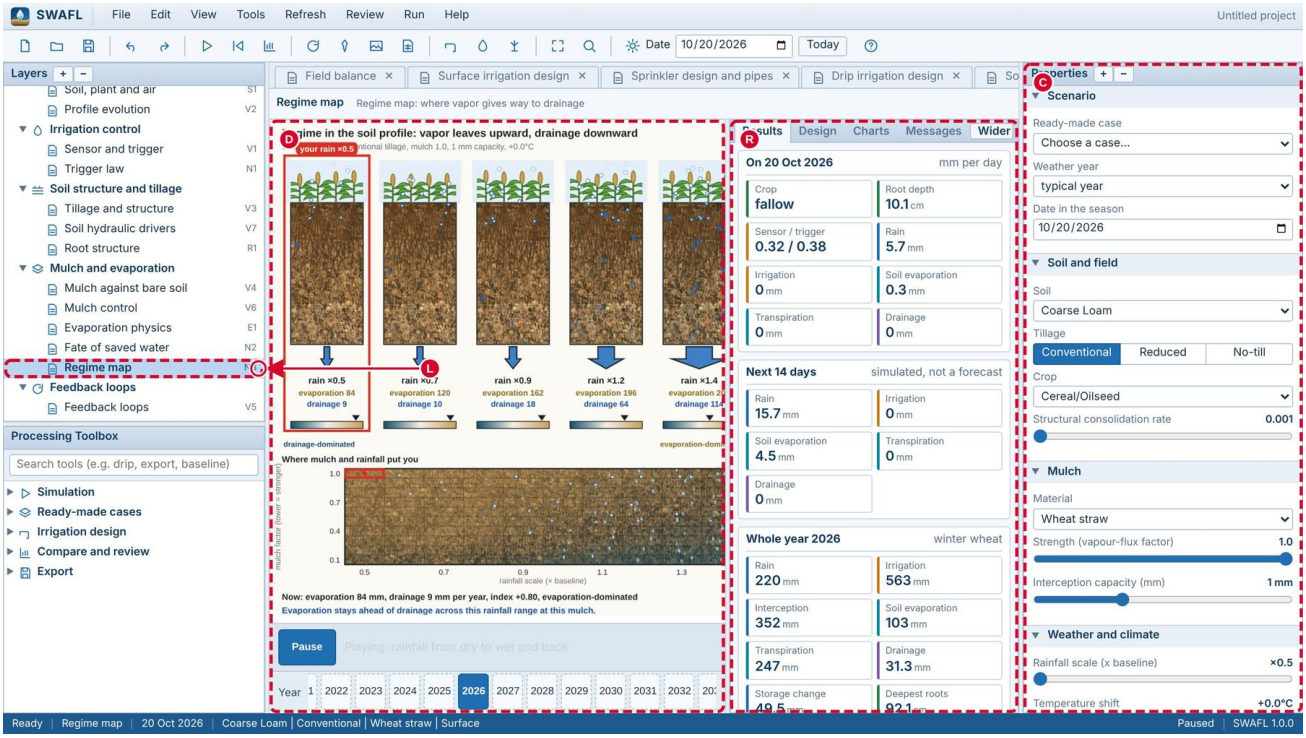


Figure 18.242: Regime map (N3): the whole workbench.

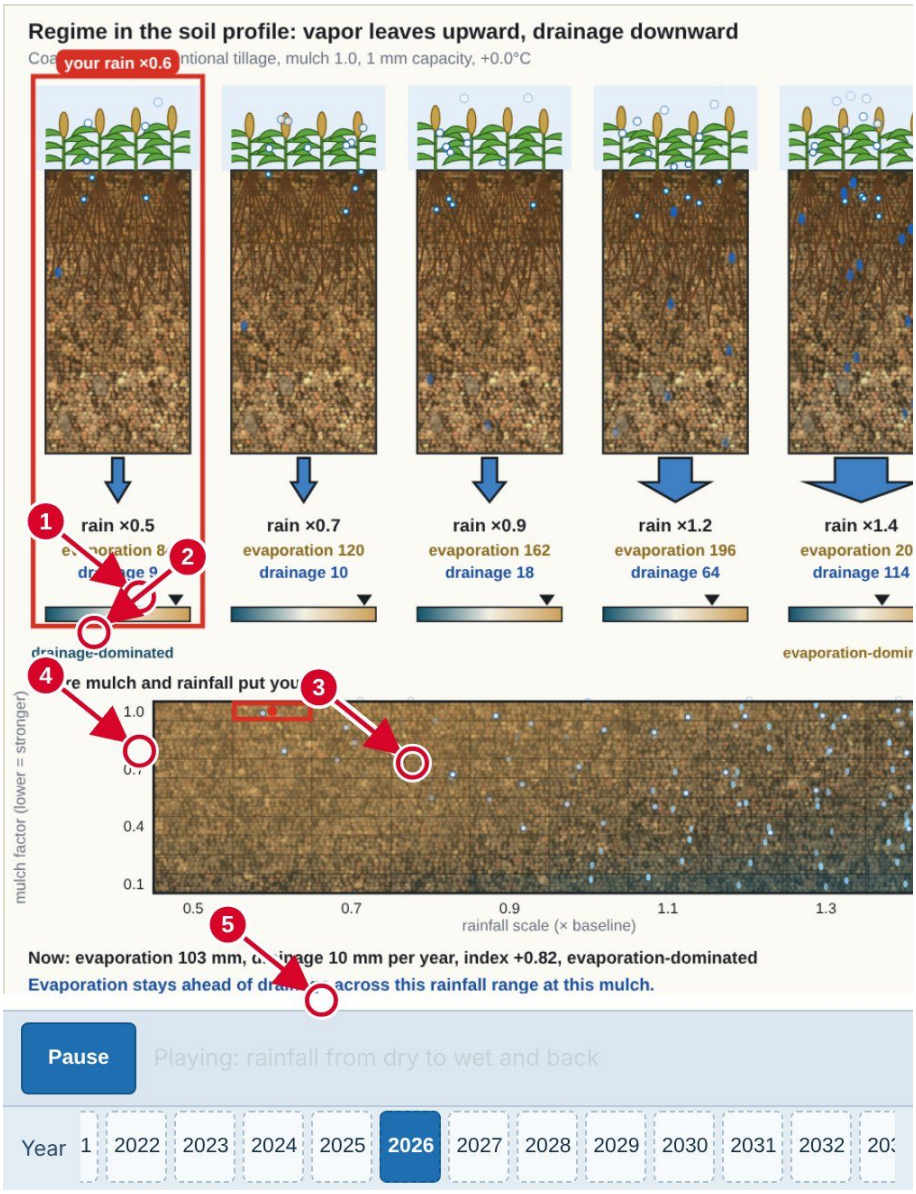


Figure 18.243: Regime map (N3): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Rainfall columns: evaporation and drainage of Engine 1 at each rainfall scale, Eqs. 5.4 and 5.9.
- **2** Colour bar under each column: regime index I = (E - D) / max(1, E + D): Eq. 9.8.
- **3** Regime map over rainfall scale and mulch factor: Eq. 9.8 in every cell.
- **4** Red box: your current setting on the map.
- **5** “Now: evaporation, drainage, index”: Eq. 9.8 at the current setting.

Equations behind this screen. Eq. 9.8.

Eq. 9.8. Where explained: chapter “Mulch and evaporation layers”, section “Regime map (N3)”.

$$\mathcal{J} = \frac{E - D}{\max\{1,\ E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{J} > +0.05 \\ \text{drainage-dominated} & \mathcal{J} < -0.05 \\ \text{on the boundary} & |\mathcal{J}| \le 0.05 \end{cases} \tag{9.8}$$

Provenance (Eq. 9.8): **[Engineered]**: index defined for the workbench.

18.17.1 The whole layer, part by part

The picture below is the complete drawing of the layer Regime map (N3), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 1 part, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

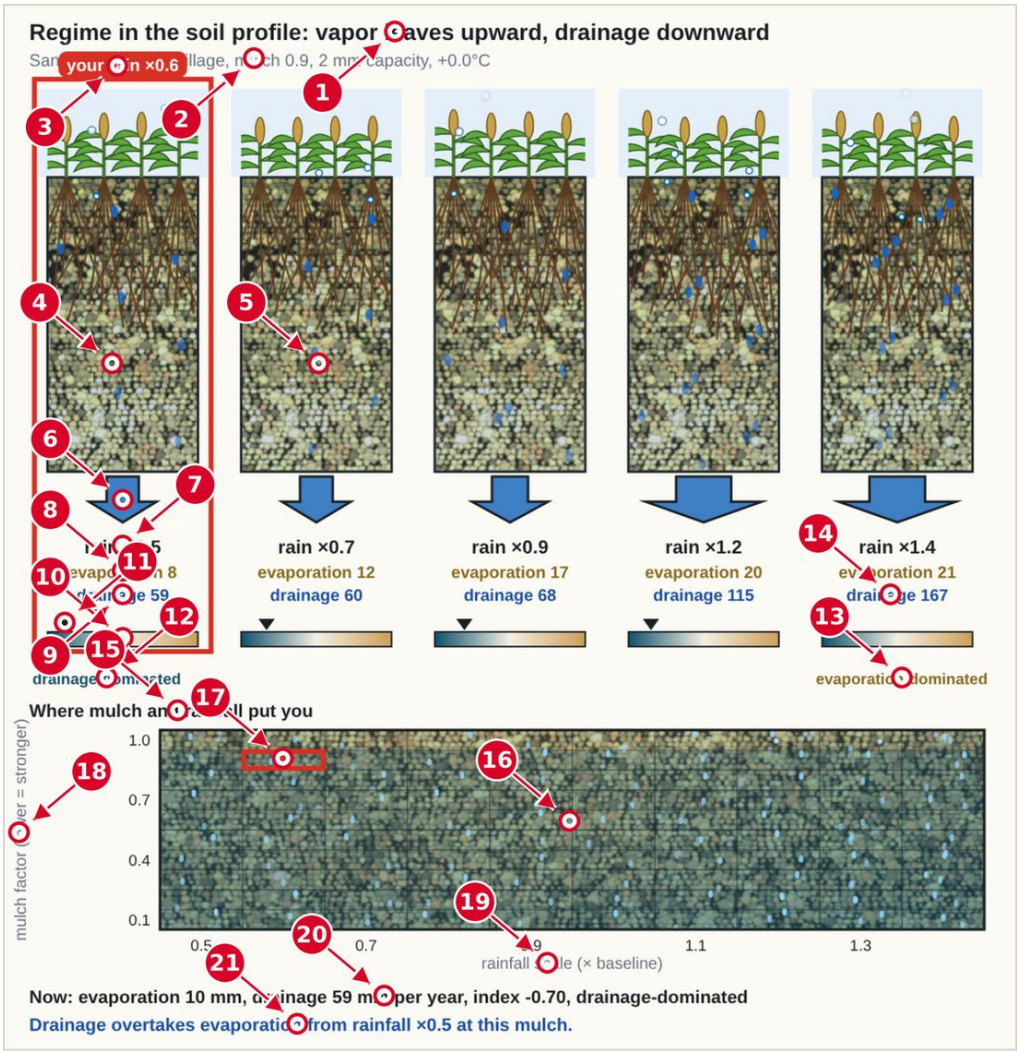


Figure 18.244: Regime map (N3), part 1 of 1.

What the arrows point to.

- **1** The title, the regime of the soil profile, vapour upward and drainage downward.
- **2** The settings of the five profiles, the soil, tillage, mulch factor, capacity and warming.
- **3** The tag of the profile that matches your own rain scaling.
- **4** The profile outlined in red, the rainfall scale in force.
- **5** The next profile, a wetter rainfall scale.

- **6** The rain arrow, thicker for more rain.
- **7** The rainfall scale of the profile.
- **8** The evaporation of the year at that rainfall scale, in mm.
- **9** The drainage of the year at that rainfall scale, in mm.
- **10** The regime scale from drainage-dominated to evaporation-dominated.
- **11** The marker of the regime index on that scale.
- **12** The label of the drainage-dominated end.
- **13** The label of the evaporation-dominated end.
- **14** The drainage of the wettest profile, far above its evaporation.
- **15** The map of where the mulch and the rainfall put you.
- **16** The regime map over the rainfall scale and the mulch factor.
- **17** The red outlined cell, your own setting on the map.
- **18** The mulch factor axis, lower means stronger.
- **19** The rainfall scale axis, as a multiple of the baseline.
- **20** The evaporation, the drainage and the regime index of your setting.
- **21** The sentence on the rainfall at which the drainage overtakes the evaporation.

18.18 Feedback loops (V5)

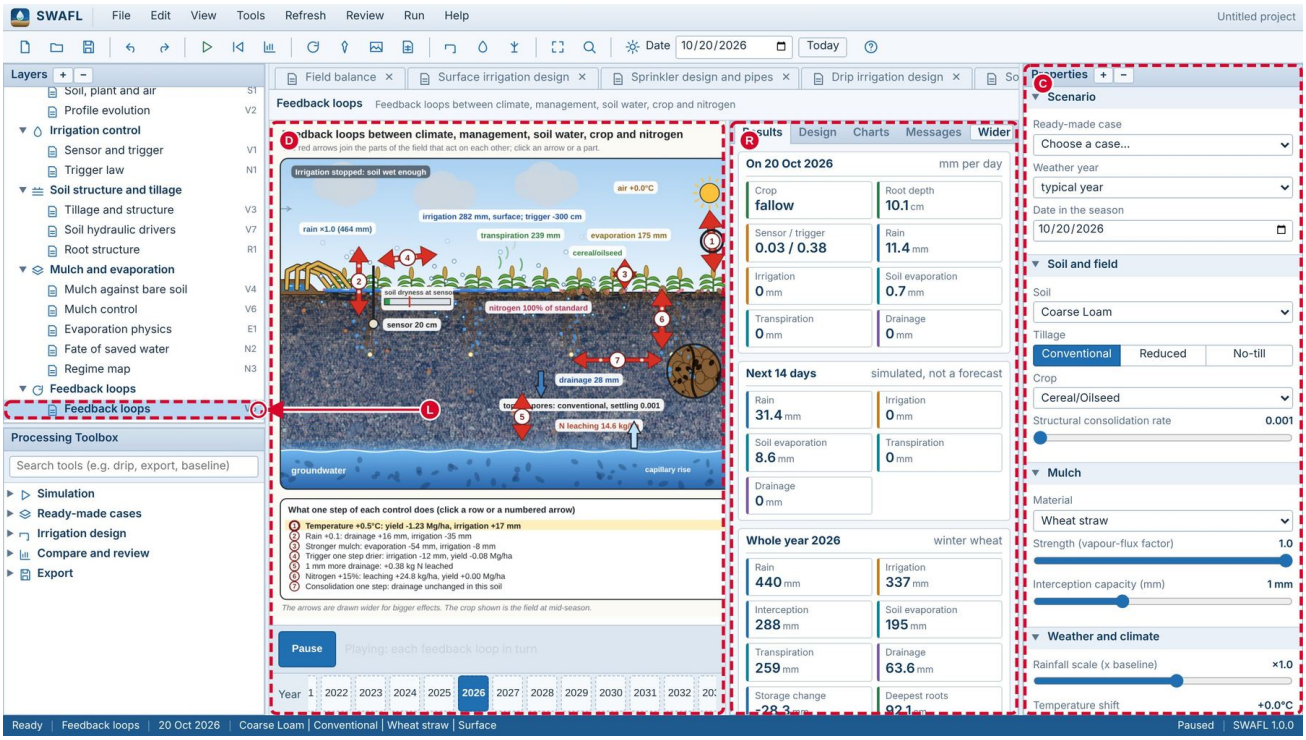


Figure 18.245: Feedback loops (V5): the whole workbench.

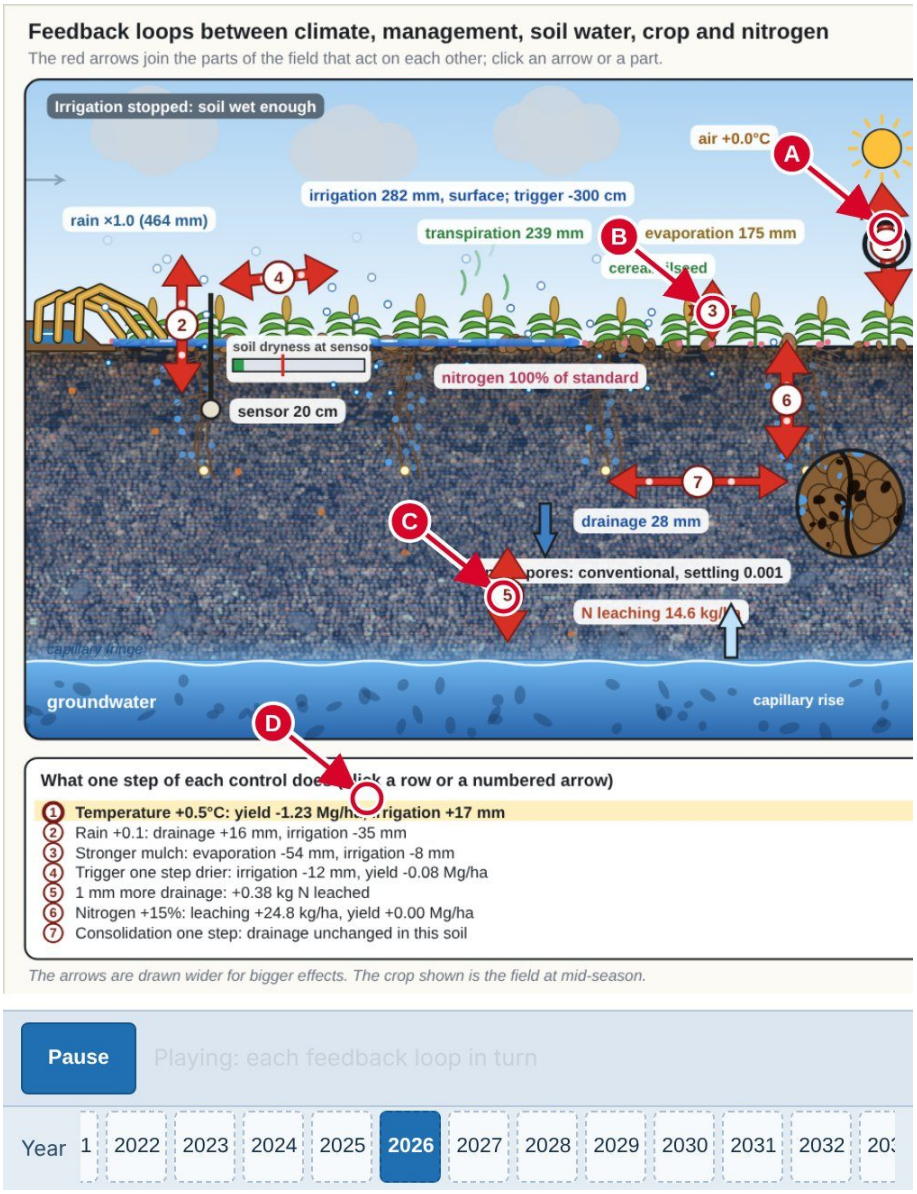


Figure 18.246: Feedback loops (V5): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **A** Arrow 1, temperature step: Engine 1 (Eqs. 5.4 to 5.12) re-evaluated; arrow width from Eq. 10.1.
- **B** Arrow 3, stronger mulch: change in evaporation and irrigation, Eqs. 5.4, 5.7 and 9.1.
- **C** Arrow 5, nitrate cost of drainage, kappa_N: Eq. 10.2.
- **D** List of the change produced by one step of each control; widths of the arrows: Eq. 10.1.

Equations behind this screen. Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 10.1, Eq. 10.2.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".

$$Q \leftarrow Q\,\text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 10.1. Where explained: chapter "Feedback loops", section "How the seven effects are computed".

$$w_k = 5 + \text{clip}(\mu_k;\ 0,\ 4). \tag{10.1}$$

Provenance (Eq. 10.1): **[Engineered]**: drawing rule.

Eq. 10.2. Where explained: chapter "Feedback loops", section "How the seven effects are computed".

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D}\ \ [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

Provenance (Eq. 10.2): **[Data-driven]**: one-step differences of Daisy results.

18.18.1 The whole layer, part by part

The picture below is the complete drawing of the layer Feedback loops (V5), which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in 1 part, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

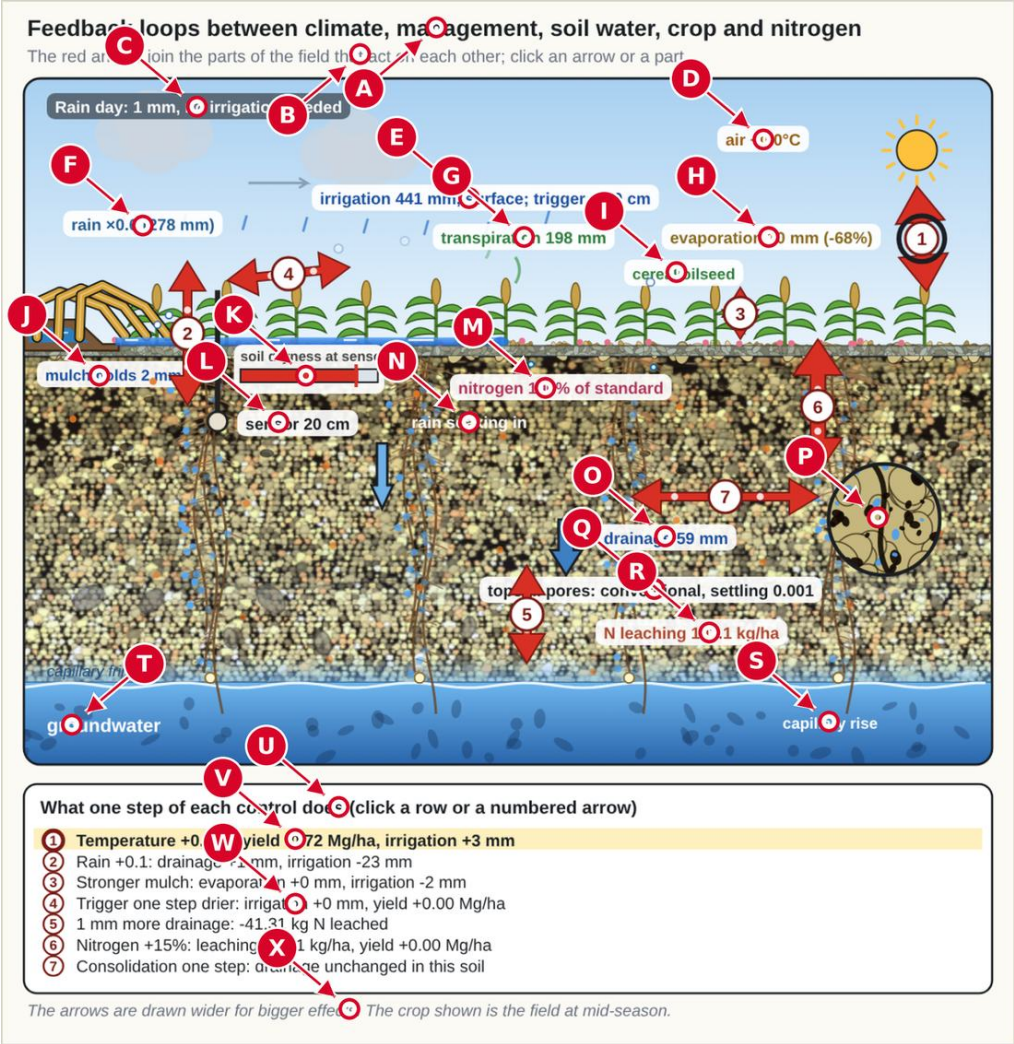


Figure 18.247: Feedback loops (V5), part 1 of 1.

What the arrows point to.

- **A** The title, the feedback loops between climate, management, soil water, crop and nitrogen.
- **B** The instruction, the red arrows join the parts that act on each other, and an arrow or a part is clicked.
- **C** The banner of the rain on the day shown and whether irrigation is needed.
- **D** The air temperature change in force.
- **E** The irrigation of the year, its method and the trigger.
- **F** The rain scaling and the rain of the year.
- **G** The transpiration of the year.
- **H** The evaporation of the year and its change against bare soil.
- **I** The crop shown, the field at mid-season.
- **J** The water held by the mulch.
- **K** The gauge of the soil dryness at the sensor.
- **L** The sensor at 20 cm.
- **M** The nitrogen supply as a share of the standard.
- **N** The rain soaking into the soil.
- **O** The drainage of the year.
- **P** The aggregate that stands for the structure of the topsoil.
- **Q** The topsoil pores, the tillage and the settling rate.

- **R** The nitrogen leaching of the year.
- **S** The capillary rise from the groundwater.
- **T** The groundwater.
- **U** The table of what one step of each control does.
- **V** The row of the loop selected, highlighted and clickable.
- **W** The other rows, each loop with its effect on the yield, irrigation and drainage.
- **X** The note that the arrows are wider for bigger effects.

18.18.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

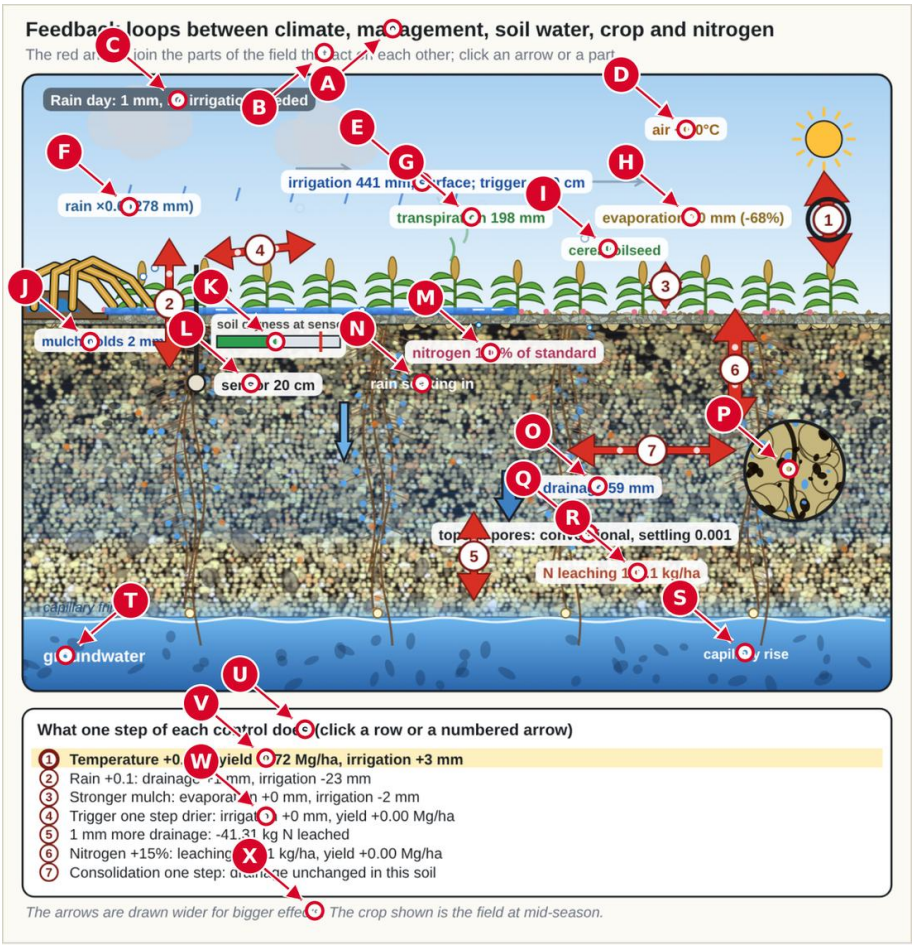


Figure 18.248: Feedback loops (V5), part 1, playback a third of the way (Paused: each feedback loop in turn).

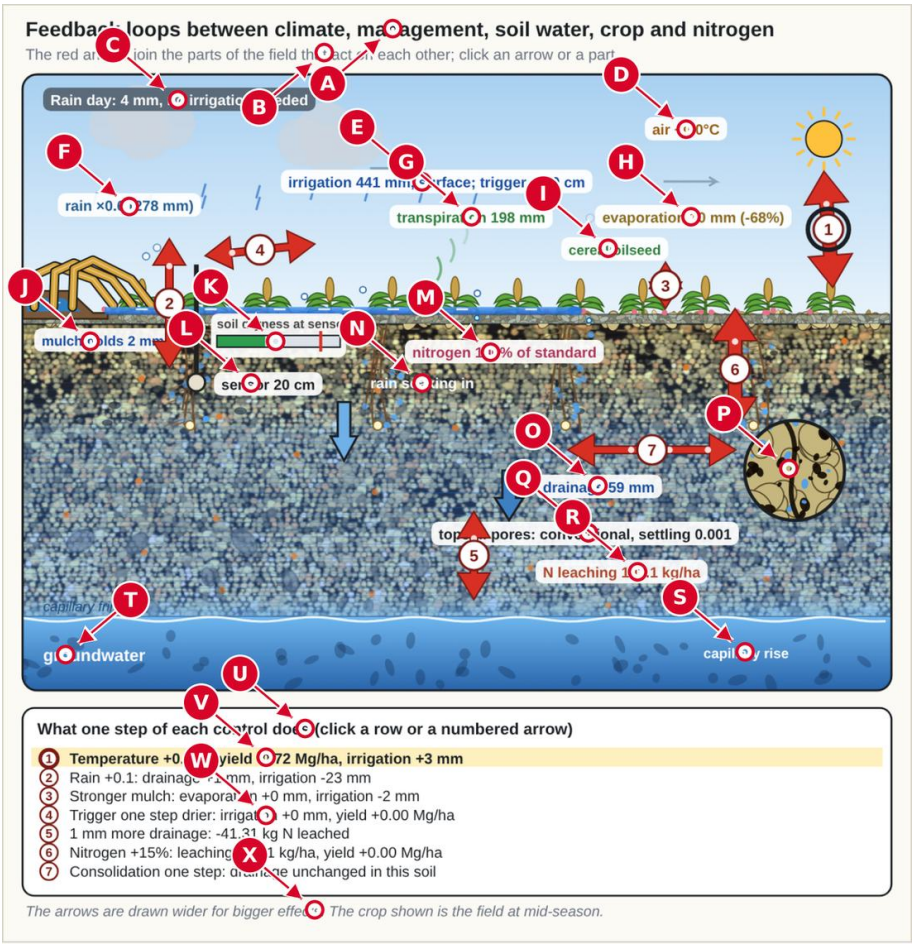


Figure 18.249: Feedback loops (V5), part 1, playback two thirds of the way (Paused: each feedback loop in turn).

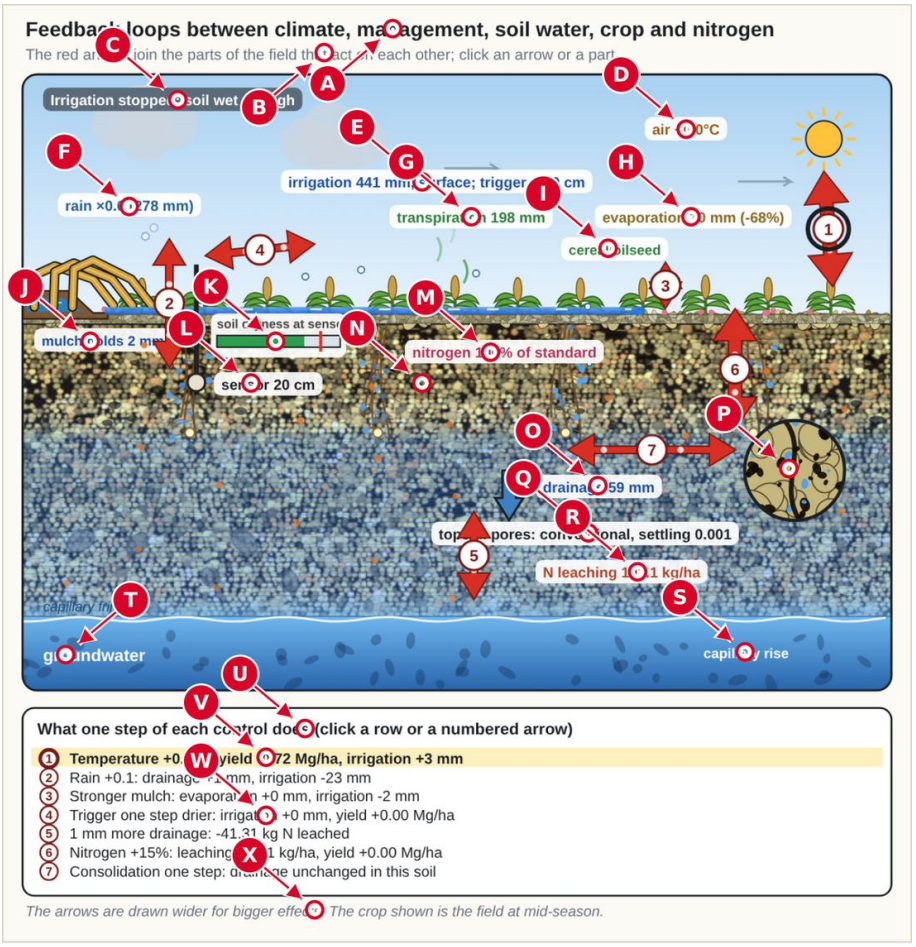


Figure 18.250: Feedback loops (V5), part 1, playback the end (Paused: each feedback loop in turn).

18.19 Reverse index: where each equation appears on screen

Equation	Explained in chapter	Screens that use it

4.1 | Dates | The toolbar; The workbench |

4.2 | Dates | The toolbar; The workbench |

5.1 | The two computing engines | Soil, plant and air (S1); Tillage and structure (V3) |

5.2 | The two computing engines | Soil, plant and air (S1) |

5.3 | The two computing engines | Soil, plant and air (S1); Trigger law (N1) |

5.4 | The two computing engines | Evaporation physics (E1); Feedback loops (V5); Field balance; Mulch against bare soil (V4); Mulch control (V6); Sensor and trigger (V1); Soil hydraulic drivers (V7) |

5.5 | The two computing engines | Feedback loops (V5); Field balance; Sensor and trigger (V1); Soil hydraulic drivers (V7) |

5.6 | The two computing engines | Feedback loops (V5); Field balance; Sensor and trigger (V1); Soil hydraulic drivers (V7) |

5.7 | The two computing engines | Feedback loops (V5); Field balance; Mulch against bare soil (V4); Sensor and trigger (V1); Soil hydraulic drivers (V7) |

5.8 | The two computing engines | Feedback loops (V5); Field balance; Sensor and trigger (V1); Soil hydraulic drivers (V7) |

5.9 | The two computing engines | Feedback loops (V5); Field balance; Mulch against bare soil (V4); Mulch control (V6); Sensor and trigger (V1); Soil hydraulic drivers (V7) |

5.10 | The two computing engines | Feedback loops (V5); Field balance; Sensor and trigger (V1); Soil hydraulic drivers (V7) |

5.11 | The two computing engines | Feedback loops (V5); Field balance; Sensor and trigger (V1); Soil hydraulic drivers (V7) |

5.12 | The two computing engines | Feedback loops (V5); Field balance; Sensor and trigger (V1); Soil hydraulic drivers (V7) |

5.13 | The two computing engines | Field balance |

5.14 | The two computing engines | Field balance |

5.15 | The two computing engines | Field balance; Profile evolution (V2) |

5.16 | The two computing engines | Field balance |

5.17 | The two computing engines | Field balance; Profile evolution (V2) |

5.18 | The two computing engines | Field balance; Root structure (R1); The Properties panel |

5.19 | The two computing engines | Field balance; Root structure (R1) |

5.20 | The two computing engines | Field balance; Root structure (R1) |

5.21 | The two computing engines | Field balance; Root structure (R1) |

5.22 | The two computing engines | Field balance; Root structure (R1) |

5.23 | The two computing engines | Field balance; Root structure (R1) |

5.24 | The two computing engines | Field balance |

5.25 | The two computing engines | Field balance; Soil, plant and air (S1) |

5.26 | The two computing engines | Evaporation physics (E1); The Properties panel |

5.27 | The two computing engines | S2 Surface irrigation design; S2 Surface irrigation design / Cross-section at the inlet; S2 Surface irrigation design / Layout in plan; S3 Drip irrigation design; S4 Sprinkler design and pipes; Trigger law (N1) |

6.1 | Irrigation control layers | Sensor and trigger (V1); The Dashboard, Results tab |

6.2 | Irrigation control layers | Sensor and trigger (V1) |

6.3 | Irrigation control layers | Trigger law (N1) |

6.4 | Irrigation control layers | Trigger law (N1) |

6.5 | Irrigation control layers | Trigger law (N1) |

7.1 | Field water balance layers | Soil, plant and air (S1) |

7.2 | Field water balance layers | Soil, plant and air (S1) |

7.3 | Field water balance layers | Soil, plant and air (S1) |

7.4 | Field water balance layers | Soil, plant and air (S1) |

7.5 | Field water balance layers | Soil, plant and air (S1) |

7.6 | Field water balance layers | Profile evolution (V2) |

8.1 | Soil structure and tillage layers | Field balance; Soil hydraulic drivers (V7); The Properties panel; Tillage and structure (V3) |

8.2 | Soil structure and tillage layers | Tillage and structure (V3) |

8.3 | Soil structure and tillage layers | Evaporation physics (E1); Tillage and structure (V3) |

8.4 | Soil structure and tillage layers | Evaporation physics (E1); Tillage and structure (V3) |

8.5 | Soil structure and tillage layers | Soil hydraulic drivers (V7) |

8.6 | Soil structure and tillage layers | Soil hydraulic drivers (V7) |

8.7 | Soil structure and tillage layers | Root structure (R1) |

9.1 | Mulch and evaporation layers | Mulch against bare soil (V4) |

9.2 | Mulch and evaporation layers | Evaporation physics (E1) |

9.3 | Mulch and evaporation layers | Evaporation physics (E1) |

9.4 | Mulch and evaporation layers | Evaporation physics (E1) |

9.5 | Mulch and evaporation layers | Evaporation physics (E1) |

9.6 | Mulch and evaporation layers | Fate of saved water (N2) |

9.7 | Mulch and evaporation layers | Fate of saved water (N2) |

9.8 | Mulch and evaporation layers | Regime map (N3) |

10.1 | Feedback loops | Feedback loops (V5) |

10.2 | Feedback loops | Feedback loops (V5) |

11.1 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Cross-section at the inlet |

11.2 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Cross-section at the inlet |

11.3 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Cross-section at the inlet; S3 Drip irrigation design; S3 Drip irrigation design / How the wetted bulb grows with the run time; S3 Drip irrigation design / Soil moisture around the emitter |

11.4 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Cross-section at the inlet |

11.5 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Cross-section at the inlet |

11.6 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Advance and opportunity time; S2 Surface irrigation design / Cross-section at the inlet; S2 Surface irrigation design / Infiltrated depth along the field against the required depth Zn; S2 Surface irrigation design / Longitudinal section; S4 Sprinkler design and pipes; S4 Sprinkler design and pipes / Along the lateral; S4 Sprinkler design and pipes / Soil intake against the application rate |

11.7 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Cross-section at the inlet |

11.8 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Layout in plan |

11.9 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Advance and opportunity time; S2 Surface irrigation design / Layout in plan; S2 Surface irrigation design / Longitudinal section |

11.10 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Advance and opportunity time; S2 Surface irrigation design / Layout in plan; S2 Surface irrigation design / Longitudinal section |

11.11 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Advance and opportunity time; S2 Surface irrigation design / Layout in plan; S2 Surface irrigation design / Longitudinal section |

11.12 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Advance and opportunity time; S2 Surface irrigation design / Layout in plan; S2 Surface irrigation design / Longitudinal section |

11.13 | Surface irrigation design | S2 Surface irrigation design; S2 Surface irrigation design / Advance and opportunity time; S2 Surface irrigation design / Layout in plan; S2 Surface irrigation design / Longitudinal section |

11.15 | Surface irrigation design | S2 Surface irrigation design / Advance and opportunity time; S2 Surface irrigation design / Infiltrated depth along the field against the required depth Zn; S2 Surface irrigation design / Longitudinal section |

11.16 | Surface irrigation design | S2 Surface irrigation design / Advance and opportunity time; S2 Surface irrigation design / Infiltrated depth along the field against the required depth Zn; S2 Surface irrigation design / Longitudinal section |

11.17 | Surface irrigation design | S2 Surface irrigation design / Advance and opportunity time; S2 Surface irrigation design / Infiltrated depth along the field against the required depth Zn; S2 Surface irrigation design / Longitudinal section |

11.18 | Surface irrigation design | S2 Surface irrigation design / Hydrograph; S2 Surface irrigation design / Performance indices and the water balance of one irrigation; The Dashboard, Design tab |

11.19 | Surface irrigation design | S2 Surface irrigation design / Performance indices and the water balance of one irrigation; The Dashboard, Design tab |

11.20 | Surface irrigation design | S2 Surface irrigation design / Hydrograph |

11.21 | Surface irrigation design | S2 Surface irrigation design / Hydrograph |

11.22 | Surface irrigation design | S2 Surface irrigation design / Design table |

11.23 | Surface irrigation design | S2 Surface irrigation design / Design table |

11.24 | Surface irrigation design | S2 Surface irrigation design / Intervals, efficiency and the advice in short; S2 Surface irrigation design / What this means for the season and the farm |

12.1 | Drip irrigation design | S3 Drip irrigation design |

12.2 | Drip irrigation design | S3 Drip irrigation design |

12.3 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / Layout and infrastructure; S3 Drip irrigation design / Uniformity along the lateral; S4 Sprinkler design and pipes / Layout and infrastructure; S4 Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.4 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / Uniformity along the lateral |

12.5 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / Uniformity along the lateral |

12.6 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / Uniformity along the lateral |

12.7 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / Uniformity

along the lateral |

12.8 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / Uniformity along the lateral |

12.9 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / How the wetted bulb grows with the run time; S3 Drip irrigation design / Soil moisture around the emitter |

12.10 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / How the wetted bulb grows with the run time; S3 Drip irrigation design / Result for this design; S3 Drip irrigation design / Soil moisture around the emitter; S3 Drip irrigation design / Uniformity along the lateral |

12.11 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / How the wetted bulb grows with the run time; S3 Drip irrigation design / Soil moisture around the emitter |

12.12 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / How the wetted bulb grows with the run time; S3 Drip irrigation design / Soil moisture around the emitter |

12.13 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / How the wetted bulb grows with the run time; S3 Drip irrigation design / Soil moisture around the emitter |

12.14 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / How the wetted bulb grows with the run time; S3 Drip irrigation design / Soil moisture around the emitter |

12.15 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / How the wetted bulb grows with the run time; S3 Drip irrigation design / Result for this design; S3 Drip irrigation design / Soil moisture around the emitter; S3 Drip irrigation design / Uniformity along the lateral |

12.16 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / Result for this design; S3 Drip irrigation design / Uniformity along the lateral |

12.17 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / Result for this design; S3 Drip irrigation design / Uniformity along the lateral |

12.18 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / Result for this design; S3 Drip irrigation design / Uniformity along the lateral |

12.19 | Drip irrigation design | S3 Drip irrigation design; S3 Drip irrigation design / Result for this design; S3 Drip irrigation design / Uniformity along the lateral; The Dashboard, Design tab |

12.20 | Drip irrigation design | S3 Drip irrigation design / Layout and infrastructure |

12.21 | Drip irrigation design | S3 Drip irrigation design / Layout and infrastructure; S4 Sprinkler design and pipes / Layout and infrastructure; S4 Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.22 | Drip irrigation design | S3 Drip irrigation design / Layout and infrastructure; S4 Sprinkler design and pipes / Layout and infrastructure; S4 Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.23 | Drip irrigation design | S3 Drip irrigation design / Layout and infrastructure; S4 Sprinkler design and pipes / Layout and infrastructure; S4 Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.24 | Drip irrigation design | S3 Drip irrigation design / Intervals, efficiency and the advice in short; S3 Drip irrigation design / Irrigation schedule; S3 Drip irrigation design / What this means for the season and the farm |

12.25 | Drip irrigation design | S3 Drip irrigation design / Intervals, efficiency and the advice in short; S3 Drip irrigation design / Irrigation schedule; S3 Drip irrigation design / What this means for the season and the farm |

12.26 | Drip irrigation design | S3 Drip irrigation design / Design table; S4 Sprinkler design and pipes / Design table; S4 Sprinkler design and pipes / What the efficiency is most sensitive to |

13.1 | Sprinkler design | S4 Sprinkler design and pipes |

13.2 | Sprinkler design | S4 Sprinkler design and pipes; S4 Sprinkler design and pipes / The jet; S4 Sprinkler design and pipes / The sprinkler |

13.3 | Sprinkler design | S4 Sprinkler design and pipes; S4 Sprinkler design and pipes / The jet; S4 Sprinkler design and pipes / The sprinkler |

13.4 | Sprinkler design | S4 Sprinkler design and pipes; S4 Sprinkler design and pipes / Catch-can test; S4 Sprinkler design and pipes / Layout in plan; S4 Sprinkler design and pipes / Layouts; S4 Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.5 | Sprinkler design | S4 Sprinkler design and pipes; S4 Sprinkler design and pipes / Catch-can test; S4 Sprinkler design and pipes / Layout in plan; S4 Sprinkler design and pipes / Layouts; S4 Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.6 | Sprinkler design | S4 Sprinkler design and pipes; S4 Sprinkler design and pipes / Catch-can test; S4 Sprinkler design and pipes / Layout in plan; S4 Sprinkler design and pipes / Layouts; S4 Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.7 | Sprinkler design | S4 Sprinkler design and pipes; S4 Sprinkler design and pipes / Catch-can test; S4 Sprinkler design and pipes / Layout in plan; S4 Sprinkler design and pipes / Layouts; S4 Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.8 | Sprinkler design | S4 Sprinkler design and pipes / The jet; S4 Sprinkler design and pipes / What wind and soil allow |

13.9 | Sprinkler design | S4 Sprinkler design and pipes; S4 Sprinkler design and pipes / Along the lateral; S4 Sprinkler design and pipes / Soil intake against the application rate; The Properties panel |

13.10 | Sprinkler design | S4 Sprinkler design and pipes; S4 Sprinkler design and pipes / Along the lateral; S4 Sprinkler design and pipes / Soil intake against the application rate |

13.11 | Sprinkler design | S4 Sprinkler design and pipes / Along the lateral; S4 Sprinkler design and pipes / Result for this design; S4 Sprinkler design and pipes / Soil water as a reservoir; The Dashboard, Design tab |

13.12 | Sprinkler design | S4 Sprinkler design and pipes / Along the lateral; S4 Sprinkler design and pipes / Result for this design; S4 Sprinkler design and pipes / Soil water as a reservoir |

13.13 | Sprinkler design | S4 Sprinkler design and pipes / Does your design pass? The course checks |

13.14 | Sprinkler design | S4 Sprinkler design and pipes / Does your design pass? The course checks |

13.15 | Sprinkler design | S4 Sprinkler design and pipes / Does your design pass? The course checks |

13.16 | Sprinkler design | S4 Sprinkler design and pipes / Does your design pass? The course checks |

13.17 | Sprinkler design | S4 Sprinkler design and pipes / Does your design pass? The course checks |

13.18 | Sprinkler design | S4 Sprinkler design and pipes / Does your design pass? The course checks |

13.19 | Sprinkler design | S4 Sprinkler design and pipes / Does your design pass? The course checks |

13.20 | Sprinkler design | S4 Sprinkler design and pipes / Does your design pass? The course checks |

13.21 | Sprinkler design | S4 Sprinkler design and pipes / Layout and infrastructure; S4 Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

13.22 | Sprinkler design | S4 Sprinkler design and pipes / Layout and infrastructure; S4 Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

13.23 | Sprinkler design | S4 Sprinkler design and pipes / Intervals, efficiency and the advice in

short; S4 Sprinkler design and pipes / Irrigation schedule; S4 Sprinkler design and pipes / Soil water as a reservoir; S4 Sprinkler design and pipes / What this means for the season and the farm |

14.1 | The Dashboard | The Dashboard, Results tab |

14.2 | The Dashboard | The Dashboard, Results tab |

14.3 | The Dashboard | The Dashboard, Results tab |

14.4 | The Dashboard | The Dashboard, Results tab |

14.5 | The Dashboard | The Dashboard, Charts tab |

Chapter 19

Release 1.1: design runs, weather, virtual farm, crops and seasonal records

Release 1.1 adds four groups of layers and three workbench functions. Every new layer follows the controls of the Properties panel (soil, tillage, crop, mulch, rainfall, warming, irrigation method, trigger, sensor depth and nitrogen) and is redrawn when any of them changes.

19.1 The layers

The Layers panel lists, in order: **Irrigation design** (surface, drip, sprinklers, each with its *Virtual farm simulation*), **Irrigation control**, **Mulch and evaporation**, **Soil structure and tillage**, **Field water balance**, **Seasonal records: Daisy data** (Y1, W1, N4), **Crops** (C1) and **Feedback loops**. The working date and the year are set only in Properties, Scenario (every year from 1980 to 2100); no year strip or date slider sits under the drawings.

19.2 Running an irrigation design

Opening a design layer does not start it. Set the inputs at the top of the layer, then press **Run simulation** (in the drawing, on the bar under it, or on the dashboard Design tab). Until the run, and after any change of an input or of a Properties control, the results are hidden behind a panel that says why, so nothing moves while the inputs are being set.

Inputs. Every number is a value box. A click opens a picker with the exact value, its unit (for example m, km or ft for lengths; L/s, m^3/h or L/min for flows; ha, m^2 or acre for areas; kPa, bar, m of water or psi for pressure) and a drop-down of values across the allowed range. The allowed range is shown; a value outside it is set to the nearest limit. Choices such as the system, intake, cutoff, emitter or sprinkler type are buttons; the crop is a drop-down of all crops.

Most efficient inputs. *Find the most efficient inputs* searches the inputs that matter most (surface: inflow, cutoff rule and cutback; drip: emitter flow, spacing and interval; sprinklers: nozzle, pressure and spacing) for the highest application efficiency with the root zone at least 95 % refilled, shows the combination on the Design tab and applies it with *Apply these inputs and run*.

Dashboard. The Design tab gives the efficiencies, uniformity, times and volumes of the last run, the best combination and a warning when a surface stream cannot reach the end of the field in a practical time. The Charts tab of a design layer draws the hydrograph (inflow, infiltration and runoff), the advance, cut-off and recession times along the field, the infiltrated depth against the

required depth, the times of the irrigation (tL, ts, tco, tr) and where the water goes; for drip the head and emitter flow along the lateral; for sprinklers the soil intake rate against the application rate and the sorted depths caught. Every dashboard block has a header: click it to hide or show the block.

Field sketch. View or Tools, *Field sketch to design*: click the corners of the field on the grid (choose the size of a grid square), mark the water source, give the slope and the system. The area, run length, width and mainline length become the inputs of the design layer, which opens ready to run. *Compare the irrigation systems on this field* lists the three systems side by side.

19.3 Weather data

File (or Tools, or Properties, Scenario) → **Import weather data** reads a daily weather file: CSV or TXT (comma, semicolon, tab or space separated), Excel (XLSX, XLS, ODS), a Daisy weather file (DWF) or a Daisy daily log (DLF). A Daisy setup file (DAI) names its weather file: import that file. The columns are found by their names: a date (or year, month and day, or year and day of year), daily rainfall (mm, or inches when the header says so) and, if present, daily reference evapotranspiration. Without it, reference ET is computed from the daily maximum and minimum temperature (Hargreaves, at the latitude given in the dialog) or from mean temperature and global radiation (Makkink). Hourly files are summed to days.

Online weather for a location finds a place by name (or takes its latitude and longitude) and gets its measured daily weather for the last twenty years and the forecast for the next sixteen days. If the service cannot be reached, the current weather stays in use and the Messages tab says so.

The weather is put on the twenty-year grid of the program (the last twenty complete years; fewer years are repeated in turn, and missing days of rain are filled from the same day of another year). Then the daily model takes the rain of each day and the evaporative demand of each day from its reference ET; the crop water balance takes the same daily rain and reference ET; and the annual model takes the rainfall and the demand of the new climate through its rainfall and warming controls, at the nearest simulated step (the report of the import says when the climate lies outside the simulated range). Properties, Scenario, **Weather source** switches between the recorded weather of the simulations, the imported file and the online weather; the choices of a file or online weather become active only after the data are loaded. The first time the File menu opens, the program offers to bring in weather data.

19.4 Virtual farm simulation

The virtual farm (under each design layer, *Virtual farm simulation*, or the *Virtual farm simulation* button on the view bar of every layer) draws a field in perspective with the water moving from the source to the plants. Six layouts are available:

Layout	Head works drawn	Water on the field
Furrow	canal from the reservoir, lifting gate, siphons	advance front along each furrow, tail-water ditch with the runoff share
Basin	canal, gate	sheet of water filling each basin
Border strip	canal, gate, siphons	sheet advancing down each strip, tail water
Sprinkler	well, pump, filters, fertiliser tank, mainline, control valve	rotating sprays with wind drift
Surface drip	well, pump, filters, fertiliser tank, mainline, control valve	emitters and wet spots along the laterals
Subsurface drip	as surface drip	buried laterals and wetted bands below the surface

The gate or valve follows the predicted schedule. The daily model (Engine 2, chapter 5)

predicts the dates and depths of irrigation for the present settings. The board at the gate lists the next four irrigations with the hours the gate or valve stays open; tap a line to jump to that date. During an irrigation the gate is lifted (or the valve opened) and water flows; between irrigations it is closed. *Farmer opens the valve* shows a farmer walking to the gate; *Automatic controller* shows a controller box wired to the soil-water sensor, whose lamp lights when the trigger is reached.

Below the farm are three panels: (i) a cross-section through the field showing the water in each soil layer of the daily model, the wetting front, the roots of the chosen crop and the sensor; (ii) the predicted schedule for the next 30 days (rain, irrigation, valve open, sensor depletion and trigger); (iii) the fate of the next irrigation (stored in the root zone, deep percolation, surface runoff, evaporation and drift), taken from the surface (S2), drip (S3) or sprinkler (S4) design of the same layout.

Record video (20 s) records the farm for 20 seconds and saves it as a WebM film.

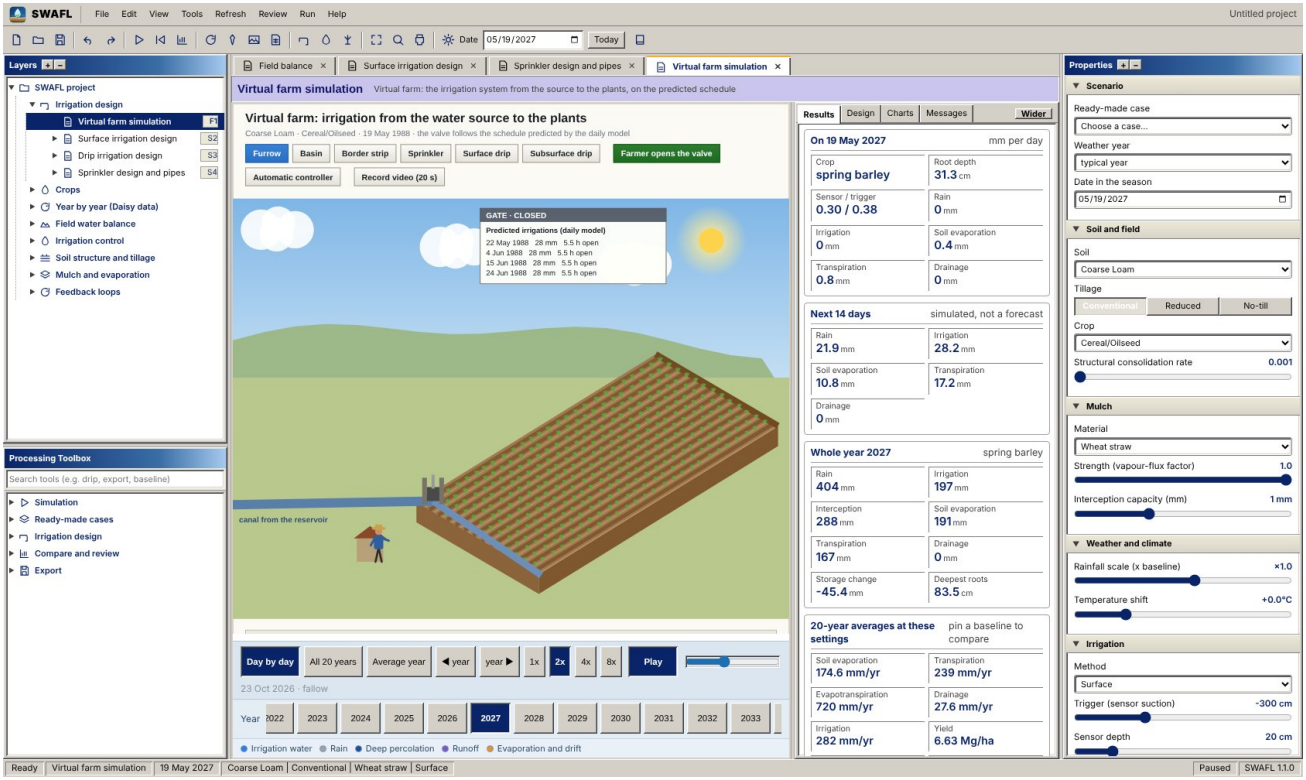


Figure 19.1: Virtual farm simulation: furrow layout, gate on the predicted schedule, the farmer at the gate.

19.5 Crops and C1 Crop water requirement

Properties → Crop lists the Daisy cereal rotation (simulated) and 48 library crops in five groups:

Group	Crops
Staple food crops	cassava, yam, rice (paddy), maize, sorghum, millet, cowpea, wheat, potato
Cash and oil crops	groundnut, soybean, cotton, mustard, green gram, chickpea, oil palm, cocoa
Vegetables	tomato, onion, cabbage, sweet pepper

Group	Crops
Trees, vines and palms	table and wine grapes, kiwi, blueberry, hop, citrus, olive, apple, pear, cherry, peach, apricot, almond, walnut, pistachio, pecan, hazelnut, pomegranate, fig, persimmon, avocado, mango, guava, date palm, coconut, papaya, cherimoya

The crop coefficients of the staple and cash crops are: cassava 0.30 / 0.80 / 0.30, yam 0.50 / 1.10 / 0.85, rice 1.05 / 1.20 / 0.90, maize 0.30 / 1.20 / 0.60, sorghum 0.30 / 1.00 / 0.55, millet 0.30 / 1.00 / 0.30, cowpea 0.40 / 1.05 / 0.50, groundnut 0.40 / 1.15 / 0.60, soybean 0.40 / 1.15 / 0.50, oil palm 0.90 / 0.95 / 0.95, cocoa 1.00 / 1.05 / 1.05 (initial / mid-season / end). The former legume and vegetable categories, which only multiplied the cereal yield, are removed; projects saved with them open with soybean and tomato. Each library crop carries crop coefficients for the initial, mid-season and late stages, stage lengths, height, maximum rooting depth, the depletion fraction p and the yield-response factor K_y; tree and vine crops have several planting densities (ground cover and height).

For a library crop the program runs a daily root-zone water balance on the same 20 weather years as the Daisy simulations: the crop water need is $ET_c = K_c\,ET_0$, the soil is refilled to field capacity when the depletion reaches the trigger, and the relative yield follows $1 - Y_a/Y_m = K_y\,(1 - ET_a/ET_c)$. The Daisy results of the cereal are then transferred to the crop by the ratio of the two balances (irrigation, transpiration, evaporation and drainage), so that every other layer responds to the crop. The bar under the drawing states which model is in use. Yield of a library crop is shown as a percentage of the potential.

The layer **C1 Crop water requirement** shows the crop's tiles (water need, water used, net and gross irrigation, number of irrigations, relative yield, water below the root zone, stress days), the K_c curve with the stage bands and the daily water need, the root-zone depletion with the irrigations and rain of the selected year, and a table of every crop on the present soil ranked by net irrigation. Tap a row to select that crop.

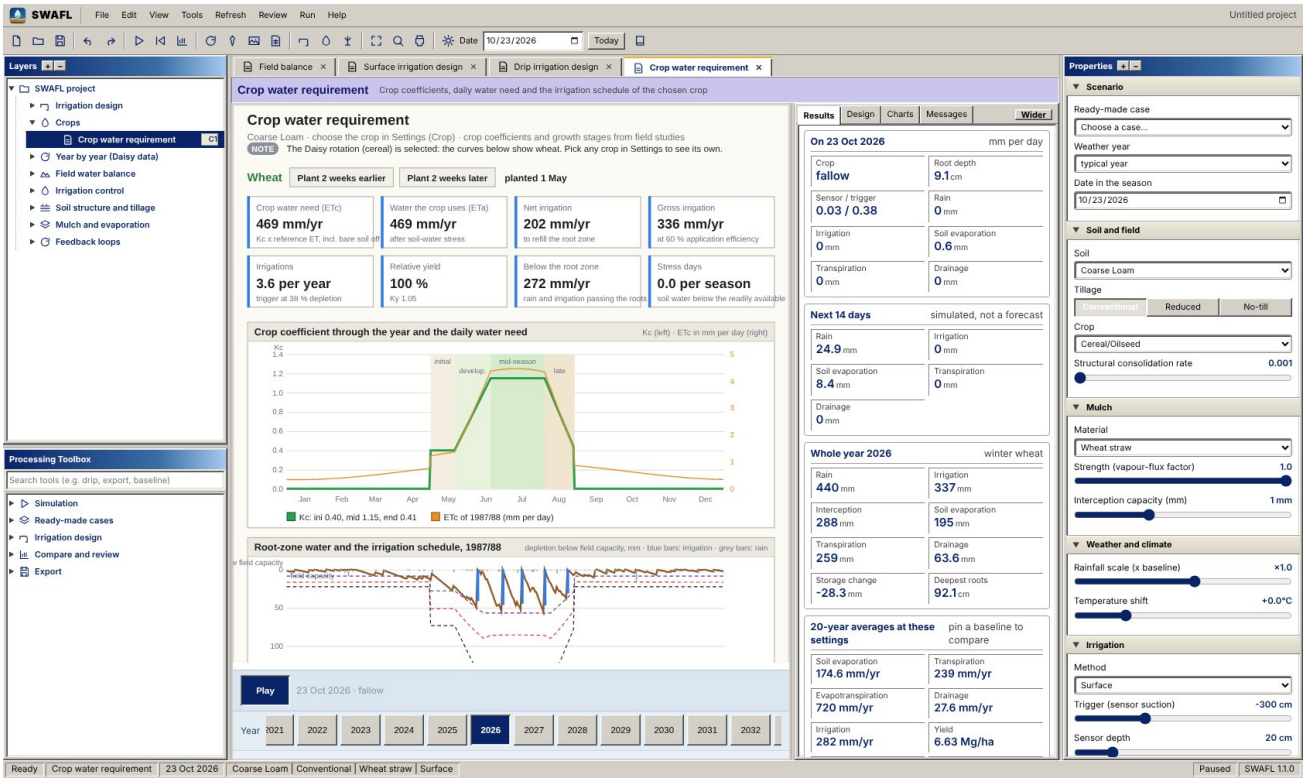


Figure 19.2: C1 Crop water requirement.

19.6 Daily records that follow the settings (V1, V2)

The 20-year sensor record (V1) and the season profiles (V2) start from your Daisy daily output, which exists for the coarse loam and the coarse sand at the standard settings (conventional tillage, bare soil, surface irrigation, trigger −300 cm at 20 cm, cereal, unscaled rain, no warming, full nitrogen). For any other soil or setting the daily model is run twice, at your settings and at the standard settings, and the difference in depletion of available water at the sensor depth is added day by day to the Daisy record, which is then converted to suction with the van Genuchten curve of your soil. The real weather sequence of the record is kept, and every control moves the traces.

19.7 Seasonal records: Daisy data (Y1, W1, N4)

These layers use the per-year output of 1 003 Daisy runs (experiments E00, E02, E03, E04, E05 and E07, 20 years each), extracted from the complete simulation output by `tools/make_peryear.py`. The run matching the present settings is selected automatically (mulch, method, trigger, sensor depth, tillage and consolidation rate).

- **Y1 Reliability over 20 seasons**: yield, irrigation, drainage, nitrogen leaching, water-stress days and water productivity of each year; a reliability table (median, 10 % and 90 % years) and exceedance curves.
- The Daisy run gives the year-to-year pattern; controls that its experiment does not vary (crop, rainfall scale, warming and nitrogen rate) are applied by the response of the annual model, so that the 20-year mean of every series equals the annual result at your settings.
- **W1 Full water balance**: the water balance of each year as stacked bars; the evapotranspiration split into transpiration, soil evaporation, canopy evaporation, litter evaporation, evaporation of ponded water and snow evaporation, which together close ET to within 1.1 mm; potential and actual evapotranspiration; the infiltration path (matrix and macropores)

and drainage.

- **N4 Nitrogen budget**: fertiliser, mineralisation, plant uptake, denitrification, volatilisation and leaching per year, with leaching against drainage.

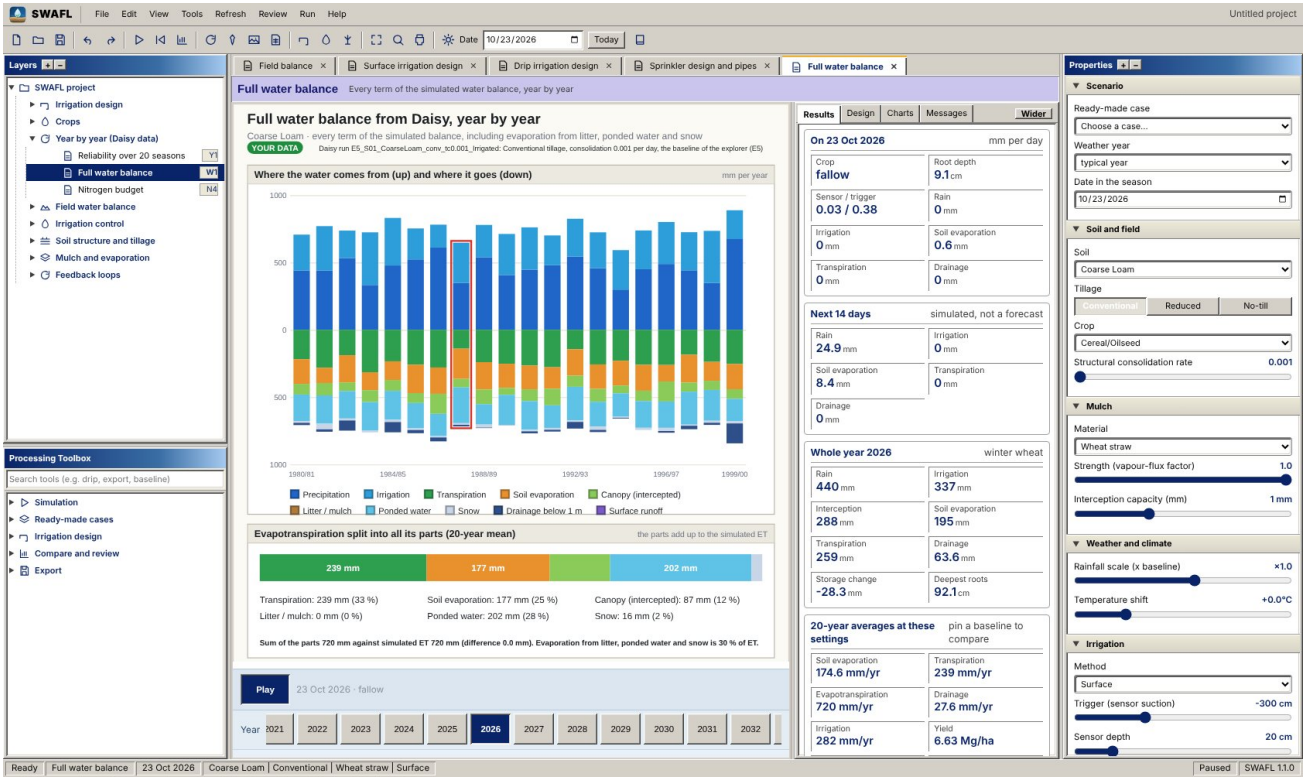


Figure 19.3: W1 Full water balance.

19.8 Consolidation rate

In your Daisy runs (E5) the consolidation rate of the tilled layer changes the results only in the five finer soils (loam to silty clay loam: up to 6 mm/yr of drainage and 14 kg N/ha of leaching); in the five coarser soils all rates give the same result, and the program shows the same.

19.9 Workbench functions

- **F1** opens this manual directly (Help → *Manual (PDF)*, or the book button on the toolbar).
- **Save full report as PDF** (File menu, *Report* toolbar button, `Ctrl+Shift+P`) builds the complete report of every layer end to end (inputs, results, key numbers and pictures of each layer) and asks where to save it. *Report of chosen layers* lets you pick the parts.
- **Speed**: every frame paints only the part of the sheet on the screen; pictures, prints and reports always paint the whole sheet.
- **Long layers** are fitted to their content: the drawing grows to show every panel and the centre window scrolls to the end of the layer.

19.10 Correction of the surface layer (release 1.1.0)

In release 1.0.0 the infiltration of the steady-intake term f_0 during advance was divided by $(1+r)$ with a fixed advance exponent $r = 0.55$ (Eq. 11.9). Release 1.1.0 uses the iterated advance

exponent of the volume-balance solution. With the correction the shipped layer reproduces the 13 advance times of the FAO-45 border examples within -0.2 to $+0.5$ % (before: -2.2 to $+2.9$ %) and the eight advance times of Example 6.1 within -0.2 to $+1.8$ % (before: up to -29 % when advance is slow). Results with the pure Kostiakov law ($f_0 = 0$) are unchanged.

Symbols and units

Table A.1. Symbols used in this manual, grouped by chapter of first use. Units are SI-derived as shown; $[-]$ is dimensionless.

Symbol	Meaning	Unit
θ, θ_r, θ_s	Volumetric water content; residual; saturated	$m^3\ m^{-3}$
θ_{fc}, θ_{wp}	Water content at field capacity; at wilting point	$m^3\ m^{-3}$
h, ψ	Soil-water suction (positive); matric potential	cm; kPa
α, n, m	van Genuchten parameters ($m = 1 - 1/n$)	cm^{-1}, [–], [–]
K_s, $K(h)$	Saturated and unsaturated hydraulic conductivity	$mm\ h^{-1}$
AWC	Plant-available water capacity	$mm\ cm^{-1}$
f, f_T	Relative depletion of the sensor layer; trigger depletion	[–]
$R_{\max}$, R_d	Maximum and present rooting depth	cm
P, I, E, T, D	Rain, irrigation, soil evaporation, transpiration, drainage	$mm\ d^{-1}$ or $mm\ a^{-1}$
E_0	Atmospheric evaporative demand	$mm\ d^{-1}$
$\mathrm{ET}_{\mathrm{pk}}$	Peak 5-day mean of $E + T$	$mm\ d^{-1}$
k_t	Calibrated transpiration coefficient of Engine 2	[–]
Z_n	Required (net) depth per irrigation	mm
$Z(\tau)$, $F(t)$	Cumulative infiltration (Kostiakov; Green–Ampt)	mm
k, A	Kostiakov intake constants	$mm\ min^{-A}$; [–]
ψ_f	Wetting-front suction	mm
L, S, w, n	Field length; slope; spacing or width; Manning roughness	m; %; m; $s\ m^{-1/3}$
q, Q	Unit and total inflow	$L\ s^{-1}\ m^{-1}$, $L\ s^{-1}$
t_x, t_L, t_{co}, τ	Advance time; time to end; cut-off time; opportunity time	min
σ_y, σ_z	Surface and subsurface shape factors	[–]
E_a, E_s	Application and storage efficiency	[–] or %
DP, RO, WD	Deep percolation, runoff, wind drift and interception fractions	[–]
CU, DU, E_u	Christiansen uniformity; distribution uniformity; emission uniformity	[–] or %
S_e, S_l, L_L, H_0, D	Emitter spacing; lateral spacing; lateral length; inlet head; inside diameter	m, m, m, m, mm
c_v, x	Emitter coefficient of variation; flow exponent	[–]
r, a, u	Bulb radius; aspect ratio; upper semi-axis ratio	mm, [–], [–]
d, P, h_s, R	Nozzle diameter; pressure; pressure head; wetted radius	mm, kPa, m, m

Symbol	Meaning	Unit
S_s, S_l	Sprinkler spacing along and between laterals	m
u	Wind speed	m s^{-1}
AR	Mean gross application rate	mm h^{-1}
C	Hazen–Williams roughness coefficient	[–]
J	Hydraulic gradient	m m^{-1}
TDH, H_{man}	Total dynamic head; manometric head	m
A_f, Q_s	Irrigated area; farm supply	ha; L s^{-1}

Control value lists

Table B.1. Index lists behind the Properties controls (defaults in bold). All other controls are described with their ranges in Chapter 3.

Control	Values
Trigger (sensor suction)	−50, −75, −100, −150, −200, −250, **−300**, −350, −400, −500, −600, −700, −800, −1000, −1200, −1500, −2000, −3000, −5000, −10000 cm
Sensor depth	5, 10, 15, …, 100 cm (20 values), default **20** cm
Nitrogen rate	10, 25, 40, 55, 70, 85, 100, 115, 130, 150
Mulch storage capacity (CAP)	0.5, 1, 2, 4 mm
Rainfall factor	0.5 to 1.4
Temperature offset	−1 to +3.5 °C
Mulch cover (VFF)	0.1 to 1.0 (1.0 = bare soil)
Thermal conductance (TC)	0.001, 0.0025, 0.005, 0.01, 0.02
Crop yield factors	cereal/oilseed 1.0, legume 0.55, vegetable 2.3

Table B.2. Mulch materials (thermal weight / storage capacity in mm): husk 0.5 / 2, straw 0.4 / 4, chips 0.3 / 2, sawdust 0.6 / 1, grass 0.7 / 4, plastic 0.1 / 0.5.

Simulation data sets

The annual meta-model (Engine 1) is built on the following Daisy 7.1 run sets of the study (Chapter 5, Table 5.1): E2, E3, E4, E5 (300 runs), E7, E11, E13 and E14. The study as a whole comprises 10,190 simulation runs in 17 experiments, available from the project repository (`Results/` and `Software/` folders). The 20-year weather record (1980–1999) is the baseline of the study; the projections to 2100 are analogue-year scenarios (Chapter 4).

Provenance ledger of the numbered equations

All 134 numbered equations are labelled in the text and listed here: 38 **[Literature]**, 18 **[Data-driven]** and 78 **[Engineered]**. A **[Literature]** label means the relation is the published one; it does not mean that SWAFL has been validated in the field. The surface, drip and sprinkler design relations marked in Paper 3 were tested against published worked examples; nothing has been tested against field measurements.

Eq.	Label	Basis
4.1	Engineered	calendar bookkeeping and choice of analogue year; no physical law
4.2	Engineered	calendar bookkeeping and choice of analogue year; no physical law
5.1	Literature	van Genuchten (1980)
5.2	Literature	Mualem (1976) with van Genuchten (1980)
5.3	Literature	standard definition of plant-available water between field capacity and wilting point
5.4	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.5	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.6	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.7	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.8	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.9	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.10	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.11	Data-driven	set-specific corrections from the Daisy run sets; clip bounds engineered
5.12	Engineered	closure constraint and bounds chosen for the workbench
5.13	Engineered	canopy curves, parameters chosen for the workbench; not a measured leaf-area record
5.14	Engineered	canopy curves, parameters chosen for the workbench; not a measured leaf-area record

Eq.	Label	Basis
5.15	Engineered	stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen
5.16	Engineered	stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen
5.17	Engineered	tipping-bucket redistribution with a chosen coefficient
5.18	Engineered	rooting rules with constants chosen for the workbench
5.19	Engineered	rooting rules with constants chosen for the workbench
5.20	Engineered	rooting rules with constants chosen for the workbench
5.21	Engineered	demand partitioning with chosen weights and a calibration factor
5.22	Engineered	demand partitioning with chosen weights and a calibration factor
5.23	Engineered	demand partitioning with chosen weights and a calibration factor
5.24	Data-driven	scaling of the daily series to the Engine 1 twenty-year means
5.25	Literature	water-balance identity
5.26	Engineered	atmosphere factor with chosen constants
5.27	Literature	readily available water as in FAO-56 practice, with an engineered clip
6.1	Data-driven	counts over the simulated daily series
6.2	Data-driven	counts over the simulated daily series
6.3	Literature	inverse of the van Genuchten curve, solved by bisection
6.4	Literature	same relation as Eq. (5.27) without the clip
6.5	Data-driven	ratio of Daisy run-set E2 yields
7.1	Literature	inverse of Eq. (5.3); the 30 cm floor is engineered
7.2	Engineered	series-resistance chain after van den Honert (1948); resistance values chosen for the workbench
7.3	Engineered	series-resistance chain after van den Honert (1948); resistance values chosen for the workbench
7.4	Engineered	series-resistance chain after van den Honert (1948); resistance values chosen for the workbench
7.5	Literature	Nobel (2009)
7.6	Engineered	indicator defined for the workbench
8.1	Engineered	exponential relaxation with chosen constants
8.2	Engineered	porosity relation with chosen coefficients
8.3	Literature	van Genuchten (1980)
8.4	Literature	Young–Laplace (Jurin) capillary law
8.5	Data-driven	differences of Daisy run-set E5 results
8.6	Data-driven	ordinary least squares applied to Daisy results
8.7	Engineered	diagnostic defined for the workbench
9.1	Data-driven	differences and ratios of Daisy run-set E14 and E11 results

Eq.	Label	Basis
9.2	Engineered	two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench
9.3	Engineered	two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench
9.4	Engineered	drying-front geometry chosen for drawing
9.5	Engineered	drying-front geometry chosen for drawing
9.6	Data-driven	differences and ratios of Daisy run-set E14 and E11 results
9.7	Data-driven	differences and ratios of Daisy run-set E14 and E11 results
9.8	Engineered	index defined for the workbench
10.1	Engineered	drawing rule
10.2	Data-driven	one-step differences of Daisy results
11.1	Literature	Saxton and Rawls (2006), implemented as written
11.2	Literature	Saxton and Rawls (2006), implemented as written
11.3	Engineered	rational-polynomial approximation of a Rawls-type suction
11.4	Engineered	initial and wetted water contents chosen for the workbench
11.5	Literature	Green and Ampt (1911)
11.6	Literature	Kostiakov (1932)
11.7	Literature	ordinary least squares on the log–log plot
11.8	Literature	Manning (1891)
11.9	Literature	volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3)
11.10	Literature	volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3)
11.11	Literature	volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3)
11.12	Engineered	alternative advance models set by worked examples and course notes
11.13	Engineered	alternative advance models set by worked examples and course notes
11.15	Literature	inversion of the Kostiakov law
11.16	Engineered	cut-off rules chosen as design options
11.17	Literature	standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997)
11.18	Literature	standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997)
11.19	Literature	standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997)
11.20	Engineered	hydrograph shapes chosen for drawing; volume conserving

Eq.	Label	Basis
11.21	Engineered	hydrograph shapes chosen for drawing; volume conserving
11.22	Engineered	search limit; the furrow stream limit follows Criddle et al. (1956)
11.23	Engineered	optimisation score and seasonal bookkeeping chosen for the workbench
11.24	Engineered	optimisation score and seasonal bookkeeping chosen for the workbench
12.1	Engineered	depth and demand rules chosen for the workbench
12.2	Engineered	depth and demand rules chosen for the workbench
12.3	Literature	Hazen and Williams (1905)
12.4	Literature	Keller and Karmeli (1974) emitter law
12.5	Engineered	pressure-compensating emitter model with chosen constants
12.6	Literature	definitions of mean and minimum discharge
12.7	Literature	Keller and Karmeli (1974)
12.8	Engineered	combined spread chosen to drive the bulb quantiles
12.9	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.10	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.11	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.12	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.13	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.14	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.15	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.16	Engineered	five-quantile delivery and run-time rules chosen for the workbench
12.17	Literature	definition of the requirement over the area served by one emitter
12.18	Engineered	five-quantile delivery and run-time rules chosen for the workbench
12.19	Engineered	five-quantile delivery and run-time rules chosen for the workbench
12.20	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.21	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.22	Engineered	Christiansen-type reduction with a chosen factor 0.4
12.23	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice

Eq.	Label	Basis
12.24	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.25	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.26	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
13.1	Engineered	depression storage reduction with slope
13.2	Literature	Torricelli orifice equation
13.3	Engineered	range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen
13.4	Engineered	pattern and wind shapes chosen to reproduce published qualitative patterns
13.5	Engineered	pattern and wind shapes chosen to reproduce published qualitative patterns
13.6	Literature	overlap by superposition of single-sprinkler patterns
13.7	Literature	Christiansen (1942)
13.8	Engineered	loss and ponding rules chosen for the workbench
13.9	Engineered	loss and ponding rules chosen for the workbench
13.10	Engineered	loss and ponding rules chosen for the workbench
13.11	Literature	efficiency and distribution uniformity of Merriam and Keller (1978)
13.12	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.13	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.14	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.15	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.16	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.17	Literature	Blasius friction law for smooth pipe
13.18	Literature	20 % rule of Keller and Bliesner (1990)
13.19	Engineered	head and sets bookkeeping
13.20	Literature	hydraulic power definition
13.21	Engineered	head and sets bookkeeping
13.22	Literature	Christiansen reduction factor with Hazen–Williams exponent
13.23	Engineered	head and sets bookkeeping

Eq.	Label	Basis
14.1	Engineered	bookkeeping of Engine 2 series
14.2	Engineered	bookkeeping of Engine 2 series
14.3	Literature	water-balance identity
14.4	Engineered	bookkeeping of Engine 2 series
14.5	Engineered	bookkeeping of Engine 2 series

References

Abrahamsen, P., and Hansen, S. (2000). Daisy: an open soil-crop-atmosphere system model. *Environmental Modelling & Software* 15(3), 313–330.

Allen, R. G., Pereira, L. S., Raes, D., and Smith, M. (1998). *Crop evapotranspiration: guidelines for computing crop water requirements*. FAO Irrigation and Drainage Paper 56. FAO, Rome.

Blasius, H. (1913). Das Ähnlichkeitsgesetz bei Reibungsvorgängen in Flüssigkeiten. *Forschungsarbeiten auf dem Gebiete des Ingenieurwesens* 131.

Burt, C. M., Clemmens, A. J., Strelkoff, T. S., Solomon, K. H., Bliesner, R. D., Hardy, L. A., Howell, T. A., and Eisenhauer, D. E. (1997). Irrigation performance measures: efficiency and uniformity. *Journal of Irrigation and Drainage Engineering* 123(6), 423–442.

Carsel, R. F., and Parrish, R. S. (1988). Developing joint probability distributions of soil water retention characteristics. *Water Resources Research* 24(5), 755–769.

Christiansen, J. E. (1942). *Irrigation by sprinkling*. University of California Agricultural Experiment Station Bulletin 670.

Criddle, W. D., Davis, S., Pair, C. H., and Shockley, D. G. (1956). *Methods for evaluating irrigation systems*. USDA Soil Conservation Service, Agriculture Handbook 82.

Elliott, R. L., and Walker, W. R. (1982). Field evaluation of furrow infiltration and advance functions. *Transactions of the ASAE* 25(2), 396–400.

Green, W. H., and Ampt, G. A. (1911). Studies on soil physics: 1. The flow of air and water through soils. *Journal of Agricultural Science* 4(1), 1–24.

Hazen, A., and Williams, G. S. (1905). *Hydraulic tables*. John Wiley & Sons, New York.

Hillel, D. (1998). *Environmental soil physics*. Academic Press, San Diego.

Jurin, J. (1718). An account of some experiments shown before the Royal Society; with an enquiry into the cause of the ascent and suspension of water in capillary tubes. *Philosophical Transactions* 30(355), 739–747.

Keller, J., and Bliesner, R. D. (1990). *Sprinkle and trickle irrigation*. Van Nostrand Reinhold, New York.

Keller, J., and Karmeli, D. (1974). Trickle irrigation design parameters. *Transactions of the ASAE* 17(4), 678–684.

Kostiakov, A. N. (1932). On the dynamics of the coefficient of water-percolation in soils and on the necessity of studying it from a dynamic point of view for purposes of amelioration. *Transactions of the 6th Committee of the International Society of Soil Science*, Part A, 17–21.

Manning, R. (1891). On the flow of water in open channels and pipes. *Transactions of the Institution of Civil Engineers of Ireland* 20, 161–207.

Merriam, J. L., and Keller, J. (1978). *Farm irrigation system evaluation: a guide for management*. Utah State University, Logan.

Mualem, Y. (1976). A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resources Research* 12(3), 513–522.

Nobel, P. S. (2009). *Physicochemical and environmental plant physiology*, 4th ed. Academic Press, Amsterdam.

Philip, J. R. (1957). Evaporation, and moisture and heat fields in the soil. *Journal of Meteorology* 14(4), 354–366.

Ritchie, J. T. (1972). Model for predicting evaporation from a row crop with incomplete cover. *Water Resources Research* 8(5), 1204–1213.

Saxton, K. E., and Rawls, W. J. (2006). Soil water characteristic estimates by texture and organic matter for hydrologic solutions. *Soil Science Society of America Journal* 70(5), 1569–1578.

van den Honert, T. H. (1948). Water transport in plants as a catenary process. *Discussions of the Faraday Society* 3, 146–153.

van Genuchten, M. Th. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal* 44(5), 892–898.

Walker, W. R., and Skogerboe, G. V. (1987). *Surface irrigation: theory and practice*. Prentice-Hall, Englewood Cliffs, NJ.